

Elements, their Atomic Number and Molar Mass

Element	Symbol	Atomic Number	Molar mass/ (g mol ⁻¹)
Actinium	Ac	89	227.03
Aluminium	Al	13	26.98
Americium	Am	95	(243)
Antimony	Sb	51	121.75
Argon	Ar	18	39.95
Arsenic	As	33	74.92
Astatine	At	85	210
Barium	Ba	56	137.34
Berkelium	Bk	97	(247)
Beryllium	Be	4	9.01
Bismuth	Bi	83	208.98
Bohrium	Bh	107	(264)
Boron	B	5	10.81
Bromine	Br	35	79.91
Cadmium	Cd	48	112.40
Caesium	Cs	55	132.91
Calcium	Ca	20	40.08
Californium	Cf	98	251.08
Carbon	C	6	12.01
Cerium	Ce	58	140.12
Chlorine	Cl	17	35.45
Chromium	Cr	24	52.00
Cobalt	Co	27	58.93
Copper	Cu	29	63.54
Curium	Cm	96	247.07
Dubnium	Db	105	(263)
Dysprosium	Dy	66	162.50
Einsteinium	Es	99	(252)
Erbium	Er	68	167.26
Europium	Eu	63	151.96
Fermium	Fm	100	(257.10)
Fluorine	F	9	19.00
Francium	Fr	87	(223)
Gadolinium	Gd	64	157.25
Gallium	Ga	31	69.72
Germanium	Ge	32	72.61
Gold	Au	79	196.97
Hafnium	Hf	72	178.49
Hassium	Hs	108	(269)
Helium	He	2	4.00
Holmium	Ho	67	164.93
Hydrogen	H	1	1.0079
Indium	In	49	114.82
Iodine	I	53	126.90
Iridium	Ir	77	192.2
Iron	Fe	26	55.85
Krypton	Kr	36	83.80
Lanthanum	La	57	138.91
Lawrencium	Lr	103	(262.1)
Lead	Pb	82	207.19
Lithium	Li	3	6.94
Lutetium	Lu	71	174.96
Magnesium	Mg	12	24.31
Manganese	Mn	25	54.94
Meitneium	Mt	109	(268)
Mendelevium	Md	101	258.10

Element	Symbol	Atomic Number	Molar mass/ (g mol ⁻¹)
Mercury	Hg	80	200.59
Molybdenum	Mo	42	95.94
Neodymium	Nd	60	144.24
Neon	Ne	10	20.18
Neptunium	Np	93	(237.05)
Nickel	Ni	28	58.71
Niobium	Nb	41	92.91
Nitrogen	N	7	14.0067
Nobelium	No	102	(259)
Osmium	Os	76	190.2
Oxygen	O	8	16.00
Palladium	Pd	46	106.4
Phosphorus	P	15	30.97
Platinum	Pt	78	195.09
Plutonium	Pu	94	(244)
Polonium	Po	84	210
Potassium	K	19	39.10
Praseodymium	Pr	59	140.91
Promethium	Pm	61	(145)
Protactinium	Pa	91	231.04
Radium	Ra	88	(226)
Radon	Rn	86	(222)
Rhenium	Re	75	186.2
Rhodium	Rh	45	102.91
Rubidium	Rb	37	85.47
Ruthenium	Ru	44	101.07
Rutherfordium	Rf	104	(261)
Samarium	Sm	62	150.35
Scandium	Sc	21	44.96
Seaborgium	Sg	106	(266)
Selenium	Se	34	78.96
Silicon	Si	14	28.08
Silver	Ag	47	107.87
Sodium	Na	11	22.99
Strontium	Sr	38	87.62
Sulphur	S	16	32.06
Tantalum	Ta	73	180.95
Technetium	Tc	43	(98.91)
Tellurium	Te	52	127.60
Terbium	Tb	65	158.92
Thallium	Tl	81	204.37
Thorium	Th	90	232.04
Thulium	Tm	69	168.93
Tin	Sn	50	118.69
Titanium	Ti	22	47.88
Tungsten	W	74	183.85
Ununbium	Uub	112	(277)
Ununnilium	Uun	110	(269)
Unununium	Uuu	111	(272)
Uranium	U	92	238.03
Vanadium	V	23	50.94
Xenon	Xe	54	131.30
Ytterbium	Yb	70	173.04
Yttrium	Y	39	88.91
Zinc	Zn	30	65.37
Zirconium	Zr	40	91.22

The value given in parenthesis is the molar mass of the isotope of largest known half-life.

Some Useful Conversion Factors

Common Unit of Mass and Weight**1 pound = 453.59 grams**

1 pound = 453.59 grams = 0.45359 kilogram
 1 kilogram = 1000 grams = 2.205 pounds
 1 gram = 10 decigrams = 100 centigrams
 = 1000 milligrams
 1 gram = 6.022×10^{23} atomic mass units or u
 1 atomic mass unit = 1.6606×10^{-24} gram
 1 metric tonne = 1000 kilograms
 = 2205 pounds

Common Unit of Volume**1 quart = 0.9463 litre****1 litre = 1.056 quarts**

1 litre = 1 cubic decimetre = 1000 cubic centimetres = 0.001 cubic metre
 1 millilitre = 1 cubic centimetre = 0.001 litre
 = 1.056×10^{-3} quart
 1 cubic foot = 28.316 litres = 29.902 quarts
 = 7.475 gallons

Common Units of Energy**1 joule = 1×10^7 ergs**

1 thermochemical calorie**
 = 4.184 joules
 = 4.184×10^7 ergs
 = 4.129×10^{-2} litre-atmospheres
 = 2.612×10^{19} electron volts
 1 ergs = 1×10^{-7} joule = 2.3901×10^{-8} calorie
 1 electron volt = 1.6022×10^{-19} joule
 = 1.6022×10^{-12} erg
 = 96.487 kJ/mol†
 1 litre-atmosphere = 24.217 calories
 = 101.32 joules
 = 1.0132×10^9 ergs
 1 British thermal unit = 1055.06 joules
 = 1.05506×10^{10} ergs
 = 252.2 calories

Common Units of Length**1 inch = 2.54 centimetres (exactly)**

1 mile = 5280 feet = 1.609 kilometres
 1 yard = 36 inches = 0.9144 metre
 1 metre = 100 centimetres = 39.37 inches
 = 3.281 feet
 = 1.094 yards
 1 kilometre = 1000 metres = 1094 yards
 = 0.6215 mile
 1 Angstrom = 1.0×10^{-8} centimetre
 = 0.10 nanometre
 = 1.0×10^{-10} metre
 = 3.937×10^{-9} inch

Common Units of Force* and Pressure

1 atmosphere = 760 millimetres of mercury
 = 1.013×10^5 pascals
 = 14.70 pounds per square inch
 1 bar = 10^5 pascals
 1 torr = 1 millimetre of mercury
 1 pascal = 1 kg/ms² = 1 N/m²

Temperature**SI Base Unit: Kelvin (K)**

K = -273.15°C
 K = °C + 273.15
 °F = 1.8(°C) + 32

$$^{\circ}\text{C} = \frac{^{\circ}\text{F} - 32}{1.8}$$

* Force: 1 newton (N) = 1 kg m/s², i.e., the force that, when applied for 1 second, gives a 1-kilogram mass a velocity of 1 metre per second.

** The amount of heat required to raise the temperature of one gram of water from 14.5°C to 15.5°C.

† Note that the other units are per particle and must be multiplied by 6.022×10^{23} to be strictly comparable.

Standard potentials at 298 K in electrochemical order

Reduction half-reaction	E°/V	Reduction half-reaction	E°/V
$H_4XeO_6 + 2H^+ + 2e^- \longrightarrow XeO_3 + 3H_2O$	+3.0	$Cu^+ + e^- \longrightarrow Cu$	+0.52
$F_2 + 2e^- \longrightarrow 2F^-$	+2.87	$NiOOH + H_2O + e^- \longrightarrow Ni(OH)_2 + OH^-$	+0.49
$O_3 + 2H^+ + 2e^- \longrightarrow O_2 + H_2O$	+2.07	$Ag_2CrO_4 + 2e^- \longrightarrow 2Ag + CrO_4^{2-}$	+0.45
$S_2O_8^{2-} + 2e^- \longrightarrow 2SO_4^{2-}$	+2.05	$O_2 + 2H_2O + 4e^- \longrightarrow 4OH^-$	+0.40
$Ag^+ + e^- \longrightarrow Ag^+$	+1.98	$ClO_3^- + H_2O + 2e^- \longrightarrow ClO_2^- + 2OH^-$	+0.36
$Co^{3+} + e^- \longrightarrow Co^{2+}$	+1.81	$[Fe(CN)_6]^{3-} + e^- \longrightarrow [Fe(CN)_6]^{4-}$	+0.36
$H_2O_2 + 2H^+ + 2e^- \longrightarrow 2H_2O$	+1.78	$Cu^{2+} + 2e^- \longrightarrow Cu$	+0.34
$Au^+ + e^- \longrightarrow Au$	+1.69	$Hg_2Cl_2 + 2e^- \longrightarrow 2Hg + 2Cl^-$	+0.27
$Pb^{4+} + 2e^- \longrightarrow Pb^{2+}$	+1.67	$AgCl + e^- \longrightarrow Ag + Cl^-$	+0.27
$2HClO + 2H^+ + 2e^- \longrightarrow Cl_2 + 2H_2O$	+1.63	$Bi^{3+} + 3e^- \longrightarrow Bi$	+0.20
$Ce^{4+} + e^- \longrightarrow Ce^{3+}$	+1.61	$SO_4^{2-} + 4H^+ + 2e^- \longrightarrow H_2SO_3 + H_2O$	+0.17
$2HBrO + 2H^+ + 2e^- \longrightarrow Br_2 + 2H_2O$	+1.60	$Cu^{2+} + e^- \longrightarrow Cu^+$	+0.16
$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$	+1.51	$Sn^{4+} + 2e^- \longrightarrow Sn^{2+}$	+0.15
$Mn^{3+} + e^- \longrightarrow Mn^{2+}$	+1.51	$AgBr + e^- \longrightarrow Ag + Br^-$	+0.07
$Au^{3+} + 3e^- \longrightarrow Au$	+1.40	$Ti^{4+} + e^- \longrightarrow Ti^{3+}$	0.00
$Cl_2 + 2e^- \longrightarrow 2Cl^-$	+1.36	$2H^+ + 2e^- \longrightarrow H_2$	0.0 by definition
$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$	+1.33	$Fe^{3+} + 3e^- \longrightarrow Fe$	-0.04
$O_3 + H_2O + 2e^- \longrightarrow O_2 + 2OH^-$	+1.24	$O_2 + H_2O + 2e^- \longrightarrow HO_2^- + OH^-$	-0.08
$O_2 + 4H^+ + 4e^- \longrightarrow 2H_2O$	+1.23	$Pb^{2+} + 2e^- \longrightarrow Pb$	-0.13
$ClO_4^- + 2H^+ + 2e^- \longrightarrow ClO_3^- + 2H_2O$	+1.23	$In^+ + e^- \longrightarrow In$	-0.14
$MnO_2 + 4H^+ + 2e^- \longrightarrow Mn^{2+} + 2H_2O$	+1.23	$Sn^{2+} + 2e^- \longrightarrow Sn$	-0.14
$Pt^{2+} + 2e^- \longrightarrow Pt$	+1.20	$AgI + e^- \longrightarrow Ag + I^-$	-0.15
$Br_2 + 2e^- \longrightarrow 2Br^-$	+1.09	$Ni^{2+} + 2e^- \longrightarrow Ni$	-0.23
$Pu^{4+} + e^- \longrightarrow Pu^{3+}$	+0.97	$V^{3+} + e^- \longrightarrow V^{2+}$	-0.26
$NO_3^- + 4H^+ + 3e^- \longrightarrow NO + 2H_2O$	+0.96	$Co^{2+} + 2e^- \longrightarrow Co$	-0.28
$2Hg^{2+} + 2e^- \longrightarrow Hg_2^{2+}$	+0.92	$In^{3+} + 3e^- \longrightarrow In$	-0.34
$ClO^- + H_2O + 2e^- \longrightarrow Cl^- + 2OH^-$	+0.89	$Tl^+ + e^- \longrightarrow Tl$	-0.34
$Hg^{2+} + 2e^- \longrightarrow Hg$	+0.86	$PbSO_4 + 2e^- \longrightarrow Pb + SO_4^{2-}$	-0.36
$NO_3^- + 2H^+ + e^- \longrightarrow NO_2 + H_2O$	+0.80	$Ti^{3+} + e^- \longrightarrow Ti^{2+}$	-0.37
$Ag^+ + e^- \longrightarrow Ag$	+0.80	$Cd^{2+} + 2e^- \longrightarrow Cd$	-0.40
$Hg_2^{2+} + 2e^- \longrightarrow 2Hg$	+0.79	$In^{2+} + e^- \longrightarrow In^+$	-0.40
$Fe^{3+} + e^- \longrightarrow Fe^{2+}$	+0.77	$Cr^{3+} + e^- \longrightarrow Cr^{2+}$	-0.41
$BrO^- + H_2O + 2e^- \longrightarrow Br^- + 2OH^-$	+0.76	$Fe^{2+} + 2e^- \longrightarrow Fe$	-0.44
$Hg_2SO_4 + 2e^- \longrightarrow 2Hg + SO_4^{2-}$	+0.62	$In^{3+} + 2e^- \longrightarrow In^+$	-0.44
$MnO_4^{2-} + 2H_2O + 2e^- \longrightarrow MnO_2 + 4OH^-$	+0.60	$S + 2e^- \longrightarrow S^{2-}$	-0.48
$MnO_4^- + e^- \longrightarrow MnO_4^{2-}$	+0.56	$In^{3+} + e^- \longrightarrow In^{2+}$	-0.49
$I_2 + 2e^- \longrightarrow 2I^-$	+0.54	$U^{4+} + e^- \longrightarrow U^{3+}$	-0.61
$I_3^- + 2e^- \longrightarrow 3I^-$	+0.53	$Cr^{3+} + 3e^- \longrightarrow Cr$	-0.74
		$Zn^{2+} + 2e^- \longrightarrow Zn$	-0.76

(continued)

APPENDIX III CONTINUED

Reduction half-reaction	E°/V	Reduction half-reaction	E°/V
$\text{Cd}(\text{OH})_2 + 2\text{e}^- \longrightarrow \text{Cd} + 2\text{OH}^-$	-0.81	$\text{La}^{3+} + 3\text{e}^- \longrightarrow \text{La}$	-2.52
$2\text{H}_2\text{O} + 2\text{e}^- \longrightarrow \text{H}_2 + 2\text{OH}^-$	-0.83	$\text{Na}^+ + \text{e}^- \longrightarrow \text{Na}$	-2.71
$\text{Cr}^{2+} + 2\text{e}^- \longrightarrow \text{Cr}$	-0.91	$\text{Ca}^{2+} + 2\text{e}^- \longrightarrow \text{Ca}$	-2.87
$\text{Mn}^{2+} + 2\text{e}^- \longrightarrow \text{Mn}$	-1.18	$\text{Sr}^{2+} + 2\text{e}^- \longrightarrow \text{Sr}$	-2.89
$\text{V}^{2+} + 2\text{e}^- \longrightarrow \text{V}$	-1.19	$\text{Ba}^{2+} + 2\text{e}^- \longrightarrow \text{Ba}$	-2.91
$\text{Ti}^{2+} + 2\text{e}^- \longrightarrow \text{Ti}$	-1.63	$\text{Ra}^{2+} + 2\text{e}^- \longrightarrow \text{Ra}$	-2.92
$\text{Al}^{3+} + 3\text{e}^- \longrightarrow \text{Al}$	-1.66	$\text{Cs}^+ + \text{e}^- \longrightarrow \text{Cs}$	-2.92
$\text{U}^{3+} + 3\text{e}^- \longrightarrow \text{U}$	-1.79	$\text{Rb}^+ + \text{e}^- \longrightarrow \text{Rb}$	-2.93
$\text{Sc}^{3+} + 3\text{e}^- \longrightarrow \text{Sc}$	-2.09	$\text{K}^+ + \text{e}^- \longrightarrow \text{K}$	-2.93
$\text{Mg}^{2+} + 2\text{e}^- \longrightarrow \text{Mg}$	-2.36	$\text{Li}^+ + \text{e}^- \longrightarrow \text{Li}$	-3.05
$\text{Ce}^{3+} + 3\text{e}^- \longrightarrow \text{Ce}$	-2.48		

Logarithms

Sometimes, a numerical expression may involve multiplication, division or rational powers of large numbers. For such calculations, logarithms are very useful. They help us in making difficult calculations easy. In Chemistry, logarithm values are required in solving problems of chemical kinetics, thermodynamics, electrochemistry, etc. We shall first introduce this concept, and discuss the laws, which will have to be followed in working with logarithms, and then apply this technique to a number of problems to show how it makes difficult calculations simple.

We know that

$$2^3 = 8, 3^2 = 9, 5^3 = 125, 7^0 = 1$$

In general, for a positive real number a , and a rational number m , let $a^m = b$,

where b is a real number. In other words

the m^{th} power of base a is b .

Another way of stating the same fact is

logarithm of b to base a is m .

If for a positive real number a , $a \neq 1$

$$a^m = b,$$

we say that m is the logarithm of b to the base a .

We write this as $\log_a b = m$,

“log” being the abbreviation of the word “logarithm”.

Thus, we have

$$\log_2 8 = 3, \quad \text{Since } 2^3 = 8$$

$$\log_3 9 = 2, \quad \text{Since } 3^2 = 9$$

$$\log_5 125 = 3, \quad \text{Since } 5^3 = 125$$

$$\log_7 1 = 0, \quad \text{Since } 7^0 = 1$$

Laws of Logarithms

In the following discussion, we shall take logarithms to any base a , ($a > 0$ and $a \neq 1$)

First Law: $\log_a (mn) = \log_a m + \log_a n$

Proof: Suppose that $\log_a m = x$ and $\log_a n = y$

Then $a^x = m$, $a^y = n$

$$\text{Hence } mn = a^x \cdot a^y = a^{x+y}$$

It now follows from the definition of logarithms that

$$\log_a (mn) = x + y = \log_a m + \log_a n$$

Second Law: $\log_a \left(\frac{m}{n} \right) = \log_a m - \log_a n$

Proof: Let $\log_a m = x$, $\log_a n = y$

Then $a^x = m$, $a^y = n$

Hence $\frac{m}{n} = \frac{a^x}{a^y} = a^{x-y}$

Therefore

$$\log_a \left(\frac{m}{n} \right) = x - y = \log_a m - \log_a n$$

Third Law : $\log_a(m^n) = n \log_a m$

Proof : As before, if $\log_a m = x$, then $a^x = m$

Then $m^n = (a^x)^n = a^{nx}$

giving $\log_a(m^n) = nx = n \log_a m$

Thus according to First Law: “the log of the product of two numbers is equal to the sum of their logs. Similarly, the Second Law says: the log of the ratio of two numbers is the difference of their logs. Thus, the use of these laws converts a problem of multiplication/division into a problem of addition/subtraction, which are far easier to perform than multiplication/division. That is why logarithms are so useful in all numerical computations.

Logarithms to Base 10

Because number 10 is the base of writing numbers, it is very convenient to use logarithms to the base 10. Some examples are:

$\log_{10} 10 = 1$,	since $10^1 = 10$
$\log_{10} 100 = 2$,	since $10^2 = 100$
$\log_{10} 10000 = 4$,	since $10^4 = 10000$
$\log_{10} 0.01 = -2$,	since $10^{-2} = 0.01$
$\log_{10} 0.001 = -3$,	since $10^{-3} = 0.001$
and $\log_{10} 1 = 0$	since $10^0 = 1$

The above results indicate that if n is an integral power of 10, i.e., 1 followed by several zeros or 1 preceded by several zeros immediately to the right of the decimal point, then $\log n$ can be easily found.

If n is not an integral power of 10, then it is not easy to calculate $\log n$. But mathematicians have made tables from which we can read off approximate value of the logarithm of any positive number between 1 and 10. And these are sufficient for us to calculate the logarithm of any number expressed in decimal form. For this purpose, we always express the given decimal as the product of an integral power of 10 and a number between 1 and 10.

Standard Form of Decimal

We can express any number in decimal form, as the product of (i) an integral power of 10, and (ii) a number between 1 and 10. Here are some examples:

(i) 25.2 lies between 10 and 100

$$25.2 = \frac{25.2}{10} \times 10 = 2.52 \times 10^1$$

(ii) 1038.4 lies between 1000 and 10000.

$$\therefore 1038.4 = \frac{1038.4}{1000} \times 10^3 = 1.0384 \times 10^3$$

(iii) 0.005 lies between 0.001 and 0.01

$$\therefore 0.005 = (0.005 \times 1000) \times 10^{-3} = 5.0 \times 10^{-3}$$

(iv) 0.00025 lies between 0.0001 and 0.001

$$\therefore 0.00025 = (0.00025 \times 10000) \times 10^{-4} = 2.5 \times 10^{-4}$$

In each case, we divide or multiply the decimal by a power of 10, to bring one non-zero digit to the left of the decimal point, and do the reverse operation by the same power of 10, indicated separately.

Thus, any positive decimal can be written in the form

$$n = m \times 10^p$$

where p is an integer (positive, zero or negative) and $1 \leq m < 10$. This is called the “standard form of n .”

Working Rule

1. Move the decimal point to the left, or to the right, as may be necessary, to bring one non-zero digit to the left of decimal point.
2. (i) If you move p places to the left, multiply by 10^p .
(ii) If you move p places to the right, multiply by 10^{-p} .
(iii) If you do not move the decimal point at all, multiply by 10^0 .
(iv) Write the new decimal obtained by the power of 10 (of step 2) to obtain the standard form of the given decimal.

Characteristic and Mantissa

Consider the standard form of n

$$n = m \times 10^p, \text{ where } 1 \leq m < 10$$

Taking logarithms to the base 10 and using the laws of logarithms

$$\begin{aligned} \log n &= \log m + \log 10^p \\ &= \log m + p \log 10 \\ &= p + \log m \end{aligned}$$

Here p is an integer and as $1 \leq m < 10$, so $0 \leq \log m < 1$, i.e., m lies between 0 and 1. When $\log n$ has been expressed as $p + \log m$, where p is an integer and $0 \leq \log m < 1$, we say that p is the “characteristic” of $\log n$ and that $\log m$ is the “mantissa of $\log n$.” Note that characteristic is always an integer – positive, negative or zero, and mantissa is never negative and is always less than 1. If we can find the characteristics and the mantissa of $\log n$, we have to just add them to get $\log n$.

Thus to find $\log n$, all we have to do is as follows:

1. Put n in the standard form, say
 $n = m \times 10^p, 1 \leq m < 10$
2. Read off the characteristic p of $\log n$ from this expression (exponent of 10).
3. Look up $\log m$ from tables, which is being explained below.
4. Write $\log n = p + \log m$

If the characteristic p of a number n is say, 2 and the mantissa is .4133, then we have $\log n = 2 + .4133$ which we can write as 2.4133. If, however, the characteristic p of a number m is say -2 and the mantissa is .4123, then we have $\log m = -2 + .4123$. We cannot write this as -2.4123. (Why?) In order to avoid this confusion we write $\bar{2}$ for -2 and thus we write $\log m = \bar{2}.4123$.

Now let us explain how to use the table of logarithms to find mantissas. A table is appended at the end of this Appendix.

Observe that in the table, every row starts with a two digit number, 10, 11, 12,... 97, 98, 99. Every column is headed by a one-digit number, 0, 1, 2, ...9. On the right, we have the section called “Mean differences” which has 9 columns headed by 1, 2...9.

	0	1	2	3	4	5	6	7	8	9		1	2	3	4	5	6	7	8	9
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
61	7853	7860	7868	7875	7882	7889	7896	7803	7810	7817		1	1	2	3	4	4	5	6	6
62	7924	7931	7935	7945	7954	7959	7966	7973	7980	7987		1	1	2	3	3	4	5	6	6
63	7993	8000	8007	8014	8021	8028	8035	8041	8048	8055		1	1	2	3	3	4	5	6	6
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..

Now suppose we wish to find $\log (6.234)$. Then look into the row starting with 62. In this row, look at the number in the column headed by 3. The number is 7945. This means that

$$\log (6.230) = 0.7945^*$$

But we want $\log (6.234)$. So our answer will be a little more than 0.7945. How much more? We look this up in the section on Mean differences. Since our fourth digit is 4, look under the column headed by 4 in the Mean difference section (in the row 62). We see the number 3 there. So add 3 to 7945. We get 7948. So we finally have

$$\log (6.234) = 0.7948.$$

Take another example. To find $\log (8.127)$, we look in the row 81 under column 2, and we find 9096. We continue in the same row and see that the mean difference under 7 is 4. Adding this to 9096, and we get 9100. So, $\log (8.127) = 0.9100$.

Finding N when $\log N$ is given

We have so far discussed the procedure for finding $\log n$ when a positive number n is given. We now turn to its converse i.e., to find n when $\log n$ is given and give a method for this purpose. If $\log n = t$, we sometimes say $n = \text{antilog } t$. Therefore our task is given t , find its antilog. For this, we use the ready-made antilog tables.

Suppose $\log n = 2.5372$.

To find n , first take just the mantissa of $\log n$. In this case it is .5372. (Make sure it is positive.) Now take up antilog of this number in the antilog table which is to be used exactly like the log table. In the antilog table, the entry under column 7 in the row .53 is 3443 and the mean difference for the last digit 2 in that row is 2, so the table gives 3445. Hence,

$$\text{antilog} (.5372) = 3.445$$

Now since $\log n = 2.5372$, the characteristic of $\log n$ is 2. So the standard form of n is given by

$$n = 3.445 \times 10^2$$

$$\text{or } n = 344.5$$

Illustration 1:

If $\log x = 1.0712$, find x .

Solution: We find that the number corresponding to 0712 is 1179. Since characteristic of $\log x$ is 1, we have

$$\begin{aligned} x &= 1.179 \times 10^1 \\ &= 11.79 \end{aligned}$$

Illustration 2:

If $\log_{10} x = \bar{2}.1352$, find x .

Solution: From antilog tables, we find that the number corresponding to 1352 is 1366. Since the characteristic is $\bar{2}$ i.e., -2, so

$$x = 1.366 \times 10^{-2} = 0.01366$$

Use of Logarithms in Numerical Calculations

Illustration 1:

Find 6.3×1.29

Solution: Let $x = 6.3 \times 1.29$

$$\text{Then } \log_{10} x = \log (6.3 \times 1.29) = \log 6.3 + \log 1.29$$

Now,

$$\log 6.3 = 0.7993$$

$$\log 1.29 = 0.1106$$

$$\therefore \log_{10} x = 0.9099,$$

* It should, however, be noted that the values given in the table are not exact. They are only approximate values, although we use the sign of equality which may give the impression that they are exact values. The same convention will be followed in respect of antilogarithm of a number.

Taking antilog

$$x = 8.127$$

Illustration 2:

Find $\frac{(1.23)^{1.5}}{11.2 \times 23.5}$

Solution: Let $x = \frac{(1.23)^{\frac{3}{2}}}{11.2 \times 23.5}$

Then $\log x = \log \frac{(1.23)^{\frac{3}{2}}}{11.2 \times 23.5}$

$$= \frac{3}{2} \log 1.23 - \log (11.2 \times 23.5)$$

$$= \frac{3}{2} \log 1.23 - \log 11.2 - \log 23.5$$

Now,

$$\log 1.23 = 0.0899$$

$$\frac{3}{2} \log 1.23 = 0.13485$$

$$\log 11.2 = 1.0492$$

$$\log 23.5 = 1.3711$$

$$\log x = 0.13485 - 1.0492 - 1.3711$$

$$= \bar{3}.71455$$

$$\therefore x = 0.005183$$

Illustration 3:

Find $\sqrt{\frac{(71.24)^5 \times \sqrt{56}}{(2.3)^7 \times \sqrt{21}}}$

Solution: Let $x = \sqrt{\frac{(71.24)^5 \times \sqrt{56}}{(2.3)^7 \times \sqrt{21}}}$

$$\begin{aligned} \text{Then } \log x &= \frac{1}{2} \log \left[\frac{(71.24)^5 \times \sqrt{56}}{(2.3)^7 \times \sqrt{21}} \right] \\ &= \frac{1}{2} [\log (71.24)^5 + \log \sqrt{56} - \log (2.3)^7 - \log \sqrt{21}] \\ &= \frac{5}{2} \log 71.24 + \frac{1}{4} \log 56 - \frac{7}{2} \log 2.3 - \frac{1}{4} \log 21 \end{aligned}$$

Now, using log tables

$$\log 71.24 = 1.8527$$

$$\log 56 = 1.7482$$

$$\log 2.3 = 0.3617$$

$$\log 21 = 1.3222$$

$$\begin{aligned} \therefore \log x &= \frac{5}{2} \log (1.8527) + \frac{1}{4} (1.7482) - \frac{7}{2} (0.3617) - \frac{1}{4} (1.3222) \\ &= 3.4723 \end{aligned}$$

$$\therefore x = 2967$$

LOGARITHMS

TABLE I

N	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
10	0000	0043	0086	0128	0170	0212	0253	0294	0334	0374	5	9	13	17	21	26	30	34	38
											4	8	12	16	20	24	28	32	36
11	0414	0453	0492	0531	0569	0607	0645	0682	0719	0755	4	8	12	16	20	23	27	31	35
											4	7	11	15	18	22	26	29	33
12	0792	0828	0864	0899	0934	0969	1004	1038	1072	1106	3	7	11	14	18	21	25	28	32
											3	7	10	14	17	20	24	27	31
13	1139	1173	1206	1239	1271	1303	1335	1367	1399	1430	3	6	10	13	16	19	23	26	29
											3	7	10	13	16	19	22	25	29
14	1461	1492	1523	1553	1584	1614	1644	1673	1703	1732	3	6	9	12	15	19	22	25	28
											3	6	9	12	14	17	20	23	26
15	1761	1790	1818	1847	1875	1903	1931	1959	1987	2014	3	6	9	11	14	17	20	23	26
											3	6	8	11	14	17	19	22	25
16	2041	2068	2095	2122	2148	2175	2201	2227	2253	2279	3	6	8	11	14	16	19	22	24
											3	5	8	10	13	16	18	21	23
17	2304	2330	2355	2380	2405	2430	2455	2480	2504	2529	3	5	8	10	13	15	18	20	23
											3	5	8	10	12	15	17	20	22
18	2553	2577	2601	2625	2648	2672	2695	2718	2742	2765	2	5	7	9	12	14	17	19	21
											2	4	7	9	11	14	16	18	21
19	2788	2810	2833	2856	2878	2900	2923	2945	2967	2989	2	4	7	9	11	13	16	18	20
											2	4	6	8	11	13	15	17	19
20	3010	3032	3054	3075	3096	3118	3139	3160	3181	3201	2	4	6	8	11	13	15	17	19
21	3222	3243	3263	3284	3304	3324	3345	3365	3385	3404	2	4	6	8	10	12	14	16	18
22	3424	3444	3464	3483	3502	3522	3541	3560	3579	3598	2	4	6	8	10	12	14	15	17
23	3617	3636	3655	3674	3692	3711	3729	3747	3766	3784	2	4	6	7	9	11	13	15	17
24	3802	3820	3838	3856	3874	3892	3909	3927	3945	3962	2	4	5	7	9	11	12	14	16
25	3979	3997	4014	4031	4048	4065	4082	4099	4116	4133	2	3	5	7	9	10	12	14	15
26	4150	4166	4183	4200	4216	4232	4249	4265	4281	4298	2	3	5	7	8	10	11	13	15
27	4314	4330	4346	4362	4378	4393	4409	4425	4440	4456	2	3	5	6	8	9	11	13	14
28	4472	4487	4502	4518	4533	4548	4564	4579	4594	4609	2	3	5	6	8	9	11	12	14
29	4624	4639	4654	4669	4683	4698	4713	4728	4742	4757	1	3	4	6	7	9	10	12	13
30	4771	4786	4800	4814	4829	4843	4857	4871	4886	4900	1	3	4	6	7	9	10	11	13
31	4914	4928	4942	4955	4969	4983	4997	5011	5024	5038	1	3	4	6	7	8	10	11	12
32	5051	5065	5079	5092	5105	5119	5132	5145	5159	5172	1	3	4	5	7	8	9	11	12
33	5185	5198	5211	5224	5237	5250	5263	5276	5289	5302	1	3	4	5	6	8	9	10	12
34	5315	5328	5340	5353	5366	5378	5391	5403	5416	5428	1	3	4	5	6	8	9	10	11
35	5441	5453	5465	5478	5490	5502	5514	5527	5539	5551	1	2	4	5	6	7	9	10	11
36	5563	5575	5587	5599	5611	5623	5635	5647	5658	5670	1	2	4	5	6	7	8	10	11
37	5682	5694	5705	5717	5729	5740	5752	5763	5775	5786	1	2	3	5	6	7	8	9	10
38	5798	5809	5821	5832	5843	5855	5866	5877	5888	5899	1	2	3	5	6	7	8	9	10
39	5911	5922	5933	5944	5955	5966	5977	5988	5999	6010	1	2	3	4	5	7	8	9	10
40	6021	6031	6042	6053	6064	6075	6085	6096	6107	6117	1	2	3	4	5	6	8	9	10
41	6128	6138	6149	6160	6170	6180	6191	6201	6212	6222	1	2	3	4	5	6	7	8	9
42	6232	6243	6253	6263	6274	6284	6294	6304	6314	6325	1	2	3	4	5	6	7	8	9
43	6335	6345	6355	6365	6375	6385	6395	6405	6415	6425	1	2	3	4	5	6	7	8	9
44	6435	6444	6454	6464	6474	6484	6493	6503	6513	6522	1	2	3	4	5	6	7	8	9
45	6532	6542	6551	6561	6471	6580	6590	6599	6609	6618	1	2	3	4	5	6	7	8	9
46	6628	6637	6646	6656	6665	6675	6684	6693	6702	6712	1	2	3	4	5	6	7	7	8
47	6721	6730	6739	6749	6758	6767	6776	6785	6794	6803	1	2	3	4	5	5	6	7	8
48	6812	6821	6830	6839	6848	6857	6866	6875	6884	6893	1	2	3	4	4	5	6	7	8
49	6902	6911	6920	6928	6937	6946	6955	6964	6972	6981	1	2	3	4	4	5	6	7	8

LOGARITHMS

TABLE 1 (Continued)

N	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
50	6990	6998	7007	7016	7024	7033	7042	7050	7059	7067	1	2	3	3	4	5	6	7	8
51	7076	7084	7093	7101	7110	7118	7126	7135	7143	7152	1	2	3	3	4	5	6	7	8
52	7160	7168	7177	7185	7193	7202	7210	7218	7226	7235	1	2	2	3	4	5	6	7	7
53	7243	7251	7259	7267	7275	7284	7292	7300	7308	7316	1	2	2	3	4	5	6	6	7
54	7324	7332	7340	7348	7356	7364	7372	7380	7388	7396	1	2	2	3	4	5	6	6	7
55	7404	7412	7419	7427	7435	7443	7451	7459	7466	7474	1	2	2	3	4	5	5	6	7
56	7482	7490	7497	7505	7513	7520	7528	7536	7543	7551	1	2	2	3	4	5	5	6	7
57	7559	7566	7574	7582	7589	7597	7604	7612	7619	7627	1	2	2	3	4	5	5	6	7
58	7634	7642	7649	7657	7664	7672	7679	7686	7694	7701	1	1	2	3	4	4	5	6	7
59	7709	7716	7723	7731	7738	7745	7752	7760	7767	7774	1	1	2	3	4	4	5	6	7
60	7782	7789	7796	7803	7810	7818	7825	7832	7839	7846	1	1	2	3	4	4	5	6	6
61	7853	7860	7868	7875	7882	7889	7896	7903	7910	7917	1	1	2	3	4	4	5	6	6
62	7924	7931	7938	7945	7952	7959	7966	7973	7980	7987	1	1	2	3	3	4	5	6	6
63	7993	8000	8007	8014	8021	8028	8035	8041	8048	8055	1	1	2	3	3	4	5	5	6
64	8062	8069	8075	8082	8089	8096	8102	8109	8116	8122	1	1	2	3	3	4	5	5	6
65	8129	8136	8142	8149	8156	8162	8169	8176	8182	8189	1	1	2	3	3	4	5	5	6
66	8195	8202	8209	8215	8222	8228	8235	8241	8248	8254	1	1	2	3	3	4	5	5	6
67	8261	8267	8274	8280	8287	8293	8299	8306	8312	8319	1	1	2	3	3	4	5	5	6
68	8325	8331	8338	8344	8351	8357	8363	8370	8376	8382	1	1	2	3	3	4	4	5	6
69	8388	8395	8401	8407	8414	8420	8426	8432	8439	8445	1	1	2	2	3	4	4	5	6
70	8451	8457	8463	8470	8476	8482	8488	8494	8500	8506	1	1	2	2	3	4	4	5	6
71	8513	8519	8525	8531	8537	8543	8549	8555	8561	8567	1	1	2	2	3	4	4	5	5
72	8573	8579	8585	8591	8597	8603	8609	8615	8621	8627	1	1	2	2	3	4	4	5	5
73	8633	8639	8645	8651	8657	8663	8669	8675	8681	8686	1	1	2	2	3	4	4	5	5
74	8692	8698	8704	8710	8716	8722	8727	8733	8739	8745	1	1	2	2	3	4	4	5	5
75	8751	8756	8762	8768	8774	8779	8785	8791	8797	8802	1	1	2	2	3	3	4	5	5
76	8808	8814	8820	8825	8831	8837	8842	8848	8854	8859	1	1	2	2	3	3	4	5	5
77	8865	8871	8876	8882	8887	8893	8899	8904	8910	8915	1	1	2	2	3	3	4	4	5
78	8921	8927	8932	8938	8943	8949	8954	8960	8965	8971	1	1	2	2	3	3	4	4	5
79	8976	8982	8987	8993	8998	9004	9009	9015	9020	9025	1	1	2	2	3	3	4	4	5
80	9031	9036	9042	9047	9053	9058	9063	9069	9074	9079	1	1	2	2	3	3	4	4	5
81	9085	9090	9096	9101	9106	9112	9117	9122	9128	9133	1	1	2	2	3	3	4	4	5
82	9138	9143	9149	9154	9159	9165	9170	9175	9180	9186	1	1	2	2	3	3	4	4	5
83	9191	9196	9201	9206	9212	9217	9222	9227	9232	9238	1	1	2	2	3	3	4	4	5
84	9243	9248	9253	9258	9263	9269	9274	9279	9284	9289	1	1	2	2	3	3	4	4	5
85	9294	9299	9304	9309	9315	9320	9325	9330	9335	9340	1	1	2	2	3	3	4	4	5
86	9345	9350	9355	9360	9365	9370	9375	9380	9385	9390	1	1	2	2	3	3	4	4	5
87	9395	9400	9405	9410	9415	9420	9425	9430	9435	9440	0	1	1	2	2	3	3	4	4
88	9445	9450	9455	9460	9465	9469	9474	9479	9484	9489	0	1	1	2	2	3	3	4	4
89	9494	9499	9504	9509	9513	9518	9523	9528	9533	9538	0	1	1	2	2	3	3	4	4
90	9542	9547	9552	9557	9562	9566	9571	9576	9581	9586	0	1	1	2	2	3	3	4	4
91	9590	9595	9600	9605	9609	9614	9619	9624	9628	9633	0	1	1	2	2	3	3	4	4
92	9638	9643	9647	9652	9657	9661	9666	9671	9675	9680	0	1	1	2	2	3	3	4	4
93	9685	9689	9694	9699	9703	9708	9713	9717	9722	9727	0	1	1	2	2	3	3	4	4
94	9731	9736	9741	9745	9750	9754	9759	9763	9768	9773	0	1	1	2	2	3	3	4	4
95	9777	9782	9786	9791	9795	9800	9805	9809	9814	9818	0	1	1	2	2	3	3	4	4
96	9823	9827	9832	9836	9841	9845	9850	9854	9859	9863	0	1	1	2	2	3	3	4	4
97	9868	9872	9877	9881	9886	9890	9894	9899	9903	9908	0	1	1	2	2	3	3	4	4
98	9912	9917	9921	9926	9930	9934	9939	9943	9948	9952	0	1	1	2	2	3	3	4	4
99	9956	9961	9965	9969	9974	9978	9983	9987	9992	9996	0	1	1	2	2	3	3	3	4

ANTILOGARITHMS

TABLE II

N	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
.00	1000	1002	1005	1007	1009	1012	1014	1016	1019	1021	0	0	1	1	1	1	2	2	2
.01	1023	1026	1028	1030	1033	1035	1038	1040	1042	1045	0	0	1	1	1	1	2	2	2
.02	1047	1050	1052	1054	1057	1059	1062	1064	1067	1069	0	0	1	1	1	1	2	2	2
.03	1072	1074	1076	1079	1081	1084	1086	1089	1091	1094	0	0	1	1	1	1	2	2	2
.04	1096	1099	1102	1104	1107	1109	1112	1114	1117	1119	0	1	1	1	1	2	2	2	2
.05	1122	1125	1127	1130	1132	1135	1138	1140	1143	1146	0	1	1	1	1	2	2	2	2
.06	1148	1151	1153	1156	1159	1161	1164	1167	1169	1172	0	1	1	1	1	2	2	2	2
.07	1175	1178	1180	1183	1186	1189	1191	1194	1197	1199	0	1	1	1	1	2	2	2	2
.08	1202	1205	1208	1211	1213	1216	1219	1222	1225	1227	0	1	1	1	1	2	2	2	3
.09	1230	1233	1236	1239	1242	1245	1247	1250	1253	1256	0	1	1	1	1	2	2	2	3
.10	1259	1262	1265	1268	1271	1274	1276	1279	1282	1285	0	1	1	1	1	2	2	2	3
.11	1288	1291	1294	1297	1300	1303	1306	1309	1312	1315	0	1	1	1	2	2	2	2	3
.12	1318	1321	1324	1327	1330	1334	1337	1340	1343	1346	0	1	1	1	2	2	2	2	3
.13	1349	1352	1355	1358	1361	1365	1368	1371	1374	1377	0	1	1	1	2	2	2	3	3
.14	1380	1384	1387	1390	1393	1396	1400	1403	1406	1409	0	1	1	1	2	2	2	3	3
.15	1413	1416	1419	1422	1426	1429	1432	1435	1439	1442	0	1	1	1	2	2	2	3	3
.16	1445	1449	1452	1455	1459	1462	1466	1469	1472	1476	0	1	1	1	2	2	2	3	3
.17	1479	1483	1486	1489	1493	1496	1500	1503	1507	1510	0	1	1	1	2	2	2	3	3
.18	1514	1517	1521	1524	1528	1531	1535	1538	1542	1545	0	1	1	1	2	2	2	3	3
.19	1549	1552	1556	1560	1563	1567	1570	1574	1578	1581	0	1	1	1	2	2	3	3	3
.20	1585	1589	1592	1596	1600	1603	1607	1611	1614	1618	0	1	1	1	2	2	3	3	3
.21	1622	1626	1629	1633	1637	1641	1644	1648	1652	1656	0	1	1	2	2	2	3	3	3
.22	1660	1663	1667	1671	1675	1679	1683	1687	1690	1694	0	1	1	2	2	2	3	3	3
.23	1698	1702	1706	1710	1714	1718	1722	1726	1730	1734	0	1	1	2	2	2	3	3	4
.24	1738	1742	1746	1750	1754	1758	1762	1766	1770	1774	0	1	1	2	2	2	3	3	4
.25	1778	1782	1786	1791	1795	1799	1803	1807	1811	1816	0	1	1	2	2	2	3	3	4
.26	1820	1824	1828	1832	1837	1841	1845	1849	1854	1858	0	1	1	2	2	3	3	3	4
.27	1862	1866	1871	1875	1879	1884	1888	1892	1897	1901	0	1	1	2	2	3	3	3	4
.28	1905	1910	1914	1919	1923	1928	1932	1936	1941	1945	0	1	1	2	2	3	3	4	4
.29	1950	1954	1959	1963	1968	1972	1977	1982	1986	1991	0	1	1	2	2	3	3	4	4
.30	1995	2000	2004	2009	2014	2018	2023	2028	2032	2037	0	1	1	2	2	3	3	4	4
.31	2042	2046	2051	2056	2061	2065	2070	2075	2080	2084	0	1	1	2	2	3	3	4	4
.32	2089	2094	2099	2104	2109	2113	2118	2123	2128	2133	0	1	1	2	2	3	3	4	4
.33	2138	2143	2148	2153	2158	2163	2168	2173	2178	2183	0	1	1	2	2	3	3	4	4
.34	2188	2193	2198	2203	2208	2213	2218	2223	2228	2234	1	1	2	2	3	3	4	4	5
.35	2239	2244	2249	2254	2259	2265	2270	2275	2280	2286	1	1	2	2	3	3	4	4	5
.36	2291	2296	2301	2307	2312	2317	2323	2328	2333	2339	1	1	2	2	3	3	4	4	5
.37	2344	2350	2355	2360	2366	2371	2377	2382	2388	2393	1	1	2	2	3	3	4	4	5
.38	2399	2404	2410	2415	2421	2427	2432	2438	2443	2449	1	1	2	2	3	3	4	4	5
.39	2455	2460	2466	2472	2477	2483	2489	2495	2500	2506	1	1	2	2	3	3	4	5	5
.40	2512	2518	2523	2529	2535	2541	2547	2553	2559	2564	1	1	2	2	3	4	4	5	5
.41	2570	2576	2582	2588	2594	2600	2606	2612	2618	2624	1	1	2	2	3	4	4	5	5
.42	2630	2636	2642	2649	2655	2661	2667	2673	2679	2685	1	1	2	2	3	4	4	5	6
.43	2692	2698	2704	2710	2716	2723	2729	2735	2742	2748	1	1	2	3	3	4	4	5	6
.44	2754	2761	2767	2773	2780	2786	2793	2799	2805	2812	1	1	2	3	3	4	4	5	6
.45	2818	2825	2831	2838	2844	2851	2858	2864	2871	2877	1	1	2	3	3	4	5	5	6
.46	2884	2891	2897	2904	2911	2917	2924	2931	2938	2944	1	1	2	3	3	4	5	5	6
.47	2951	2958	2965	2972	2979	2985	2992	2999	3006	3013	1	1	2	3	3	4	5	5	6
.48	3020	3027	3034	3041	3048	3055	3062	3069	3076	3083	1	1	2	3	3	4	5	6	6
.49	3090	3097	3105	3112	3119	3126	3133	3141	3148	3155	1	1	2	3	3	4	5	6	6

ANTILOGARITHMS

TABLE II (Continued)

N	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
.50	3162	3170	3177	3184	3192	3199	3206	3214	3221	3228	1	1	2	3	4	4	5	6	7
.51	3236	3243	3251	3258	3266	3273	3281	3289	3296	3304	1	2	2	3	4	5	5	6	7
.52	3311	3319	3327	3334	3342	3350	3357	3365	3373	3381	1	2	2	3	4	5	5	6	7
.53	3388	3396	3404	3412	3420	3428	3436	3443	3451	3459	1	2	2	3	4	5	6	6	7
.54	3467	3475	3483	3491	3499	3508	3516	3524	3532	3540	1	2	2	3	4	5	6	6	7
.55	3548	3556	3565	3573	3581	3589	3597	3606	3614	3622	1	2	2	3	4	5	6	7	7
.56	3631	3639	3648	3656	3664	3673	3681	3690	3698	3707	1	2	3	3	4	5	6	7	8
.57	3715	3724	3733	3741	3750	3758	3767	3776	3784	3793	1	2	3	3	4	5	6	7	8
.58	3802	3811	3819	3828	3837	3846	3855	3864	3873	3882	1	2	3	4	4	5	6	7	8
.59	3890	3899	3908	3917	3926	3936	3945	3954	3963	3972	1	2	3	4	5	5	6	7	8
.60	3981	3990	3999	4009	4018	4027	4036	4046	4055	4064	1	2	3	4	5	6	6	7	8
.61	4074	4083	4093	4102	4111	4121	4130	4140	4150	4159	1	2	3	4	5	6	7	8	9
.62	4169	4178	4188	4198	4207	4217	4227	4236	4246	4256	1	2	3	4	5	6	7	8	9
.63	4266	4276	4285	4295	4305	4315	4325	4335	4345	4355	1	2	3	4	5	6	7	8	9
.64	4365	4375	4385	4395	4406	4416	4426	4436	4446	4457	1	2	3	4	5	6	7	8	9
.65	4467	4477	4487	4498	4508	4519	4529	4539	4550	4560	1	2	3	4	5	6	7	8	9
.66	4571	4581	4592	4603	4613	4624	4634	4645	4656	4667	1	2	3	4	5	6	7	9	10
.67	4677	4688	4699	4710	4721	4732	4742	4753	4764	4775	1	2	3	4	5	7	8	9	10
.68	4786	4797	4808	4819	4831	4842	4853	4864	4875	4887	1	2	3	4	6	7	8	9	10
.69	4898	4909	4920	4932	4943	4955	4966	4977	4989	5000	1	2	3	5	6	7	8	9	10
.70	5012	5023	5035	5047	5058	5070	5082	5093	5105	5117	1	2	4	5	6	7	8	9	11
.71	5129	5140	5152	5164	5176	5188	5200	5212	5224	5236	1	2	4	5	6	7	8	10	11
.72	5248	5260	5272	5284	5297	5309	5321	5333	5346	5358	1	2	4	5	6	7	9	10	11
.73	5370	5383	5395	5408	5420	5433	5445	5458	5470	5483	1	3	4	5	6	8	9	10	11
.74	5495	5508	5521	5534	5546	5559	5572	5585	5598	5610	1	3	4	5	6	8	9	10	12
.75	5623	5636	5649	5662	5675	5689	5702	5715	5728	5741	1	3	4	5	7	8	9	10	12
.76	5754	5768	5781	5794	5808	5821	5834	5848	5861	5875	1	3	4	5	7	8	9	11	12
.77	5888	5902	5916	5929	5943	5957	5970	5984	5998	6012	1	3	4	5	7	8	10	11	12
.78	6026	6039	6053	6067	6081	6095	6109	6124	6138	6152	1	3	4	6	7	8	10	11	13
.79	6166	6180	6194	6209	6223	6237	6252	6266	6281	6295	1	3	4	6	7	9	10	11	13
.80	6310	6324	6339	6353	6368	6383	6397	6412	6427	6442	1	3	4	6	7	9	10	12	13
.81	6457	6471	6486	6501	6516	6531	6546	6561	6577	6592	2	3	5	6	8	9	11	12	14
.82	6607	6622	6637	6653	6668	6683	6699	6714	6730	6745	2	3	5	6	8	9	11	12	14
.83	6761	6776	6792	6808	6823	6839	6855	6871	6887	6902	2	3	5	6	8	9	11	13	14
.84	6918	6934	6950	6966	6982	6998	7015	7031	7047	7063	2	3	5	6	8	10	11	13	15
.85	7079	7096	7112	7129	7145	7161	7178	7194	7211	7228	2	3	5	7	8	10	12	13	15
.86	7244	7261	7278	7295	7311	7328	7345	7362	7379	7396	2	3	5	7	8	10	12	13	15
.87	7413	7430	7447	7464	7482	7499	7516	7534	7551	7568	2	3	5	7	9	10	12	14	16
.88	7586	7603	7621	7638	7656	7674	7691	7709	7727	7745	2	4	5	7	9	11	12	14	16
.89	7762	7780	7798	7816	7834	7852	7870	7889	7907	7925	2	4	5	7	9	11	13	14	16
.90	7943	7962	7980	7998	8017	8035	8054	8072	8091	8110	2	4	6	7	9	11	13	15	17
.91	8128	8147	8166	8185	8204	8222	8241	8260	8279	8299	2	4	6	8	9	11	13	15	17
.92	8318	8337	8356	8375	8395	8414	8433	8453	8472	8492	2	4	6	8	10	12	14	15	17
.93	8511	8531	8551	8570	8590	8610	8630	8650	8670	8690	2	4	6	8	10	12	14	16	18
.94	8710	8730	8750	8770	8790	8810	8831	8851	8872	8892	2	4	6	8	10	12	14	16	18
.95	8913	8933	8954	8974	8995	9016	9036	9057	9078	9099	2	4	6	8	10	12	15	17	19
.96	9120	9141	9162	9183	9204	9226	9247	9268	9290	9311	2	4	6	8	11	13	15	17	19
.97	9333	9354	9376	9397	9419	9441	9462	9484	9506	9528	2	4	7	9	11	13	15	17	20
.98	9550	9572	9594	9616	9638	9661	9683	9705	9727	9750	2	4	7	9	11	13	16	18	20
.99	9772	9795	9817	9840	9863	9886	9908	9931	9954	9977	2	5	7	9	11	14	16	18	20

Answers to Some Questions in Exercises

UNIT 1

- | | |
|---|---|
| <p>1.4 16.23 M</p> <p>1.6 157.8 mL</p> <p>1.8 17.95 m and 9.10 M</p> <p>1.15 40.907 g mol⁻¹</p> <p>1.17 12.08 kPa</p> <p>1.19 23 g mol⁻¹, 3.53 kPa</p> <p>1.21 A = 25.58 u and B = 42.64 u</p> <p>1.24 KCl, CH₃OH, CH₃CN, Cyclohexane</p> <p>1.25 Toluene, chloroform; Phenol, Pentanol; Formic acid, ethylene glycol</p> <p>1.26 5 m</p> <p>1.28 1.424%</p> <p>1.30 4.575 g</p> <p>1.33 $i = 1.0753$, $K_a = 3.07 \times 10^{-3}$</p> <p>1.35 178×10^{-5}</p> <p>1.38 0.6 and 0.4</p> <p>1.40 0.03 mol of CaCl₂</p> | <p>1.5 0.617 m, 0.01 and 0.99, 0.67</p> <p>1.7 33.5%</p> <p>1.9 $1.5 \times 10^{-3}\%$, 1.25×10^{-4} m</p> <p>1.16 73.58 kPa</p> <p>1.18 10 g</p> <p>1.20 269.07 K</p> <p>1.22 0.061 M</p> <p>1.27 2.45×10^{-8} M</p> <p>1.29 3.2 g of water</p> <p>1.32 0.65°</p> <p>1.34 17.44 mm Hg</p> <p>1.36 280.7 torr, 32 torr</p> <p>1.39 $x(\text{O}_2) 4.6 \times 10^{-5}$, $x(\text{N}_2) 9.22 \times 10^{-5}$</p> <p>1.41 5.27×10^{-3} atm.</p> |
|---|---|

UNIT 2

- 2.4** (i) $E^\circ = 0.34\text{V}$, $\Delta_r G^\circ = -196.86 \text{ kJ mol}^{-1}$, $K = 3.124 \times 10^{34}$
 (ii) $E^\circ = 0.03\text{V}$, $\Delta_r G^\circ = -2.895 \text{ kJ mol}^{-1}$, $K = 3.2$
- 2.5** (i) 2.68 V, (ii) 0.53 V, (iii) 0.08 V, (iv) -1.298 V
- 2.6** 1.56 V
- 2.8** $124.0 \text{ S cm}^2 \text{ mol}^{-1}$
- 2.9** 0.219 cm^{-1}
- 2.11** 1.85×10^{-5}
- 2.12** 3F, 2F, 5F
- 2.13** 1F, 4.44F
- 2.14** 2F, 1F
- 2.15** 1.8258g
- 2.16** 14.40 min, Copper 0.427g, Zinc 0.437 g

UNIT 3

- 3.2** (i) $8.0 \times 10^{-9} \text{ mol L}^{-1} \text{ s}^{-1}$; $3.89 \times 10^{-9} \text{ mol L}^{-1} \text{ s}^{-1}$
- 3.4** $\text{bar}^{-1/2} \text{ s}^{-1}$
- 3.6** (i) 4 times (ii) $\frac{1}{4}$ times
- 3.8** (i) $4.67 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$ (ii) $1.98 \times 10^{-2} \text{ s}^{-1}$
- 3.9** (i) $\text{rate} = k[\text{A}][\text{B}]^2$ (ii) 9 times

- 3.10** Orders with respect to A is 1.5 and order with respect to B is zero.
- 3.11** rate law = $k[A][B]^2$; rate constant = $6.0 \text{ M}^2\text{min}^{-1}$
- 3.13** (i) 3.47×10^{-3} seconds (ii) 0.35 minutes (iii) 0.173 years
- 3.14** 1845 years **3.16** $4.6 \times 10^{-2} \text{ s}$
- 3.17** $0.7814 \mu\text{g}$ and $0.227 \mu\text{g}$. **3.19** 77.7 minutes
- 3.20** $2.20 \times 10^{-3} \text{ s}^{-1}$ **3.21** $2.23 \times 10^{-3} \text{ s}^{-1}$, $7.8 \times 10^{-4} \text{ atm s}^{-1}$
- 3.23** $3.9 \times 10^{12} \text{ s}^{-1}$ **3.24** 0.135 M
- 3.25** 0.158 M **3.26** $232.79 \text{ kJ mol}^{-1}$
- 3.27** $239.339 \text{ kJ mol}^{-1}$ **3.28** 24°C
- 3.29** $E_a = 76.750 \text{ kJ mol}^{-1}$, $k = 0.9965 \times 10^{-2} \text{ s}^{-1}$
- 3.30** 52.8 kJ mol^{-1}

UNIT 4

- 4.2** It is because Mn^{2+} has $3d^5$ configuration which has extra stability.
- 4.5** Stable oxidation states.
 $3d^3$ (Vanadium): +2, +3, +4, and +5
 $3d^5$ (Chromium): +3, +4, +6
 $3d^5$ (Manganese): +2, +4, +6, +7
 $3d^8$ (Nickel): +2, +3 (in complexes)
 $3d^4$ There is no d^4 configuration in the ground state.
- 4.6** Vanadate VO_3^- , chromate CrO_4^{2-} , permanganate MnO_4^-
- 4.10** +3 is the common oxidation state of the lanthanoids
 In addition to +3, oxidation states +2 and +4 are also exhibited by some of the lanthanoids.
- 4.13** In transition elements the oxidation states vary from +1 to any highest oxidation state by one
 For example, for manganese it may vary as +2, +3, +4, +5, +6, +7. In the nontransition elements the variation is selective, always differing by 2, e.g. +2, +4, or +3, +5 or +4, +6 etc.
- 4.18** Except Sc^{3+} , all others will be coloured in aqueous solution because of incompletely filled $3d$ -orbitals, will give rise to $d-d$ transitions.
- 4.21** (i) Cr^{2+} is reducing as it involves change from d^4 to d^3 , the latter is more stable configuration
 (t_{2g}^3) Mn(III) to Mn(II) is from $3d^4$ to $3d^5$ again $3d^5$ is an extra stable configuration.
 (ii) Due to CFSE, which more than compensates the 3rd IE.
 (iii) The hydration or lattice energy more than compensates the ionisation enthalpy involved in removing electron from d^1 .
- 4.23** Copper, because with +1 oxidation state an extra stable configuration, $3d^{10}$ results.
- 4.24** Unpaired electrons $\text{Mn}^{3+} = 4$, $\text{Cr}^{3+} = 3$, $\text{V}^{3+} = 2$, $\text{Ti}^{3+} = 1$. Most stable Cr^{3+}
- 4.28** Second part 59, 95, 102.
- 4.30** Lawrencium, 103, +3
- 4.36** $\text{Ti}^{2+} = 2$, $\text{V}^{2+} = 3$, $\text{Cr}^{3+} = 3$, $\text{Mn}^{2+} = 5$, $\text{Fe}^{2+} = 6$, $\text{Fe}^{3+} = 5$, $\text{Co}^{2+} = 7$, $\text{Ni}^{2+} = 8$, $\text{Cu}^{2+} = 9$
- 4.38** $M\sqrt{n(n+2)} = 2.2$, $n \approx 1$, $d^2 \text{ sp}^3$, CN^- strong ligand
 $= 5.3$, $n \approx 4$, sp^3 , d^2 , H_2O weak ligand
 $= 5.9$, $n \approx 5$, sp^3 , Cl^- weak ligand.

UNIT 5

- 5.5** (i) +3 (ii) +3 (iii) +2 (iv) +3 (v) +3
- 5.6** (i) $[\text{Zn}(\text{OH})_4]^{2-}$ (ii) $\text{K}_2[\text{PdCl}_4]$ (iii) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ (iv) $\text{K}_2[\text{Ni}(\text{CN})_4]$
 (v) $[\text{Co}(\text{NH}_3)_5(\text{ONO})]^{2+}$ (vi) $[\text{Co}(\text{NH}_3)_6]_2(\text{SO}_4)_3$ (vii) $\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$ (viii) $[\text{Pt}(\text{NH}_3)_6]^{4+}$
 (ix) $[\text{CuBr}_4]^{2-}$ (x) $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]^{2+}$
- 5.9** (i) $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$ - Nil
 (ii) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ - Two (*fac*- and *mer*-)
- 5.12** Three (two *cis* and one *trans*)
- 5.13** Aqueous CuSO_4 solution exists as $[\text{Cu}(\text{H}_2\text{O})_4]\text{SO}_4$ which has blue colour due to $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ ions.
 (i) When KF is added, the weak H_2O ligands are replaced by F^- ligands, forming $[\text{CuF}_4]^{2-}$ ions which is a green precipitate.

$$[\text{Cu}(\text{H}_2\text{O})_4]^{2+} + 4\text{F}^- \rightarrow [\text{CuF}_4]^{2-} + 4\text{H}_2\text{O}$$

 (ii) When KCl is added, Cl^- ligands replace the weak H_2O ligands forming $[\text{CuCl}_4]^{2-}$ ions which has bright green colour.

$$[\text{Cu}(\text{H}_2\text{O})_4]^{2+} + 4\text{Cl}^- \rightarrow [\text{CuCl}_4]^{2-} + 4\text{H}_2\text{O}$$
- 5.14** $[\text{Cu}(\text{H}_2\text{O})_4]^{2+} + 4\text{CN}^- \rightarrow [\text{Cu}(\text{CN})_4]^{2-} + 4\text{H}_2\text{O}$
 As CN is a strong ligand, it forms a highly stable complex with Cu^{2+} ion. On passing H_2S , free Cu^{2+} ions are not available to form the precipitate of CuS .
- 5.23** (i) OS = +3, CN = 6, d-orbital occupation is $t_{2g}^6 e_g^0$,
 (ii) OS = +3, CN = 6, $d^3 (t_{2g}^3)$,
 (iii) OS = +2, CN = 4, $d^7 (t_{2g}^5 e_g^2)$,
 (iv) OS = +2, CN = 6, $d^5 (t_{2g}^3 e_g^2)$.
- 5.28** (iii)
- 5.29** (ii)
- 5.30** (iii)
- 5.31** (iii)
- 5.32** (i) The order of the ligand in the spectrochemical series :
 $\text{H}_2\text{O} < \text{NH}_3 < \text{NO}_2^-$
 Hence the energy of the observed light will be in the order :
 $[\text{Ni}(\text{H}_2\text{O})_6]^{2+} < [\text{Ni}(\text{NH}_3)_6]^{2+} < [\text{Ni}(\text{NO}_2)_6]^{4-}$
 Thus, wavelengths absorbed ($E = hc/\lambda$) will be in the opposite order.

NOTES

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Chemistry

Part I

Textbook for Class XII



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FOREWORD

The National Curriculum Framework (NCF), 2005 recommends that children's life at school must be linked to their life outside the school. This principle marks a departure from the legacy of bookish learning which continues to shape our system and causes a gap between the school, home and community. The syllabi and textbooks developed on the basis of NCF signify an attempt to implement this basic idea. They also attempt to discourage rote learning and the maintenance of sharp boundaries between different subject areas. We hope these measures will take us significantly further in the direction of a child-centred system of education outlined in the National Policy on Education (1986).

The success of this effort depends on the steps that school principals and teachers will take to encourage children to reflect on their own learning and to pursue imaginative activities and questions. We must recognise that, given space, time and freedom, children generate new knowledge by engaging with the information passed on to them by adults. Treating the prescribed textbook as the sole basis of examination is one of the key reasons why other resources and sites of learning are ignored. Inculcating creativity and initiative is possible if we perceive and treat children as participants in learning, not as receivers of a fixed body of knowledge.

These aims imply considerable change in school routines and mode of functioning. Flexibility in the daily time-table is as necessary as rigour in implementing the annual calendar so that the required number of teaching days are actually devoted to teaching. The methods used for teaching and evaluation will also determine how effective this textbook proves for making children's life at school a happy experience, rather than a source of stress or boredom. Syllabus designers have tried to address the problem of curricular burden by restructuring and reorienting knowledge at different stages with greater consideration for child psychology and the time available for teaching. The textbook attempts to enhance this endeavour by giving higher priority and space to opportunities for contemplation and wondering, discussion in small groups, and activities requiring hands-on experience.

The National Council of Educational Research and Training (NCERT) appreciates the hard work done by the textbook development committee



responsible for this book. We wish to thank the Chairperson of the advisory group in science and mathematics, Professor J.V. Narlikar and the Chief Advisor for this book, Professor B. L. Khandelwal for guiding the work of this committee. Several teachers contributed to the development of this textbook; we are grateful to their principals for making this possible. We are indebted to the institutions and organisations which have generously permitted us to draw upon their resources, material and personnel. As an organisation committed to systemic reform and continuous improvement in the quality of its products, NCERT welcomes comments and suggestions which will enable us to undertake further revision and refinement.

New Delhi
20 November 2006

Director
National Council of Educational
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RATIONALISATION OF CONTENT IN THE TEXTBOOKS

In view of the COVID-19 pandemic, it is imperative to reduce content load on students. The National Education Policy 2020, also emphasises reducing the content load and providing opportunities for experiential learning with creative mindset. In this background, the NCERT has undertaken the exercise to rationalise the textbooks across all classes. Learning Outcomes already developed by the NCERT across classes have been taken into consideration in this exercise.

Contents of the textbooks have been rationalised in view of the following:

- Overlapping with similar content included in other subject areas in the same class
- Similar content included in the lower or higher class in the same subject
- Difficulty level
- Content, which is easily accessible to students without much interventions from teachers and can be learned by children through self-learning or peer-learning
- Content, which is irrelevant in the present context

This present edition, is a reformatted version after carrying out the changes given above.

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PREFACE

Chemistry has made a profound impact on the society. It is intimately linked to the well-being of human kind. The rate of advancements in chemistry is so high that curriculum developers continuously look for strategies to cope with these advancements. Also, the students have to be inspired to be the future leaders who would make fundamental contributions. The present textbook is a sincere effort in this direction.

The structure of the textbook, presented in two parts, comprises of sixteen Units. Although the titles of various Units indicate a sort of compartmentalisation into physical, inorganic and organic chemistry, readers will find that these sub-disciplines have been intermingled, at least to a certain extent, to have a unified approach to the subject. The approach of presentation of the subject matter discourages students from rote memorisation. The subject has in fact, been organised around the laws and principles of chemistry. As students master these laws and principles, they will soon get to the point where they can predict much of what will come.

Efforts have been directed towards making the subject stimulating and exciting by references to the historical developments and its usefulness to our lives, wherever appropriate. The text is well illustrated with examples from surrounding environment to facilitate grasping of the qualitative and quantitative aspects of the concept easily. Physical data are given in SI units throughout the book to make comparison of various properties easier. IUPAC system of nomenclature has been followed along with the common system. Structural formulae of chemical compounds showing functional/coordinating groups in different colours are drawn using electronic system. Each Unit has a good number of examples, as illustrations, with their solutions and some intext questions, the answers of some of which are given at the end of the Unit. The end of Unit exercises are designed to apply important principles and provoke thinking process to solve them. Answers of some of these exercises are given at the end of the book.

A variety of materials, e.g., biographical sketches of some scientists, additional information related to a particular topic, etc., is given in boxes with a deep yellow coloured bar. This boxed material with a 'deep yellow bar' is to bring additional life to the topic. However, it is non-evaluative. The structures of some of the more complex compounds incorporated in the book are for understanding their chemistry. As their reproduction would lead to memorisation, it is also a non-evaluative portion of the text.

The information part has been significantly reduced and, wherever possible, it has been substantiated with facts. However, it is necessary for students to

be aware of commercially important chemicals, their process of manufacture and sources of raw materials. This leads to descriptive material in the book. Attempts have been made to make descriptions of such compounds interesting by considering their structures and reactivity. Thermodynamics, kinetics and electrochemical aspects have been applied to chemical reactions which should be beneficial to students for understanding why a particular reaction happened and why a particular property is exhibited by the product. There is currently great awareness of environmental and energy issues which are directly related to chemistry. Such issues have been highlighted and dealt with at appropriate places in the book.

A team of experts constituted by the NCERT has developed the manuscript of the book. It gives me great pleasure to acknowledge the valuable contribution of all the members of this team. I also acknowledge the valuable and relentless contribution of the editors in bringing the book to the present shape. I also acknowledge with thanks the dedicated efforts and valuable contribution of Professor Brahm Parkash, who not only coordinated the entire programme but also actively involved in writing and editing of this book. Thanks are also due to the participating teachers and subject experts of the review workshop for their contribution, which has helped us to make the book learner friendly. Also, I thank the technical and administrative staff of the NCERT for their support in the entire process.

The team of this textbook development programme hopes that the book stimulates its readers and makes them feel the excitement and fascination for this subject. Efforts have been made to bring out this book error-free. Nevertheless, it is recognised that in such a book of complexity, there could inevitably be occasional errors. It will always be a pleasure to hear about them from readers to take necessary steps to rectify them.

B.L. KHANDELWAL

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THE CONSTITUTION OF INDIA

PREAMBLE

WE, THE PEOPLE OF INDIA, having solemnly resolved to constitute India into a ¹**[SOVEREIGN SOCIALIST SECULAR DEMOCRATIC REPUBLIC]** and to secure to all its citizens :

JUSTICE, social, economic and political;

LIBERTY of thought, expression, belief, faith and worship;

EQUALITY of status and of opportunity; and to promote among them all

FRATERNITY assuring the dignity of the individual and the ²[unity and integrity of the Nation];

IN OUR CONSTITUENT ASSEMBLY this twenty-sixth day of November, 1949 do **HEREBY ADOPT, ENACT AND GIVE TO OURSELVES THIS CONSTITUTION.**

1. Subs. by the Constitution (Forty-second Amendment) Act, 1976, Sec.2, for "Sovereign Democratic Republic" (w.e.f. 3.1.1977)
2. Subs. by the Constitution (Forty-second Amendment) Act, 1976, Sec.2, for "Unity of the Nation" (w.e.f. 3.1.1977)

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Unit

1

Solutions

Objectives

After studying this Unit, you will be able to

- describe the formation of different types of solutions;
- express concentration of solution in different units;
- state and explain Henry's law and Raoult's law;
- distinguish between ideal and non-ideal solutions;
- explain deviations of real solutions from Raoult's law;
- describe colligative properties of solutions and correlate these with molar masses of the solutes;
- explain abnormal colligative properties exhibited by some solutes in solutions.

Almost all processes in body occur in some kind of liquid solutions.

In normal life we rarely come across pure substances. Most of these are mixtures containing two or more pure substances. Their utility or importance in life depends on their composition. For example, the properties of brass (mixture of copper and zinc) are quite different from those of German silver (mixture of copper, zinc and nickel) or bronze (mixture of copper and tin); 1 part per million (ppm) of fluoride ions in water prevents tooth decay, while 1.5 ppm causes the tooth to become mottled and high concentrations of fluoride ions can be poisonous (for example, sodium fluoride is used in rat poison); intravenous injections are always dissolved in water containing salts at particular ionic concentrations that match with blood plasma concentrations and so on.

In this Unit, we will consider mostly liquid solutions and their formation. This will be followed by studying the properties of the solutions, like vapour pressure and colligative properties. We will begin with types of solutions and then various alternatives in which concentrations of a solute can be expressed in liquid solution.

1.1 Types of Solutions

Solutions are **homogeneous** mixtures of two or more than two components. By homogenous mixture we mean that its composition and properties are uniform throughout the mixture. Generally, the component that is present in the largest quantity is known as **solvent**. Solvent determines the physical state in which solution exists. One or more components present in the solution other than solvent are called **solutes**. In this Unit we shall consider only **binary solutions** (i.e.,

consisting of two components). Here each component may be solid, liquid or in gaseous state and are summarised in Table 1.1.

Table 1.1: Types of Solutions

Type of Solution	Solute	Solvent	Common Examples
<i>Gaseous Solutions</i>	Gas	Gas	Mixture of oxygen and nitrogen gases
	Liquid	Gas	Chloroform mixed with nitrogen gas
	Solid	Gas	Camphor in nitrogen gas
<i>Liquid Solutions</i>	Gas	Liquid	Oxygen dissolved in water
	Liquid	Liquid	Ethanol dissolved in water
	Solid	Liquid	Glucose dissolved in water
<i>Solid Solutions</i>	Gas	Solid	Solution of hydrogen in palladium
	Liquid	Solid	Amalgam of mercury with sodium
	Solid	Solid	Copper dissolved in gold

1.2 Expressing Concentration of Solutions

Composition of a solution can be described by expressing its concentration. The latter can be expressed either qualitatively or quantitatively. For example, qualitatively we can say that the solution is dilute (i.e., relatively very small quantity of solute) or it is concentrated (i.e., relatively very large quantity of solute). But in real life these kinds of description can add to lot of confusion and thus the need for a quantitative description of the solution.

There are several ways by which we can describe the concentration of the solution quantitatively.

- (i) *Mass percentage (w/w)*: The mass percentage of a component of a solution is defined as:

Mass % of a component

$$= \frac{\text{Mass of the component in the solution}}{\text{Total mass of the solution}} \times 100 \quad (1.1)$$

For example, if a solution is described by 10% glucose in water by mass, it means that 10 g of glucose is dissolved in 90 g of water resulting in a 100 g solution. Concentration described by mass percentage is commonly used in industrial chemical applications. For example, commercial bleaching solution contains 3.62 mass percentage of sodium hypochlorite in water.

- (ii) *Volume percentage (V/V)*: The volume percentage is defined as:

$$\text{Volume \% of a component} = \frac{\text{Volume of the component}}{\text{Total volume of solution}} \times 100 \quad (1.2)$$

For example, 10% ethanol solution in water means that 10 mL of ethanol is dissolved in water such that the total volume of the solution is 100 mL. Solutions containing liquids are commonly expressed in this unit. For example, a 35% (v/v) solution of ethylene glycol, an antifreeze, is used in cars for cooling the engine. At this concentration the antifreeze lowers the freezing point of water to 255.4K (-17.6°C).

(iii) *Mass by volume percentage (w/V)*: Another unit which is commonly used in medicine and pharmacy is mass by volume percentage. It is the mass of solute dissolved in 100 mL of the solution.

(iv) *Parts per million*: When a solute is present in **trace** quantities, it is convenient to express concentration in **parts per million (ppm)** and is defined as:

Parts per million =

$$\frac{\text{Number of parts of the component}}{\text{Total number of parts of all components of the solution}} \times 10^6 \quad (1.3)$$

As in the case of percentage, concentration in parts per million can also be expressed as mass to mass, volume to volume and mass to volume. A litre of sea water (which weighs 1030 g) contains about 6×10^{-3} g of dissolved oxygen (O_2). Such a small concentration is also expressed as 5.8 g per 10^6 g (5.8 ppm) of sea water. The concentration of pollutants in water or atmosphere is often expressed in terms of $\mu\text{g mL}^{-1}$ or ppm.

(v) *Mole fraction*: Commonly used symbol for mole fraction is x and subscript used on the right hand side of x denotes the component. It is defined as:

Mole fraction of a component =

$$\frac{\text{Number of moles of the component}}{\text{Total number of moles of all the components}} \quad (1.4)$$

For example, in a binary mixture, if the number of moles of A and B are n_A and n_B respectively, the mole fraction of A will be

$$x_A = \frac{n_A}{n_A + n_B} \quad (1.5)$$

For a solution containing i number of components, we have:

$$x_i = \frac{n_i}{n_1 + n_2 + \dots + n_i} = \frac{n_i}{\sum n_i} \quad (1.6)$$

It can be shown that in a given solution sum of all the mole fractions is unity, i.e.

$$x_1 + x_2 + \dots + x_i = 1 \quad (1.7)$$

Mole fraction unit is very useful in relating some physical properties of solutions, say vapour pressure with the concentration of the solution and quite useful in describing the calculations involving gas mixtures.

Example 1.1

Calculate the mole fraction of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) in a solution containing 20% of $\text{C}_2\text{H}_6\text{O}_2$ by mass.

Solution

Assume that we have 100 g of solution (one can start with any amount of solution because the results obtained will be the same). Solution will contain 20 g of ethylene glycol and 80 g of water.

Molar mass of $\text{C}_2\text{H}_6\text{O}_2 = 12 \times 2 + 1 \times 6 + 16 \times 2 = 62 \text{ g mol}^{-1}$.

$$\text{Moles of } \text{C}_2\text{H}_6\text{O}_2 = \frac{20 \text{ g}}{62 \text{ g mol}^{-1}} = 0.322 \text{ mol}$$

$$\text{Moles of water} = \frac{80 \text{ g}}{18 \text{ g mol}^{-1}} = 4.444 \text{ mol}$$

$$\begin{aligned} x_{\text{glycol}} &= \frac{\text{moles of } \text{C}_2\text{H}_6\text{O}_2}{\text{moles of } \text{C}_2\text{H}_6\text{O}_2 + \text{moles of } \text{H}_2\text{O}} \\ &= \frac{0.322 \text{ mol}}{0.322 \text{ mol} + 4.444 \text{ mol}} = 0.068 \end{aligned}$$

$$\text{Similarly, } x_{\text{water}} = \frac{4.444 \text{ mol}}{0.322 \text{ mol} + 4.444 \text{ mol}} = 0.932$$

Mole fraction of water can also be calculated as: $1 - 0.068 = 0.932$

(vi) *Molarity*: Molarity (M) is defined as number of moles of solute dissolved in one litre (or one cubic decimetre) of solution,

$$\text{Molarity} = \frac{\text{Moles of solute}}{\text{Volume of solution in litre}} \quad (1.8)$$

For example, 0.25 mol L^{-1} (or 0.25 M) solution of NaOH means that 0.25 mol of NaOH has been dissolved in one litre (or one cubic decimetre).

Example 1.2

Calculate the molarity of a solution containing 5 g of NaOH in 450 mL solution.

Solution

$$\text{Moles of } \text{NaOH} = \frac{5 \text{ g}}{40 \text{ g mol}^{-1}} = 0.125 \text{ mol}$$

$$\text{Volume of the solution in litres} = 450 \text{ mL} / 1000 \text{ mL L}^{-1}$$

Using equation (2.8),

$$\text{Molarity} = \frac{0.125 \text{ mol} \times 1000 \text{ mL L}^{-1}}{450 \text{ mL}} = 0.278 \text{ M}$$

$$= 0.278 \text{ mol L}^{-1}$$

$$= 0.278 \text{ mol dm}^{-3}$$

(vii) **Molality:** Molality (m) is defined as the number of moles of the solute per kilogram (kg) of the solvent and is expressed as:

$$\text{Molality (m)} = \frac{\text{Moles of solute}}{\text{Mass of solvent in kg}} \quad (1.9)$$

For example, 1.00 mol kg^{-1} (or 1.00 m) solution of KCl means that 1 mol (74.5 g) of KCl is dissolved in 1 kg of water.

Each method of expressing concentration of the solutions has its own merits and demerits. Mass %, ppm, mole fraction and molality are independent of temperature, whereas molarity is a function of temperature. This is because volume depends on temperature and the mass does not.

Calculate molality of 2.5 g of ethanoic acid (CH_3COOH) in 75 g of benzene. **Example 1.3**

Molar mass of $\text{C}_2\text{H}_4\text{O}_2$: $12 \times 2 + 1 \times 4 + 16 \times 2 = 60 \text{ g mol}^{-1}$

$$\text{Moles of } \text{C}_2\text{H}_4\text{O}_2 = \frac{2.5 \text{ g}}{60 \text{ g mol}^{-1}} = 0.0417 \text{ mol}$$

$$\text{Mass of benzene in kg} = 75 \text{ g} / 1000 \text{ g kg}^{-1} = 75 \times 10^{-3} \text{ kg}$$

$$\begin{aligned} \text{Molality of } \text{C}_2\text{H}_4\text{O}_2 &= \frac{\text{Moles of } \text{C}_2\text{H}_4\text{O}_2}{\text{kg of benzene}} = \frac{0.0417 \text{ mol} \times 1000 \text{ g kg}^{-1}}{75 \text{ g}} \\ &= 0.556 \text{ mol kg}^{-1} \end{aligned}$$

Solution

Intext Questions

- 1.1** Calculate the mass percentage of benzene (C_6H_6) and carbon tetrachloride (CCl_4) if 22 g of benzene is dissolved in 122 g of carbon tetrachloride.
- 1.2** Calculate the mole fraction of benzene in solution containing 30% by mass in carbon tetrachloride.
- 1.3** Calculate the molarity of each of the following solutions: (a) 30 g of $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in 4.3 L of solution (b) 30 mL of 0.5 M H_2SO_4 diluted to 500 mL .
- 1.4** Calculate the mass of urea (NH_2CONH_2) required in making 2.5 kg of 0.25 molal aqueous solution.
- 1.5** Calculate (a) molality (b) molarity and (c) mole fraction of KI if the density of 20% (mass/mass) aqueous KI is 1.202 g mL^{-1} .

1.3 Solubility

Solubility of a substance is its maximum amount that can be dissolved in a specified amount of solvent at a specified temperature. It depends upon the nature of solute and solvent as well as temperature and pressure. Let us consider the effect of these factors in solution of a solid or a gas in a liquid.

1.3.1 Solubility of a Solid in a Liquid

Every solid does not dissolve in a given liquid. While sodium chloride and sugar dissolve readily in water, naphthalene and anthracene do not. On the other hand, naphthalene and anthracene dissolve readily in benzene but sodium chloride and sugar do not. It is observed that polar solutes dissolve in polar solvents and non polar solutes in non-polar solvents. In general, a solute dissolves in a solvent if the intermolecular interactions are similar in the two or we may say **like dissolves like**.

When a solid solute is added to the solvent, some solute dissolves and its concentration increases in solution. This process is known as dissolution. Some solute particles in solution collide with the solid solute particles and get separated out of solution. This process is known as crystallisation. A stage is reached when the two processes occur at the same rate. Under such conditions, number of solute particles going into solution will be equal to the solute particles separating out and a state of dynamic equilibrium is reached.



At this stage the concentration of solute in solution will remain constant under the given conditions, i.e., temperature and pressure. Similar process is followed when gases are dissolved in liquid solvents. Such a solution in which no more solute can be dissolved at the same temperature and pressure is called a **saturated solution**. An *unsaturated solution* is one in which more solute can be dissolved at the same temperature. The solution which is in dynamic equilibrium with undissolved solute is the saturated solution and contains the maximum amount of solute dissolved in a given amount of solvent. Thus, the concentration of solute in such a solution is its solubility.

Earlier we have observed that solubility of one substance into another depends on the nature of the substances. In addition to these variables, two other parameters, i.e., temperature and pressure also control this phenomenon.

Effect of temperature

The solubility of a solid in a liquid is significantly affected by temperature changes. Consider the equilibrium represented by equation 1.10. This, being dynamic equilibrium, must follow **Le Chateliers Principle**. In general, if in a *nearly saturated solution*, the dissolution process is endothermic ($\Delta_{\text{sol}} H > 0$), the solubility should increase with rise in temperature and if it is exothermic ($\Delta_{\text{sol}} H < 0$) the solubility should decrease. These trends are also observed experimentally.

Effect of pressure

Pressure does not have any significant effect on solubility of solids in liquids. It is so because solids and liquids are highly incompressible and practically remain unaffected by changes in pressure.

1.3.2 Solubility of a Gas in a Liquid

Many gases dissolve in water. Oxygen dissolves only to a small extent in water. It is this dissolved oxygen which sustains all aquatic life. On the other hand, hydrogen chloride gas (HCl) is highly soluble in water. Solubility of gases in liquids is greatly affected by pressure and

temperature. The solubility of gases increase with increase of pressure. For solution of gases in a solvent, consider a system as shown in Fig. 1.1 (a). The lower part is solution and the upper part is gaseous system at pressure p and temperature T . Assume this system to be in a state of dynamic equilibrium, i.e., under these conditions rate of gaseous particles entering and leaving the solution phase is the same. Now increase the pressure over the solution phase by compressing the gas to a smaller volume [Fig. 1.1 (b)]. This will increase the number of gaseous particles per unit volume over the solution and also the rate at which the gaseous particles are striking the surface of solution to enter it. The solubility of the gas will increase until a new equilibrium is reached resulting in an increase in the pressure of a gas above the solution and thus its solubility increases.

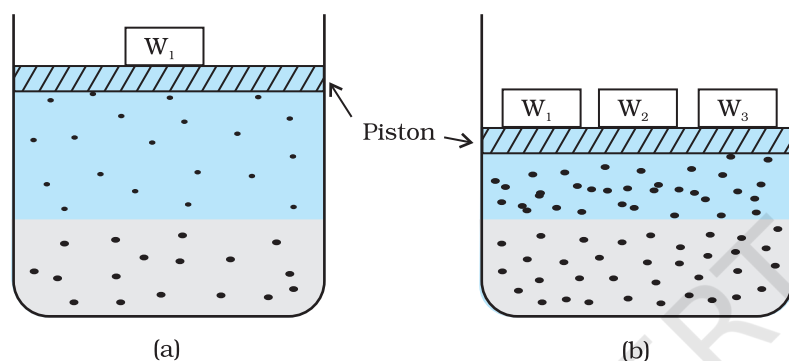


Fig. 1.1: Effect of pressure on the solubility of a gas. The concentration of dissolved gas is proportional to the pressure on the gas above the solution.

Henry was the first to give a quantitative relation between pressure and solubility of a gas in a solvent which is known as **Henry's law**. The law states that at a constant temperature, **the solubility of a gas in a liquid is directly proportional to the partial pressure of the gas present above the surface of liquid or solution**. Dalton, a contemporary of Henry, also concluded independently that the solubility of a gas in a liquid solution is a function of partial pressure of the gas. If we use the mole fraction of a gas in the solution as a measure of its solubility, then it can be said that the **mole fraction of gas in the solution is proportional to the partial pressure of the gas over the solution**. The most commonly used form of Henry's law states that "**the partial pressure of the gas in vapour phase (p) is proportional to the mole fraction of the gas (x) in the solution**" and is expressed as:

$$p = K_H x \quad (1.11)$$

Here K_H is the Henry's law constant. If we draw a graph between partial pressure of the gas versus mole fraction of the gas in solution, then we should get a plot of the type as shown in Fig. 1.2.

Different gases have different K_H values at the same temperature (Table 1.2). This suggests that K_H is a function of the nature of the gas.

It is obvious from equation (1.11) that higher the value of K_H at a given pressure, the lower is the solubility of the gas in the liquid. It can be seen from Table 1.2 that K_H values for both N_2 and O_2 increase with increase of temperature indicating that the solubility of gases

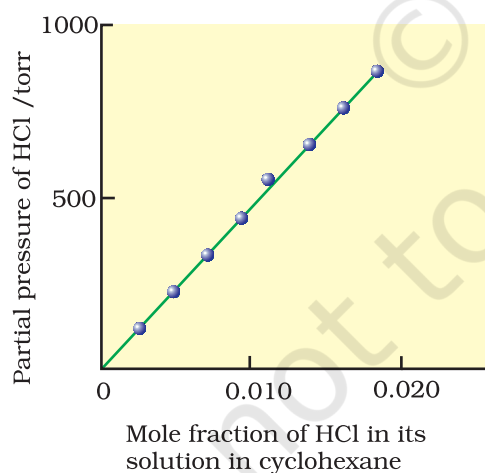


Fig. 1.2: Experimental results for the solubility of HCl gas in cyclohexane at 293 K. The slope of the line is the Henry's Law constant, K_H .

Table 1.2: Values of Henry's Law Constant for Some Selected Gases in Water

Gas	Temperature/K	K_H /kbar	Gas	Temperature/K	K_H /kbar
He	293	144.97	Argon	298	40.3
H ₂	293	69.16	CO ₂	298	1.67
N ₂	293	76.48	Formaldehyde	298	1.83×10 ⁻⁵
N ₂	303	88.84	Methane	298	0.413
O ₂	293	34.86	Vinyl chloride	298	0.611
O ₂	303	46.82			

increases with decrease of temperature. It is due to this reason that aquatic species are more comfortable in cold waters rather than in warm waters.

Example 1.4

If N₂ gas is bubbled through water at 293 K, how many millimoles of N₂ gas would dissolve in 1 litre of water? Assume that N₂ exerts a partial pressure of 0.987 bar. Given that Henry's law constant for N₂ at 293 K is 76.48 kbar.

Solution

The solubility of gas is related to the mole fraction in aqueous solution. The mole fraction of the gas in the solution is calculated by applying Henry's law. Thus:

$$x(\text{Nitrogen}) = \frac{p(\text{nitrogen})}{K_H} = \frac{0.987 \text{ bar}}{76,480 \text{ bar}} = 1.29 \times 10^{-5}$$

As 1 litre of water contains 55.5 mol of it, therefore if n represents number of moles of N₂ in solution,

$$x(\text{Nitrogen}) = \frac{n \text{ mol}}{n \text{ mol} + 55.5 \text{ mol}} = \frac{n}{55.5} = 1.29 \times 10^{-5}$$

(n in denominator is neglected as it is $\ll 55.5$)

Thus $n = 1.29 \times 10^{-5} \times 55.5 \text{ mol} = 7.16 \times 10^{-4} \text{ mol}$

$$= \frac{7.16 \times 10^{-4} \text{ mol} \times 1000 \text{ mmol}}{1 \text{ mol}} = 0.716 \text{ mmol}$$

Henry's law finds several applications in industry and explains some biological phenomena. Notable among these are:

- To increase the solubility of CO₂ in soft drinks and soda water, the bottle is sealed under high pressure.
- Scuba divers must cope with high concentrations of dissolved gases while breathing air at high pressure underwater. Increased pressure increases the solubility of atmospheric gases in blood. When the divers come towards surface, the pressure gradually decreases. This releases the dissolved gases and leads to the formation of bubbles of nitrogen in the blood. This blocks capillaries and creates a medical condition known as *bends*, which are painful and dangerous to life.

To avoid bends, as well as, the toxic effects of high concentrations of nitrogen in the blood, the tanks used by scuba divers are filled with air diluted with helium (11.7% helium, 56.2% nitrogen and 32.1% oxygen).

- At high altitudes the partial pressure of oxygen is less than that at the ground level. This leads to low concentrations of oxygen in the blood and tissues of people living at high altitudes or climbers. Low blood oxygen causes climbers to become weak and unable to think clearly, symptoms of a condition known as *anoxia*.

Effect of Temperature

Solubility of gases in liquids decreases with rise in temperature. When dissolved, the gas molecules are present in liquid phase and the process of dissolution can be considered similar to condensation and heat is evolved in this process. We have learnt in the last Section that dissolution process involves dynamic equilibrium and thus must follow **Le Chatelier's Principle**. As dissolution is an exothermic process, the solubility should decrease with increase of temperature.

Intext Questions

- 1.6** H_2S , a toxic gas with rotten egg like smell, is used for the qualitative analysis. If the solubility of H_2S in water at STP is 0.195 m, calculate Henry's law constant.
- 1.7** Henry's law constant for CO_2 in water is 1.67×10^8 Pa at 298 K. Calculate the quantity of CO_2 in 500 mL of soda water when packed under 2.5 atm CO_2 pressure at 298 K.

1.4 Vapour Pressure of Liquid Solutions

Liquid solutions are formed when solvent is a liquid. The solute can be a gas, a liquid or a solid. Solutions of gases in liquids have already been discussed in Section 1.3.2. In this Section, we shall discuss the solutions of liquids and solids in a liquid. Such solutions may contain one or more volatile components. Generally, the liquid solvent is volatile. The solute may or may not be volatile. We shall discuss the properties of only binary solutions, that is, the solutions containing two components, namely, the solutions of (i) liquids in liquids and (ii) solids in liquids.

1.4.1 Vapour Pressure of Liquid-Liquid Solutions

Let us consider a binary solution of two volatile liquids and denote the two components as 1 and 2. When taken in a closed vessel, both the components would evaporate and eventually an equilibrium would be established between vapour phase and the liquid phase. Let the total vapour pressure at this stage be p_{total} and p_1 and p_2 be the partial vapour pressures of the two components 1 and 2 respectively. These partial pressures are related to the mole fractions x_1 and x_2 of the two components 1 and 2 respectively.

The French chemist, Francois Marte Raoult (1886) gave the quantitative relationship between them. The relationship is known as the **Raoult's law** which states that **for a solution of volatile liquids,**

the partial vapour pressure of each component of the solution is directly proportional to its mole fraction present in solution.

Thus, for component 1

$$p_1 \propto x_1$$

$$\text{and } p_1 = p_1^0 x_1 \quad (1.12)$$

where p_1^0 is the vapour pressure of pure component 1 at the same temperature.

Similarly, for component 2

$$p_2 = p_2^0 x_2 \quad (1.13)$$

where p_2^0 represents the vapour pressure of the pure component 2.

According to **Dalton's law of partial pressures**, the total pressure (p_{total}) over the solution phase in the container will be the sum of the partial pressures of the components of the solution and is given as:

$$p_{\text{total}} = p_1 + p_2 \quad (1.14)$$

Substituting the values of p_1 and p_2 , we get

$$\begin{aligned} p_{\text{total}} &= x_1 p_1^0 + x_2 p_2^0 \\ &= (1 - x_2) p_1^0 + x_2 p_2^0 \end{aligned} \quad (1.15)$$

$$= p_1^0 + (p_2^0 - p_1^0) x_2 \quad (1.16)$$

Following conclusions can be drawn from equation (1.16).

- (i) Total vapour pressure over the solution can be related to the mole fraction of any one component.
- (ii) Total vapour pressure over the solution varies linearly with the mole fraction of component 2.

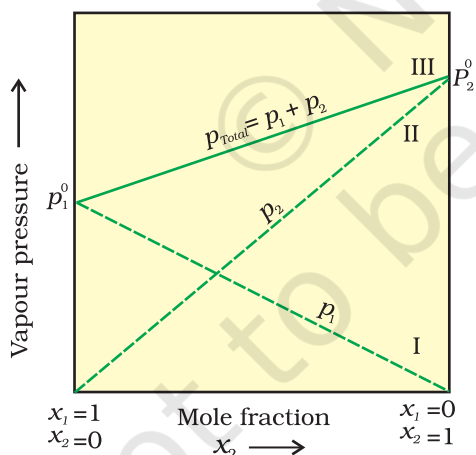


Fig. 1.3: The plot of vapour pressure and mole fraction of an ideal solution at constant temperature. The dashed lines I and II represent the partial pressure of the components. (It can be seen from the plot that p_1 and p_2 are directly proportional to x_1 and x_2 , respectively). The total vapour pressure is given by line marked III in the figure.

- (iii) Depending on the vapour pressures of the pure components 1 and 2, total vapour pressure over the solution decreases or increases with the increase of the mole fraction of component 1.

A plot of p_1 or p_2 versus the mole fractions x_1 and x_2 for a solution gives a linear plot as shown in Fig. 1.3. These lines (I and II) pass through the points for which x_1 and x_2 are equal to unity. Similarly the plot (line III) of p_{total} versus x_2 is also linear (Fig. 1.3). The minimum value of p_{total} is p_1^0 and the maximum value is p_2^0 , assuming that component 1 is less volatile than component 2, i.e., $p_1^0 < p_2^0$.

The composition of vapour phase in equilibrium with the solution is determined by the partial pressures of the components. If y_1 and y_2 are the mole fractions of the

components 1 and 2 respectively in the vapour phase then, using Dalton's law of partial pressures:

$$p_1 = y_1 p_{\text{total}} \quad (1.17)$$

$$p_2 = y_2 p_{\text{total}} \quad (1.18)$$

In general

$$p_i = y_i p_{\text{total}} \quad (1.19)$$

Vapour pressure of chloroform (CHCl_3) and dichloromethane (CH_2Cl_2) at 298 K are 200 mm Hg and 415 mm Hg respectively. (i) Calculate the vapour pressure of the solution prepared by mixing 25.5 g of CHCl_3 and 40 g of CH_2Cl_2 at 298 K and, (ii) mole fractions of each component in vapour phase. Example 1.5

Solution

(i) Molar mass of $\text{CH}_2\text{Cl}_2 = 12 \times 1 + 1 \times 2 + 35.5 \times 2 = 85 \text{ g mol}^{-1}$

Molar mass of $\text{CHCl}_3 = 12 \times 1 + 1 \times 1 + 35.5 \times 3 = 119.5 \text{ g mol}^{-1}$

Moles of $\text{CH}_2\text{Cl}_2 = \frac{40 \text{ g}}{85 \text{ g mol}^{-1}} = 0.47 \text{ mol}$

Moles of $\text{CHCl}_3 = \frac{25.5 \text{ g}}{119.5 \text{ g mol}^{-1}} = 0.213 \text{ mol}$

Total number of moles = $0.47 + 0.213 = 0.683 \text{ mol}$

$x_{\text{CH}_2\text{Cl}_2} = \frac{0.47 \text{ mol}}{0.683 \text{ mol}} = 0.688$

$x_{\text{CHCl}_3} = 1.00 - 0.688 = 0.312$

Using equation (2.16),

$$\begin{aligned} p_{\text{total}} &= p_1^0 + (p_2^0 - p_1^0) x_2 = 200 + (415 - 200) \times 0.688 \\ &= 200 + 147.9 = 347.9 \text{ mm Hg} \end{aligned}$$

(ii) Using the relation (2.19), $y_i = p_i/p_{\text{total}}$, we can calculate the mole fraction of the components in gas phase (y_i).

$p_{\text{CH}_2\text{Cl}_2} = 0.688 \times 415 \text{ mm Hg} = 285.5 \text{ mm Hg}$

$p_{\text{CHCl}_3} = 0.312 \times 200 \text{ mm Hg} = 62.4 \text{ mm Hg}$

$y_{\text{CH}_2\text{Cl}_2} = 285.5 \text{ mm Hg} / 347.9 \text{ mm Hg} = 0.82$

$y_{\text{CHCl}_3} = 62.4 \text{ mm Hg} / 347.9 \text{ mm Hg} = 0.18$

Note: Since, CH_2Cl_2 is a more volatile component than CHCl_3 , [$p_{\text{CH}_2\text{Cl}_2}^0 = 415 \text{ mm Hg}$ and $p_{\text{CHCl}_3}^0 = 200 \text{ mm Hg}$] and the vapour phase is also richer in CH_2Cl_2 [$y_{\text{CH}_2\text{Cl}_2} = 0.82$ and $y_{\text{CHCl}_3} = 0.18$], it may thus be concluded that **at equilibrium, vapour phase will be always rich in the component which is more volatile.**

1.4.2 Raoult's Law as a special case of Henry's Law

According to Raoult's law, the vapour pressure of a volatile component in a given solution is given by $p_i = x_i p_i^0$. In the solution of a gas in a liquid, one of the components is so volatile that it exists as a gas and we have already seen that its solubility is given by Henry's law which states that

$$p = K_H x.$$

If we compare the equations for Raoult's law and Henry's law, it can be seen that the partial pressure of the volatile component or gas is directly proportional to its mole fraction in solution. Only the proportionality constant K_H differs from p_i^0 . Thus, Raoult's law becomes a special case of Henry's law in which K_H becomes equal to p_i^0 .

1.4.3 Vapour Pressure of Solutions of Solids in Liquids

Another important class of solutions consists of solids dissolved in liquid, for example, sodium chloride, glucose, urea and cane sugar in water and iodine and sulphur dissolved in carbon disulphide. Some physical properties of these solutions are quite different from those of pure solvents. For example, vapour pressure. Liquids at a given

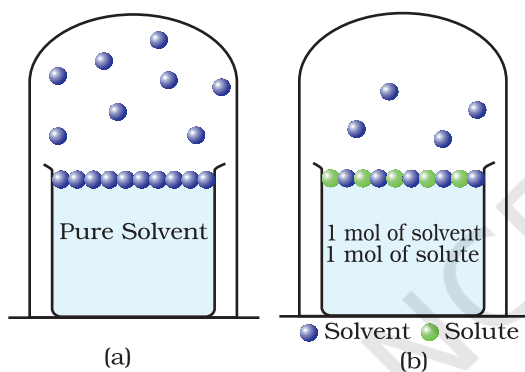


Fig. 1.4: Decrease in the vapour pressure of the solvent on account of the presence of solute in the solvent (a) evaporation of the molecules of the solvent from its surface is denoted by ●, (b) in a solution, solute particles have been denoted by ● and they also occupy part of the surface area.

temperature vapourise and under equilibrium conditions the pressure exerted by the vapours of the liquid over the liquid phase is called vapour pressure [Fig. 1.4 (a)]. In a pure liquid the entire surface is occupied by the molecules of the liquid. If a non-volatile solute is added to a solvent to give a solution [Fig. 1.4.(b)], the vapour pressure of the solution is solely from the solvent alone. This vapour pressure of the solution at a given temperature is found to be lower than the vapour pressure of the pure solvent at the same temperature. In the solution, the surface has both solute and solvent molecules; thereby the fraction of the surface covered by the solvent molecules gets reduced. Consequently, the number of solvent molecules escaping from the surface is correspondingly reduced, thus, the vapour pressure is also reduced.

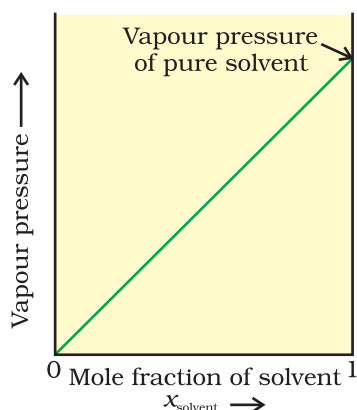
The decrease in the vapour pressure of solvent depends on the quantity of non-volatile solute present in the solution, irrespective of its nature. For example, decrease in the vapour pressure of water by adding 1.0 mol of sucrose to one kg of water is nearly similar to that produced by adding 1.0 mol of urea to the same quantity of water at the same temperature.

Raoult's law in its general form can be stated as, **for any solution the partial vapour pressure of each volatile component in the solution is directly proportional to its mole fraction.**

In a binary solution, let us denote the solvent by 1 and solute by 2. When the solute is non-volatile, only the solvent molecules are present in vapour phase and contribute to vapour pressure. Let p_1 be

Fig. 1.5

If a solution obeys Raoult's law for all concentrations, its vapour pressure would vary linearly from zero to the vapour pressure of the pure solvent.



the vapour pressure of the solvent, x_1 be its mole fraction, p_1^0 be its vapour pressure in the pure state. Then according to Raoult's law

$$p_1 \propto x_1$$
$$\text{and } p_1 = x_1 p_1^0 \quad (1.20)$$

The proportionality constant is equal to the vapour pressure of pure solvent, p_1^0 . A plot between the vapour pressure and the mole fraction of the solvent is linear (Fig. 1.5).

1.5 Ideal and Non-ideal Solutions

1.5.1 Ideal Solutions

Liquid-liquid solutions can be classified into ideal and non-ideal solutions on the basis of Raoult's law.

The solutions which obey Raoult's law over the entire range of concentration are known as *ideal solutions*. The ideal solutions have two other important properties. The enthalpy of mixing of the pure components to form the solution is zero and the volume of mixing is also zero, i.e.,

$$\Delta_{\text{mix}}H = 0, \quad \Delta_{\text{mix}}V = 0 \quad (1.21)$$

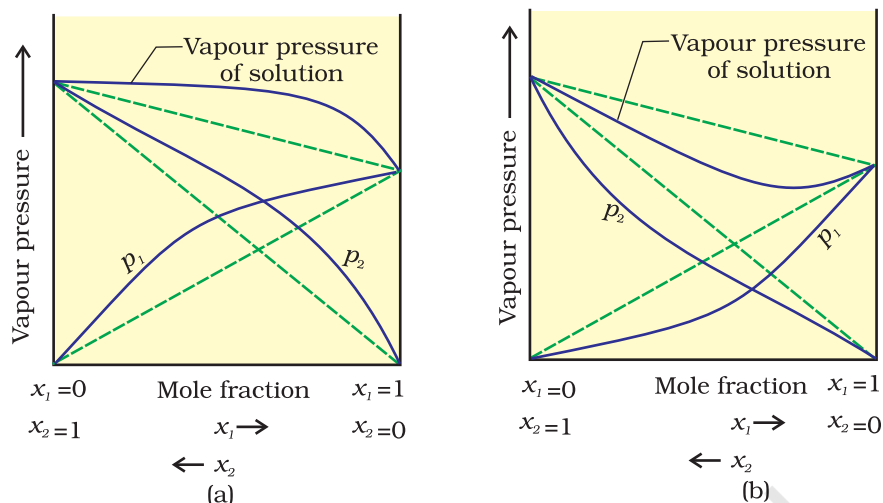
It means that no heat is absorbed or evolved when the components are mixed. Also, the volume of solution would be equal to the sum of volumes of the two components. At molecular level, ideal behaviour of the solutions can be explained by considering two components A and B. In pure components, the intermolecular attractive interactions will be of types A-A and B-B, whereas in the binary solutions in addition to these two interactions, A-B type of interactions will also be present. If the intermolecular attractive forces between the A-A and B-B are nearly equal to those between A-B, this leads to the formation of ideal solution. A perfectly ideal solution is rare but some solutions are nearly ideal in behaviour. Solution of n-hexane and n-heptane, bromoethane and chloroethane, benzene and toluene, etc. fall into this category.

1.5.2 Non-ideal Solutions

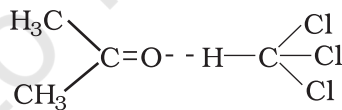
When a solution does not obey Raoult's law over the entire range of concentration, then it is called *non-ideal solution*. The vapour pressure of such a solution is either higher or lower than that predicted by Raoult's law (equation 1.16). If it is higher, the solution exhibits **positive deviation** and if it is lower, it exhibits **negative deviation** from Raoult's law. The plots of vapour pressure as a function of mole fractions for such solutions are shown in Fig. 1.6.

The cause for these deviations lie in the nature of interactions at the molecular level. In case of positive deviation from Raoult's law, A-B interactions are weaker than those between A-A or B-B, i.e., in this case the intermolecular attractive forces between the solute-solvent molecules are weaker than those between the solute-solute and solvent-solvent molecules. This means that in such solutions, molecules of A (or B) will find it easier to escape than in pure state. This will increase the vapour

The vapour pressures of two component systems as a function of composition (a) a solution that shows positive deviation from Raoult's law and (b) a solution that shows negative deviation from Raoult's law.



In case of negative deviations from Raoult's law, the intermolecular attractive forces between A-A and B-B are weaker than those between A-B and leads to decrease in vapour pressure resulting in negative deviations. An example of this type is a mixture of phenol and aniline. In this case the intermolecular hydrogen bonding between phenolic proton and lone pair on nitrogen atom of aniline is stronger than the respective intermolecular hydrogen bonding between similar molecules. Similarly, a mixture of chloroform and acetone forms a solution with negative deviation from Raoult's law. This is because chloroform molecule is able to form hydrogen bond with acetone molecule as shown.



This decreases the escaping tendency of molecules for each component and consequently the vapour pressure decreases resulting in negative deviation from Raoult's law [Fig. 1.6. (b)].

Some liquids on mixing, form **azeotropes** which are binary mixtures having the same composition in liquid and vapour phase and boil at a constant temperature. In such cases, it is not possible to separate the components by fractional distillation. There are two types of azeotropes called **minimum boiling azeotrope** and **maximum boiling azeotrope**. The solutions which show a large positive deviation from Raoult's law form minimum boiling azeotrope at a specific composition.

For example, ethanol-water mixture (obtained by fermentation of sugars) on fractional distillation gives a solution containing approximately 95% by volume of ethanol. Once this composition, known as azeotrope composition, has been achieved, the liquid and vapour have the same composition, and no further separation occurs.

The solutions that show large negative deviation from Raoult's law form maximum boiling azeotrope at a specific composition. Nitric acid and water is an example of this class of azeotrope. This azeotrope has the approximate composition, 68% nitric acid and 32% water by mass, with a boiling point of 393.5 K.

Intext Question

- 1.8** The vapour pressure of pure liquids A and B are 450 and 700 mm Hg respectively, at 350 K. Find out the composition of the liquid mixture if total vapour pressure is 600 mm Hg. Also find the composition of the vapour phase.

1.6 Colligative Properties and Determination of Molar Mass

We have learnt in Section 1.4.3 that the vapour pressure of solution decreases when a non-volatile solute is added to a volatile solvent. There are many properties of solutions which are connected with this decrease of vapour pressure. These are: (1) relative lowering of vapour pressure of the solvent (2) depression of freezing point of the solvent (3) elevation of boiling point of the solvent and (4) osmotic pressure of the solution. **All these properties depend on the number of solute particles irrespective of their nature relative to the total number of particles present in the solution. Such properties are called colligative properties** (colligative: from Latin: co means together, ligare means to bind). In the following Sections we will discuss these properties one by one.

1.6.1 Relative Lowering of Vapour Pressure

We have learnt in Section 1.4.3 that the vapour pressure of a solvent in solution is less than that of the pure solvent. Raoult established that the lowering of vapour pressure depends only on the concentration of the solute particles and it is independent of their identity. The equation (1.20) given in Section 1.4.3 establishes a relation between vapour pressure of the solution, mole fraction and vapour pressure of the solvent, i.e.,

$$p_1 = x_1 p_1^0 \quad (1.22)$$

The reduction in the vapour pressure of solvent (Δp_1) is given as:

$$\begin{aligned} \Delta p_1 &= p_1^0 - p_1 = p_1^0 - p_1^0 x_1 \\ &= p_1^0 (1 - x_1) \end{aligned} \quad (1.23)$$

Knowing that $x_2 = 1 - x_1$, equation (1.23) reduces to

$$\Delta p_1 = x_2 p_1^0 \quad (1.24)$$

In a solution containing several non-volatile solutes, the lowering of the vapour pressure depends on the sum of the mole fraction of different solutes.

Equation (1.24) can be written as

$$\frac{\Delta p_1}{p_1^0} = \frac{p_1^0 - p_1}{p_1^0} = x_2 \quad (1.25)$$

The expression on the left hand side of the equation as mentioned earlier is called **relative lowering of vapour pressure and is equal to the mole fraction of the solute**. The above equation can be written as:

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{n_2}{n_1 + n_2} \quad \left(\text{since } x_2 = \frac{n_2}{n_1 + n_2} \right) \quad (1.26)$$

Here n_1 and n_2 are the number of moles of solvent and solute respectively present in the solution. For dilute solutions $n_2 \ll n_1$, hence neglecting n_2 in the denominator we have

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{n_2}{n_1} \quad (1.27)$$

$$\text{or } \frac{p_1^0 - p_1}{p_1^0} = \frac{w_2 \times M_1}{M_2 \times w_1} \quad (1.28)$$

Here w_1 and w_2 are the masses and M_1 and M_2 are the molar masses of the solvent and solute respectively.

From this equation (1.28), knowing all other quantities, the molar mass of solute (M_2) can be calculated.

Example 1.6

The vapour pressure of pure benzene at a certain temperature is 0.850 bar. A non-volatile, non-electrolyte solid weighing 0.5 g when added to 39.0 g of benzene (molar mass 78 g mol⁻¹). Vapour pressure of the solution, then, is 0.845 bar. What is the molar mass of the solid substance?

Solution

The various quantities known to us are as follows:

$p_1^0 = 0.850$ bar; $p = 0.845$ bar; $M_1 = 78$ g mol⁻¹; $w_2 = 0.5$ g; $w_1 = 39$ g

Substituting these values in equation (2.28), we get

$$\frac{0.850 \text{ bar} - 0.845 \text{ bar}}{0.850 \text{ bar}} = \frac{0.5 \text{ g} \times 78 \text{ g mol}^{-1}}{M_2 \times 39 \text{ g}}$$

Therefore, $M_2 = 170$ g mol⁻¹

1.6.2 Elevation of Boiling Point

The vapour pressure of a liquid increases with increase of temperature. It boils at the temperature at which its vapour pressure is equal to the atmospheric pressure. For example, water boils at 373.15 K (100° C) because at this temperature the vapour pressure of water is 1.013 bar (1 atmosphere). We have also learnt in the last section that vapour pressure of the solvent decreases in the presence of non-volatile solute. Fig. 1.7 depicts the variation of vapour pressure of the pure solvent and solution as a function of temperature. For example, the vapour pressure of an aqueous solution of sucrose is less than 1.013 bar at 373.15 K. In order to make this solution boil, its vapour pressure must be increased to 1.013 bar by raising the temperature above the boiling temperature of the pure solvent (water). Thus, the boiling point of a solution is

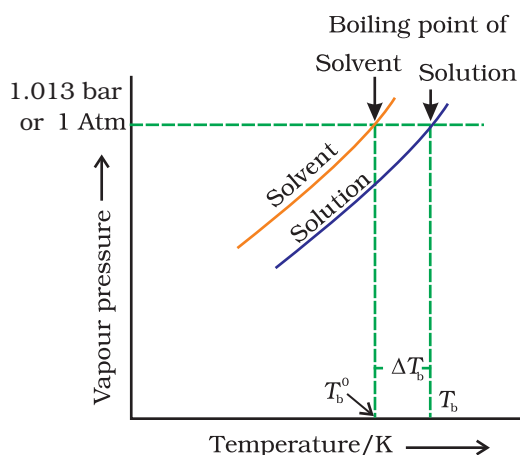


Fig. 1.7: The vapour pressure curve for solution lies below the curve for pure water. The diagram shows that ΔT_b denotes the elevation of boiling point of a solvent in solution.

always higher than that of the boiling point of the pure solvent in which the solution is prepared as shown in Fig. 1.7. Similar to lowering of vapour pressure, the elevation of boiling point also depends on the number of solute molecules rather than their nature. A solution of 1 mol of sucrose in 1000 g of water boils at 373.52 K at one atmospheric pressure.

Let T_b^0 be the boiling point of pure solvent and T_b be the boiling point of solution. The increase in the boiling point $\Delta T_b = T_b - T_b^0$ is known as **elevation of boiling point**.

Experiments have shown that for **dilute solutions** the elevation of boiling point (ΔT_b) is directly proportional to the molal concentration of the solute in a solution. Thus

$$\Delta T_b \propto m \quad (1.29)$$

$$\text{or } \Delta T_b = K_b m \quad (1.30)$$

Here m (molality) is the number of moles of solute dissolved in 1 kg of solvent and the constant of proportionality, K_b is called **Boiling Point Elevation Constant or Molal Elevation Constant (Ebullioscopic Constant)**. The unit of K_b is K kg mol^{-1} . Values of K_b for some common solvents are given in Table 1.3. If w_2 gram of solute of molar mass M_2 is dissolved in w_1 gram of solvent, then molality, m of the solution is given by the expression:

$$m = \frac{w_2/M_2}{w_1/1000} = \frac{1000 \times w_2}{M_2 \times w_1} \quad (1.31)$$

Substituting the value of molality in equation (1.30) we get

$$\Delta T_b = \frac{K_b \times 1000 \times w_2}{M_2 \times w_1} \quad (1.32)$$

$$M_2 = \frac{1000 \times w_2 \times K_b}{\Delta T_b \times w_1} \quad (1.33)$$

Thus, in order to determine M_2 , molar mass of the solute, known mass of solute in a known mass of the solvent is taken and ΔT_b is determined experimentally for a known solvent whose K_b value is known.

18 g of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is dissolved in 1 kg of water in a saucepan. At what temperature will water boil at 1.013 bar? K_b for water is $0.52 \text{ K kg mol}^{-1}$.

Example 1.7

Moles of glucose = $18 \text{ g} / 180 \text{ g mol}^{-1} = 0.1 \text{ mol}$
 Number of kilograms of solvent = 1 kg
 Thus molality of glucose solution = 0.1 mol kg^{-1}
 For water, change in boiling point

Solution

$$\Delta T_b = K_b \times m = 0.52 \text{ K kg mol}^{-1} \times 0.1 \text{ mol kg}^{-1} = 0.052 \text{ K}$$

Since water boils at 373.15 K at 1.013 bar pressure, therefore, the boiling point of solution will be $373.15 + 0.052 = 373.202 \text{ K}$.

Example 1.8

The boiling point of benzene is 353.23 K. When 1.80 g of a non-volatile solute was dissolved in 90 g of benzene, the boiling point is raised to 354.11 K. Calculate the molar mass of the solute. K_b for benzene is $2.53 \text{ K kg mol}^{-1}$

Solution

The elevation (ΔT_b) in the boiling point = $354.11 \text{ K} - 353.23 \text{ K} = 0.88 \text{ K}$
Substituting these values in expression (2.33) we get

$$M_2 = \frac{2.53 \text{ K kg mol}^{-1} \times 1.8 \text{ g} \times 1000 \text{ g kg}^{-1}}{0.88 \text{ K} \times 90 \text{ g}} = 58 \text{ g mol}^{-1}$$

Therefore, molar mass of the solute, $M_2 = 58 \text{ g mol}^{-1}$

1.6.3 Depression of Freezing Point

The lowering of vapour pressure of a solution causes a lowering of the freezing point compared to that of the pure solvent (Fig. 1.8). We know that at the freezing point of a substance, the solid phase is in dynamic equilibrium with the liquid phase. Thus, the freezing point of a substance may be defined as the temperature at which the vapour pressure of the substance in its liquid phase is equal to its vapour pressure in the solid phase. A solution will freeze when its vapour pressure equals the vapour pressure of the pure solid solvent as is clear from Fig. 1.8. According to Raoult's law, when a non-volatile solid is added to the solvent its vapour pressure decreases and now it would become equal to that of solid solvent at lower temperature. Thus, the freezing point of the solvent decreases.

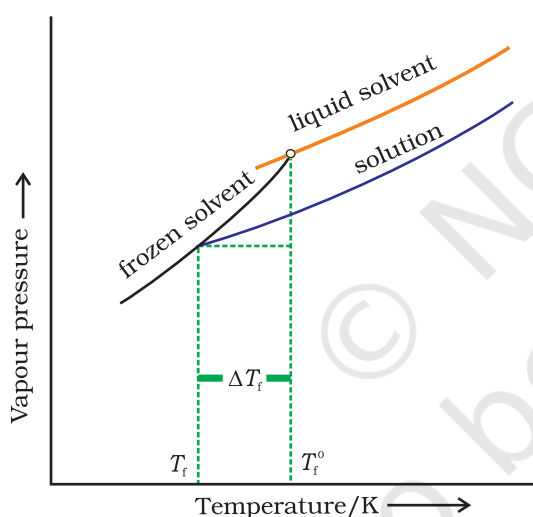


Fig. 1.8: Diagram showing ΔT_f , depression of the freezing point of a solvent in a solution.

Let T_f^0 be the freezing point of pure solvent and T_f be its freezing point when non-volatile solute is dissolved in it. The decrease in freezing point.

$\Delta T_f = T_f^0 - T_f$ is known as depression in freezing point.

Similar to elevation of boiling point, depression of freezing point (ΔT_f) for **dilute solution** (ideal solution) is directly proportional to molality, m of the solution. Thus,

$$\Delta T_f \propto m$$

$$\text{or } \Delta T_f = K_f m \quad (1.34)$$

The proportionality constant, K_f , which depends on the nature of the solvent is known as **Freezing Point Depression Constant or Molal**

Depression Constant or Cryoscopic Constant. The unit of K_f is K kg mol^{-1} . Values of K_f for some common solvents are listed in Table 1.3.

If w_2 gram of the solute having molar mass as M_2 , present in w_1 gram of solvent, produces the depression in freezing point ΔT_f of the solvent then molality of the solute is given by the equation (1.31).

$$m = \frac{w_2 / M_2}{w_1 / 1000} \quad (1.31)$$

Substituting this value of molality in equation (1.34) we get:

$$\Delta T_f = \frac{K_f \times w_2 / M_2}{w_1 / 1000}$$

$$\Delta T_f = \frac{K_f \times w_2 \times 1000}{M_2 \times w_1} \quad (1.35)$$

$$M_2 = \frac{K_f \times w_2 \times 1000}{\Delta T_f \times w_1} \quad (1.36)$$

Thus for determining the molar mass of the solute we should know the quantities w_1 , w_2 , ΔT_f , along with the molal freezing point depression constant.

The values of K_f and K_b , which depend upon the nature of the solvent, can be ascertained from the following relations.

$$K_f = \frac{R \times M_1 \times T_f^2}{1000 \times \Delta_{\text{fus}} H} \quad (1.37)$$

$$K_b = \frac{R \times M_1 \times T_b^2}{1000 \times \Delta_{\text{vap}} H} \quad (1.38)$$

Here the symbols R and M_1 stand for the gas constant and molar mass of the solvent, respectively and T_f and T_b denote the freezing point and the boiling point of the pure solvent respectively in kelvin. Further, $\Delta_{\text{fus}} H$ and $\Delta_{\text{vap}} H$ represent the enthalpies for the fusion and vapourisation of the solvent, respectively.

Table 1.3: Molal Boiling Point Elevation and Freezing Point Depression Constants for Some Solvents

Solvent	b. p./K	$K_b/\text{K kg mol}^{-1}$	f. p./K	$K_f/\text{K kg mol}^{-1}$
Water	373.15	0.52	273.0	1.86
Ethanol	351.5	1.20	155.7	1.99
Cyclohexane	353.74	2.79	279.55	20.00
Benzene	353.3	2.53	278.6	5.12
Chloroform	334.4	3.63	209.6	4.79
Carbon tetrachloride	350.0	5.03	250.5	31.8
Carbon disulphide	319.4	2.34	164.2	3.83
Diethyl ether	307.8	2.02	156.9	1.79
Acetic acid	391.1	2.93	290.0	3.90

Example 1.9

45 g of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) is mixed with 600 g of water. Calculate (a) the freezing point depression and (b) the freezing point of the solution.

Solution

Depression in freezing point is related to the molality, therefore, the molality of the solution with respect to ethylene glycol = $\frac{\text{moles of ethylene glycol}}{\text{mass of water in kilogram}}$

$$\text{Moles of ethylene glycol} = \frac{45 \text{ g}}{62 \text{ g mol}^{-1}} = 0.73 \text{ mol}$$

$$\text{Mass of water in kg} = \frac{600 \text{ g}}{1000 \text{ g kg}^{-1}} = 0.6 \text{ kg}$$

$$\text{Hence molality of ethylene glycol} = \frac{0.73 \text{ mol}}{0.60 \text{ kg}} = 1.2 \text{ mol kg}^{-1}$$

Therefore freezing point depression,

$$\Delta T_f = 1.86 \text{ K kg mol}^{-1} \times 1.2 \text{ mol kg}^{-1} = 2.2 \text{ K}$$

$$\text{Freezing point of the aqueous solution} = 273.15 \text{ K} - 2.2 \text{ K} = 270.95 \text{ K}$$

Example 1.10

1.00 g of a non-electrolyte solute dissolved in 50 g of benzene lowered the freezing point of benzene by 0.40 K. The freezing point depression constant of benzene is $5.12 \text{ K kg mol}^{-1}$. Find the molar mass of the solute.

Solution

Substituting the values of various terms involved in equation (1.36) we get,

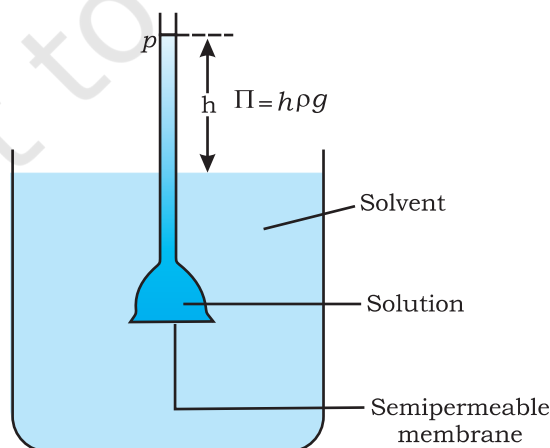
$$M_2 = \frac{5.12 \text{ K kg mol}^{-1} \times 1.00 \text{ g} \times 1000 \text{ g kg}^{-1}}{0.40 \times 50 \text{ g}} = 256 \text{ g mol}^{-1}$$

$$\text{Thus, molar mass of the solute} = 256 \text{ g mol}^{-1}$$

1.6.4 Osmosis and Osmotic Pressure

There are many phenomena which we observe in nature or at home. For example, raw mangoes shrivel when pickled in brine (salt water); wilted flowers revive when placed in fresh water, blood cells collapse when suspended in saline water, etc. If we look into these processes we

find one thing common in all, that is, all these substances are bound by membranes. These membranes can be of animal or vegetable origin and these occur naturally such as pig's bladder or parchment or can be synthetic such as cellophane. These membranes appear to be continuous sheets or films, yet they contain a network of submicroscopic holes or pores. Small solvent

**Fig. 1.9**

Level of solution rises in the thistle funnel due to osmosis of solvent.

molecules, like water, can pass through these holes but the passage of bigger molecules like solute is hindered. Membranes having this kind of properties are known as *semipermeable membranes* (SPM).

Assume that only solvent molecules can pass through these semi-permeable membranes. If this membrane is placed between the solvent and solution as shown in Fig. 1.9, the solvent molecules will flow through the membrane from pure solvent to the solution. **This process of flow of the solvent is called osmosis.**

The flow will continue till the equilibrium is attained. The flow of the solvent from its side to solution side across a semipermeable membrane can be stopped if some extra pressure is applied on the solution. **This pressure that just stops the flow of solvent is called osmotic pressure of the solution.** The flow of solvent from dilute solution to the concentrated solution across a semipermeable membrane is due to osmosis. The important point to be kept in mind is that solvent molecules always flow from lower concentration to higher concentration of solution. The osmotic pressure has been found to depend on the concentration of the solution.

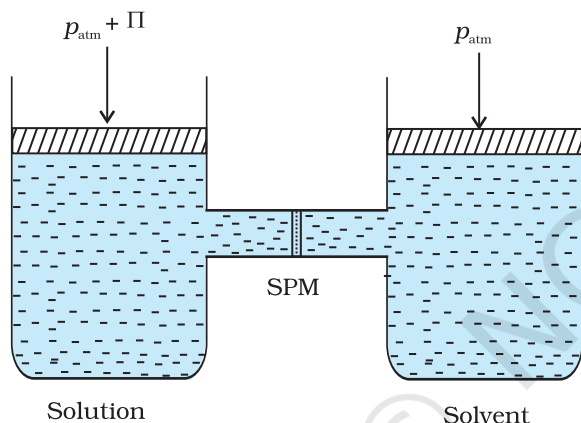


Fig. 1.10: The excess pressure equal to the osmotic pressure must be applied on the solution side to prevent osmosis.

The osmotic pressure of a solution is the excess pressure that must be applied to a solution to prevent osmosis, i.e., to stop the passage of solvent molecules through a semipermeable membrane into the solution. This is illustrated in Fig. 1.10. Osmotic pressure is a colligative property as it depends on the number of solute molecules and not on their identity. For dilute solutions, it has been found experimentally that **osmotic pressure is proportional to the molarity, C of the solution at a given temperature T .** Thus:

$$\Pi = C R T \quad (1.39)$$

Here Π is the osmotic pressure and R is the gas constant.

$$\Pi = (n_2 / V) R T \quad (1.40)$$

Here V is volume of a solution in litres containing n_2 moles of solute. If w_2 grams of solute, of molar mass, M_2 is present in the solution, then $n_2 = w_2 / M_2$ and we can write,

$$\Pi V = \frac{w_2 R T}{M_2} \quad (1.41)$$

$$\text{or } M_2 = \frac{w_2 R T}{\Pi V} \quad (1.42)$$

Thus, knowing the quantities w_2 , T , Π and V we can calculate the molar mass of the solute.

Measurement of osmotic pressure provides another method of determining molar masses of solutes. This method is widely used to determine molar masses of proteins, polymers and other

macromolecules. The osmotic pressure method has the advantage over other methods as pressure measurement is around the room temperature and the molarity of the solution is used instead of molality. As compared to other colligative properties, its magnitude is large even for very dilute solutions. The technique of osmotic pressure for determination of molar mass of solutes is particularly useful for biomolecules as they are generally not stable at higher temperatures and polymers have poor solubility.

Two solutions having same osmotic pressure at a given temperature are called isotonic solutions. When such solutions are separated by semipermeable membrane no osmosis occurs between them. For example, the osmotic pressure associated with the fluid inside the blood cell is equivalent to that of 0.9% (mass/volume) sodium chloride solution, called normal saline solution and it is safe to inject intravenously. On the other hand, if we place the cells in a solution containing more than 0.9% (mass/volume) sodium chloride, water will flow out of the cells and they would shrink. Such a solution is called **hypertonic**. If the salt concentration is less than 0.9% (mass/volume), the solution is said to be **hypotonic**. In this case, water will flow into the cells if placed in this solution and they would swell.

Example 1.11

200 cm³ of an aqueous solution of a protein contains 1.26 g of the protein. The osmotic pressure of such a solution at 300 K is found to be 2.57×10^{-3} bar. Calculate the molar mass of the protein.

Solution

The various quantities known to us are as follows: $\Pi = 2.57 \times 10^{-3}$ bar,

$$V = 200 \text{ cm}^3 = 0.200 \text{ litre}$$

$$T = 300 \text{ K}$$

$$R = 0.083 \text{ L bar mol}^{-1} \text{ K}^{-1}$$

Substituting these values in equation (2.42) we get

$$M_2 = \frac{1.26 \text{ g} \times 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1} \times 300 \text{ K}}{2.57 \times 10^{-3} \text{ bar} \times 0.200 \text{ L}} = 61,022 \text{ g mol}^{-1}$$

The phenomena mentioned in the beginning of this section can be explained on the basis of osmosis. A raw mango placed in concentrated salt solution loses water via osmosis and shrivel into pickle. Wilted flowers revive when placed in fresh water. A carrot that has become limp because of water loss into the atmosphere can be placed into the water making it firm once again. Water will move into its cells through osmosis. When placed in water containing less than 0.9% (mass/volume) salt, blood cells swell due to flow of water in them by osmosis. People taking a lot of salt or salty food experience water retention in tissue cells and intercellular spaces because of osmosis. The resulting

puffiness or swelling is called **edema**. Water movement from soil into plant roots and subsequently into upper portion of the plant is partly due to osmosis. The preservation of meat by salting and of fruits by adding sugar protects against bacterial action. Through the process of osmosis, a bacterium on salted meat or candid fruit loses water, shrivels and dies.

1.6.5 Reverse Osmosis and Water Purification

The direction of osmosis can be reversed if a pressure larger than the osmotic pressure is applied to the solution side. That is, now the pure solvent flows out of the solution through the semi permeable membrane. This phenomenon is called **reverse osmosis** and is of great practical utility. Reverse osmosis is used in desalination of sea water. A schematic set up for the process is shown in Fig. 1.11.

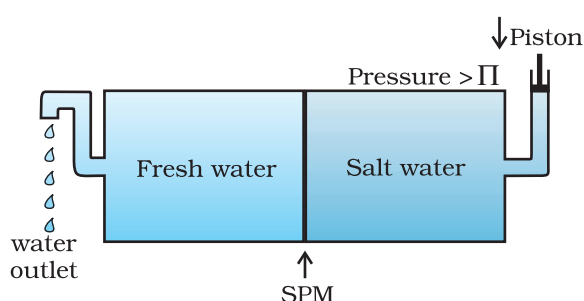


Fig. 1.11: Reverse osmosis occurs when a pressure larger than the osmotic pressure is applied to the solution.

When pressure more than osmotic pressure is applied, pure water is squeezed out of the sea water through the membrane. A variety of polymer membranes are available for this purpose.

The pressure required for the reverse osmosis is quite high. A workable porous membrane is a film of cellulose acetate placed over a suitable support. Cellulose acetate is permeable to water but impermeable to impurities and ions present in sea water. These days many countries use desalination plants to meet their potable water requirements.

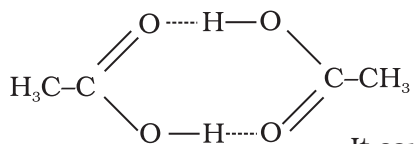
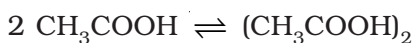
Intext Questions

- 1.9** Vapour pressure of pure water at 298 K is 23.8 mm Hg. 50 g of urea (NH_2CONH_2) is dissolved in 850 g of water. Calculate the vapour pressure of water for this solution and its relative lowering.
- 1.10** Boiling point of water at 750 mm Hg is 99.63°C. How much sucrose is to be added to 500 g of water such that it boils at 100°C.
- 1.11** Calculate the mass of ascorbic acid (Vitamin C, $\text{C}_6\text{H}_8\text{O}_6$) to be dissolved in 75 g of acetic acid to lower its melting point by 1.5°C. $K_f = 3.9 \text{ K kg mol}^{-1}$.
- 1.12** Calculate the osmotic pressure in pascals exerted by a solution prepared by dissolving 1.0 g of polymer of molar mass 185,000 in 450 mL of water at 37°C.

1.7 Abnormal Molar Masses

We know that ionic compounds when dissolved in water dissociate into cations and anions. For example, if we dissolve one mole of KCl (74.5 g) in water, we expect one mole each of K^+ and Cl^- ions to be released in the solution. If this happens, there would be two moles of particles in the solution. If we ignore interionic attractions, one mole of KCl in one kg of water would be expected to increase the boiling point by $2 \times 0.52 \text{ K} = 1.04 \text{ K}$. Now if we did not know about the degree of

dissociation, we could be led to conclude that the mass of 2 mol particles is 74.5 g and the mass of one mole of KCl would be 37.25 g. This brings into light the rule that, when there is dissociation of solute into ions, the experimentally determined molar mass is always lower than the true value.



Molecules of ethanoic acid (acetic acid) dimerise in benzene due to hydrogen bonding. This normally happens in solvents of low dielectric constant. In this case the number of particles is reduced due to dimerisation. Association of molecules is depicted as follows:

It can be undoubtedly stated here that if all the molecules of ethanoic acid associate in benzene, then ΔT_b or ΔT_f for ethanoic acid will be half of the normal value. The molar mass calculated on the basis of this ΔT_b or ΔT_f will, therefore, be twice the expected value. Such a molar mass that is either lower or higher than the expected or normal value is called as **abnormal molar mass**.

In 1880 van't Hoff introduced a factor i , known as the van't Hoff factor, to account for the extent of dissociation or association. This factor i is defined as:

$$i = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$$

$$= \frac{\text{Observed colligative property}}{\text{Calculated colligative property}}$$

$$i = \frac{\text{Total number of moles of particles after association/dissociation}}{\text{Number of moles of particles before association/dissociation}}$$

Here abnormal molar mass is the experimentally determined molar mass and calculated **colligative properties** are obtained by assuming that the non-volatile solute is neither associated nor dissociated. In case of association, value of i is less than unity while for dissociation it is greater than unity. For example, the value of i for aqueous KCl solution is close to 2, while the value for ethanoic acid in benzene is nearly 0.5.

Inclusion of van't Hoff factor modifies the equations for colligative properties as follows:

Relative lowering of vapour pressure of solvent,

$$\frac{p_1^0 - p_1}{p_1^0} = i \cdot \frac{n_2}{n_1}$$

Elevation of Boiling point, $\Delta T_b = i K_b m$

Depression of Freezing point, $\Delta T_f = i K_f m$

Osmotic pressure of solution, $\Pi = i n_2 R T / V$

Table 1.4 depicts values of the factor, i for several strong electrolytes. For KCl, NaCl and MgSO_4 , i values approach 2 as the solution becomes very dilute. As expected, the value of i gets close to 3 for K_2SO_4 .

Table 1.4: Values of van't Hoff factor, i , at Various Concentrations for NaCl, KCl, MgSO_4 and K_2SO_4 .

Salt	*Values of i			van't Hoff Factor i for complete dissociation of solute
	0.1 m	0.01 m	0.001 m	
NaCl	1.87	1.94	1.97	2.00
KCl	1.85	1.94	1.98	2.00
MgSO_4	1.21	1.53	1.82	2.00
K_2SO_4	2.32	2.70	2.84	3.00

* represent i values for incomplete dissociation.

2 g of benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$) dissolved in 25 g of benzene shows a depression in freezing point equal to 1.62 K. Molal depression constant for benzene is $4.9 \text{ K kg mol}^{-1}$. What is the percentage association of acid if it forms dimer in solution? **Example 1.12**

The given quantities are: $w_2 = 2 \text{ g}$; $K_f = 4.9 \text{ K kg mol}^{-1}$; $w_1 = 25 \text{ g}$,

$$\Delta T_f = 1.62 \text{ K}$$

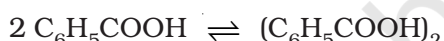
Substituting these values in equation (1.36) we get:

$$M_2 = \frac{4.9 \text{ K kg mol}^{-1} \times 2 \text{ g} \times 1000 \text{ g kg}^{-1}}{25 \text{ g} \times 1.62 \text{ K}} = 241.98 \text{ g mol}^{-1}$$

Thus, experimental molar mass of benzoic acid in benzene is

$$= 241.98 \text{ g mol}^{-1}$$

Now consider the following equilibrium for the acid:



If x represents the degree of association of the solute then we would have $(1 - x)$ mol of benzoic acid left in unassociated form and

correspondingly $\frac{x}{2}$ as associated moles of benzoic acid at equilibrium.

Therefore, total number of moles of particles at equilibrium is:

$$1 - x + \frac{x}{2} = 1 - \frac{x}{2}$$

Thus, total number of moles of particles at equilibrium equals van't Hoff factor i .

$$\text{But } i = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$$

Solution

$$= \frac{122 \text{ g mol}^{-1}}{241.98 \text{ g mol}^{-1}}$$

$$\text{or } \frac{x}{2} = 1 - \frac{122}{241.98} = 1 - 0.504 = 0.496$$

$$\text{or } x = 2 \times 0.496 = 0.992$$

Therefore, degree of association of benzoic acid in benzene is 99.2 %.

Example 1.13

0.6 mL of acetic acid (CH_3COOH), having density 1.06 g mL^{-1} , is dissolved in 1 litre of water. The depression in freezing point observed for this strength of acid was 0.0205°C . Calculate the van't Hoff factor and the dissociation constant of acid.

Solution

$$\begin{aligned} \text{Number of moles of acetic acid} &= \frac{0.6 \text{ mL} \times 1.06 \text{ g mL}^{-1}}{60 \text{ g mol}^{-1}} \\ &= 0.0106 \text{ mol} = n \end{aligned}$$

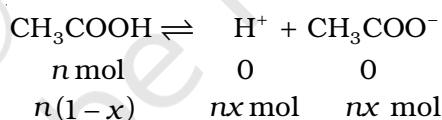
$$\text{Molality} = \frac{0.0106 \text{ mol}}{1000 \text{ mL} \times 1 \text{ g mL}^{-1}} = 0.0106 \text{ mol kg}^{-1}$$

Using equation (1.35)

$$\Delta T_f = 1.86 \text{ K kg mol}^{-1} \times 0.0106 \text{ mol kg}^{-1} = 0.0197 \text{ K}$$

$$\text{van't Hoff Factor } (i) = \frac{\text{Observed freezing point}}{\text{Calculated freezing point}} = \frac{0.0205 \text{ K}}{0.0197 \text{ K}} = 1.041$$

Acetic acid is a weak electrolyte and will dissociate into two ions: acetate and hydrogen ions per molecule of acetic acid. If x is the degree of dissociation of acetic acid, then we would have $n(1-x)$ moles of undissociated acetic acid, nx moles of CH_3COO^- and nx moles of H^+ ions,



Thus total moles of particles are: $n(1-x+x+x) = n(1+x)$

$$i = \frac{n(1+x)}{n} = 1+x = 1.041$$

Thus degree of dissociation of acetic acid $= x = 1.041 - 1.000 = 0.041$

Then $[\text{CH}_3\text{COOH}] = n(1-x) = 0.0106(1-0.041)$,

$[\text{CH}_3\text{COO}^-] = nx = 0.0106 \times 0.041$, $[\text{H}^+] = nx = 0.0106 \times 0.041$.

$$\begin{aligned} K_a &= \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]} = \frac{0.0106 \times 0.041 \times 0.0106 \times 0.041}{0.0106(1.00 - 0.041)} \\ &= 1.86 \times 10^{-5} \end{aligned}$$

Summary

A solution is a homogeneous mixture of two or more substances. Solutions are classified as solid, liquid and gaseous solutions. The concentration of a solution is expressed in terms of mole fraction, molarity, molality and in percentages. The dissolution of a gas in a liquid is governed by **Henry's law**, according to which, at a given temperature, the **solubility of a gas in a liquid is directly proportional to the partial pressure of the gas**. The vapour pressure of the solvent is lowered by the presence of a non-volatile solute in the solution and this lowering of vapour pressure of the solvent is governed by Raoult's law, according to which the **relative lowering of vapour pressure of the solvent over a solution is equal to the mole fraction of a non-volatile solute present in the solution**. However, in a binary liquid solution, if both the components of the solution are volatile then another form of Raoult's law is used. Mathematically, this form of the Raoult's law is stated as: $p_{\text{total}} = p_1^0 x_1 + p_2^0 x_2$. **Solutions which obey Raoult's law over the entire range of concentration are called ideal solutions**. Two types of deviations from Raoult's law, called positive and negative deviations are observed. Azeotropes arise due to very large deviations from Raoult's law.

The properties of solutions which depend on the number of solute particles and are independent of their chemical identity are called colligative properties. These are lowering of vapour pressure, elevation of boiling point, depression of freezing point and osmotic pressure. The process of osmosis can be reversed if a pressure higher than the osmotic pressure is applied to the solution. Colligative properties have been used to determine the molar mass of solutes. Solutes which dissociate in solution exhibit molar mass lower than the actual molar mass and those which associate show higher molar mass than their actual values.

Quantitatively, the extent to which a solute is dissociated or associated can be expressed by van't Hoff factor i . This factor has been defined as ratio of normal molar mass to experimentally determined molar mass or as the ratio of observed colligative property to the calculated colligative property.

Exercises

- 1.1 Define the term solution. How many types of solutions are formed? Write briefly about each type with an example.
- 1.2 Give an example of a solid solution in which the solute is a gas.
- 1.3 Define the following terms:
(i) Mole fraction (ii) Molality (iii) Molarity (iv) Mass percentage.
- 1.4 Concentrated nitric acid used in laboratory work is 68% nitric acid by mass in aqueous solution. What should be the molarity of such a sample of the acid if the density of the solution is 1.504 g mL^{-1} ?

- 1.5** A solution of glucose in water is labelled as 10% w/w, what would be the molality and mole fraction of each component in the solution? If the density of solution is 1.2 g mL^{-1} , then what shall be the molarity of the solution?
- 1.6** How many mL of 0.1 M HCl are required to react completely with 1 g mixture of Na_2CO_3 and NaHCO_3 containing equimolar amounts of both?
- 1.7** A solution is obtained by mixing 300 g of 25% solution and 400 g of 40% solution by mass. Calculate the mass percentage of the resulting solution.
- 1.8** An antifreeze solution is prepared from 222.6 g of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) and 200 g of water. Calculate the molality of the solution. If the density of the solution is 1.072 g mL^{-1} , then what shall be the molarity of the solution?
- 1.9** A sample of drinking water was found to be severely contaminated with chloroform (CHCl_3) supposed to be a carcinogen. The level of contamination was 15 ppm (by mass):
- express this in percent by mass
 - determine the molality of chloroform in the water sample.
- 1.10** What role does the molecular interaction play in a solution of alcohol and water?
- 1.11** Why do gases always tend to be less soluble in liquids as the temperature is raised?
- 1.12** State Henry's law and mention some important applications.
- 1.13** The partial pressure of ethane over a solution containing $6.56 \times 10^{-3} \text{ g}$ of ethane is 1 bar. If the solution contains $5.00 \times 10^{-2} \text{ g}$ of ethane, then what shall be the partial pressure of the gas?
- 1.14** What is meant by positive and negative deviations from Raoult's law and how is the sign of $\Delta_{\text{mix}}H$ related to positive and negative deviations from Raoult's law?
- 1.15** An aqueous solution of 2% non-volatile solute exerts a pressure of 1.004 bar at the normal boiling point of the solvent. What is the molar mass of the solute?
- 1.16** Heptane and octane form an ideal solution. At 373 K, the vapour pressures of the two liquid components are 105.2 kPa and 46.8 kPa respectively. What will be the vapour pressure of a mixture of 26.0 g of heptane and 35 g of octane?
- 1.17** The vapour pressure of water is 12.3 kPa at 300 K. Calculate vapour pressure of 1 molal solution of a non-volatile solute in it.
- 1.18** Calculate the mass of a non-volatile solute (molar mass 40 g mol^{-1}) which should be dissolved in 114 g octane to reduce its vapour pressure to 80%.
- 1.19** A solution containing 30 g of non-volatile solute exactly in 90 g of water has a vapour pressure of 2.8 kPa at 298 K. Further, 18 g of water is then added to the solution and the new vapour pressure becomes 2.9 kPa at 298 K. Calculate:
- molar mass of the solute
 - vapour pressure of water at 298 K.
- 1.20** A 5% solution (by mass) of cane sugar in water has freezing point of 271 K. Calculate the freezing point of 5% glucose in water if freezing point of pure water is 273.15 K.
- 1.21** Two elements A and B form compounds having formula AB_2 and AB_4 . When dissolved in 20 g of benzene (C_6H_6), 1 g of AB_2 lowers the freezing point by 2.3 K whereas 1.0 g of AB_4 lowers it by 1.3 K. The molar depression constant for benzene is $5.1 \text{ K kg mol}^{-1}$. Calculate atomic masses of A and B.

- 1.22** At 300 K, 36 g of glucose present in a litre of its solution has an osmotic pressure of 4.98 bar. If the osmotic pressure of the solution is 1.52 bars at the same temperature, what would be its concentration?
- 1.23** Suggest the most important type of intermolecular attractive interaction in the following pairs.
- (i) n-hexane and n-octane
 - (ii) I_2 and CCl_4
 - (iii) $NaClO_4$ and water
 - (iv) methanol and acetone
 - (v) acetonitrile (CH_3CN) and acetone (C_3H_6O).
- 1.24** Based on solute-solvent interactions, arrange the following in order of increasing solubility in n-octane and explain. Cyclohexane, KCl, CH_3OH , CH_3CN .
- 1.25** Amongst the following compounds, identify which are insoluble, partially soluble and highly soluble in water?
- (i) phenol
 - (ii) toluene
 - (iii) formic acid
 - (iv) ethylene glycol
 - (v) chloroform
 - (vi) pentanol.
- 1.26** If the density of some lake water is 1.25 g mL^{-1} and contains 92 g of Na^+ ions per kg of water, calculate the molarity of Na^+ ions in the lake.
- 1.27** If the solubility product of CuS is 6×10^{-16} , calculate the maximum molarity of CuS in aqueous solution.
- 1.28** Calculate the mass percentage of aspirin ($C_9H_8O_4$) in acetonitrile (CH_3CN) when 6.5 g of $C_9H_8O_4$ is dissolved in 450 g of CH_3CN .
- 1.29** Nalorphene ($C_{19}H_{21}NO_3$), similar to morphine, is used to combat withdrawal symptoms in narcotic users. Dose of nalorphene generally given is 1.5 mg. Calculate the mass of $1.5 \times 10^{-3} \text{ m}$ aqueous solution required for the above dose.
- 1.30** Calculate the amount of benzoic acid (C_6H_5COOH) required for preparing 250 mL of 0.15 M solution in methanol.
- 1.31** The depression in freezing point of water observed for the same amount of acetic acid, trichloroacetic acid and trifluoroacetic acid increases in the order given above. Explain briefly.
- 1.32** Calculate the depression in the freezing point of water when 10 g of $CH_3CH_2CHClCOOH$ is added to 250 g of water. $K_a = 1.4 \times 10^{-3}$, $K_f = 1.86 \text{ K kg mol}^{-1}$.
- 1.33** 19.5 g of CH_2FCOOH is dissolved in 500 g of water. The depression in the freezing point of water observed is 1.0°C . Calculate the van't Hoff factor and dissociation constant of fluoroacetic acid.
- 1.34** Vapour pressure of water at 293 K is 17.535 mm Hg. Calculate the vapour pressure of water at 293 K when 25 g of glucose is dissolved in 450 g of water.
- 1.35** Henry's law constant for the molality of methane in benzene at 298 K is $4.27 \times 10^5 \text{ mm Hg}$. Calculate the solubility of methane in benzene at 298 K under 760 mm Hg.
- 1.36** 100 g of liquid A (molar mass 140 g mol^{-1}) was dissolved in 1000 g of liquid B (molar mass 180 g mol^{-1}). The vapour pressure of pure liquid B was found to be 500 torr. Calculate the vapour pressure of pure liquid A and its vapour pressure in the solution if the total vapour pressure of the solution is 475 Torr.

- 1.37** Vapour pressures of pure acetone and chloroform at 328 K are 741.8 mm Hg and 632.8 mm Hg respectively. Assuming that they form ideal solution over the entire range of composition, plot p_{total} , $p_{\text{chloroform}}$, and p_{acetone} as a function of x_{acetone} . The experimental data observed for different compositions of mixture is:

$100 \times x_{\text{acetone}}$	0	11.8	23.4	36.0	50.8	58.2	64.5	72.1
p_{acetone} /mm Hg	0	54.9	110.1	202.4	322.7	405.9	454.1	521.1
$p_{\text{chloroform}}$ /mm Hg	632.8	548.1	469.4	359.7	257.7	193.6	161.2	120.7

Plot this data also on the same graph paper. Indicate whether it has positive deviation or negative deviation from the ideal solution.

- 1.38** Benzene and toluene form ideal solution over the entire range of composition. The vapour pressure of pure benzene and toluene at 300 K are 50.71 mm Hg and 32.06 mm Hg respectively. Calculate the mole fraction of benzene in vapour phase if 80 g of benzene is mixed with 100 g of toluene.
- 1.39** The air is a mixture of a number of gases. The major components are oxygen and nitrogen with approximate proportion of 20% is to 79% by volume at 298 K. The water is in equilibrium with air at a pressure of 10 atm. At 298 K if the Henry's law constants for oxygen and nitrogen at 298 K are 3.30×10^7 mm and 6.51×10^7 mm respectively, calculate the composition of these gases in water.
- 1.40** Determine the amount of CaCl_2 ($i = 2.47$) dissolved in 2.5 litre of water such that its osmotic pressure is 0.75 atm at 27° C.
- 1.41** Determine the osmotic pressure of a solution prepared by dissolving 25 mg of K_2SO_4 in 2 litre of water at 25° C, assuming that it is completely dissociated.

Answers to Some Intext Questions

- 1.1** $\text{C}_6\text{H}_6 = 15.28\%$, $\text{CCl}_4 = 84.72\%$
- 1.2** 0.459, 0.541
- 1.3** 0.024 M, 0.03 M
- 1.4** 36.946 g
- 1.5** 1.5 mol kg^{-1} , 1.45 mol L^{-1} 0.0263
- 1.9** 23.4 mm Hg
- 1.10** 121.67 g
- 1.11** 5.077 g
- 1.12** 30.96 Pa



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Unit

2

Electrochemistry

Objectives

After studying this Unit, you will be able to

- describe an electrochemical cell and differentiate between galvanic and electrolytic cells;
- apply Nernst equation for calculating the emf of galvanic cell and define standard potential of the cell;
- derive relation between standard potential of the cell, Gibbs energy of cell reaction and its equilibrium constant;
- define resistivity (ρ), conductivity (κ) and molar conductivity (Λ_m) of ionic solutions;
- differentiate between ionic (electrolytic) and electronic conductivity;
- describe the method for measurement of conductivity of electrolytic solutions and calculation of their molar conductivity;
- justify the variation of conductivity and molar conductivity of solutions with change in their concentration and define Λ_m° (molar conductivity at zero concentration or infinite dilution);
- enunciate Kohlrausch law and learn its applications;
- understand quantitative aspects of electrolysis;
- describe the construction of some primary and secondary batteries and fuel cells;
- explain corrosion as an electrochemical process.

Chemical reactions can be used to produce electrical energy, conversely, electrical energy can be used to carry out chemical reactions that do not proceed spontaneously.

Electrochemistry is the study of production of electricity from energy released during spontaneous chemical reactions and the use of electrical energy to bring about non-spontaneous chemical transformations. The subject is of importance both for theoretical and practical considerations. A large number of metals, sodium hydroxide, chlorine, fluorine and many other chemicals are produced by electrochemical methods. Batteries and fuel cells convert chemical energy into electrical energy and are used on a large scale in various instruments and devices. The reactions carried out electrochemically can be energy efficient and less polluting. Therefore, study of electrochemistry is important for creating new technologies that are ecofriendly. The transmission of sensory signals through cells to brain and vice versa and communication between the cells are known to have electrochemical origin. Electrochemistry, is therefore, a very vast and interdisciplinary subject. In this Unit, we will cover only some of its important elementary aspects.

2.1 Electrochemical Cells

We had studied the construction and functioning of **Daniell cell** (Fig. 2.1). This cell converts the chemical energy liberated during the redox reaction

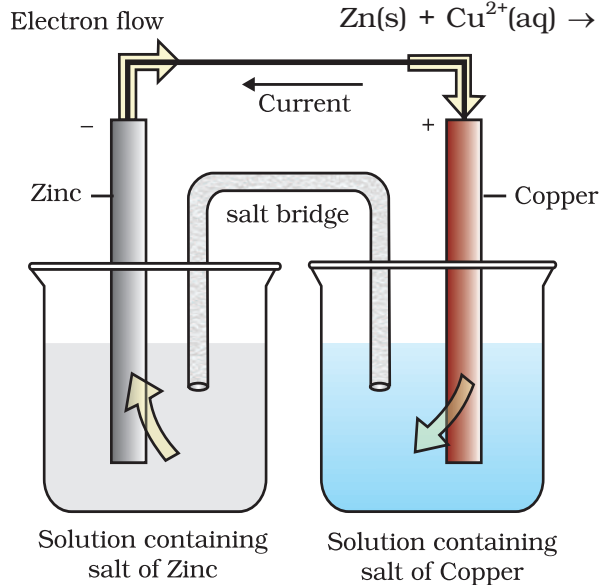
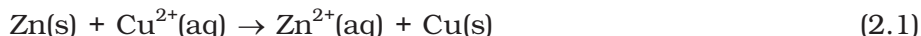


Fig. 2.1: Daniell cell having electrodes of zinc and copper dipping in the solutions of their respective salts.

to electrical energy and has an electrical potential equal to 1.1 V when concentration of Zn^{2+} and Cu^{2+} ions is unity (1 mol dm^{-3})*. Such a device is called a **galvanic** or a **voltaic** cell.

If an external opposite potential is applied in the galvanic cell [Fig. 2.2(a)] and increased slowly, we find that the reaction continues to take place till the opposing voltage reaches the value 1.1 V [Fig. 2.2(b)] when, the reaction stops altogether and no current flows through the cell. Any further increase in the external potential again starts the reaction but in the opposite direction [Fig. 2.2(c)]. It now functions as an **electrolytic cell**, a device for using electrical energy to carry non-spontaneous chemical reactions. Both types of cells are quite important and we shall study some of their salient features in the following pages.

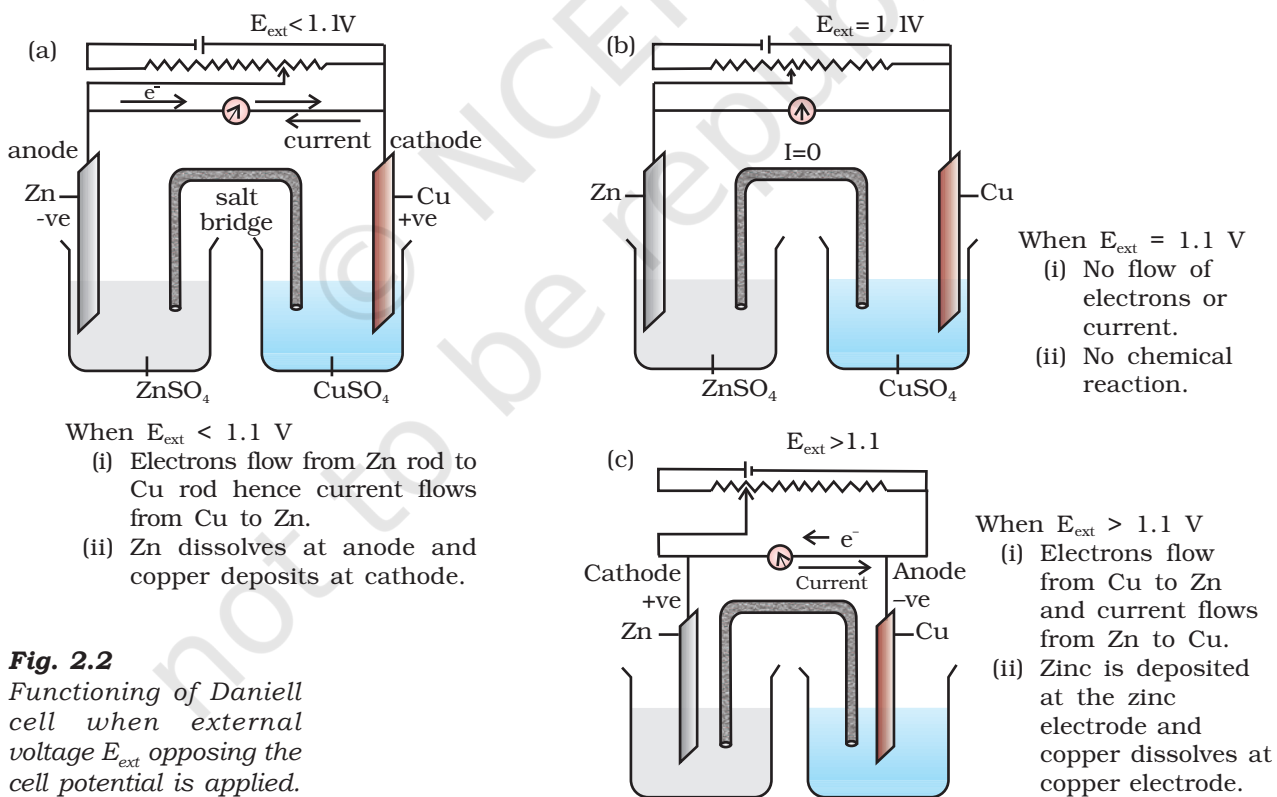


Fig. 2.2

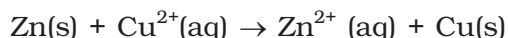
Functioning of Daniell cell when external voltage E_{ext} opposing the cell potential is applied.

* Strictly speaking activity should be used instead of concentration. It is directly proportional to concentration. In dilute solutions, it is equal to concentration. You will study more about it in higher classes.

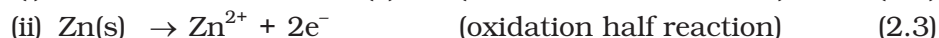
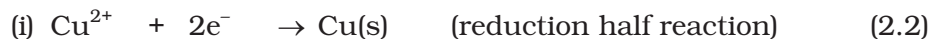
2.2 Galvanic Cells

As mentioned earlier a galvanic cell is an electrochemical cell that converts the chemical energy of a spontaneous redox reaction into electrical energy. In this device the **Gibbs energy** of the spontaneous redox reaction is converted into electrical work which may be used for running a motor or other electrical gadgets like heater, fan, geyser, etc.

Daniell cell discussed earlier is one such cell in which the following redox reaction occurs.



This reaction is a combination of two half reactions whose addition gives the overall cell reaction:



These reactions occur in two different portions of the Daniell cell. The reduction half reaction occurs on the copper electrode while the oxidation half reaction occurs on the zinc electrode. These two portions of the cell are also called **half-cells** or **redox couples**. The copper electrode may be called the reduction half cell and the zinc electrode, the oxidation half-cell.

We can construct innumerable number of galvanic cells on the pattern of Daniell cell by taking combinations of different half-cells. Each half-cell consists of a metallic electrode dipped into an electrolyte. The two half-cells are connected by a metallic wire through a voltmeter and a switch externally. The electrolytes of the two half-cells are connected internally through a salt bridge as shown in Fig. 2.1. Sometimes, both the electrodes dip in the same electrolyte solution and in such cases we do not require a salt bridge.

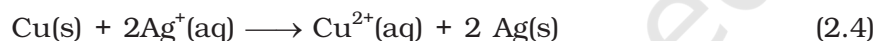
At each electrode-electrolyte interface there is a tendency of metal ions from the solution to deposit on the metal electrode trying to make it positively charged. At the same time, metal atoms of the electrode have a tendency to go into the solution as ions and leave behind the electrons at the electrode trying to make it negatively charged. At equilibrium, there is a separation of charges and depending on the tendencies of the two opposing reactions, the electrode may be positively or negatively charged with respect to the solution. A potential difference develops between the electrode and the electrolyte which is called **electrode potential**. When the concentrations of all the species involved in a half-cell is unity then the electrode potential is known as **standard electrode potential**. According to IUPAC convention, standard reduction potentials are now called standard electrode potentials. In a galvanic cell, the half-cell in which oxidation takes place is called **anode** and it has a negative potential with respect to the solution. The other half-cell in which reduction takes place is called **cathode** and it has a positive potential with respect to the solution. Thus, there exists a potential difference between the two electrodes and as soon as the switch is in the *on* position the electrons flow from negative electrode to positive electrode. The direction of current flow is opposite to that of electron flow.

The potential difference between the two electrodes of a galvanic cell is called the *cell potential* and is measured in volts. The **cell potential** is the difference between the electrode potentials (reduction potentials) of the cathode and anode. It is called the **cell electromotive force (emf)** of the cell when no current is drawn through the cell. It is now an accepted convention that we keep the anode on the left and the cathode on the right while representing the galvanic cell. A galvanic cell is generally represented by putting a vertical line between metal and electrolyte solution and putting a double vertical line between the two electrolytes connected by a salt bridge. Under this convention the emf of the cell is positive and is given by the potential of the half-cell on the right hand side minus the potential of the half-cell on the left hand side i.e.,

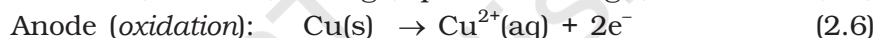
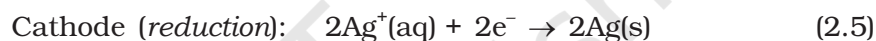
$$E_{\text{cell}} = E_{\text{right}} - E_{\text{left}}$$

This is illustrated by the following example:

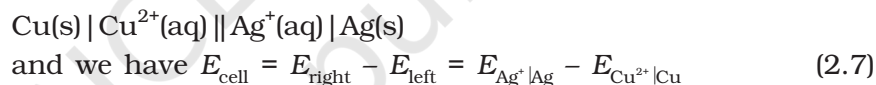
Cell reaction:



Half-cell reactions:



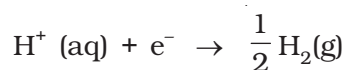
It can be seen that the sum of (3.5) and (3.6) leads to overall reaction (2.4) in the cell and that silver electrode acts as a cathode and copper electrode acts as an anode. The cell can be represented as:



2.2.1 Measurement of Electrode Potential

The potential of individual half-cell cannot be measured. We can measure only the difference between the two half-cell potentials that gives the emf of the cell. If we arbitrarily choose the potential of one electrode (half-cell) then that of the other can be determined with respect

to this. According to convention, a half-cell called standard hydrogen electrode (Fig.3.3) represented by $\text{Pt(s)} | \text{H}_2(\text{g}) | \text{H}^+(\text{aq})$, is assigned a zero potential at all temperatures corresponding to the reaction



The standard hydrogen electrode consists of a platinum electrode coated with platinum black. The electrode is dipped in an acidic solution and pure hydrogen gas is bubbled through it. The concentration of both the reduced and oxidised forms of hydrogen is maintained at unity (Fig. 2.3). This implies that the pressure of hydrogen gas is one bar and the concentration of hydrogen ion in the solution is one molar.

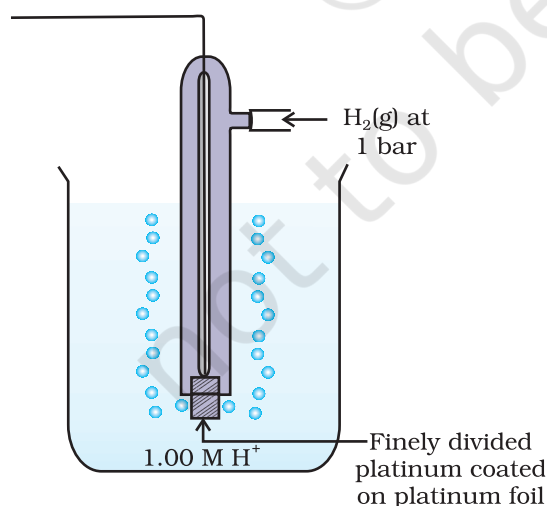


Fig. 2.3: Standard Hydrogen Electrode (SHE).

At 298 K the emf of the cell, standard hydrogen electrode || second half-cell constructed by taking standard hydrogen electrode as anode (reference half-cell) and the other half-cell as cathode, gives the reduction potential of the other half-cell. If the concentrations of the oxidised and the reduced forms of the species in the right hand half-cell are unity, then the cell potential is equal to standard electrode potential, E°_R of the given half-cell.

$$E^\circ = E^\circ_R - E^\circ_L$$

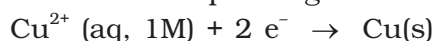
As E°_L for standard hydrogen electrode is zero.

$$E^\circ = E^\circ_R - 0 = E^\circ_R$$

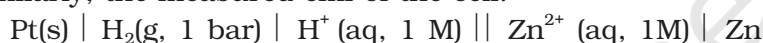
The measured emf of the cell:



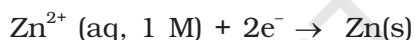
is 0.34 V and it is also the value for the standard electrode potential of the half-cell corresponding to the reaction:



Similarly, the measured emf of the cell:

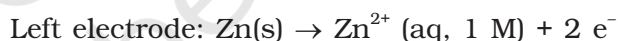


is -0.76 V corresponding to the standard electrode potential of the half-cell reaction:

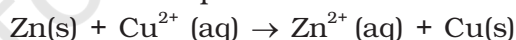


The positive value of the standard electrode potential in the first case indicates that Cu^{2+} ions get reduced more easily than H^+ ions. The reverse process cannot occur, that is, hydrogen ions cannot oxidise Cu (or alternatively we can say that hydrogen gas can reduce copper ion) under the standard conditions described above. Thus, Cu does not dissolve in HCl. In nitric acid it is oxidised by nitrate ion and not by hydrogen ion. The negative value of the standard electrode potential in the second case indicates that hydrogen ions can oxidise zinc (or zinc can reduce hydrogen ions).

In view of this convention, the half reaction for the Daniell cell in Fig. 2.1 can be written as:

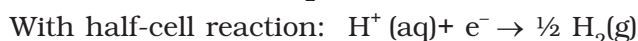
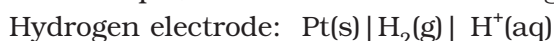


The overall reaction of the cell is the sum of above two reactions and we obtain the equation:



$$\begin{aligned} \text{emf of the cell} &= E^\circ_{\text{cell}} = E^\circ_R - E^\circ_L \\ &= 0.34 \text{ V} - (-0.76) \text{ V} = 1.10 \text{ V} \end{aligned}$$

Sometimes metals like platinum or gold are used as inert electrodes. They do not participate in the reaction but provide their surface for oxidation or reduction reactions and for the conduction of electrons. For example, Pt is used in the following half-cells:



With half-cell reaction: $\frac{1}{2} \text{Br}_2(\text{aq}) + \text{e}^- \rightarrow \text{Br}^-(\text{aq})$

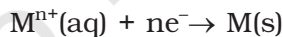
The standard electrode potentials are very important and we can extract a lot of useful information from them. The values of standard electrode potentials for some selected half-cell reduction reactions are given in Table 2.1. If the standard electrode potential of an electrode is greater than zero then its reduced form is more stable compared to hydrogen gas. Similarly, if the standard electrode potential is negative then hydrogen gas is more stable than the reduced form of the species. It can be seen that the standard electrode potential for fluorine is the highest in the Table indicating that fluorine gas (F_2) has the maximum tendency to get reduced to fluoride ions (F^-) and therefore fluorine gas is the strongest oxidising agent and fluoride ion is the weakest reducing agent. Lithium has the lowest electrode potential indicating that lithium ion is the weakest oxidising agent while lithium metal is the most powerful reducing agent in an aqueous solution. It may be seen that as we go from top to bottom in Table 2.1 the standard electrode potential decreases and with this, decreases the oxidising power of the species on the left and increases the reducing power of the species on the right hand side of the reaction. Electrochemical cells are extensively used for determining the pH of solutions, solubility product, equilibrium constant and other thermodynamic properties and for potentiometric titrations.

Intext Questions

- 2.1** How would you determine the standard electrode potential of the system $\text{Mg}^{2+} | \text{Mg}$?
- 2.2** Can you store copper sulphate solutions in a zinc pot?
- 2.3** Consult the table of standard electrode potentials and suggest three substances that can oxidise ferrous ions under suitable conditions.

2.3 Nernst Equation

We have assumed in the previous section that the concentration of all the species involved in the electrode reaction is unity. This need not be always true. Nernst showed that for the electrode reaction:



the electrode potential at any concentration measured with respect to standard hydrogen electrode can be represented by:

$$E_{(\text{M}^{n+}/\text{M})} = E_{(\text{M}^{n+}/\text{M})}^{\circ} - \frac{RT}{nF} \ln \frac{[\text{M}]}{[\text{M}^{n+}]}$$

but concentration of solid M is taken as unity and we have

$$E_{(\text{M}^{n+}/\text{M})} = E_{(\text{M}^{n+}/\text{M})}^{\circ} - \frac{RT}{nF} \ln \frac{1}{[\text{M}^{n+}]} \quad (2.8)$$

$E_{(\text{M}^{n+}/\text{M})}^{\circ}$ has already been defined, R is gas constant ($8.314 \text{ JK}^{-1} \text{ mol}^{-1}$), F is Faraday constant (96487 C mol^{-1}), T is temperature in kelvin and $[\text{M}^{n+}]$ is the concentration of the species, M^{n+} .

Table 2.1: Standard Electrode Potentials at 298 K

Ions are present as aqueous species and H₂O as liquid; gases and solids are shown by g and s.

Reaction (Oxidised form + ne ⁻ → Reduced form)		E ⁰ /V
 Increasing strength of oxidising agent	F ₂ (g) + 2e ⁻ → 2F ⁻	2.87
	Co ³⁺ + e ⁻ → Co ²⁺	1.81
	H ₂ O ₂ + 2H ⁺ + 2e ⁻ → 2H ₂ O	1.78
	MnO ₄ ⁻ + 8H ⁺ + 5e ⁻ → Mn ²⁺ + 4H ₂ O	1.51
	Au ³⁺ + 3e ⁻ → Au(s)	1.40
	Cl ₂ (g) + 2e ⁻ → 2Cl ⁻	1.36
	Cr ₂ O ₇ ²⁻ + 14H ⁺ + 6e ⁻ → 2Cr ³⁺ + 7H ₂ O	1.33
	O ₂ (g) + 4H ⁺ + 4e ⁻ → 2H ₂ O	1.23
	MnO ₂ (s) + 4H ⁺ + 2e ⁻ → Mn ²⁺ + 2H ₂ O	1.23
	Br ₂ + 2e ⁻ → 2Br ⁻	1.09
	NO ₃ ⁻ + 4H ⁺ + 3e ⁻ → NO(g) + 2H ₂ O	0.97
	2Hg ²⁺ + 2e ⁻ → Hg ₂ ²⁺	0.92
	Ag ⁺ + e ⁻ → Ag(s)	0.80
	Fe ³⁺ + e ⁻ → Fe ²⁺	0.77
	O ₂ (g) + 2H ⁺ + 2e ⁻ → H ₂ O ₂	0.68
	I ₂ + 2e ⁻ → 2I ⁻	0.54
	Cu ⁺ + e ⁻ → Cu(s)	0.52
	Cu ²⁺ + 2e ⁻ → Cu(s)	0.34
	AgCl(s) + e ⁻ → Ag(s) + Cl ⁻	0.22
	AgBr(s) + e ⁻ → Ag(s) + Br ⁻	0.10
	2H⁺ + 2e⁻ → H₂(g)	0.00
	Pb ²⁺ + 2e ⁻ → Pb(s)	-0.13
	Sn ²⁺ + 2e ⁻ → Sn(s)	-0.14
	Ni ²⁺ + 2e ⁻ → Ni(s)	-0.25
	Fe ²⁺ + 2e ⁻ → Fe(s)	-0.44
	Cr ³⁺ + 3e ⁻ → Cr(s)	-0.74
	Zn ²⁺ + 2e ⁻ → Zn(s)	-0.76
	2H ₂ O + 2e ⁻ → H ₂ (g) + 2OH ⁻ (aq)	-0.83
	Al ³⁺ + 3e ⁻ → Al(s)	-1.66
	Mg ²⁺ + 2e ⁻ → Mg(s)	-2.36
	Na ⁺ + e ⁻ → Na(s)	-2.71
	Ca ²⁺ + 2e ⁻ → Ca(s)	-2.87
	K ⁺ + e ⁻ → K(s)	-2.93
	Li ⁺ + e ⁻ → Li(s)	-3.05
 Increasing strength of reducing agent		

1. A negative E⁰ means that the redox couple is a stronger reducing agent than the H⁺/H₂ couple.
2. A positive E⁰ means that the redox couple is a weaker reducing agent than the H⁺/H₂ couple.

In Daniell cell, the electrode potential for any given concentration of Cu^{2+} and Zn^{2+} ions, we write

For Cathode:

$$E_{(\text{Cu}^{2+}/\text{Cu})} = E_{(\text{Cu}^{2+}/\text{Cu})}^{\circ} - \frac{RT}{2F} \ln \frac{1}{[\text{Cu}^{2+}(\text{aq})]} \quad (2.9)$$

For Anode:

$$E_{(\text{Zn}^{2+}/\text{Zn})} = E_{(\text{Zn}^{2+}/\text{Zn})}^{\circ} - \frac{RT}{2F} \ln \frac{1}{[\text{Zn}^{2+}(\text{aq})]} \quad (2.10)$$

The cell potential, $E_{(\text{cell})} = E_{(\text{Cu}^{2+}/\text{Cu})} - E_{(\text{Zn}^{2+}/\text{Zn})}$

$$\begin{aligned} &= E_{(\text{Cu}^{2+}/\text{Cu})}^{\circ} - \frac{RT}{2F} \ln \frac{1}{[\text{Cu}^{2+}(\text{aq})]} - E_{(\text{Zn}^{2+}/\text{Zn})}^{\circ} + \frac{RT}{2F} \ln \frac{1}{[\text{Zn}^{2+}(\text{aq})]} \\ &= E_{(\text{Cu}^{2+}/\text{Cu})}^{\circ} - E_{(\text{Zn}^{2+}/\text{Zn})}^{\circ} - \frac{RT}{2F} \ln \frac{1}{[\text{Cu}^{2+}(\text{aq})]} - \ln \frac{1}{[\text{Zn}^{2+}(\text{aq})]} \end{aligned}$$

$$E_{(\text{cell})} = E_{(\text{cell})}^{\circ} - \frac{RT}{2F} \ln \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} \quad (2.11)$$

It can be seen that $E_{(\text{cell})}$ depends on the concentration of both Cu^{2+} and Zn^{2+} ions. It increases with increase in the concentration of Cu^{2+} ions and decrease in the concentration of Zn^{2+} ions.

By converting the natural logarithm in Eq. (2.11) to the base 10 and substituting the values of R , F and $T = 298 \text{ K}$, it reduces to

$$E_{(\text{cell})} = E_{(\text{cell})}^{\circ} - \frac{0.059}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} \quad (2.12)$$

We should use the same number of electrons (n) for both the electrodes and thus for the following cell

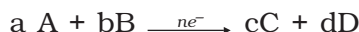


The cell reaction is $\text{Ni(s)} + 2\text{Ag}^{+}(\text{aq}) \rightarrow \text{Ni}^{2+}(\text{aq}) + 2\text{Ag(s)}$

The **Nernst equation** can be written as

$$E_{(\text{cell})} = E_{(\text{cell})}^{\circ} - \frac{RT}{2F} \ln \frac{[\text{Ni}^{2+}]}{[\text{Ag}^{+}]^2}$$

and for a general electrochemical reaction of the type:



Nernst equation can be written as:

$$\begin{aligned} E_{(\text{cell})} &= E_{(\text{cell})}^{\circ} - \frac{RT}{nF} \ln Q \\ &= E_{(\text{cell})}^{\circ} - \frac{RT}{nF} \ln \frac{[\text{C}]^c [\text{D}]^d}{[\text{A}]^a [\text{B}]^b} \end{aligned} \quad (2.13)$$

Represent the cell in which the following reaction takes place
 $\text{Mg(s)} + 2\text{Ag}^+(0.0001\text{M}) \rightarrow \text{Mg}^{2+}(0.130\text{M}) + 2\text{Ag(s)}$

Example 2.1

Calculate its $E_{(\text{cell})}$ if $E_{(\text{cell})}^{\circ} = 3.17 \text{ V}$.

The cell can be written as $\text{Mg} | \text{Mg}^{2+}(0.130\text{M}) || \text{Ag}^+(0.0001\text{M}) | \text{Ag}$

Solution

$$\begin{aligned} E_{(\text{cell})} &= E_{(\text{cell})}^{\circ} - \frac{RT}{2F} \ln \frac{\text{Mg}^{2+}}{\text{Ag}^+{}^2} \\ &= 3.17 \text{ V} - \frac{0.059\text{V}}{2} \log \frac{0.130}{(0.0001)^2} = 3.17 \text{ V} - 0.21\text{V} = 2.96 \text{ V}. \end{aligned}$$

2.3.1 Equilibrium Constant from Nernst Equation

If the circuit in Daniell cell (Fig. 2.1) is closed then we note that the reaction



takes place and as time passes, the concentration of Zn^{2+} keeps on increasing while the concentration of Cu^{2+} keeps on decreasing. At the same time voltage of the cell as read on the voltmeter keeps on decreasing. After some time, we shall note that there is no change in the concentration of Cu^{2+} and Zn^{2+} ions and at the same time, voltmeter gives zero reading. This indicates that equilibrium has been attained. In this situation the Nernst equation may be written as:

$$E_{(\text{cell})} = 0 = E_{(\text{cell})}^{\circ} - \frac{2.303RT}{2F} \log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$$

$$\text{or } E_{(\text{cell})}^{\circ} = \frac{2.303RT}{2F} \log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$$

But at equilibrium,

$$\frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} = K_c \text{ for the reaction 2.1}$$

and at $T = 298\text{K}$ the above equation can be written as

$$E_{(\text{cell})}^{\circ} = \frac{0.059 \text{ V}}{2} \log K_c = 1.1 \text{ V} \quad (E_{(\text{cell})}^{\circ} = 1.1\text{V})$$

$$\log K_c = \frac{(1.1\text{V} \times 2)}{0.059 \text{ V}} = 37.288$$

$$K_c = 2 \times 10^{37} \text{ at } 298\text{K}.$$

In general,

$$E_{(\text{cell})}^{\circ} = \frac{2.303RT}{nF} \log K_c \quad (2.14)$$

Thus, Eq. (2.14) gives a relationship between equilibrium constant of the reaction and standard potential of the cell in which that reaction takes place. Thus, equilibrium constants of the reaction, difficult to measure otherwise, can be calculated from the corresponding E° value of the cell.

Example 2.2 Calculate the equilibrium constant of the reaction:
 $\text{Cu(s)} + 2\text{Ag}^+(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{Ag(s)}$

$$E_{(\text{cell})}^{\circ} = 0.46 \text{ V}$$

Solution $E_{(\text{cell})}^{\circ} = \frac{0.059 \text{ V}}{2} \log K_C = 0.46 \text{ V}$ or

$$\log K_C = \frac{0.46 \text{ V} \times 2}{0.059 \text{ V}} = 15.6$$

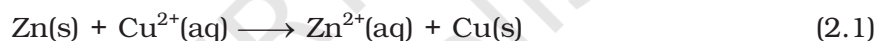
$$K_C = 3.92 \times 10^{15}$$

2.3.2 Electro-chemical Cell and Gibbs Energy of the Reaction

Electrical work done in one second is equal to electrical potential multiplied by total charge passed. If we want to obtain maximum work from a galvanic cell then charge has to be passed reversibly. The reversible work done by a galvanic cell is equal to decrease in its Gibbs energy and therefore, if the emf of the cell is E and nF is the amount of charge passed and $\Delta_r G$ is the Gibbs energy of the reaction, then

$$\Delta_r G = - nFE_{(\text{cell})} \quad (2.15)$$

It may be remembered that $E_{(\text{cell})}$ is an intensive parameter but $\Delta_r G$ is an extensive thermodynamic property and the value depends on n . Thus, if we write the reaction



$$\Delta_r G = - 2FE_{(\text{cell})}$$

but when we write the reaction



$$\Delta_r G = - 4FE_{(\text{cell})}$$

If the concentration of all the reacting species is unity, then

$$E_{(\text{cell})} = E_{(\text{cell})}^{\circ} \text{ and we have}$$

$$\Delta_r G^{\circ} = - nFE_{(\text{cell})}^{\circ} \quad (2.16)$$

Thus, from the measurement of $E_{(\text{cell})}^{\circ}$ we can obtain an important thermodynamic quantity, $\Delta_r G^{\circ}$, standard Gibbs energy of the reaction. From the latter we can calculate equilibrium constant by the equation:

$$\Delta_r G^{\circ} = -RT \ln K.$$

Example 2.3 The standard electrode potential for Daniell cell is 1.1V. Calculate the standard Gibbs energy for the reaction:



Solution $\Delta_r G^{\circ} = - nFE_{(\text{cell})}^{\circ}$

n in the above equation is 2, $F = 96487 \text{ C mol}^{-1}$ and $E_{(\text{cell})}^{\circ} = 1.1 \text{ V}$

$$\begin{aligned} \text{Therefore, } \Delta_r G^{\circ} &= - 2 \times 1.1\text{V} \times 96487 \text{ C mol}^{-1} \\ &= - 21227 \text{ J mol}^{-1} \\ &= - 212.27 \text{ kJ mol}^{-1} \end{aligned}$$

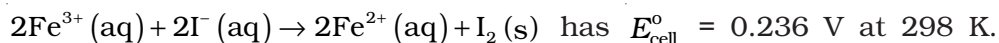
2.4 Calculate the potential of hydrogen electrode in contact with a solution whose pH is 10.

2.5 Calculate the emf of the cell in which the following reaction takes place:



Given that $E_{\text{cell}}^{\circ} = 1.05 \text{ V}$

2.6 The cell in which the following reaction occurs:



Calculate the standard Gibbs energy and the equilibrium constant of the cell reaction.

2.4 Conductance of Electrolytic Solutions

It is necessary to define a few terms before we consider the subject of conductance of electricity through electrolytic solutions. The electrical resistance is represented by the symbol 'R' and it is measured in ohm (Ω) which in terms of SI base units is equal to $(\text{kg m}^2)/(\text{S}^3 \text{ A}^2)$. It can be measured with the help of a **Wheatstone bridge** with which you are familiar from your study of physics. The electrical resistance of any object is directly proportional to its length, l , and inversely proportional to its area of cross section, A . That is,

$$R \propto \frac{l}{A} \text{ or } R = \rho \frac{l}{A} \quad (2.17)$$

The constant of proportionality, ρ (Greek, rho), is called **resistivity** (specific resistance). Its SI units are ohm metre ($\Omega \text{ m}$) and quite often its submultiple, ohm centimetre ($\Omega \text{ cm}$) is also used. IUPAC recommends the use of the term resistivity over specific resistance and hence in the rest of the book we shall use the term resistivity. Physically, the resistivity for a substance is its resistance when it is one metre long and its area of cross section is one m^2 . It can be seen that:

$$1 \Omega \text{ m} = 100 \Omega \text{ cm} \text{ or } 1 \Omega \text{ cm} = 0.01 \Omega \text{ m}$$

The inverse of resistance, R , is called **conductance**, G , and we have the relation:

$$G = \frac{1}{R} = \frac{A}{\rho l} = \kappa \frac{A}{l} \quad (2.18)$$

The SI unit of conductance is siemens, represented by the symbol 'S' and is equal to ohm^{-1} (also known as mho) or Ω^{-1} . The inverse of resistivity, called **conductivity** (specific conductance) is represented by the symbol, κ (Greek, kappa). IUPAC has recommended the use of term conductivity over specific conductance and hence we shall use the term conductivity in the rest of the book. The SI units of conductivity are S m^{-1} but quite often, κ is expressed in S cm^{-1} . Conductivity of a material in S m^{-1} is its conductance when it is 1 m long and its area of cross section is 1 m^2 . It may be noted that $1 \text{ S cm}^{-1} = 100 \text{ S m}^{-1}$.

Table 2.2: The values of Conductivity of some Selected Materials at 298.15 K

Material	Conductivity/ S m^{-1}	Material	Conductivity/ S m^{-1}
Conductors		Aqueous Solutions	
Sodium	2.1×10^3	Pure water	3.5×10^{-5}
Copper	5.9×10^3	0.1 M HCl	3.91
Silver	6.2×10^3	0.01M KCl	0.14
Gold	4.5×10^3	0.01M NaCl	0.12
Iron	1.0×10^3	0.1 M HAc	0.047
Graphite	1.2×10	0.01M HAc	0.016
Insulators		Semiconductors	
Glass	1.0×10^{-16}	CuO	1×10^{-7}
Teflon	1.0×10^{-18}	Si	1.5×10^{-2}
		Ge	2.0

It can be seen from Table 2.2 that the magnitude of conductivity varies a great deal and depends on the nature of the material. It also depends on the temperature and pressure at which the measurements are made. Materials are classified into conductors, insulators and semiconductors depending on the magnitude of their conductivity. Metals and their alloys have very large conductivity and are known as conductors. Certain non-metals like carbon-black, graphite and some organic polymers* are also electronically conducting. Substances like glass, ceramics, etc., having very low conductivity are known as insulators. Substances like silicon, doped silicon and gallium arsenide having conductivity between conductors and insulators are called semiconductors and are important electronic materials. Certain materials called superconductors by definition have zero resistivity or infinite conductivity. Earlier, only metals and their alloys at very low temperatures (0 to 15 K) were known to behave as superconductors, but nowadays a number of ceramic materials and mixed oxides are also known to show superconductivity at temperatures as high as 150 K.

Electrical conductance through metals is called metallic or electronic conductance and is due to the movement of electrons. The electronic conductance depends on

- the nature and structure of the metal
- the number of valence electrons per atom
- temperature (it decreases with increase of temperature).

* Electronically conducting polymers – In 1977 MacDiarmid, Heeger and Shirakawa discovered that acetylene gas can be polymerised to produce a polymer, polyacetylene when exposed to vapours of iodine acquires metallic lustre and conductivity. Since then several organic conducting polymers have been made such as polyaniline, polypyrrole and polythiophene. These organic polymers which have properties like metals, being composed wholly of elements like carbon, hydrogen and occasionally nitrogen, oxygen or sulphur, are much lighter than normal metals and can be used for making light-weight batteries. Besides, they have the mechanical properties of polymers such as flexibility so that one can make electronic devices such as transistors that can bend like a sheet of plastic. For the discovery of conducting polymers, MacDiarmid, Heeger and Shirakawa were awarded the Nobel Prize in Chemistry for the year 2000.

As the electrons enter at one end and go out through the other end, the composition of the metallic conductor remains unchanged. The mechanism of conductance through semiconductors is more complex.

We already know that even very pure water has small amounts of hydrogen and hydroxyl ions ($\sim 10^{-7}\text{M}$) which lend it very low conductivity ($3.5 \times 10^{-5} \text{ S m}^{-1}$). When electrolytes are dissolved in water, they furnish their own ions in the solution hence its conductivity also increases. The conductance of electricity by ions present in the solutions is called electrolytic or **ionic conductance**. The conductivity of electrolytic (ionic) solutions depends on:

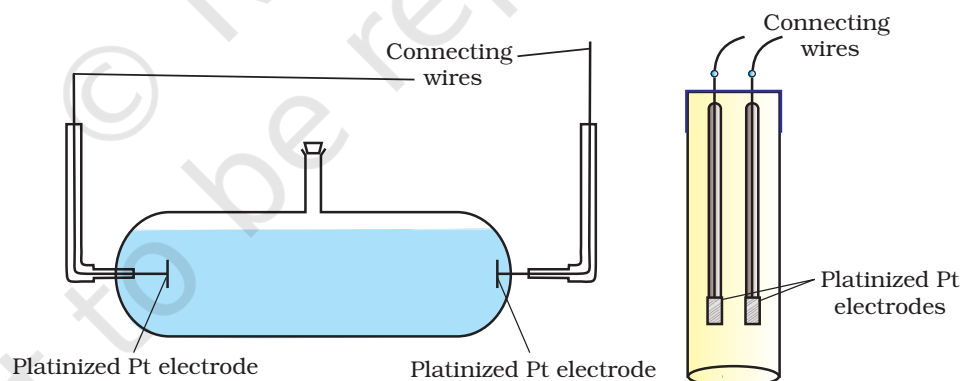
- (i) the nature of the electrolyte added
- (ii) size of the ions produced and their solvation
- (iii) the nature of the solvent and its viscosity
- (iv) concentration of the electrolyte
- (v) temperature (it increases with the increase of temperature).

Passage of direct current through ionic solution over a prolonged period can lead to change in its composition due to electrochemical reactions (Section 2.4.1).

2.4.1 Measurement of the Conductivity of Ionic Solutions

We know that accurate measurement of an unknown resistance can be performed on a Wheatstone bridge. However, for measuring the resistance of an ionic solution we face two problems. Firstly, passing direct current (DC) changes the composition of the solution. Secondly, a solution cannot be connected to the bridge like a metallic wire or other solid conductor. The first difficulty is resolved by using an alternating current (AC) source of power. The second problem is solved by using a specially designed vessel called **conductivity cell**. It is available in several designs and two simple ones are shown in Fig. 2.4.

Fig. 2.4
Two different types of conductivity cells.



Basically it consists of two platinum electrodes coated with platinum black (finely divided metallic Pt is deposited on the electrodes electrochemically). These have area of cross section equal to 'A' and are separated by distance 'l'. Therefore, solution confined between these electrodes is a column of length l and area of cross section A. The resistance of such a column of solution is then given by the equation:

$$R = \rho \frac{l}{A} = \frac{l}{\kappa A} \quad (2.17)$$

The quantity l/A is called cell constant denoted by the symbol, G^* . It depends on the distance between the electrodes and their area of cross-section and has the dimension of length^{-1} and can be calculated if we know l and A . Measurement of l and A is not only inconvenient but also unreliable. The cell constant is usually determined by measuring the resistance of the cell containing a solution whose conductivity is already known. For this purpose, we generally use KCl solutions whose conductivity is known accurately at various concentrations (Table 2.3) and at different temperatures. The cell constant, G^* , is then given by the equation:

$$G^* = \frac{l}{A} = R \kappa \quad (2.18)$$

Table 2.3: Conductivity and Molar conductivity of KCl solutions at 298.15K

Concentration/Molarity		Conductivity		Molar Conductivity	
mol L^{-1}	mol m^{-3}	S cm^{-1}	S m^{-1}	$\text{S cm}^2\text{mol}^{-1}$	$\text{S m}^2 \text{mol}^{-1}$
1.000	1000	0.1113	11.13	111.3	111.3×10^{-4}
0.100	100.0	0.0129	1.29	129.0	129.0×10^{-4}
0.010	10.00	0.00141	0.141	141.0	141.0×10^{-4}

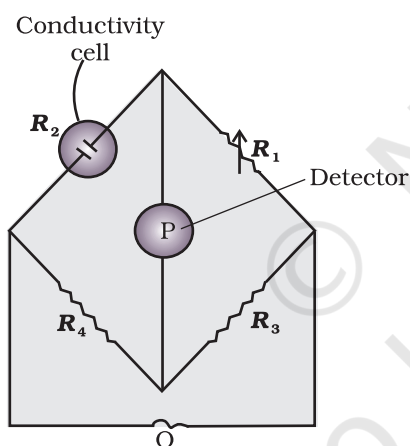


Fig. 2.5: Arrangement for measurement of resistance of a solution of an electrolyte.

Once the cell constant is determined, we can use it for measuring the resistance or conductivity of any solution. The set up for the measurement of the resistance is shown in Fig. 2.5.

It consists of two resistances R_3 and R_4 , a variable resistance R_1 and the conductivity cell having the unknown resistance R_2 . The **Wheatstone bridge** is fed by an oscillator O (a source of a.c. power in the audio frequency range 550 to 5000 cycles per second). P is a suitable detector (a headphone or other electronic device) and the bridge is balanced when no current passes through the detector. Under these conditions:

$$\text{Unknown resistance } R_2 = \frac{R_1 R_4}{R_3} \quad (2.19)$$

These days, inexpensive conductivity meters are available which can directly read the conductance or resistance of the solution in the conductivity cell. Once the cell constant and the resistance of the solution in the cell is determined, the conductivity of the solution is given by the equation:

$$\kappa = \frac{\text{cell constant}}{R} = \frac{G^*}{R} \quad (2.20)$$

The conductivity of solutions of different electrolytes in the same solvent and at a given temperature differs due to charge and size of the

ions in which they dissociate, the concentration of ions or ease with which the ions move under a potential gradient. It, therefore, becomes necessary to define a physically more meaningful quantity called **molar conductivity** denoted by the symbol Λ_m (Greek, lambda). It is related to the conductivity of the solution by the equation:

$$\text{Molar conductivity} = \Lambda_m = \frac{\kappa}{c} \quad (2.21)$$

In the above equation, if κ is expressed in S m^{-1} and the concentration, c in mol m^{-3} then the units of Λ_m are in $\text{S m}^2 \text{mol}^{-1}$. It may be noted that:

$$1 \text{ mol m}^{-3} = 1000(\text{L/m}^3) \times \text{molarity (mol/L)}, \text{ and hence}$$

$$\Lambda_m (\text{S cm}^2 \text{mol}^{-1}) = \frac{\kappa (\text{S cm}^{-1})}{1000 \text{ L m}^{-3} \times \text{molarity (mol L}^{-1}\text{)}}$$

If we use S cm^{-1} as the units for κ and mol cm^{-3} , the units of concentration, then the units for Λ_m are $\text{S cm}^2 \text{mol}^{-1}$. It can be calculated by using the equation:

$$\Lambda_m (\text{S cm}^2 \text{mol}^{-1}) = \frac{\kappa (\text{S cm}^{-1}) \times 1000 (\text{cm}^3 / \text{L})}{\text{molarity (mol/L)}}$$

Both type of units are used in literature and are related to each other by the equations:

$$\begin{aligned} 1 \text{ S m}^2 \text{mol}^{-1} &= 10^4 \text{ S cm}^2 \text{mol}^{-1} \quad \text{or} \\ 1 \text{ S cm}^2 \text{mol}^{-1} &= 10^{-4} \text{ S m}^2 \text{mol}^{-1}. \end{aligned}$$

Resistance of a conductivity cell filled with 0.1 mol L^{-1} KCl solution is 100Ω . If the resistance of the same cell when filled with 0.02 mol L^{-1} KCl solution is 520Ω , calculate the conductivity and molar conductivity of 0.02 mol L^{-1} KCl solution. The conductivity of 0.1 mol L^{-1} KCl solution is 1.29 S/m .

Example 2.4

Solution

The cell constant is given by the equation:

$$\text{Cell constant} = G^* = \text{conductivity} \times \text{resistance}$$

$$= 1.29 \text{ S/m} \times 100 \Omega = 129 \text{ m}^{-1} = 1.29 \text{ cm}^{-1}$$

$$\text{Conductivity of } 0.02 \text{ mol L}^{-1} \text{ KCl solution} = \text{cell constant} / \text{resistance}$$

$$= \frac{G^*}{R} = \frac{129 \text{ m}^{-1}}{520 \Omega} = 0.248 \text{ S m}^{-1}$$

Concentration

$$= 0.02 \text{ mol L}^{-1}$$

$$= 1000 \times 0.02 \text{ mol m}^{-3} = 20 \text{ mol m}^{-3}$$

$$\text{Molar conductivity} = \Lambda_m = \frac{\kappa}{c}$$

$$= \frac{248 \times 10^{-3} \text{ S m}^{-1}}{20 \text{ mol m}^{-3}} = 124 \times 10^{-4} \text{ S m}^2 \text{mol}^{-1}$$

Alternatively,

$$\kappa = \frac{1.29 \text{ cm}^{-1}}{520 \Omega} = 0.248 \times 10^{-2} \text{ S cm}^{-1}$$

$$\begin{aligned}
 \text{and} \quad \Lambda_m &= \kappa \times 1000 \text{ cm}^3 \text{ L}^{-1} \text{ molarity}^{-1} \\
 &= \frac{0.248 \times 10^{-2} \text{ S cm}^{-1} \times 1000 \text{ cm}^3 \text{ L}^{-1}}{0.02 \text{ mol L}^{-1}} \\
 &= 124 \text{ S cm}^2 \text{ mol}^{-1}
 \end{aligned}$$

Example 2.5 The electrical resistance of a column of 0.05 mol L^{-1} NaOH solution of diameter 1 cm and length 50 cm is $5.55 \times 10^3 \text{ ohm}$. Calculate its resistivity, conductivity and molar conductivity.

Solution $A = \pi r^2 = 3.14 \times 0.5^2 \text{ cm}^2 = 0.785 \text{ cm}^2 = 0.785 \times 10^{-4} \text{ m}^2$
 $l = 50 \text{ cm} = 0.5 \text{ m}$

$$R = \frac{\rho l}{A} \quad \text{or} \quad \rho = \frac{RA}{l} = \frac{5.55 \times 10^3 \Omega \times 0.785 \text{ cm}^2}{50 \text{ cm}} = 87.135 \Omega \text{ cm}$$

$$\begin{aligned}
 \text{Conductivity} = \kappa &= \frac{1}{\rho} = \left(\frac{1}{87.135} \right) \text{ S cm}^{-1} \\
 &= 0.01148 \text{ S cm}^{-1}
 \end{aligned}$$

$$\begin{aligned}
 \text{Molar conductivity, } \Lambda_m &= \frac{\kappa \times 1000}{c} \text{ cm}^3 \text{ L}^{-1} \\
 &= \frac{0.01148 \text{ S cm}^{-1} \times 1000 \text{ cm}^3 \text{ L}^{-1}}{0.05 \text{ mol L}^{-1}} \\
 &= 229.6 \text{ S cm}^2 \text{ mol}^{-1}
 \end{aligned}$$

If we want to calculate the values of different quantities in terms of 'm' instead of 'cm',

$$\begin{aligned}
 \rho &= \frac{RA}{l} \\
 &= \frac{5.55 \times 10^3 \Omega \times 0.785 \times 10^{-4} \text{ m}^2}{0.5 \text{ m}} = 87.135 \times 10^{-2} \Omega \text{ m}
 \end{aligned}$$

$$\kappa = \frac{1}{\rho} = \frac{100}{87.135} \Omega \text{ m} = 1.148 \text{ S m}^{-1}$$

$$\text{and } \Lambda_m = \frac{\kappa}{c} = \frac{1.148 \text{ S m}^{-1}}{50 \text{ mol m}^{-3}} = 229.6 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}.$$

2.4.2 Variation of Conductivity and Molar Conductivity with Concentration

Both conductivity and molar conductivity change with the concentration of the electrolyte. Conductivity always decreases with decrease in concentration both, for weak and strong electrolytes. This can be explained by the fact that the number of ions per unit volume that carry the current in a solution decreases on dilution. The conductivity of a solution at any given concentration is the conductance of one unit volume of solution kept between two

platinum electrodes with unit area of cross section and at a distance of unit length. This is clear from the equation:

$G = \frac{\kappa A}{l} = \kappa$ (both A and l are unity in their appropriate units in m or cm)

Molar conductivity of a solution at a given concentration is the conductance of the volume V of solution containing one mole of electrolyte kept between two electrodes with area of cross section A and distance of unit length. Therefore,

$$\Lambda_m = \frac{\kappa A}{l} = \kappa$$

Since $l = 1$ and $A = V$ (volume containing 1 mole of electrolyte)

$$\Lambda_m = \kappa V \quad (2.22)$$

Molar conductivity increases with decrease in concentration. This is because the total volume, V , of solution containing one mole of electrolyte also increases. It has been found that decrease in κ on dilution of a solution is more than compensated by increase in its volume. Physically, it means that at a given concentration, Λ_m can be defined as the conductance of the electrolytic solution kept between the electrodes of a conductivity cell at unit distance but having area of cross section large enough to accommodate sufficient volume of solution that contains one mole of the electrolyte. When concentration approaches zero, the molar conductivity is known as **limiting molar conductivity** and is represented by the symbol Λ_m° . The variation in Λ_m with concentration is different (Fig. 2.6) for strong and weak electrolytes.

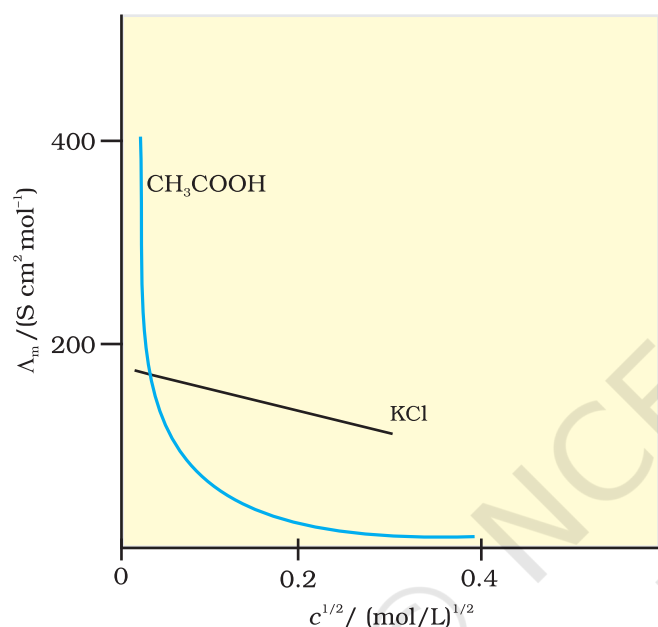


Fig. 2.6: Molar conductivity versus $c^{1/2}$ for acetic acid (weak electrolyte) and potassium chloride (strong electrolyte) in aqueous solutions.

Strong Electrolytes

For strong electrolytes, Λ_m increases slowly with dilution and can be represented by the equation:

$$\Lambda_m = \Lambda_m^\circ - A c^{1/2} \quad (2.23)$$

It can be seen that if we plot (Fig. 2.6) Λ_m against $c^{1/2}$, we obtain a straight line with intercept equal to Λ_m° and slope equal to $-A$. The value of the constant A for a given solvent and temperature depends on the type of electrolyte i.e., the charges on the cation and anion produced on the dissociation of the electrolyte in the solution. Thus, NaCl , CaCl_2 , MgSO_4 are known as 1-1, 2-1 and 2-2 electrolytes respectively. All electrolytes of a particular type have the same value for A .

Example 2.6 The molar conductivity of KCl solutions at different concentrations at 298 K are given below:

$c/\text{mol L}^{-1}$	$\Lambda_m/\text{S cm}^2 \text{mol}^{-1}$
0.000198	148.61
0.000309	148.29
0.000521	147.81
0.000989	147.09

Show that a plot between Λ_m and $c^{1/2}$ is a straight line. Determine the values of Λ_m° and A for KCl.

Solution Taking the square root of concentration we obtain:

$c^{1/2}/(\text{mol L}^{-1})^{1/2}$	$\Lambda_m/\text{S cm}^2 \text{mol}^{-1}$
0.01407	148.61
0.01758	148.29
0.02283	147.81
0.03145	147.09

A plot of Λ_m (y-axis) and $c^{1/2}$ (x-axis) is shown in (Fig. 3.7).

It can be seen that it is nearly a straight line. From the intercept ($c^{1/2} = 0$), we find that

$$\Lambda_m^\circ = 150.0 \text{ S cm}^2 \text{mol}^{-1} \text{ and}$$

$$A = -\text{slope} = 87.46 \text{ S cm}^2 \text{mol}^{-1}/(\text{mol/L}^{-1})^{1/2}.$$

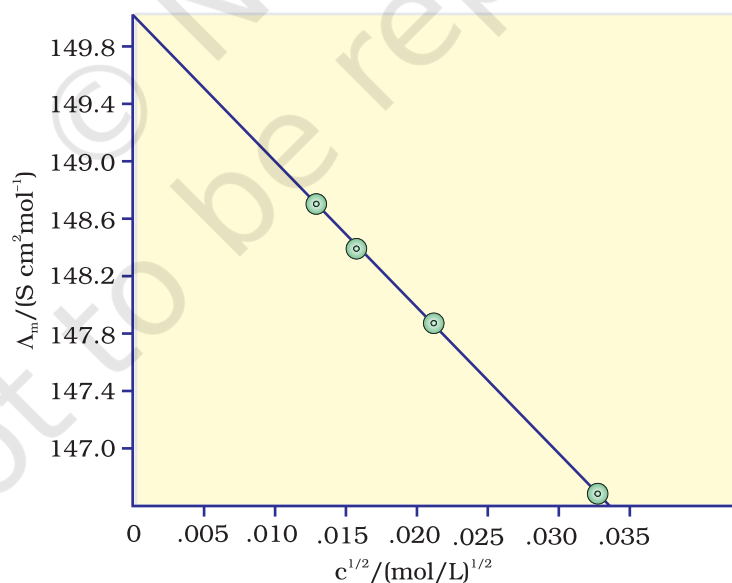


Fig. 2.7: Variation of Λ_m against $c^{1/2}$.

Kohlrausch examined Λ_m° values for a number of strong electrolytes and observed certain regularities. He noted that the difference in Λ_m° of the electrolytes NaX and KX for any X is nearly constant. For example at 298 K:

$$\begin{aligned}\Lambda_m^\circ(\text{KCl}) - \Lambda_m^\circ(\text{NaCl}) &= \Lambda_m^\circ(\text{KBr}) - \Lambda_m^\circ(\text{NaBr}) \\ &= \Lambda_m^\circ(\text{KI}) - \Lambda_m^\circ(\text{NaI}) \simeq 23.4 \text{ S cm}^2 \text{ mol}^{-1}\end{aligned}$$

and similarly it was found that

$$\Lambda_m^\circ(\text{NaBr}) - \Lambda_m^\circ(\text{NaCl}) = \Lambda_m^\circ(\text{KBr}) - \Lambda_m^\circ(\text{KCl}) \simeq 1.8 \text{ S cm}^2 \text{ mol}^{-1}$$

On the basis of the above observations he enunciated **Kohlrausch law of independent migration of ions**. The law states that *limiting molar conductivity of an electrolyte can be represented as the sum of the individual contributions of the anion and cation of the electrolyte*. Thus, if $\lambda_{\text{Na}^+}^\circ$ and $\lambda_{\text{Cl}^-}^\circ$ are limiting molar conductivity of the sodium and chloride ions respectively, then the limiting molar conductivity for sodium chloride is given by the equation:

$$\Lambda_m^\circ(\text{NaCl}) = \lambda_{\text{Na}^+}^\circ + \lambda_{\text{Cl}^-}^\circ \quad (2.24)$$

In general, if an electrolyte on dissociation gives ν_+ cations and ν_- anions then its limiting molar conductivity is given by:

$$\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ \quad (2.25)$$

Here, λ_+° and λ_-° are the limiting molar conductivities of the cation and anion respectively. The values of λ° for some cations and anions at 298 K are given in Table 2.4.

Table 2.4: Limiting Molar Conductivity for some Ions in Water at 298 K

Ion	$\lambda^\circ / (\text{S cm}^2 \text{ mol}^{-1})$	Ion	$\lambda^\circ / (\text{S cm}^2 \text{ mol}^{-1})$
H ⁺	349.6	OH ⁻	199.1
Na ⁺	50.1	Cl ⁻	76.3
K ⁺	73.5	Br ⁻	78.1
Ca ²⁺	119.0	CH ₃ COO ⁻	40.9
Mg ²⁺	106.0	SO ₄ ²⁻	160.0

Weak Electrolytes

Weak electrolytes like acetic acid have lower degree of dissociation at higher concentrations and hence for such electrolytes, the change in Λ_m with dilution is due to increase in the degree of dissociation and consequently the number of ions in total volume of solution that contains 1 mol of electrolyte. In such cases Λ_m increases steeply (Fig. 2.6) on dilution, especially near lower concentrations. Therefore, Λ_m° cannot be obtained by extrapolation of Λ_m to zero concentration. At infinite dilution (i.e., concentration $c \rightarrow$ zero) electrolyte dissociates completely ($\alpha = 1$), but at such low concentration the conductivity of the solution is so low that it cannot be measured accurately. Therefore, Λ_m° for weak electrolytes is obtained by using Kohlrausch law of independent migration of ions (Example 2.8). At any concentration c , if α is the degree of dissociation

then it can be approximated to the ratio of molar conductivity Λ_m at the concentration c to limiting molar conductivity, Λ_m^0 . Thus we have:

$$\alpha = \frac{\Lambda_m}{\Lambda_m^0} \quad (2.26)$$

But we know that for a weak electrolyte like acetic acid (Class XI, Unit 7),

$$K_a = \frac{c\alpha^2}{(1-\alpha)} = \frac{c\Lambda_m^2}{\Lambda_m^0{}^2 \left(1 - \frac{\Lambda_m}{\Lambda_m^0}\right)} = \frac{c\Lambda_m^2}{\Lambda_m^0(\Lambda_m^0 - \Lambda_m)} \quad (2.27)$$

Applications of Kohlrausch law

Using Kohlrausch law of independent migration of ions, it is possible to calculate Λ_m^0 for any electrolyte from the λ^0 of individual ions. Moreover, for weak electrolytes like acetic acid it is possible to determine the value of its dissociation constant once we know the Λ_m^0 and Λ_m at a given concentration c .

Example 2.7 Calculate Λ_m^0 for CaCl_2 and MgSO_4 from the data given in Table 3.4.

Solution We know from Kohlrausch law that

$$\begin{aligned} \Lambda_{m(\text{CaCl}_2)}^0 &= \lambda_{\text{Ca}^{2+}}^0 + 2\lambda_{\text{Cl}^-}^0 = 119.0 \text{ S cm}^2 \text{ mol}^{-1} + 2(76.3) \text{ S cm}^2 \text{ mol}^{-1} \\ &= (119.0 + 152.6) \text{ S cm}^2 \text{ mol}^{-1} \\ &= 271.6 \text{ S cm}^2 \text{ mol}^{-1} \end{aligned}$$

$$\begin{aligned} \Lambda_{m(\text{MgSO}_4)}^0 &= \lambda_{\text{Mg}^{2+}}^0 + \lambda_{\text{SO}_4^{2-}}^0 = 106.0 \text{ S cm}^2 \text{ mol}^{-1} + 160.0 \text{ S cm}^2 \text{ mol}^{-1} \\ &= 266 \text{ S cm}^2 \text{ mol}^{-1} \end{aligned}$$

Example 2.8 Λ_m^0 for NaCl , HCl and NaAc are 126.4, 425.9 and 91.0 $\text{S cm}^2 \text{ mol}^{-1}$ respectively. Calculate Λ^0 for HAc .

Solution

$$\begin{aligned} \Lambda_{m(\text{HAc})}^0 &= \lambda_{\text{H}^+}^0 + \lambda_{\text{Ac}^-}^0 = \lambda_{\text{H}^+}^0 + \lambda_{\text{Cl}^-}^0 + \lambda_{\text{Ac}^-}^0 + \lambda_{\text{Na}^+}^0 - \lambda_{\text{Cl}^-}^0 - \lambda_{\text{Na}^+}^0 \\ &= \Lambda_{m(\text{HCl})}^0 + \Lambda_{m(\text{NaAc})}^0 - \Lambda_{m(\text{NaCl})}^0 \\ &= (425.9 + 91.0 - 126.4) \text{ S cm}^2 \text{ mol}^{-1} \\ &= 390.5 \text{ S cm}^2 \text{ mol}^{-1} \end{aligned}$$

Example 2.9 The conductivity of $0.001028 \text{ mol L}^{-1}$ acetic acid is $4.95 \times 10^{-5} \text{ S cm}^{-1}$. Calculate its dissociation constant if Λ_m^0 for acetic acid is $390.5 \text{ S cm}^2 \text{ mol}^{-1}$.

Solution

$$\begin{aligned} \Lambda_m &= \frac{\kappa}{c} = \frac{4.95 \times 10^{-5} \text{ S cm}^{-1}}{0.001028 \text{ mol L}^{-1}} \times \frac{1000 \text{ cm}^3}{\text{L}} = 48.15 \text{ S cm}^2 \text{ mol}^{-1} \\ \alpha &= \frac{\Lambda_m}{\Lambda_m^0} = \frac{48.15 \text{ S cm}^2 \text{ mol}^{-1}}{390.5 \text{ S cm}^2 \text{ mol}^{-1}} = 0.1233 \\ k &= \frac{c\alpha^2}{(1-\alpha)} = \frac{0.001028 \text{ mol L}^{-1} \times (0.1233)^2}{1-0.1233} = 1.78 \times 10^{-5} \text{ mol L}^{-1} \end{aligned}$$

- 2.7** Why does the conductivity of a solution decrease with dilution?
- 2.8** Suggest a way to determine the Λ_m° value of water.
- 2.9** The molar conductivity of 0.025 mol L⁻¹ methanoic acid is 46.1 S cm² mol⁻¹. Calculate its degree of dissociation and dissociation constant. Given $\lambda^\circ(\text{H}^+) = 349.6 \text{ S cm}^2 \text{ mol}^{-1}$ and $\lambda^\circ(\text{HCOO}^-) = 54.6 \text{ S cm}^2 \text{ mol}^{-1}$.

2.5 Electrolytic Cells and Electrolysis

In an **electrolytic cell** external source of voltage is used to bring about a chemical reaction. The electrochemical processes are of great importance in the laboratory and the chemical industry. One of the simplest electrolytic cell consists of two copper strips dipping in an aqueous solution of copper sulphate. If a DC voltage is applied to the two electrodes, then Cu²⁺ ions discharge at the cathode (negatively charged) and the following reaction takes place:



Copper metal is deposited on the cathode. At the anode, copper is converted into Cu²⁺ ions by the reaction:



Thus copper is dissolved (oxidised) at anode and deposited (reduced) at cathode. This is the basis for an industrial process in which impure copper is converted into copper of high purity. The impure copper is made an anode that dissolves on passing current and pure copper is deposited at the cathode. Many metals like Na, Mg, Al, etc. are produced on large scale by electrochemical reduction of their respective cations where no suitable chemical reducing agents are available for this purpose.

Sodium and magnesium metals are produced by the electrolysis of their fused chlorides and aluminium is produced by electrolysis of aluminium oxide in presence of cryolite.

Quantitative Aspects of Electrolysis

Michael Faraday was the first scientist who described the quantitative aspects of electrolysis. Now Faraday's laws also flow from what has been discussed earlier.

Faraday's Laws of Electrolysis

After his extensive investigations on electrolysis of solutions and melts of electrolytes, Faraday published his results during 1833-34 in the form of the following well known Faraday's two laws of electrolysis:

- (i) *First Law*: The amount of chemical reaction which occurs at any electrode during electrolysis by a current is proportional to the quantity of electricity passed through the electrolyte (solution or melt).
- (ii) *Second Law*: The amounts of different substances liberated by the same quantity of electricity passing through the electrolytic solution are proportional to their chemical equivalent weights (Atomic Mass of Metal ÷ Number of electrons required to reduce the cation).

There were no constant current sources available during Faraday's times. The general practice was to put a coulometer (a standard electrolytic cell) for determining the quantity of electricity passed from the amount of metal (generally silver or copper) deposited or consumed. However, coulometers are now obsolete and we now have constant current (I) sources available and the quantity of electricity Q , passed is given by

$$Q = It$$

Q is in coulombs when I is in ampere and t is in second.

The amount of electricity (or charge) required for oxidation or reduction depends on the stoichiometry of the electrode reaction. For example, in the reaction:



One mole of the electron is required for the reduction of one mole of silver ions.

We know that charge on one electron is equal to $1.6021 \times 10^{-19} \text{C}$.

Therefore, the charge on one mole of electrons is equal to:

$$\begin{aligned} N_A \times 1.6021 \times 10^{-19} \text{ C} &= 6.02 \times 10^{23} \text{ mol}^{-1} \times 1.6021 \times 10^{-19} \\ &= 96487 \text{ C mol}^{-1} \end{aligned}$$

This quantity of electricity is called **Faraday** and is represented by the symbol **F**.

For approximate calculations we use $1\text{F} \approx 96500 \text{ C mol}^{-1}$.

For the electrode reactions:



It is obvious that one mole of Mg^{2+} and Al^{3+} require 2 mol of electrons (2F) and 3 mol of electrons (3F) respectively. The charge passed through the electrolytic cell during electrolysis is equal to the product of current in amperes and time in seconds. In commercial production of metals, current as high as 50,000 amperes are used that amounts to about 0.518 F per second.

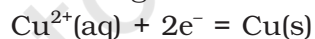
Example 2.10

A solution of CuSO_4 is electrolysed for 10 minutes with a current of 1.5 amperes. What is the mass of copper deposited at the cathode?

Solution

$t = 600 \text{ s}$ charge = current $\times$ time = $1.5 \text{ A} \times 600 \text{ s} = 900 \text{ C}$

According to the reaction:



We require 2F or $2 \times 96487 \text{ C}$ to deposit 1 mol or 63 g of Cu.

For 900 C, the mass of Cu deposited

$$= (63 \text{ g mol}^{-1} \times 900 \text{ C}) / (2 \times 96487 \text{ C mol}^{-1}) = 0.2938 \text{ g.}$$

2.5.1 Products of Electrolysis

Products of electrolysis depend on the nature of material being electrolysed and the type of electrodes being used. If the electrode is inert (e.g., platinum or gold), it does not participate in the chemical reaction and acts only as source or sink for electrons. On the other hand, if the electrode is reactive, it participates in the electrode reaction. Thus, the products of electrolysis may be different for reactive and inert

electrodes. The products of electrolysis depend on the different oxidising and reducing species present in the electrolytic cell and their standard electrode potentials. Moreover, some of the electrochemical processes although feasible, are so slow kinetically that at lower voltages these do not seem to take place and extra potential (called *overpotential*) has to be applied, which makes such process more difficult to occur.

For example, if we use molten NaCl, the products of electrolysis are sodium metal and Cl₂ gas. Here we have only one cation (Na⁺) which is reduced at the cathode (Na⁺ + e⁻ → Na) and one anion (Cl⁻) which is oxidised at the anode (Cl⁻ → ½Cl₂ + e⁻). During the electrolysis of aqueous sodium chloride solution, the products are NaOH, Cl₂ and H₂. In this case besides Na⁺ and Cl⁻ ions we also have H⁺ and OH⁻ ions along with the solvent molecules, H₂O.

At the cathode there is competition between the following reduction reactions:



The reaction with higher value of E° is preferred and therefore, the reaction at the cathode during electrolysis is:



but H⁺ (aq) is produced by the dissociation of H₂O, i.e.,



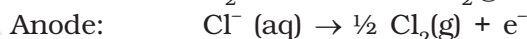
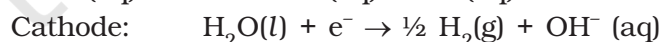
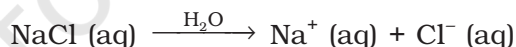
Therefore, the net reaction at the cathode may be written as the sum of (2.33) and (2.34) and we have



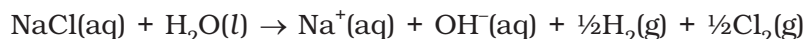
At the anode the following oxidation reactions are possible:



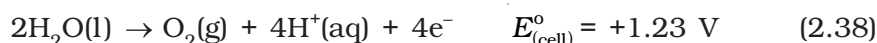
The reaction at anode with lower value of E° is preferred and therefore, water should get oxidised in preference to Cl⁻ (aq). However, on account of overpotential of oxygen, reaction (2.36) is preferred. Thus, the net reactions may be summarised as:

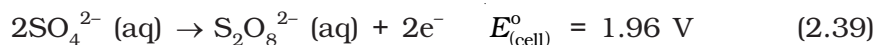


Net reaction:



The standard electrode potentials are replaced by electrode potentials given by Nernst equation (Eq. 2.8) to take into account the concentration effects. During the electrolysis of sulphuric acid, the following processes are possible at the anode:





For dilute sulphuric acid, reaction (2.38) is preferred but at higher concentrations of H_2SO_4 , reaction (2.39) is preferred.

Intext Questions

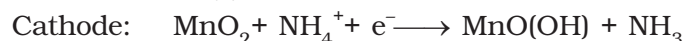
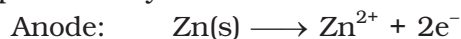
- 2.10** If a current of 0.5 ampere flows through a metallic wire for 2 hours, then how many electrons would flow through the wire?
- 2.11** Suggest a list of metals that are extracted electrolytically.
- 2.12** Consider the reaction: $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$
What is the quantity of electricity in coulombs needed to reduce 1 mol of $\text{Cr}_2\text{O}_7^{2-}$?

2.6 Batteries

Any battery (actually it may have one or more than one cell connected in series) or cell that we use as a source of electrical energy is basically a galvanic cell where the chemical energy of the redox reaction is converted into electrical energy. However, for a battery to be of practical use it should be reasonably light, compact and its voltage should not vary appreciably during its use. There are mainly two types of batteries.

2.6.1 Primary Batteries

In the primary batteries, the reaction occurs only once and after use over a period of time battery becomes dead and cannot be reused again. The most familiar example of this type is the dry cell (known as Leclanche cell after its discoverer) which is used commonly in our transistors and clocks. The cell consists of a zinc container that also acts as anode and the cathode is a carbon (graphite) rod surrounded by powdered manganese dioxide and carbon (Fig.2.8). The space between the electrodes is filled by a moist paste of ammonium chloride (NH_4Cl) and zinc chloride (ZnCl_2). The electrode reactions are complex, but they can be written approximately as follows :



In the reaction at cathode, manganese is reduced from the + 4 oxidation state to the +3 state. Ammonia produced in the reaction forms a complex with Zn^{2+} to give $[\text{Zn}(\text{NH}_3)_4]^{2+}$. The cell has a potential of nearly 1.5 V.

Mercury cell, (Fig. 2.9) suitable for low current devices like hearing aids, watches, etc. consists of zinc – mercury amalgam as anode and a paste of HgO and carbon as the cathode. The electrolyte is a paste of KOH and ZnO . The electrode reactions for the cell are given below:

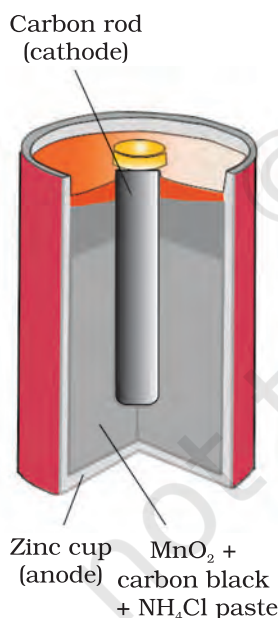
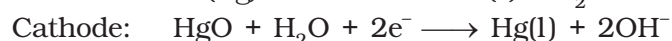
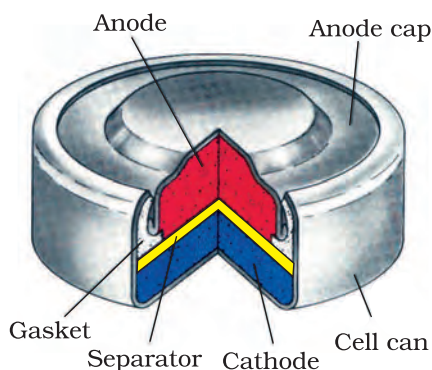


Fig. 2.8: A commercial dry cell consists of a graphite (carbon) cathode in a zinc container; the latter acts as the anode.

Fig. 2.9

Commonly used mercury cell. The reducing agent is zinc and the oxidising agent is mercury (II) oxide.



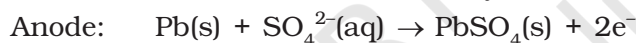
The overall reaction is represented by
$$\text{Zn(Hg)} + \text{HgO(s)} \longrightarrow \text{ZnO(s)} + \text{Hg(l)}$$

The cell potential is approximately 1.35 V and remains constant during its life as the overall reaction does not involve any ion in solution whose concentration can change during its life time.

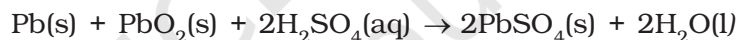
2.6.2 Secondary Batteries

A secondary cell after use can be recharged by passing current through it in the opposite direction so that it can be used again. A good secondary cell can undergo a large number of discharging and charging cycles. The most important secondary cell is the lead storage battery (Fig. 2.10) commonly used in automobiles and invertors. It consists of a lead anode and a grid of lead packed with lead dioxide (PbO_2) as cathode. A 38% solution of sulphuric acid is used as an electrolyte.

The cell reactions when the battery is in use are given below:



i.e., overall cell reaction consisting of cathode and anode reactions is:



On charging the battery the reaction is reversed and $\text{PbSO}_4(\text{s})$ on anode and cathode is converted into Pb and PbO_2 , respectively.

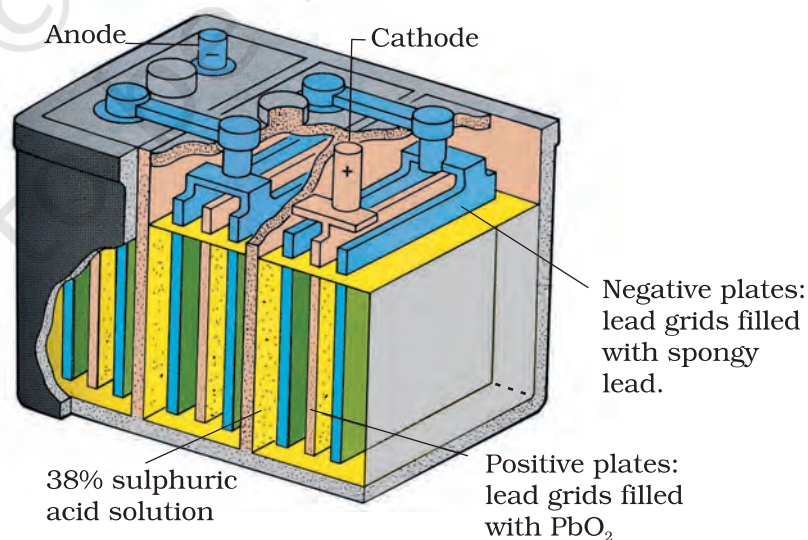
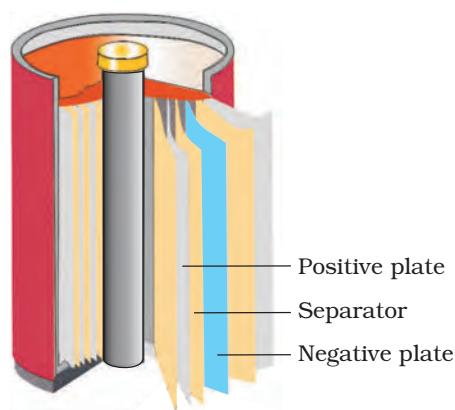


Fig. 2.10: The Lead storage battery.

Fig. 2.11

A rechargeable nickel-cadmium cell in a jelly roll arrangement and separated by a layer soaked in moist sodium or potassium hydroxide.



Another important secondary cell is the nickel-cadmium cell (Fig. 2.11) which has longer life than the lead storage cell but more expensive to manufacture. We shall not go into details of working of the cell and the electrode reactions during charging and discharging. The overall reaction during discharge is:



2.7 Fuel Cells

Production of electricity by thermal plants is not a very efficient method and is a major source of pollution. In such plants, the chemical energy (heat of combustion) of fossil fuels (coal, gas or oil) is first used for converting water into high pressure steam. This is then used to run a turbine to produce electricity. We know that a galvanic cell directly converts chemical energy into electricity and is highly efficient. It is now possible to make such cells in which reactants are fed continuously to the electrodes and products are removed continuously from the electrolyte compartment. **Galvanic cells** that are designed to convert the energy of combustion of fuels like hydrogen, methane, methanol, etc. directly into electrical energy are called **fuel cells**.

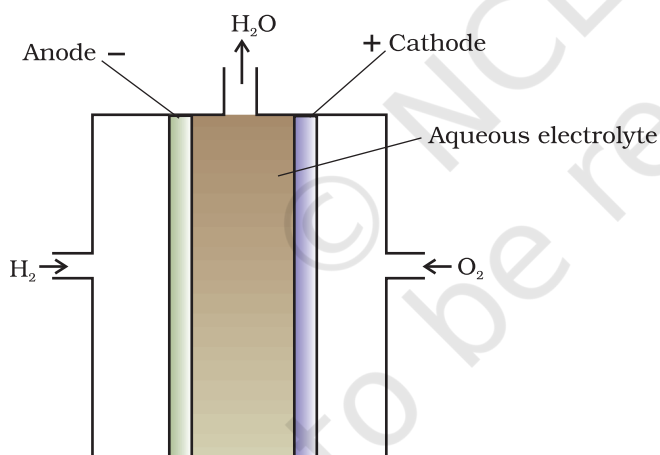


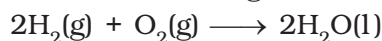
Fig. 2.12: Fuel cell using H₂ and O₂ produces electricity.

One of the most successful fuel cells uses the reaction of hydrogen with oxygen to form water (Fig. 2.12). The cell was used for providing electrical power in the Apollo space programme. The water vapours produced during the reaction were condensed and added to the drinking water supply for the astronauts. In the cell, hydrogen and oxygen are bubbled through porous carbon electrodes into concentrated aqueous sodium hydroxide solution. Catalysts like finely divided platinum or palladium metal are incorporated into the electrodes for increasing the rate of electrode reactions. The electrode reactions are

given below:



Overall reaction being:



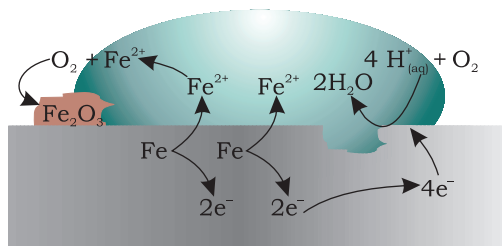
The cell runs continuously as long as the reactants are supplied. Fuel cells produce electricity with an efficiency of about 70 % compared

to thermal plants whose efficiency is about 40%. There has been tremendous progress in the development of new electrode materials, better catalysts and electrolytes for increasing the efficiency of fuel cells. These have been used in automobiles on an experimental basis. Fuel cells are pollution free and in view of their future importance, a variety of fuel cells have been fabricated and tried.

2.8 Corrosion

Corrosion slowly coats the surfaces of metallic objects with oxides or other salts of the metal. The rusting of iron, tarnishing of silver, development of green coating on copper and bronze are some of the examples of corrosion. It causes enormous damage to buildings, bridges, ships and to all objects made of metals especially that of iron. We lose crores of rupees every year on account of corrosion.

In corrosion, a metal is oxidised by loss of electrons to oxygen and formation of oxides. Corrosion of iron (commonly known as rusting) occurs in presence of water and air. The chemistry of corrosion is quite complex but it may be considered essentially as an electrochemical phenomenon. At a particular spot (Fig. 2.13) of an object made of iron, oxidation takes place and that spot behaves as anode and we can write the reaction



Oxidation: $\text{Fe (s)} \rightarrow \text{Fe}^{2+}(\text{aq}) + 2\text{e}^-$

Reduction: $\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}(\text{l})$

Atmospheric

oxidation: $2\text{Fe}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{Fe}_2\text{O}_3(\text{s}) + 4\text{H}^+(\text{aq})$

Fig. 2.13: Corrosion of iron in atmosphere

Anode: $2\text{Fe (s)} \longrightarrow 2\text{Fe}^{2+} + 4\text{e}^- \quad E_{(\text{Fe}^{2+}/\text{Fe})}^{\circ} = -0.44\text{ V}$

Electrons released at anodic spot move through the metal and go to another spot on the metal and reduce oxygen in the presence of H^+ (which is believed to be available from H_2CO_3 formed due to dissolution of carbon dioxide from air into water. Hydrogen ion in water may also be available due to dissolution of other acidic oxides from the atmosphere). This spot behaves as cathode with the reaction

Cathode: $\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \longrightarrow 2\text{H}_2\text{O (l)} \quad E_{\text{H}^+|\text{O}_2|\text{H}_2\text{O}}^{\circ} = 1.23\text{ V}$

The overall reaction being:

$2\text{Fe(s)} + \text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) \longrightarrow 2\text{Fe}^{2+}(\text{aq}) + 2\text{H}_2\text{O (l)} \quad E_{(\text{cell})}^{\circ} = 1.67\text{ V}$

The ferrous ions are further oxidised by atmospheric oxygen to ferric ions which come out as rust in the form of hydrated ferric oxide ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$) and with further production of hydrogen ions.

Prevention of corrosion is of prime importance. It not only saves money but also helps in preventing accidents such as a bridge collapse or failure of a key component due to corrosion. One of the simplest methods of preventing corrosion is to prevent the surface of the metallic object to come in contact with atmosphere. This can be done by covering the surface with paint or by some chemicals (e.g. bisphenol). Another simple method is to cover the surface by other metals (Sn, Zn, etc.) that are inert or react to save the object. An electrochemical method is to provide a sacrificial electrode of another metal (like Mg, Zn, etc.) which corrodes itself but saves the object.

Intext Questions

- 2.13** Write the chemistry of recharging the lead storage battery, highlighting all the materials that are involved during recharging.
- 2.14** Suggest two materials other than hydrogen that can be used as fuels in fuel cells.
- 2.15** Explain how rusting of iron is envisaged as setting up of an electrochemical cell.

The Hydrogen Economy

At present the main source of energy that is driving our economy is fossil fuels such as coal, oil and gas. As more people on the planet aspire to improve their standard of living, their energy requirement will increase. In fact, the per capita consumption of energy used is a measure of development. Of course, it is assumed that energy is used for productive purpose and not merely wasted. We are already aware that carbon dioxide produced by the combustion of fossil fuels is resulting in the 'Greenhouse Effect'. This is leading to a rise in the temperature of the Earth's surface, causing polar ice to melt and ocean levels to rise. This will flood low-lying areas along the coast and some island nations such as Maldives face total submergence. In order to avoid such a catastrophe, we need to limit our use of carbonaceous fuels. Hydrogen provides an ideal alternative as its combustion results in water only. Hydrogen production must come from splitting water using solar energy. Therefore, hydrogen can be used as a renewable and non polluting source of energy. This is the vision of the Hydrogen Economy. Both the production of hydrogen by electrolysis of water and hydrogen combustion in a fuel cell will be important in the future. And both these technologies are based on electrochemical principles.

Summary

An **electrochemical cell** consists of two metallic electrodes dipping in electrolytic solution(s). Thus an important component of the electrochemical cell is the ionic conductor or electrolyte. Electrochemical cells are of two types. In **galvanic cell**, the **chemical energy** of a **spontaneous redox reaction** is converted into electrical work, whereas in an electrolytic cell, electrical energy is used to carry out a **non-spontaneous redox reaction**. The **standard electrode potential** for any electrode dipping in an appropriate solution is defined with respect to standard electrode potential of **hydrogen electrode** taken as zero. The standard potential of the cell can be obtained by taking the difference of the standard potentials of cathode and anode ($E_{\text{(cell)}}^{\circ} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}}$). The standard potential of the cells are related to standard Gibbs energy ($\Delta_r G^{\circ} = -nF E_{\text{(cell)}}^{\circ}$) and **equilibrium constant** ($\Delta_r G^{\circ} = -RT \ln K$) of the reaction taking place in the cell. Concentration dependence of the potentials of the electrodes and the cells are given by Nernst equation.

The **conductivity**, κ , of an electrolytic solution depends on the concentration of the electrolyte, nature of solvent and temperature. **Molar conductivity**, Λ_m , is defined by $\Lambda_m = \kappa/c$ where c is the concentration. Conductivity decreases but molar conductivity increases with decrease in concentration. It increases slowly with decrease in concentration for strong electrolytes while the increase is very steep for weak electrolytes in very dilute solutions. Kohlrausch found that molar conductivity at infinite dilution, for an electrolyte is sum of the contribution of the

molar conductivity of the ions in which it dissociates. It is known as **law of independent migration of ions** and has many applications. Ions conduct electricity through the solution but oxidation and reduction of the ions take place at the electrodes in an electrochemical cell. **Batteries** and **fuel cells** are very useful forms of galvanic cell. **Corrosion** of metals is essentially an **electrochemical phenomenon**. Electrochemical principles are relevant to the **Hydrogen Economy**.

Exercises

- 2.1** Arrange the following metals in the order in which they displace each other from the solution of their salts.
Al, Cu, Fe, Mg and Zn.
- 2.2** Given the standard electrode potentials,
 $K^+/K = -2.93V$, $Ag^+/Ag = 0.80V$,
 $Hg^{2+}/Hg = 0.79V$
 $Mg^{2+}/Mg = -2.37 V$, $Cr^{3+}/Cr = -0.74V$
 Arrange these metals in their increasing order of reducing power.
- 2.3** Depict the galvanic cell in which the reaction
 $Zn(s) + 2Ag^+(aq) \rightarrow Zn^{2+}(aq) + 2Ag(s)$ takes place. Further show:
 (i) Which of the electrode is negatively charged?
 (ii) The carriers of the current in the cell.
 (iii) Individual reaction at each electrode.
- 2.4** Calculate the standard cell potentials of galvanic cell in which the following reactions take place:
 (i) $2Cr(s) + 3Cd^{2+}(aq) \rightarrow 2Cr^{3+}(aq) + 3Cd$
 (ii) $Fe^{2+}(aq) + Ag^+(aq) \rightarrow Fe^{3+}(aq) + Ag(s)$
 Calculate the $\Delta_r G^\circ$ and equilibrium constant of the reactions.
- 2.5** Write the Nernst equation and emf of the following cells at 298 K:
 (i) $Mg(s) | Mg^{2+}(0.001M) || Cu^{2+}(0.0001 M) | Cu(s)$
 (ii) $Fe(s) | Fe^{2+}(0.001M) || H^+(1M) | H_2(g)(1bar) | Pt(s)$
 (iii) $Sn(s) | Sn^{2+}(0.050 M) || H^+(0.020 M) | H_2(g) (1 bar) | Pt(s)$
 (iv) $Pt(s) | Br^-(0.010 M) | Br_2(l) || H^+(0.030 M) | H_2(g) (1 bar) | Pt(s)$.
- 2.6** In the button cells widely used in watches and other devices the following reaction takes place:
 $Zn(s) + Ag_2O(s) + H_2O(l) \rightarrow Zn^{2+}(aq) + 2Ag(s) + 2OH^-(aq)$
 Determine $\Delta_r G^\circ$ and E° for the reaction.
- 2.7** Define conductivity and molar conductivity for the solution of an electrolyte. Discuss their variation with concentration.
- 2.8** The conductivity of 0.20 M solution of KCl at 298 K is $0.0248 S cm^{-1}$. Calculate its molar conductivity.
- 2.9** The resistance of a conductivity cell containing 0.001M KCl solution at 298 K is 1500Ω . What is the cell constant if conductivity of 0.001M KCl solution at 298 K is $0.146 \times 10^{-3} S cm^{-1}$.

- 2.10** The conductivity of sodium chloride at 298 K has been determined at different concentrations and the results are given below:

Concentration/M	0.001	0.010	0.020	0.050	0.100
$10^2 \times \kappa/\text{S m}^{-1}$	1.237	11.85	23.15	55.53	106.74

Calculate Λ_m for all concentrations and draw a plot between Λ_m and $c^{1/2}$. Find the value of Λ_m^0 .

- 2.11** Conductivity of 0.00241 M acetic acid is $7.896 \times 10^{-5} \text{ S cm}^{-1}$. Calculate its molar conductivity. If Λ_m^0 for acetic acid is $390.5 \text{ S cm}^2 \text{ mol}^{-1}$, what is its dissociation constant?
- 2.12** How much charge is required for the following reductions:
 (i) 1 mol of Al^{3+} to Al?
 (ii) 1 mol of Cu^{2+} to Cu?
 (iii) 1 mol of MnO_4^- to Mn^{2+} ?
- 2.13** How much electricity in terms of Faraday is required to produce
 (i) 20.0 g of Ca from molten CaCl_2 ?
 (ii) 40.0 g of Al from molten Al_2O_3 ?
- 2.14** How much electricity is required in coulomb for the oxidation of
 (i) 1 mol of H_2O to O_2 ?
 (ii) 1 mol of FeO to Fe_2O_3 ?
- 2.15** A solution of $\text{Ni}(\text{NO}_3)_2$ is electrolysed between platinum electrodes using a current of 5 amperes for 20 minutes. What mass of Ni is deposited at the cathode?
- 2.16** Three electrolytic cells A,B,C containing solutions of ZnSO_4 , AgNO_3 and CuSO_4 , respectively are connected in series. A steady current of 1.5 amperes was passed through them until 1.45 g of silver deposited at the cathode of cell B. How long did the current flow? What mass of copper and zinc were deposited?
- 2.17** Using the standard electrode potentials given in Table 3.1, predict if the reaction between the following is feasible:
 (i) $\text{Fe}^{3+}(\text{aq})$ and $\text{I}^-(\text{aq})$
 (ii) $\text{Ag}^+(\text{aq})$ and $\text{Cu}(\text{s})$
 (iii) $\text{Fe}^{3+}(\text{aq})$ and $\text{Br}^-(\text{aq})$
 (iv) $\text{Ag}(\text{s})$ and $\text{Fe}^{3+}(\text{aq})$
 (v) $\text{Br}_2(\text{aq})$ and $\text{Fe}^{2+}(\text{aq})$.
- 2.18** Predict the products of electrolysis in each of the following:
 (i) An aqueous solution of AgNO_3 with silver electrodes.
 (ii) An aqueous solution of AgNO_3 with platinum electrodes.
 (iii) A dilute solution of H_2SO_4 with platinum electrodes.
 (iv) An aqueous solution of CuCl_2 with platinum electrodes.

Answers to Some Intext Questions

2.5 $E_{(\text{cell})} = 0.91 \text{ V}$

2.6 $\Delta_r G^\circ = -45.54 \text{ kJ mol}^{-1}$, $K_c = 9.62 \times 10^7$

2.9 0.114, $3.67 \times 10^{-4} \text{ mol L}^{-1}$



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Unit

3

Chemical Kinetics

Objectives

After studying this Unit, you will be able to

- define the average and instantaneous rate of a reaction;
- express the rate of a reaction in terms of change in concentration of either of the reactants or products with time;
- distinguish between elementary and complex reactions;
- differentiate between the molecularity and order of a reaction;
- define rate constant;
- discuss the dependence of rate of reactions on concentration, temperature and catalyst;
- derive integrated rate equations for the zero and first order reactions;
- determine the rate constants for zeroth and first order reactions;
- describe collision theory.

Chemical Kinetics helps us to understand how chemical reactions occur.

Chemistry, by its very nature, is concerned with change. Substances with well defined properties are converted by chemical reactions into other substances with different properties. For any chemical reaction, chemists try to find out

- (a) the feasibility of a chemical reaction which can be predicted by thermodynamics (as you know that a reaction with $\Delta G < 0$, at constant temperature and pressure is feasible);
- (b) extent to which a reaction will proceed can be determined from chemical equilibrium;
- (c) speed of a reaction i.e. time taken by a reaction to reach equilibrium.

Along with feasibility and extent, it is equally important to know the rate and the factors controlling the rate of a chemical reaction for its complete understanding. For example, which parameters determine as to how rapidly food gets spoiled? How to design a rapidly setting material for dental filling? Or what controls the rate at which fuel burns in an auto engine? All these questions can be answered by the branch of chemistry, which deals with the study of reaction rates and their mechanisms, called **chemical kinetics**. The word kinetics is derived from the Greek word 'kinesis' meaning movement. Thermodynamics tells only about the feasibility of a reaction whereas chemical kinetics tells about the rate of a reaction. For example, thermodynamic data indicate that diamond shall convert to graphite but in reality the conversion rate is so slow that the change is not perceptible at all. Therefore, most people think

3.1 Rate of a Chemical Reaction

that diamond is forever. Kinetic studies not only help us to determine the speed or rate of a chemical reaction but also describe the conditions by which the reaction rates can be altered. The factors such as concentration, temperature, pressure and catalyst affect the rate of a reaction. At the macroscopic level, we are interested in amounts reacted or formed and the rates of their consumption or formation. At the molecular level, the reaction mechanisms involving orientation and energy of molecules undergoing collisions, are discussed.

In this Unit, we shall be dealing with average and instantaneous rate of reaction and the factors affecting these. Some elementary ideas about the collision theory of reaction rates are also given. However, in order to understand all these, let us first learn about the reaction rate.

Some reactions such as ionic reactions occur very fast, for example, precipitation of silver chloride occurs instantaneously by mixing of aqueous solutions of silver nitrate and sodium chloride. On the other hand, some reactions are very slow, for example, rusting of iron in the presence of air and moisture. Also there are reactions like inversion of cane sugar and hydrolysis of starch, which proceed with a moderate speed. Can you think of more examples from each category?

You must be knowing that speed of an automobile is expressed in terms of change in the position or distance covered by it in a certain period of time. Similarly, the speed of a reaction or the rate of a reaction can be defined as the change in concentration of a reactant or product in unit time. To be more specific, it can be expressed in terms of:

- (i) the rate of decrease in concentration of any one of the reactants, or
- (ii) the rate of increase in concentration of any one of the products.

Consider a hypothetical reaction, assuming that the volume of the system remains constant.



One mole of the reactant R produces one mole of the product P. If $[R]_1$ and $[P]_1$ are the concentrations of R and P respectively at time t_1 and $[R]_2$ and $[P]_2$ are their concentrations at time t_2 then,

$$\Delta t = t_2 - t_1$$

$$\Delta[R] = [R]_2 - [R]_1$$

$$\Delta[P] = [P]_2 - [P]_1$$

The square brackets in the above expressions are used to express molar concentration.

Rate of disappearance of R

$$= \frac{\text{Decrease in concentration of R}}{\text{Time taken}} = - \frac{\Delta[R]}{\Delta t} \quad (3.1)$$

Rate of appearance of P

$$= \frac{\text{Increase in concentration of P}}{\text{Time taken}} = + \frac{\Delta [P]}{\Delta t} \quad (3.2)$$

Since, $\Delta[R]$ is a negative quantity (as concentration of reactants is decreasing), it is multiplied with -1 to make the rate of the reaction a positive quantity.

Equations (3.1) and (3.2) given above represent the **average rate of a reaction**, r_{av} .

Average rate depends upon the change in concentration of reactants or products and the time taken for that change to occur (Fig. 3.1).

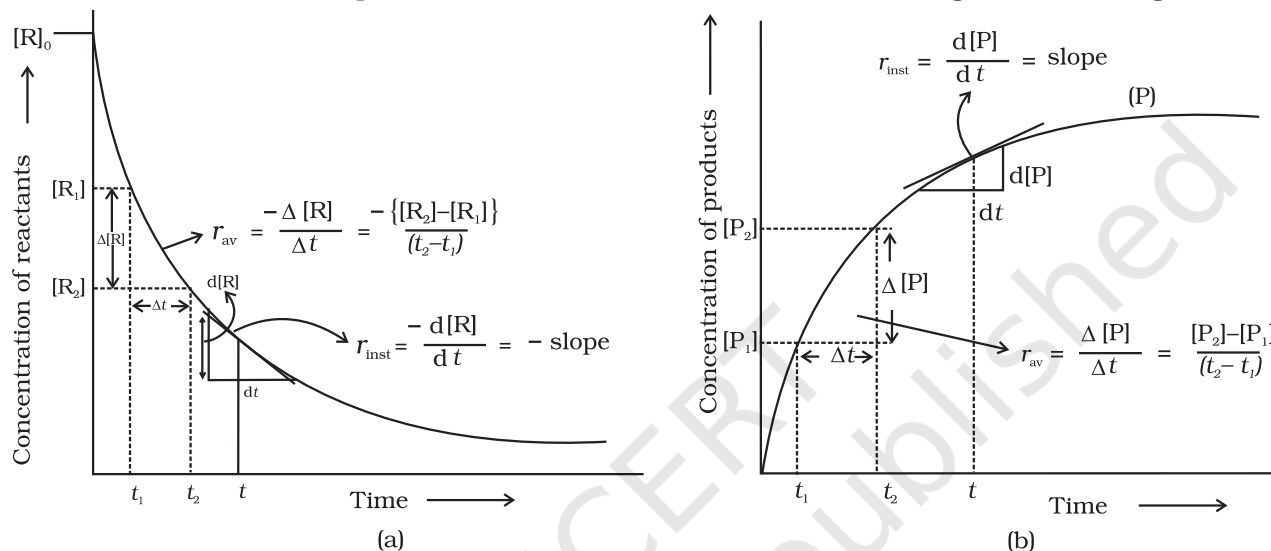


Fig. 3.1: Instantaneous and average rate of a reaction

Units of rate of a reaction

From equations (3.1) and (3.2), it is clear that units of rate are concentration time⁻¹. For example, if concentration is in mol L⁻¹ and time is in seconds then the units will be mol L⁻¹s⁻¹. However, in gaseous reactions, when the concentration of gases is expressed in terms of their partial pressures, then the units of the rate equation will be atm s⁻¹.

From the concentrations of C₄H₉Cl (butyl chloride) at different times given below, calculate the average rate of the reaction: **Example 3.1**



during different intervals of time.

t/s	0	50	100	150	200	300	400	700	800
$[\text{C}_4\text{H}_9\text{Cl}]/\text{mol L}^{-1}$	0.100	0.0905	0.0820	0.0741	0.0671	0.0549	0.0439	0.0210	0.017

We can determine the difference in concentration over different intervals of time and thus determine the average rate by dividing $\Delta[R]$ by Δt (Table 3.1). **Solution**

Table 3.1: Average rates of hydrolysis of butyl chloride

$[C_4H_9Cl]_{t_1} / \text{mol L}^{-1}$	$[C_4H_9Cl]_{t_2} / \text{mol L}^{-1}$	t_1/s	t_2/s	$r_{av} \times 10^4 / \text{mol L}^{-1}\text{s}^{-1}$ $= - \{ [C_4H_9Cl]_{t_2} - [C_4H_9Cl]_{t_1} \} / (t_2 - t_1) \} \times 10^4$
0.100	0.0905	0	50	1.90
0.0905	0.0820	50	100	1.70
0.0820	0.0741	100	150	1.58
0.0741	0.0671	150	200	1.40
0.0671	0.0549	200	300	1.22
0.0549	0.0439	300	400	1.10
0.0439	0.0335	400	500	1.04
0.0210	0.017	700	800	0.4

It can be seen (Table 3.1) that the average rate falls from $1.90 \times 10^{-4} \text{ mol L}^{-1}\text{s}^{-1}$ to $0.4 \times 10^{-4} \text{ mol L}^{-1}\text{s}^{-1}$. However, average rate cannot be used to predict the rate of a reaction at a particular instant as it would be constant for the time interval for which it is calculated. So, to express the rate at a particular moment of time we determine the **instantaneous rate**. It is obtained when we consider the average rate at the smallest time interval say dt (i.e. when Δt approaches zero). Hence, mathematically for an infinitesimally small dt instantaneous rate is given by

$$r_{av} = \frac{-\Delta[R]}{\Delta t} = \frac{\Delta[P]}{\Delta t} \quad (3.3)$$

$$\text{As } \Delta t \rightarrow 0 \quad \text{or} \quad r_{inst} = \frac{-d[R]}{dt} = \frac{d[P]}{dt}$$

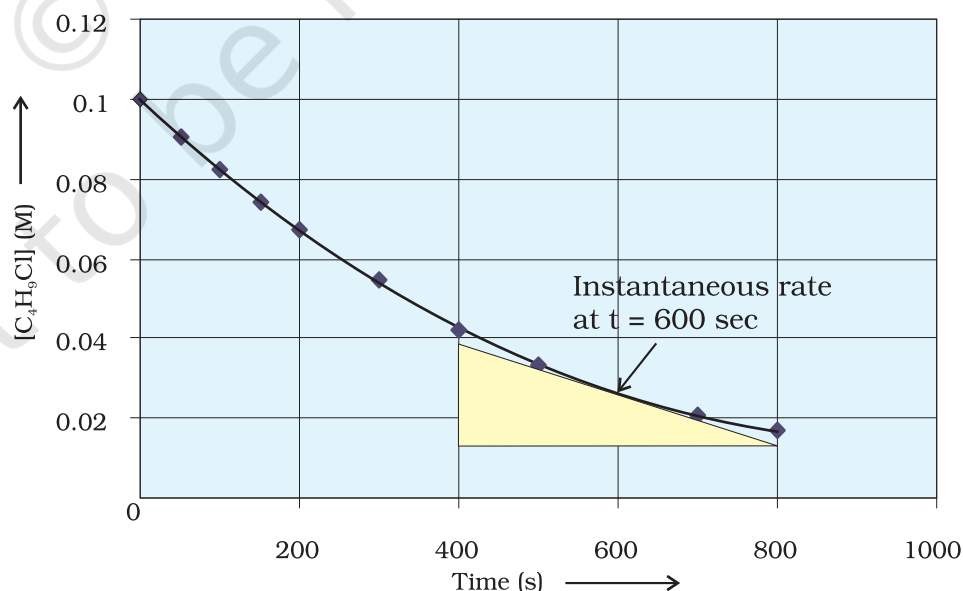


Fig 3.2
Instantaneous rate
of hydrolysis of butyl
chloride(C_4H_9Cl)

It can be determined graphically by drawing a tangent at time t on either of the curves for concentration of R and P vs time t and calculating its slope (Fig. 3.1). So in problem 3.1, r_{inst} at 600s for example, can be calculated by plotting concentration of butyl chloride as a function of time. A tangent is drawn that touches the curve at $t = 600$ s (Fig. 3.2).

The slope of this tangent gives the instantaneous rate.

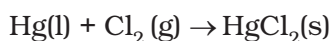
$$\text{So, } r_{\text{inst}} \text{ at } 600 \text{ s} = - \left(\frac{0.0165 - 0.037}{(800 - 400)\text{s}} \right) \text{ mol L}^{-1} = 5.12 \times 10^{-5} \text{ mol L}^{-1}\text{s}^{-1}$$

$$\text{At } t = 250 \text{ s } \quad r_{\text{inst}} = 1.22 \times 10^{-4} \text{ mol L}^{-1}\text{s}^{-1}$$

$$t = 350 \text{ s } \quad r_{\text{inst}} = 1.0 \times 10^{-4} \text{ mol L}^{-1}\text{s}^{-1}$$

$$t = 450 \text{ s } \quad r_{\text{inst}} = 6.4 \times 10^{-5} \text{ mol L}^{-1}\text{s}^{-1}$$

Now consider a reaction



Where stoichiometric coefficients of the reactants and products are same, then rate of the reaction is given as

$$\text{Rate of reaction} = - \frac{\Delta[\text{Hg}]}{\Delta t} = - \frac{\Delta[\text{Cl}_2]}{\Delta t} = \frac{\Delta[\text{HgCl}_2]}{\Delta t}$$

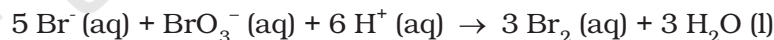
i.e., rate of disappearance of any of the reactants is same as the rate of appearance of the products. But in the following reaction, two moles of HI decompose to produce one mole each of H_2 and I_2 .



For expressing the rate of such a reaction where stoichiometric coefficients of reactants or products are not equal to one, rate of disappearance of any of the reactants or the rate of appearance of products is divided by their respective stoichiometric coefficients. Since rate of consumption of HI is twice the rate of formation of H_2 or I_2 , to make them equal, the term $\Delta[\text{HI}]$ is divided by 2. The rate of this reaction is given by

$$\text{Rate of reaction} = - \frac{1}{2} \frac{\Delta[\text{HI}]}{\Delta t} = \frac{\Delta[\text{H}_2]}{\Delta t} = \frac{\Delta[\text{I}_2]}{\Delta t}$$

Similarly, for the reaction



$$\text{Rate} = - \frac{1}{5} \frac{\Delta[\text{Br}^-]}{\Delta t} = - \frac{\Delta[\text{BrO}_3^-]}{\Delta t} = - \frac{1}{6} \frac{\Delta[\text{H}^+]}{\Delta t} = \frac{1}{3} \frac{\Delta[\text{Br}_2]}{\Delta t} = \frac{1}{3} \frac{\Delta[\text{H}_2\text{O}]}{\Delta t}$$

For a gaseous reaction at constant temperature, concentration is directly proportional to the partial pressure of a species and hence, rate can also be expressed as rate of change in partial pressure of the reactant or the product.

Example 3.2 The decomposition of N_2O_5 in CCl_4 at 318K has been studied by monitoring the concentration of N_2O_5 in the solution. Initially the concentration of N_2O_5 is 2.33 mol L^{-1} and after 184 minutes, it is reduced to 2.08 mol L^{-1} . The reaction takes place according to the equation



Calculate the average rate of this reaction in terms of hours, minutes and seconds. What is the rate of production of NO_2 during this period?

Solution

$$\begin{aligned}\text{Average Rate} &= \frac{1}{2} \left\{ -\frac{\Delta[\text{N}_2\text{O}_5]}{\Delta t} \right\} = -\frac{1}{2} \left[\frac{(2.08 - 2.33) \text{ mol L}^{-1}}{184 \text{ min}} \right] \\&= 6.79 \times 10^{-4} \text{ mol L}^{-1} / \text{min} = (6.79 \times 10^{-4} \text{ mol L}^{-1} \text{ min}^{-1}) \times (60 \text{ min} / 1 \text{ h}) \\&= 4.07 \times 10^{-2} \text{ mol L}^{-1} / \text{h} \\&= 6.79 \times 10^{-4} \text{ mol L}^{-1} \times 1 \text{ min} / 60 \text{ s} \\&= 1.13 \times 10^{-5} \text{ mol L}^{-1} \text{ s}^{-1}\end{aligned}$$

It may be remembered that

$$\begin{aligned}\text{Rate} &= \frac{1}{4} \left\{ \frac{\Delta[\text{NO}_2]}{\Delta t} \right\} \\ \frac{\Delta[\text{NO}_2]}{\Delta t} &= 6.79 \times 10^{-4} \times 4 \text{ mol L}^{-1} \text{ min}^{-1} = 2.72 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1}\end{aligned}$$

Intext Questions

- 3.1** For the reaction $\text{R} \rightarrow \text{P}$, the concentration of a reactant changes from 0.03M to 0.02M in 25 minutes. Calculate the average rate of reaction using units of time both in minutes and seconds.
- 3.2** In a reaction, $2\text{A} \rightarrow \text{Products}$, the concentration of A decreases from 0.5 mol L^{-1} to 0.4 mol L^{-1} in 10 minutes. Calculate the rate during this interval?

3.2 Factors Influencing Rate of a Reaction

Rate of reaction depends upon the experimental conditions such as concentration of reactants (pressure in case of gases), temperature and catalyst.

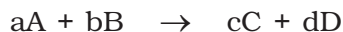
3.2.1 Dependence of Rate on Concentration

The rate of a chemical reaction at a given temperature may depend on the concentration of one or more reactants and products. The representation of rate of reaction in terms of concentration of the reactants is known as **rate law**. It is also called as rate equation or rate expression.

3.2.2 Rate Expression and Rate Constant

The results in Table 3.1 clearly show that rate of a reaction decreases with the passage of time as the concentration of reactants decrease. Conversely, rates generally increase when reactant concentrations increase. So, rate of a reaction depends upon the concentration of reactants.

Consider a general reaction



where a, b, c and d are the stoichiometric coefficients of reactants and products.

The rate expression for this reaction is

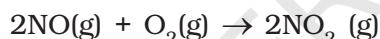
$$\text{Rate} \propto [A]^x [B]^y \quad (3.4)$$

where exponents x and y may or may not be equal to the stoichiometric coefficients (a and b) of the reactants. Above equation can also be written as

$$\text{Rate} = k [A]^x [B]^y \quad (3.4a)$$

$$-\frac{d[R]}{dt} = k[A]^x [B]^y \quad (3.4b)$$

This form of equation (3.4 b) is known as differential rate equation, where *k* is a proportionality constant called **rate constant**. The equation like (3.4), which relates the rate of a reaction to concentration of reactants is called rate law or rate expression. Thus, **rate law is the expression in which reaction rate is given in terms of molar concentration of reactants with each term raised to some power, which may or may not be same as the stoichiometric coefficient of the reacting species in a balanced chemical equation**. For example:



We can measure the rate of this reaction as a function of initial concentrations either by keeping the concentration of one of the reactants constant and changing the concentration of the other reactant or by changing the concentration of both the reactants. The following results are obtained (Table 3.2).

Table 3.2: Initial rate of formation of NO₂

Experiment	Initial [NO]/ mol L ⁻¹	Initial [O ₂]/ mol L ⁻¹	Initial rate of formation of NO ₂ / mol L ⁻¹ s ⁻¹
1.	0.30	0.30	0.096
2.	0.60	0.30	0.384
3.	0.30	0.60	0.192
4.	0.60	0.60	0.768

It is obvious, after looking at the results, that when the concentration of NO is doubled and that of O₂ is kept constant then the initial rate increases by a factor of four from 0.096 to 0.384 mol L⁻¹s⁻¹. This indicates that the rate depends upon the square of the concentration of NO. When concentration of NO is kept constant and concentration of O₂ is doubled the rate also gets doubled indicating that rate depends on concentration of O₂ to the first power. Hence, the rate equation for this reaction will be

$$\text{Rate} = k [\text{NO}]^2 [\text{O}_2]$$

The differential form of this rate expression is given as

$$-\frac{d[R]}{dt} = k[\text{NO}]^2 [\text{O}_2]$$

Now, we observe that for this reaction in the rate equation derived from the experimental data, the exponents of the concentration terms are the same as their stoichiometric coefficients in the balanced chemical equation.

Some other examples are given below:

Reaction	Experimental rate expression
1. $\text{CHCl}_3 + \text{Cl}_2 \rightarrow \text{CCl}_4 + \text{HCl}$	$\text{Rate} = k [\text{CHCl}_3] [\text{Cl}_2]^{1/2}$
2. $\text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH}$	$\text{Rate} = k [\text{CH}_3\text{COOC}_2\text{H}_5]^1 [\text{H}_2\text{O}]^0$

In these reactions, the exponents of the concentration terms are not the same as their stoichiometric coefficients. Thus, we can say that:

Rate law for any reaction cannot be predicted by merely looking at the balanced chemical equation, i.e., theoretically but must be determined experimentally.

3.2.3 Order of a Reaction

In the rate equation (3.4)

$$\text{Rate} = k [\text{A}]^x [\text{B}]^y$$

x and y indicate how sensitive the rate is to the change in concentration of A and B. Sum of these exponents, i.e., $x + y$ in (3.4) gives the overall order of a reaction whereas x and y represent the order with respect to the reactants A and B respectively.

Hence, **the sum of powers of the concentration of the reactants in the rate law expression is called the order of that chemical reaction.**

Order of a reaction can be 0, 1, 2, 3 and even a fraction. A zero order reaction means that the rate of reaction is independent of the concentration of reactants.

Example 3.3

Calculate the overall order of a reaction which has the rate expression

(a) $\text{Rate} = k [\text{A}]^{1/2} [\text{B}]^{3/2}$

(b) $\text{Rate} = k [\text{A}]^{3/2} [\text{B}]^{-1}$

Solution

(a) $\text{Rate} = k [\text{A}]^x [\text{B}]^y$

order = $x + y$

So order = $1/2 + 3/2 = 2$, i.e., second order

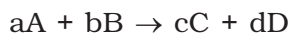
(b) order = $3/2 + (-1) = 1/2$, i.e., half order.

A balanced chemical equation never gives us a true picture of how a reaction takes place since rarely a reaction gets completed in one step. The reactions taking place in one step are called **elementary reactions**. When a sequence of elementary reactions (called mechanism) gives us the products, the reactions are called **complex reactions**.

These may be consecutive reactions (e.g., oxidation of ethane to CO₂ and H₂O passes through a series of intermediate steps in which alcohol, aldehyde and acid are formed), reverse reactions and side reactions (e.g., nitration of phenol yields *o*-nitrophenol and *p*-nitrophenol).

Units of rate constant

For a general reaction



$$\text{Rate} = k [A]^x [B]^y$$

Where $x + y = n = \text{order of the reaction}$

$$k = \frac{\text{Rate}}{[A]^x [B]^y} \\ = \frac{\text{concentration}}{\text{time}} \times \frac{1}{(\text{concentration})^n} \quad (\text{where } [A]=[B])$$

Taking SI units of concentration, mol L⁻¹ and time, s, the units of k for different reaction order are listed in Table 3.3

Table 3.3: Units of rate constant

Reaction	Order	Units of rate constant
Zero order reaction	0	$\frac{\text{mol L}^{-1}}{\text{s}} \times \frac{1}{(\text{mol L}^{-1})^0} = \text{mol L}^{-1} \text{s}^{-1}$
First order reaction	1	$\frac{\text{mol L}^{-1}}{\text{s}} \times \frac{1}{(\text{mol L}^{-1})^1} = \text{s}^{-1}$
Second order reaction	2	$\frac{\text{mol L}^{-1}}{\text{s}} \times \frac{1}{(\text{mol L}^{-1})^2} = \text{mol}^{-1} \text{L s}^{-1}$

Identify the reaction order from each of the following rate constants.

Example 3.4

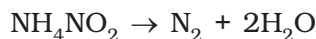
- (i) $k = 2.3 \times 10^{-5} \text{ L mol}^{-1} \text{ s}^{-1}$
 (ii) $k = 3 \times 10^{-4} \text{ s}^{-1}$

- (i) The unit of second order rate constant is L mol⁻¹ s⁻¹, therefore $k = 2.3 \times 10^{-5} \text{ L mol}^{-1} \text{ s}^{-1}$ represents a second order reaction.
 (ii) The unit of a first order rate constant is s⁻¹ therefore $k = 3 \times 10^{-4} \text{ s}^{-1}$ represents a first order reaction.

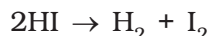
Solution

3.2.4 Molecularity of a Reaction

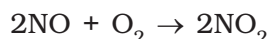
Another property of a reaction called molecularity helps in understanding its mechanism. **The number of reacting species (atoms, ions or molecules) taking part in an elementary reaction, which must collide simultaneously in order to bring about a chemical reaction is called molecularity of a reaction.** The reaction can be unimolecular when one reacting species is involved, for example, decomposition of ammonium nitrite.



Bimolecular reactions involve simultaneous collision between two species, for example, dissociation of hydrogen iodide.

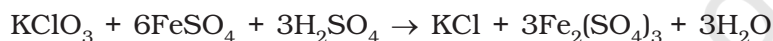


Trimolecular or termolecular reactions involve simultaneous collision between three reacting species, for example,

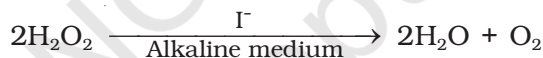


The probability that more than three molecules can collide and react simultaneously is very small. Hence, reactions with the molecularity three are very rare and slow to proceed.

It is, therefore, evident that complex reactions involving more than three molecules in the stoichiometric equation must take place in more than one step.



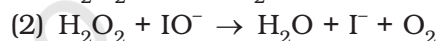
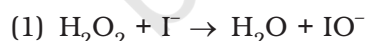
This reaction which apparently seems to be of tenth order is actually a second order reaction. This shows that this reaction takes place in several steps. Which step controls the rate of the overall reaction? The question can be answered if we go through the mechanism of reaction, for example, chances to win the relay race competition by a team depend upon the slowest person in the team. Similarly, the overall rate of the reaction is controlled by the slowest step in a reaction called the **rate determining step**. Consider the decomposition of hydrogen peroxide which is catalysed by iodide ion in an alkaline medium.



The rate equation for this reaction is found to be

$$\text{Rate} = \frac{-d[\text{H}_2\text{O}_2]}{dt} = k[\text{H}_2\text{O}_2][\text{I}^-]$$

This reaction is first order with respect to both H_2O_2 and I^- . Evidences suggest that this reaction takes place in two steps



Both the steps are bimolecular elementary reactions. Species IO^- is called as an intermediate since it is formed during the course of the reaction but not in the overall balanced equation. The first step, being slow, is the rate determining step. Thus, the rate of formation of intermediate will determine the rate of this reaction.

Thus, from the discussion, till now, we conclude the following:

- (i) Order of a reaction is an experimental quantity. It can be zero and even a fraction but molecularity cannot be zero or a non integer.
- (ii) Order is applicable to elementary as well as complex reactions whereas molecularity is applicable only for elementary reactions. For complex reaction molecularity has no meaning.

- (iii) For complex reaction, order is given by the slowest step and molecularity of the slowest step is same as the order of the overall reaction.

Intext Questions

- 3.3** For a reaction, $A + B \rightarrow \text{Product}$; the rate law is given by, $r = k [A]^{1/2} [B]^2$. What is the order of the reaction?
- 3.4** The conversion of molecules X to Y follows second order kinetics. If concentration of X is increased to three times how will it affect the rate of formation of Y?

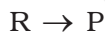
3.3 Integrated Rate Equations

We have already noted that the concentration dependence of rate is called differential rate equation. It is not always convenient to determine the instantaneous rate, as it is measured by determination of slope of the tangent at point 't' in concentration vs time plot (Fig. 3.1). This makes it difficult to determine the rate law and hence the order of the reaction. In order to avoid this difficulty, we can integrate the differential rate equation to give a relation between directly measured experimental data, i.e., concentrations at different times and rate constant.

The integrated rate equations are different for the reactions of different reaction orders. We shall determine these equations only for zero and first order chemical reactions.

3.3.1 Zero Order Reactions

Zero order reaction means that the rate of the reaction is proportional to zero power of the concentration of reactants. Consider the reaction,



$$\text{Rate} = -\frac{d[R]}{dt} = k[R]^0$$

As any quantity raised to power zero is unity

$$\text{Rate} = -\frac{d[R]}{dt} = k \times 1$$

$$d[R] = -k dt$$

Integrating both sides

$$[R] = -k t + I \quad (3.5)$$

where, I is the constant of integration.

At $t = 0$, the concentration of the reactant $R = [R]_0$, where $[R]_0$ is initial concentration of the reactant.

Substituting in equation (3.5)

$$[R]_0 = -k \times 0 + I$$

$$[R]_0 = I$$

Substituting the value of I in the equation (3.5)

$$[R] = -kt + [R]_0 \quad (3.6)$$

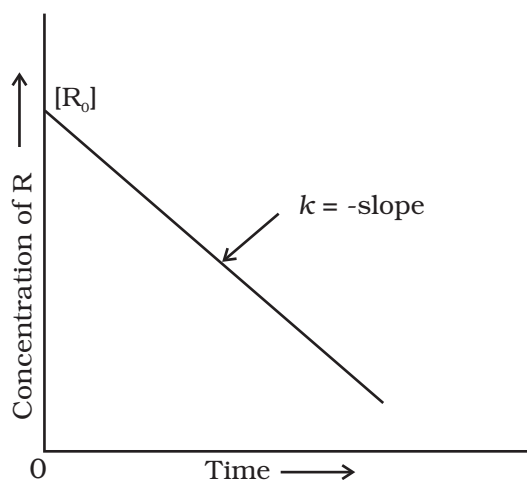


Fig. 3.3: Variation in the concentration vs time plot for a zero order reaction

Comparing (3.6) with equation of a straight line, $y = mx + c$, if we plot $[R]$ against t , we get a straight line (Fig. 3.3) with slope $= -k$ and intercept equal to $[R]_0$.

Further simplifying equation (3.6), we get the rate constant, k as

$$k = \frac{[R]_0 - [R]}{t} \quad (3.7)$$

Zero order reactions are relatively uncommon but they occur under special conditions. Some enzyme catalysed reactions and reactions which occur on metal surfaces are a few examples of zero order reactions. The decomposition of gaseous ammonia on a hot platinum surface is a zero order reaction at high pressure.



$$\text{Rate} = k [\text{NH}_3]^0 = k$$

In this reaction, platinum metal acts as a catalyst. At high pressure, the metal surface gets saturated with gas molecules. So, a further change in reaction conditions is unable to alter the amount of ammonia on the surface of the catalyst making rate of the reaction independent of its concentration. The thermal decomposition of HI on gold surface is another example of zero order reaction.

3.3.2 First Order Reactions

In this class of reactions, the rate of the reaction is proportional to the first power of the concentration of the reactant R . For example,



$$\text{Rate} = -\frac{d[R]}{dt} = k[R]$$

$$\text{or } \frac{d[R]}{[R]} = -kdt$$

Integrating this equation, we get

$$\ln [R] = -kt + I \quad (3.8)$$

Again, I is the constant of integration and its value can be determined easily.

When $t = 0$, $R = [R]_0$, where $[R]_0$ is the initial concentration of the reactant.

Therefore, equation (3.8) can be written as

$$\ln [R]_0 = -k \times 0 + I$$

$$\ln [R]_0 = I$$

Substituting the value of I in equation (3.8)

$$\ln [R] = -kt + \ln [R]_0 \quad (3.9)$$

Rearranging this equation

$$\ln \frac{[R]}{[R]_0} = -kt$$
$$\text{or } k = \frac{1}{t} \ln \frac{[R]_0}{[R]} \quad (3.10)$$

At time t_1 from equation (3.8)

$$*\ln[R]_1 = -kt_1 + *\ln[R]_0 \quad (3.11)$$

At time t_2

$$\ln[R]_2 = -kt_2 + \ln[R]_0 \quad (3.12)$$

where, $[R]_1$ and $[R]_2$ are the concentrations of the reactants at time t_1 and t_2 respectively.

Subtracting (3.12) from (3.11)

$$\ln[R]_1 - \ln[R]_2 = -kt_1 - (-kt_2)$$
$$\ln \frac{[R]_1}{[R]_2} = k(t_2 - t_1)$$
$$k = \frac{1}{(t_2 - t_1)} \ln \frac{[R]_1}{[R]_2} \quad (3.13)$$

Equation (3.9) can also be written as

$$\ln \frac{[R]}{[R]_0} = -kt$$

Taking antilog of both sides

$$[R] = [R]_0 e^{-kt} \quad (3.14)$$

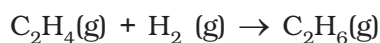
Comparing equation (3.9) with $y = mx + c$, if we plot $\ln [R]$ against t (Fig. 3.4) we get a straight line with slope = $-k$ and intercept equal to $\ln [R]_0$

The first order rate equation (3.10) can also be written in the form

$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]} \quad (3.15)$$
$$*\log \frac{[R]_0}{[R]} = \frac{kt}{2.303}$$

If we plot a graph between $\log [R]_0/[R]$ vs t , (Fig. 3.5), the slope = $k/2.303$

Hydrogenation of ethene is an example of first order reaction.



$$\text{Rate} = k [\text{C}_2\text{H}_4]$$

All natural and artificial radioactive decay of unstable nuclei take place by first order kinetics.

* Refer to Appendix-IV for $\ln$ and $\log$ (logarithms).

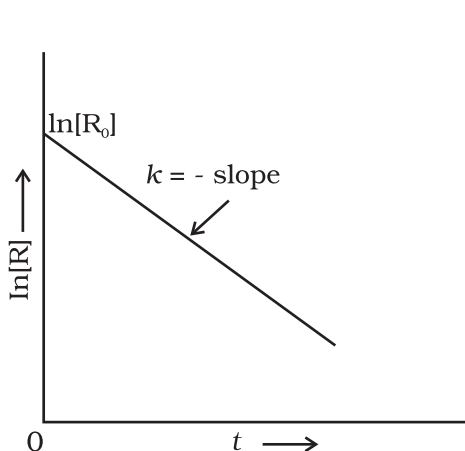


Fig. 3.4: A plot between $\ln[R]$ and t for a first order reaction

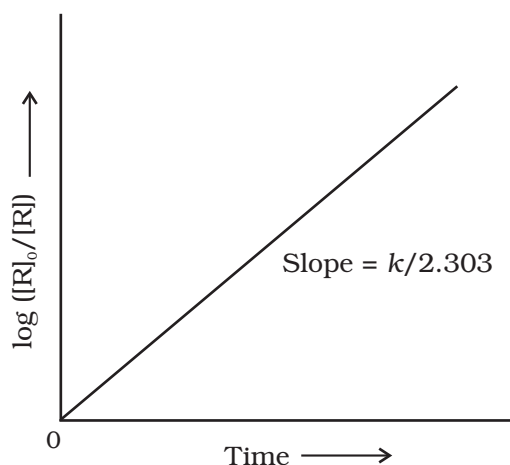
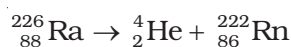


Fig. 3.5: Plot of $\log [R]_0/[R]$ vs time for a first order reaction



$$\text{Rate} = k [\text{Ra}]$$

Decomposition of N_2O_5 and N_2O are some more examples of first order reactions.

Example 3.5 The initial concentration of N_2O_5 in the following first order reaction $\text{N}_2\text{O}_5(\text{g}) \rightarrow 2 \text{NO}_2(\text{g}) + 1/2 \text{O}_2(\text{g})$ was $1.24 \times 10^{-2} \text{ mol L}^{-1}$ at 318 K. The concentration of N_2O_5 after 60 minutes was $0.20 \times 10^{-2} \text{ mol L}^{-1}$. Calculate the rate constant of the reaction at 318 K.

Solution For a first order reaction

$$\begin{aligned} \log \frac{[R]_1}{[R]_2} &= \frac{k(t_2 - t_1)}{2.303} \\ k &= \frac{2.303}{(t_2 - t_1)} \log \frac{[R]_1}{[R]_2} \\ &= \frac{2.303}{(60 \text{ min} - 0 \text{ min})} \log \frac{1.24 \times 10^{-2} \text{ mol L}^{-1}}{0.20 \times 10^{-2} \text{ mol L}^{-1}} \\ &= \frac{2.303}{60} \log 6.2 \text{ min}^{-1} \\ k &= 0.0304 \text{ min}^{-1} \end{aligned}$$

Let us consider a typical first order gas phase reaction



Let p_i be the initial pressure of A and p_t the total pressure at time ' t '. Integrated rate equation for such a reaction can be derived as

$$\text{Total pressure } p_t = p_A + p_B + p_C \text{ (pressure units)}$$

p_A , p_B and p_C are the partial pressures of A, B and C, respectively.

If x atm be the decrease in pressure of A at time t and one mole each of B and C is being formed, the increase in pressure of B and C will also be x atm each.

	A(g)	→	B(g)	+	C(g)
At $t = 0$	p_i atm		0 atm		0 atm
At time t	$(p_i - x)$ atm		x atm		x atm

where, p_i is the initial pressure at time $t = 0$.

$$p_t = (p_i - x) + x + x = p_i + x$$

$$x = (p_t - p_i)$$

where, $p_A = p_i - x = p_i - (p_t - p_i)$

$$= 2p_i - p_t$$

$$k = \left(\frac{2.303}{t} \right) \left(\log \frac{p_i}{p_A} \right) \quad (3.16)$$

$$= \frac{2.303}{t} \log \frac{p_i}{(2p_i - p_t)}$$

The following data were obtained during the first order thermal decomposition of N_2O_5 (g) at constant volume: **Example 3.6**



S.No.	Time/s	Total Pressure/(atm)
1.	0	0.5
2.	100	0.512

Calculate the rate constant.

Let the pressure of N_2O_5 (g) decrease by $2x$ atm. As two moles of N_2O_5 decompose to give two moles of N_2O_4 (g) and one mole of O_2 (g), the pressure of N_2O_4 (g) increases by $2x$ atm and that of O_2 (g) increases by x atm. **Solution**

	$2N_2O_5(g)$	→	$2N_2O_4(g)$	+	$O_2(g)$
Start $t = 0$	0.5 atm		0 atm		0 atm
At time t	$(0.5 - 2x)$ atm		$2x$ atm		x atm

$$p_t = p_{N_2O_5} + p_{N_2O_4} + p_{O_2}$$

$$= (0.5 - 2x) + 2x + x = 0.5 + x$$

$$x = p_t - 0.5$$

$$p_{N_2O_5} = 0.5 - 2x$$

$$= 0.5 - 2(p_t - 0.5) = 1.5 - 2p_t$$

$$\text{At } t = 100 \text{ s; } p_t = 0.512 \text{ atm}$$

$$p_{\text{N}_2\text{O}_5} = 1.5 - 2 \times 0.512 = 0.476 \text{ atm}$$

Using equation (3.16)

$$\begin{aligned} k &= \frac{2.303}{t} \log \frac{p_i}{p_A} = \frac{2.303}{100 \text{ s}} \log \frac{0.5 \text{ atm}}{0.476 \text{ atm}} \\ &= \frac{2.303}{100 \text{ s}} \times 0.0216 = 4.98 \times 10^{-4} \text{ s}^{-1} \end{aligned}$$

3.3.3 Half-Life of a Reaction

The half-life of a reaction is the time in which the concentration of a reactant is reduced to one half of its initial concentration. It is represented as $t_{1/2}$.

For a zero order reaction, rate constant is given by equation 3.7.

$$k = \frac{[\text{R}]_0 - [\text{R}]}{t}$$

$$\text{At } t = t_{1/2}, [\text{R}] = \frac{1}{2} [\text{R}]_0$$

The rate constant at $t_{1/2}$ becomes

$$k = \frac{[\text{R}]_0 - 1/2 [\text{R}]_0}{t_{1/2}}$$

$$t_{1/2} = \frac{[\text{R}]_0}{2k}$$

It is clear that $t_{1/2}$ for a zero order reaction is directly proportional to the initial concentration of the reactants and inversely proportional to the rate constant.

For the first order reaction,

$$k = \frac{2.303}{t} \log \frac{[\text{R}]_0}{[\text{R}]} \quad (3.15)$$

$$\text{at } t_{1/2} \quad [\text{R}] = \frac{[\text{R}]_0}{2} \quad (3.16)$$

So, the above equation becomes

$$k = \frac{2.303}{t_{1/2}} \log \frac{[\text{R}]_0}{[\text{R}]/2}$$

$$\text{or } t_{1/2} = \frac{2.303}{k} \log 2$$

$$t_{1/2} = \frac{2.303}{k} \times 0.301$$

$$t_{1/2} = \frac{0.693}{k} \quad (3.17)$$

It can be seen that for a first order reaction, half-life period is constant, i.e., it is independent of initial concentration of the reacting species. The half-life of a first order equation is readily calculated from the rate constant and vice versa.

For zero order reaction $t_{1/2} \propto [R]_0$. For first order reaction $t_{1/2}$ is independent of $[R]_0$.

A first order reaction is found to have a rate constant, $k = 5.5 \times 10^{-14} \text{ s}^{-1}$. [Example 3.7](#)
Find the half-life of the reaction.

Half-life for a first order reaction is

[Solution](#)

$$t_{1/2} = \frac{0.693}{k}$$

$$t_{1/2} = \frac{0.693}{5.5 \times 10^{-14} \text{ s}^{-1}} = 1.26 \times 10^{13} \text{ s}$$

Show that in a first order reaction, time required for completion of 99.9% is 10 times of half-life ($t_{1/2}$) of the reaction.

When reaction is completed 99.9%, $[R]_n = [R]_0 - 0.999[R]_0$

[Example 3.8](#)

[Solution](#)

$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$$

$$= \frac{2.303}{t} \log \frac{[R]_0}{[R]_0 - 0.999[R]_0} = \frac{2.303}{t} \log 10^3$$

$$t = 6.909/k$$

For half-life of the reaction

$$t_{1/2} = 0.693/k$$

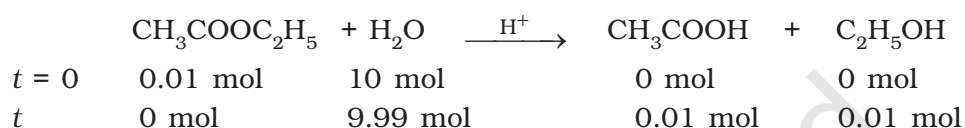
$$\frac{t}{t_{1/2}} = \frac{6.909}{k} \times \frac{k}{0.693} = 10$$

Table 3.4 summarises the mathematical features of integrated laws of zero and first order reactions.

Table 3.4: Integrated Rate Laws for the Reactions of Zero and First Order

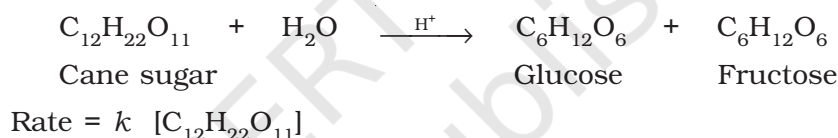
Order	Reaction type	Differential rate law	Integrated rate law	Straight line plot	Half-life	Units of k
0	$R \rightarrow P$	$d[R]/dt = -k$	$kt = [R]_0 - [R]$	$[R]$ vs t	$[R]_0/2k$	conc time ⁻¹ or mol L ⁻¹ s ⁻¹
1	$R \rightarrow P$	$d[R]/dt = -k[R]$	$[R] = [R]_0 e^{-kt}$ or $kt = \ln\{[R]_0/[R]\}$	$\ln[R]$ vs t	$\ln 2/k$	time ⁻¹ or s ⁻¹

The order of a reaction is sometimes altered by conditions. There are many reactions which obey first order rate law although they are higher order reactions. Consider the hydrolysis of ethyl acetate which is a chemical reaction between ethyl acetate and water. In reality, it is a second order reaction and concentration of both ethyl acetate and water affect the rate of the reaction. But water is taken in large excess for hydrolysis, therefore, concentration of water is not altered much during the reaction. Thus, the rate of reaction is affected by concentration of ethyl acetate only. For example, during the hydrolysis of 0.01 mol of ethyl acetate with 10 mol of water, amounts of the reactants and products at the beginning ($t = 0$) and completion (t) of the reaction are given as under.



The concentration of water does not get altered much during the course of the reaction. So, the reaction behaves as first order reaction. Such reactions are called **pseudo first order reactions**.

Inversion of cane sugar is another pseudo first order reaction.



Intext Questions

- 3.5** A first order reaction has a rate constant $1.15 \times 10^{-3} \text{ s}^{-1}$. How long will 5 g of this reactant take to reduce to 3 g?
- 3.6** Time required to decompose SO_2Cl_2 to half of its initial amount is 60 minutes. If the decomposition is a first order reaction, calculate the rate constant of the reaction.

3.4 Temperature Dependence of the Rate of a Reaction

Most of the chemical reactions are accelerated by increase in temperature. For example, in decomposition of N_2O_5 , the time taken for half of the original amount of material to decompose is 12 min at 50°C , 5 h at 25°C and 10 days at 0°C . You also know that in a mixture of potassium permanganate (KMnO_4) and oxalic acid ($\text{H}_2\text{C}_2\text{O}_4$), potassium permanganate gets decolourised faster at a higher temperature than that at a lower temperature.

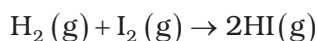
It has been found that **for a chemical reaction with rise in temperature by 10° , the rate constant is nearly doubled**.

The temperature dependence of the rate of a chemical reaction can be accurately explained by **Arrhenius equation** (3.18). It was first proposed by Dutch chemist, J.H. van't Hoff but Swedish chemist, Arrhenius provided its physical justification and interpretation.

$$k = A e^{-E_a / RT} \quad (3.18)$$

where A is the Arrhenius factor or the **frequency factor**. It is also called pre-exponential factor. It is a constant specific to a particular reaction. R is gas constant and E_a is activation energy measured in joules/mole (J mol^{-1}).

It can be understood clearly using the following simple reaction



According to Arrhenius, this reaction can take place only when a molecule of hydrogen and a molecule of iodine collide to form an unstable intermediate (Fig. 3.6). It exists for a very short time and then breaks up to form two molecules of hydrogen iodide.

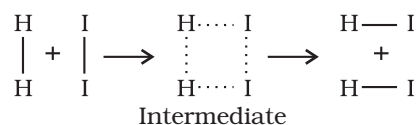


Fig. 3.6: Formation of HI through the intermediate

The energy required to form this intermediate, called **activated complex (C)**, is known as **activation energy (E_a)**. Fig. 3.7 is obtained by plotting potential energy vs reaction coordinate. Reaction coordinate represents the profile of energy change when reactants change into products.

Some energy is released when the complex decomposes to form products. So, the final enthalpy of the reaction depends upon the nature of reactants and products.

All the molecules in the reacting species do not have the same kinetic energy. Since it is difficult to predict the behaviour of any one molecule with precision, Ludwig Boltzmann and James Clark Maxwell used statistics to predict the behaviour of large number of molecules. According to them, the distribution of kinetic energy may be described by plotting the fraction of molecules (N_E/N_T) with a given kinetic energy (E) vs kinetic energy (Fig. 3.8). Here, N_E is the number of molecules with energy E and N_T is total number of molecules.

The peak of the curve corresponds to the **most probable kinetic energy**, i.e., kinetic energy of maximum fraction of molecules. There are decreasing number of molecules with energies higher or lower than this value. When the

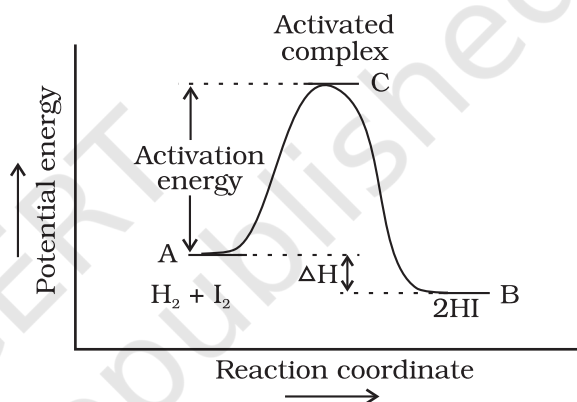


Fig. 3.7: Diagram showing plot of potential energy vs reaction coordinate

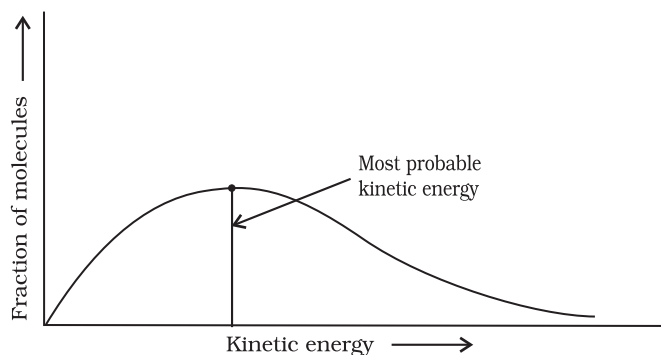


Fig. 3.8: Distribution curve showing energies among gaseous molecules

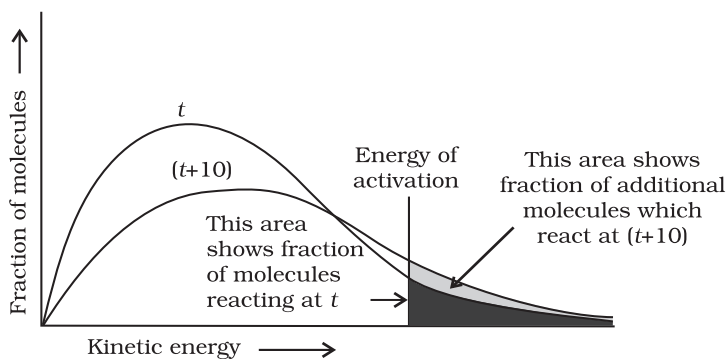


Fig. 3.9: Distribution curve showing temperature dependence of rate of a reaction

temperature is raised, the maximum of the curve moves to the higher energy value (Fig. 3.9) and the curve broadens out, i.e., spreads to the right such that there is a greater proportion of molecules with much higher energies. The area under the curve must be constant since total probability must be one at all times. We can mark the position of E_a on Maxwell Boltzmann distribution curve (Fig. 3.9).

Increasing the temperature of the substance increases the fraction of molecules, which collide with energies greater than E_a . It is clear from the diagram that in the curve at $(t + 10)$, the area showing the fraction of molecules having energy equal to or greater than activation energy gets doubled leading to doubling the rate of a reaction.

In the Arrhenius equation (3.18) the factor $e^{-E_a/RT}$ corresponds to the fraction of molecules that have kinetic energy greater than E_a . Taking natural logarithm of both sides of equation (3.18)

$$\ln k = -\frac{E_a}{RT} + \ln A \quad (3.19)$$

The plot of $\ln k$ vs $1/T$ gives a straight line according to the equation (3.19) as shown in Fig. 3.10.

Thus, it has been found from Arrhenius equation (3.18) that increasing the temperature or decreasing the activation energy will result in an increase in the rate of the reaction and an exponential increase in the rate constant.

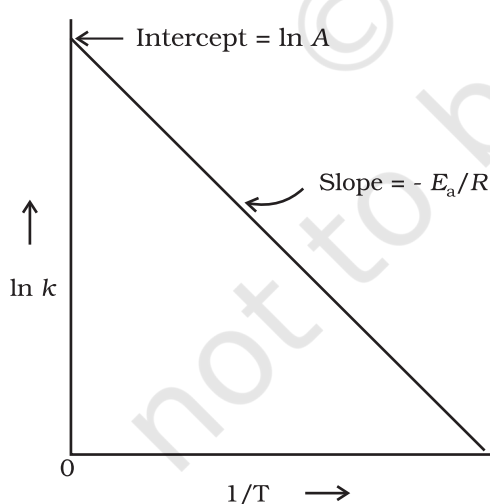


Fig. 3.10: A plot between $\ln k$ and $1/T$

In Fig. 3.10, slope = $-\frac{E_a}{R}$ and intercept = $\ln A$. So we can calculate E_a and A using these values.

At temperature T_1 , equation (3.19) is

$$\ln k_1 = -\frac{E_a}{RT_1} + \ln A \quad (3.20)$$

At temperature T_2 , equation (3.19) is

$$\ln k_2 = -\frac{E_a}{RT_2} + \ln A \quad (3.21)$$

(since A is constant for a given reaction)

k_1 and k_2 are the values of rate constants at temperatures T_1 and T_2 respectively.

Subtracting equation (3.20) from (3.21), we obtain

$$\begin{aligned}\ln k_2 - \ln k_1 &= \frac{E_a}{RT_1} - \frac{E_a}{RT_2} \\ \ln \frac{k_2}{k_1} &= \frac{E_a}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right] \\ \log \frac{k_2}{k_1} &= \frac{E_a}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right] \\ \log \frac{k_2}{k_1} &= \frac{E_a}{2.303R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]\end{aligned}\quad (3.22)$$

Example 3.9 The rate constants of a reaction at 500K and 700K are 0.02s^{-1} and 0.07s^{-1} respectively. Calculate the values of E_a and A .

Solution

$$\begin{aligned}\log \frac{k_2}{k_1} &= \frac{E_a}{2.303R} \left[\frac{T_2 - T_1}{T_1 T_2} \right] \\ \log \frac{0.07}{0.02} &= \left(\frac{E_a}{2.303 \times 8.314 \text{ JK}^{-1} \text{ mol}^{-1}} \right) \left[\frac{700 - 500}{700 \times 500} \right] \\ 0.544 &= E_a \times 5.714 \times 10^{-4} / 19.15 \\ E_a &= 0.544 \times 19.15 / 5.714 \times 10^{-4} = 18230.8 \text{ J} \\ \text{Since } k &= Ae^{-E_a/RT} \\ 0.02 &= Ae^{-18230.8/8.314 \times 500} \\ A &= 0.02/0.012 = 1.61\end{aligned}$$

Example 3.10 The first order rate constant for the decomposition of ethyl iodide by the reaction



at 600K is $1.60 \times 10^{-5} \text{ s}^{-1}$. Its energy of activation is 209 kJ/mol. Calculate the rate constant of the reaction at 700K.

Solution

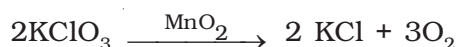
We know that

$$\log k_2 - \log k_1 = \frac{E_a}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$\begin{aligned}\log k_2 &= \log k_1 + \frac{E_a}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right] \\ &= \log(1.60 \times 10^{-5}) + \frac{209000 \text{ J mol}^{-1}}{2.303 \times 8.314 \text{ J mol}^{-1} \text{ K}^{-1}} \left[\frac{1}{600 \text{ K}} - \frac{1}{700 \text{ K}} \right] \\ \log k_2 &= -4.796 + 2.599 = -2.197 \\ k_2 &= 6.36 \times 10^{-3} \text{ s}^{-1}\end{aligned}$$

3.4.1 Effect of Catalyst

A catalyst is a substance which increases the rate of a reaction without itself undergoing any permanent chemical change. For example, MnO_2 catalyses the following reaction so as to increase its rate considerably.



The word catalyst should not be used when the added substance reduces the rate of reaction. The substance is then called inhibitor. The action of the catalyst can be explained by intermediate complex theory. According to this theory, a catalyst participates in a chemical reaction by forming temporary bonds with the reactants resulting in an intermediate complex. This has a transitory existence and decomposes to yield products and the catalyst.

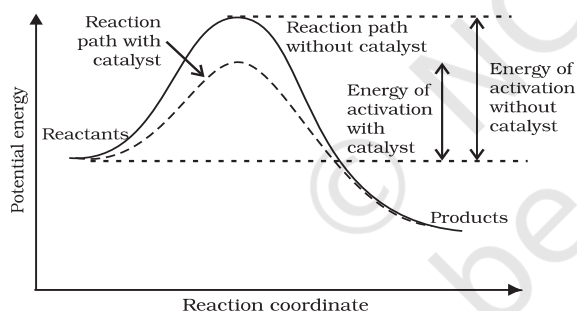


Fig. 3.11: Effect of catalyst on activation energy

It is believed that the catalyst provides an alternate pathway or reaction mechanism by reducing the activation energy between reactants and products and hence lowering the potential energy barrier as shown in Fig. 3.11.

It is clear from Arrhenius equation (3.18) that lower the value of activation energy faster will be the rate of a reaction.

A small amount of the catalyst can catalyse a large amount of reactants. A catalyst does not alter **Gibbs energy**, ΔG of a reaction. It catalyses the spontaneous reactions but does not catalyse non-spontaneous reactions. It is also found that a catalyst does not change the equilibrium constant of a reaction rather, it helps in attaining the equilibrium faster, that is, it catalyses the forward as well as the backward reactions to the same extent so that the equilibrium state remains same but is reached earlier.

3.5 Collision Theory of Chemical Reactions

Though Arrhenius equation is applicable under a wide range of circumstances, collision theory, which was developed by Max Trautz and William Lewis in 1916-18, provides a greater insight into the energetic and mechanistic aspects of reactions. It is based on kinetic theory of gases. According to this theory, the reactant molecules are

assumed to be hard spheres and reaction is postulated to occur when molecules collide with each other. **The number of collisions per second per unit volume of the reaction mixture is known as collision frequency (Z).** Another factor which affects the rate of chemical reactions is activation energy (as we have already studied). For a bimolecular elementary reaction



rate of reaction can be expressed as

$$\text{Rate} = Z_{AB} e^{-E_a / RT} \quad (3.23)$$

where Z_{AB} represents the collision frequency of reactants, A and B and $e^{-E_a / RT}$ represents the fraction of molecules with energies equal to or greater than E_a . Comparing (3.23) with Arrhenius equation, we can say that A is related to collision frequency.

Equation (3.23) predicts the value of rate constants fairly accurately for the reactions that involve atomic species or simple molecules but for complex molecules significant deviations are observed. The reason could be that all collisions do not lead to the formation of products. The collisions in which molecules collide with sufficient kinetic energy (called threshold energy*) and proper orientation, so as to facilitate breaking of bonds between reacting species and formation of new bonds to form products are called as **effective collisions**.

For example, formation of methanol from bromoethane depends upon the orientation of reactant molecules as shown in Fig. 3.12. The proper orientation of reactant molecules lead to bond formation whereas improper orientation makes them simply bounce back and no products are formed.

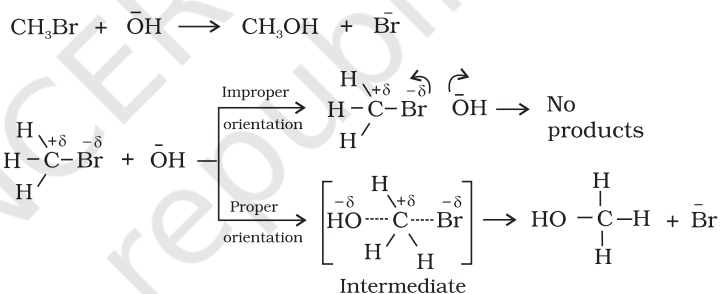


Fig. 3.12: Diagram showing molecules having proper and improper orientation

To account for effective collisions, another factor P , called the probability or steric factor is introduced. It takes into account the fact that in a collision, molecules must be properly oriented i.e.,

$$\text{Rate} = PZ_{AB} e^{-E_a / RT}$$

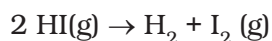
Thus, in collision theory activation energy and proper orientation of the molecules together determine the criteria for an effective collision and hence the rate of a chemical reaction.

Collision theory also has certain drawbacks as it considers atoms/molecules to be hard spheres and ignores their structural aspect. You will study details about this theory and more on other theories in your higher classes.

* Threshold energy = Activation Energy + energy possessed by reacting species.

Intext Questions

- 3.7** What will be the effect of temperature on rate constant ?
- 3.8** The rate of the chemical reaction doubles for an increase of 10K in absolute temperature from 298K. Calculate E_a .
- 3.9** The activation energy for the reaction



is $209.5 \text{ kJ mol}^{-1}$ at 581K. Calculate the fraction of molecules of reactants having energy equal to or greater than activation energy?

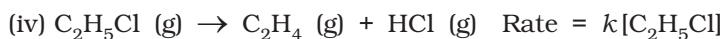
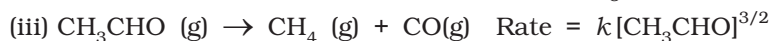
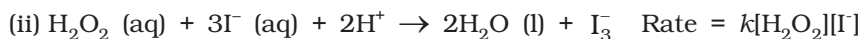
Summary

Chemical kinetics is the study of chemical reactions with respect to reaction rates, effect of various variables, rearrangement of atoms and formation of intermediates. The rate of a reaction is concerned with decrease in concentration of reactants or increase in the concentration of products per unit time. It can be expressed as instantaneous rate at a particular instant of time and average rate over a large interval of time. A number of factors such as temperature, concentration of reactants, catalyst, affect the rate of a reaction. Mathematical representation of rate of a reaction is given by **rate law**. It has to be determined experimentally and cannot be predicted. **Order of a reaction** with respect to a reactant is the power of its concentration which appears in the rate law equation. The order of a reaction is the sum of all such powers of concentration of terms for different reactants. **Rate constant** is the proportionality factor in the rate law. Rate constant and order of a reaction can be determined from rate law or its integrated rate equation. **Molecularity** is defined only for an elementary reaction. Its values are limited from 1 to 3 whereas order can be 0, 1, 2, 3 or even a fraction. Molecularity and order of an elementary reaction are same.

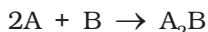
Temperature dependence of rate constants is described by Arrhenius equation ($k = Ae^{-E_a/RT}$). E_a corresponds to the **activation energy** and is given by the energy difference between activated complex and the reactant molecules, and A (Arrhenius factor or pre-exponential factor) corresponds to the collision frequency. The equation clearly shows that increase of temperature or lowering of E_a will lead to an increase in the rate of reaction and presence of a catalyst lowers the activation energy by providing an alternate path for the reaction. According to collision theory, another factor P called steric factor which refers to the orientation of molecules which collide, is important and contributes to effective collisions, thus, modifying the Arrhenius equation to $k = PZ_{AB}e^{-E_a/RT}$.

Exercises

- 3.1** From the rate expression for the following reactions, determine their order of reaction and the dimensions of the rate constants.



- 3.2** For the reaction:



the rate = $k[\text{A}][\text{B}]^2$ with $k = 2.0 \times 10^{-6} \text{ mol}^{-2} \text{ L}^2 \text{ s}^{-1}$. Calculate the initial rate of the reaction when $[\text{A}] = 0.1 \text{ mol L}^{-1}$, $[\text{B}] = 0.2 \text{ mol L}^{-1}$. Calculate the rate of reaction after $[\text{A}]$ is reduced to 0.06 mol L^{-1} .

- 3.3** The decomposition of NH_3 on platinum surface is zero order reaction. What are the rates of production of N_2 and H_2 if $k = 2.5 \times 10^{-4} \text{ mol l}^{-1} \text{ s}^{-1}$?

- 3.4** The decomposition of dimethyl ether leads to the formation of CH_4 , H_2 and CO and the reaction rate is given by

$$\text{Rate} = k [\text{CH}_3\text{OCH}_3]^{3/2}$$

The rate of reaction is followed by increase in pressure in a closed vessel, so the rate can also be expressed in terms of the partial pressure of dimethyl ether, i.e.,

$$\text{Rate} = k(p_{\text{CH}_3\text{OCH}_3})^{3/2}$$

If the pressure is measured in bar and time in minutes, then what are the units of rate and rate constants?

- 3.5** Mention the factors that affect the rate of a chemical reaction.

- 3.6** A reaction is second order with respect to a reactant. How is the rate of reaction affected if the concentration of the reactant is

- (i) doubled (ii) reduced to half ?

- 3.7** What is the effect of temperature on the rate constant of a reaction? How can this effect of temperature on rate constant be represented quantitatively?

- 3.8** In a pseudo first order reaction in water, the following results were obtained:

t/s	0	30	60	90
$[\text{A}]/\text{mol L}^{-1}$	0.55	0.31	0.17	0.085

Calculate the average rate of reaction between the time interval 30 to 60 seconds.

- 3.9** A reaction is first order in A and second order in B.

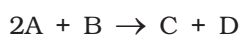
- (i) Write the differential rate equation.
 (ii) How is the rate affected on increasing the concentration of B three times?
 (iii) How is the rate affected when the concentrations of both A and B are doubled?

- 3.10** In a reaction between A and B, the initial rate of reaction (r_0) was measured for different initial concentrations of A and B as given below:

A/ mol L ⁻¹	0.20	0.20	0.40
B/ mol L ⁻¹	0.30	0.10	0.05
r_0 /mol L ⁻¹ s ⁻¹	5.07×10^{-5}	5.07×10^{-5}	1.43×10^{-4}

What is the order of the reaction with respect to A and B?

- 3.11** The following results have been obtained during the kinetic studies of the reaction:



Experiment	[A]/mol L ⁻¹	[B]/mol L ⁻¹	Initial rate of formation of D/mol L ⁻¹ min ⁻¹
I	0.1	0.1	6.0×10^{-3}
II	0.3	0.2	7.2×10^{-2}
III	0.3	0.4	2.88×10^{-1}
IV	0.4	0.1	2.40×10^{-2}

Determine the rate law and the rate constant for the reaction.

- 3.12** The reaction between A and B is first order with respect to A and zero order with respect to B. Fill in the blanks in the following table:

Experiment	[A]/ mol L ⁻¹	[B]/ mol L ⁻¹	Initial rate/ mol L ⁻¹ min ⁻¹
I	0.1	0.1	2.0×10^{-2}
II	–	0.2	4.0×10^{-2}
III	0.4	0.4	–
IV	–	0.2	2.0×10^{-2}

- 3.13** Calculate the half-life of a first order reaction from their rate constants given below:

(i) 200 s^{-1} (ii) 2 min^{-1} (iii) 4 years^{-1}

- 3.14** The half-life for radioactive decay of ^{14}C is 5730 years. An archaeological artifact containing wood had only 80% of the ^{14}C found in a living tree. Estimate the age of the sample.

- 3.15** The experimental data for decomposition of N_2O_5



in gas phase at 318K are given below:

t/s	0	400	800	1200	1600	2000	2400	2800	3200
$10^2 \times [\text{N}_2\text{O}_5]/\text{mol L}^{-1}$	1.63	1.36	1.14	0.93	0.78	0.64	0.53	0.43	0.35

- Plot $[\text{N}_2\text{O}_5]$ against t .
- Find the half-life period for the reaction.
- Draw a graph between $\log[\text{N}_2\text{O}_5]$ and t .
- What is the rate law ?

- (v) Calculate the rate constant.
 (vi) Calculate the half-life period from k and compare it with (ii).
- 3.16** The rate constant for a first order reaction is 60 s^{-1} . How much time will it take to reduce the initial concentration of the reactant to its $1/16^{\text{th}}$ value?
- 3.17** During nuclear explosion, one of the products is ^{90}Sr with half-life of 28.1 years. If $1 \mu\text{g}$ of ^{90}Sr was absorbed in the bones of a newly born baby instead of calcium, how much of it will remain after 10 years and 60 years if it is not lost metabolically.
- 3.18** For a first order reaction, show that time required for 99% completion is twice the time required for the completion of 90% of reaction.
- 3.19** A first order reaction takes 40 min for 30% decomposition. Calculate $t_{1/2}$.
- 3.20** For the decomposition of azoisopropane to hexane and nitrogen at 543 K, the following data are obtained.

t (sec)	P (mm of Hg)
0	35.0
360	54.0
720	63.0

Calculate the rate constant.

- 3.21** The following data were obtained during the first order thermal decomposition of SO_2Cl_2 at a constant volume.



Experiment	Time/ s^{-1}	Total pressure/atm
1	0	0.5
2	100	0.6

Calculate the rate of the reaction when total pressure is 0.65 atm.

- 3.22** The rate constant for the decomposition of N_2O_5 at various temperatures is given below:

$T/^\circ\text{C}$	0	20	40	60	80
$10^5 \times k/\text{s}^{-1}$	0.0787	1.70	25.7	178	2140

Draw a graph between $\ln k$ and $1/T$ and calculate the values of A and E_a . Predict the rate constant at 30° and 50°C .

- 3.23** The rate constant for the decomposition of hydrocarbons is $2.418 \times 10^{-5}\text{s}^{-1}$ at 546 K. If the energy of activation is 179.9 kJ/mol, what will be the value of pre-exponential factor.
- 3.24** Consider a certain reaction $A \rightarrow \text{Products}$ with $k = 2.0 \times 10^{-2}\text{s}^{-1}$. Calculate the concentration of A remaining after 100 s if the initial concentration of A is 1.0 mol L^{-1} .
- 3.25** Sucrose decomposes in acid solution into glucose and fructose according to the first order rate law, with $t_{1/2} = 3.00$ hours. What fraction of sample of sucrose remains after 8 hours ?
- 3.26** The decomposition of hydrocarbon follows the equation

$$k = (4.5 \times 10^{11}\text{s}^{-1}) e^{-28000\text{K}/T}$$

Calculate E_a .

- 3.27** The rate constant for the first order decomposition of H_2O_2 is given by the following equation:
 $\log k = 14.34 - 1.25 \times 10^4 K/T$
 Calculate E_a for this reaction and at what temperature will its half-period be 256 minutes?
- 3.28** The decomposition of A into product has value of k as $4.5 \times 10^3 \text{ s}^{-1}$ at 10°C and energy of activation 60 kJ mol^{-1} . At what temperature would k be $1.5 \times 10^4 \text{ s}^{-1}$?
- 3.29** The time required for 10% completion of a first order reaction at 298K is equal to that required for its 25% completion at 308K . If the value of A is $4 \times 10^{10} \text{ s}^{-1}$. Calculate k at 318K and E_a .
- 3.30** The rate of a reaction quadruples when the temperature changes from 293 K to 313 K . Calculate the energy of activation of the reaction assuming that it does not change with temperature.

Answers to Some Intext Questions

- 3.1** $r_{av} = 6.66 \times 10^{-6} \text{ Ms}^{-1}$
- 3.2** Rate of reaction = rate of disappearance of A
 $= 0.005 \text{ mol litre}^{-1}\text{min}^{-1}$
- 3.3** Order of the reaction is 2.5
- 3.4** $\text{X} \rightarrow \text{Y}$
 $\text{Rate} = k[\text{X}]^2$
 The rate will increase 9 times
- 3.5** $t = 444 \text{ s}$
- 3.6** $1.925 \times 10^{-4} \text{ s}^{-1}$
- 3.8** $E_a = 52.897 \text{ kJ mol}^{-1}$
- 3.9** 1.471×10^{-19}



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Unit

4

The *d*- and *f*-Block Elements

Objectives

After studying this Unit, you will be able to

- learn the positions of the *d*- and *f*-block elements in the periodic table;
- know the electronic configurations of the transition (*d*-block) and the inner transition (*f*-block) elements;
- appreciate the relative stability of various oxidation states in terms of electrode potential values;
- describe the preparation, properties, structures and uses of some important compounds such as $\text{K}_2\text{Cr}_2\text{O}_7$ and KMnO_4 ;
- understand the general characteristics of the *d*- and *f*-block elements and the general horizontal and group trends in them;
- describe the properties of the *f*-block elements and give a comparative account of the lanthanoids and actinoids with respect to their electronic configurations, oxidation states and chemical behaviour.

Iron, copper, silver and gold are among the transition elements that have played important roles in the development of human civilisation. The inner transition elements such as Th, Pa and U are proving excellent sources of nuclear energy in modern times.

The *d*-block of the periodic table contains the elements of the groups 3-12 in which the *d* orbitals are progressively filled in each of the four long periods. The *f*-block consists of elements in which 4 *f* and 5 *f* orbitals are progressively filled. They are placed in a separate panel at the bottom of the periodic table. The names *transition metals* and *inner transition metals* are often used to refer to the elements of *d*- and *f*-blocks respectively.

There are mainly four series of the transition metals, 3*d* series (Sc to Zn), 4*d* series (Y to Cd), 5*d* series (La and Hf to Hg) and 6*d* series which has Ac and elements from Rf to Cn. The two series of the inner transition metals; 4*f* (Ce to Lu) and 5*f* (Th to Lr) are known as *lanthanoids* and *actinoids* respectively.

Originally the name transition metals was derived from the fact that their chemical properties were transitional between those of *s* and *p*-block elements. Now according to IUPAC, transition metals are defined as metals which have incomplete *d* subshell either in neutral atom or in their ions. Zinc, cadmium and mercury of group 12 have full d^{10} configuration in their ground state as well as in their common oxidation states and hence, are not regarded as transition metals. However, being the end members of the 3*d*, 4*d* and 5*d* transition series, respectively, their chemistry is studied along with the chemistry of the transition metals.

The presence of partly filled *d* or *f* orbitals in their atoms makes transition elements different from that of

the non-transition elements. Hence, transition elements and their compounds are studied separately. However, the usual theory of valence as applicable to the non-transition elements can be applied successfully to the transition elements also.

Various precious metals such as silver, gold and platinum and industrially important metals like iron, copper and titanium belong to the transition metals series.

In this Unit, we shall first deal with the electronic configuration, occurrence and general characteristics of transition elements with special emphasis on the trends in the properties of the first row (3d) transition metals along with the preparation and properties of some important compounds. This will be followed by consideration of certain general aspects such as electronic configurations, oxidation states and chemical reactivity of the inner transition metals.

THE TRANSITION ELEMENTS (d-BLOCK)

4.1 Position in the Periodic Table

The *d*-block occupies the large middle section of the periodic table flanked between *s*- and *p*- blocks in the periodic table. The *d*-orbitals of the penultimate energy level of atoms receive electrons giving rise to four rows of the transition metals, i.e., 3d, 4d, 5d and 6d. All these series of transition elements are shown in Table 4.1.

4.2 Electronic Configurations of the *d*-Block Elements

In general the electronic configuration of outer orbitals of these elements is $(n-1)d^{1-10}ns^{1-2}$ except for Pd where its electronic configuration is $4d^{10}5s^0$. The $(n-1)$ stands for the inner *d* orbitals which may have one to ten electrons and the outermost *ns* orbital may have one or two electrons. However, this generalisation has several exceptions because of very little energy difference between $(n-1)d$ and *ns* orbitals. Furthermore, half and completely filled sets of orbitals are relatively more stable. A consequence of this factor is reflected in the electronic configurations of Cr and Cu in the 3d series. For example, consider the case of Cr, which has $3d^5 4s^1$ configuration instead of $3d^4 4s^2$; the energy gap between the two sets (3d and 4s) of orbitals is small enough to prevent electron entering the 3d orbitals. Similarly in case of Cu, the configuration is $3d^{10} 4s^1$ and not $3d^9 4s^2$. The ground state electronic configurations of the outer orbitals of transition elements are given in Table 4.1.

Table 4.1: Electronic Configurations of outer orbitals of the Transition Elements (ground state)

1st Series										
	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
<i>Z</i>	21	22	23	24	25	26	27	28	29	30
4s	2	2	2	1	2	2	2	2	1	2
3d	1	2	3	5	5	6	7	8	10	10

2nd Series										
	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
Z	39	40	41	42	43	44	45	46	47	48
5s	2	2	1	1	1	1	1	0	1	2
4d	1	2	4	5	6	7	8	10	10	10

3rd Series										
	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
Z	57	72	73	74	75	76	77	78	79	80
6s	2	2	2	2	2	2	2	1	1	2
5d	1	2	3	4	5	6	7	9	10	10

4th Series										
	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn
Z	89	104	105	106	107	108	109	110	111	112
7s	2	2	2	2	2	2	2	2	1	2
6d	1	2	3	4	5	6	7	8	10	10

The electronic configurations of outer orbitals of Zn, Cd, Hg and Cn are represented by the general formula $(n-1)d^{10}ns^2$. The orbitals in these elements are completely filled in the ground state as well as in their common oxidation states. Therefore, they are not regarded as transition elements.

The d orbitals of the transition elements protrude to the periphery of an atom more than the other orbitals (i.e., s and p), hence, they are more influenced by the surroundings as well as affect the atoms or molecules surrounding them. In some respects, ions of a given d^n configuration ($n = 1 - 9$) have similar magnetic and electronic properties. With partly filled d orbitals these elements exhibit certain characteristic properties such as display of a variety of oxidation states, formation of coloured ions and entering into complex formation with a variety of ligands.

The transition metals and their compounds also exhibit catalytic property and paramagnetic behaviour. All these characteristics have been discussed in detail later in this Unit.

There are greater similarities in the properties of the transition elements of a horizontal row in contrast to the non-transition elements. However, some group similarities also exist. We shall first study the general characteristics and their trends in the horizontal rows (particularly $3d$ row) and then consider some group similarities.

On what ground can you say that scandium ($Z = 21$) is a transition element but zinc ($Z = 30$) is not? **Example 4.1**

On the basis of incompletely filled $3d$ orbitals in case of scandium atom in its ground state ($3d^1$), it is regarded as a transition element. On the other hand, zinc atom has completely filled d orbitals ($3d^{10}$) in its ground state as well as in its oxidised state, hence it is not regarded as a transition element. **Solution**

Intext Question

- 4.1 Silver atom has completely filled d orbitals ($4d^{10}$) in its ground state. How can you say that it is a transition element?

We will discuss the properties of elements of first transition series only in the following sections.

4.3 General Properties of the Transition Elements (d -Block)

4.3.1 Physical Properties

Nearly all the transition elements display typical metallic properties such as high tensile strength, ductility, malleability, high thermal and electrical conductivity and metallic lustre. With the exceptions of Zn, Cd, Hg and Mn, they have one or more typical metallic structures at normal temperatures.

Lattice Structures of Transition Metals

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
hcp (bcc)	hcp (bcc)	bcc	bcc	X (bcc, ccp)	bcc (hcp)	ccp (hcp)	ccp	ccp	X (hcp)
Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
hcp (bcc)	hcp (bcc)	bcc	bcc	hcp	hcp	ccp	ccp	ccp	X (hcp)
La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
hcp (ccp, bcc)	hcp (bcc)	bcc	bcc	hcp	hcp	ccp	ccp	ccp	X

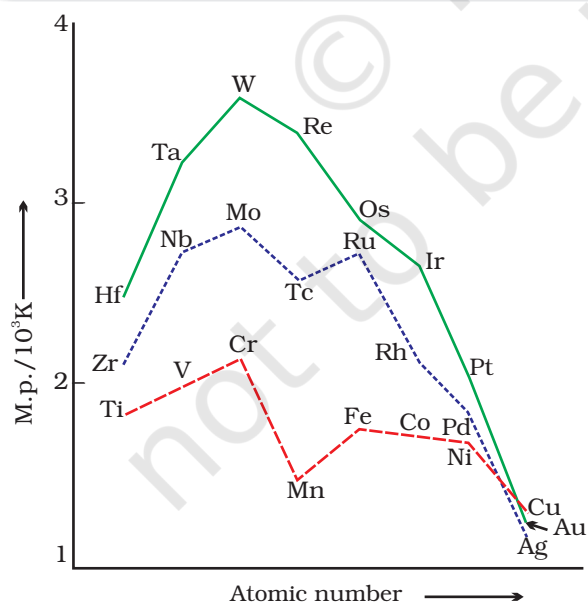


Fig. 4.1: Trends in melting points of transition elements

(bcc = body centred cubic; hcp = hexagonal close packed; ccp = cubic close packed; X = a typical metal structure).

The transition metals (with the exception of Zn, Cd and Hg) are very hard and have low volatility. Their melting and boiling points are high. Fig. 4.1 depicts the melting points of transition metals belonging to $3d$, $4d$ and $5d$ series. The high melting points of these metals are attributed to the involvement of greater number of electrons from $(n-1)d$ in addition to the ns electrons in the interatomic metallic bonding. In any row the melting points of these metals rise to a maximum at d^5 except for anomalous values of Mn and Tc and fall regularly as the atomic number increases. They have high enthalpies of atomisation which are shown in Fig. 4.2. The maxima at about the middle of each series indicate that one unpaired electron per d orbital is particularly

favourable for strong interatomic interaction. In general, greater the number of valence electrons, stronger is the resultant bonding. Since the enthalpy of atomisation is an important factor in determining the standard electrode potential of a metal, metals with very high enthalpy of atomisation (i.e., very high boiling point) tend to be noble in their reactions (see later for electrode potentials).

Another generalisation that may be drawn from Fig. 4.2 is that the metals of the second and third series have greater enthalpies of atomisation than the corresponding elements of the first series; this is an important factor in accounting for the occurrence of much more frequent metal – metal bonding in compounds of the heavy transition metals.

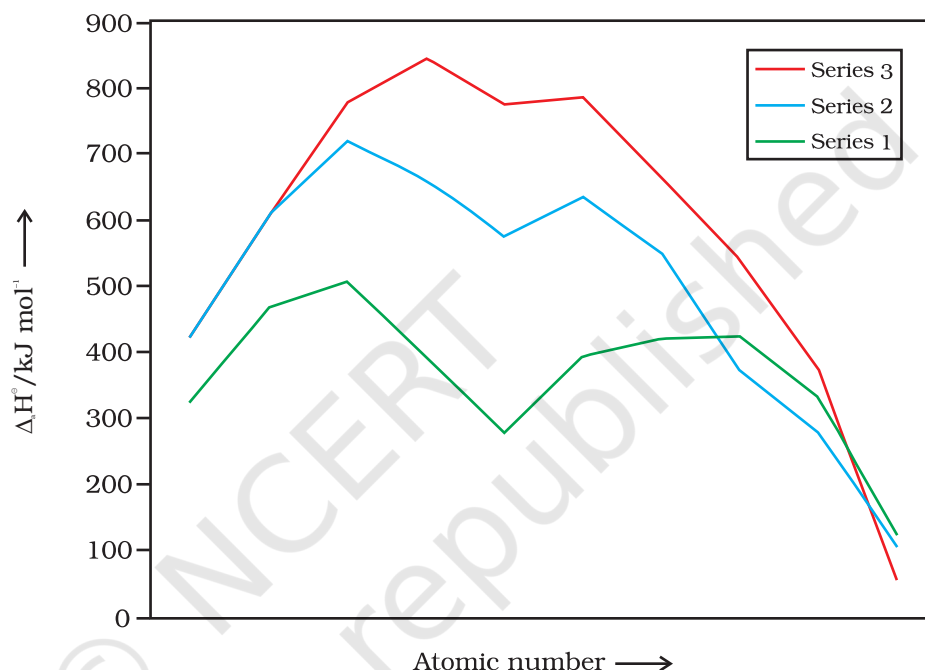


Fig. 4.2
Trends in enthalpies
of atomisation of
transition elements

4.3.2 Variation in Atomic and Ionic Sizes of Transition Metals

In general, ions of the same charge in a given series show progressive decrease in radius with increasing atomic number. This is because the new electron enters a d orbital each time the nuclear charge increases by unity. It may be recalled that the shielding effect of a d electron is not that effective, hence the net electrostatic attraction between the nuclear charge and the outermost electron increases and the ionic radius decreases. The same trend is observed in the atomic radii of a given series. However, the variation within a series is quite small. An interesting point emerges when atomic sizes of one series are compared with those of the corresponding elements in the other series. The curves in Fig. 4.3 show an increase from the first ($3d$) to the second ($4d$) series of the elements but the radii of the third ($5d$) series are virtually the same as those of the corresponding members of the second series. This phenomenon is associated with the intervention of the $4f$ orbitals which must be filled before the $5d$ series of elements begin. The filling of $4f$ before $5d$ orbital results in a regular decrease in atomic radii called **Lanthanoid contraction** which essentially compensates for the expected

increase in atomic size with increasing atomic number. The net result of the lanthanoid contraction is that the second and the third *d* series exhibit similar radii (e.g., Zr 160 pm, Hf 159 pm) and have very similar physical and chemical properties much more than that expected on the basis of usual family relationship.

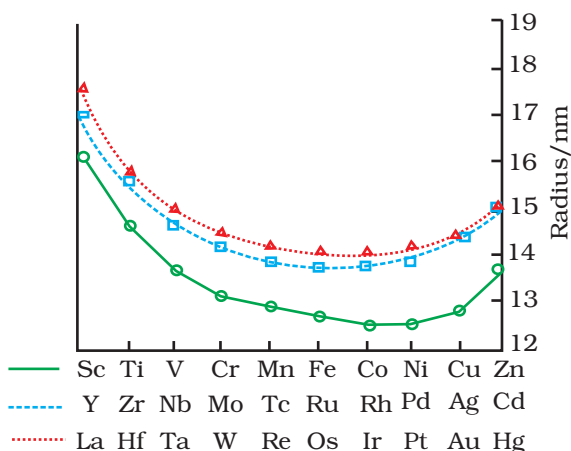


Fig. 4.3: Trends in atomic radii of transition elements

The factor responsible for the lanthanoid contraction is somewhat similar to that observed in an ordinary transition series and is attributed to similar cause, i.e., the imperfect shielding of one electron by another in the same set of orbitals. However, the shielding of one *4f* electron by another is less than that of one *d* electron by another, and as the nuclear charge increases along the series, there is fairly regular decrease in the size of the entire *4fⁿ* orbitals.

The decrease in metallic radius coupled with increase in atomic mass results in a general increase in the density of these elements. Thus, from titanium (*Z* = 22) to copper (*Z* = 29) the significant increase in the density may be noted (Table 4.2).

Table 4.2: Electronic Configurations and some other Properties of the First Series of Transition Elements

Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Atomic number	21	22	23	24	25	26	27	28	29	30
Electronic configuration										
M	$3d^1 4s^2$	$3d^2 4s^2$	$3d^3 4s^2$	$3d^5 4s^1$	$3d^5 4s^2$	$3d^6 4s^2$	$3d^7 4s^2$	$3d^8 4s^2$	$3d^{10} 4s^1$	$3d^{10} 4s^2$
M^+	$3d^1 4s^1$	$3d^2 4s^1$	$3d^3 4s^1$	$3d^5$	$3d^5 4s^1$	$3d^6 4s^1$	$3d^7 4s^1$	$3d^8 4s^1$	$3d^{10}$	$3d^{10} 4s^1$
M^{2+}	$3d^1$	$3d^2$	$3d^3$	$3d^4$	$3d^5$	$3d^6$	$3d^7$	$3d^8$	$3d^9$	$3d^{10}$
M^{3+}	[Ar]	$3d^1$	$3d^2$	$3d^3$	$3d^4$	$3d^5$	$3d^6$	$3d^7$	–	–
Enthalpy of atomisation, $\Delta_a H^\circ / \text{kJ mol}^{-1}$	326	473	515	397	281	416	425	430	339	126
Ionisation enthalpy/$\Delta_i H^\circ / \text{kJ mol}^{-1}$										
$\Delta_i H^\circ$ I	631	656	650	653	717	762	758	736	745	906
$\Delta_i H^\circ$ II	1235	1309	1414	1592	1509	1561	1644	1752	1958	1734
$\Delta_i H^\circ$ III	2393	2657	2833	2990	3260	2962	3243	3402	3556	3837
Metallic/ionic radii/pm										
M	164	147	135	129	137	126	125	125	128	137
M^{2+}	–	–	79	82	82	77	74	70	73	75
M^{3+}	73	67	64	62	65	65	61	60	–	–
Standard electrode potential E° / V										
M^{2+}/M	–	–1.63	–1.18	–0.90	–1.18	–0.44	–0.28	–0.25	+0.34	–0.76
M^{3+}/M^{2+}	–	–0.37	–0.26	–0.41	+1.57	+0.77	+1.97	–	–	–
Density/g cm^{-3}	3.43	4.1	6.07	7.19	7.21	7.8	8.7	8.9	8.9	7.1

Why do the transition elements exhibit higher enthalpies of atomisation?

Example 4.2

Because of large number of unpaired electrons in their atoms they have stronger interatomic interaction and hence stronger bonding between atoms resulting in higher enthalpies of atomisation.

Solution

Intext Question

4.2 In the series Sc ($Z = 21$) to Zn ($Z = 30$), the enthalpy of atomisation of zinc is the lowest, i.e., 126 kJ mol^{-1} . Why?

4.3.3 Ionisation Enthalpies

There is an increase in ionisation enthalpy along each series of the transition elements from left to right due to an increase in nuclear charge which accompanies the filling of the inner d orbitals. Table 4.2 gives the values of the first three ionisation enthalpies of the first series of transition elements. These values show that the successive enthalpies of these elements do not increase as steeply as in the case of non-transition elements. The variation in ionisation enthalpy along a series of transition elements is much less in comparison to the variation along a period of non-transition elements. The first ionisation enthalpy, in general, increases, but the magnitude of the increase in the second and third ionisation enthalpies for the successive elements, is much higher along a series.

The irregular trend in the first ionisation enthalpy of the metals of $3d$ series, though of little chemical significance, can be accounted for by considering that the removal of one electron alters the relative energies of $4s$ and $3d$ orbitals. You have learnt that when d -block elements form ions, ns electrons are lost before $(n - 1) d$ electrons. As we move along the period in $3d$ series, we see that nuclear charge increases from scandium to zinc but electrons are added to the orbital of inner subshell, i.e., $3d$ orbitals. These $3d$ electrons shield the $4s$ electrons from the increasing nuclear charge somewhat more effectively than the outer shell electrons can shield one another. Therefore, the atomic radii decrease less rapidly. Thus, ionization energies increase only slightly along the $3d$ series. The doubly or more highly charged ions have d^n configurations with no $4s$ electrons. A general trend of increasing values of second ionisation enthalpy is expected as the effective nuclear charge increases because one d electron does not shield another electron from the influence of nuclear charge because d -orbitals differ in direction. However, the trend of steady increase in second and third ionisation enthalpy breaks for the formation of Mn^{2+} and Fe^{3+} respectively. In both the cases, ions have d^5 configuration. Similar breaks occur at corresponding elements in the later transition series.

The interpretation of variation in ionisation enthalpy for an electronic configuration d^n is as follows:

The three terms responsible for the value of ionisation enthalpy are attraction of each electron towards nucleus, repulsion between the

electrons and the exchange energy. Exchange energy is responsible for the stabilisation of energy state. Exchange energy is approximately proportional to the total number of possible pairs of parallel spins in the degenerate orbitals. When several electrons occupy a set of degenerate orbitals, the lowest energy state corresponds to the maximum possible extent of single occupation of orbital and parallel spins (Hunds rule). The loss of exchange energy increases the stability. As the stability increases, the ionisation becomes more difficult. There is no loss of exchange energy at d^6 configuration. Mn^+ has $3d^5 4s^1$ configuration and configuration of Cr^+ is d^5 , therefore, ionisation enthalpy of Mn^+ is lower than Cr^+ . In the same way, Fe^{2+} has d^6 configuration and Mn^{2+} has $3d^5$ configuration. Hence, ionisation enthalpy of Fe^{2+} is lower than the Mn^{2+} . In other words, we can say that the third ionisation enthalpy of Fe is lower than that of Mn.

The lowest common oxidation state of these metals is +2. To form the M^{2+} ions from the gaseous atoms, the sum of the first and second ionisation enthalpy is required in addition to the enthalpy of atomisation. The dominant term is the second ionisation enthalpy which shows unusually high values for Cr and Cu where M^+ ions have the d^5 and d^{10} configurations respectively. The value for Zn is correspondingly low as the ionisation causes the removal of one 4s electron which results in the formation of stable d^{10} configuration. The trend in the third ionisation enthalpies is not complicated by the 4s orbital factor and shows the greater difficulty of removing an electron from the d^5 (Mn^{2+}) and d^{10} (Zn^{2+}) ions. In general, the third ionisation enthalpies are quite high. Also the high values for third ionisation enthalpies of copper, nickel and zinc indicate why it is difficult to obtain oxidation state greater than two for these elements.

Although ionisation enthalpies give some guidance concerning the relative stabilities of oxidation states, this problem is very complex and not amenable to ready generalisation.

4.3.4 Oxidation States

One of the notable features of a transition elements is the great variety of oxidation states these may show in their compounds. Table 4.3 lists the common oxidation states of the first row transition elements.

Table 4.3: Oxidation States of the first row Transition Metal (the most common ones are in bold types)

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
	+2	+2	+2	+2	+2	+2	+2	+1	+2
+3	+3	+3	+3	+3	+3	+3	+3	+2	
	+4	+4	+4	+4	+4	+4	+4		
		+5	+5	+5					
			+6	+6	+6				
				+7					

The elements which give the greatest number of oxidation states occur in or near the middle of the series. Manganese, for example, exhibits all the oxidation states from +2 to +7. The lesser number of oxidation states at the extreme ends stems from either too few electrons to lose or share (Sc, Ti) or too many *d* electrons (hence fewer orbitals available in which to share electrons with others) for higher valence (Cu, Zn). Thus, early in the series scandium(II) is virtually unknown and titanium (IV) is more stable than Ti(III) or Ti(II). At the other end, the only oxidation state of zinc is +2 (no *d* electrons are involved). The maximum oxidation states of reasonable stability correspond in value to the sum of the *s* and *d* electrons upto manganese ($\text{Ti}^{\text{IV}}\text{O}_2$, $\text{V}^{\text{V}}\text{O}_2^+$, $\text{Cr}^{\text{VI}}\text{O}_4^{2-}$, $\text{Mn}^{\text{VII}}\text{O}_4^-$) followed by a rather abrupt decrease in stability of higher oxidation states, so that the typical species to follow are $\text{Fe}^{\text{II,III}}$, $\text{Co}^{\text{II,III}}$, Ni^{II} , $\text{Cu}^{\text{I,II}}$, Zn^{II} .

The variability of oxidation states, a characteristic of transition elements, arises out of incomplete filling of *d* orbitals in such a way that their oxidation states differ from each other by unity, e.g., V^{II} , V^{III} , V^{IV} , V^{V} . This is in contrast with the variability of oxidation states of non transition elements where oxidation states normally differ by a unit of two.

An interesting feature in the variability of oxidation states of the *d*-block elements is noticed among the groups (groups 4 through 10). Although in the *p*-block the lower oxidation states are favoured by the heavier members (due to inert pair effect), the opposite is true in the groups of *d*-block. For example, in group 6, Mo(VI) and W(VI) are found to be more stable than Cr(VI). Thus Cr(VI) in the form of dichromate in acidic medium is a strong oxidising agent, whereas MoO_3 and WO_3 are not.

Low oxidation states are found when a complex compound has ligands capable of π -acceptor character in addition to the σ -bonding. For example, in $\text{Ni}(\text{CO})_4$ and $\text{Fe}(\text{CO})_5$, the oxidation state of nickel and iron is zero.

Name a transition element which does not exhibit variable oxidation states.

[Example 4.3](#)

Scandium ($Z = 21$) does not exhibit variable oxidation states.

[Solution](#)

Intext Question

4.3 Which of the 3*d* series of the transition metals exhibits the largest number of oxidation states and why?

4.3.5 Trends in the M^{2+}/M Standard Electrode Potentials

Table 4.4 contains the thermochemical parameters related to the transformation of the solid metal atoms to M^{2+} ions in solution and their standard electrode potentials. The observed values of $E^\ominus$ and those calculated using the data of Table 4.4 are compared in Fig. 4.4.

The unique behaviour of Cu, having a positive $E^\ominus$, accounts for its inability to liberate H_2 from acids. Only oxidising acids (nitric and hot concentrated sulphuric) react with Cu, the acids being reduced. The high energy to transform $Cu(s)$ to $Cu^{2+}(aq)$ is not balanced by its hydration enthalpy. The general trend towards less negative $E^\ominus$ values across the

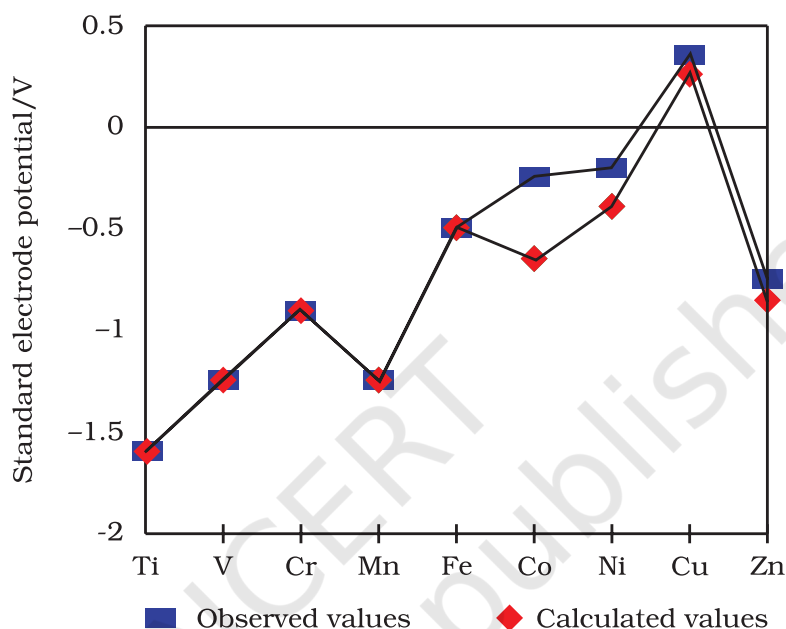


Fig. 4.4: Observed and calculated values for the standard electrode potentials ($M^{2+} \rightarrow M^\ominus$) of the elements Ti to Zn

series is related to the general increase in the sum of the first and second ionisation enthalpies. It is interesting to note that the value of $E^\ominus$ for Mn, Ni and Zn are more negative than expected from the trend.

Why is Cr^{2+} reducing and Mn^{3+} oxidising when both have d^4 configuration? [Example 4.4](#)

Cr^{2+} is reducing as its configuration changes from d^4 to d^3 , the latter having a half-filled t_{2g} level (see Unit 5). On the other hand, the change from Mn^{3+} to Mn^{2+} results in the half-filled (d^5) configuration which has extra stability. [Solution](#)

Intext Question

4.4 The $E^\ominus(M^{2+}/M)$ value for copper is positive (+0.34V). What is possible reason for this? (Hint: consider its high $\Delta_a H^\ominus$ and low $\Delta_{hyd} H^\ominus$)

Table 4.4: Thermochemical data (kJ mol⁻¹) for the first row Transition Elements and the Standard Electrode Potentials for the Reduction of M^{II} to M.

Element (M)	$\Delta_a H^\circ$ (M)	$\Delta_i H_1^\circ$	$\Delta_i H_2^\circ$	$\Delta_{\text{hyd}} H^\circ(\text{M}^{2+})$	E°/V
Ti	469	656	1309	-1866	-1.63
V	515	650	1414	-1895	-1.18
Cr	398	653	1592	-1925	-0.90
Mn	279	717	1509	-1862	-1.18
Fe	418	762	1561	-1998	-0.44
Co	427	758	1644	-2079	-0.28
Ni	431	736	1752	-2121	-0.25
Cu	339	745	1958	-2121	0.34
Zn	130	906	1734	-2059	-0.76

The stability of the half-filled d sub-shell in Mn^{2+} and the completely filled d^{10} configuration in Zn^{2+} are related to their E° values, whereas E° for Ni is related to the highest negative $\Delta_{\text{hyd}} H^\circ$.

4.3.6 Trends in the $\text{M}^{3+}/\text{M}^{2+}$ Standard Electrode Potentials

An examination of the $E^\circ(\text{M}^{3+}/\text{M}^{2+})$ values (Table 4.2) shows the varying trends. The low value for Sc reflects the stability of Sc^{3+} which has a noble gas configuration. The highest value for Zn is due to the removal of an electron from the stable d^{10} configuration of Zn^{2+} . The comparatively high value for Mn shows that $\text{Mn}^{2+}(d^5)$ is particularly stable, whereas comparatively low value for Fe shows the extra stability of $\text{Fe}^{3+}(d^5)$. The comparatively low value for V is related to the stability of V^{2+} (half-filled t_{2g} level, Unit 5).

4.3.7 Trends in Stability of Higher Oxidation States

Table 4.5 shows the stable halides of the 3d series of transition metals. The highest oxidation numbers are achieved in TiX_4 (tetrahalides), VF_5 and CrF_6 . The +7 state for Mn is not represented in simple halides but MnO_3F is known, and beyond Mn no metal has a trihalide except FeX_3 and CoF_3 . The ability of fluorine to stabilise the highest oxidation state is due to either higher lattice energy as in the case of CoF_3 , or higher bond enthalpy terms for the higher covalent compounds, e.g., VF_5 and CrF_6 .

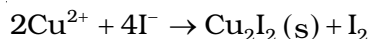
Although V^{+5} is represented only by VF_5 , the other halides, however, undergo hydrolysis to give oxohalides, VOX_3 . Another feature of fluorides is their instability in the low oxidation states e.g., VX_2 ($\text{X} = \text{Cl}, \text{Br}$ or I)

Table 4.5: Formulas of Halides of 3d Metals

Oxidation Number									
+ 6			CrF_6						
+ 5		VF_5	CrF_5						
+ 4	TiX_4	VX_4^{I}	CrX_4	MnF_4					
+ 3	TiX_3	VX_3	CrX_3	MnF_3	FeX_3^{I}	CoF_3			
+ 2	$\text{TiX}_2^{\text{III}}$	VX_2	CrX_2	MnX_2	FeX_2	CoX_2	NiX_2	CuX_2^{II}	ZnX_2
+ 1								CuX^{III}	

Key: $\text{X} = \text{F} \rightarrow \text{I}$; $\text{X}^{\text{I}} = \text{F} \rightarrow \text{Br}$; $\text{X}^{\text{II}} = \text{F}, \text{Cl}$; $\text{X}^{\text{III}} = \text{Cl} \rightarrow \text{I}$

and the same applies to CuX. On the other hand, all Cu^{II} halides are known except the iodide. In this case, Cu²⁺ oxidises I⁻ to I₂:



However, many copper (I) compounds are unstable in aqueous solution and undergo disproportionation.



The stability of Cu²⁺ (aq) rather than Cu⁺(aq) is due to the much more negative $\Delta_{\text{hyd}}H^{\circ}$ of Cu²⁺ (aq) than Cu⁺, which more than compensates for the second ionisation enthalpy of Cu.

The ability of oxygen to stabilise the highest oxidation state is demonstrated in the oxides. The highest oxidation number in the oxides (Table 4.6) coincides with the group number and is attained in Sc₂O₃ to Mn₂O₇. Beyond Group 7, no higher oxides of Fe above Fe₂O₃, are known, although ferrates (VI)(FeO₄)²⁻, are formed in alkaline media but they readily decompose to Fe₂O₃ and O₂. Besides the oxides, oxocations stabilise V^V as VO₂⁺, V^{IV} as VO²⁺ and Ti^{IV} as TiO²⁺. The ability of oxygen to stabilise these high oxidation states exceeds that of fluorine. Thus the highest Mn fluoride is MnF₄ whereas the highest oxide is Mn₂O₇. The ability of oxygen to form multiple bonds to metals explains its superiority. In the covalent oxide Mn₂O₇, each Mn is tetrahedrally surrounded by O's including a Mn–O–Mn bridge. The tetrahedral [MO₄]ⁿ⁻ ions are known for V^V, Cr^{VI}, Mn^V, Mn^{VI} and Mn^{VII}.

Table 4.6: Oxides of 3d Metals

Oxidation Number	3	4	5	6	7	8	9	10	11	12
+ 7					Mn ₂ O ₇					
+ 6				CrO ₃						
+ 5			V ₂ O ₅							
+ 4		TiO ₂	V ₂ O ₄	CrO ₂	MnO ₂					
+ 3	Sc ₂ O ₃	Ti ₂ O ₃	V ₂ O ₃	Cr ₂ O ₃	Mn ₂ O ₃	Fe ₂ O ₃				
					Mn ₃ O ₄ *	Fe ₃ O ₄ *	Co ₃ O ₄ *			
+ 2		TiO	VO	(CrO)	MnO	FeO	CoO	NiO	CuO	ZnO
+ 1									Cu ₂ O	

* mixed oxides

How would you account for the increasing oxidising power in the [Example 4.5](#) series VO₂⁺ < Cr₂O₇²⁻ < MnO₄⁻?

This is due to the increasing stability of the lower species to which they [Solution](#) are reduced.

Intext Question

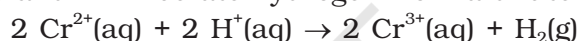
4.5 How would you account for the irregular variation of ionisation enthalpies (first and second) in the first series of the transition elements?

4.3.8 Chemical Reactivity and E° Values

Transition metals vary widely in their chemical reactivity. Many of them are sufficiently electropositive to dissolve in mineral acids, although a few are 'noble'—that is, they are unaffected by single acids.

The metals of the first series with the exception of copper are relatively more reactive and are oxidised by 1M H^+ , though the actual rate at which these metals react with oxidising agents like hydrogen ion (H^+) is sometimes slow. For example, titanium and vanadium, in practice, are passive to dilute non oxidising acids at room temperature. The E° values for M^{2+}/M (Table 4.2) indicate a decreasing tendency to form divalent cations across the series. This general trend towards less negative E° values is related to the increase in the sum of the first and second ionisation enthalpies. It is interesting to note that the E° values for Mn, Ni and Zn are more negative than expected from the general trend. Whereas the stabilities of half-filled d subshell (d^5) in Mn^{2+} and completely filled d subshell (d^{10}) in zinc are related to their E° values; for nickel, E° value is related to the highest negative enthalpy of hydration.

An examination of the E° values for the redox couple M^{3+}/M^{2+} (Table 4.2) shows that Mn^{3+} and Co^{3+} ions are the strongest oxidising agents in aqueous solutions. The ions Ti^{2+} , V^{2+} and Cr^{2+} are strong reducing agents and will liberate hydrogen from a dilute acid, e.g.,



Example 4.6

For the first row transition metals the E° values are:

E° (M^{2+}/M)	V	Cr	Mn	Fe	Co	Ni	Cu
	-1.18	-0.91	-1.18	-0.44	-0.28	-0.25	+0.34

Explain the irregularity in the above values.

Solution

The E° (M^{2+}/M) values are not regular which can be explained from the irregular variation of ionisation enthalpies ($\Delta_1 H_1 + \Delta_1 H_2$) and also the sublimation enthalpies which are relatively much less for manganese and vanadium.

Example 4.7

Why is the E° value for the Mn^{3+}/Mn^{2+} couple much more positive than that for Cr^{3+}/Cr^{2+} or Fe^{3+}/Fe^{2+} ? Explain.

Solution

Much larger third ionisation energy of Mn (where the required change is d^5 to d^4) is mainly responsible for this. This also explains why the +3 state of Mn is of little importance.

Intext Questions

- 4.6 Why is the highest oxidation state of a metal exhibited in its oxide or fluoride only?
- 4.7 Which is a stronger reducing agent Cr^{2+} or Fe^{2+} and why ?

4.3.9 Magnetic Properties

When a magnetic field is applied to substances, mainly two types of magnetic behaviour are observed: *diamagnetism* and *paramagnetism*. Diamagnetic substances are repelled by the applied field while the paramagnetic substances are attracted. Substances which are

attracted very strongly are said to be **ferromagnetic**. In fact, ferromagnetism is an extreme form of paramagnetism. Many of the transition metal ions are paramagnetic.

Paramagnetism arises from the presence of unpaired electrons, each such electron having a magnetic moment associated with its spin angular momentum and orbital angular momentum. For the compounds of the first series of transition metals, the contribution of the orbital angular momentum is effectively quenched and hence is of no significance. For these, the magnetic moment is determined by the number of unpaired electrons and is calculated by using the 'spin-only' formula, i.e.,

$$\mu = \sqrt{n(n+2)}$$

where n is the number of unpaired electrons and μ is the magnetic moment in units of **Bohr magneton (BM)**. A single unpaired electron has a magnetic moment of 1.73 Bohr magnetons (BM).

The magnetic moment increases with the increasing number of unpaired electrons. Thus, the observed magnetic moment gives a useful indication about the number of unpaired electrons present in the atom, molecule or ion. The magnetic moments calculated from the 'spin-only' formula and those derived experimentally for some ions of the first row transition elements are given in Table 4.7. The experimental data are mainly for hydrated ions in solution or in the solid state.

Table 4.7: Calculated and Observed Magnetic Moments (BM)

Ion	Configuration	Unpaired electron(s)	Magnetic moment	
			Calculated	Observed
Sc ³⁺	3d ⁰	0	0	0
Ti ³⁺	3d ¹	1	1.73	1.75
Ti ²⁺	3d ²	2	2.84	2.76
V ²⁺	3d ³	3	3.87	3.86
Cr ²⁺	3d ⁴	4	4.90	4.80
Mn ²⁺	3d ⁵	5	5.92	5.96
Fe ²⁺	3d ⁶	4	4.90	5.3 – 5.5
Co ²⁺	3d ⁷	3	3.87	4.4 – 5.2
Ni ²⁺	3d ⁸	2	2.84	2.9 – 3, 4
Cu ²⁺	3d ⁹	1	1.73	1.8 – 2.2
Zn ²⁺	3d ¹⁰	0	0	

Calculate the magnetic moment of a divalent ion in aqueous solution if its atomic number is 25.

Example 4.8

With atomic number 25, the divalent ion in aqueous solution will have d⁵ configuration (five unpaired electrons). The magnetic moment, μ is

Solution

$$\mu = \sqrt{5(5+2)} = 5.92 \text{ BM}$$

Intext Question

4.8 Calculate the 'spin only' magnetic moment of $M^{2+}_{(aq)}$ ion ($Z = 27$).

4.3.10 Formation of Coloured Ions

When an electron from a lower energy d orbital is excited to a higher energy d orbital, the energy of excitation corresponds to the frequency of light absorbed (Unit 5). This frequency generally lies in the visible region. The colour observed corresponds to the complementary colour of the light absorbed. The frequency of the light absorbed is determined by the nature of the ligand. In aqueous solutions where water molecules are the ligands, the colours of the ions observed are listed in Table 4.8. A few coloured solutions of d -block elements are illustrated in Fig. 4.5.



Fig. 4.5: Colours of some of the first row transition metal ions in aqueous solutions. From left to right: V^{4+} , V^{3+} , Mn^{2+} , Fe^{3+} , Co^{2+} , Ni^{2+} and Cu^{2+} .

Table 4.8: Colours of Some of the First Row (aquated) Transition Metal Ions

Configuration	Example	Colour
$3d^0$	Sc^{3+}	colourless
$3d^0$	Ti^{4+}	colourless
$3d^1$	Ti^{3+}	purple
$3d^1$	V^{4+}	blue
$3d^2$	V^{3+}	green
$3d^3$	V^{2+}	violet
$3d^3$	Cr^{3+}	violet
$3d^4$	Mn^{3+}	violet
$3d^4$	Cr^{2+}	blue
$3d^5$	Mn^{2+}	pink
$3d^5$	Fe^{3+}	yellow
$3d^6$	Fe^{2+}	green
$3d^6 3d^7$	$Co^{3+}Co^{2+}$	bluepink
$3d^8$	Ni^{2+}	green
$3d^9$	Cu^{2+}	blue
$3d^{10}$	Zn^{2+}	colourless

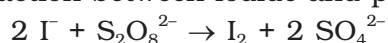
4.3.11 Formation of Complex Compounds

Complex compounds are those in which the metal ions bind a number of anions or neutral molecules giving complex species with characteristic properties. A few examples are: $[Fe(CN)_6]^{3-}$, $[Fe(CN)_6]^{4-}$, $[Cu(NH_3)_4]^{2+}$ and $[PtCl_4]^{2-}$. (The chemistry of complex compounds is

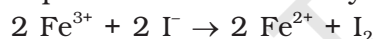
dealt with in detail in Unit 5). The transition metals form a large number of complex compounds. This is due to the comparatively smaller sizes of the metal ions, their high ionic charges and the availability of *d* orbitals for bond formation.

4.3.12 Catalytic Properties

The transition metals and their compounds are known for their catalytic activity. This activity is ascribed to their ability to adopt multiple oxidation states and to form complexes. Vanadium(V) oxide (in Contact Process), finely divided iron (in Haber's Process), and nickel (in Catalytic Hydrogenation) are some of the examples. Catalysts at a solid surface involve the formation of bonds between reactant molecules and atoms of the surface of the catalyst (first row transition metals utilise 3*d* and 4*s* electrons for bonding). This has the effect of increasing the concentration of the reactants at the catalyst surface and also weakening of the bonds in the reacting molecules (the activation energy is lowering). Also because the transition metal ions can change their oxidation states, they become more effective as catalysts. For example, iron(III) catalyses the reaction between iodide and persulphate ions.



An explanation of this catalytic action can be given as:



4.3.13 Formation of Interstitial Compounds

Interstitial compounds are those which are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals. They are usually non stoichiometric and are neither typically ionic nor covalent, for example, TiC, Mn₄N, Fe₃H, VH_{0.56} and TiH_{1.7}, etc. The formulas quoted do not, of course, correspond to any normal oxidation state of the metal. Because of the nature of their composition, these compounds are referred to as *interstitial* compounds. The principal physical and chemical characteristics of these compounds are as follows:

- (i) They have high melting points, higher than those of pure metals.
- (ii) They are very hard, some borides approach diamond in hardness.
- (iii) They retain metallic conductivity.
- (iv) They are chemically inert.

4.3.14 Alloy Formation

An alloy is a blend of metals prepared by mixing the components. Alloys may be homogeneous solid solutions in which the atoms of one metal are distributed randomly among the atoms of the other. Such alloys are formed by atoms with metallic radii that are within about 15 percent of each other. Because of similar radii and other characteristics of transition metals, alloys are readily formed by these metals. The alloys so formed are hard and have often high melting points. The best known are ferrous alloys: chromium, vanadium, tungsten, molybdenum and manganese are used for the production of a variety of steels and stainless steel. Alloys of transition metals with non transition metals such as brass (copper-zinc) and bronze (copper-tin), are also of considerable industrial importance.

Example 4.9 What is meant by 'disproportionation' of an oxidation state? Give an example.

Solution When a particular oxidation state becomes less stable relative to other oxidation states, one lower, one higher, it is said to undergo disproportionation. For example, manganese (VI) becomes unstable relative to manganese(VII) and manganese (IV) in acidic solution.



Intext Question

4.9 Explain why Cu^+ ion is not stable in aqueous solutions?

4.4 Some Important Compounds of Transition Elements

4.4.1 Oxides and Oxoanions of Metals

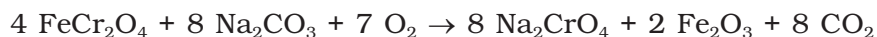
These oxides are generally formed by the reaction of metals with oxygen at high temperatures. All the metals except scandium form MO oxides which are ionic. The highest oxidation number in the oxides, coincides with the group number and is attained in Sc_2O_3 to Mn_2O_7 . Beyond group 7, no higher oxides of iron above Fe_2O_3 are known. Besides the oxides, the oxocations stabilise V^{V} as VO_2^+ , V^{IV} as VO^{2+} and Ti^{IV} as TiO^{2+} .

As the oxidation number of a metal increases, ionic character decreases. In the case of Mn, Mn_2O_7 is a covalent green oil. Even CrO_3 and V_2O_5 have low melting points. In these higher oxides, the acidic character is predominant.

Thus, Mn_2O_7 gives HMnO_4 and CrO_3 gives H_2CrO_4 and $\text{H}_2\text{Cr}_2\text{O}_7$. V_2O_5 is, however, amphoteric though mainly acidic and it gives VO_4^{3-} as well as VO_2^+ salts. In vanadium there is gradual change from the basic V_2O_3 to less basic V_2O_4 and to amphoteric V_2O_5 . V_2O_4 dissolves in acids to give VO^{2+} salts. Similarly, V_2O_5 reacts with alkalis as well as acids to give VO_4^{3-} and VO_2^+ respectively. The well characterised CrO is basic but Cr_2O_3 is amphoteric.

Potassium dichromate $\text{K}_2\text{Cr}_2\text{O}_7$

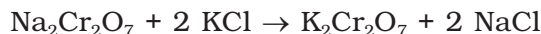
Potassium dichromate is a very important chemical used in leather industry and as an oxidant for preparation of many azo compounds. Dichromates are generally prepared from chromate, which in turn are obtained by the fusion of chromite ore (FeCr_2O_4) with sodium or potassium carbonate in free access of air. The reaction with sodium carbonate occurs as follows:



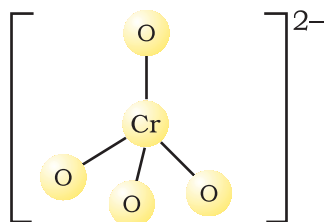
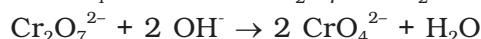
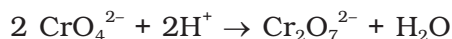
The yellow solution of sodium chromate is filtered and acidified with sulphuric acid to give a solution from which orange sodium dichromate, $\text{Na}_2\text{Cr}_2\text{O}_7 \cdot 2\text{H}_2\text{O}$ can be crystallised.



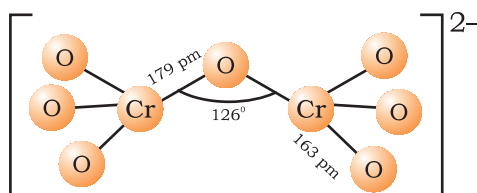
Sodium dichromate is more soluble than potassium dichromate. The latter is therefore, prepared by treating the solution of sodium dichromate with potassium chloride.



Orange crystals of potassium dichromate crystallise out. The chromates and dichromates are interconvertible in aqueous solution depending upon pH of the solution. The oxidation state of chromium in chromate and dichromate is the same.



Chromate ion



Dichromate ion

The structures of chromate ion, CrO_4^{2-} and the dichromate ion, $\text{Cr}_2\text{O}_7^{2-}$ are shown below. The chromate ion is tetrahedral whereas the dichromate ion consists of two tetrahedra sharing one corner with Cr–O–Cr bond angle of 126° .

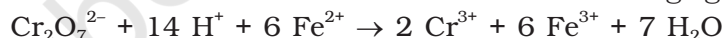
Sodium and potassium dichromates are strong oxidising agents; the sodium salt has a greater solubility in water and is extensively used as an oxidising agent in organic chemistry. Potassium dichromate is used as a primary standard in volumetric analysis. In acidic solution, its oxidising action can be represented as follows:



Thus, acidified potassium dichromate will oxidise iodides to iodine, sulphides to sulphur, tin(II) to tin(IV) and iron(II) salts to iron(III). The half-reactions are noted below:

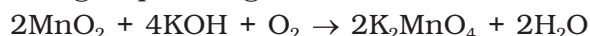


The full ionic equation may be obtained by adding the half-reaction for potassium dichromate to the half-reaction for the reducing agent, for e.g.,

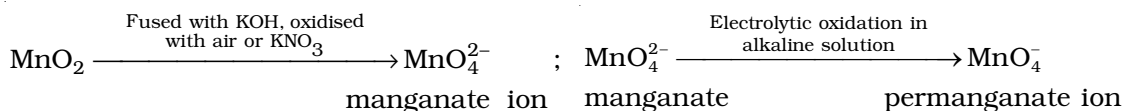


Potassium permanganate KMnO_4

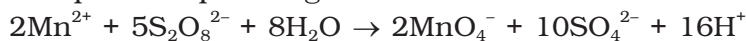
Potassium permanganate is prepared by fusion of MnO_2 with an alkali metal hydroxide and an oxidising agent like KNO_3 . This produces the dark green K_2MnO_4 which disproportionates in a neutral or acidic solution to give permanganate.



Commercially it is prepared by the alkaline oxidative fusion of MnO_2 followed by the electrolytic oxidation of manganate (VI).



In the laboratory, a manganese (II) ion salt is oxidised by peroxodisulphate to permanganate.



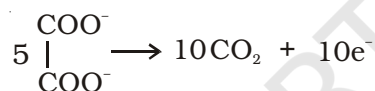
Potassium permanganate forms dark purple (almost black) crystals which are isostructural with those of KClO_4 . The salt is not very soluble in water (6.4 g/100 g of water at 293 K), but when heated it decomposes at 513 K.



It has two physical properties of considerable interest: its intense colour and its diamagnetism along with temperature-dependent weak paramagnetism. These can be explained by the use of molecular orbital theory which is beyond the present scope.

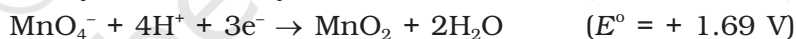
The manganate and permanganate ions are tetrahedral; the π -bonding takes place by overlap of p orbitals of oxygen with d orbitals of manganese. The green manganate is paramagnetic because of one unpaired electron but the permanganate is diamagnetic due to the absence of unpaired electron.

Acidified permanganate solution oxidises oxalates to carbon dioxide, iron(II) to iron(III), nitrites to nitrates and iodides to free iodine. The half-reactions of reductants are:



The full reaction can be written by adding the half-reaction for KMnO_4 to the half-reaction of the reducing agent, balancing wherever necessary.

If we represent the reduction of permanganate to manganate, manganese dioxide and manganese(II) salt by half-reactions,

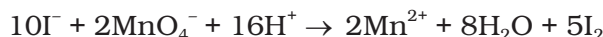


We can very well see that the hydrogen ion concentration of the solution plays an important part in influencing the reaction. Although many reactions can be understood by consideration of redox potential, kinetics of the reaction is also an important factor. Permanganate at $[\text{H}^+] = 1$ should oxidise water but in practice the reaction is extremely slow unless either manganese(II) ions are present or the temperature is raised.

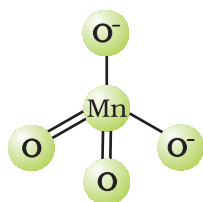
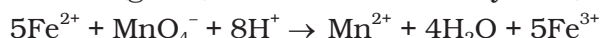
A few important oxidising reactions of KMnO_4 are given below:

1. In acid solutions:

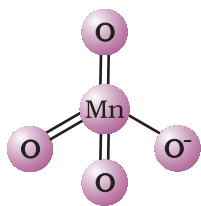
(a) Iodine is liberated from potassium iodide :



(b) Fe^{2+} ion (green) is converted to Fe^{3+} (yellow):



Tetrahedral manganate ion (green)

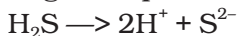


Tetrahedral permanganate ion (purple)

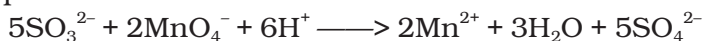
(c) Oxalate ion or oxalic acid is oxidised at 333 K:



(d) Hydrogen sulphide is oxidised, sulphur being precipitated:



(e) Sulphurous acid or sulphite is oxidised to a sulphate or sulphuric acid:

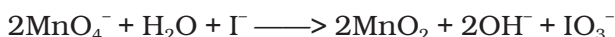


(f) Nitrite is oxidised to nitrate:



2. In neutral or faintly alkaline solutions:

(a) A notable reaction is the oxidation of iodide to iodate:



(b) Thiosulphate is oxidised almost quantitatively to sulphate:



(c) Manganous salt is oxidised to MnO_2 ; the presence of zinc sulphate or zinc oxide catalyses the oxidation:



Note: Permanganate titrations in presence of hydrochloric acid are unsatisfactory since hydrochloric acid is oxidised to chlorine.

Uses: Besides its use in analytical chemistry, potassium permanganate is used as a favourite oxidant in preparative organic chemistry. Its uses for the bleaching of wool, cotton, silk and other textile fibres and for the decolourisation of oils are also dependent on its strong oxidising power.

THE INNER TRANSITION ELEMENTS (*f*-BLOCK)

The *f*-block consists of the two series, lanthanoids (the fourteen elements following lanthanum) and actinoids (the fourteen elements following actinium). Because lanthanum closely resembles the lanthanoids, it is usually included in any discussion of the lanthanoids for which the general symbol Ln is often used. Similarly, a discussion of the actinoids includes actinium besides the fourteen elements constituting the series. The lanthanoids resemble one another more closely than do the members of ordinary transition elements in any series. They have only one stable oxidation state and their chemistry provides an excellent opportunity to examine the effect of small changes in size and nuclear charge along a series of otherwise similar elements. The chemistry of the actinoids is, on the other hand, much more complicated. The complication arises partly owing to the occurrence of a wide range of oxidation states in these elements and partly because their radioactivity creates special problems in their study; the two series will be considered separately here.

4.5 The Lanthanoids

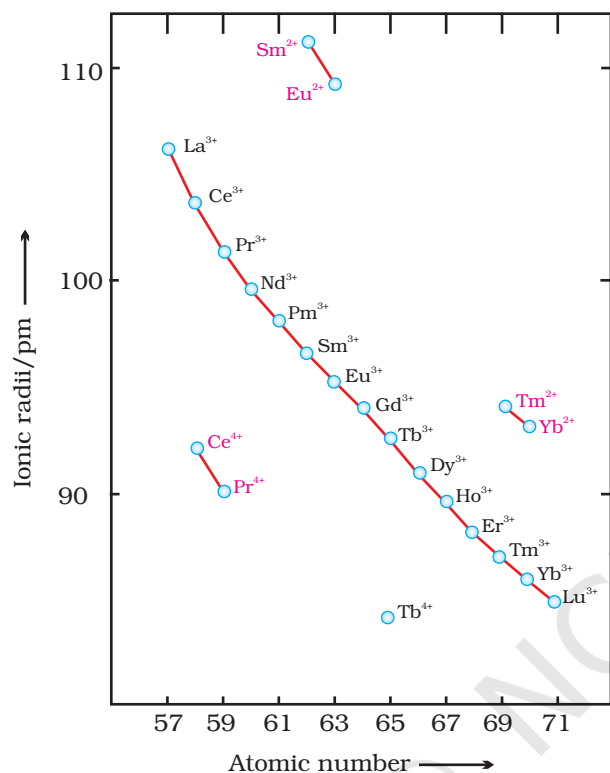
The names, symbols, electronic configurations of atomic and some ionic states and atomic and ionic radii of lanthanum and lanthanoids (for which the general symbol Ln is used) are given in Table 4.9.

4.5.1 Electronic Configurations

It may be noted that atoms of these elements have electronic configuration with $6s^2$ common but with variable occupancy of $4f$ level (Table 4.9). However, the electronic configurations of all the tripositive ions (the most stable oxidation state of all the lanthanoids) are of the form $4f^n$ ($n = 1$ to 14 with increasing atomic number).

4.5.2 Atomic and Ionic Sizes

The overall decrease in atomic and ionic radii from lanthanum to lutetium (the **lanthanoid contraction**) is a unique feature in the chemistry of the lanthanoids. It has far reaching



consequences in the chemistry of the third transition series of the elements. The decrease in atomic radii (derived from the structures of metals) is not quite regular as it is regular in M^{3+} ions (Fig. 4.6). This contraction is, of course, similar to that observed in an ordinary transition series and is attributed to the same cause, the imperfect shielding of one electron by another in the same sub-shell. However, the shielding of one $4f$ electron by another is less than one d electron by another with the increase in nuclear charge along the series. There is fairly regular decrease in the sizes with increasing atomic number.

The cumulative effect of the contraction of the lanthanoid series, known as *lanthanoid contraction*, causes the radii of the members of the third transition series to be very similar to those of the corresponding members of the second series. The almost identical radii of Zr (160 pm) and Hf (159 pm), a consequence of the lanthanoid contraction, account for their occurrence together in nature and for the difficulty faced in their separation.

Fig. 4.6: Trends in ionic radii of lanthanoids

4.5.3 Oxidation States

In the lanthanoids, La(II) and Ln(III) compounds are predominant species. However, occasionally +2 and +4 ions in solution or in solid compounds are also obtained. This irregularity (as in ionisation enthalpies) arises mainly from the extra stability of empty, half-filled or filled f subshell. Thus, the formation of Ce^{IV} is favoured by its noble gas configuration, but it is a strong oxidant reverting to the common +3 state. The E° value for Ce^{4+}/Ce^{3+} is + 1.74 V which suggests that it can oxidise water. However, the reaction rate is very slow and hence Ce(IV) is a good analytical reagent. Pr, Nd, Tb and Dy also exhibit +4 state but only in oxides, MO_2 . Eu^{2+} is formed by losing the two s electrons and its f^7 configuration accounts for the formation of this ion. However, Eu^{2+} is a strong reducing agent changing to the common +3 state. Similarly Yb^{2+} which has f^{14} configuration is a reductant. Tb^{IV} has half-filled f -orbitals and is an oxidant. The behaviour of samarium is very much like europium, exhibiting both +2 and +3 oxidation states.

Table 4.9: Electronic Configurations and Radii of Lanthanum and Lanthanoids

Atomic Number	Name	Symbol	Electronic configurations*			Radii/pm		
			Ln	Ln ²⁺	Ln ³⁺	Ln ⁴⁺	Ln	Ln ³⁺
57	Lanthanum	La	5d ¹ 6s ²	5d ¹	4f ⁰		187	106
58	Cerium	Ce	4f ¹ 5d ¹ 6s ²	4f ²	4f ¹	4f ⁰	183	103
59	Praseodymium	Pr	4f ³ 6s ²	4f ³	4f ²	4f ¹	182	101
60	Neodymium	Nd	4f ⁴ 6s ²	4f ⁴	4f ³	4f ²	181	99
61	Promethium	Pm	4f ⁵ 6s ²	4f ⁵	4f ⁴		181	98
62	Samarium	Sm	4f ⁶ 6s ²	4f ⁶	4f ⁵		180	96
63	Europium	Eu	4f ⁷ 6s ²	4f ⁷	4f ⁶		199	95
64	Gadolinium	Gd	4f ⁷ 5d ¹ 6s ²	4f ⁷ 5d ¹	4f ⁷		180	94
65	Terbium	Tb	4f ⁹ 6s ²	4f ⁹	4f ⁸	4f ⁷	178	92
66	Dysprosium	Dy	4f ¹⁰ 6s ²	4f ¹⁰	4f ⁹	4f ⁸	177	91
67	Holmium	Ho	4f ¹¹ 6s ²	4f ¹¹	4f ¹⁰		176	89
68	Erbium	Er	4f ¹² 6s ²	4f ¹²	4f ¹¹		175	88
69	Thulium	Tm	4f ¹³ 6s ²	4f ¹³	4f ¹²		174	87
70	Ytterbium	Yb	4f ¹⁴ 6s ²	4f ¹⁴	4f ¹³		173	86
71	Lutetium	Lu	4f ¹⁴ 5d ¹ 6s ²	4f ¹⁴ 5d ¹	4f ¹⁴	–	–	–

* Only electrons outside [Xe] core are indicated

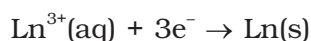
4.5.4 General Characteristics

All the lanthanoids are silvery white soft metals and tarnish rapidly in air. The hardness increases with increasing atomic number, samarium being steel hard. Their melting points range between 1000 to 1200 K but samarium melts at 1623 K. They have typical metallic structure and are good conductors of heat and electricity. Density and other properties change smoothly except for Eu and Yb and occasionally for Sm and Tm.

Many trivalent lanthanoid ions are coloured both in the solid state and in aqueous solutions. Colour of these ions may be attributed to the presence of *f* electrons. Neither La³⁺ nor Lu³⁺ ion shows any colour but the rest do so. However, absorption bands are narrow, probably because of the excitation within *f* level. The lanthanoid ions other than the *f*⁰ type (La³⁺ and Ce⁴⁺) and the *f*¹⁴ type (Yb²⁺ and Lu³⁺) are all paramagnetic.

The first ionisation enthalpies of the lanthanoids are around 600 kJ mol⁻¹, the second about 1200 kJ mol⁻¹ comparable with those of calcium. A detailed discussion of the variation of the third ionisation enthalpies indicates that the exchange enthalpy considerations (as in 3*d* orbitals of the first transition series), appear to impart a certain degree of stability to empty, half-filled and completely filled orbitals *f* level. This is indicated from the abnormally low value of the third ionisation enthalpy of lanthanum, gadolinium and lutetium.

In their chemical behaviour, in general, the earlier members of the series are quite reactive similar to calcium but, with increasing atomic number, they behave more like aluminium. Values for *E*^o for the half-reaction:



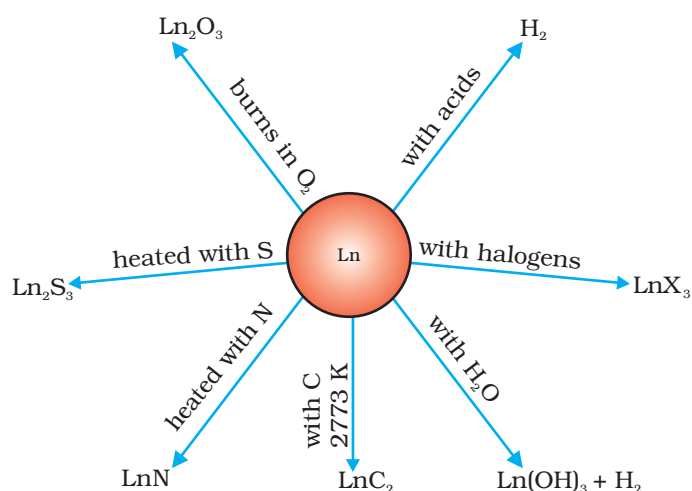


Fig 4.7: Chemical reactions of the lanthanoids.

are in the range of -2.2 to -2.4 V except for Eu for which the value is -2.0 V. This is, of course, a small variation. The metals combine with hydrogen when gently heated in the gas. The carbides, Ln_3C , Ln_2C_3 and LnC_2 are formed when the metals are heated with carbon. They liberate hydrogen from dilute acids and burn in halogens to form halides. They form oxides M_2O_3 and hydroxides $\text{M}(\text{OH})_3$. The hydroxides are definite compounds, not just hydrated oxides. They are basic like alkaline earth metal oxides and hydroxides. Their general reactions are depicted in Fig. 4.7.

The best single use of the lanthanoids is for the production of alloy steels for plates and pipes. A well known alloy is *mischmetall* which consists of a lanthanoid metal ($\sim 95\%$) and iron ($\sim 5\%$) and traces of S, C, Ca and Al. A good deal of **mischmetall** is used in Mg-based alloy to produce bullets, shell and lighter flint. Mixed oxides of lanthanoids are employed as catalysts in petroleum cracking. Some individual Ln oxides are used as phosphors in television screens and similar fluorescing surfaces.

4.6 The Actinoids

The actinoids include the fourteen elements from Th to Lr. The names, symbols and some properties of these elements are given in Table 4.10.

Table 4.10: Some Properties of Actinium and Actinoids

Atomic Number	Name	Symbol	Electronic configurations*			Radii/pm	
			M	M^{3+}	M^{4+}	M^{3+}	M^{4+}
89	Actinium	Ac	$6d^1 7s^2$	$5f^0$		111	
90	Thorium	Th	$6d^2 7s^2$	$5f^1$	$5f^0$		99
91	Protactinium	Pa	$5f^2 6d^1 7s^2$	$5f^2$	$5f^1$		96
92	Uranium	U	$5f^3 6d^1 7s^2$	$5f^3$	$5f^2$	103	93
93	Neptunium	Np	$5f^4 6d^1 7s^2$	$5f^4$	$5f^3$	101	92
94	Plutonium	Pu	$5f^6 7s^2$	$5f^5$	$5f^4$	100	90
95	Americium	Am	$5f^7 7s^2$	$5f^6$	$5f^5$	99	89
96	Curium	Cm	$5f^7 6d^1 7s^2$	$5f^7$	$5f^6$	99	88
97	Berkelium	Bk	$5f^9 7s^2$	$5f^8$	$5f^7$	98	87
98	Californium	Cf	$5f^{10} 7s^2$	$5f^9$	$5f^8$	98	86
99	Einsteinium	Es	$5f^{11} 7s^2$	$5f^{10}$	$5f^9$	–	–
100	Fermium	Fm	$5f^{12} 7s^2$	$5f^{11}$	$5f^{10}$	–	–
101	Mendelevium	Md	$5f^{13} 7s^2$	$5f^{12}$	$5f^{11}$	–	–
102	Nobelium	No	$5f^{14} 7s^2$	$5f^{13}$	$5f^{12}$	–	–
103	Lawrencium	Lr	$5f^{14} 6d^1 7s^2$	$5f^{14}$	$5f^{13}$	–	–

The actinoids are radioactive elements and the earlier members have relatively long half-lives, the latter ones have half-life values ranging from a day to 3 minutes for lawrencium ($Z=103$). The latter members could be prepared only in nanogram quantities. These facts render their study more difficult.

4.6.1 Electronic Configurations

All the actinoids are believed to have the electronic configuration of $7s^2$ and variable occupancy of the $5f$ and $6d$ subshells. The fourteen electrons are formally added to $5f$, though not in thorium ($Z = 90$) but from Pa onwards the $5f$ orbitals are complete at element 103. The irregularities in the electronic configurations of the actinoids, like those in the lanthanoids are related to the stabilities of the f^0 , f^7 and f^{14} occupancies of the $5f$ orbitals. Thus, the configurations of Am and Cm are $[Rn] 5f^7 7s^2$ and $[Rn] 5f^7 6d^1 7s^2$. Although the $5f$ orbitals resemble the $4f$ orbitals in their angular part of the wave-function, they are not as buried as $4f$ orbitals and hence $5f$ electrons can participate in bonding to a far greater extent.

4.6.2 Ionic Sizes

The general trend in lanthanoids is observable in the actinoids as well. There is a gradual decrease in the size of atoms or M^{3+} ions across the series. This may be referred to as the **actinoid contraction** (like lanthanoid contraction). The contraction is, however, greater from element to element in this series resulting from poor shielding by $5f$ electrons.

4.6.3 Oxidation States

There is a greater range of oxidation states, which is in part attributed to the fact that the $5f$, $6d$ and $7s$ levels are of comparable energies. The known oxidation states of actinoids are listed in Table 4.11.

The actinoids show in general +3 oxidation state. The elements, in the first half of the series frequently exhibit higher oxidation states. For example, the maximum oxidation state increases from +4 in Th to +5, +6 and +7 respectively in Pa, U and Np but decreases in succeeding elements (Table 4.11). The actinoids resemble the lanthanoids in having more compounds in +3 state than in the +4 state. However, +3 and +4 ions tend to hydrolyse. Because the distribution of oxidation states among the actinoids is so uneven and so different for the former and later elements, it is unsatisfactory to review their chemistry in terms of oxidation states.

Table 4.11: Oxidation States of Actinium and Actinoids

Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
3		3	3	3	3	3	3	3	3	3	3	3	3	3
	4	4	4	4	4	4	4	4						
		5	5	5	5	5								
			6	6	6	6								
				7	7									

4.6.4 General Characteristics and Comparison with Lanthanoids

The actinoid metals are all silvery in appearance but display a variety of structures. The structural variability is obtained due to irregularities in metallic radii which are far greater than in lanthanoids.

The actinoids are highly reactive metals, especially when finely divided. The action of boiling water on them, for example, gives a mixture of oxide and hydride and combination with most non metals takes place at moderate temperatures. Hydrochloric acid attacks all metals but most are slightly affected by nitric acid owing to the formation of protective oxide layers; alkalies have no action.

The magnetic properties of the actinoids are more complex than those of the lanthanoids. Although the variation in the magnetic susceptibility of the actinoids with the number of unpaired $5f$ electrons is roughly parallel to the corresponding results for the lanthanoids, the latter have higher values.

It is evident from the behaviour of the actinoids that the ionisation enthalpies of the early actinoids, though not accurately known, but are lower than for the early lanthanoids. This is quite reasonable since it is to be expected that when $5f$ orbitals are beginning to be occupied, they will penetrate less into the inner core of electrons. The $5f$ electrons, will therefore, be more effectively shielded from the nuclear charge than the $4f$ electrons of the corresponding lanthanoids. Because the outer electrons are less firmly held, they are available for bonding in the actinoids.

A comparison of the actinoids with the lanthanoids, with respect to different characteristics as discussed above, reveals that behaviour similar to that of the lanthanoids is not evident until the second half of the actinoid series. However, even the early actinoids resemble the lanthanoids in showing close similarities with each other and in gradual variation in properties which do not entail change in oxidation state. The lanthanoid and actinoid contractions, have extended effects on the sizes, and therefore, the properties of the elements succeeding them in their respective periods. The lanthanoid contraction is more important because the chemistry of elements succeeding the actinoids are much less known at the present time.

Example 4.10 Name a member of the lanthanoid series which is well known to exhibit +4 oxidation state.

Solution Cerium ($Z = 58$)

Intext Question

4.10 Actinoid contraction is greater from element to element than lanthanoid contraction. Why?

4.7 Some Applications of d - and f -Block Elements

Iron and steels are the most important construction materials. Their production is based on the reduction of iron oxides, the removal of impurities and the addition of carbon and alloying metals such as Cr, Mn and Ni. Some compounds are manufactured for special purposes such as TiO for the pigment industry and MnO_2 for use in dry battery cells. The battery industry also requires Zn and Ni/Cd. The elements of Group 11 are still worthy of being called the coinage metals, although Ag and Au

are restricted to collection items and the contemporary UK 'copper' coins are copper-coated steel. The 'silver' UK coins are a Cu/Ni alloy. Many of the metals and/or their compounds are essential catalysts in the chemical industry. V_2O_5 catalyses the oxidation of SO_2 in the manufacture of sulphuric acid. $TiCl_4$ with $Al(CH_3)_3$ forms the basis of the Ziegler catalysts used to manufacture polyethylene (polythene). Iron catalysts are used in the Haber process for the production of ammonia from N_2/H_2 mixtures. Nickel catalysts enable the hydrogenation of fats to proceed. In the Wacker process the oxidation of ethyne to ethanal is catalysed by $PdCl_2$. Nickel complexes are useful in the polymerisation of alkynes and other organic compounds such as benzene. The photographic industry relies on the special light-sensitive properties of AgBr.

Summary

The **d-block** consisting of **Groups 3-12** occupies the large middle section of the **periodic table**. In these elements the inner *d* orbitals are progressively filled. The **f-block** is placed **outside** at the **bottom** of the **periodic table** and in the elements of this block, 4*f* and 5*f* orbitals are progressively filled.

Corresponding to the filling of 3*d*, 4*d* and 5*d* orbitals, three series of transition elements are well recognised. All the transition elements exhibit typical metallic properties such as –high tensile strength, ductility, malleability, thermal and electrical conductivity and metallic character. Their melting and boiling points are high which are attributed to the involvement of (*n*–1) *d* electrons resulting into **strong interatomic bonding**. In many of these properties, the maxima occur at about the middle of each series which indicates that one unpaired electron per *d* orbital is particularly a favourable configuration for strong interatomic interaction.

Successive ionisation enthalpies do not increase as steeply as in the main group elements with increasing atomic number. Hence, the loss of variable number of electrons from (*n*–1) *d* orbitals is not energetically unfavourable. The involvement of (*n*–1) ***d* electrons** in the behaviour of transition elements impart certain distinct characteristics to these elements. Thus, in addition to variable oxidation states, they exhibit paramagnetic behaviour, catalytic properties and tendency for the formation of coloured ions, interstitial compounds and complexes.

The **transition elements** vary widely in their chemical behaviour. Many of them are sufficiently electropositive to dissolve in mineral acids, although a few are 'noble'. Of the first series, with the exception of copper, all the metals are relatively reactive.

The transition metals react with a number of non-metals like oxygen, nitrogen, sulphur and halogens to form binary compounds. The first series transition metal oxides are generally formed from the reaction of metals with oxygen at high temperatures. These oxides dissolve in acids and bases to form oxometallic salts. Potassium dichromate and potassium permanganate are common examples. Potassium dichromate is prepared from the chromite ore by fusion with alkali in presence of air and acidifying the extract. Pyrolusite ore (MnO_2) is used for the preparation of potassium permanganate. Both the dichromate and the permanganate ions are strong oxidising agents.

The two series of **inner transition elements**, **lanthanoids** and **actinoids** constitute the **f-block** of the periodic table. With the successive filling of the inner orbitals, 4*f*, there is a gradual decrease in the atomic and ionic sizes of these metals along the series (**lanthanoid contraction**). This has far reaching consequences in the chemistry of the elements succeeding them. Lanthanum and all the lanthanoids are rather soft white metals. They react easily with water to give solutions giving +3 ions. The principal oxidation state is +3, although +4 and +2 oxidation states are also exhibited by some

occasionally. The chemistry of the **actinoids** is more complex in view of their ability to exist in different oxidation states. Furthermore, many of the actinoid elements are radioactive which make the study of these elements rather difficult.

There are many useful applications of the *d*- and *f*-block elements and their compounds, notable among them being in varieties of steels, catalysts, complexes, organic syntheses, etc.

Exercises

- 4.1 Write down the electronic configuration of:
- | | | | |
|-----------------------|-----------------------|-----------------------|-------------------------|
| (i) Cr^{3+} | (iii) Cu^+ | (v) Co^{2+} | (vii) Mn^{2+} |
| (ii) Pm^{3+} | (iv) Ce^{4+} | (vi) Lu^{2+} | (viii) Th^{4+} |
- 4.2 Why are Mn^{2+} compounds more stable than Fe^{2+} towards oxidation to their +3 state?
- 4.3 Explain briefly how +2 state becomes more and more stable in the first half of the first row transition elements with increasing atomic number?
- 4.4 To what extent do the electronic configurations decide the stability of oxidation states in the first series of the transition elements? Illustrate your answer with examples.
- 4.5 What may be the stable oxidation state of the transition element with the following *d* electron configurations in the ground state of their atoms : $3d^3$, $3d^5$, $3d^8$ and $3d^4$?
- 4.6 Name the oxometal anions of the first series of the transition metals in which the metal exhibits the oxidation state equal to its group number.
- 4.7 What is lanthanoid contraction? What are the consequences of lanthanoid contraction?
- 4.8 What are the characteristics of the transition elements and why are they called transition elements? Which of the *d*-block elements may not be regarded as the transition elements?
- 4.9 In what way is the electronic configuration of the transition elements different from that of the non transition elements?
- 4.10 What are the different oxidation states exhibited by the lanthanoids?
- 4.11 Explain giving reasons:
- (i) Transition metals and many of their compounds show paramagnetic behaviour.
 - (ii) The enthalpies of atomisation of the transition metals are high.
 - (iii) The transition metals generally form coloured compounds.
 - (iv) Transition metals and their many compounds act as good catalyst.
- 4.12 What are interstitial compounds? Why are such compounds well known for transition metals?
- 4.13 How is the variability in oxidation states of transition metals different from that of the non transition metals? Illustrate with examples.
- 4.14 Describe the preparation of potassium dichromate from iron chromite ore. What is the effect of increasing pH on a solution of potassium dichromate?
- 4.15 Describe the oxidising action of potassium dichromate and write the ionic equations for its reaction with:
- | | | |
|------------|----------------------------|----------------------------|
| (i) iodide | (ii) iron(II) solution and | (iii) H_2S |
|------------|----------------------------|----------------------------|

- 4.16** Describe the preparation of potassium permanganate. How does the acidified permanganate solution react with (i) iron(II) ions (ii) SO_2 and (iii) oxalic acid? Write the ionic equations for the reactions.
- 4.17** For M^{2+}/M and $\text{M}^{3+}/\text{M}^{2+}$ systems the E° values for some metals are as follows:
- | | | | |
|----------------------------|-------|---------------------------------|--------|
| Cr^{2+}/Cr | -0.9V | $\text{Cr}^{3+}/\text{Cr}^{2+}$ | -0.4 V |
| Mn^{2+}/Mn | -1.2V | $\text{Mn}^{3+}/\text{Mn}^{2+}$ | +1.5 V |
| Fe^{2+}/Fe | -0.4V | $\text{Fe}^{3+}/\text{Fe}^{2+}$ | +0.8 V |
- Use this data to comment upon:
- the stability of Fe^{3+} in acid solution as compared to that of Cr^{3+} or Mn^{3+} and
 - the ease with which iron can be oxidised as compared to a similar process for either chromium or manganese metal.
- 4.18** Predict which of the following will be coloured in aqueous solution? Ti^{3+} , V^{3+} , Cu^+ , Sc^{3+} , Mn^{2+} , Fe^{3+} and Co^{2+} . Give reasons for each.
- 4.19** Compare the stability of +2 oxidation state for the elements of the first transition series.
- 4.20** Compare the chemistry of actinoids with that of the lanthanoids with special reference to:
- electronic configuration
 - atomic and ionic sizes and
 - oxidation state
 - chemical reactivity.
- 4.21** How would you account for the following:
- Of the d^4 species, Cr^{2+} is strongly reducing while manganese(III) is strongly oxidising.
 - Cobalt(II) is stable in aqueous solution but in the presence of complexing reagents it is easily oxidised.
 - The d^1 configuration is very unstable in ions.
- 4.22** What is meant by 'disproportionation'? Give two examples of disproportionation reaction in aqueous solution.
- 4.23** Which metal in the first series of transition metals exhibits +1 oxidation state most frequently and why?
- 4.24** Calculate the number of unpaired electrons in the following gaseous ions: Mn^{3+} , Cr^{3+} , V^{3+} and Ti^{3+} . Which one of these is the most stable in aqueous solution?
- 4.25** Give examples and suggest reasons for the following features of the transition metal chemistry:
- The lowest oxide of transition metal is basic, the highest is amphoteric/acidic.
 - A transition metal exhibits highest oxidation state in oxides and fluorides.
 - The highest oxidation state is exhibited in oxoanions of a metal.
- 4.26** Indicate the steps in the preparation of:
- $\text{K}_2\text{Cr}_2\text{O}_7$ from chromite ore.
 - KMnO_4 from pyrolusite ore.
- 4.27** What are alloys? Name an important alloy which contains some of the lanthanoid metals. Mention its uses.
- 4.28** What are inner transition elements? Decide which of the following atomic numbers are the atomic numbers of the inner transition elements : 29, 59, 74, 95, 102, 104.
- 4.29** The chemistry of the actinoid elements is not so smooth as that of the lanthanoids. Justify this statement by giving some examples from the oxidation state of these elements.
- 4.30** Which is the last element in the series of the actinoids? Write the electronic configuration of this element. Comment on the possible oxidation state of this element.

- 4.31** Use Hund's rule to derive the electronic configuration of Ce^{3+} ion, and calculate its magnetic moment on the basis of 'spin-only' formula.
- 4.32** Name the members of the lanthanoid series which exhibit +4 oxidation states and those which exhibit +2 oxidation states. Try to correlate this type of behaviour with the electronic configurations of these elements.
- 4.33** Compare the chemistry of the actinoids with that of lanthanoids with reference to: (i) electronic configuration (ii) oxidation states and (iii) chemical reactivity.
- 4.34** Write the electronic configurations of the elements with the atomic numbers 61, 91, 101, and 109.
- 4.35** Compare the general characteristics of the first series of the transition metals with those of the second and third series metals in the respective vertical columns. Give special emphasis on the following points:
(i) electronic configurations (ii) oxidation states (iii) ionisation enthalpies and (iv) atomic sizes.
- 4.36** Write down the number of 3d electrons in each of the following ions: Ti^{2+} , V^{2+} , Cr^{3+} , Mn^{2+} , Fe^{2+} , Fe^{3+} , Co^{2+} , Ni^{2+} and Cu^{2+} . Indicate how would you expect the five 3d orbitals to be occupied for these hydrated ions (octahedral).
- 4.37** Comment on the statement that elements of the first transition series possess many properties different from those of heavier transition elements.
- 4.38** What can be inferred from the magnetic moment values of the following complex species ?

Example	Magnetic Moment (BM)
$\text{K}_4[\text{Mn}(\text{CN})_6]$	2.2
$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	5.3
$\text{K}_2[\text{MnCl}_4]$	5.9

Answers to Some Intext Questions

- 4.1** Silver ($Z = 47$) can exhibit +2 oxidation state wherein it will have incompletely filled d -orbitals ($4d$), hence a transition element.
- 4.2** In the formation of metallic bonds, no electrons from 3d-orbitals are involved in case of zinc, while in all other metals of the 3d series, electrons from the d -orbitals are always involved in the formation of metallic bonds.
- 4.3** Manganese ($Z = 25$), as its atom has the maximum number of unpaired electrons.
- 4.5** Irregular variation of ionisation enthalpies is mainly attributed to varying degree of stability of different 3d-configurations (e.g., d^0 , d^5 , d^{10} are exceptionally stable).
- 4.6** Because of small size and high electronegativity oxygen or fluorine can oxidise the metal to its highest oxidation state.
- 4.7** Cr^{2+} is stronger reducing agent than Fe^{2+}
Reason: $d^4 \rightarrow d^3$ occurs in case of Cr^{2+} to Cr^{3+}
But $d^6 \rightarrow d^5$ occurs in case of Fe^{2+} to Fe^{3+}
In a medium (like water) d^3 is more stable as compared to d^5 (see CFSE)
- 4.9** Cu^+ in aqueous solution undergoes disproportionation, i.e.,
 $2\text{Cu}^+(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + \text{Cu}(\text{s})$
The E^0 value for this is favourable.
- 4.10** The 5f electrons are more effectively shielded from nuclear charge. In other words the 5f electrons themselves provide poor shielding from element to element in the series.



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Objectives

After studying this Unit, you will be able to

- appreciate the postulates of Werner's theory of coordination compounds;
- know the meaning of the terms: coordination entity, central atom/ion, ligand, coordination number, coordination sphere, coordination polyhedron, oxidation number, homoleptic and heteroleptic;
- learn the rules of nomenclature of coordination compounds;
- write the formulas and names of mononuclear coordination compounds;
- define different types of isomerism in coordination compounds;
- understand the nature of bonding in coordination compounds in terms of the Valence Bond and Crystal Field theories;
- appreciate the importance and applications of coordination compounds in our day to day life.

5.1 Werner's Theory of Coordination Compounds

Alfred Werner (1866-1919), a Swiss chemist was the first to formulate his ideas about the structures of coordination compounds. He prepared and characterised a large number of coordination compounds and studied their physical and chemical behaviour by simple experimental techniques. Werner proposed the concept of a **primary valence** and a **secondary valence** for a metal ion. Binary compounds such as CrCl_3 , CoCl_2 or PdCl_2 have primary valence of 3, 2 and 2 respectively. In a series of compounds of cobalt(III) chloride with ammonia, it was found that some of the chloride ions could be precipitated as AgCl on adding excess silver nitrate solution in cold but some remained in solution.

Unit

5

Coordination Compounds

Coordination Compounds are the backbone of modern inorganic and bio-inorganic chemistry and chemical industry.

In the previous Unit we learnt that the transition metals form a large number of **complex compounds** in which the metal atoms are bound to a number of anions or neutral molecules by sharing of electrons. In modern terminology such compounds are called **coordination compounds**. The chemistry of coordination compounds is an important and challenging area of modern inorganic chemistry. New concepts of chemical bonding and molecular structure have provided insights into the functioning of these compounds as vital components of biological systems. Chlorophyll, haemoglobin and vitamin B_{12} are coordination compounds of magnesium, iron and cobalt respectively. Variety of metallurgical processes, industrial catalysts and analytical reagents involve the use of coordination compounds. Coordination compounds also find many applications in electroplating, textile dyeing and medicinal chemistry.

1 mol	$\text{CoCl}_3 \cdot 6\text{NH}_3$ (Yellow)	gave	3 mol AgCl
1 mol	$\text{CoCl}_3 \cdot 5\text{NH}_3$ (Purple)	gave	2 mol AgCl
1 mol	$\text{CoCl}_3 \cdot 4\text{NH}_3$ (Green)	gave	1 mol AgCl
1 mol	$\text{CoCl}_3 \cdot 4\text{NH}_3$ (Violet)	gave	1 mol AgCl

These observations, together with the results of conductivity measurements in solution can be explained if (i) six groups in all, either chloride ions or ammonia molecules or both, remain bonded to the cobalt ion during the reaction and (ii) the compounds are formulated as shown in Table 5.1, where the atoms within the square brackets form a single entity which does not dissociate under the reaction conditions. Werner proposed the term **secondary valence** for the number of groups bound directly to the metal ion; in each of these examples the secondary valences are six.

Table 5.1: Formulation of Cobalt(III) Chloride-Ammonia Complexes

Colour	Formula	Solution conductivity corresponds to
Yellow	$[\text{Co}(\text{NH}_3)_6]^{3+} 3\text{Cl}^-$	1:3 electrolyte
Purple	$[\text{CoCl}(\text{NH}_3)_5]^{2+} 2\text{Cl}^-$	1:2 electrolyte
Green	$[\text{CoCl}_2(\text{NH}_3)_4]^+ \text{Cl}^-$	1:1 electrolyte
Violet	$[\text{CoCl}_2(\text{NH}_3)_4]^+ \text{Cl}^-$	1:1 electrolyte

Note that the last two compounds in Table 5.1 have identical empirical formula, $\text{CoCl}_3 \cdot 4\text{NH}_3$, but distinct properties. Such compounds are termed as isomers. Werner in 1898, propounded his theory of coordination compounds. The main postulates are:

1. In coordination compounds metals show two types of linkages (valences)-primary and secondary.
2. The primary valences are normally ionisable and are satisfied by negative ions.
3. The secondary valences are non ionisable. These are satisfied by neutral molecules or negative ions. The secondary valence is equal to the coordination number and is fixed for a metal.
4. The ions/groups bound by the secondary linkages to the metal have characteristic spatial arrangements corresponding to different coordination numbers.

In modern formulations, such spatial arrangements are called coordination *polyhedra*. The species within the square bracket are coordination entities or complexes and the ions outside the square bracket are called counter ions.

He further postulated that octahedral, tetrahedral and square planar geometrical shapes are more common in coordination compounds of transition metals. Thus, $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{CoCl}(\text{NH}_3)_5]^{2+}$ and $[\text{CoCl}_2(\text{NH}_3)_4]^+$ are octahedral entities, while $[\text{Ni}(\text{CO})_4]$ and $[\text{PtCl}_4]^{2-}$ are tetrahedral and square planar, respectively.

On the basis of the following observations made with aqueous solutions, [Example 5.1](#) assign secondary valences to metals in the following compounds:

Formula	Moles of AgCl precipitated per mole of the compounds with excess AgNO_3
(i) $\text{PdCl}_2 \cdot 4\text{NH}_3$	2
(ii) $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$	2
(iii) $\text{PtCl}_4 \cdot 2\text{HCl}$	0
(iv) $\text{CoCl}_3 \cdot 4\text{NH}_3$	1
(v) $\text{PtCl}_2 \cdot 2\text{NH}_3$	0

(i) Secondary 4
(iii) Secondary 6

(ii) Secondary 6
(iv) Secondary 6

(v) Secondary 4

[Solution](#)

Difference between a double salt and a complex

Both double salts as well as complexes are formed by the combination of two or more stable compounds in stoichiometric ratio. However, they differ in the fact that double salts such as carnallite, $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$, Mohr's salt, $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$, potash alum, $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$, etc. dissociate into simple ions completely when dissolved in water. However, complex ions such as $[\text{Fe}(\text{CN})_6]^{4-}$ of $\text{K}_4[\text{Fe}(\text{CN})_6]$ do not dissociate into Fe^{2+} and CN^- ions.



(1866-1919)

Werner was born on December 12, 1866, in Mülhouse, a small community in the French province of Alsace. His study of chemistry began in Karlsruhe (Germany) and continued in Zurich (Switzerland), where in his doctoral thesis in 1890, he explained the difference in properties of certain nitrogen containing organic substances on the basis of isomerism. He extended van Hoff's theory of tetrahedral carbon atom and modified it for nitrogen. Werner showed optical and electrical differences between complex compounds based on physical measurements. In fact, Werner was the first to discover optical activity in certain coordination compounds.

He, at the age of 29 years became a full professor at Technische Hochschule in Zurich in 1895. Alfred Werner was a chemist and educationist. His accomplishments included the development of the theory of coordination compounds. This theory, in which Werner proposed revolutionary ideas about how atoms and molecules are linked together, was formulated in a span of only three years, from 1890 to 1893. The remainder of his career was spent gathering the experimental support required to validate his new ideas. Werner became the first Swiss chemist to win the Nobel Prize in 1913 for his work on the linkage of atoms and the coordination theory.

5.2 Definitions of (a) Coordination entity

Some Important Terms Pertaining to Coordination Compounds

A coordination entity constitutes a central metal atom or ion bonded to a fixed number of ions or molecules. For example, $[\text{CoCl}_3(\text{NH}_3)_3]$ is a coordination entity in which the cobalt ion is surrounded by three ammonia molecules and three chloride ions. Other examples are $[\text{Ni}(\text{CO})_4]$, $[\text{PtCl}_2(\text{NH}_3)_2]$, $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$.

(b) Central atom/ion

In a coordination entity, the atom/ion to which a fixed number of ions/groups are bound in a definite geometrical arrangement around it, is called the central atom or ion. For example, the central atom/ion in the coordination entities: $[\text{NiCl}_2(\text{H}_2\text{O})_4]$, $[\text{CoCl}(\text{NH}_3)_5]^{2+}$ and $[\text{Fe}(\text{CN})_6]^{3-}$ are Ni^{2+} , Co^{3+} and Fe^{3+} , respectively. These central atoms/ions are also referred to as **Lewis acids**.

(c) Ligands

The ions or molecules bound to the central atom/ion in the coordination entity are called ligands. These may be simple ions such as Cl^- , small molecules such as H_2O or NH_3 , larger molecules such as $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$ or $\text{N}(\text{CH}_2\text{CH}_2\text{NH}_2)_3$ or even macromolecules, such as proteins.

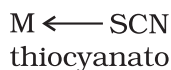
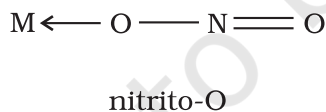
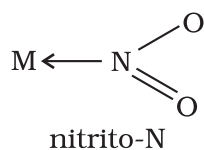
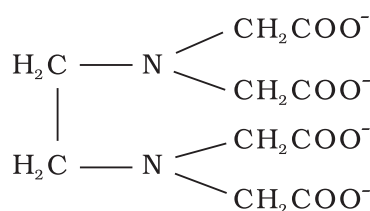
When a ligand is bound to a metal ion through a single donor atom, as with Cl^- , H_2O or NH_3 , the ligand is said to be **unidentate**.

When a ligand can bind through two donor atoms as in $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$ (ethane-1,2-diamine) or $\text{C}_2\text{O}_4^{2-}$ (oxalate), the ligand is said to be **didentate** and when several donor atoms are present in a single ligand as in $\text{N}(\text{CH}_2\text{CH}_2\text{NH}_2)_3$, the ligand is said to be **polydentate**. Ethylenediaminetetraacetate ion (EDTA^{4-}) is an important hexadentate ligand. It can bind through two nitrogen and four oxygen atoms to a central metal ion.

When a di- or polydentate ligand uses its two or more donor atoms simultaneously to bind a single metal ion, it is said to be a **chelate** ligand. The number of such ligating groups is called the **denticity** of the ligand. Such complexes, called chelate complexes tend to be more stable than similar complexes containing unidentate ligands. Ligand which has two different donor atoms and either of

the two ligates in the complex is called **ambidentate ligand**. Examples of such ligands are the NO_2^- and SCN^- ions. NO_2^- ion can coordinate either through nitrogen or through oxygen to a central metal atom/ion.

Similarly, SCN^- ion can coordinate through the sulphur or nitrogen atom.



(d) Coordination number

The coordination number (CN) of a metal ion in a complex can be defined as the number of ligand donor atoms to which the metal is directly bonded. For example, in the complex ions, $[\text{PtCl}_6]^{2-}$ and $[\text{Ni}(\text{NH}_3)_4]^{2+}$, the coordination number of Pt and Ni are 6 and 4 respectively. Similarly, in the complex ions, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ and $[\text{Co}(\text{en})_3]^{3+}$, the coordination number of both, Fe and Co, is 6 because $\text{C}_2\text{O}_4^{2-}$ and en (ethane-1,2-diamine) are didentate ligands.

It is important to note here that coordination number of the central atom/ion is determined only by the number of sigma bonds formed by the ligand with the central atom/ion. Pi bonds, if formed between the ligand and the central atom/ion, are not counted for this purpose.

(e) Coordination sphere

The central atom/ion and the ligands attached to it are enclosed in square bracket and is collectively termed as the **coordination sphere**. The ionisable groups are written outside the bracket and are called counter ions. For example, in the complex $K_4[Fe(CN)_6]$, the coordination sphere is $[Fe(CN)_6]^{4-}$ and the counter ion is K^+ .

(f) Coordination polyhedron

The spatial arrangement of the ligand atoms which are directly attached to the central atom/ion defines a coordination polyhedron about the central atom. The most common coordination polyhedra are octahedral, square planar and tetrahedral. For example, $[Co(NH_3)_6]^{3+}$ is octahedral, $[Ni(CO)_4]$ is tetrahedral and $[PtCl_4]^{2-}$ is square planar. Fig. 5.1 shows the shapes of different coordination polyhedra.

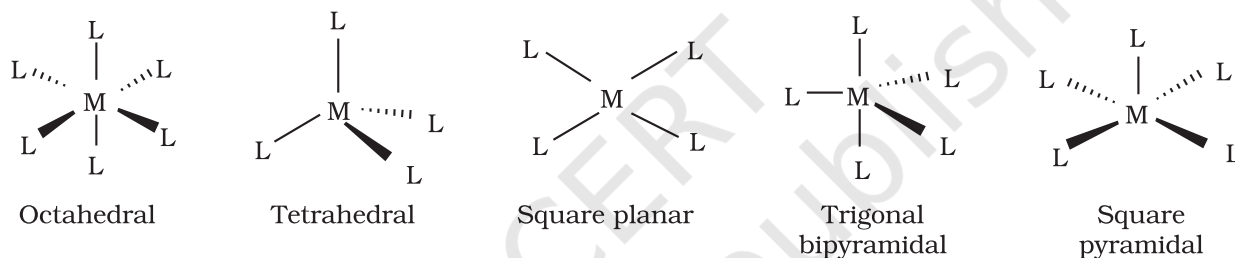


Fig. 5.1: Shapes of different coordination polyhedra. M represents the central atom/ion and L , a unidentate ligand.

(g) Oxidation number of central atom

The oxidation number of the central atom in a complex is defined as the charge it would carry if all the ligands are removed along with the electron pairs that are shared with the central atom. The oxidation number is represented by a Roman numeral in parenthesis following the name of the coordination entity. For example, oxidation number of copper in $[Cu(CN)_4]^{3-}$ is +1 and it is written as Cu(I).

(h) Homoleptic and heteroleptic complexes

Complexes in which a metal is bound to only one kind of donor groups, *e.g.*, $[Co(NH_3)_6]^{3+}$, are known as homoleptic. Complexes in which a metal is bound to more than one kind of donor groups, *e.g.*, $[Co(NH_3)_4Cl_2]^+$, are known as heteroleptic.

5.3 Nomenclature of Coordination Compounds

Nomenclature is important in Coordination Chemistry because of the need to have an unambiguous method of describing formulas and writing systematic names, particularly when dealing with isomers. The formulas and names adopted for coordination entities are based on the recommendations of the International Union of Pure and Applied Chemistry (IUPAC).

5.3.1 Formulas of Mononuclear Coordination Entities

The formula of a compound is a shorthand tool used to provide basic information about the constitution of the compound in a concise and convenient manner. Mononuclear coordination entities contain a single central metal atom. The following rules are applied while writing the formulas:

- (i) The central atom is listed first.
- (ii) The ligands are then listed in alphabetical order. The placement of a ligand in the list does not depend on its charge.
- (iii) Polydentate ligands are also listed alphabetically. In case of abbreviated ligand, the first letter of the abbreviation is used to determine the position of the ligand in the alphabetical order.
- (iv) The formula for the entire coordination entity, whether charged or not, is enclosed in square brackets. When ligands are polyatomic, their formulas are enclosed in parentheses. Ligand abbreviations are also enclosed in parentheses.
- (v) There should be no space between the ligands and the metal within a coordination sphere.
- (vi) When the formula of a charged coordination entity is to be written without that of the counter ion, the charge is indicated outside the square brackets as a right superscript with the number before the sign. For example, $[\text{Co}(\text{CN})_6]^{3-}$, $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$, etc.
- (vii) The charge of the cation(s) is balanced by the charge of the anion(s).

Note: The 2004 IUPAC draft recommends that ligands will be sorted alphabetically, irrespective of charge.

5.3.2 Naming of Mononuclear Coordination Compounds

The names of coordination compounds are derived by following the principles of additive nomenclature. Thus, the groups that surround the central atom must be identified in the name. They are listed as prefixes to the name of the central atom along with any appropriate multipliers. The following rules are used when naming coordination compounds:

- (i) The cation is named first in both positively and negatively charged coordination entities.
- (ii) The ligands are named in an alphabetical order before the name of the central atom/ion. (This procedure is reversed from writing formula).
- (iii) Names of the anionic ligands end in -o, those of neutral and cationic ligands are the same except aqua for H_2O , ammine for NH_3 , carbonyl for CO and nitrosyl for NO . While writing the formula of coordination entity, these are enclosed in brackets ().
- (iv) Prefixes mono, di, tri, etc., are used to indicate the number of the individual ligands in the coordination entity. When the names of the ligands include a numerical prefix, then the terms, *bis*, *tris*, *tetrakis* are used, the ligand to which they refer being placed in parentheses. For example, $[\text{NiCl}_2(\text{PPh}_3)_2]$ is named as dichloridobis(triphenylphosphine)nickel(II).
- (v) Oxidation state of the metal in cation, anion or neutral coordination entity is indicated by Roman numeral in parenthesis.
- (vi) If the complex ion is a cation, the metal is named same as the element. For example, Co in a complex cation is called cobalt and Pt is called platinum. If the complex ion is an anion, the name of the metal ends with the suffix -ate. For example, Co in a complex anion, $[\text{Co}(\text{SCN})_4]^{2-}$ is called cobaltate. For some metals, the Latin names are used in the complex anions, e.g., ferrate for Fe.

Note: The 2004 IUPAC draft recommends that anionic ligands will end with -ido so that chloro would become chlorido, etc.

(vii) The neutral complex molecule is named similar to that of the complex cation.

The following examples illustrate the nomenclature for coordination compounds.

1. $[\text{Cr}(\text{NH}_3)_3(\text{H}_2\text{O})_3]\text{Cl}_3$ is named as:

triamminetriaquachromium(III) chloride

Explanation: The complex ion is inside the square bracket, which is a cation. The amine ligands are named before the aqua ligands according to alphabetical order. Since there are three chloride ions in the compound, the charge on the complex ion must be +3 (since the compound is electrically neutral). From the charge on the complex ion and the charge on the ligands, we can calculate the oxidation number of the metal. In this example, all the ligands are neutral molecules. Therefore, the oxidation number of chromium must be the same as the charge of the complex ion, +3.

2. $[\text{Co}(\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2)_3]_2(\text{SO}_4)_3$ is named as:

tris(ethane-1,2-diamine)cobalt(III) sulphate

Explanation: The sulphate is the counter anion in this molecule. Since it takes 3 sulphates to bond with two complex cations, the charge on each complex cation must be +3. Further, ethane-1,2-diamine is a neutral molecule, so the oxidation number of cobalt in the complex ion must be +3. Remember that you never have to indicate the number of cations and anions in the name of an ionic compound.

3. $[\text{Ag}(\text{NH}_3)_2][\text{Ag}(\text{CN})_2]$ is named as:

diamminesilver(I)dicyanidoargentate(I)

Notice how the name of the metal differs in cation and anion even though they contain the same metal ions.

Example 5.2

Write the formulas for the following coordination compounds:

- (a) tetraammineaquachloridocobalt(III) chloride
- (b) potassium tetrahydroxidozincate(II)
- (c) potassium trioxalatoaluminate(III)
- (d) dichloridobis(ethane-1,2-diamine)cobalt(III)
- (e) tetracarbonylnickel(0)

Solution

- (a) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$
- (b) $\text{K}_2[\text{Zn}(\text{OH})_4]$
- (c) $\text{K}_3[\text{Al}(\text{C}_2\text{O}_4)_3]$
- (d) $[\text{CoCl}_2(\text{en})_2]^+$
- (e) $[\text{Ni}(\text{CO})_4]$

Example 5.3

Write the IUPAC names of the following coordination compounds:

- (a) $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NO}_2)]$
- (b) $\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$
- (c) $[\text{CoCl}_2(\text{en})_2]\text{Cl}$
- (d) $[\text{Co}(\text{NH}_3)_5(\text{CO}_3)]\text{Cl}$
- (e) $\text{Hg}[\text{Co}(\text{SCN})_4]$

Solution

- (a) diamminechloridonitrito-N-platinum(II)
- (b) potassium trioxalatochromate(III)
- (c) dichloridobis(ethane-1,2-diamine)cobalt(III) chloride
- (d) pentaamminecarbonatocobalt(III) chloride
- (e) mercury (I) tetrathiocyanato-S-cobaltate(III)

Intext Questions

5.1 Write the formulas for the following coordination compounds:

- tetraamminediaquacobalt(III) chloride
- potassium tetracyanonickelate(II)
- tris(ethane-1,2-diamine) chromium(III) chloride
- amminebromidochloridonitrito-N-platinate(II)
- dichloridobis(ethane-1,2-diamine)platinum(IV) nitrate
- iron(III) hexacyanidoferrate(II)

5.2 Write the IUPAC names of the following coordination compounds:

- | | | |
|--|---|---|
| (i) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ | (ii) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ | (iii) $\text{K}_3[\text{Fe}(\text{CN})_6]$ |
| (iv) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ | (v) $\text{K}_2[\text{PdCl}_4]$ | (vi) $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NH}_2\text{CH}_3)]\text{Cl}$ |

5.4 Isomerism in Coordination Compounds

Isomers are two or more compounds that have the same chemical formula but a different arrangement of atoms. Because of the different arrangement of atoms, they differ in one or more physical or chemical properties. Two principal types of isomerism are known among coordination compounds. Each of which can be further subdivided.

(a) Stereoisomerism

- | | |
|---------------------------|------------------------|
| (i) Geometrical isomerism | (ii) Optical isomerism |
|---------------------------|------------------------|

(b) Structural isomerism

- | | |
|----------------------------|-----------------------------|
| (i) Linkage isomerism | (ii) Coordination isomerism |
| (iii) Ionisation isomerism | (iv) Solvate isomerism |

Stereoisomers have the same chemical formula and chemical bonds but they have different spatial arrangement. Structural isomers have different bonds. A detailed account of these isomers are given below.

5.4.1 Geometric Isomerism

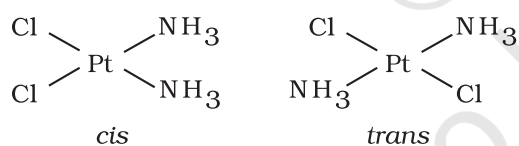


Fig. 5.2: Geometrical isomers (*cis* and *trans*) of $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$

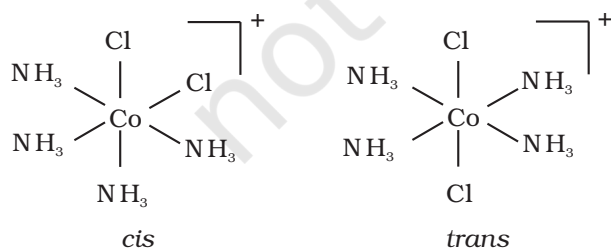


Fig. 5.3: Geometrical isomers (*cis* and *trans*) of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$

This type of isomerism arises in heteroleptic complexes due to different possible geometric arrangements of the ligands. Important examples of this behaviour are found with coordination numbers 4 and 6. In a square planar complex of formula $[\text{MX}_2\text{L}_2]$ (X and L are unidentate), the two ligands X may be arranged adjacent to each other in a *cis* isomer, or opposite to each other in a *trans* isomer as depicted in Fig. 5.2.

Other square planar complex of the type MABXL (where A, B, X, L are unidentates) shows three isomers—two *cis* and one *trans*. You may attempt to draw these structures. Such isomerism is not possible for a tetrahedral geometry but similar behaviour is possible in octahedral complexes of formula $[\text{MX}_2\text{L}_4]$ in which the two ligands X may be oriented *cis* or *trans* to each other (Fig. 5.3).

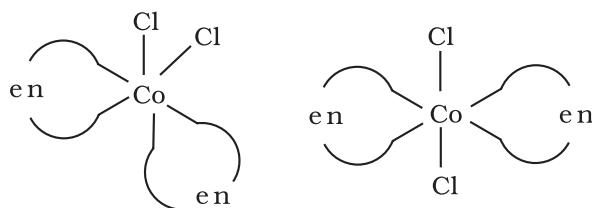
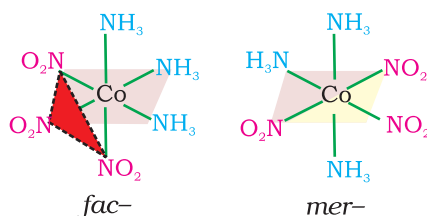


Fig. 5.4: Geometrical isomers (cis and trans) of $[\text{CoCl}_2(\text{en})_2]$

Fig. 5.5

The facial (fac) and meridional (mer) isomers of $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$



This type of isomerism also arises when didentate ligands L-L [e.g., $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$ (en)] are present in complexes of formula $[\text{MX}_2(\text{L-L})_2]$ (Fig. 5.4).

Another type of geometrical isomerism occurs in octahedral coordination entities of the type $[\text{Ma}_3\text{b}_3]$ like $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$. If three donor atoms of the same ligands occupy adjacent positions at the corners of an octahedral face, we have the **facial (fac) isomer**. When the positions are around the meridian of the octahedron, we get the **meridional (mer) isomer** (Fig. 5.5).

Why is geometrical isomerism not possible in tetrahedral complexes having two different types of unidentate ligands coordinated with the central metal ion ?

Example 5.4

Tetrahedral complexes do not show geometrical isomerism because the relative positions of the unidentate ligands attached to the central metal atom are the same with respect to each other.

Solution

5.4.2 Optical Isomerism

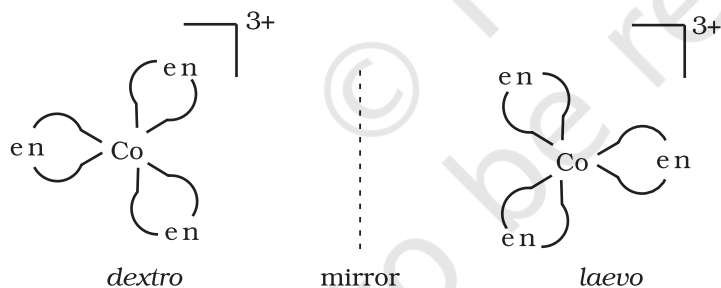
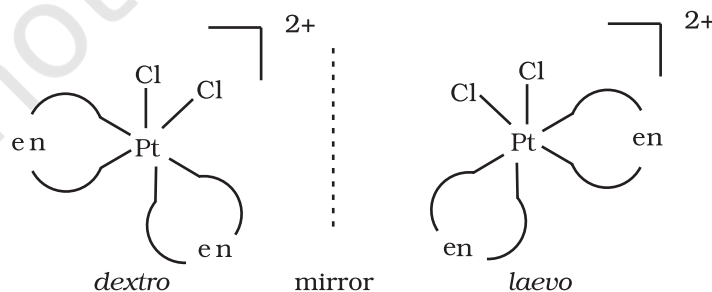


Fig. 5.6: Optical isomers (d and l) of $[\text{Co}(\text{en})_3]^{3+}$

Optical isomers are mirror images that cannot be superimposed on one another. These are called as *enantiomers*. The molecules or ions that cannot be superimposed are called *chiral*. The two forms are called *dextro* (d) and *laevo* (l) depending upon the direction they rotate the plane of polarised light in a polarimeter (d rotates to the right, l to the left). Optical isomerism is common in octahedral complexes involving didentate ligands (Fig. 5.6).

Fig. 5.7

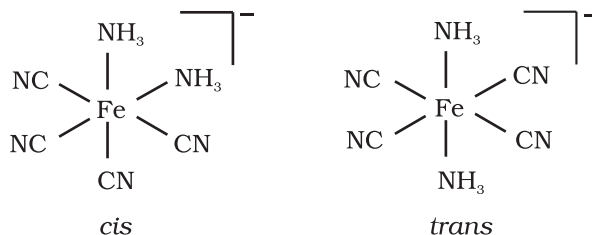
Optical isomers (d and l) of cis- $[\text{PtCl}_2(\text{en})_2]^{2+}$



In a coordination entity of the type $[\text{PtCl}_2(\text{en})_2]^{2+}$, only the cis-isomer shows optical activity (Fig. 5.7).

Example 5.5 Draw structures of geometrical isomers of $[\text{Fe}(\text{NH}_3)_2(\text{CN})_4]^-$

Solution

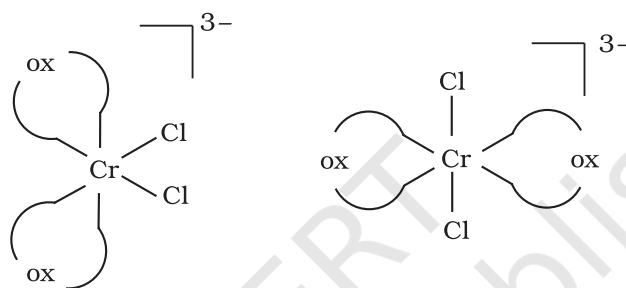


Example 5.6 Out of the following two coordination entities which is chiral (optically active)?

- (a) $\text{cis}-[\text{CrCl}_2(\text{ox})_2]^{3-}$ (b) $\text{trans}-[\text{CrCl}_2(\text{ox})_2]^{3-}$

Solution

The two entities are represented as



- (a) $\text{cis}-[\text{CrCl}_2(\text{ox})_2]^{3-}$ (b) $\text{trans}-[\text{CrCl}_2(\text{ox})_2]^{3-}$

Out of the two, (a) $\text{cis}-[\text{CrCl}_2(\text{ox})_2]^{3-}$ is chiral (optically active).

5.4.3 Linkage Isomerism

Linkage isomerism arises in a coordination compound containing ambidentate ligand. A simple example is provided by complexes containing the thiocyanate ligand, NCS^- , which may bind through the nitrogen to give $\text{M}-\text{NCS}$ or through sulphur to give $\text{M}-\text{SCN}$. Jørgensen discovered such behaviour in the complex $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$, which is obtained as the red form, in which the nitrite ligand is bound through oxygen ($-\text{ONO}$), and as the yellow form, in which the nitrite ligand is bound through nitrogen ($-\text{NO}_2$).

5.4.4 Coordination Isomerism

This type of isomerism arises from the interchange of ligands between cationic and anionic entities of different metal ions present in a complex. An example is provided by $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$, in which the NH_3 ligands are bound to Co^{3+} and the CN^- ligands to Cr^{3+} . In its coordination isomer $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$, the NH_3 ligands are bound to Cr^{3+} and the CN^- ligands to Co^{3+} .

5.4.5 Ionisation Isomerism

This form of isomerism arises when the counter ion in a complex salt is itself a potential ligand and can displace a ligand which can then become the counter ion. An example is provided by the ionisation isomers $[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$.

5.4.6 Solvate Isomerism

This form of isomerism is known as '**hydrate isomerism**' in case where water is involved as a solvent. This is similar to ionisation isomerism. Solvate isomers differ by whether or not a solvent molecule is directly bonded to the metal ion or merely present as free solvent molecules in the crystal lattice. An example is provided by the aqua complex $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (violet) and its solvate isomer $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ (grey-green).

Intext Questions

5.3 Indicate the types of isomerism exhibited by the following complexes and draw the structures for these isomers:

- (i) $\text{K}[\text{Cr}(\text{H}_2\text{O})_2(\text{C}_2\text{O}_4)_2]$ (ii) $[\text{Co}(\text{en})_3]\text{Cl}_3$
(iii) $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)](\text{NO}_3)_2$ (iv) $[\text{Pt}(\text{NH}_3)(\text{H}_2\text{O})\text{Cl}_2]$

5.4 Give evidence that $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Cl}$ are ionisation isomers.

5.5 Bonding in Coordination Compounds

Werner was the first to describe the bonding features in coordination compounds. But his theory could not answer basic questions like:

- Why only certain elements possess the remarkable property of forming coordination compounds?
- Why the bonds in coordination compounds have directional properties?
- Why coordination compounds have characteristic magnetic and optical properties?

Many approaches have been put forth to explain the nature of bonding in coordination compounds *viz.* Valence Bond Theory (VBT), Crystal Field Theory (CFT), **Ligand Field Theory** (LFT) and Molecular Orbital Theory (MOT). We shall focus our attention on elementary treatment of the application of VBT and CFT to coordination compounds.

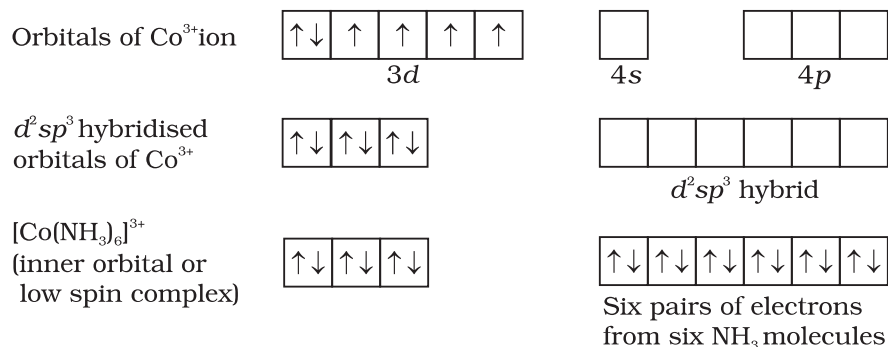
5.5.1 Valence Bond Theory

According to this theory, the metal atom or ion under the influence of ligands can use its $(n-1)d$, ns , np or ns , np , nd orbitals for hybridisation to yield a set of equivalent orbitals of definite geometry such as octahedral, tetrahedral, square planar and so on (Table 5.2). These hybridised orbitals are allowed to overlap with ligand orbitals that can donate electron pairs for bonding. This is illustrated by the following examples.

Table 5.2: Number of Orbitals and Types of Hybridisations

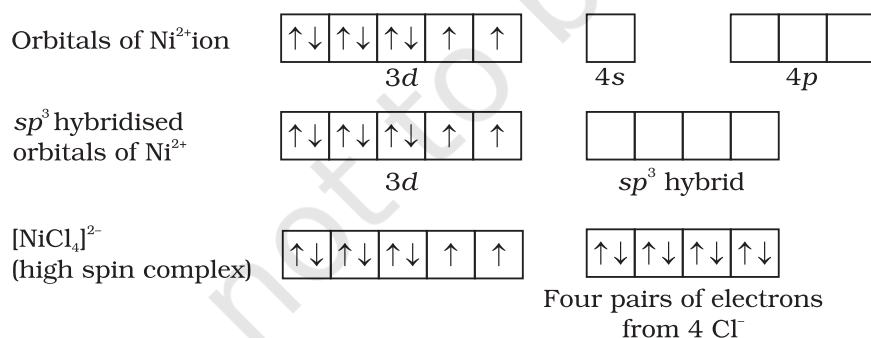
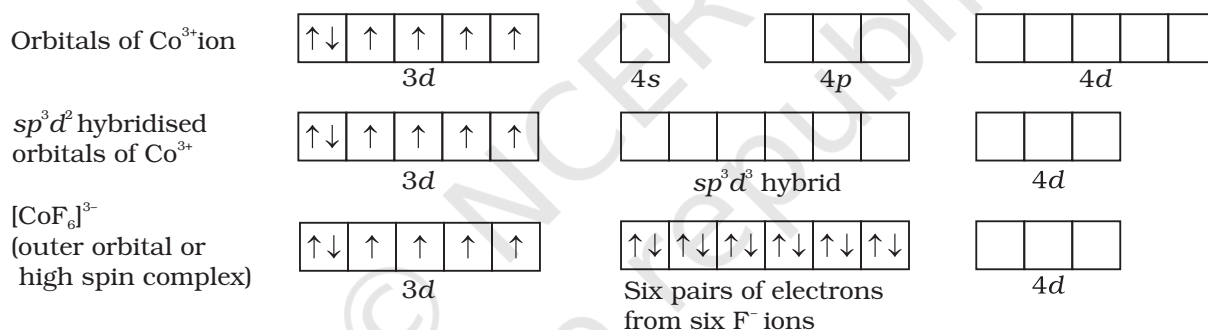
Coordination number	Type of hybridisation	Distribution of hybrid orbitals in space
4	sp^3	Tetrahedral
4	dsp^2	Square planar
5	sp^3d	Trigonal bipyramidal
6	sp^3d^2	Octahedral
6	d^2sp^3	Octahedral

It is usually possible to predict the geometry of a complex from the knowledge of its magnetic behaviour on the basis of the valence bond theory.



In the diamagnetic octahedral complex, $[\text{Co}(\text{NH}_3)_6]^{3+}$, the cobalt ion is in +3 oxidation state and has the electronic configuration $3d^6$. The hybridisation scheme is as shown in diagram.

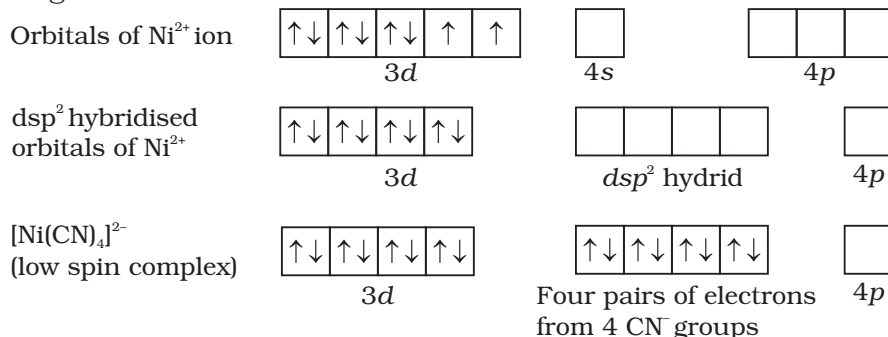
Six pairs of electrons, one from each NH_3 molecule, occupy the six hybrid orbitals. Thus, the complex has octahedral geometry and is diamagnetic because of the absence of unpaired electron. In the formation of this complex, since the inner d orbital ($3d$) is used in hybridisation, the complex, $[\text{Co}(\text{NH}_3)_6]^{3+}$ is called an **inner orbital** or **low spin** or **spin paired complex**. The paramagnetic octahedral complex, $[\text{CoF}_6]^{3-}$ uses outer orbital ($4d$) in hybridisation (sp^3d^2). It is thus called **outer orbital** or **high spin** or **spin free complex**. Thus:



In tetrahedral complexes one s and three p orbitals are hybridised to form four equivalent orbitals oriented tetrahedrally. This is illustrated below for $[\text{NiCl}_4]^{2-}$. Here nickel is in +2 oxidation state and the ion has the electronic configuration $3d^8$. The hybridisation scheme is as shown in diagram.

Each Cl^- ion donates a pair of electrons. The compound is paramagnetic since it contains two unpaired electrons. Similarly, $[\text{Ni}(\text{CO})_4]$ has tetrahedral geometry but is diamagnetic since nickel is in zero oxidation state and contains no unpaired electron.

In the square planar complexes, the hybridisation involved is dsp^2 . An example is $[\text{Ni}(\text{CN})_4]^{2-}$. Here nickel is in +2 oxidation state and has the electronic configuration $3d^8$. The hybridisation scheme is as shown in diagram:



Each of the hybridised orbitals receives a pair of electrons from a cyanide ion. The compound is diamagnetic as evident from the absence of unpaired electron.

It is important to note that the hybrid orbitals do not actually exist. In fact, hybridisation is a mathematical manipulation of wave equation for the atomic orbitals involved.

5.5.2 Magnetic Properties of Coordination Compounds

The magnetic moment of coordination compounds can be measured by the magnetic susceptibility experiments. The results can be used to obtain information about the number of unpaired electrons and hence structures adopted by metal complexes.

A critical study of the magnetic data of coordination compounds of metals of the first transition series reveals some complications. For metal ions with upto three electrons in the d orbitals, like Ti^{3+} (d^1); V^{3+} (d^2); Cr^{3+} (d^3); two vacant d orbitals are available for octahedral hybridisation with $4s$ and $4p$ orbitals. The magnetic behaviour of these free ions and their coordination entities is similar. When more than three $3d$ electrons are present, the required pair of $3d$ orbitals for octahedral hybridisation is not directly available (as a consequence of Hund's rule). Thus, for d^4 (Cr^{2+} , Mn^{3+}), d^5 (Mn^{2+} , Fe^{3+}), d^6 (Fe^{2+} , Co^{3+}) cases, a vacant pair of d orbitals results only by pairing of $3d$ electrons which leaves two, one and zero unpaired electrons, respectively.

The magnetic data agree with maximum spin pairing in many cases, especially with coordination compounds containing d^6 ions. However, with species containing d^4 and d^5 ions there are complications. $[\text{Mn}(\text{CN})_6]^{3-}$ has magnetic moment of two unpaired electrons while $[\text{MnCl}_6]^{3-}$ has a paramagnetic moment of four unpaired electrons. $[\text{Fe}(\text{CN})_6]^{3-}$ has magnetic moment of a single unpaired electron while $[\text{FeF}_6]^{3-}$ has a paramagnetic moment of five unpaired electrons. $[\text{CoF}_6]^{3-}$ is paramagnetic with four unpaired electrons while $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ is diamagnetic. This apparent anomaly is explained by valence bond theory in terms of formation of inner orbital and outer orbital coordination entities. $[\text{Mn}(\text{CN})_6]^{3-}$, $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ are inner orbital complexes involving d^2sp^3 hybridisation, the former two complexes are paramagnetic and the latter diamagnetic. On the other hand, $[\text{MnCl}_6]^{3-}$, $[\text{FeF}_6]^{3-}$ and $[\text{CoF}_6]^{3-}$ are outer orbital complexes involving sp^3d^2 hybridisation and are paramagnetic corresponding to four, five and four unpaired electrons.

Example 5.7

The spin only magnetic moment of $[\text{MnBr}_4]^{2-}$ is 5.9 BM. Predict the geometry of the complex ion ?

Solution

Since the coordination number of Mn^{2+} ion in the complex ion is 4, it will be either tetrahedral (sp^3 hybridisation) or square planar (dsp^2 hybridisation). But the fact that the magnetic moment of the complex ion is 5.9 BM, it should be tetrahedral in shape rather than square planar because of the presence of five unpaired electrons in the d orbitals.

5.5.3 Limitations of Valence Bond Theory

While the VB theory, to a larger extent, explains the formation, structures and magnetic behaviour of coordination compounds, it suffers from the following shortcomings:

- It involves a number of assumptions.
- It does not give quantitative interpretation of magnetic data.
- It does not explain the colour exhibited by coordination compounds.
- It does not give a quantitative interpretation of the thermodynamic or kinetic stabilities of coordination compounds.
- It does not make exact predictions regarding the tetrahedral and square planar structures of 4-coordinate complexes.
- It does not distinguish between weak and strong ligands.

5.5.4 Crystal Field Theory

The crystal field theory (CFT) is an electrostatic model which considers the metal-ligand bond to be ionic arising purely from electrostatic interactions between the metal ion and the ligand. Ligands are treated as point charges in case of anions or point dipoles in case of neutral molecules. The five d orbitals in an isolated gaseous metal atom/ion have same energy, i.e., they are degenerate. This degeneracy is maintained if a spherically symmetrical field of negative charges surrounds the metal atom/ion. However, when this negative field is due to ligands (either anions or the negative ends of dipolar molecules like NH_3 and H_2O) in a complex, it becomes asymmetrical and the degeneracy of the d orbitals is lifted. It results in splitting of the d orbitals. The pattern of splitting depends upon the nature of the crystal field. Let us explain this splitting in different crystal fields.

(a) Crystal field splitting in octahedral coordination entities

In an octahedral coordination entity with six ligands surrounding the metal atom/ion, there will be repulsion between the electrons in metal d orbitals and the electrons (or negative charges) of the ligands. Such a repulsion is more when the metal d orbital is directed towards the ligand than when it is away from the ligand. Thus, the $d_{x^2-y^2}$ and d_{z^2} orbitals which point towards the axes along the direction of the ligand will experience more repulsion and will be raised in energy; and the d_{xy} , d_{yz} and d_{zx} orbitals which are directed between the axes will be lowered in energy relative to the average energy in the spherical crystal field. Thus, the degeneracy of the d orbitals has been removed due to ligand electron-metal electron repulsions in the octahedral complex to yield three orbitals of lower energy, t_{2g} set and two orbitals of higher energy, e_g set. This splitting of the

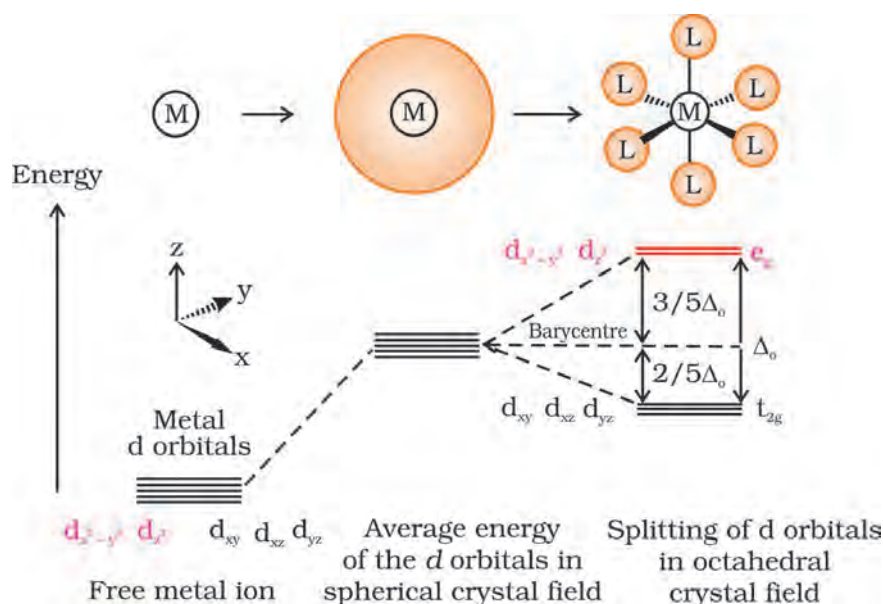
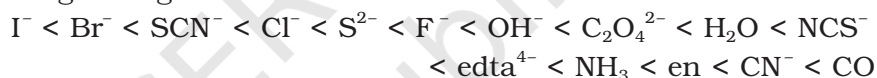


Fig.5.8: *d* orbital splitting in an octahedral crystal field

degenerate levels due to the presence of ligands in a definite geometry is termed as **crystal field splitting** and the energy separation is denoted by Δ_o (the subscript o is for octahedral) (Fig.5.8). Thus, the energy of the two e_g orbitals will increase by $(3/5)\Delta_o$ and that of the three t_{2g} will decrease by $(2/5)\Delta_o$.

The crystal field splitting, Δ_o , depends upon the field produced by the ligand and charge on the metal ion. Some ligands are able to produce strong fields in which case, the splitting will be large whereas others produce weak fields and consequently result in small splitting of *d* orbitals.

In general, ligands can be arranged in a series in the order of increasing field strength as given below:



Such a series is termed as **spectrochemical series**. It is an experimentally determined series based on the absorption of light by complexes with different ligands. Let us assign electrons in the *d* orbitals of metal ion in octahedral coordination entities. Obviously, the single *d* electron occupies one of the lower energy t_{2g} orbitals. In d^2 and d^3 coordination entities, the *d* electrons occupy the t_{2g} orbitals singly in accordance with the Hund's rule. For d^4 ions, two possible patterns of electron distribution arise: (i) the fourth electron could either enter the t_{2g} level and pair with an existing electron, or (ii) it could avoid paying the price of the pairing energy by occupying the e_g level. Which of these possibilities occurs, depends on the relative magnitude of the crystal field splitting, Δ_o and the pairing energy, *P* (*P* represents the energy required for electron pairing in a single orbital). The two options are:

- (i) If $\Delta_o < P$, the fourth electron enters one of the e_g orbitals giving the configuration $t_{2g}^3 e_g^1$. Ligands for which $\Delta_o < P$ are known as *weak field ligands* and form high spin complexes.
- (ii) If $\Delta_o > P$, it becomes more energetically favourable for the fourth electron to occupy a t_{2g} orbital with configuration $t_{2g}^4 e_g^0$. Ligands which produce this effect are known as *strong field ligands* and form low spin complexes.

Calculations show that d^4 to d^7 coordination entities are more stable for strong field as compared to weak field cases.

(b) Crystal field splitting in tetrahedral coordination entities

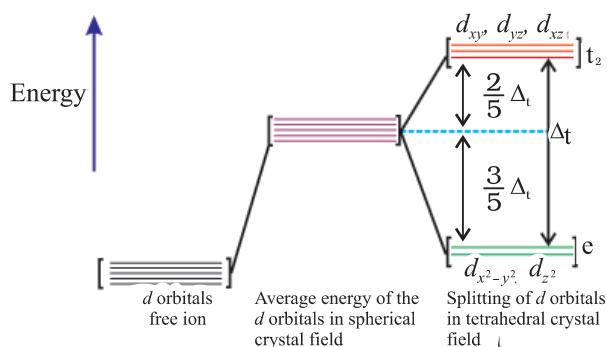


Fig.5.9: *d* orbital splitting in a tetrahedral crystal field.

In tetrahedral coordination entity formation, the *d* orbital splitting (Fig. 5.9) is inverted and is smaller as compared to the octahedral field splitting. For the same metal, the same ligands and metal-ligand distances, it can be shown that $\Delta_t = (4/9) \Delta_o$. Consequently, the orbital splitting energies are not sufficiently large for forcing pairing and, therefore, low spin configurations are rarely observed. The 'g' subscript is used for the octahedral and square planar complexes which have centre of symmetry. Since tetrahedral complexes lack symmetry, 'g' subscript is not used with energy levels.

5.5.5 Colour in Coordination Compounds

In the previous Unit, we learnt that one of the most distinctive properties of transition metal complexes is their wide range of colours. This means that some of the visible spectrum is being removed from white light as it passes through the sample, so the light that emerges is no longer white. The colour of the complex is complementary to that which is absorbed. The complementary colour is the colour generated from the wavelength left over; if green light is absorbed by the complex, it appears red. Table 5.3 gives the relationship of the different wavelength absorbed and the colour observed.

Table 5.3: Relationship between the Wavelength of Light absorbed and the Colour observed in some Coordination Entities

Coordination entity	Wavelength of light absorbed (nm)	Colour of light absorbed	Colour of coordination entity
$[\text{CoCl}(\text{NH}_3)_5]^{2+}$	535	Yellow	Violet
$[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$	500	Blue Green	Red
$[\text{Co}(\text{NH}_3)_6]^{3+}$	475	Blue	Yellow Orange
$[\text{Co}(\text{CN})_6]^{3-}$	310	Ultraviolet Not in visible region	Pale Yellow
$[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$	600	Red	Blue
$[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$	498	Blue Green	Violet

The colour in the coordination compounds can be readily explained in terms of the crystal field theory. Consider, for example, the complex $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$, which is violet in colour. This is an octahedral complex where the single electron (Ti^{3+} is a $3d^1$ system) in the metal *d* orbital is in the t_{2g} level in the ground state of the complex. The next higher state available for the electron is the empty e_g level. If light corresponding to the energy of blue-green region is absorbed by the complex, it would excite the electron from t_{2g} level to the e_g level ($t_{2g}^1 e_g^0 \rightarrow t_{2g}^0 e_g^1$). Consequently, the complex appears violet in colour (Fig. 5.10). The crystal field theory attributes the colour of the coordination compounds to *d-d* transition of the electron.

In emerald [Fig.5.12(b)], Cr^{3+} ions occupy octahedral sites in the mineral beryl ($\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$). The absorption bands seen in the ruby shift to longer wavelength, namely yellow-red and blue, causing emerald to transmit light in the green region.

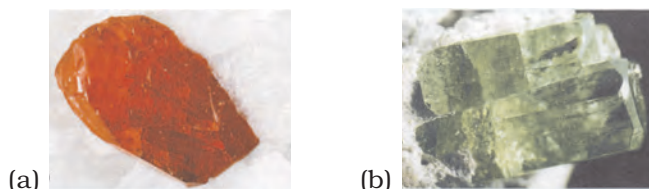


Fig.5.12: (a) Ruby: this gemstone was found in marble from Mogok, Myanmar; (b) Emerald: this gemstone was found in Muzo, Columbia.

5.5.6 Limitations of Crystal Field Theory

The crystal field model is successful in explaining the formation, structures, colour and magnetic properties of coordination compounds to a large extent. However, from the assumptions that the ligands are point charges, it follows that anionic ligands should exert the greatest splitting effect. The anionic ligands actually are found at the low end of the spectrochemical series. Further, it does not take into account the covalent character of bonding between the ligand and the central atom. These are some of the weaknesses of CFT, which are explained by ligand field theory (LFT) and molecular orbital theory which are beyond the scope of the present study.

Intext Questions

- 5.5 Explain on the basis of valence bond theory that $[\text{Ni}(\text{CN})_4]^{2-}$ ion with square planar structure is diamagnetic and the $[\text{NiCl}_4]^{2-}$ ion with tetrahedral geometry is paramagnetic.
- 5.6 $[\text{NiCl}_4]^{2-}$ is paramagnetic while $[\text{Ni}(\text{CO})_4]$ is diamagnetic though both are tetrahedral. Why?
- 5.7 $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ is strongly paramagnetic whereas $[\text{Fe}(\text{CN})_6]^{3-}$ is weakly paramagnetic. Explain.
- 5.8 Explain $[\text{Co}(\text{NH}_3)_6]^{3+}$ is an inner orbital complex whereas $[\text{Ni}(\text{NH}_3)_6]^{2+}$ is an outer orbital complex.
- 5.9 Predict the number of unpaired electrons in the square planar $[\text{Pt}(\text{CN})_4]^{2-}$ ion.
- 5.10 The hexaquo manganese(II) ion contains five unpaired electrons, while the hexacyanoion contains only one unpaired electron. Explain using Crystal Field Theory.

5.6 Bonding in Metal Carbonyls

The homoleptic carbonyls (compounds containing carbonyl ligands only) are formed by most of the transition metals. These carbonyls have simple, well defined structures. Tetracarbonylnickel(0) is tetrahedral, pentacarbonyliron(0) is trigonalbipyramidal while hexacarbonyl chromium(0) is octahedral.

Decacarbonyldimanganese(0) is made up of two square pyramidal $\text{Mn}(\text{CO})_5$ units joined by a Mn – Mn bond. Octacarbonyldicobalt(0) has a Co – Co bond bridged by two CO groups (Fig.5.13).

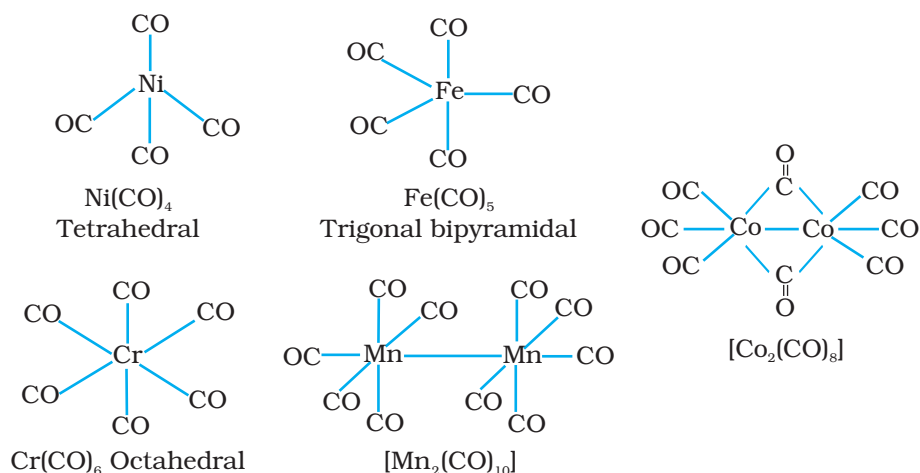


Fig. 5.13
Structures of some representative homoleptic metal carbonyls.

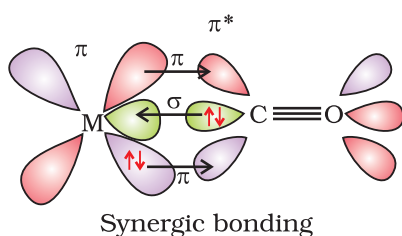


Fig. 5.14: Example of synergic bonding interactions in a carbonyl complex.

The metal-carbon bond in metal carbonyls possess both σ and π character. The M–C σ bond is formed by the donation of lone pair of electrons on the carbonyl carbon into a vacant orbital of the metal. The M–C π bond is formed by the donation of a pair of electrons from a filled d orbital of metal into the vacant antibonding π^* orbital of carbon monoxide. The metal to ligand bonding creates a synergic effect which strengthens the bond between CO and the metal (Fig.5.14).

5.7 Importance and Applications of Coordination Compounds

The coordination compounds are of great importance. These compounds are widely present in the mineral, plant and animal worlds and are known to play many important functions in the area of analytical chemistry, metallurgy, biological systems, industry and medicine. These are described below:

- Coordination compounds find use in many qualitative and quantitative chemical analysis. The familiar colour reactions given by metal ions with a number of ligands (especially chelating ligands), as a result of formation of coordination entities, form the basis for their detection and estimation by classical and instrumental methods of analysis. Examples of such reagents include EDTA, DMG (dimethylglyoxime), α -nitroso- β -naphthol, cupron, etc.
- Hardness of water is estimated by simple titration with Na_2EDTA . The Ca^{2+} and Mg^{2+} ions form stable complexes with EDTA. The selective estimation of these ions can be done due to difference in the stability constants of calcium and magnesium complexes.
- Some important extraction processes of metals, like those of silver and gold, make use of complex formation. Gold, for example, combines with cyanide in the presence of oxygen and water to form the coordination entity $[\text{Au}(\text{CN})_2]^-$ in aqueous solution. Gold can be separated in metallic form from this solution by the addition of zinc.
- Similarly, purification of metals can be achieved through formation and subsequent decomposition of their coordination compounds.

For example, impure nickel is converted to $[\text{Ni}(\text{CO})_4]$, which is decomposed to yield pure nickel.

- Coordination compounds are of great importance in biological systems. The pigment responsible for photosynthesis, chlorophyll, is a coordination compound of magnesium. Haemoglobin, the red pigment of blood which acts as oxygen carrier is a coordination compound of iron. Vitamin B_{12} , cyanocobalamin, the anti-pernicious anaemia factor, is a coordination compound of cobalt. Among the other compounds of biological importance with coordinated metal ions are the enzymes like, carboxypeptidase A and carbonic anhydrase (catalysts of biological systems).
- Coordination compounds are used as catalysts for many industrial processes. Examples include rhodium complex, $[(\text{Ph}_3\text{P})_3\text{RhCl}]$, a Wilkinson catalyst, is used for the hydrogenation of alkenes.
- Articles can be electroplated with silver and gold much more smoothly and evenly from solutions of the complexes, $[\text{Ag}(\text{CN})_2]^-$ and $[\text{Au}(\text{CN})_2]^-$ than from a solution of simple metal ions.
- In black and white photography, the developed film is fixed by washing with hypo solution which dissolves the undecomposed AgBr to form a complex ion, $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$.
- There is growing interest in the use of chelate therapy in medicinal chemistry. An example is the treatment of problems caused by the presence of metals in toxic proportions in plant/animal systems. Thus, excess of copper and iron are removed by the chelating ligands D-penicillamine and desferrioxime B via the formation of coordination compounds. EDTA is used in the treatment of lead poisoning. Some coordination compounds of platinum effectively inhibit the growth of tumours. Examples are: *cis*-platin and related compounds.

Summary

The **chemistry of coordination compounds** is an important and challenging area of modern inorganic chemistry. During the last fifty years, advances in this area, have provided development of new concepts and models of bonding and molecular structure, novel breakthroughs in **chemical industry** and vital insights into the functioning of critical components of **biological systems**.

The first systematic attempt at explaining the formation, reactions, structure and bonding of a coordination compound was made by **A. Werner**. His theory postulated the use of two types of **linkages** (**primary** and **secondary**) by a metal atom/ion in a coordination compound. In the modern language of chemistry these linkages are recognised as the ionisable (ionic) and non-ionisable (covalent) bonds, respectively. Using the property of isomerism, Werner predicted the geometrical shapes of a large number of coordination entities.

The Valence Bond Theory (VBT) explains with reasonable success, the formation, magnetic behaviour and geometrical shapes of coordination compounds. It, however, fails to provide a quantitative interpretation of magnetic behaviour and has nothing to say about the optical properties of these compounds.

The Crystal Field Theory (CFT) to coordination compounds is based on the effect of different crystal fields (provided by the ligands taken as point charges),

on the degeneracy of d orbital energies of the central metal atom/ion. The splitting of the d orbitals provides different electronic arrangements in strong and weak crystal fields. The treatment provides for quantitative estimations of orbital separation energies, magnetic moments and spectral and stability parameters. However, the assumption that ligands constitute point charges creates many theoretical difficulties.

The metal-carbon bond in **metal carbonyls** possesses both σ and π character. The ligand to metal is σ bond and metal to ligand is π bond. This unique synergic bonding provides stability to metal carbonyls.

Coordination compounds are of great importance. These compounds provide critical insights into the functioning and structures of vital components of biological systems. Coordination compounds also find extensive applications in **metallurgical processes, analytical and medicinal chemistry**.

Exercises

- 5.1 Explain the bonding in coordination compounds in terms of Werner's postulates.
- 5.2 FeSO_4 solution mixed with $(\text{NH}_4)_2\text{SO}_4$ solution in 1:1 molar ratio gives the test of Fe^{2+} ion but CuSO_4 solution mixed with aqueous ammonia in 1:4 molar ratio does not give the test of Cu^{2+} ion. Explain why?
- 5.3 Explain with two examples each of the following: coordination entity, ligand, coordination number, coordination polyhedron, homoleptic and heteroleptic.
- 5.4 What is meant by unidentate, didentate and ambidentate ligands? Give two examples for each.
- 5.5 Specify the oxidation numbers of the metals in the following coordination entities:

(i) $[\text{Co}(\text{H}_2\text{O})(\text{CN})(\text{en})_2]^{2+}$	(iii) $[\text{PtCl}_4]^{2-}$	(v) $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$
(ii) $[\text{CoBr}_2(\text{en})_2]^+$	(iv) $\text{K}_3[\text{Fe}(\text{CN})_6]$	
- 5.6 Using IUPAC norms write the formulas for the following:

(i) Tetrahydroxidozincate(II)	(vi) Hexaamminecobalt(III) sulphate
(ii) Potassium tetrachloridopalladate(II)	(vii) Potassium tri(oxalato)chromate(III)
(iii) Diamminedichloridoplatinum(II)	(viii) Hexaammineplatinum(IV)
(iv) Potassium tetracyanonickelate(II)	(ix) Tetrabromidocuprate(II)
(v) Pentaamminenitrito-O-cobalt(III)	(x) Pentaamminenitrito-N-cobalt(III)
- 5.7 Using IUPAC norms write the systematic names of the following:

(i) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$	(iv) $[\text{Co}(\text{NH}_3)_4\text{Cl}(\text{NO}_2)]\text{Cl}$	(vii) $[\text{Ni}(\text{NH}_3)_6]\text{Cl}_2$
(ii) $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NH}_2\text{CH}_3)]\text{Cl}$	(v) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$	(viii) $[\text{Co}(\text{en})_3]^{3+}$
(iii) $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$	(vi) $[\text{NiCl}_4]^{2-}$	(ix) $[\text{Ni}(\text{CO})_4]$
- 5.8 List various types of isomerism possible for coordination compounds, giving an example of each.
- 5.9 How many geometrical isomers are possible in the following coordination entities?

(i) $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$	(ii) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
--	--
- 5.10 Draw the structures of optical isomers of:

(i) $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$	(ii) $[\text{PtCl}_2(\text{en})_2]^{2+}$	(iii) $[\text{Cr}(\text{NH}_3)_2\text{Cl}_2(\text{en})]^+$
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- 5.11** Draw all the isomers (geometrical and optical) of:
 (i) $[\text{CoCl}_2(\text{en})_2]^+$ (ii) $[\text{Co}(\text{NH}_3)\text{Cl}(\text{en})_2]^{2+}$ (iii) $[\text{Co}(\text{NH}_3)_2\text{Cl}_2(\text{en})]^+$
- 5.12** Write all the geometrical isomers of $[\text{Pt}(\text{NH}_3)(\text{Br})(\text{Cl})(\text{py})]$ and how many of these will exhibit optical isomers?
- 5.13** Aqueous copper sulphate solution (blue in colour) gives:
 (i) a green precipitate with aqueous potassium fluoride and
 (ii) a bright green solution with aqueous potassium chloride. Explain these experimental results.
- 5.14** What is the coordination entity formed when excess of aqueous KCN is added to an aqueous solution of copper sulphate? Why is it that no precipitate of copper sulphide is obtained when $\text{H}_2\text{S}(\text{g})$ is passed through this solution?
- 5.15** Discuss the nature of bonding in the following coordination entities on the basis of valence bond theory:
 (i) $[\text{Fe}(\text{CN})_6]^{4-}$ (ii) $[\text{FeF}_6]^{3-}$ (iii) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ (iv) $[\text{CoF}_6]^{3-}$
- 5.16** Draw figure to show the splitting of d orbitals in an octahedral crystal field.
- 5.17** What is spectrochemical series? Explain the difference between a weak field ligand and a strong field ligand.
- 5.18** What is crystal field splitting energy? How does the magnitude of Δ_o decide the actual configuration of d orbitals in a coordination entity?
- 5.19** $[\text{Cr}(\text{NH}_3)_6]^{3+}$ is paramagnetic while $[\text{Ni}(\text{CN})_4]^{2-}$ is diamagnetic. Explain why?
- 5.20** A solution of $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ is green but a solution of $[\text{Ni}(\text{CN})_4]^{2-}$ is colourless. Explain.
- 5.21** $[\text{Fe}(\text{CN})_6]^{4-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ are of different colours in dilute solutions. Why?
- 5.22** Discuss the nature of bonding in metal carbonyls.
- 5.23** Give the oxidation state, d orbital occupation and coordination number of the central metal ion in the following complexes:
 (i) $\text{K}_3[\text{Co}(\text{C}_2\text{O}_4)_3]$ (iii) $(\text{NH}_4)_2[\text{CoF}_4]$
 (ii) $\text{cis}-[\text{CrCl}_2(\text{en})_2]\text{Cl}$ (iv) $[\text{Mn}(\text{H}_2\text{O})_6]\text{SO}_4$
- 5.24** Write down the IUPAC name for each of the following complexes and indicate the oxidation state, electronic configuration and coordination number. Also give stereochemistry and magnetic moment of the complex:
 (i) $\text{K}[\text{Cr}(\text{H}_2\text{O})_2(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O}$ (iii) $[\text{CrCl}_3(\text{py})_3]$ (v) $\text{K}_4[\text{Mn}(\text{CN})_6]$
 (ii) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ (iv) $\text{Cs}[\text{FeCl}_4]$
- 5.25** Explain the violet colour of the complex $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ on the basis of crystal field theory.
- 5.26** What is meant by the *chelate effect*? Give an example.
- 5.27** Discuss briefly giving an example in each case the role of coordination compounds in:
 (i) biological systems (iii) analytical chemistry
 (ii) medicinal chemistry and (iv) extraction/metallurgy of metals.
- 5.28** How many ions are produced from the complex $\text{Co}(\text{NH}_3)_6\text{Cl}_2$ in solution?
 (i) 6 (ii) 4 (iii) 3 (iv) 2

- 5.29** Amongst the following ions which one has the highest magnetic moment value?
 (i) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ (ii) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (iii) $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$
- 5.30** Amongst the following, the most stable complex is
 (i) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ (ii) $[\text{Fe}(\text{NH}_3)_6]^{3+}$ (iii) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ (iv) $[\text{FeCl}_6]^{3-}$
- 5.31** What will be the correct order for the wavelengths of absorption in the visible region for the following:
 $[\text{Ni}(\text{NO}_2)_6]^{4-}$, $[\text{Ni}(\text{NH}_3)_6]^{2+}$, $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$?

Answers to Some Intext Questions

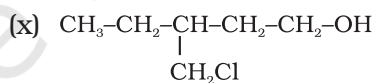
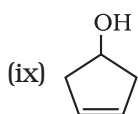
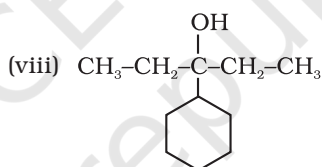
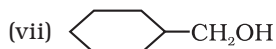
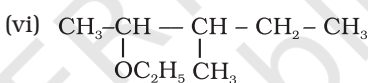
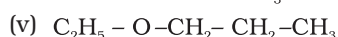
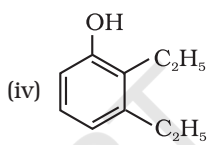
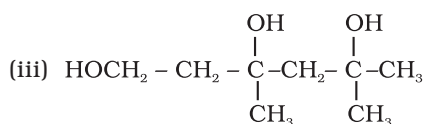
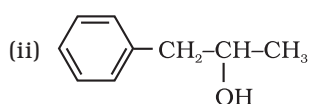
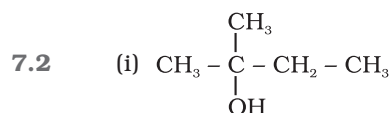
- 5.1** (i) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Cl}_3$ (iv) $[\text{Pt}(\text{NH}_3)\text{BrCl}(\text{NO}_2)]^-$
 (ii) $\text{K}_2[\text{Ni}(\text{CN})_4]$ (v) $[\text{PtCl}_2(\text{en})_2](\text{NO}_3)_2$
 (iii) $[\text{Cr}(\text{en})_3]\text{Cl}_3$ (vi) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
- 5.2** (i) Hexaamminecobalt(III) chloride
 (ii) Pentaamminechloridocobalt(III) chloride
 (iii) Potassium hexacyanidoferrate(III)
 (iv) Potassium trioxalatoferrate(III)
 (v) Potassium tetrachloridopalladate(II)
 (vi) Diamminechlorido(methanamine)platinum(II) chloride
- 5.3** (i) Both geometrical (*cis*-, *trans*-) and optical isomers for *cis* can exist.
 (ii) Two optical isomers can exist.
 (iii) There are 10 possible isomers. (Hint: There are geometrical, ionisation and linkage isomers possible).
 (iv) Geometrical (*cis*-, *trans*-) isomers can exist.
- 5.4** The ionisation isomers dissolve in water to yield different ions and thus react differently to various reagents:
 $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4 + \text{Ba}^{2+} \rightarrow \text{BaSO}_4 (\text{s})$
 $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br} + \text{Ba}^{2+} \rightarrow \text{No reaction}$
 $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4 + \text{Ag}^+ \rightarrow \text{No reaction}$
 $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br} + \text{Ag}^+ \rightarrow \text{AgBr} (\text{s})$
- 5.6** In $\text{Ni}(\text{CO})_4$, Ni is in zero oxidation state whereas in NiCl_4^{2-} , it is in +2 oxidation state. In the presence of CO ligand, the unpaired *d* electrons of Ni pair up but Cl^- being a weak ligand is unable to pair up the unpaired electrons.
- 5.7** In presence of CN^- , (a strong ligand) the 3*d* electrons pair up leaving only one unpaired electron. The hybridisation is d^2sp^3 forming inner orbital complex. In the presence of H_2O , (a weak ligand), 3*d* electrons do not pair up. The hybridisation is sp^3d^2 forming an outer orbital complex containing five unpaired electrons, it is strongly paramagnetic.
- 5.8** In the presence of NH_3 , the 3*d* electrons pair up leaving two *d* orbitals empty to be involved in d^2sp^3 hybridisation forming inner orbital complex in case of $[\text{Co}(\text{NH}_3)_6]^{3+}$. In $[\text{Ni}(\text{NH}_3)_6]^{2+}$, Ni is in +2 oxidation state and has d^8 configuration, the hybridisation involved is sp^3d^2 forming outer orbital complex.
- 5.9** For square planar shape, the hybridisation is dsp^2 . Hence the unpaired electrons in 5*d* orbital pair up to make one *d* orbital empty for dsp^2 hybridisation. Thus there is no unpaired electron.

Answers to Some Questions in Exercises

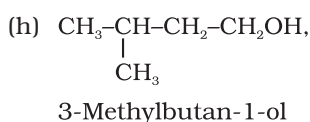
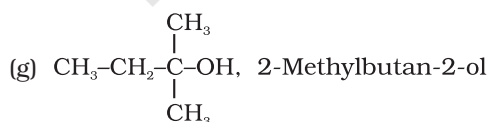
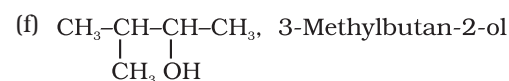
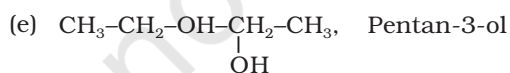
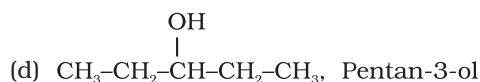
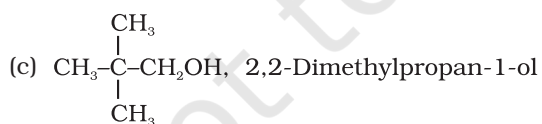
UNIT 7

- 7.1** (i) 2,2,4-Trimethylpentan-3-ol
 (iii) Butane-2,3-diol
 (v) 2-Methylphenol
 (vii) 2,5-Dimethylphenol
 (ix) 1-Methoxy-2-methylpropane
 (xi) 1-phenoxyheptane

- (ii) 5-Ethylheptan-2,4-diol
 (iv) Propan-1,2,3-triol
 (vi) 4-Methylphenol
 (viii) 2,6-Dimethylphenol
 (x) Ethoxybenzene
 (xii) 2-Ethoxybutane



- 7.3 (i)** (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$, Pentan-1-ol;
 (b) $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_2 - \text{OH}$, 2-Methylbutan-1-ol;

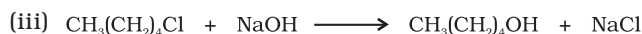
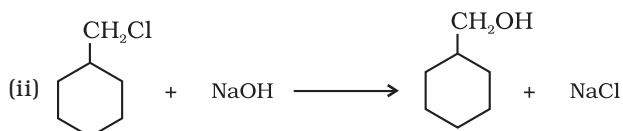
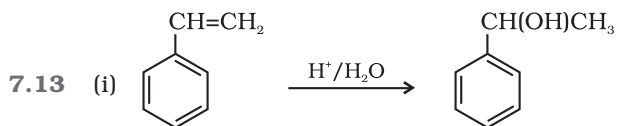


- 7.4** Hydrogen bonding in propanol.

7.5 Hydrogen bonding between alcohol and water molecules.

7.8 o-Nitrophenol is steam volatile because of intramolecular hydrogen bonding.

7.12 **Hint:** Carryout sulphonation followed by nucleophilic substitution.

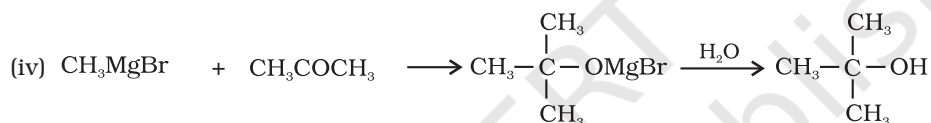
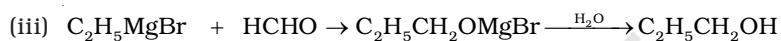


7.14 Reaction with (i) sodium and (ii) sodium hydroxide

7.15 Due to electron withdrawing effect of nitro group and electron releasing effect of methoxy group.

7.20 (i) Hydration of Propene.

(ii) By nucleophilic substitution of $-\text{Cl}$ in benzyl chloride using dilute NaOH .



7.23 (i) 1-Ethoxy-2-methylpropane.

(ii) 2-Chloro-1-methoxyethane.

(iii) 4-Nitroanisole.

(iv) 1-Methoxypropane.

(v) 1-Ethoxy-4,4-dimethylcyclohexane.

(vi) Ethoxybenzene.

UNIT 8

8.2 (i) 4-Methylpentanal

(iii) But-2-enal

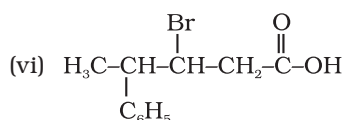
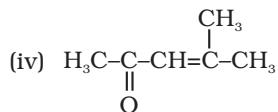
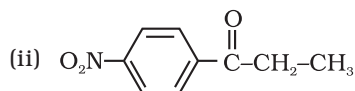
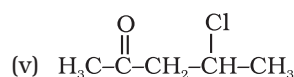
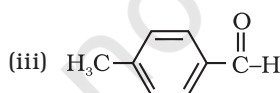
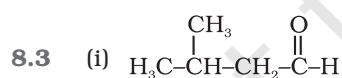
(v) 3,3,5-Trimethylhexan-2-one

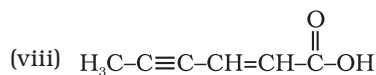
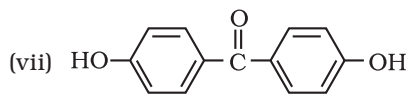
(vii) Benzene -1,4-dicarbaldehyde

(ii) 6-Chloro-4-ethylhexan-3-one

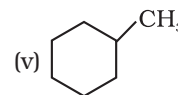
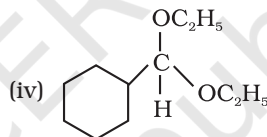
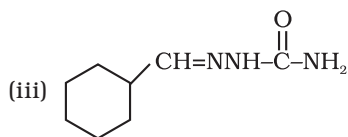
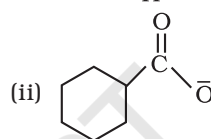
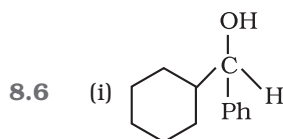
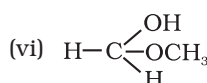
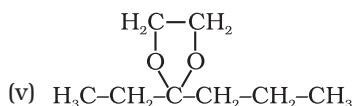
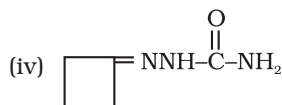
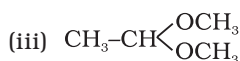
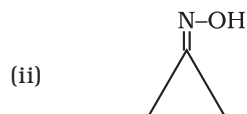
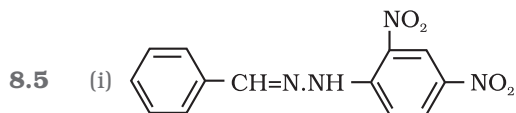
(iv) Pentane-2,4-dione

(vi) 3,3-Dimethylbutanoic acid





- 8.4 (i) Heptan-2-one (ii) 4-Bromo-2-methylhexanal (iii) Heptanal
(iv) 3-Phenylprop-2-enal (v) Cyclopentanecarbaldehyde (vi) Diphenylmethanone



- 8.7 (ii), (v), (vi), (vii): Aldol condensation. (i), (iii), (ix) Cannizzaro reaction. (iv), (viii) Neither.

8.10 2-Ethylbenzaldehyde (draw the structure yourself).

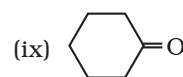
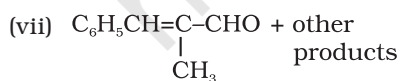
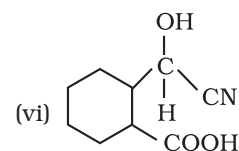
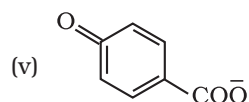
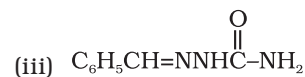
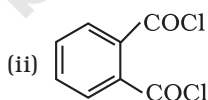
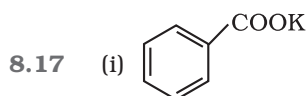
8.11 (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$, butyl butanoate.

(B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ (C) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$. Write equation yourself.

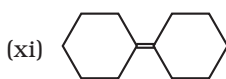
8.12 (i) Di-tert-butyl ketone < Methyl tert-butyl ketone < Acetone < Acetaldehyde

(ii) $(\text{CH}_3)_2\text{CHCOOH}$ < $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ < $\text{CH}_3\text{CH}(\text{Br})\text{CH}_2\text{COOH}$ < $\text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{COOH}$

(iii) 4-Methoxybenzoic acid < Benzoic acid < 4-Nitrobenzoic acid < 3,4-Dinitrobenzoic acid.



(x) 1. BH_3 ; 2. $\text{H}_2\text{O}_2/\text{OH}^-$; 3. PCC



8.19 The compound is methyl ketone and its structure would be: $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$

UNIT 9

- 9.1** (i) 1-methylethylamine or propan-2-amine (ii) Propan-1-amine
(iii) N-methyl-2-methylethylamine or N-methylpropan-2-amine (iv) 2-methylpropan-2-amine
(v) N-methylbenzenamine or N-methylaniline (vi) N-Ethyl-N-methylethanamine
(vii) 3-Bromoaniline or 3-Bromobenzenamine
- 9.4** (i) $\text{C}_6\text{H}_5\text{NH}_2 < \text{C}_6\text{H}_5\text{NHCH}_3 < \text{C}_2\text{H}_5\text{NH}_2 < (\text{C}_2\text{H}_5)_2\text{NH}$
(ii) $\text{C}_6\text{H}_5\text{NH}_2 < \text{C}_6\text{H}_5\text{N}(\text{CH}_3)_2 < \text{CH}_3\text{NH}_2 < (\text{C}_2\text{H}_5)_2\text{NH}$
(iii) (a) p-nitroaniline < aniline < p-toluidine
(b) $\text{C}_6\text{H}_5\text{NH}_2 < \text{C}_6\text{H}_5\text{NHCH}_3 < \text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$
(iv) $(\text{C}_2\text{H}_5)_3\text{N} > (\text{C}_2\text{H}_5)_2\text{NH} > \text{C}_2\text{H}_5\text{NH}_2 > \text{NH}_3$ (v) $(\text{CH}_3)_2\text{NH} < \text{C}_2\text{H}_5\text{NH}_2 < \text{C}_2\text{H}_5\text{OH}$
(vi) $\text{C}_6\text{H}_5\text{NH}_2 < (\text{C}_2\text{H}_5)_2\text{NH} < \text{C}_2\text{H}_5\text{NH}_2$

Notes

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Notes

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Chemistry

Part II
Textbook for Class XII



12086



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NATIONAL COUNCIL OF EDUCATIONAL RESEARCH AND TRAINING

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FOREWORD

The National Curriculum Framework (NCF), 2005 recommends that children's life at school must be linked to their life outside the school. This principle marks a departure from the legacy of bookish learning which continues to shape our system and causes a gap between the school, home and community. The syllabi and textbooks developed on the basis of NCF signify an attempt to implement this basic idea. They also attempt to discourage rote learning and the maintenance of sharp boundaries between different subject areas. We hope these measures will take us significantly further in the direction of a child-centred system of education outlined in the National Policy on Education (1986).

The success of this effort depends on the steps that school principals and teachers will take to encourage children to reflect on their own learning and to pursue imaginative activities and questions. We must recognise that, given space, time and freedom, children generate new knowledge by engaging with the information passed on to them by adults. Treating the prescribed textbook as the sole basis of examination is one of the key reasons why other resources and sites of learning are ignored. Inculcating creativity and initiative is possible if we perceive and treat children as participants in learning, not as receivers of a fixed body of knowledge.

These aims imply considerable change in school routines and mode of functioning. Flexibility in the daily time-table is as necessary as rigour in implementing the annual calendar so that the required number of teaching days are actually devoted to teaching. The methods used for teaching and evaluation will also determine how effective this textbook proves for making children's life at school a happy experience, rather than a source of stress or boredom. Syllabus designers have tried to address the problem of curricular burden by restructuring and reorienting knowledge at different stages with greater consideration for child psychology and the time available for teaching. The textbook attempts to enhance this endeavour by giving higher priority and space to opportunities for contemplation and wondering, discussion in small groups, and activities requiring hands-on experience.

The National Council of Educational Research and Training (NCERT) appreciates the hard work done by the textbook development committee responsible for this book. We wish to thank the Chairperson of the advisory group in science and mathematics, Professor J.V. Narlikar and the Chief Advisor for this book, Professor B. L. Khandelwal for guiding the work of this committee.

Several teachers contributed to the development of this textbook; we are grateful to their principals for making this possible. We are indebted to the institutions and organisations which have generously permitted us to draw upon their resources, material and personnel. As an organisation committed to systemic reform and continuous improvement in the quality of its products, NCERT welcomes comments and suggestions which will enable us to undertake further revision and refinement.

New Delhi
20 November 2006

Director
National Council of Educational
Research and Training

RATIONALISATION OF CONTENT IN THE TEXTBOOK

In view of the COVID-19 pandemic, it is imperative to reduce content load on students. The National Education Policy 2020, also emphasises reducing the content load and providing opportunities for experiential learning with creative mindset. In this background, the NCERT has undertaken the exercise to rationalise the textbooks across all classes. Learning Outcomes already developed by the NCERT across classes have been taken into consideration in this exercise.

Contents of the textbooks have been rationalised in view of the following:

- Overlapping with similar content included in other subject areas in the same class
- Similar content included in the lower or higher class in the same subject
- Difficulty level
- Content, which is easily accessible to students without much interventions from teachers and can be learned by children through self-learning or peer-learning
- Content, which is irrelevant in the present context

This present edition, is a reformatted version after carrying out the changes given above.

THE CONSTITUTION OF INDIA

PREAMBLE

WE, THE PEOPLE OF INDIA, having solemnly resolved to constitute India into a ¹**[SOVEREIGN SOCIALIST SECULAR DEMOCRATIC REPUBLIC]** and to secure to all its citizens :

JUSTICE, social, economic and political;

LIBERTY of thought, expression, belief, faith and worship;

EQUALITY of status and of opportunity; and to promote among them all

FRATERNITY assuring the dignity of the individual and the ²[unity and integrity of the Nation];

IN OUR CONSTITUENT ASSEMBLY this twenty-sixth day of November, 1949 do **HEREBY ADOPT, ENACT AND GIVE TO OURSELVES THIS CONSTITUTION.**

1. Subs. by the Constitution (Forty-second Amendment) Act, 1976, Sec.2, for "Sovereign Democratic Republic" (w.e.f. 3.1.1977)
2. Subs. by the Constitution (Forty-second Amendment) Act, 1976, Sec.2, for "Unity of the Nation" (w.e.f. 3.1.1977)

PREFACE

Chemistry has made a profound impact on the society. It is intimately linked to the well-being of human kind. The rate of advancements in chemistry is so high that curriculum developers continuously look for strategies to cope with these advancements. Also, the students have to be inspired to be the future leaders who would make fundamental contributions. The present textbook is a sincere effort in this direction.

The textbook, presented in two parts, comprises of sixteen Units. Although the titles of various Units indicate a sort of compartmentalisation into physical, inorganic and organic chemistry, readers will find that these sub-disciplines have been intermingled, at least to a certain extent, to have a unified approach to the subject. First nine Units covering physical and inorganic chemistry portions are included in Part I while organic chemistry portion comprising of seven Units is included in Part II of the book. The approach of presentation of the subject matter discourages students from rote memorisation. The subject has in fact, been organised around the laws and principles of chemistry. As students master these laws and principles, they will soon get to the point where they can predict much of what will come.

Efforts have been directed towards making the subject stimulating and exciting by references to the historical developments and its usefulness to our lives, wherever appropriate. The text is well illustrated with examples from surrounding environment to facilitate grasping of the qualitative and quantitative aspects of the concept easily. Physical data are given in SI units throughout the book to make comparison of various properties easier. IUPAC system of nomenclature has been followed along with the common names. Structural formulae of chemical compounds showing functional/coordinating groups in different colours are drawn using electronic system. Each Unit has a good number of examples, as illustrations, with their solutions and some intext questions, the answers of some of which are given at the end of the Unit. The end of Unit exercises are designed to apply important principles and provoke thinking process to solve them. Answers of some of these exercises are given at the end of the book.

A variety of materials, e.g., biographical sketches of some scientists, additional information related to a particular topic, etc., is given in boxes with a deep yellow coloured bar. This boxed material with a 'deep yellow bar' is to bring additional life to the topic. However, it is non-evaluative. The structures of some of the more complex compounds incorporated in the book are for understanding their chemistry. As their reproduction would lead to memorisation, it is also a non-evaluative portion of the text.

The information part has been significantly reduced and, wherever possible, it has been substantiated with facts. However, it is necessary for students to be aware of commercially important chemicals, their processes of manufacture and sources of raw materials. This leads to descriptive material in the book. Attempts have been made to make descriptions of such compounds interesting by considering their structures and reactivity. Thermodynamics, kinetics and electrochemical aspects have been applied to a few chemical reactions which should be beneficial to students for understanding why a particular reaction happened and why a particular property is exhibited by the product. There is currently great awareness of environmental and energy issues which are directly related to chemistry. Such issues have been highlighted and dealt with at appropriate places in the book.

A team of experts constituted by the NCERT has developed the manuscript of the book. It gives me great pleasure to acknowledge the valuable contribution of all the members of this team. I also acknowledge the valuable and relentless contribution of the editors in bringing the book to the present shape. I also acknowledge with thanks the dedicated efforts and valuable contribution of Professor Brahm Parkash, who not only coordinated the entire programme but also actively involved in writing and editing of this book. Thanks are also due to the participating teachers and subject experts of the review workshop for their contribution, which has helped us to make the book learner friendly. Also, I thank the technical and administrative staff of the NCERT for their support in the entire process.

The team of this textbook development programme hopes that the book stimulates its readers and makes them feel the excitement and fascination for this subject. Efforts have been made to bring out this book error-free. Nevertheless, it is recognised that in a book of this complexity, there could inevitably be occasional errors. It will always be a pleasure to hear about them from readers to take necessary steps to rectify them.

B.L. KHANDELWAL

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Objectives

After studying this Unit, you will be able to

- name haloalkanes and haloarenes according to the IUPAC system of nomenclature from their given structures;
- describe the reactions involved in the preparation of haloalkanes and haloarenes and understand various reactions that they undergo;
- correlate the structures of haloalkanes and haloarenes with various types of reactions;
- use stereochemistry as a tool for understanding the reaction mechanism;
- appreciate the applications of organo-metallic compounds;
- highlight the environmental effects of polyhalogen compounds.

Unit

6

Haloalkanes and Haloarenes

Halogenated compounds persist in the environment due to their resistance to breakdown by soil bacteria.

The replacement of hydrogen atom(s) in an aliphatic or aromatic hydrocarbon by halogen atom(s) results in the formation of alkyl halide (haloalkane) and aryl halide (haloarene), respectively. Haloalkanes contain halogen atom(s) attached to the sp^3 hybridised carbon atom of an alkyl group whereas haloarenes contain halogen atom(s) attached to sp^2 hybridised carbon atom(s) of an aryl group. Many halogen containing organic compounds occur in nature and some of these are clinically useful. These classes of compounds find wide applications in industry as well as in day-to-day life. They are used as solvents for relatively non-polar compounds and as starting materials for the synthesis of wide range of organic compounds. Chlorine containing antibiotic, *chloramphenicol*, produced by microorganisms is very effective for the treatment of typhoid fever. Our body produces iodine containing hormone, *thyroxine*, the deficiency of which causes a disease called *goiter*. Synthetic halogen compounds, *viz.* chloroquine is used for the treatment of malaria; halothane is used as an anaesthetic during surgery. Certain fully fluorinated compounds are being considered as potential blood substitutes in surgery.

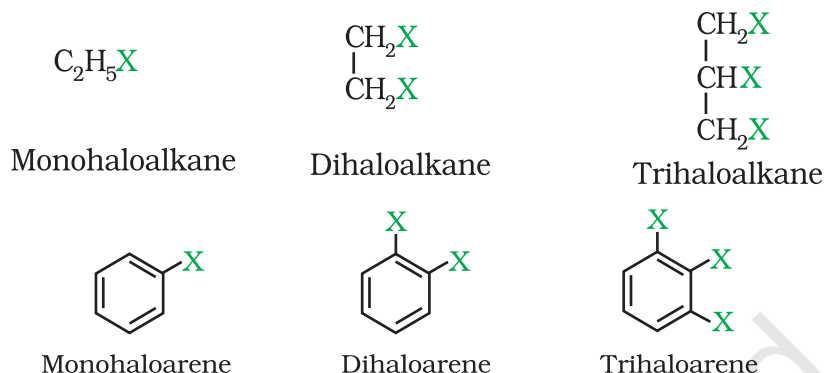
In this Unit, you will study the important methods of preparation, physical and chemical properties and uses of organohalogen compounds.

6.1 Classification

6.1.1 On the Basis of Number of Halogen Atoms

Haloalkanes and haloarenes may be classified as follows:

These may be classified as mono, di, or polyhalogen (tri-,tetra-, etc.) compounds depending on whether they contain one, two or more halogen atoms in their structures. For example,



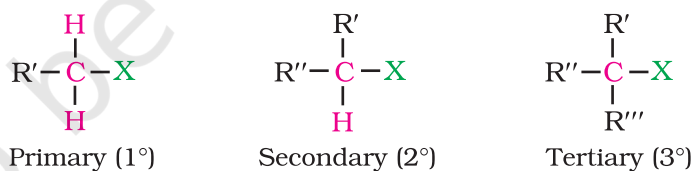
Monohalocompounds may further be classified according to the hybridisation of the carbon atom to which the halogen is bonded, as discussed below.

6.1.2 Compounds Containing $\text{sp}^3\text{C}-\text{X}$ Bond ($\text{X} = \text{F}, \text{Cl}, \text{Br}, \text{I}$)

This class includes

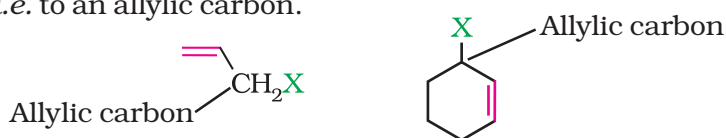
(a) Alkyl halides or haloalkanes ($\text{R}-\text{X}$)

In alkyl halides, the halogen atom is bonded to an alkyl group (R). They form a homologous series represented by $\text{C}_n\text{H}_{2n+1}\text{X}$. They are further classified as primary, secondary or tertiary according to the nature of carbon to which halogen is attached. If halogen is attached to a primary carbon atom in an alkyl halide, the alkyl halide is called primary alkyl halide or 1° alkyl halide. Similarly, if halogen is attached to secondary or tertiary carbon atom, the alkyl halide is called secondary alkyl halide (2°) and tertiary (3°) alkyl halide, respectively.



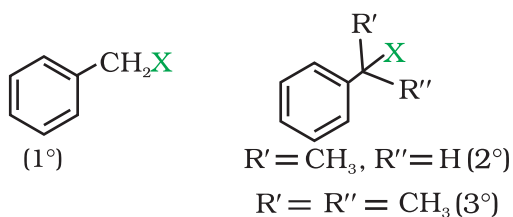
(b) Allylic halides

These are the compounds in which the halogen atom is bonded to an sp^3 -hybridised carbon atom adjacent to carbon-carbon double bond ($\text{C}=\text{C}$) i.e. to an allylic carbon.



(c) Benzylic halides

These are the compounds in which the halogen atom is bonded to an sp^3 -hybridised carbon atom attached to an aromatic ring.



6.1.3 Compounds Containing $\text{sp}^2\text{C-X}$ Bond

This class includes:

(a) Vinylic halides

These are the compounds in which the halogen atom is bonded to a sp^2 -hybridised carbon atom of a carbon-carbon double bond ($\text{C} = \text{C}$).



(b) Aryl halides

These are the compounds in which the halogen atom is directly bonded to the sp^2 -hybridised carbon atom of an aromatic ring.

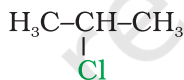


6.2 Nomenclature

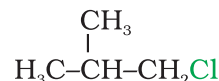
Having learnt the classification of halogenated compounds, let us now learn how these are named. The common names of alkyl halides are derived by naming the alkyl group followed by the name of halide. In the IUPAC system of nomenclature, alkyl halides are named as halosubstituted hydrocarbons. For mono halogen substituted derivatives of benzene, common and IUPAC names are the same. For dihalogen derivatives, the prefixes *o*-, *m*-, *p*- are used in common system but in IUPAC system, as you have learnt in Class XI, the numerals 1,2; 1,3 and 1,4 are used.



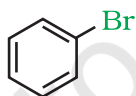
Common name: n-Propyl bromide
 IUPAC name: 1-Bromopropane



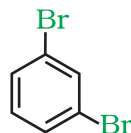
Isopropyl chloride
 2-Chloropropane



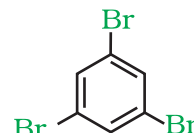
Isobutyl chloride
 1-Chloro-2-methylpropane



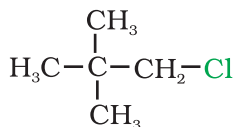
Common name: Bromobenzene
 IUPAC name: Bromobenzene



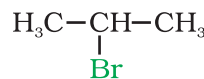
m-Dibromobenzene
 1,3-Dibromobenzene



sym-Tribromobenzene
 1,3,5-Tribromobenzene



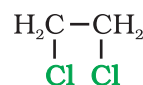
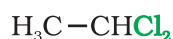
IUPAC name: 1-Chloro-2,2-dimethylpropane



2-Bromopropane

The dihaloalkanes having the same type of halogen atoms are named as alkylidene or alkylene dihalides. The dihalo-compounds having both the halogen atoms are further classified as geminal halides or gem-dihalides when both the halogen atoms are present on the same carbon atom of the

chain and vicinal halides or vic-dihalides when halogen atoms are present on adjacent carbon atoms. In common name system, gem-dihalides are named as alkylidene halides and vic-dihalides are named as alkylene dihalides. In IUPAC system, they are named as dihaloalkanes.



Common name: Ethylidene chloride
(gem-dihalide)

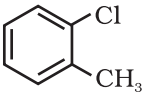
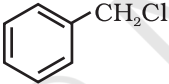
Ethylene dichloride
(vic-dihalide)

IUPAC name: 1, 1-Dichloroethane

1, 2-Dichloroethane

Some common examples of halocompounds are mentioned in Table 6.1.

Table 6.1: Common and IUPAC Names of some Halides

Structure	Common name	IUPAC name
$\text{CH}_3\text{CH}_2\text{CH}(\text{Cl})\text{CH}_3$	sec-Butyl chloride	2-Chlorobutane
$(\text{CH}_3)_3\text{CCH}_2\text{Br}$	neo-Pentyl bromide	1-Bromo-2,2-dimethylpropane
$(\text{CH}_3)_3\text{CBr}$	tert-Butyl bromide	2-Bromo-2-methylpropane
$\text{CH}_2=\text{CHCl}$	Vinyl chloride	Chloroethene
$\text{CH}_2=\text{CHCH}_2\text{Br}$	Allyl bromide	3-Bromopropene
	<i>o</i> -Chlorotoluene	1-Chloro-2-methylbenzene or 2-Chlorotoluene
	Benzyl chloride	Chlorophenylmethane
CH_2Cl_2	Methylene chloride	Dichloromethane
CHCl_3	Chloroform	Trichloromethane
CHBr_3	Bromoform	Tribromomethane
CCl_4	Carbon tetrachloride	Tetrachloromethane
$\text{CH}_3\text{CH}_2\text{CH}_2\text{F}$	n-Propyl fluoride	1-Fluoropropane

Example 6.1

Draw the structures of all the eight structural isomers that have the molecular formula $\text{C}_5\text{H}_{11}\text{Br}$. Name each isomer according to IUPAC system and classify them as primary, secondary or tertiary bromide.

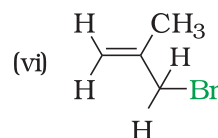
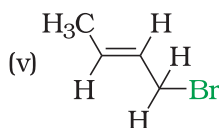
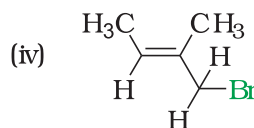
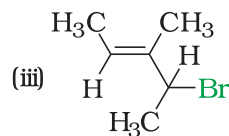
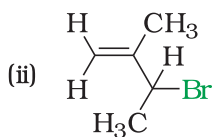
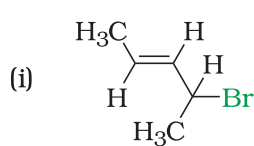
Solution

- | | |
|--|--------------------------------------|
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ | 1-Bromopentane (1°) |
| $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{Br})\text{CH}_3$ | 2-Bromopentane (2°) |
| $\text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{CH}_2\text{CH}_3$ | 3-Bromopentane (2°) |
| $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{Br}$ | 1-Bromo-3-methylbutane (1°) |

$(\text{CH}_3)_2\text{CHCHBrCH}_3$	2-Bromo-3-methylbutane(2°)
$(\text{CH}_3)_2\text{CBrCH}_2\text{CH}_3$	2-Bromo-2-methylbutane (3°)
$\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{Br}$	1-Bromo-2-methylbutane(1°)
$(\text{CH}_3)_3\text{CCH}_2\text{Br}$	1-Bromo-2,2-dimethylpropane (1°)

Write IUPAC names of the following:

Example 6.2



- (i) 4-Bromopent-2-ene (ii) 3-Bromo-2-methylbut-1-ene
 (iii) 4-Bromo-3-methylpent-2-ene (iv) 1-Bromo-2-methylbut-2-ene
 (v) 1-Bromobut-2-ene (vi) 3-Bromo-2-methylpropene

Solution

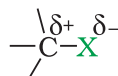
Intext Question

6.1 Write structures of the following compounds:

- 2-Chloro-3-methylpentane
- 1-Chloro-4-ethylcyclohexane
- 4-tert. Butyl-3-iodoheptane
- 1,4-Dibromobut-2-ene
- 1-Bromo-4-sec. butyl-2-methylbenzene.

6.3 Nature of C-X Bond

Halogen atoms are more electronegative than carbon, therefore, carbon-halogen bond of alkyl halide is polarised; the carbon atom bears a partial positive charge whereas the halogen atom bears a partial negative charge.



As we go down the group in the periodic table, the size of halogen atom increases. Fluorine atom is the smallest and iodine atom is the largest. Consequently the carbon-halogen bond length also increases from C—F to C—I. Some typical bond lengths, bond enthalpies and dipole moments are given in Table 6.2.

Alkyl halides are best prepared from alcohols, which are easily accessible.

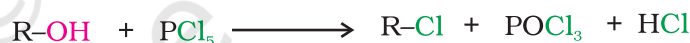
Table 6.2: Carbon-Halogen (C—X) Bond Lengths, Bond Enthalpies and Dipole Moments

Bond	Bond length/pm	C-X Bond enthalpies/ kJmol ⁻¹	Dipole moment/Debye
CH ₃ —F	139	452	1.847
CH ₃ —Cl	178	351	1.860
CH ₃ —Br	193	293	1.830
CH ₃ —I	214	234	1.636

6.4 Methods of Preparation of Haloalkanes

6.4.1 From Alcohols

The hydroxyl group of an alcohol is replaced by halogen on reaction with concentrated halogen acids, phosphorus halides or thionyl chloride. Thionyl chloride is preferred because in this reaction alkyl halide is formed along with gases SO₂ and HCl. The two gaseous products are escapable, hence, the reaction gives pure alkyl halides. The reactions of primary and secondary alcohols with HCl require the presence of a catalyst, ZnCl₂. With tertiary alcohols, the reaction is conducted by simply shaking the alcohol with concentrated HCl at room temperature. Constant boiling with HBr (48%) is used for preparing alkyl bromide. Good yields of R—I may be obtained by heating alcohols with sodium or potassium iodide in 95% orthophosphoric acid. The order of reactivity of alcohols with a given haloacid is 3° > 2° > 1°. Phosphorus tribromide and triiodide are usually generated *in situ* (produced in the reaction mixture) by the reaction of red phosphorus with bromine and iodine respectively.



The preparation of alkyl chloride is carried out either by passing dry hydrogen chloride gas through a solution of alcohol or by heating a mixture of alcohol and concentrated aqueous halogen acid.

The above methods are not applicable for the preparation of aryl halides because the carbon-oxygen bond in phenols has a partial double bond character and is difficult to break being stronger than a single bond.

6.4.2 From Hydrocarbons

(I) From alkanes by free radical halogenation

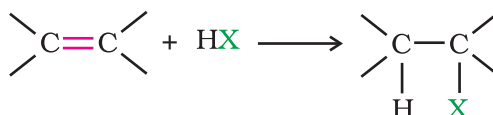
Free radical chlorination or bromination of alkanes gives a complex mixture of isomeric mono- and polyhaloalkanes, which is difficult to

separate as pure compounds. Consequently, the yield of any single compound is low.

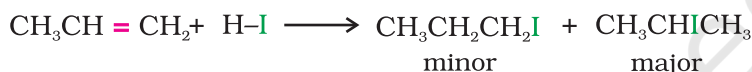


(II) From alkenes

- (i) *Addition of hydrogen halides:* An alkene is converted to corresponding alkyl halide by reaction with hydrogen chloride, hydrogen bromide or hydrogen iodide.



Propene yields two products, however only one predominates as per Markovnikov's rule. (Unit 13, Class XI)



- (ii) *Addition of halogens:* In the laboratory, addition of bromine in CCl_4 to an alkene resulting in discharge of reddish brown colour of bromine constitutes an important method for the detection of double bond in a molecule. The addition results in the synthesis of *vic*-dibromides, which are colourless (Unit 9, Class XI).



Identify all the possible monochloro structural isomers expected to be formed on free radical monochlorination of $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_3$.

Example 6.3

In the given molecule, there are four different types of hydrogen atoms. Replacement of these hydrogen atoms will give the following

Solution



6.4.3 Halogen Exchange

Alkyl iodides are often prepared by the reaction of alkyl chlorides/bromides with NaI in dry acetone. This reaction is known as **Finkelstein** reaction.



NaCl or NaBr thus formed is precipitated in dry acetone. It facilitates the forward reaction according to Le Chatelier's Principle.

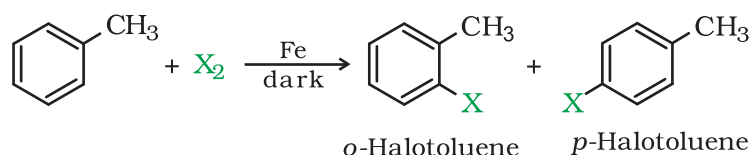
The synthesis of alkyl fluorides is best accomplished by heating an alkyl chloride/bromide in the presence of a metallic fluoride such as

AgF, Hg₂F₂, CoF₂ or SbF₃. The reaction is termed as **Swarts** reaction.



6.5 Preparation of (i) **From hydrocarbons by electrophilic substitution** Haloarenes

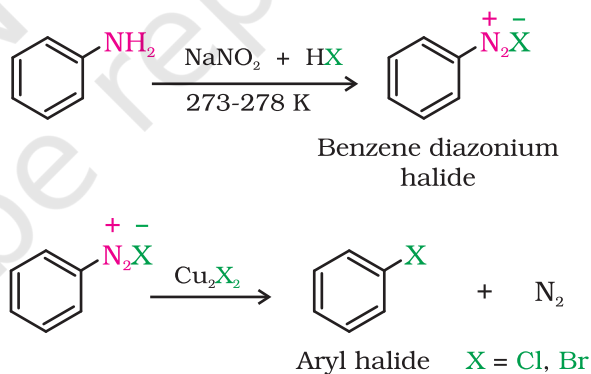
Aryl chlorides and bromides can be easily prepared by electrophilic substitution of arenes with chlorine and bromine respectively in the presence of Lewis acid catalysts like iron or iron(III) chloride.



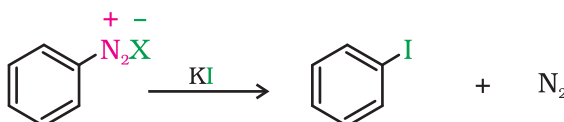
The *ortho* and *para* isomers can be easily separated due to large difference in their melting points. Reactions with iodine are reversible in nature and require the presence of an oxidising agent (HNO₃, HIO₄) to oxidise the HI formed during iodination. Fluoro compounds are not prepared by this method due to high reactivity of fluorine.

(ii) **From amines by Sandmeyer's reaction**

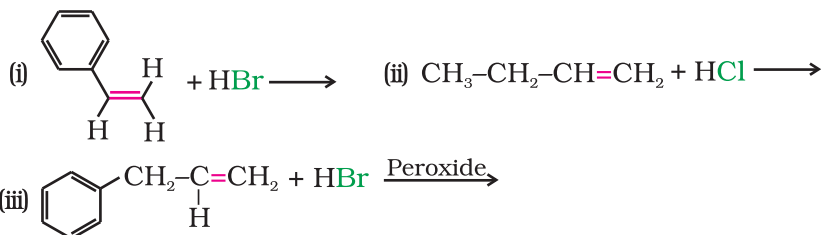
When a primary aromatic amine, dissolved or suspended in cold aqueous mineral acid, is treated with sodium nitrite, a diazonium salt is formed. Mixing the solution of freshly prepared diazonium salt with cuprous chloride or cuprous bromide results in the replacement of the diazonium group by -Cl or -Br.



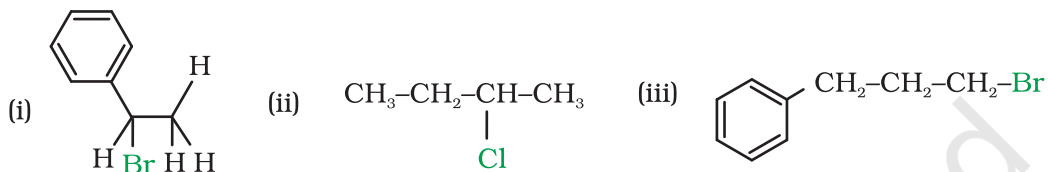
Replacement of the diazonium group by iodine does not require the presence of cuprous halide and is done simply by shaking the diazonium salt with potassium iodide.



Example 6.4 Write the products of the following reactions:

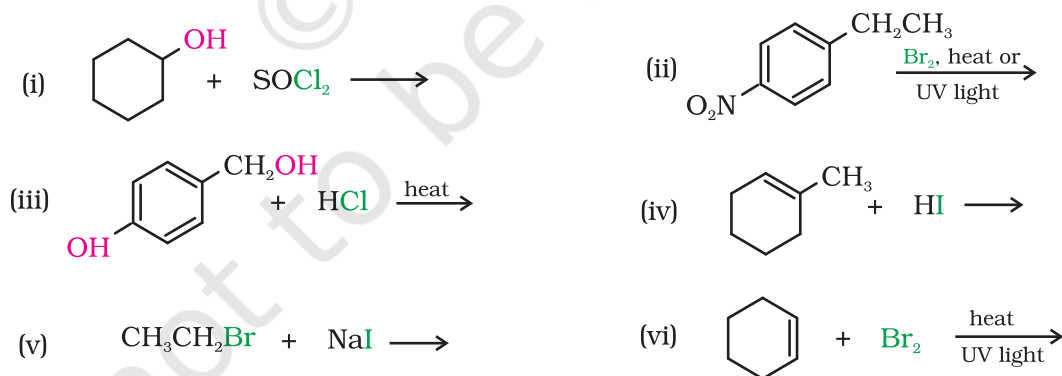


Solution



Intext Questions

- 6.2** Why is sulphuric acid not used during the reaction of alcohols with KI?
- 6.3** Write structures of different dihalogen derivatives of propane.
- 6.4** Among the isomeric alkanes of molecular formula C_5H_{12} , identify the one that on photochemical chlorination yields
- A single monochloride.
 - Three isomeric monochlorides.
 - Four isomeric monochlorides.
- 6.5** Draw the structures of major monohalo products in each of the following reactions:



6.6 Physical Properties

Alkyl halides are colourless when pure. However, bromides and iodides develop colour when exposed to light. Many volatile halogen compounds have sweet smell.

Melting and boiling points

Methyl chloride, methyl bromide, ethyl chloride and some chlorofluoromethanes are gases at room temperature. Higher members are liquids or solids. As we have already learnt, molecules of organic halogen compounds are generally polar. Due to greater polarity as well as higher molecular mass as compared to the parent hydrocarbon, the intermolecular forces of attraction (dipole-dipole and van der Waals) are stronger in the halogen derivatives. That is why the boiling points of chlorides, bromides and iodides are considerably higher than those of the hydrocarbons of comparable molecular mass.

The attractions get stronger as the molecules get bigger in size and have more electrons. The pattern of variation of boiling points of different halides is depicted in Fig. 6.1. For the same alkyl group, the boiling points of alkyl halides decrease in the order: RI > RBr > RCl > RF. This is because with the increase in size and mass of halogen atom, the magnitude of van der Waals forces increases.

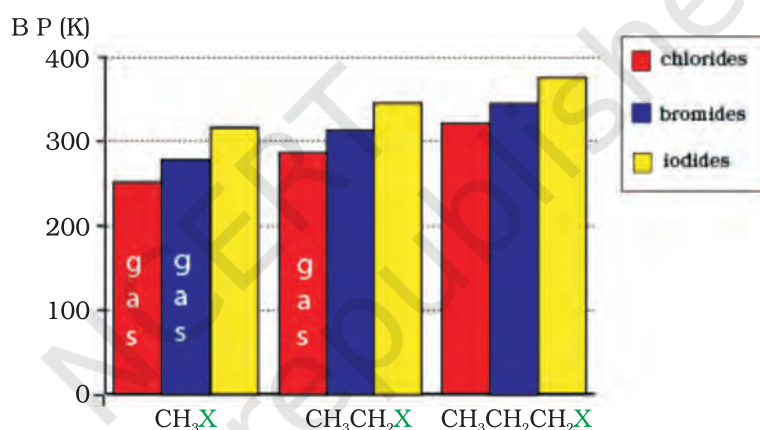
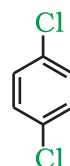
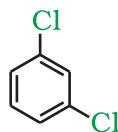
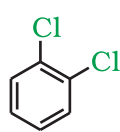


Fig. 6.1: Comparison of boiling points of some alkyl halides

The boiling points of isomeric haloalkanes decrease with increase in branching. For example, 2-bromo-2-methylpropane has the lowest boiling point among the three isomers.

$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$	$\text{CH}_3\text{CH}_2\underset{\text{Br}}{\text{CH}}\text{CH}_3$	$\text{H}_3\text{C}-\overset{\text{CH}_3}{\underset{\text{Br}}{\text{C}}}-\text{CH}_3$
b.p./K 375	364	346

Boiling points of isomeric dihalobenzenes are very nearly the same. However, the *para*-isomers are high melting as compared to their *ortho*- and *meta*-isomers. It is due to symmetry of *para*-isomers that fits in crystal lattice better as compared to *ortho*- and *meta*-isomers.



b.p /K	453	446	448
m.p/K	256	249	323

Density

Bromo, iodo and polychloro derivatives of hydrocarbons are heavier than water. The density increases with increase in number of carbon atoms, halogen atoms and atomic mass of the halogen atoms (Table 6.3).

Table 6.3: Density of Some Haloalkanes

Compound	Density (g/mL)	Compound	Density (g/mL)
n-C ₃ H ₇ Cl	0.89	CH ₂ Cl ₂	1.336
n-C ₃ H ₇ Br	1.335	CHCl ₃	1.489
n-C ₃ H ₇ I	1.747	CCl ₄	1.595

Solubility

The haloalkanes are very slightly soluble in water. In order to dissolve haloalkane in water, energy is required to overcome the attractions between the haloalkane molecules and break the hydrogen bonds between water molecules. Less energy is released when new attractions are set up between the haloalkane and the water molecules as these are not as strong as the original hydrogen bonds in water. As a result, the solubility of haloalkanes in water is low. However, haloalkanes tend to dissolve in organic solvents because the new intermolecular attractions between haloalkanes and solvent molecules have much the same strength as the ones being broken in the separate haloalkane and solvent molecules.

Intext Question

- 6.6** Arrange each set of compounds in order of increasing boiling points.
- Bromomethane, Bromoform, Chloromethane, Dibromomethane.
 - 1-Chloropropane, Isopropyl chloride, 1-Chlorobutane.

6.7 Chemical Reactions

The reactions of haloalkanes may be divided into the following categories:

- Nucleophilic substitution
- Elimination reactions
- Reaction with metals.

(1) Nucleophilic substitution reactions

You have learnt in Class XI that nucleophiles are electron rich species. Therefore, they attack at that part of the substrate molecule which is electron deficient. The reaction in which a nucleophile replaces

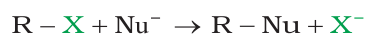
6.7.1 Reactions of Haloalkanes

already existing nucleophile in a molecule is called nucleophilic substitution reaction. Haloalkanes are substrate in these reactions. In this type of reaction, a nucleophile reacts with haloalkane (the substrate) having a partial positive charge on the carbon atom bonded to halogen. A substitution reaction takes place and halogen atom, called leaving group departs as halide ion. Since the substitution reaction is initiated by a nucleophile, it is called nucleophilic substitution reaction.



It is one of the most useful classes of organic reactions of alkyl halides in which halogen is bonded to sp^3 hybridised carbon. The products formed by the reaction of haloalkanes with some common nucleophiles are given in Table 6.4.

Table 6.4: Nucleophilic Substitution of Alkyl Halides (R-X)



Reagent	Nucleophile (Nu ⁻)	Substitution product R-Nu	Class of main product
NaOH (KOH)	HO ⁻	ROH	Alcohol
H ₂ O	H ₂ O	ROH	Alcohol
NaOR'	R'O ⁻	ROR'	Ether
NaI	I ⁻	R-I	Alkyl iodide
NH ₃	NH ₃	RNH ₂	Primary amine
R'NH ₂	R'NH ₂	RNHR'	Sec. amine
R'R''NH	R'R''NH	RNR'R''	Tert. amine
KCN	$\text{C}\equiv\text{N:}^-$	RCN	Nitrile (cyanide)
AgCN	Ag-CN:	RNC (isocyanide)	Isonitrile
KNO ₂	O=N-O ⁻	R-O-N=O	Alkyl nitrite
AgNO ₂	Ag- $\ddot{\text{O}}$ -N=O	R-NO ₂	Nitroalkane
R'COOAg	R'COO ⁻	R'COOR	Ester
LiAlH ₄	H ⁻	RH	Hydrocarbon
R ⁻ M ⁺	R ⁻	RR'	Alkane

Groups like cyanides and nitrites possess two nucleophilic centres and are called **ambident nucleophiles**. Actually cyanide group is a hybrid of two contributing structures and therefore can act as a nucleophile in two different ways [$\text{C}\equiv\text{N}^- \leftrightarrow \text{:C}=\text{N}^\ominus$], i.e., linking through

Haloalkanes react with KCN to form alkyl cyanides as main product while AgCN forms isocyanides as the chief product. Explain.

Example 6.5

KCN is predominantly ionic and provides cyanide ions in solution. Although both carbon and nitrogen atoms are in a position to donate electron pairs, the attack takes place mainly through carbon atom and not through nitrogen atom since C—C bond is more stable than C—N bond. However, AgCN is mainly covalent in nature and nitrogen is free to donate electron pair forming isocyanide as the main product.

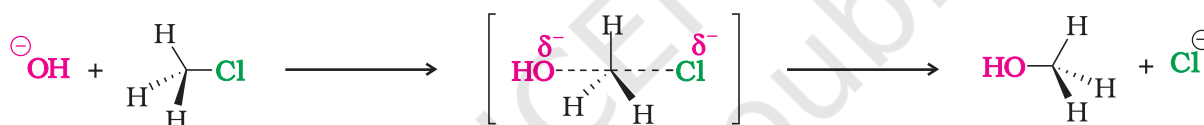
Solution

carbon atom resulting in alkyl cyanides and through nitrogen atom leading to isocyanides. Similarly nitrite ion also represents an ambident nucleophile with two different points of linkage [$\text{O}=\text{N}-\text{O}$]. The linkage through oxygen results in alkyl nitrites while through nitrogen atom, it leads to nitroalkanes.

Mechanism: This reaction has been found to proceed by two different mechanisms which are described below:

(a) Substitution nucleophilic bimolecular ($\text{S}_{\text{N}}2$)

The reaction between CH_3Cl and hydroxide ion to yield methanol and chloride ion follows a second order kinetics, i.e., the rate depends upon the concentration of both the reactants.



The solid wedge represents the bond coming out of the paper, dashed line going down the paper and a straight line representing bond in the plane of the paper.

The above reaction can be represented diagrammatically as shown in Fig. 6.2.

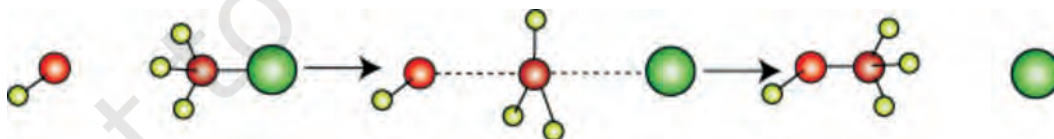


Fig. 6.2: Red ball represents the incoming hydroxide ion and green ball represents the outgoing halide ion

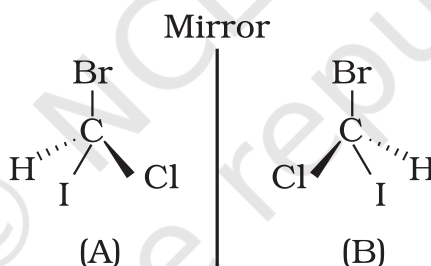
In the year 1937, Edward Davies Hughes and Sir Christopher Ingold proposed a mechanism for an $\text{S}_{\text{N}}2$ reaction.

It depicts a bimolecular nucleophilic substitution ($\text{S}_{\text{N}}2$) reaction; the incoming nucleophile interacts with alkyl halide causing the carbon-halide bond to break and a new bond is formed between carbon and attacking nucleophile. Here it is C—O bond formed between C and $-\text{OH}$. These two processes take place simultaneously in a

single step and no intermediate is formed. As the reaction progresses and the bond between the incoming nucleophile and the carbon atom starts forming, the bond between carbon atom and leaving group weakens. As this happens, the three carbon-hydrogen bonds of the substrate start moving away from the attacking nucleophile. In transition state all the three C-H bonds are in the same plane and the attacking and leaving nucleophiles are partially attached to the carbon. As the attacking nucleophile approaches closer to the carbon, C-H bonds still keep on moving in the same direction till the attacking nucleophile attaches to carbon and leaving group leaves the carbon. As a result configuration is inverted, the configuration (See box) of carbon atom under attack inverts in much the same way as an umbrella is turned inside out when caught in a strong wind. This process is called as **inversion of configuration**. In the transition state, the carbon atom is simultaneously bonded to incoming nucleophile and the outgoing leaving group. Such structures are unstable and cannot be isolated. Thus, in the transition state, carbon is simultaneously bonded to five atoms.

Configuration

Spatial arrangement of functional groups around carbon is called its configuration. See the structures (A) and (B) given below carefully.



These are the two structures of the same compound. They differ in spatial arrangement of functional groups attached to carbon. Structure (A) is mirror image of Structure (B). We say configuration of carbon in structure (A) is mirror image of the configuration of carbon in structure (B).

Hughes worked under Ingold and earned a D.Sc. degree from the University of London.

Since this reaction requires the approach of the nucleophile to the carbon bearing the leaving group, the presence of bulky substituents on or near the carbon atom have a dramatic inhibiting effect. Of the simple alkyl halides, methyl halides react most rapidly in S_N2 reactions because there are only three small hydrogen atoms. Tertiary halides are the least reactive because bulky groups hinder the approaching

nucleophiles. Thus the order of reactivity followed is:
Primary halide > Secondary halide > Tertiary halide.

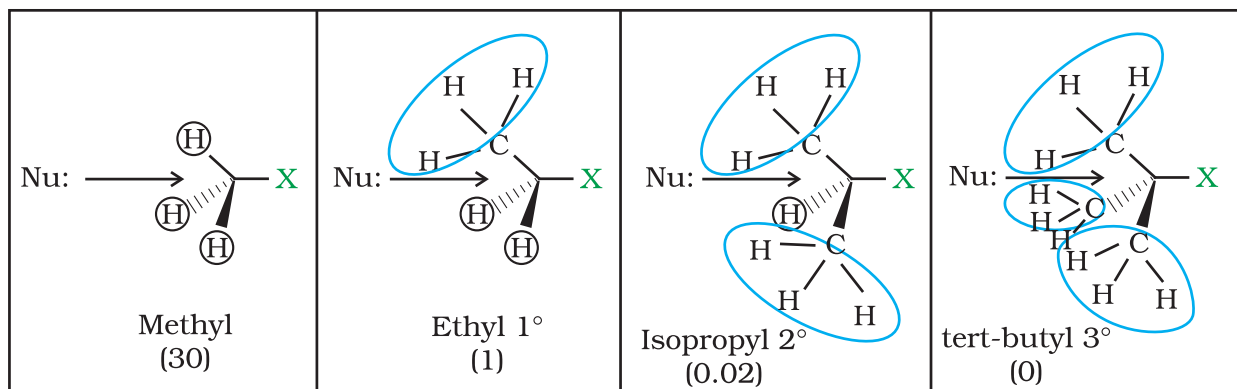


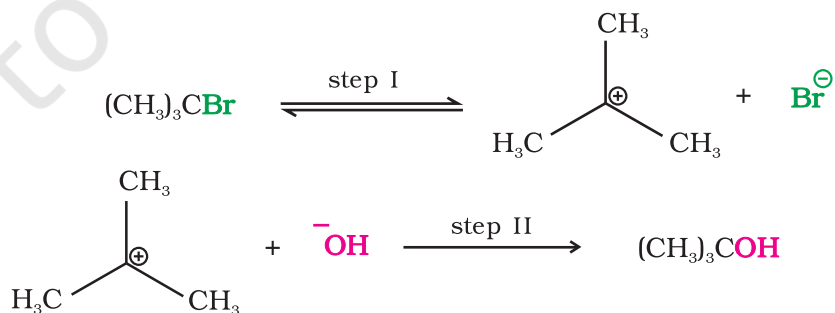
Fig.6.3: Steric effects in S_N2 reaction. The relative rate of S_N2 reaction is given in parenthesis

(b) Substitution nucleophilic unimolecular (S_N1)

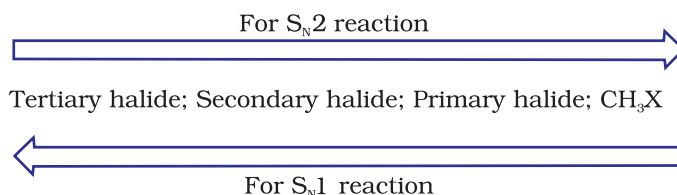
S_N1 reactions are generally carried out in polar protic solvents (like water, alcohol, acetic acid, etc.). The reaction between *tert*-butyl bromide and hydroxide ion yields *tert*-butyl alcohol and follows the first order kinetics, *i.e.*, the rate of reaction depends upon the concentration of only one reactant, which is *tert*-butyl bromide.



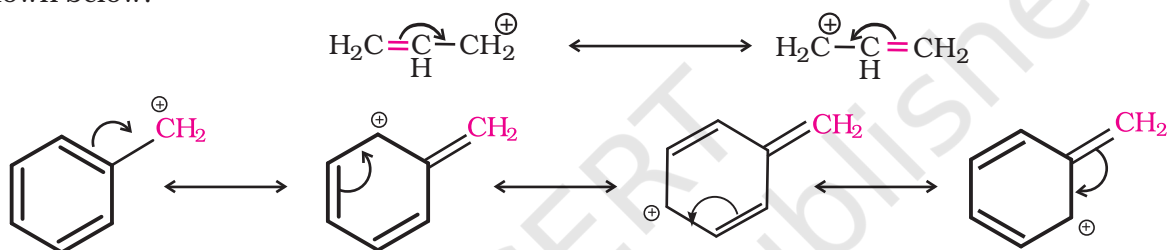
It occurs in two steps. In step I, the polarised C—Br bond undergoes slow cleavage to produce a carbocation and a bromide ion. The carbocation thus formed is then attacked by nucleophile in step II to complete the substitution reaction.



Step I is the slowest and reversible. It involves the C–Br bond breaking for which the energy is obtained through solvation of halide ion with the proton of protic solvent. Since the rate of reaction depends upon the slowest step, the rate of reaction depends only on the concentration of alkyl halide and not on the concentration of hydroxide ion. Further, greater the stability of carbocation, greater will be its ease of formation from alkyl halide and faster will be the rate of reaction. In case of alkyl halides, 3° alkyl halides undergo S_N1 reaction very fast because of the high stability of 3° carbocations. We can sum up the order of reactivity of alkyl halides towards S_N1 and S_N2 reactions as follows:



For the same reasons, allylic and benzylic halides show high reactivity towards the S_N1 reaction. The carbocation thus formed gets stabilised through resonance (Unit 8, Class XI) as shown below:



For a given alkyl group, the reactivity of the halide, $R-X$, follows the same order in both the mechanisms $R-I > R-Br > R-Cl >> R-F$.

In the following pairs of halogen compounds, which would undergo S_N2 reaction faster? Example 6.6



C1CCCCC1CH2Cl It is primary halide and therefore undergoes S_N2 reaction faster. Solution

CCCCI As iodine is a better leaving group because of its large size, it will be released at a faster rate in the presence of incoming nucleophile.

Predict the order of reactivity of the following compounds in S_N1 and S_N2 reactions: Example 6.7

(i) The four isomeric bromobutanes

(ii) C6H5CH2Br, C6H5CH(C6H5)Br, C6H5CH(CH3)Br, C6H5C(CH3)(C6H5)Br

Solution (i) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br} < (\text{CH}_3)_2\text{CHCH}_2\text{Br} < \text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{CH}_3 < (\text{CH}_3)_3\text{CBr}$ ($\text{S}_{\text{N}}1$)
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br} > (\text{CH}_3)_2\text{CHCH}_2\text{Br} > \text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{CH}_3 > (\text{CH}_3)_3\text{CBr}$ ($\text{S}_{\text{N}}2$)

Of the two primary bromides, the carbocation intermediate derived from $(\text{CH}_3)_2\text{CHCH}_2\text{Br}$ is more stable than derived from $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ because of greater electron donating inductive effect of $(\text{CH}_3)_2\text{CH}$ - group. Therefore, $(\text{CH}_3)_2\text{CHCH}_2\text{Br}$ is more reactive than $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ in $\text{S}_{\text{N}}1$ reactions. $\text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{CH}_3$ is a secondary bromide and $(\text{CH}_3)_3\text{CBr}$ is a tertiary bromide. Hence the above order is followed in $\text{S}_{\text{N}}1$. The reactivity in $\text{S}_{\text{N}}2$ reactions follows the reverse order as the steric hinderance around the electrophilic carbon increases in that order.

(ii) $\text{C}_6\text{H}_5\text{C}(\text{CH}_3)(\text{C}_6\text{H}_5)\text{Br} > \text{C}_6\text{H}_5\text{CH}(\text{C}_6\text{H}_5)\text{Br} > \text{C}_6\text{H}_5\text{CH}(\text{CH}_3)\text{Br} > \text{C}_6\text{H}_5\text{CH}_2\text{Br}$ ($\text{S}_{\text{N}}1$)

$\text{C}_6\text{H}_5\text{C}(\text{CH}_3)(\text{C}_6\text{H}_5)\text{Br} < \text{C}_6\text{H}_5\text{CH}(\text{C}_6\text{H}_5)\text{Br} < \text{C}_6\text{H}_5\text{CH}(\text{CH}_3)\text{Br} < \text{C}_6\text{H}_5\text{CH}_2\text{Br}$ ($\text{S}_{\text{N}}2$)
 Of the two secondary bromides, the carbocation intermediate obtained from $\text{C}_6\text{H}_5\text{CH}(\text{C}_6\text{H}_5)\text{Br}$ is more stable than obtained from $\text{C}_6\text{H}_5\text{CH}(\text{CH}_3)\text{Br}$ because it is stabilised by two phenyl groups due to resonance. Therefore, the former bromide is more reactive than the latter in $\text{S}_{\text{N}}1$ reactions. A phenyl group is bulkier than a methyl group. Therefore, $\text{C}_6\text{H}_5\text{CH}(\text{C}_6\text{H}_5)\text{Br}$ is less reactive than $\text{C}_6\text{H}_5\text{CH}(\text{CH}_3)\text{Br}$ in $\text{S}_{\text{N}}2$ reactions.

(c) Stereochemical aspects of nucleophilic substitution reactions

In order to understand the stereochemical aspects of substitution reactions, we need to learn some basic stereochemical principles and notations (**optical activity, chirality, retention, inversion, racemisation**, etc.).

(i) **Optical activity:** Plane of plane polarised light produced by passing ordinary light through Nicol prism is rotated when it is passed through the solutions of certain compounds. Such compounds are called **optically active** compounds. The angle by which the plane polarised light is rotated is measured by an instrument called polarimeter. If the compound rotates the plane of plane polarised light to the right, i.e., clockwise direction, it is called *dextrorotatory* (Greek for right rotating) or the *d*-form and is indicated by placing a positive (+) sign before the degree of rotation. If the light is rotated towards left (anticlockwise direction), the compound is said to be laevo-rotatory or the *l*-form and a negative (-) sign is placed before the degree of rotation. Such (+) and (-) isomers of a compound are called **optical isomers** and the phenomenon is termed as **optical isomerism**.

(ii) **Molecular asymmetry, chirality and enantiomers:** The observation of Louis Pasteur (1848) that crystals of certain compounds exist in the form of mirror images laid the foundation of modern stereochemistry. He demonstrated that aqueous solutions of both types of crystals showed optical rotation, equal in magnitude (for solution of equal concentration) but opposite in direction. He believed that this difference in optical activity was associated with the three dimensional arrangements of atoms in the molecules (**configurations**) of

William Nicol (1768-1851) developed the first prism that produced plane polarised light.

Jacobus Hendricus Van't Hoff (1852-1911) received the first Nobel Prize in Chemistry in 1901 for his work on solutions.

two types of crystals. Dutch scientist, J. Van't Hoff and French scientist, C. Le Bel in the same year (1874), independently argued that the spatial arrangement of four groups (valencies) around a central carbon is tetrahedral and if all the substituents attached to that carbon are different, the mirror image of the molecule is not superimposed (overlapped) on the molecule; such a carbon is called **asymmetric carbon** or **stereocentre**. The resulting molecule would lack symmetry and is referred to as asymmetric molecule. The asymmetry of the molecule along with non superimposability of mirror images is responsible for the optical activity in such organic compounds.

The symmetry and asymmetry are also observed in many day to day

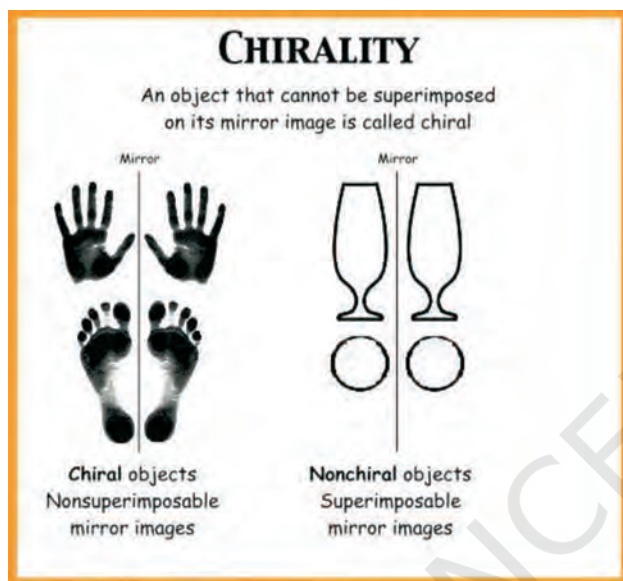


Fig 6.4: Some common examples of chiral and achiral objects

objects: a sphere, a cube, a cone, are all identical to their mirror images and can be superimposed. However, many objects are non superimposable on their mirror images. For example, your left and right hand look similar but if you put your left hand on your right hand by moving them in the same plane, they do not coincide. The objects which are non-superimposable on their mirror image (like a pair of hands) are said to be **chiral** and this property is known as **chirality**. Chiral molecules are optically active, while the objects, which are, superimposable on their mirror images are called **achiral**. These molecules are optically inactive.

The above test of molecular chirality can be applied to organic molecules by constructing models and its mirror images or by drawing three dimensional structures and attempting to superimpose them in our minds. There are other aids, however, that can assist us in recognising chiral molecules. One such aid is the presence of

a single asymmetric carbon atom. Let us consider two simple molecules propan-2-ol (Fig.6.5) and butan-2-ol (Fig.6.6) and their mirror images.

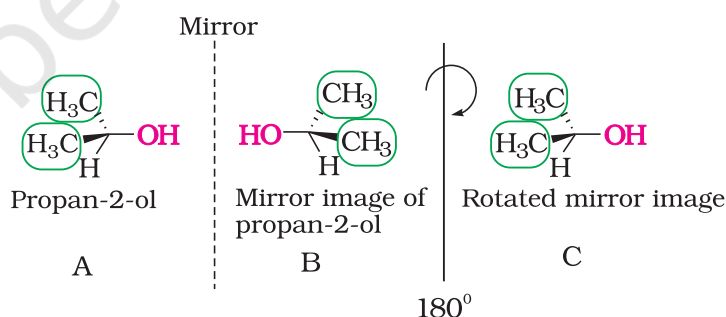
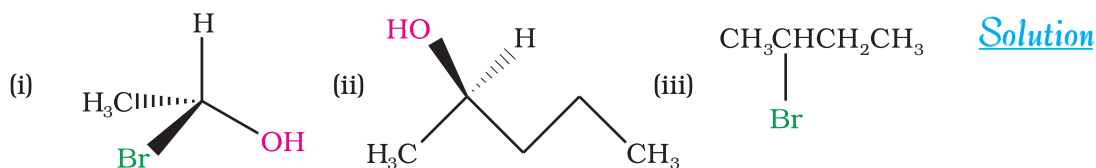


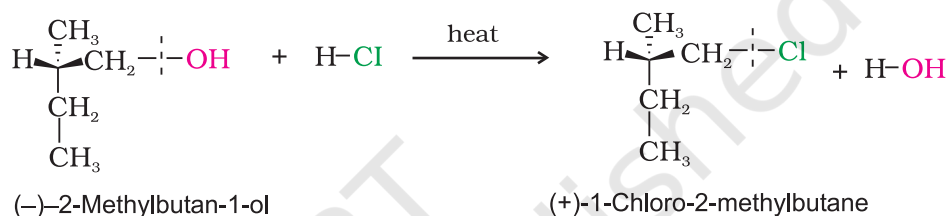
Fig 6.5: B is mirror image of A; B is rotated by 180° and C is obtained; C is superimposable on A.

As you can see very clearly, propan-2-ol (A) does not contain an asymmetric carbon, as all the four groups attached to the tetrahedral carbon are not different. We rotate the mirror image (B) of the molecule by 180° (structure C) and try to overlap the structure (C) with the structure (A), these structures completely overlap. Thus propan-2-ol is an **achiral** molecule.



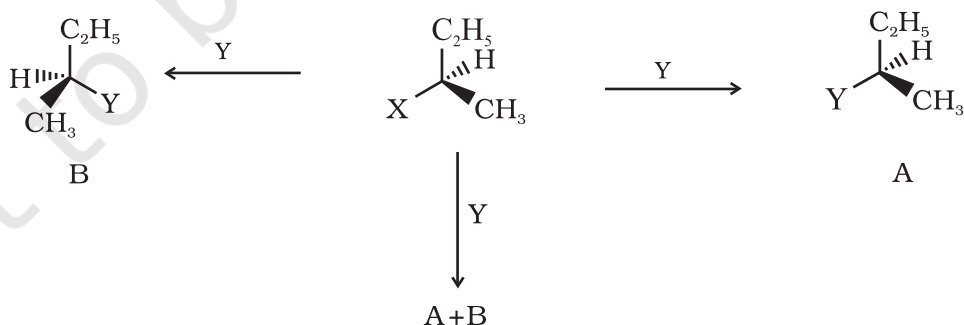
(iii) *Retention*: Retention of configuration is the preservation of the spatial arrangement of bonds to an asymmetric centre during a chemical reaction or transformation.

In general, if during a reaction, no bond to the stereocentre is broken, the product will have the same general configuration of groups around the stereocentre as that of reactant. Such a reaction is said to proceed with retention of the configuration. Consider as an example, the reaction that takes place when (–)-2-methylbutan-1-ol is heated with concentrated hydrochloric acid.



It is important to note that configuration at a symmetric centre in the reactant and product is same but the sign of optical rotation has changed in the product. This is so because two different compounds with same configuration at asymmetric centre may have different optical rotation. One may be dextrorotatory (plus sign of optical rotation) while other may be laevorotatory (negative sign of optical rotation).

(iv) *Inversion, retention and racemisation*: There are three outcomes for a reaction at an asymmetric carbon atom, when a bond directly linked to an asymmetric carbon atom is broken. Consider the replacement of a group X by Y in the following reaction;



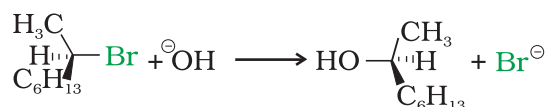
If (A) is the only compound obtained, the process is called retention of configuration. Note that configuration has been rotated in A.

If (B) is the only compound obtained, the process is called inversion of configuration. Configuration has been inverted in B.

If a 50:50 mixture of A and B is obtained then the process is called racemisation and the product is optically inactive, as one isomer will rotate the plane polarised light in the direction opposite to another.

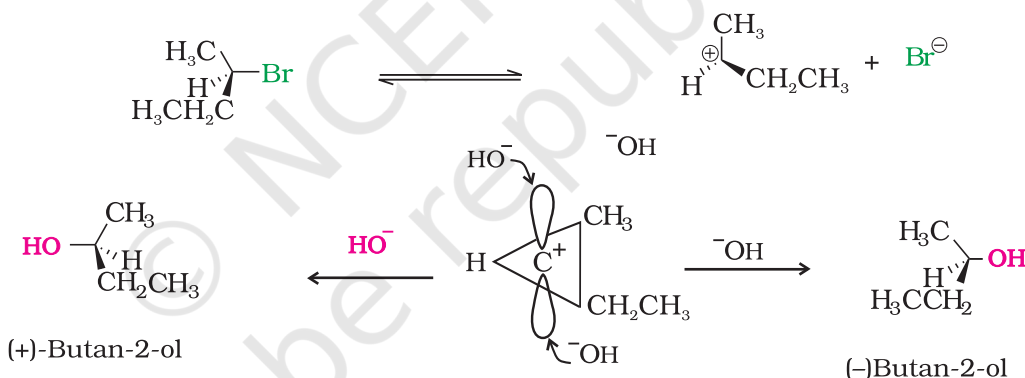
Now let us have a fresh look at S_N1 and S_N2 mechanisms by taking examples of optically active alkyl halides.

In case of optically active alkyl halides, the product formed as a result of S_N2 mechanism has the inverted configuration as compared to the reactant. This is because the nucleophile attaches itself on the side opposite to the one where the halogen atom is present. When (–)-2-bromooctane is allowed to react with sodium hydroxide, (+)-octan-2-ol is formed with the –OH group occupying the position opposite to what bromide had occupied.



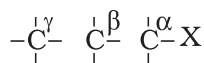
Thus, S_N2 reactions of optically active halides are accompanied by inversion of configuration.

In case of optically active alkyl halides, S_N1 reactions are accompanied by racemisation. Can you think of the reason why it happens? Actually the carbocation formed in the slow step being sp² hybridised is planar (achiral). The attack of the nucleophile may be accomplished from either side of the plane of carbocation resulting in a mixture of products, one having the same configuration (the –OH



Location of α and β carbon in a molecule

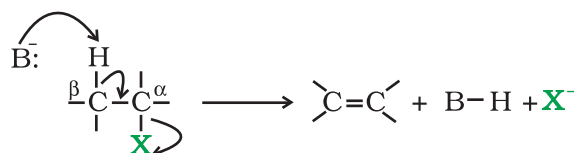
Carbon on which halogen atom is directly attached is called α-carbon and the carbon atom adjacent to this carbon is called β-carbon.



attaching on the same position as halide ion) and the other having opposite configuration (the –OH attaching on the side opposite to halide ion). This may be illustrated by hydrolysis of optically active 2-bromobutane, which results in the formation of (±)-butan-2-ol.

2. Elimination reactions

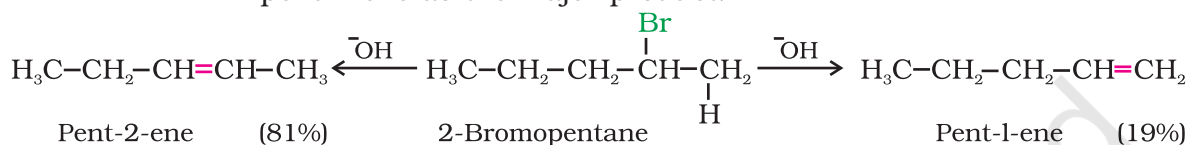
When a haloalkane with β-hydrogen atom is heated with alcoholic solution of potassium hydroxide, there is elimination of hydrogen atom from β-carbon and a halogen atom from the α-carbon atom.



B=Base ; X=Leaving group

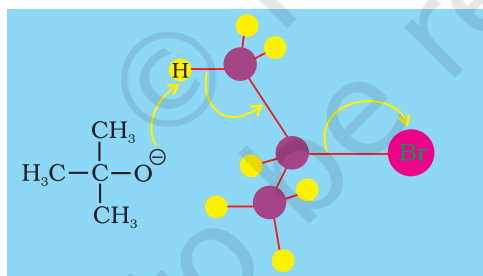
As a result, an alkene is formed as a product. Since β -hydrogen atom is involved in elimination, it is often called **β -elimination**.

If there is possibility of formation of more than one alkene due to the availability of more than one β -hydrogen atoms, usually one alkene is formed as the major product. These form part of a pattern first observed by Russian chemist, Alexander Zaitsev (also pronounced as Saytzeff) who in 1875 formulated a rule which can be summarised as “*in dehydrohalogenation reactions, the preferred product is that alkene which has the greater number of alkyl groups attached to the doubly bonded carbon atoms.*” Thus, 2-bromopentane gives pent-2-ene as the major product.



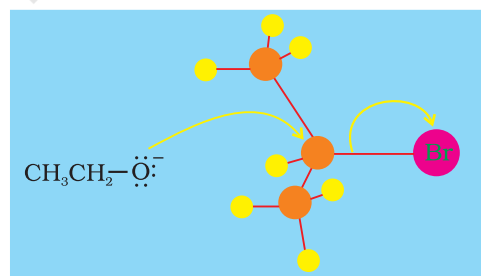
Elimination versus substitution

A chemical reaction is the result of competition; it is a race that is won by the fastest runner. A collection of molecules tend to do, by and large, what is easiest for them. An alkyl halide with β -hydrogen atoms when reacted with a base or a nucleophile has two competing routes: substitution ($\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$) and elimination. Which route will be taken up depends upon the nature of alkyl halide, strength and size of base/nucleophile and reaction conditions. Thus, a bulkier nucleophile will prefer to act as a base and abstracts a proton rather than approach a tetravalent carbon atom (steric reasons) and *vice versa*. Similarly, a primary alkyl halide will prefer a $\text{S}_{\text{N}}2$ reaction, a secondary halide- $\text{S}_{\text{N}}2$ or elimination depending upon the strength of base/nucleophile and a tertiary halide- $\text{S}_{\text{N}}1$ or elimination depending upon the stability of carbocation or the more substituted alkene.



Elimination

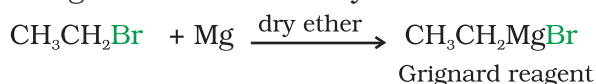
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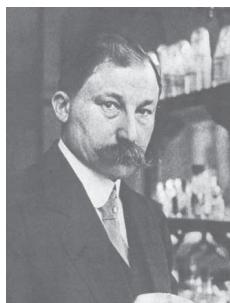


Substitution

3. Reaction with metals

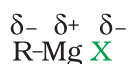
Most organic chlorides, bromides and iodides react with certain metals to give compounds containing carbon-metal bonds. Such compounds are known as **organo-metallic compounds**. An important class of organo-metallic compounds discovered by Victor Grignard in 1900 is alkyl magnesium halide, RMgX , referred as **Grignard Reagents**. These reagents are obtained by the reaction of haloalkanes with magnesium metal in dry ether.





Victor Grignard had a strange start in academic life for a chemist - he took a maths degree. When he eventually switched to chemistry, it was not to the mathematical province of physical chemistry but to organic chemistry. While attempting to find an efficient catalyst for the process of methylation, he noted that Zn in diethyl ether had been used for this purpose and wondered whether the Mg/ether combination might be successful. Grignard reagents were first reported in 1900 and Grignard used this work for his doctoral thesis in 1901. In 1910, Grignard obtained a professorship at the University of Nancy and in 1912, he was awarded the Nobel prize for Chemistry which he shared with Paul Sabatier who had made advances in nickel catalysed hydrogenation.

In the Grignard reagent, the carbon-magnesium bond is covalent but highly polar, with carbon pulling electrons from electropositive magnesium; the magnesium halogen bond is essentially ionic.



Grignard reagents are highly reactive and react with any source of proton to give hydrocarbons. Even water, alcohols, amines are sufficiently acidic to convert them to corresponding hydrocarbons.



It is therefore necessary to avoid even traces of moisture from a Grignard reagent. That is why reaction is carried out in dry ether. On the other hand, this could be considered as one of the methods for converting halides to hydrocarbons.

Wurtz reaction

Alkyl halides react with sodium in dry ether to give hydrocarbons containing double the number of carbon atoms present in the halide. This reaction is known as Wurtz reaction.

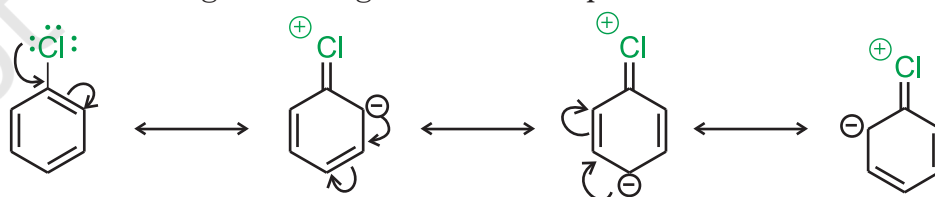


6.7.2 Reactions of Haloarenes

1. Nucleophilic substitution

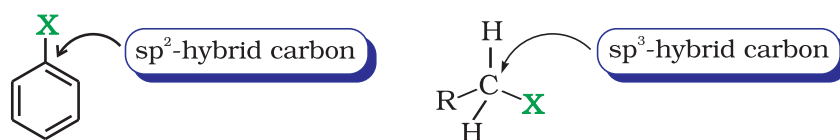
Aryl halides are extremely less reactive towards nucleophilic substitution reactions due to the following reasons:

- (i) *Resonance effect* : In haloarenes, the electron pairs on halogen atom are in conjugation with π -electrons of the ring and the following resonating structures are possible.



C—Cl bond acquires a partial double bond character due to resonance. As a result, the bond cleavage in haloarene is difficult than haloalkane and therefore, they are less reactive towards nucleophilic substitution reaction.

- (ii) *Difference in hybridisation of carbon atom in C—X bond:* In haloalkane, the carbon atom attached to halogen is sp^3 hybridised while in case of haloarene, the carbon atom attached to halogen is sp^2 -hybridised.

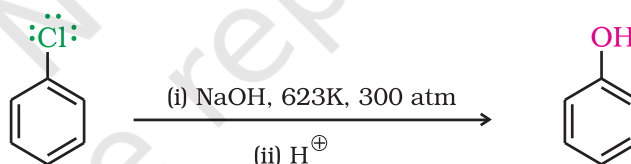


The sp^2 hybridised carbon with a greater s-character is more electronegative and can hold the electron pair of C—X bond more tightly than sp^3 -hybridised carbon in haloalkane with less s-character. Thus, C—Cl bond length in haloalkane is 177pm while in haloarene is 169 pm. Since it is difficult to break a shorter bond than a longer bond, therefore, haloarenes are less reactive than haloalkanes towards nucleophilic substitution reaction.

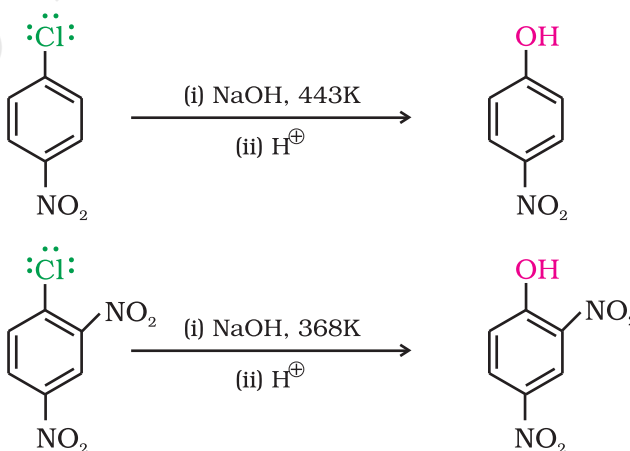
- (iii) *Instability of phenyl cation:* In case of haloarenes, the phenyl cation formed as a result of self-ionisation will not be stabilised by resonance and therefore, S_N1 mechanism is ruled out.
- (iv) Because of the possible repulsion, it is less likely for the electron rich nucleophile to approach electron rich arenes.

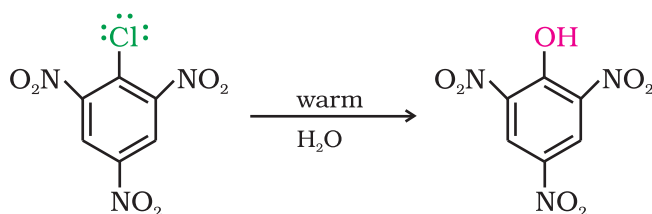
Replacement by hydroxyl group

Chlorobenzene can be converted into phenol by heating in aqueous sodium hydroxide solution at a temperature of 623K and a pressure of 300 atmospheres.

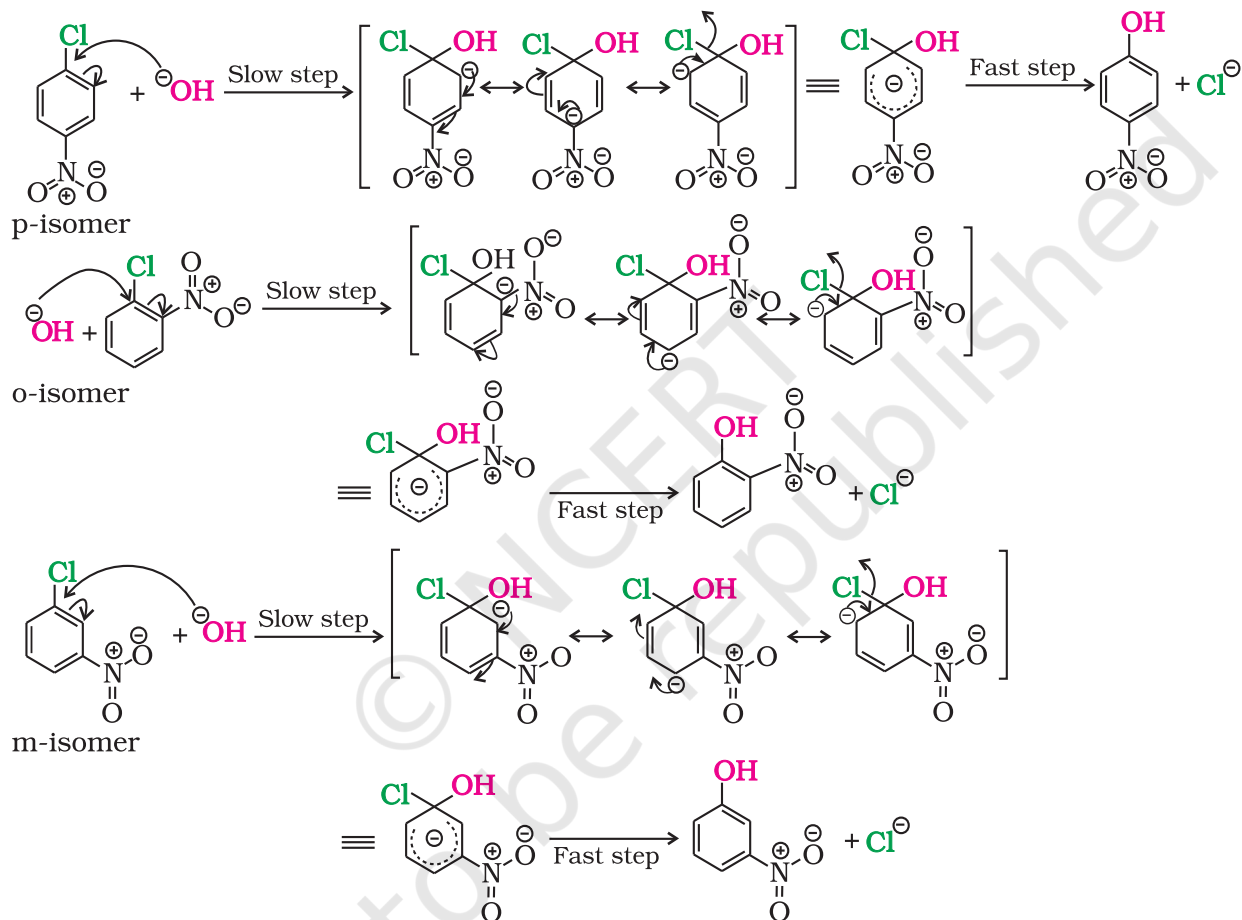


The presence of an electron withdrawing group ($-NO_2$) at *ortho*- and *para*-positions increases the reactivity of haloarenes.





The effect is pronounced when ($-\text{NO}_2$) group is introduced at *ortho*- and *para*- positions. However, no effect on reactivity of haloarenes is observed by the presence of electron withdrawing group at *meta*-position. Mechanism of the reaction is as depicted:

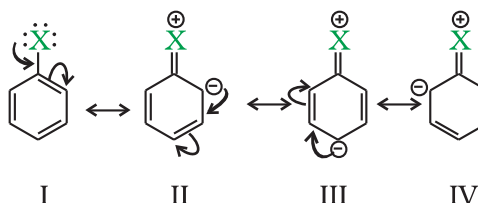


Can you think why does NO_2 group show its effect only at *ortho*- and *para*- positions and not at *meta*- position?

As shown, the presence of nitro group at *ortho*- and *para*-positions withdraws the electron density from the benzene ring and thus facilitates the attack of the nucleophile on haloarene. The carbanion thus formed is stabilised through resonance. The negative charge appeared at *ortho*- and *para*- positions with respect to the halogen substituent is stabilised by $-\text{NO}_2$ group while in case of *meta*-nitrobenzene, none of the resonating structures bear the negative charge on carbon atom bearing the $-\text{NO}_2$ group. Therefore, the presence of nitro group at *meta*- position does not stabilise the negative charge and no effect on reactivity is observed by the presence of $-\text{NO}_2$ group at *meta*-position.

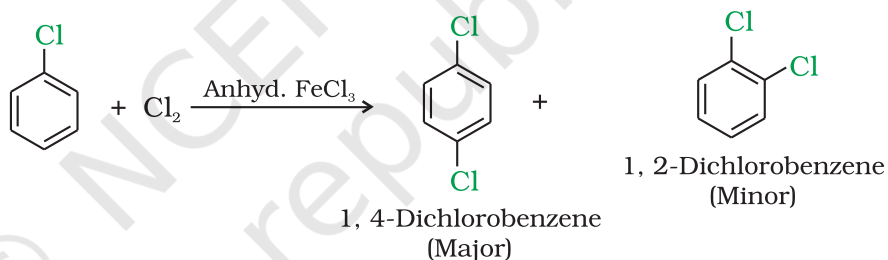
2. Electrophilic substitution reactions

Haloarenes undergo the usual electrophilic reactions of the benzene ring such as halogenation, nitration, sulphonation and Friedel-Crafts reactions. Halogen atom besides being slightly deactivating is *o*, *p*-directing; therefore, further substitution occurs at *ortho*- and *para*-positions with respect to the halogen atom. The *o*, *p*-directing influence of halogen atom can be easily understood if we consider the resonating structures of halobenzene as shown:

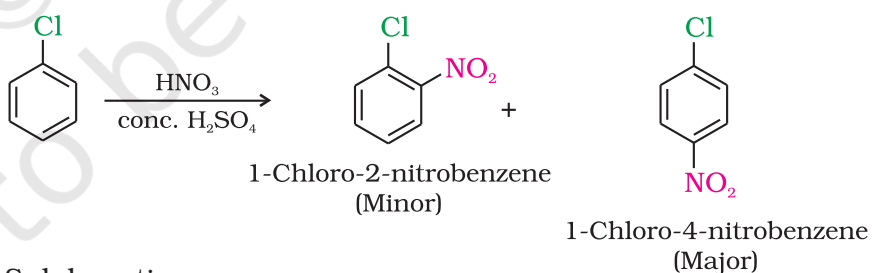


Due to resonance, the electron density increases more at *ortho*- and *para*-positions than at *meta*-positions. Further, the halogen atom because of its $-I$ effect has some tendency to withdraw electrons from the benzene ring. As a result, the ring gets somewhat deactivated as compared to benzene and hence the electrophilic substitution reactions in haloarenes occur slowly and require more drastic conditions as compared to those in benzene.

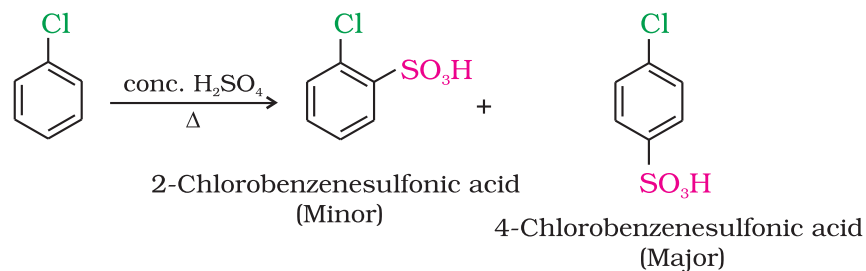
(i) Halogenation



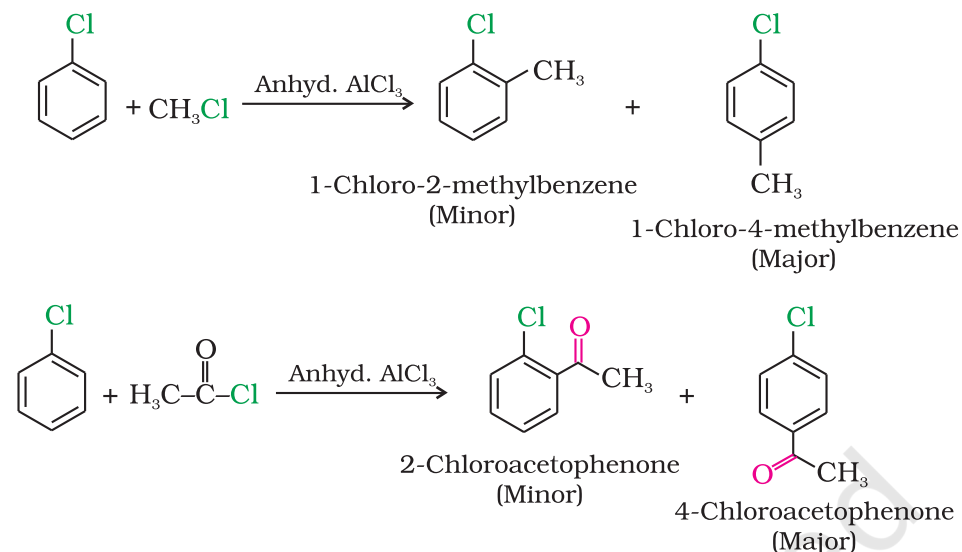
(ii) Nitration



(iii) Sulphonation



(iv) Friedel-Crafts reaction

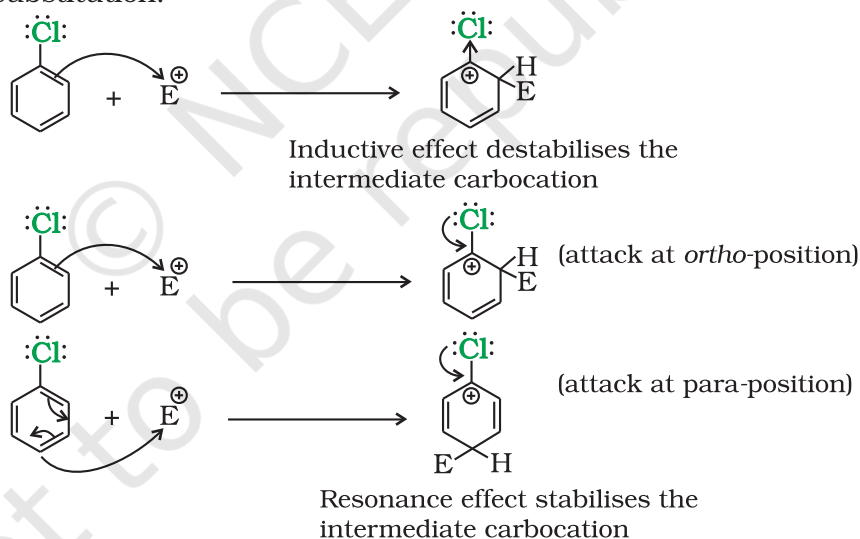


Example 6.9

Solution

Although chlorine is an electron withdrawing group, yet it is *ortho*-, *para*- directing in electrophilic aromatic substitution reactions. Why?

Chlorine withdraws electrons through inductive effect and releases electrons through resonance. Through inductive effect, chlorine destabilises the intermediate carbocation formed during the electrophilic substitution.

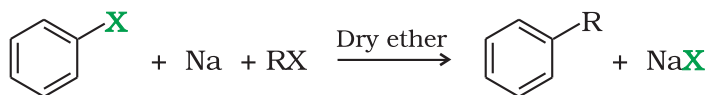


Through resonance, halogen tends to stabilise the carbocation and the effect is more pronounced at *ortho*- and *para*- positions. The inductive effect is stronger than resonance and causes net electron withdrawal and thus causes net deactivation. The resonance effect tends to oppose the inductive effect for the attack at *ortho*- and *para*-positions and hence makes the deactivation less for *ortho*- and *para*-attack. Reactivity is thus controlled by the stronger inductive effect and orientation is controlled by resonance effect.

3. Reaction with metals

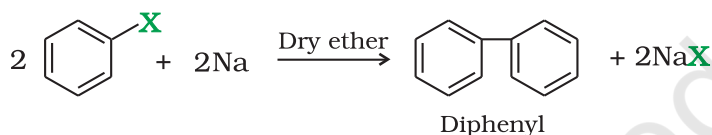
Wurtz-Fittig reaction

A mixture of an alkyl halide and aryl halide gives an alkylarene when treated with sodium in dry ether and is called Wurtz-Fittig reaction.



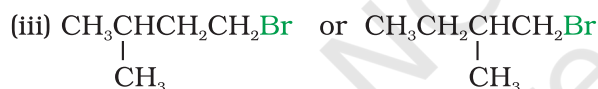
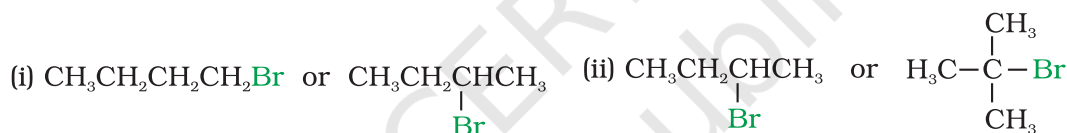
Fittig reaction

Aryl halides also give analogous compounds when treated with sodium in dry ether, in which two aryl groups are joined together. It is called Fittig reaction.

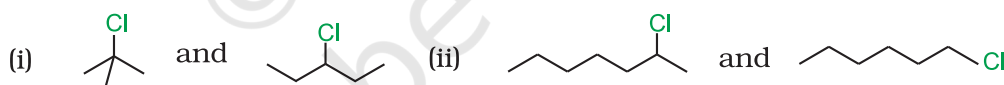


Intext Questions

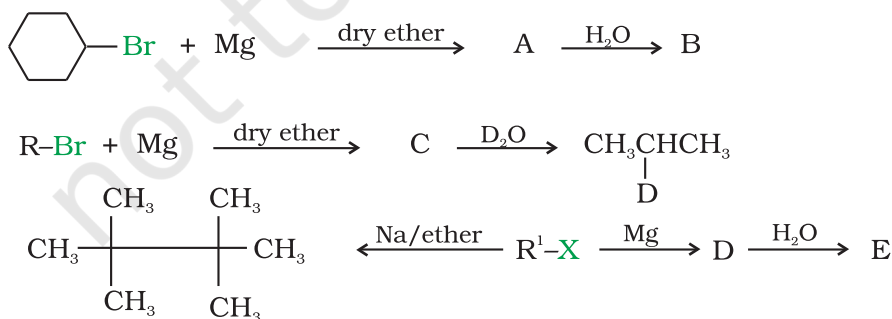
- 6.7 Which alkyl halide from the following pairs would you expect to react more rapidly by an $\text{S}_{\text{N}}2$ mechanism? Explain your answer.



- 6.8 In the following pairs of halogen compounds, which compound undergoes faster $\text{S}_{\text{N}}1$ reaction?



- 6.9 Identify A, B, C, D, E, R and R^1 in the following:



6.8 Polyhalogen Compounds

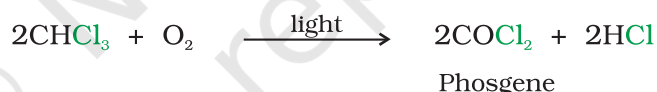
Carbon compounds containing more than one halogen atom are usually referred to as polyhalogen compounds. Many of these compounds are useful in industry and agriculture. Some polyhalogen compounds are described in this section.

6.8.1 Dichloromethane (Methylene chloride)

Dichloromethane is widely used as a solvent as a paint remover, as a propellant in aerosols, and as a process solvent in the manufacture of drugs. It is also used as a metal cleaning and finishing solvent. Methylene chloride harms the human central nervous system. Exposure to lower levels of methylene chloride in air can lead to slightly impaired hearing and vision. Higher levels of methylene chloride in air cause dizziness, nausea, tingling and numbness in the fingers and toes. In humans, direct skin contact with methylene chloride causes intense burning and mild redness of the skin. Direct contact with the eyes can burn the cornea.

6.8.2 Trichloromethane (Chloroform)

Chemically, chloroform is employed as a solvent for fats, alkaloids, iodine and other substances. The major use of chloroform today is in the production of the freon refrigerant R-22. It was once used as a general anaesthetic in surgery but has been replaced by less toxic, safer anaesthetics, such as ether. As might be expected from its use as an anaesthetic, inhaling chloroform vapours depresses the central nervous system. Breathing about 900 parts of chloroform per million parts of air (900 parts per million) for a short time can cause dizziness, fatigue, and headache. Chronic chloroform exposure may cause damage to the liver (where chloroform is metabolised to phosgene) and to the kidneys, and some people develop sores when the skin is immersed in chloroform. Chloroform is slowly oxidised by air in the presence of light to an extremely poisonous gas, carbonyl chloride, also known as phosgene. It is therefore stored in closed dark coloured bottles completely filled so that air is kept out.



6.8.3 Triiodomethane (Iodoform)

It was used earlier as an antiseptic but the antiseptic properties are due to the liberation of free iodine and not due to iodoform itself. Due to its objectionable smell, it has been replaced by other formulations containing iodine.

6.8.4 Tetrachloromethane (Carbon tetrachloride)

It is produced in large quantities for use in the manufacture of refrigerants and propellants for aerosol cans. It is also used as feedstock in the synthesis of chlorofluorocarbons and other chemicals, pharmaceutical manufacturing, and general solvent use. Until the mid 1960s, it was also widely used as a cleaning fluid, both in industry, as a degreasing agent, and in the home, as a spot remover and as fire extinguisher. There is some evidence that exposure to carbon tetrachloride causes liver cancer in humans. The most common effects are dizziness, light headedness, nausea and vomiting, which can cause permanent damage to nerve cells. In severe cases, these effects can lead rapidly to stupor, coma, unconsciousness or death. Exposure to CCl_4 can make the heart beat irregularly or stop. The chemical may irritate the eyes on contact. When carbon tetrachloride is released into the air, it rises to the atmosphere and depletes the ozone layer. Depletion of the

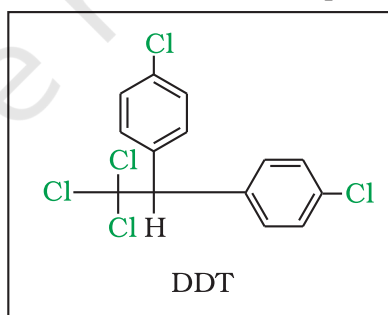
ozone layer is believed to increase human exposure to ultraviolet rays, leading to increased skin cancer, eye diseases and disorders, and possible disruption of the immune system.

6.8.5 Freons

The chlorofluorocarbon compounds of methane and ethane are collectively known as freons. They are extremely stable, unreactive, non-toxic, non-corrosive and easily liquefiable gases. Freon 12 (CCl_2F_2) is one of the most common freons in industrial use. It is manufactured from tetrachloromethane by **Swarts reaction**. These are usually produced for aerosol propellants, refrigeration and air conditioning purposes. By 1974, total freon production in the world was about 2 billion pounds annually. Most freon, even that used in refrigeration, eventually makes its way into the atmosphere where it diffuses unchanged into the stratosphere. In stratosphere, freon is able to initiate radical chain reactions that can upset the natural ozone balance.

6.8.6 *p,p'*-Dichlorodiphenyl-trichloroethane (DDT)

DDT, the first chlorinated organic insecticides, was originally prepared in 1873, but it was not until 1939 that Paul Muller of Geigy Pharmaceuticals in Switzerland discovered the effectiveness of DDT as an insecticide. Paul Muller was awarded the Nobel Prize in Medicine and Physiology in 1948 for this discovery. The use of DDT increased enormously on a worldwide basis after World War II, primarily because of its effectiveness against the mosquito that spreads malaria and lice that carry typhus. However, problems related to extensive use of DDT began to appear in the late 1940s. Many species of insects developed resistance to DDT, and it was also discovered to have a high toxicity towards fish. The chemical stability of DDT and its fat solubility compounded the problem. DDT is not metabolised very rapidly by animals; instead, it is deposited and stored in the fatty tissues. If ingestion continues at a steady rate, DDT builds up within the animal over time. The use of DDT was banned in the United States in 1973, although it is still in use in some other parts of the world.



Summary

Alkyl/ Aryl halides may be classified as mono, di, or polyhalogen (tri-, tetra-, etc.) compounds depending on whether they contain one, two or more halogen atoms in their structures. Since halogen atoms are more electronegative than carbon, the carbon-halogen bond of alkyl halide is polarised; the carbon atom bears a partial positive charge, and the halogen atom bears a partial negative charge.

Alkyl halides are prepared by the **free radical halogenation** of alkanes, addition of halogen acids to alkenes, replacement of $-\text{OH}$ group of alcohols with halogens using

phosphorus halides, thionyl chloride or halogen acids. Aryl halides are prepared by **electrophilic substitution** to arenes. Fluorides and iodides are best prepared by halogen exchange method.

The boiling points of organohalogen compounds are comparatively higher than the corresponding hydrocarbons because of strong dipole-dipole and van der Waals forces of attraction. These are slightly soluble in water but completely soluble in organic solvents.

The polarity of carbon-halogen bond of alkyl halides is responsible for their **nucleophilic substitution, elimination** and their reaction with metal atoms to form **organometallic compounds**. Nucleophilic substitution reactions are categorised into **S_N1** and **S_N2** on the basis of their kinetic properties. **Chirality** has a profound role in understanding the reaction mechanisms of S_N1 and S_N2 reactions. S_N2 reactions of chiral alkyl halides are characterised by the inversion of configuration while S_N1 reactions are characterised by racemisation.

A number of polyhalogen compounds e.g., **dichloromethane, chloroform, iodoform, carbon tetrachloride, freon** and **DDT** have many industrial applications. However, some of these compounds cannot be easily decomposed and even cause depletion of ozone layer and are proving **environmental hazards**.

Exercises

- 6.1** Name the following halides according to IUPAC system and classify them as alkyl, allyl, benzyl (primary, secondary, tertiary), vinyl or aryl halides:
- $(\text{CH}_3)_2\text{CHCH}(\text{Cl})\text{CH}_3$
 - $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}(\text{C}_2\text{H}_5)\text{Cl}$
 - $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)_2\text{CH}_2\text{I}$
 - $(\text{CH}_3)_3\text{CCH}_2\text{CH}(\text{Br})\text{C}_6\text{H}_5$
 - $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}(\text{Br})\text{CH}_3$
 - $\text{CH}_3\text{C}(\text{C}_2\text{H}_5)_2\text{CH}_2\text{Br}$
 - $\text{CH}_3\text{C}(\text{Cl})(\text{C}_2\text{H}_5)\text{CH}_2\text{CH}_3$
 - $\text{CH}_3\text{CH}=\text{C}(\text{Cl})\text{CH}_2\text{CH}(\text{CH}_3)_2$
 - $\text{CH}_3\text{CH}=\text{CHC}(\text{Br})(\text{CH}_3)_2$
 - $p\text{-ClC}_6\text{H}_4\text{CH}_2\text{CH}(\text{CH}_3)_2$
 - $m\text{-ClCH}_2\text{C}_6\text{H}_4\text{CH}_2\text{C}(\text{CH}_3)_3$
 - $o\text{-Br-C}_6\text{H}_4\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$
- 6.2** Give the IUPAC names of the following compounds:
- $\text{CH}_3\text{CH}(\text{Cl})\text{CH}(\text{Br})\text{CH}_3$
 - $\text{CHF}_2\text{CBrClF}$
 - $\text{ClCH}_2\text{C}\equiv\text{CCH}_2\text{Br}$
 - $(\text{CCl}_3)_3\text{CCl}$
 - $\text{CH}_3\text{C}(p\text{-ClC}_6\text{H}_4)_2\text{CH}(\text{Br})\text{CH}_3$
 - $(\text{CH}_3)_3\text{CCH}=\text{CClC}_6\text{H}_4\text{I-}p$
- 6.3** Write the structures of the following organic halogen compounds.
- 2-Chloro-3-methylpentane
 - p*-Bromochlorobenzene
 - 1-Chloro-4-ethylcyclohexane
 - 2-(2-Chlorophenyl)-1-iodooctane
 - 2-Bromobutane
 - 4-tert-Butyl-3-iodoheptane
 - 1-Bromo-4-sec-butyl-2-methylbenzene
 - 1,4-Dibromobut-2-ene
- 6.4** Which one of the following has the highest dipole moment?
- CH_2Cl_2
 - CHCl_3
 - CCl_4
- 6.5** A hydrocarbon C_5H_{10} does not react with chlorine in dark but gives a single monochloro compound $\text{C}_5\text{H}_9\text{Cl}$ in bright sunlight. Identify the hydrocarbon.
- 6.6** Write the isomers of the compound having formula $\text{C}_4\text{H}_9\text{Br}$.
- 6.7** Write the equations for the preparation of 1-iodobutane from
- 1-butanol
 - 1-chlorobutane
 - but-1-ene.
- 6.8** What are ambident nucleophiles? Explain with an example.

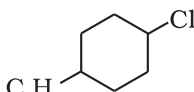
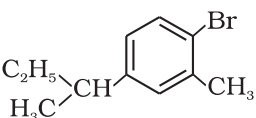
- 6.9** Which compound in each of the following pairs will react faster in S_N2 reaction with ^-OH ?
 (i) CH_3Br or CH_3I (ii) $(CH_3)_3CCl$ or CH_3Cl
- 6.10** Predict all the alkenes that would be formed by dehydrohalogenation of the following halides with sodium ethoxide in ethanol and identify the major alkene:
 (i) 1-Bromo-1-methylcyclohexane (ii) 2-Chloro-2-methylbutane
 (iii) 2,2,3-Trimethyl-3-bromopentane.
- 6.11** How will you bring about the following conversions?
 (i) Ethanol to but-1-yne (ii) Ethane to bromoethene (iii) Propene to 1-nitropropane (iv) Toluene to benzyl alcohol (v) Propene to propyne
 (vi) Ethanol to ethyl fluoride (vii) Bromomethane to propanone (viii) But-1-ene to but-2-ene (ix) 1-Chlorobutane to n-octane (x) Benzene to biphenyl.
- 6.12** Explain why
 (i) the dipole moment of chlorobenzene is lower than that of cyclohexyl chloride?
 (ii) alkyl halides, though polar, are immiscible with water?
 (iii) Grignard reagents should be prepared under anhydrous conditions?
- 6.13** Give the uses of freon 12, DDT, carbon tetrachloride and iodoform.
- 6.14** Write the structure of the major organic product in each of the following reactions:
 (i) $CH_3CH_2CH_2Cl + NaI \xrightarrow[\text{heat}]{\text{acetone}}$
 (ii) $(CH_3)_3CBr + KOH \xrightarrow[\text{heat}]{\text{ethanol}}$
 (iii) $CH_3CH(Br)CH_2CH_3 + NaOH \xrightarrow{\text{water}}$
 (iv) $CH_3CH_2Br + KCN \xrightarrow{\text{aq. ethanol}}$
 (v) $C_6H_5ONa + C_2H_5Cl \longrightarrow$
 (vi) $CH_3CH_2CH_2OH + SOCl_2 \longrightarrow$
 (vii) $CH_3CH_2CH=CH_2 + HBr \xrightarrow{\text{peroxide}}$
 (viii) $CH_3CH=C(CH_3)_2 + HBr \longrightarrow$
- 6.15** Write the mechanism of the following reaction:
 $nBuBr + KCN \xrightarrow{EtOH-H_2O} nBuCN$
- 6.16** Arrange the compounds of each set in order of reactivity towards S_N2 displacement:
 (i) 2-Bromo-2-methylbutane, 1-Bromopentane, 2-Bromopentane
 (ii) 1-Bromo-3-methylbutane, 2-Bromo-2-methylbutane, 2-Bromo-3-methylbutane
 (iii) 1-Bromobutane, 1-Bromo-2,2-dimethylpropane, 1-Bromo-2-methylbutane, 1-Bromo-3-methylbutane.
- 6.17** Out of $C_6H_5CH_2Cl$ and $C_6H_5CHClC_6H_5$, which is more easily hydrolysed by aqueous KOH.
- 6.18** *p*-Dichlorobenzene has higher m.p. than those of *o*- and *m*-isomers. Discuss.
- 6.19** How the following conversions can be carried out?
 (i) Propene to propan-1-ol
 (ii) Ethanol to but-1-yne
 (iii) 1-Bromopropane to 2-bromopropane

- (iv) Toluene to benzyl alcohol
- (v) Benzene to 4-bromonitrobenzene
- (vi) Benzyl alcohol to 2-phenylethanoic acid
- (vii) Ethanol to propanenitrile
- (viii) Aniline to chlorobenzene
- (ix) 2-Chlorobutane to 3, 4-dimethylhexane
- (x) 2-Methyl-1-propene to 2-chloro-2-methylpropane
- (xi) Ethyl chloride to propanoic acid
- (xii) But-1-ene to n-butyliodide
- (xiii) 2-Chloropropane to 1-propanol
- (xiv) Isopropyl alcohol to iodoform
- (xv) Chlorobenzene to *p*-nitrophenol
- (xvi) 2-Bromopropane to 1-bromopropane
- (xvii) Chloroethane to butane
- (xviii) Benzene to diphenyl
- (xix) *tert*-Butyl bromide to isobutyl bromide
- (xx) Aniline to phenylisocyanide

- 6.20** The treatment of alkyl chlorides with aqueous KOH leads to the formation of alcohols but in the presence of alcoholic KOH, alkenes are major products. Explain.
- 6.21** Primary alkyl halide C_4H_9Br (a) reacted with alcoholic KOH to give compound (b). Compound (b) is reacted with HBr to give (c) which is an isomer of (a). When (a) is reacted with sodium metal it gives compound (d), C_8H_{18} which is different from the compound formed when n-butyl bromide is reacted with sodium. Give the structural formula of (a) and write the equations for all the reactions.
- 6.22** What happens when
- (i) n-butyl chloride is treated with alcoholic KOH,
 - (ii) bromobenzene is treated with Mg in the presence of dry ether,
 - (iii) chlorobenzene is subjected to hydrolysis,
 - (iv) ethyl chloride is treated with aqueous KOH,
 - (v) methyl bromide is treated with sodium in the presence of dry ether,
 - (vi) methyl chloride is treated with KCN?

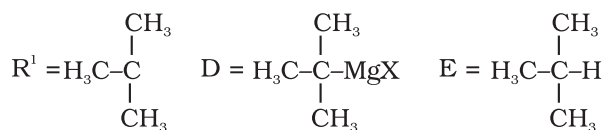
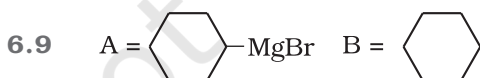
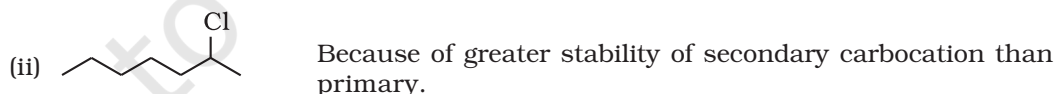
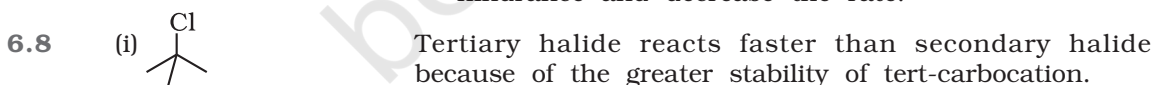
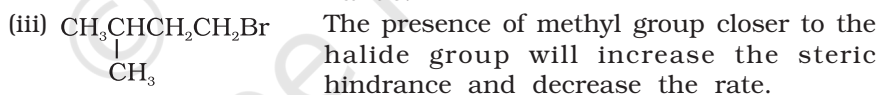
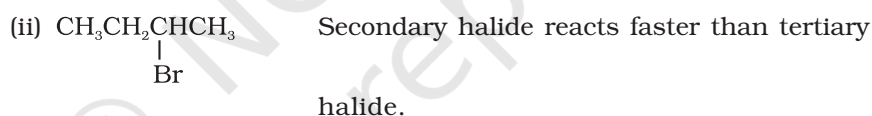
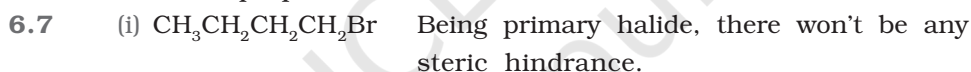
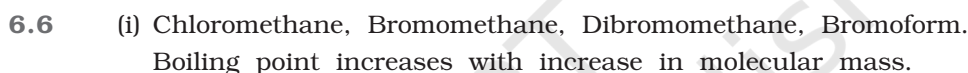
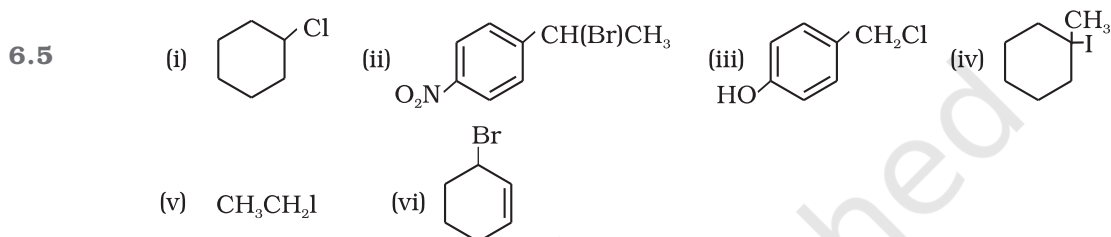
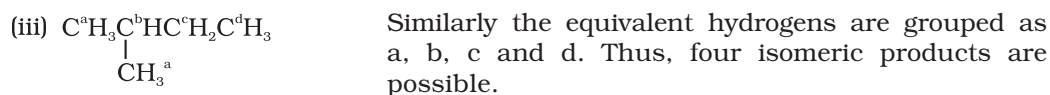
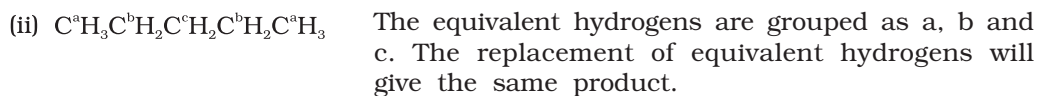
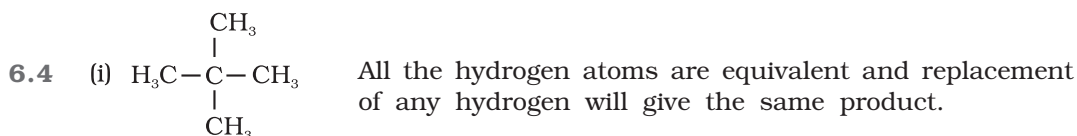
Answers to Some Intext Questions

6.1

- (i) $CH_3CH_2CH(CH_3)CHClCH_3$
- (ii) 
- (iii) $CH_3CH_2CH_2CH(CH_3)CH_2CH_3$
- (iv) $BrCH_2CH=CHCH_2Br$
- (v) 

- 6.2** (i) H_2SO_4 cannot be used along with KI in the conversion of an alcohol to an alkyl iodide as it converts KI to corresponding acid, HI which is then oxidised by it to I_2 .

- 6.3** (i) $ClCH_2CH_2CH_2Cl$ (ii) $ClCH_2CHClCH_3$ (iii) $Cl_2CHCH_2CH_3$ (iv) $CH_3CCl_2CH_3$





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Unit 7

Alcohols, Phenols and Ethers

Objectives

After studying this Unit, you will be able to

- name alcohols, phenols and ethers according to the IUPAC system of nomenclature;
- discuss the reactions involved in the preparation of alcohols from alkenes, aldehydes, ketones and carboxylic acids;
- discuss the reactions involved in the preparation of phenols from haloarenes, benzene sulphonic acids, diazonium salts and cumene;
- discuss the reactions for preparation of ethers from (i) alcohols and (ii) alkyl halides and sodium alkoxides/aryloxides;
- correlate physical properties of alcohols, phenols and ethers with their structures;
- discuss chemical reactions of the three classes of compounds on the basis of their functional groups.

Alcohols, phenols and ethers are the basic compounds for the formation of detergents, antiseptics and fragrances, respectively.

You have learnt that substitution of one or more hydrogen atom(s) from a hydrocarbon by another atom or a group of atoms result in the formation of an entirely new compound having altogether different properties and applications. **Alcohols** and **phenols** are formed when a hydrogen atom in a hydrocarbon, aliphatic and aromatic respectively, is replaced by -OH group. These classes of compounds find wide applications in industry as well as in day-to-day life. For instance, have you ever noticed that ordinary spirit used for polishing wooden furniture is chiefly a compound containing hydroxyl group, ethanol. The sugar we eat, the cotton used for fabrics, the paper we use for writing, are all made up of compounds containing -OH groups. Just think of life without paper; no note-books, books, newspapers, currency notes, cheques, certificates, etc. The magazines carrying beautiful photographs and interesting stories would disappear from our life. It would have been really a different world.

An alcohol contains one or more hydroxyl (OH) group(s) directly attached to carbon atom(s), of an aliphatic system (CH_3OH) while a phenol contains -OH group(s) directly attached to carbon atom(s) of an aromatic system ($\text{C}_6\text{H}_5\text{OH}$).

The substitution of a hydrogen atom in a hydrocarbon by an alkoxy or aryloxy group (R-O/Ar-O) yields another class of compounds known as 'ethers', for example, CH_3OCH_3 (dimethyl ether). You may also visualise ethers as compounds formed by

substituting the hydrogen atom of hydroxyl group of an alcohol or phenol by an alkyl or aryl group.

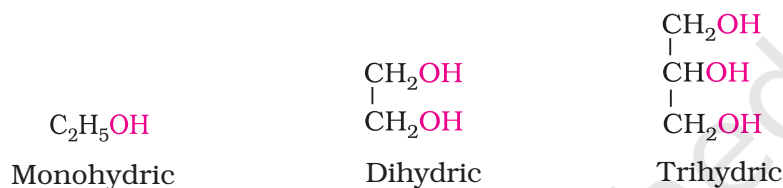
In this unit, we shall discuss the chemistry of three classes of compounds, namely — alcohols, phenols and ethers.

7.1 Classification

The classification of compounds makes their study systematic and hence simpler. Therefore, let us first learn how are alcohols, phenols and ethers classified?

7.1.1 Alcohols— Mono, Di, Tri or Polyhydric alcohols

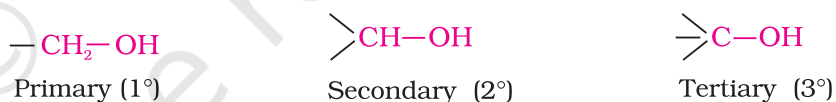
Alcohols and phenols may be classified as mono-, di-, tri- or polyhydric compounds depending on whether they contain one, two, three or many hydroxyl groups respectively in their structures as given below:



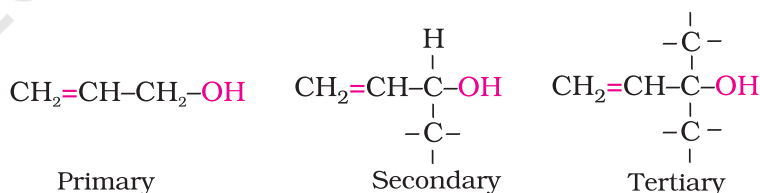
Monohydric alcohols may be further classified according to the hybridisation of the carbon atom to which the hydroxyl group is attached.

(i) *Compounds containing $\text{C}_{sp^3}\text{—OH}$ bond:* In this class of alcohols, the —OH group is attached to an sp^3 hybridised carbon atom of an alkyl group. They are further classified as follows:

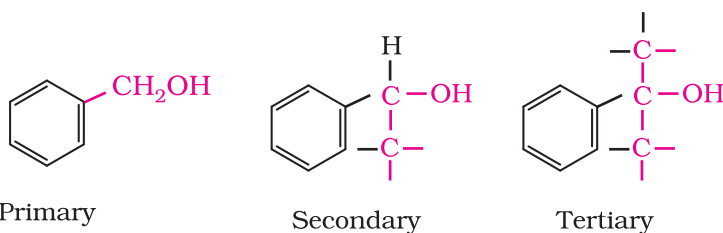
Primary, secondary and tertiary alcohols: In these three types of alcohols, the —OH group is attached to primary, secondary and tertiary carbon atom, respectively as depicted below:



Allylic alcohols: In these alcohols, the —OH group is attached to a sp^3 hybridised carbon adjacent to the carbon-carbon double bond, that is to an allylic carbon. For example



Benzylic alcohols: In these alcohols, the —OH group is attached to a sp^3 —hybridised carbon atom next to an aromatic ring. For example.

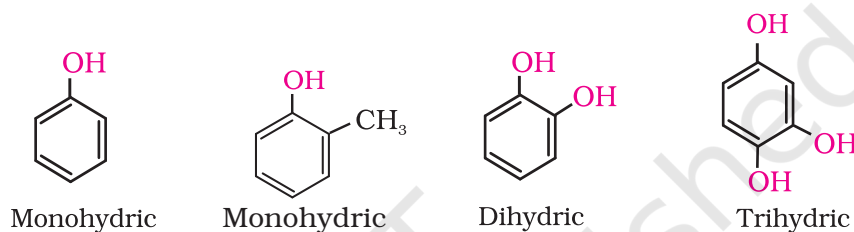


Allylic and benzylic alcohols may be primary, secondary or tertiary.

(ii) *Compounds containing $C_{sp^2}-OH$ bond:* These alcohols contain $-OH$ group bonded to a carbon-carbon double bond, i.e., to a vinylic carbon or to an aryl carbon. These alcohols are also known as vinylic alcohols.

Vinylic alcohol: $CH_2 = CH - OH$

7.1.2 Phenols— Mono, Di and trihydric phenols

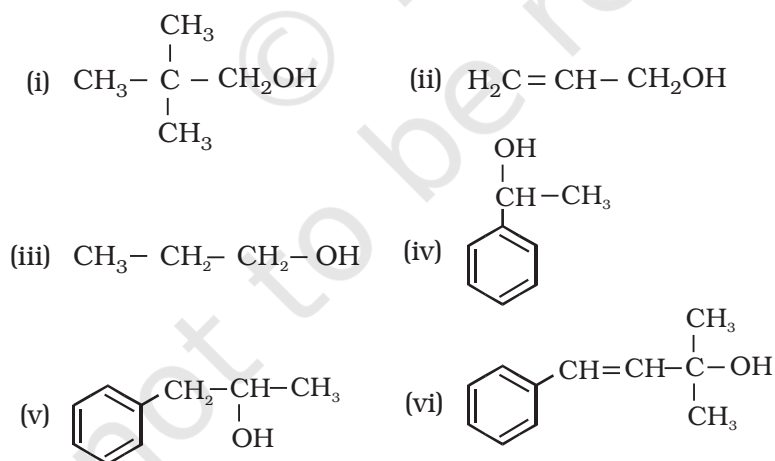


7.1.3 Ethers

Ethers are classified as **simple** or **symmetrical**, if the alkyl or aryl groups attached to the oxygen atom are the same, and **mixed** or **unsymmetrical**, if the two groups are different. Diethyl ether, $C_2H_5OC_2H_5$, is a symmetrical ether whereas $C_2H_5OCH_3$ and $C_2H_5OC_6H_5$ are unsymmetrical ethers.

Intext Questions

7.1 Classify the following as primary, secondary and tertiary alcohols:



7.2 Identify allylic alcohols in the above examples.

7.2 Nomenclature

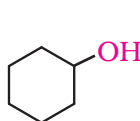
(a) **Alcohols:** The common name of an alcohol is derived from the common name of the alkyl group and adding the word alcohol to it. For example, CH_3OH is methyl alcohol.

According to IUPAC system, the name of an alcohol is derived from the name of the alkane from which the alcohol is derived, by substituting 'e' of alkane with the suffix 'ol'. The position of substituents are indicated by numerals. For this, the longest carbon chain (parent chain) is numbered starting at the end nearest to the hydroxyl group. The positions of the -OH group and other substituents are indicated by using the numbers of carbon atoms to which these are attached. For naming polyhydric alcohols, the 'e' of alkane is retained and the ending 'ol' is added. The number of -OH groups is indicated by adding the multiplicative prefix, di, tri, etc., before 'ol'. The positions of -OH groups are indicated by appropriate locants, e.g., HO-CH₂-CH₂-OH is named as ethane-1, 2-diol. Table 7.1 gives common and IUPAC names of a few alcohols as examples.

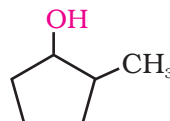
Table 7.1: Common and IUPAC Names of Some Alcohols

Compound	Common name	IUPAC name
CH ₃ - OH	Methyl alcohol	Methanol
CH ₃ - CH ₂ - CH ₂ - OH	<i>n</i> -Propyl alcohol	Propan-1-ol
$\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH}_3 \\ \\ \text{OH} \end{array}$	Isopropyl alcohol	Propan-2-ol
CH ₃ - CH ₂ - CH ₂ - CH ₂ - OH	<i>n</i> -Butyl alcohol	Butan-1-ol
$\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH}_2 - \text{CH}_3 \\ \\ \text{OH} \end{array}$	<i>sec</i> -Butyl alcohol	Butan-2-ol
$\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH}_2 - \text{OH} \\ \\ \text{CH}_3 \end{array}$	Isobutyl alcohol	2-Methylpropan-1-ol
$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3 - \text{C} - \text{OH} \\ \\ \text{CH}_3 \end{array}$	<i>tert</i> -Butyl alcohol	2-Methylpropan-2-ol
HO - H ₂ C - CH ₂ - OH	Ethylene glycol	Ethane-1,2-diol
$\begin{array}{ccc} \text{CH}_2 & - \text{CH} & - \text{CH}_2 \\ & & \\ \text{OH} & \text{OH} & \text{OH} \end{array}$	Glycerol	Propane -1, 2, 3-triol

Cyclic alcohols are named using the prefix cyclo and considering the -OH group attached to C-1.

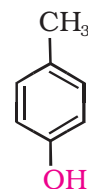
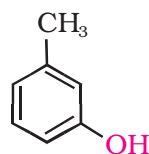
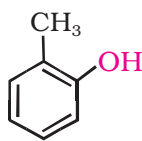
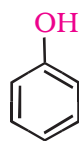


Cyclohexanol



2-Methylcyclopentanol

(b) Phenols: The simplest hydroxy derivative of benzene is phenol. It is its common name and also an accepted IUPAC name. As structure of phenol involves a benzene ring, in its substituted compounds the terms *ortho* (1,2- disubstituted), *meta* (1,3-disubstituted) and *para* (1,4-disubstituted) are often used in the common names.



Common name
IUPAC name

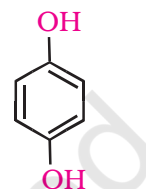
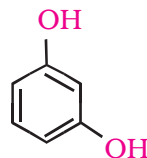
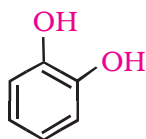
Phenol
Phenol

o-Cresol
2-Methylphenol

m-Cresol
3-Methylphenol

p-Cresol
4-Methylphenol

Dihydroxy derivatives of benzene are known as 1, 2-, 1, 3- and 1, 4-benzenediol.



Common name
IUPAC name

Catechol
Benzene-1,2-diol

Resorcinol
Benzene-1,3-diol

Hydroquinone or quinol
Benzene-1,4-diol

(c) Ethers: Common names of ethers are derived from the names of alkyl/aryl groups written as separate words in alphabetical order and adding the word 'ether' at the end. For example, $\text{CH}_3\text{OC}_2\text{H}_5$ is ethylmethyl ether.

Table 7.2: Common and IUPAC Names of Some Ethers

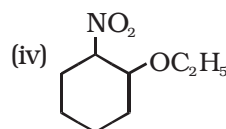
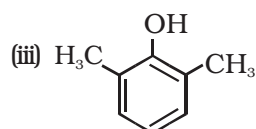
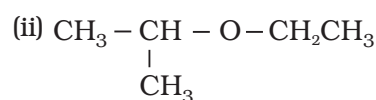
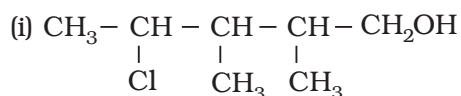
Compound	Common name	IUPAC name
CH_3OCH_3	Dimethyl ether	Methoxymethane
$\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$	Diethyl ether	Ethoxyethane
$\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_3$	Methyl n-propyl ether	1-Methoxypropane
$\text{C}_6\text{H}_5\text{OCH}_3$	Methyl phenyl ether (Anisole)	Methoxybenzene (Anisole)
$\text{C}_6\text{H}_5\text{OCH}_2\text{CH}_3$	Ethyl phenyl ether (Phenetole)	Ethoxybenzene
$\text{C}_6\text{H}_5\text{O}(\text{CH}_2)_6 - \text{CH}_3$	Heptyl phenyl ether	1-Phenoxyheptane
$\text{CH}_3\text{O}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$	Methyl isopropyl ether	2-Methoxypropane
$\text{C}_6\text{H}_5-\text{O}-\text{CH}_2-\text{CH}_2-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$	Phenyl isopentyl ether	3- Methylbutoxybenzene
$\text{CH}_3-\text{O}-\text{CH}_2-\text{CH}_2-\text{OCH}_3$	—	1,2-Dimethoxyethane
	—	2-Ethoxy- -1,1-dimethylcyclohexane

If both the alkyl groups are the same, the prefix 'di' is added before the alkyl group. For example, $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$ is diethyl ether.

According to IUPAC system of nomenclature, ethers are regarded as hydrocarbon derivatives in which a hydrogen atom is replaced by an $-\text{OR}$ or $-\text{OAr}$ group, where R and Ar represent alkyl and aryl groups, respectively. The larger (R) group is chosen as the parent hydrocarbon. The names of a few ethers are given as examples in Table 7.2.

Example 7.1

Give IUPAC names of the following compounds:



Solution

(i) 4-Chloro-2,3-dimethylpentan-1-ol

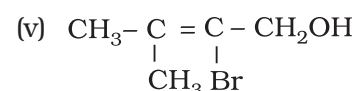
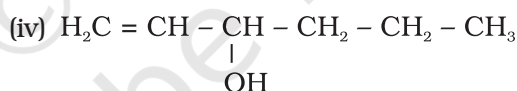
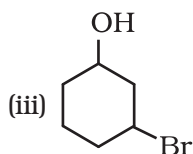
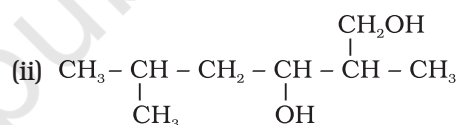
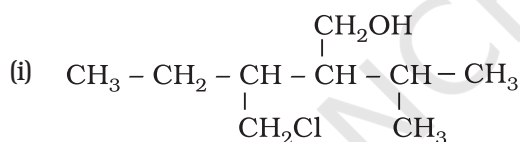
(ii) 2-Ethoxypropane

(iii) 2,6-Dimethylphenol

(iv) 1-Ethoxy-2-nitrocyclohexane

Intext Question

7.3 Name the following compounds according to IUPAC system.



7.3 Structures of Functional Groups

In alcohols, the oxygen of the $-\text{OH}$ group is attached to carbon by a sigma (σ) bond formed by the overlap of a sp^3 hybridised orbital of carbon with a sp^3 hybridised orbital of oxygen. Fig. 7.1 depicts structural aspects of methanol, phenol and methoxymethane.

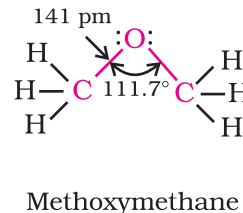
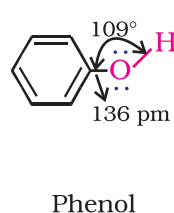
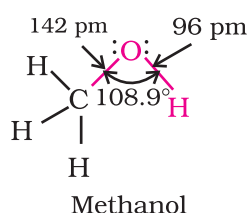


Fig. 7.1: Structures of methanol, phenol and methoxymethane

7.4 Alcohols and Phenols

The bond angle  in alcohols is slightly less than the tetrahedral angle (109°-28'). It is due to the repulsion between the unshared electron pairs of oxygen. In phenols, the -OH group is attached to sp^2 hybridised carbon of an aromatic ring. The carbon-oxygen bond length (136 pm) in phenol is slightly less than that in methanol. This is due to (i) partial double bond character on account of the conjugation of unshared electron pair of oxygen with the aromatic ring (Section 7.4.4) and (ii) sp^2 hybridised state of carbon to which oxygen is attached.

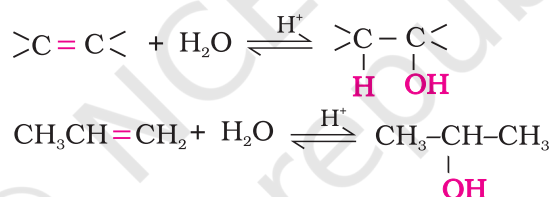
In ethers, the four electron pairs, i.e., the two bond pairs and two lone pairs of electrons on oxygen are arranged approximately in a tetrahedral arrangement. The bond angle is slightly greater than the tetrahedral angle due to the repulsive interaction between the two bulky (-R) groups. The C-O bond length (141 pm) is almost the same as in alcohols.

7.4.1 Preparation of Alcohols

Alcohols are prepared by the following methods:

1. From alkenes

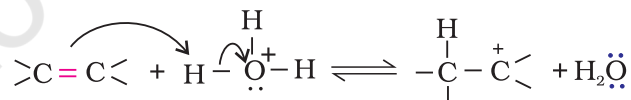
- (i) *By acid catalysed hydration:* Alkenes react with water in the presence of acid as catalyst to form alcohols. In case of unsymmetrical alkenes, the addition reaction takes place in accordance with Markovnikov's rule.



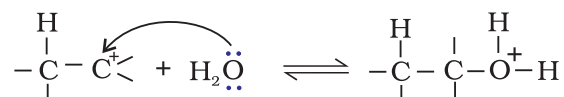
Mechanism

The mechanism of the reaction involves the following three steps:

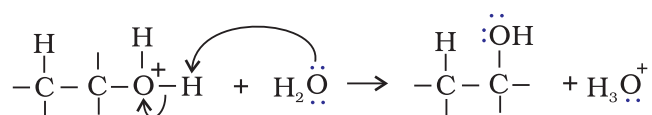
Step 1: Protonation of alkene to form carbocation by electrophilic attack of H_3O^+ .



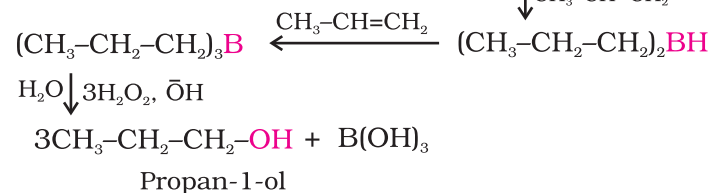
Step 2: Nucleophilic attack of water on carbocation.



Step 3: Deprotonation to form an alcohol.



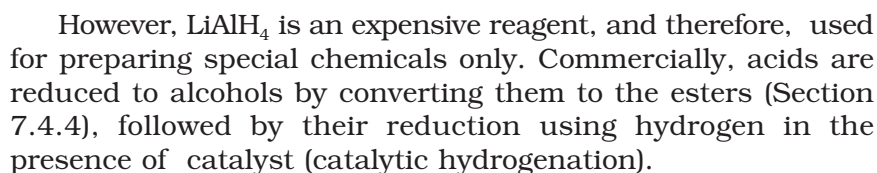
(ii) *By hydroboration-oxidation:* Diborane (BH_3)₂ reacts with alkenes to give trialkyl boranes as addition product. This is oxidised to alcohol by hydrogen peroxide in the presence of aqueous sodium hydroxide.



2. From carbonyl compounds

$$\begin{array}{l} \text{RCHO} + \text{H}_2 \xrightarrow{\text{Pd}} \text{RCH}_2\text{OH} \\ \text{RCOR}' \xrightarrow{\text{NaBH}_4} \text{R}-\underset{\text{OH}}{\text{CH}}-\text{R}' \end{array}$$

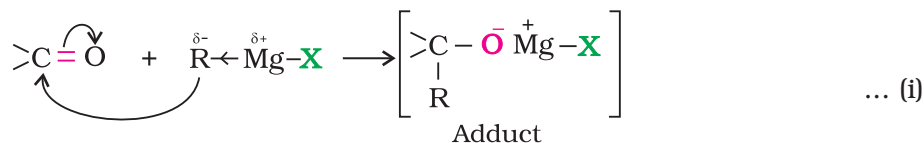
(ii) *By reduction of carboxylic acids and esters:* Carboxylic acids are reduced to primary alcohols in excellent yields by lithium aluminium hydride, a strong reducing agent.



3. From Grignard reagents

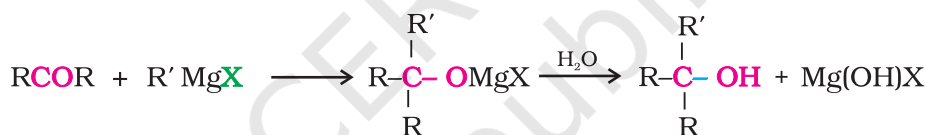
Alcohols are produced by the reaction of Grignard reagents (Unit 6, Class XII) with aldehydes and ketones.

The first step of the reaction is the nucleophilic addition of Grignard reagent to the carbonyl group to form an adduct. Hydrolysis of the adduct yields an alcohol.



The reaction of Grignard reagents with methanal produces a primary alcohol, with other aldehydes, secondary alcohols and with ketones, tertiary alcohols.

The overall reactions using different aldehydes and ketones are as follows:

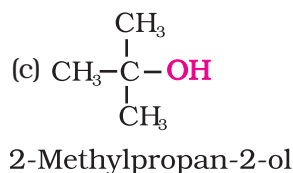
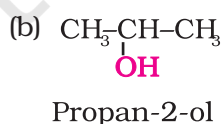
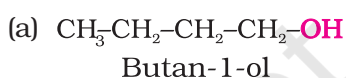


You will notice that the reaction produces a primary alcohol with methanal, a secondary alcohol with other aldehydes and tertiary alcohol with ketones.

Give the structures and IUPAC names of the products expected from the following reactions:

Example 7.2

- Catalytic reduction of butanal.
- Hydration of propene in the presence of dilute sulphuric acid.
- Reaction of propanone with methylmagnesium bromide followed by hydrolysis.



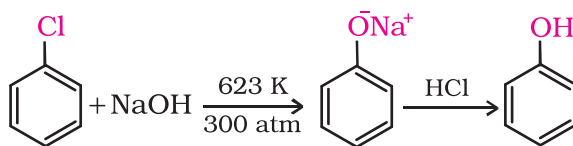
Solution

7.4.2 Preparation of Phenols

Phenol, also known as carboic acid, was first isolated in the early nineteenth century from coal tar. Nowadays, phenol is commercially produced synthetically. In the laboratory, phenols are prepared from benzene derivatives by any of the following methods:

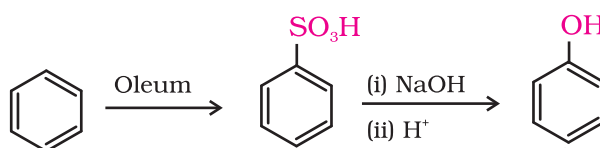
1. From haloarenes

Chlorobenzene is fused with NaOH at 623K and 320 atmospheric pressure. Phenol is obtained by acidification of sodium phenoxide so produced (Unit 6, Class XII).



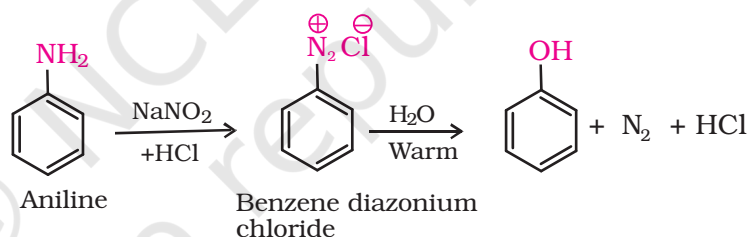
2. From benzenesulphonic acid

Benzene is sulphonated with oleum and benzene sulphonic acid so formed is converted to sodium phenoxide on heating with molten sodium hydroxide. Acidification of the sodium salt gives phenol.



3. From diazonium salts

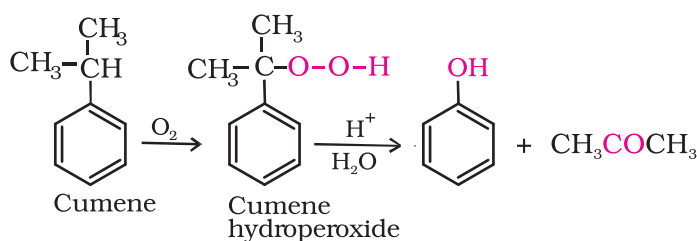
A diazonium salt is formed by treating an aromatic primary amine with nitrous acid ($\text{NaNO}_2 + \text{HCl}$) at 273-278 K. Diazonium salts are hydrolysed to phenols by warming with water or by treating with dilute acids (Unit 9, Class XII).



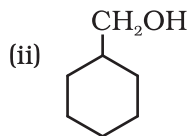
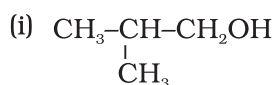
Most of the worldwide production of phenol is from cumene.

4. From cumene

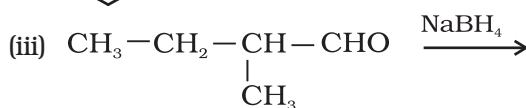
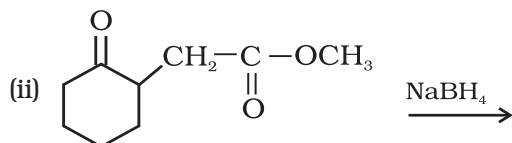
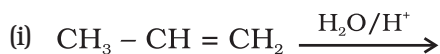
Phenol is manufactured from the hydrocarbon, cumene. Cumene (isopropylbenzene) is oxidised in the presence of air to cumene hydroperoxide. It is converted to phenol and acetone by treating it with dilute acid. Acetone, a by-product of this reaction, is also obtained in large quantities by this method.



7.4 Show how are the following alcohols prepared by the reaction of a suitable Grignard reagent on methanal ?



7.5 Write structures of the products of the following reactions:



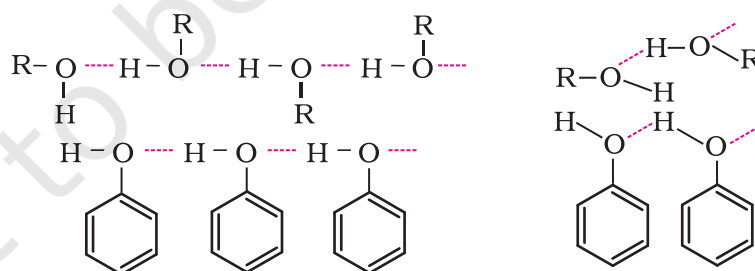
7.4.3 Physical Properties

Alcohols and phenols consist of two parts, an alkyl/aryl group and a hydroxyl group. The properties of alcohols and phenols are chiefly due to the hydroxyl group. The nature of alkyl and aryl groups simply modify these properties.

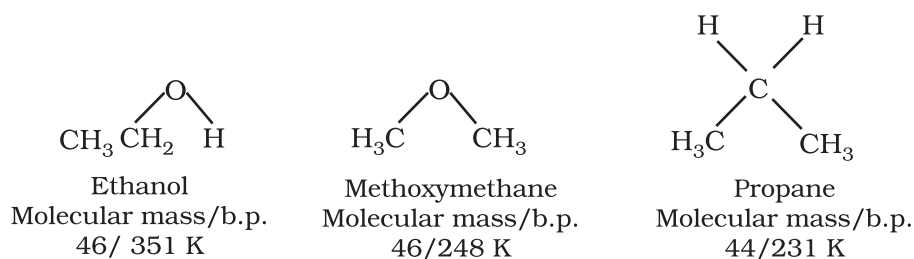
Boiling Points

The boiling points of alcohols and phenols increase with increase in the number of carbon atoms (increase in van der Waals forces). In alcohols, the boiling points decrease with increase of branching in carbon chain (because of decrease in van der Waals forces with decrease in surface area).

The -OH group in alcohols and phenols is involved in intermolecular hydrogen bonding as shown below:



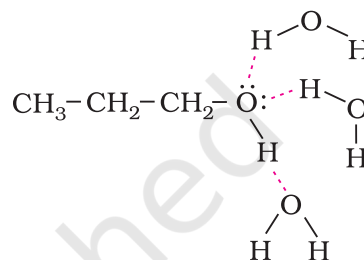
It is interesting to note that boiling points of alcohols and phenols are higher in comparison to other classes of compounds, namely hydrocarbons, ethers, haloalkanes and haloarenes of comparable molecular masses. For example, ethanol and propane have comparable molecular masses but their boiling points differ widely. The boiling point of methoxymethane is intermediate of the two boiling points.



The high boiling points of alcohols are mainly due to the presence of intermolecular hydrogen bonding in them which is lacking in ethers and hydrocarbons.

Solubility

Solubility of alcohols and phenols in water is due to their ability to form hydrogen bonds with water molecules as shown. The solubility decreases with increase in size of alkyl/aryl (hydrophobic) groups. Several of the lower molecular mass alcohols are miscible with water in all proportions.



Example 7.3

Arrange the following sets of compounds in order of their increasing boiling points:

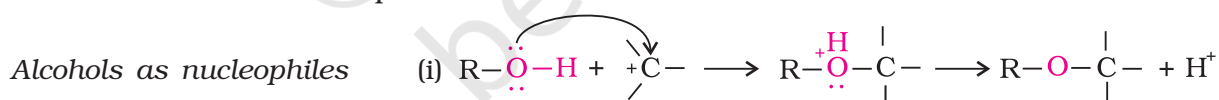
- (a) Pentan-1-ol, butan-1-ol, butan-2-ol, ethanol, propan-1-ol, methanol.
 (b) Pentan-1-ol, n-butane, pentanal, ethoxyethane.

Solution

- (a) Methanol, ethanol, propan-1-ol, butan-2-ol, butan-1-ol, pentan-1-ol.
 (b) n-Butane, ethoxyethane, pentanal and pentan-1-ol.

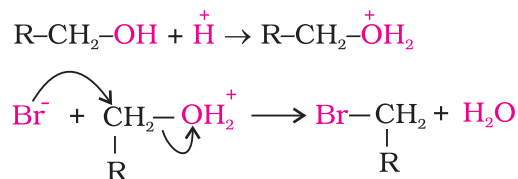
7.4.4 Chemical Reactions

Alcohols are versatile compounds. They react both as nucleophiles and electrophiles. The bond between O-H is broken when alcohols react as nucleophiles.



- (ii) The bond between C-O is broken when they react as electrophiles. Protonated alcohols react in this manner.

Protonated alcohols as electrophiles

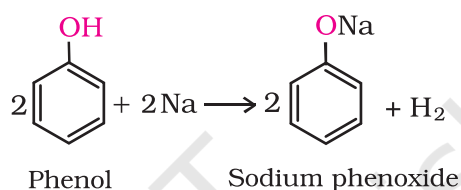
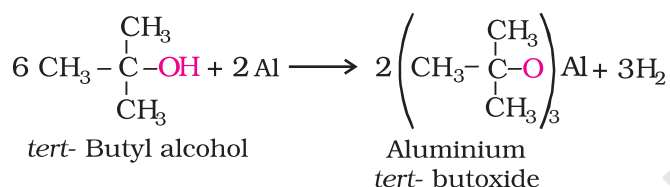


Based on the cleavage of O-H and C-O bonds, the reactions of alcohols and phenols may be divided into two groups:

1. Acidity of alcohols and phenols

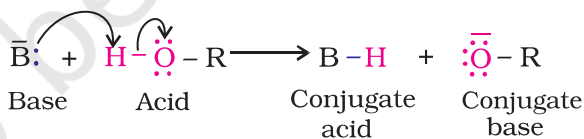
$$2\text{R}-\text{O}-\text{H} + 2\text{Na} \longrightarrow 2\text{R}-\text{O}-\text{Na} + \text{H}_2$$

Sodium
alkoxide

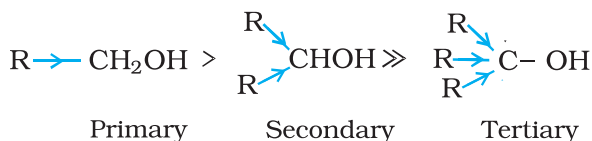



Oc1ccccc1.[Na]OH>>[Na]Oc1ccccc1.O

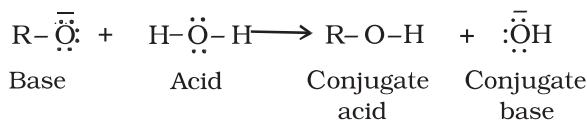
The above reactions show that alcohols and phenols are acidic in nature. In fact, alcohols and phenols are Brönsted acids i.e., they can donate a proton to a stronger base (B:).



(ii) *Acidity of alcohols:* The acidic character of alcohols is due to the polar nature of O-H bond. An electron-releasing group ($-\text{CH}_3$, $-\text{C}_2\text{H}_5$) increases electron density on oxygen tending to decrease the polarity of O-H bond. This decreases the acid strength. For this reason, the acid strength of alcohols decreases in the following order:



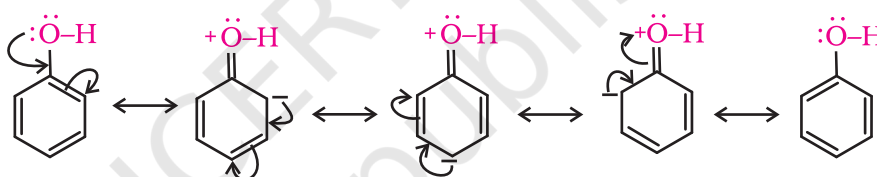
Alcohols are, however, weaker acids than water. This can be illustrated by the reaction of water with an alkoxide.



This reaction shows that water is a better proton donor (i.e., stronger acid) than alcohol. Also, in the above reaction, we note that an alkoxide ion is a better proton acceptor than hydroxide ion, which suggests that alkoxides are stronger bases (sodium ethoxide is a stronger base than sodium hydroxide).

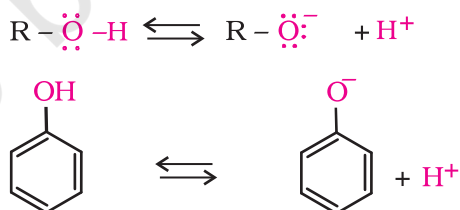
Alcohols act as Bronsted bases as well. It is due to the presence of unshared electron pairs on oxygen, which makes them proton acceptors.

- (iii) *Acidity of phenols:* The reactions of phenol with metals (e.g., sodium, aluminium) and sodium hydroxide indicate its acidic nature. The hydroxyl group, in phenol is directly attached to the sp^2 hybridised carbon of benzene ring which acts as an electron withdrawing group. Due to this, the charge distribution in phenol molecule, as depicted in its resonance structures, causes the oxygen of -OH group to be positive.



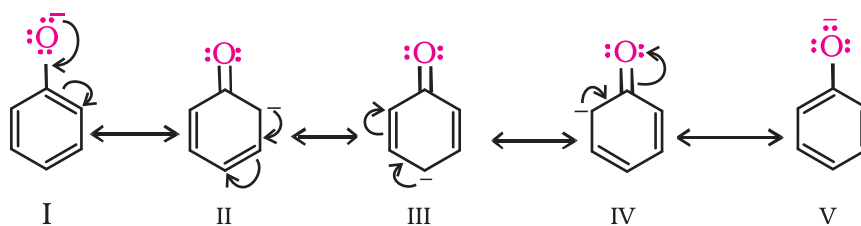
The reaction of phenol with aqueous sodium hydroxide indicates that phenols are stronger acids than alcohols and water. Let us examine how a compound in which hydroxyl group attached to an aromatic ring is more acidic than the one in which hydroxyl group is attached to an alkyl group.

The ionisation of an alcohol and a phenol takes place as follows:



Due to the higher electronegativity of sp^2 hybridised carbon of phenol to which -OH is attached, electron density decreases on oxygen. This increases the polarity of O-H bond and results in an increase in ionisation of phenols than that of alcohols. Now let us examine the stabilities of alkoxide and phenoxide ions. In alkoxide ion, the negative charge is localised on oxygen while in phenoxide ion, the charge is delocalised. The delocalisation of negative charge (structures I-V) makes

phenoxide ion more stable and favours the ionisation of phenol. Although there is also charge delocalisation in phenol, its resonance structures have charge separation due to which the phenol molecule is less stable than phenoxide ion.



In substituted phenols, the presence of electron withdrawing groups such as nitro group, enhances the acidic strength of phenol. This effect is more pronounced when such a group is present at *ortho* and *para* positions. It is due to the effective delocalisation of negative charge in phenoxide ion when substituent is at *ortho* or *para* position. On the other hand, electron releasing groups, such as alkyl groups, in general, do not favour the formation of phenoxide ion resulting in decrease in acid strength. Cresols, for example, are less acidic than phenol.

The greater the pK_a value, the weaker the acid.

Table 7.3: pK_a Values of some Phenols and Ethanol

Compound	Formula	pK_a
<i>o</i> -Nitrophenol	<i>o</i> -O ₂ N-C ₆ H ₄ -OH	7.2
<i>m</i> -Nitrophenol	<i>m</i> -O ₂ N-C ₆ H ₄ -OH	8.3
<i>p</i> -Nitrophenol	<i>p</i> -O ₂ N-C ₆ H ₄ -OH	7.1
Phenol	C ₆ H ₅ -OH	10.0
<i>o</i> -Cresol	<i>o</i> -CH ₃ -C ₆ H ₄ -OH	10.2
<i>m</i> -Cresol	<i>m</i> -CH ₃ -C ₆ H ₄ -OH	10.1
<i>p</i> -Cresol	<i>p</i> -CH ₃ -C ₆ H ₄ -OH	10.2
Ethanol	C ₂ H ₅ OH	15.9

From the above data, you will note that phenol is million times more acidic than ethanol.

Arrange the following compounds in increasing order of their acid strength: **Example 7.4**

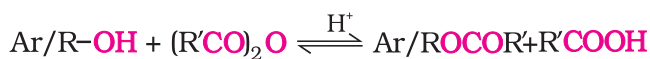
Propan-1-ol, 2,4,6-trinitrophenol, 3-nitrophenol, 3,5-dinitrophenol, phenol, 4-methylphenol.

Propan-1-ol, 4-methylphenol, phenol, 3-nitrophenol, 3,5-dinitrophenol, 2,4, 6-trinitrophenol. **Solution**

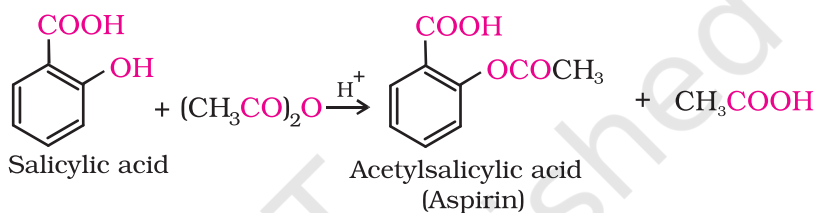
2. Esterification

Alcohols and phenols react with carboxylic acids, acid chlorides and acid anhydrides to form esters.

Aspirin possesses analgesic, anti-inflammatory and antipyretic properties.



The reaction with carboxylic acid and acid anhydride is carried out in the presence of a small amount of concentrated sulphuric acid. The reaction is reversible, and therefore, water is removed as soon as it is formed. The reaction with acid chloride is carried out in the presence of a base (pyridine) so as to neutralise HCl which is formed during the reaction. It shifts the equilibrium to the right hand side. The introduction of acetyl (CH_3CO) group in alcohols or phenols is known as acetylation. Acetylation of salicylic acid produces aspirin.



(b) Reactions involving cleavage of carbon – oxygen (C–O) bond in alcohols

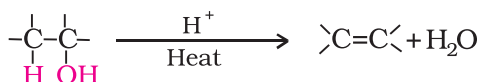
The reactions involving cleavage of C–O bond take place only in alcohols. Phenols show this type of reaction only with zinc.

- 1. Reaction with hydrogen halides:** Alcohols react with hydrogen halides to form alkyl halides (Refer Unit 6, Class XII).

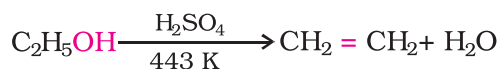


The difference in reactivity of three classes of alcohols with HCl distinguishes them from one another (**Lucas test**). Alcohols are soluble in Lucas reagent (conc. HCl and ZnCl_2) while their halides are immiscible and produce turbidity in solution. In case of tertiary alcohols, turbidity is produced immediately as they form the halides easily. Primary alcohols do not produce turbidity at room temperature.

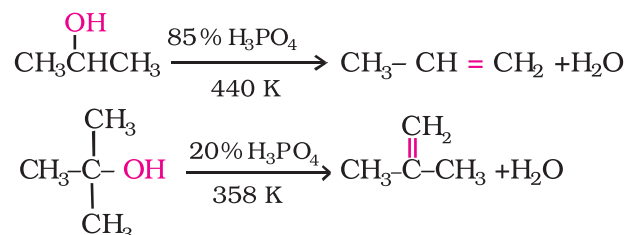
- 2. Reaction with phosphorus trihalides:** Alcohols are converted to alkyl bromides by reaction with phosphorus tribromide (Refer Unit 6, Class XII).
- 3. Dehydration:** Alcohols undergo dehydration (removal of a molecule of water) to form alkenes on treating with a protic acid e.g., concentrated H_2SO_4 or H_3PO_4 , or catalysts such as anhydrous zinc chloride or alumina.



Ethanol undergoes dehydration by heating it with concentrated H_2SO_4 at 443 K.



Secondary and tertiary alcohols are dehydrated under milder conditions. For example



Thus, the relative ease of dehydration of alcohols follows the following order:

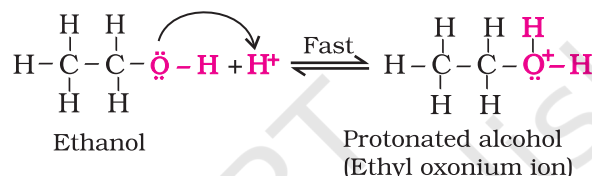


The mechanism of dehydration of ethanol involves the following steps:

Tertiary carbocations are more stable and therefore are easier to form than secondary and primary carbocations; tertiary alcohols are the easiest to dehydrate.

Mechanism

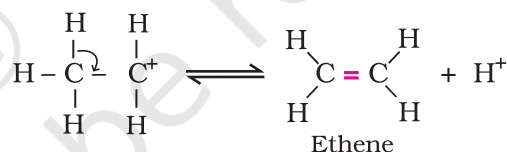
Step 1: Formation of protonated alcohol.



Step 2: Formation of carbocation: It is the slowest step and hence, the rate determining step of the reaction.

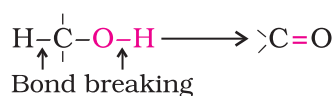


Step 3: Formation of ethene by elimination of a proton.

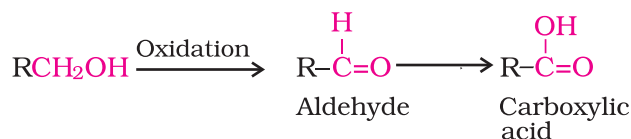


The acid used in step 1 is released in step 3. To drive the equilibrium to the right, ethene is removed as it is formed.

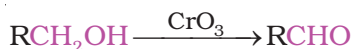
4. Oxidation: Oxidation of alcohols involves the formation of a carbon-oxygen double bond with cleavage of an O-H and C-H bonds.



Such a cleavage and formation of bonds occur in oxidation reactions. These are also known as **dehydrogenation** reactions as these involve loss of dihydrogen from an alcohol molecule. Depending on the oxidising agent used, a primary alcohol is oxidised to an aldehyde which in turn is oxidised to a carboxylic acid.



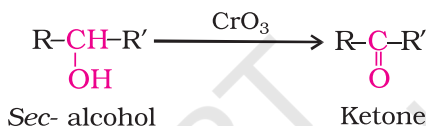
Strong oxidising agents such as acidified potassium permanganate are used for getting carboxylic acids from alcohols directly. CrO_3 in anhydrous medium is used as the oxidising agent for the isolation of aldehydes.



A better reagent for oxidation of primary alcohols to aldehydes in good yield is pyridinium chlorochromate (PCC), a complex of chromium trioxide with pyridine and HCl.

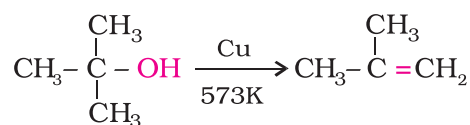
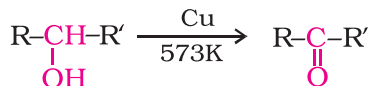
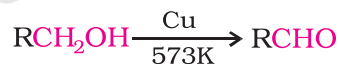


Secondary alcohols are oxidised to ketones by chromic anhydride (CrO_3).



Tertiary alcohols do not undergo oxidation reaction. Under strong reaction conditions such as strong oxidising agents (KMnO_4) and elevated temperatures, cleavage of various C-C bonds takes place and a mixture of carboxylic acids containing lesser number of carbon atoms is formed.

When the vapours of a primary or a secondary alcohol are passed over heated copper at 573 K, dehydrogenation takes place and an aldehyde or a ketone is formed while tertiary alcohols undergo dehydration.



Biological oxidation of methanol and ethanol in the body produces the corresponding aldehyde followed by the acid. At times the alcoholics, by mistake, drink ethanol, mixed with methanol also called denatured alcohol. In the body, methanol is oxidised first to methanal and then to methanoic acid, which may cause blindness and death. A methanol poisoned patient is treated by giving intravenous infusions of diluted ethanol. The enzyme responsible for oxidation of aldehyde (HCHO) to acid is swamped allowing time for kidneys to excrete methanol.

(c) Reactions of phenols

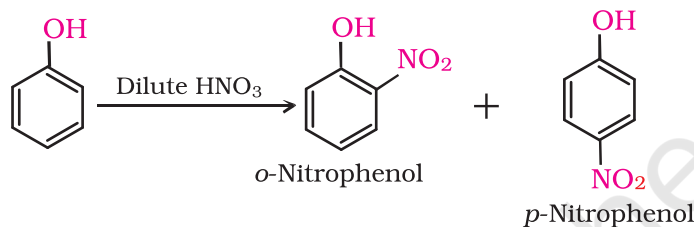
Following reactions are shown by phenols only.

1. Electrophilic aromatic substitution

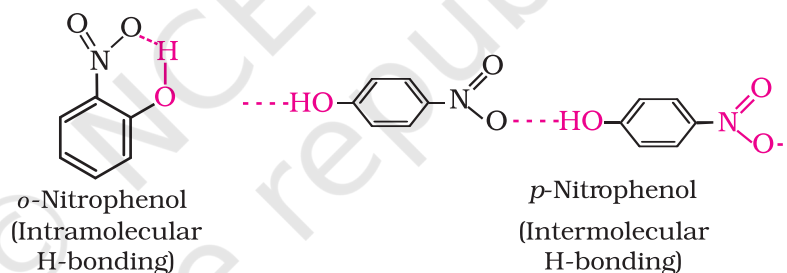
In phenols, the reactions that take place on the aromatic ring are electrophilic substitution reactions (Unit 9, Class XI). The -OH group attached to the benzene ring activates it towards electrophilic substitution. Also, it directs the incoming group to *ortho* and *para* positions in the ring as these positions become electron rich due to the resonance effect caused by -OH group. The resonance structures are shown under acidity of phenols.

Common electrophilic aromatic substitution reactions taking place in phenol are as follows:

- (i) *Nitration*: With dilute nitric acid at low temperature (298 K), phenol yields a mixture of *ortho* and *para* nitrophenols.

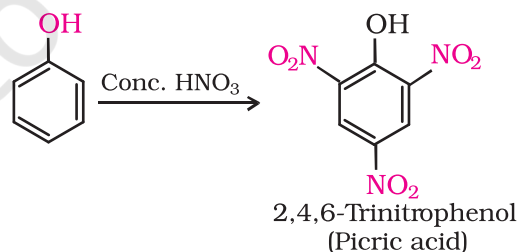


The *ortho* and *para* isomers can be separated by steam distillation. *o*-Nitrophenol is steam volatile due to intramolecular hydrogen bonding while *p*-nitrophenol is less volatile due to intermolecular hydrogen bonding which causes the association of molecules.



2, 4, 6 - Trinitrophenol is a strong acid due to the presence of three electron withdrawing -NO_2 groups which facilitate the release of hydrogen ion.

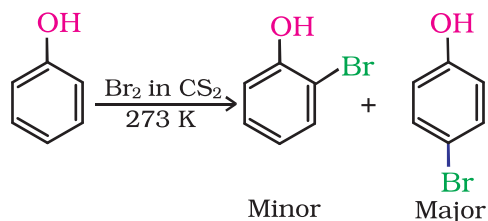
With concentrated nitric acid, phenol is converted to 2,4,6-trinitrophenol. The product is commonly known as picric acid. The yield of the reaction product is poor.



Nowadays picric acid is prepared by treating phenol first with concentrated sulphuric acid which converts it to phenol-2,4-disulphonic acid, and then with concentrated nitric acid to get 2,4,6-trinitrophenol. Can you write the equations of the reactions involved?

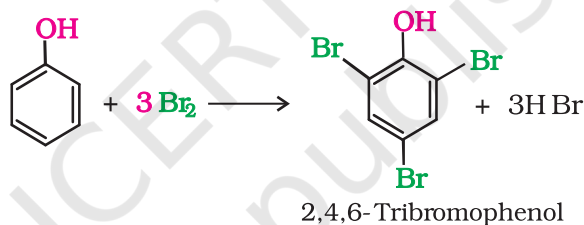
(ii) **Halogenation:** On treating phenol with bromine, different reaction products are formed under different experimental conditions.

- (a) When the reaction is carried out in solvents of low polarity such as CHCl_3 or CS_2 and at low temperature, monobromophenols are formed.



The usual halogenation of benzene takes place in the presence of a Lewis acid, such as FeBr_3 (Unit 6, Class XII), which polarises the halogen molecule. In case of phenol, the polarisation of bromine molecule takes place even in the absence of Lewis acid. It is due to the highly activating effect of $-\text{OH}$ group attached to the benzene ring.

- (b) When phenol is treated with bromine water, 2,4,6-tribromophenol is formed as white precipitate.



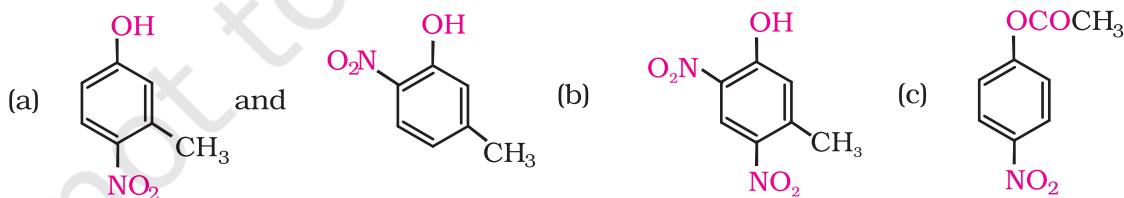
Example 7.5

Write the structures of the major products expected from the following reactions:

- (a) Mononitration of 3-methylphenol
(b) Dinitration of 3-methylphenol
(c) Mononitration of phenyl methanoate.

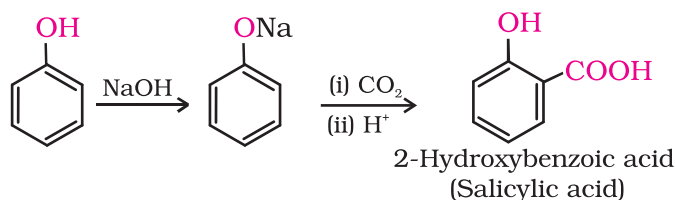
Solution

The combined influence of $-\text{OH}$ and $-\text{CH}_3$ groups determine the position of the incoming group.



2. Kolbe's reaction

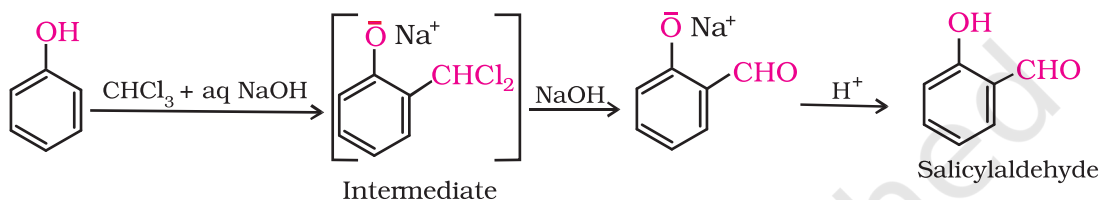
Phenoxide ion generated by treating phenol with sodium hydroxide is even more reactive than phenol towards electrophilic aromatic substitution. Hence, it undergoes electrophilic substitution with carbon dioxide, a weak electrophile. *Ortho* hydroxybenzoic acid is formed as the main reaction product.



3. Reimer-Tiemann reaction

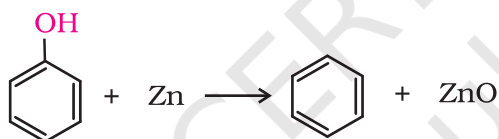
On treating phenol with chloroform in the presence of sodium hydroxide, a -CHO group is introduced at *ortho* position of benzene ring. This reaction is known as *Reimer - Tiemann reaction*.

The intermediate substituted benzal chloride is hydrolysed in the presence of alkali to produce salicylaldehyde.



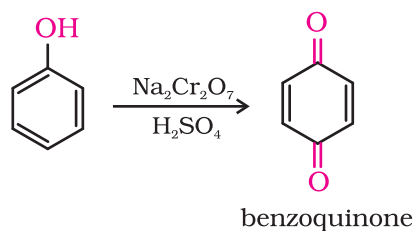
4. Reaction of phenol with zinc dust

Phenol is converted to benzene on heating with zinc dust.



5. Oxidation

Oxidation of phenol with chromic acid produces a conjugated diketone known as benzoquinone. In the presence of air, phenols are slowly oxidised to dark coloured mixtures containing quinones.



Intext Questions

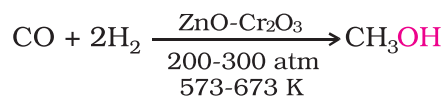
- 7.6** Give structures of the products you would expect when each of the following alcohol reacts with (a) HCl - ZnCl_2 (b) HBr and (c) SOCl_2 .
 (i) Butan-1-ol (ii) 2-Methylbutan-2-ol
- 7.7** Predict the major product of acid catalysed dehydration of
 (i) 1-methylcyclohexanol and (ii) butan-1-ol
- 7.8** *Ortho* and *para* nitrophenols are more acidic than phenol. Draw the resonance structures of the corresponding phenoxide ions.
- 7.9** Write the equations involved in the following reactions:
 (i) Reimer - Tiemann reaction (ii) Kolbe's reaction

7.5 Some Commercially Important Alcohols

Methanol and ethanol are among the two commercially important alcohols.

1. Methanol

Methanol, CH_3OH , also known as 'wood spirit', was produced by destructive distillation of wood. Today, most of the methanol is produced by catalytic hydrogenation of carbon monoxide at high pressure and temperature and in the presence of $\text{ZnO} - \text{Cr}_2\text{O}_3$ catalyst.



Methanol is a colourless liquid and boils at 337 K. It is highly poisonous in nature. Ingestion of even small quantities of methanol can cause blindness and large quantities causes even death. Methanol is used as a solvent in paints, varnishes and chiefly for making formaldehyde.

2. Ethanol

Ethanol, $\text{C}_2\text{H}_5\text{OH}$, is obtained commercially by fermentation, the oldest method is from sugars. The sugar in molasses, sugarcane or fruits such as grapes is converted to glucose and fructose, (both of which have the formula $\text{C}_6\text{H}_{12}\text{O}_6$), in the presence of an enzyme, invertase. Glucose and fructose undergo fermentation in the presence of another enzyme, zymase, which is found in yeast.



Ingestion of ethanol acts on the central nervous system. In moderate amounts, it affects judgment and lowers inhibitions. Higher concentrations cause nausea and loss of consciousness. Even at higher concentrations, it interferes with spontaneous respiration and can be fatal.

In wine making, grapes are the source of sugars and yeast. As grapes ripen, the quantity of sugar increases and yeast grows on the outer skin. When grapes are crushed, sugar and the enzyme come in contact and fermentation starts. Fermentation takes place in anaerobic conditions i.e. in absence of air. Carbon dioxide is released during fermentation.

The action of zymase is inhibited once the percentage of alcohol formed exceeds 14 percent. If air gets into fermentation mixture, the oxygen of air oxidises ethanol to ethanoic acid which in turn destroys the taste of alcoholic drinks.

Ethanol is a colourless liquid with boiling point 351 K. It is used as a solvent in paint industry and in the preparation of a number of carbon compounds. The commercial alcohol is made unfit for drinking by mixing in it some copper sulphate (to give it a colour) and pyridine (a foul smelling liquid). It is known as **denaturation** of alcohol.

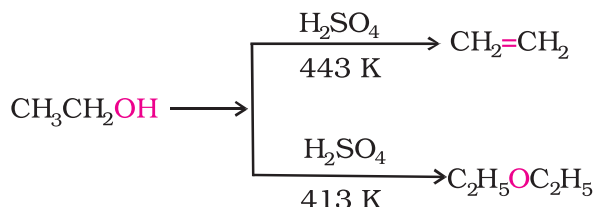
Nowadays, large quantities of ethanol are obtained by hydration of ethene (Section 7.4).

7.6 Ethers

7.6.1 Preparation of Ethers

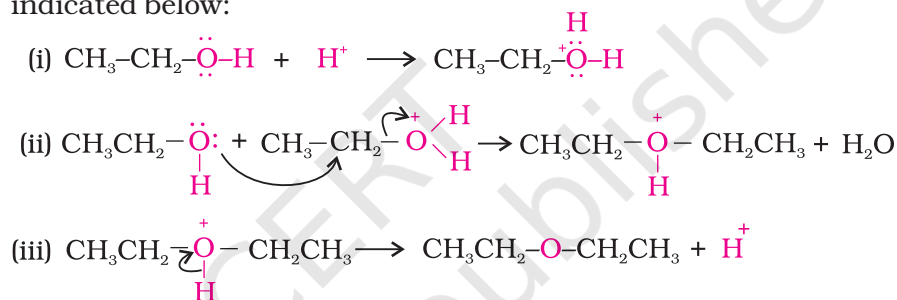
1. By dehydration of alcohols

Alcohols undergo dehydration in the presence of protic acids (H_2SO_4 , H_3PO_4). The formation of the reaction product, alkene or ether depends on the reaction conditions. For example, ethanol is dehydrated to ethene in the presence of sulphuric acid at 443 K. At 413 K, ethoxyethane is the main product.



Diethyl ether has been used widely as an inhalation anaesthetic. But due to its slow effect and an unpleasant recovery period, it has been replaced, as an anaesthetic, by other compounds.

The formation of ether is a nucleophilic bimolecular reaction ($\text{S}_{\text{N}}2$) involving the attack of alcohol molecule on a protonated alcohol, as indicated below:



Acidic dehydration of alcohols, to give an alkene is also associated with substitution reaction to give an ether.

The method is suitable for the preparation of ethers having primary alkyl groups only. The alkyl group should be unhindered and the temperature be kept low. Otherwise the reaction favours the formation of alkene. The reaction follows $\text{S}_{\text{N}}1$ pathway when the alcohol is secondary or tertiary about which you will learn in higher classes. However, the dehydration of secondary and tertiary alcohols to give corresponding ethers is unsuccessful as elimination competes over substitution and as a consequence, alkenes are easily formed.

Can you explain why is bimolecular dehydration not appropriate for the preparation of ethyl methyl ether?

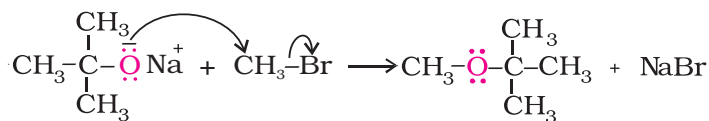
2. Williamson synthesis

Alexander William Williamson (1824–1904) was born in London of Scottish parents. In 1849, he became Professor of Chemistry at University College, London.

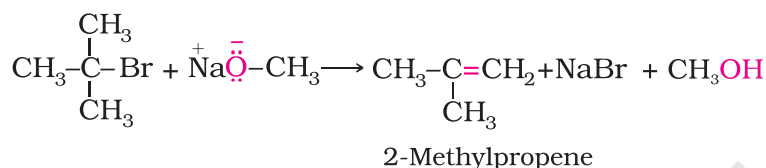
It is an important laboratory method for the preparation of symmetrical and unsymmetrical ethers. In this method, an alkyl halide is allowed to react with sodium alkoxide.



Ethers containing substituted alkyl groups (secondary or tertiary) may also be prepared by this method. The reaction involves $\text{S}_{\text{N}}2$ attack of an alkoxide ion on primary alkyl halide.



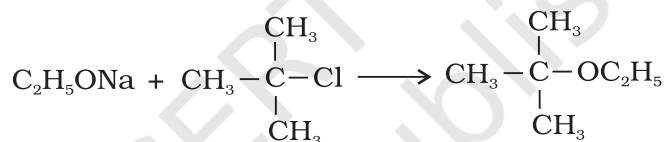
Better results are obtained if the alkyl halide is primary. In case of secondary and tertiary alkyl halides, elimination competes over substitution. If a tertiary alkyl halide is used, an alkene is the only reaction product and no ether is formed. For example, the reaction of CH_3ONa with $(\text{CH}_3)_3\text{C}-\text{Br}$ gives exclusively 2-methylpropene.



It is because alkoxides are not only nucleophiles but strong bases as well. They react with alkyl halides leading to elimination reactions.

Example 7.6

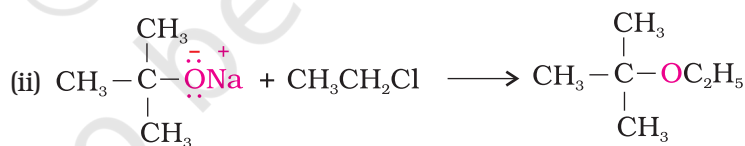
The following is not an appropriate reaction for the preparation of t-butyl ethyl ether.



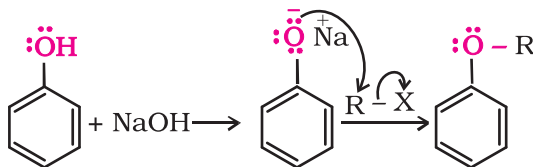
- (i) What would be the major product of this reaction ?
- (ii) Write a suitable reaction for the preparation of t-butylethyl ether.

Solution

- (i) The major product of the given reaction is 2-methylprop-1-ene. It is because sodium ethoxide is a strong nucleophile as well as a strong base. Thus elimination reaction predominates over substitution.



Phenols are also converted to ethers by this method. In this, phenol is used as the phenoxide moiety.



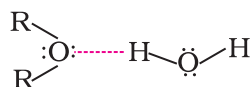
7.6.2 Physical Properties

The C-O bonds in ethers are polar and thus, ethers have a net dipole moment. The weak polarity of ethers do not appreciably affect their boiling points which are comparable to those of the alkanes of comparable molecular masses but are much lower than the boiling points of alcohols as shown in the following cases:

Formula	$\text{CH}_3(\text{CH}_2)_3\text{CH}_3$ n-Pentane	$\text{C}_2\text{H}_5\text{-O-C}_2\text{H}_5$ Ethoxyethane	$\text{CH}_3(\text{CH}_2)_3\text{-OH}$ Butan-1-ol
b.p./K	309.1	307.6	390

The large difference in boiling points of alcohols and ethers is due to the presence of hydrogen bonding in alcohols.

The miscibility of ethers with water resembles those of alcohols of the same molecular mass. Both ethoxyethane and butan-1-ol are miscible to almost the same extent i.e., 7.5 and 9 g per 100 mL water, respectively while pentane is essentially immiscible with water. Can you explain this observation? This is due to the fact that just like alcohols, oxygen of ether can also form hydrogen bonds with water molecule as shown:



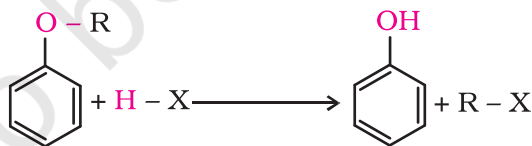
7.6.3 Chemical Reactions

1. Cleavage of C-O bond in ethers

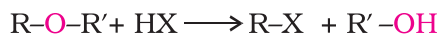
Ethers are the least reactive of the functional groups. The cleavage of C-O bond in ethers takes place under drastic conditions with excess of hydrogen halides. The reaction of dialkyl ether gives two alkyl halide molecules.



Alkyl aryl ethers are cleaved at the alkyl-oxygen bond due to the more stable aryl-oxygen bond. The reaction yields phenol and alkyl halide.



Ethers with two different alkyl groups are also cleaved in the same manner.

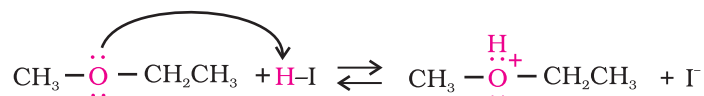


The order of reactivity of hydrogen halides is as follows: $\text{HI} > \text{HBr} > \text{HCl}$. The cleavage of ethers takes place with concentrated HI or HBr at high temperature.

Mechanism

The reaction of an ether with concentrated HI starts with protonation of ether molecule.

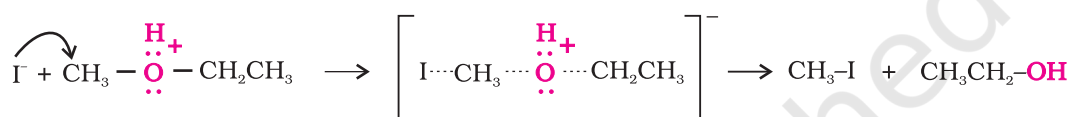
Step 1:



The reaction takes place with HBr or HI because these reagents are sufficiently acidic.

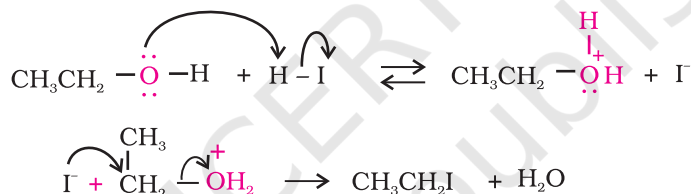
Step 2:

Iodide is a good nucleophile. It attacks the least substituted carbon of the oxonium ion formed in step 1 and displaces an alcohol molecule by $\text{S}_{\text{N}}2$ mechanism. Thus, in the cleavage of mixed ethers with two different alkyl groups, the alcohol and alkyl iodide formed, depend on the nature of alkyl groups. When primary or secondary alkyl groups are present, it is the lower alkyl group that forms alkyl iodide ($\text{S}_{\text{N}}2$ reaction).

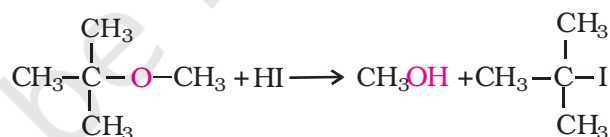


When HI is in excess and the reaction is carried out at high temperature, ethanol reacts with another molecule of HI and is converted to ethyl iodide.

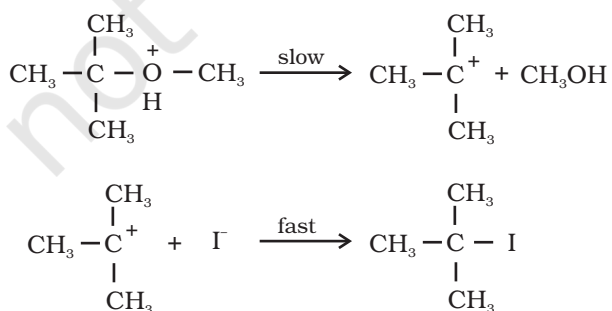
Step 3:



However, when one of the alkyl group is a tertiary group, the halide formed is a tertiary halide.



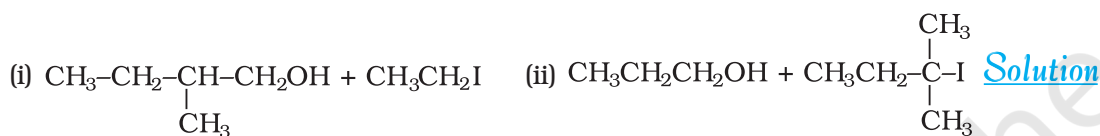
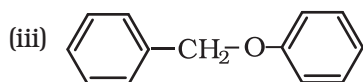
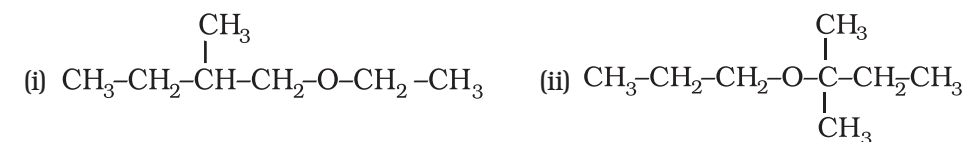
It is because in step 2 of the reaction, the departure of leaving group ($\text{HO}-\text{CH}_3$) creates a more stable carbocation $[(\text{CH}_3)_3\text{C}^+]$, and the reaction follows $\text{S}_{\text{N}}1$ mechanism.



In case of anisole, methylphenyl oxonium ion, $\text{C}_6\text{H}_5-\overset{\text{H}}{\underset{\cdot\cdot}{\text{O}}^+}-\text{CH}_3$ is formed by protonation of ether. The bond between $\text{O}-\text{CH}_3$ is weaker than the bond between $\text{O}-\text{C}_6\text{H}_5$ because the carbon of phenyl group is sp^2 hybridised and there is a partial double bond character.

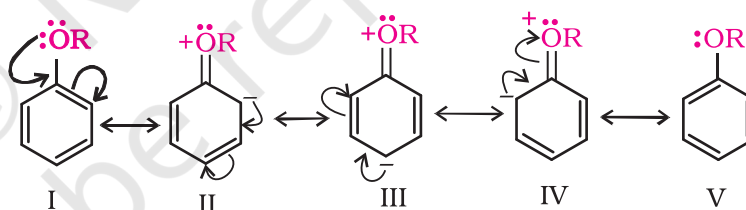
Therefore the attack by I^- ion breaks $\text{O}-\text{CH}_3$ bond to form CH_3I . Phenols do not react further to give halides because the sp^2 hybridised carbon of phenol cannot undergo nucleophilic substitution reaction needed for conversion to the halide.

Give the major products that are formed by heating each of the following ethers with HI. [Example 7.7](#)

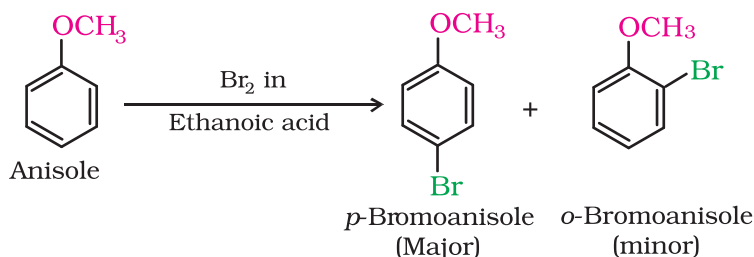


2. Electrophilic substitution

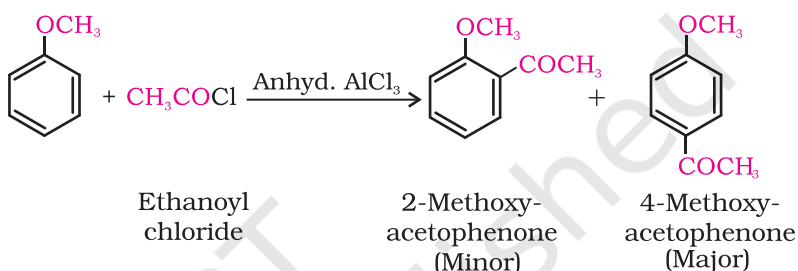
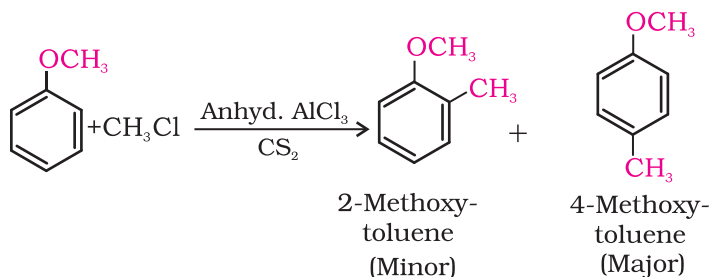
The alkoxy group ($-\text{OR}$) is *ortho*, *para* directing and activates the aromatic ring towards electrophilic substitution in the same way as in phenol.



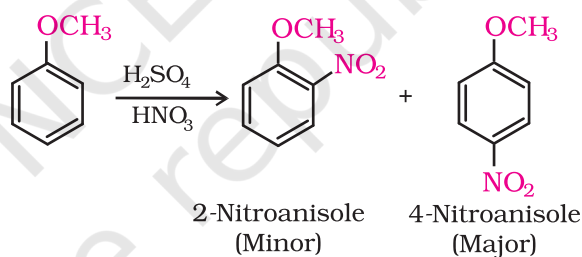
(i) **Halogenation:** Phenylalkyl ethers undergo usual halogenation in the benzene ring, *e.g.*, anisole undergoes bromination with bromine in ethanoic acid even in the absence of iron (III) bromide catalyst. It is due to the activation of benzene ring by the methoxy group. *Para* isomer is obtained in 90% yield.



- (ii) *Friedel-Crafts reaction*: Anisole undergoes Friedel-Crafts reaction, i.e., the alkyl and acyl groups are introduced at *ortho* and *para* positions by reaction with alkyl halide and acyl halide in the presence of anhydrous aluminium chloride (a Lewis acid) as catalyst.



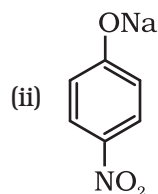
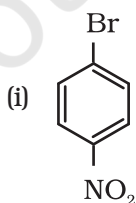
- (iii) *Nitration*: Anisole reacts with a mixture of concentrated sulphuric and nitric acids to yield a mixture of *ortho* and *para* nitroanisole.



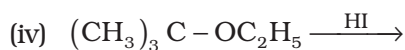
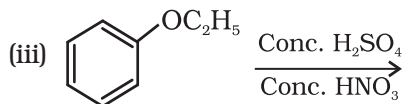
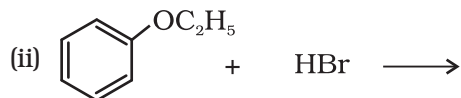
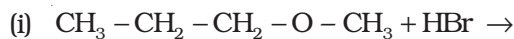
Intext Questions

7.10 Write the reactions of Williamson synthesis of 2-ethoxy-3-methylpentane starting from ethanol and 3-methylpentan-2-ol.

7.11 Which of the following is an appropriate set of reactants for the preparation of 1-methoxy-4-nitrobenzene and why?



7.12 Predict the products of the following reactions:



Summary

Alcohols and **phenols** are classified (i) on the basis of the number of hydroxyl groups and (ii) according to the hybridisation of the carbon atom, sp^3 or sp^2 to which the $-\text{OH}$ group is attached. **Ethers** are classified on the basis of groups attached to the oxygen atom.

Alcohols may be prepared (1) by hydration of alkenes (i) in presence of an acid and (ii) by hydroboration-oxidation reaction (2) from carbonyl compounds by (i) catalytic reduction and (ii) the action of Grignard reagents. Phenols may be prepared by (1) substitution of (i) halogen atom in haloarenes and (ii) sulphonic acid group in aryl sulphonic acids, by $-\text{OH}$ group (2) by hydrolysis of diazonium salts and (3) industrially from cumene.

Alcohols are higher boiling than other classes of compounds, namely hydrocarbons, ethers and haloalkanes of comparable molecular masses. The ability of alcohols, phenols and ethers to form intermolecular hydrogen bonding with water makes them soluble in it.

Alcohols and phenols are acidic in nature. **Electron withdrawing groups** in phenol increase its acidic strength and **electron releasing groups** decrease it.

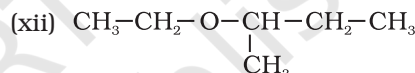
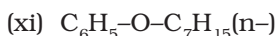
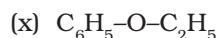
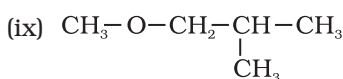
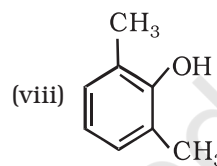
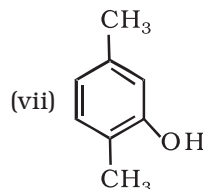
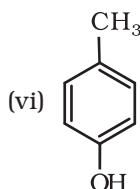
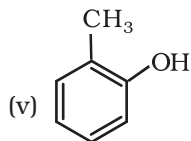
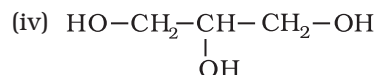
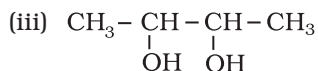
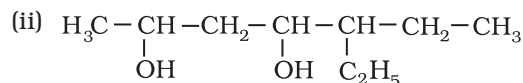
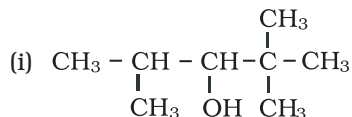
Alcohols undergo nucleophilic substitution with hydrogen halides to yield alkyl halides. Dehydration of alcohols gives alkenes. On oxidation, primary alcohols yield aldehydes with mild oxidising agents and carboxylic acids with strong oxidising agents while secondary alcohols yield ketones. Tertiary alcohols are resistant to oxidation.

The presence of $-\text{OH}$ group in phenols activates the aromatic ring towards **electrophilic substitution** and directs the incoming group to *ortho* and *para* positions due to resonance effect. **Reimer-Tiemann reaction** of phenol yields salicylaldehyde. In presence of sodium hydroxide, phenol generates phenoxide ion which is even more reactive than phenol. Thus, in alkaline medium, phenol undergoes **Kolbe's reaction**.

Ethers may be prepared by (i) dehydration of alcohols and (ii) **Williamson synthesis**. The boiling points of ethers resemble those of alkanes while their solubility is comparable to those of alcohols having same molecular mass. The $\text{C}-\text{O}$ bond in ethers can be cleaved by hydrogen halides. In electrophilic substitution, the alkoxy group activates the aromatic ring and directs the incoming group to *ortho* and *para* positions.

Exercises

7.1 Write IUPAC names of the following compounds:



7.2 Write structures of the compounds whose IUPAC names are as follows:

- | | |
|---|---------------------------------|
| (i) 2-Methylbutan-2-ol | (ii) 1-Phenylpropan-2-ol |
| (iii) 3,5-Dimethylhexane -1, 3, 5-triol | (iv) 2,3 - Diethylphenol |
| (v) 1 - Ethoxypropane | (vi) 2-Ethoxy-3-methylpentane |
| (vii) Cyclohexylmethanol | (viii) 3-Cyclohexylpentan-3-ol |
| (ix) Cyclopent-3-en-1-ol | (x) 4-Chloro-3-ethylbutan-1-ol. |

7.3 (i) Draw the structures of all isomeric alcohols of molecular formula $\text{C}_5\text{H}_{12}\text{O}$ and give their IUPAC names.

(ii) Classify the isomers of alcohols in question 11.3 (i) as primary, secondary and tertiary alcohols.

7.4 Explain why propanol has higher boiling point than that of the hydrocarbon, butane?

7.5 Alcohols are comparatively more soluble in water than hydrocarbons of comparable molecular masses. Explain this fact.

7.6 What is meant by hydroboration-oxidation reaction? Illustrate it with an example.

7.7 Give the structures and IUPAC names of monohydric phenols of molecular formula, $\text{C}_7\text{H}_8\text{O}$.

7.8 While separating a mixture of *ortho* and *para* nitrophenols by steam distillation, name the isomer which will be steam volatile. Give reason.

7.9 Give the equations of reactions for the preparation of phenol from cumene.

7.10 Write chemical reaction for the preparation of phenol from chlorobenzene.

7.11 Write the mechanism of hydration of ethene to yield ethanol.

7.12 You are given benzene, conc. H_2SO_4 and NaOH . Write the equations for the preparation of phenol using these reagents.

7.13 Show how will you synthesise:

- (i) 1-phenylethanol from a suitable alkene.
- (ii) cyclohexylmethanol using an alkyl halide by an S_N2 reaction.
- (iii) pentan-1-ol using a suitable alkyl halide?

7.14 Give two reactions that show the acidic nature of phenol. Compare acidity of phenol with that of ethanol.

7.15 Explain why is *ortho* nitrophenol more acidic than *ortho* methoxyphenol ?

7.16 Explain how does the -OH group attached to a carbon of benzene ring activate it towards electrophilic substitution?

7.17 Give equations of the following reactions:

- (i) Oxidation of propan-1-ol with alkaline $KMnO_4$ solution.
- (ii) Bromine in CS_2 with phenol.
- (iii) Dilute HNO_3 with phenol.
- (iv) Treating phenol with chloroform in presence of aqueous NaOH.

7.18 Explain the following with an example.

- (i) Kolbe's reaction.
- (ii) Reimer-Tiemann reaction.
- (iii) Williamson ether synthesis.
- (iv) Unsymmetrical ether.

7.19 Write the mechanism of acid dehydration of ethanol to yield ethene.

7.20 How are the following conversions carried out?

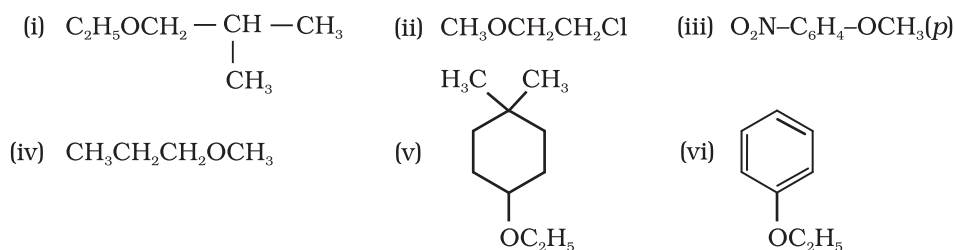
- (i) Propene $\rightarrow$ Propan-2-ol.
- (ii) Benzyl chloride $\rightarrow$ Benzyl alcohol.
- (iii) Ethyl magnesium chloride $\rightarrow$ Propan-1-ol.
- (iv) Methyl magnesium bromide $\rightarrow$ 2-Methylpropan-2-ol.

7.21 Name the reagents used in the following reactions:

- (i) Oxidation of a primary alcohol to carboxylic acid.
- (ii) Oxidation of a primary alcohol to aldehyde.
- (iii) Bromination of phenol to 2,4,6-tribromophenol.
- (iv) Benzyl alcohol to benzoic acid.
- (v) Dehydration of propan-2-ol to propene.
- (vi) Butan-2-one to butan-2-ol.

7.22 Give reason for the higher boiling point of ethanol in comparison to methoxymethane.

7.23 Give IUPAC names of the following ethers:



7.24 Write the names of reagents and equations for the preparation of the following ethers by Williamson's synthesis:

- (i) 1-Propoxypropane (ii) Ethoxybenzene
(iii) 2-Methoxy-2-methylpropane (iv) 1-Methoxyethane

7.25 Illustrate with examples the limitations of Williamson synthesis for the preparation of certain types of ethers.

7.26 How is 1-propoxypropane synthesised from propan-1-ol? Write mechanism of this reaction.

7.27 Preparation of ethers by acid dehydration of secondary or tertiary alcohols is not a suitable method. Give reason.

7.28 Write the equation of the reaction of hydrogen iodide with:

- (i) 1-propoxypropane (ii) methoxybenzene and (iii) benzyl ethyl ether.

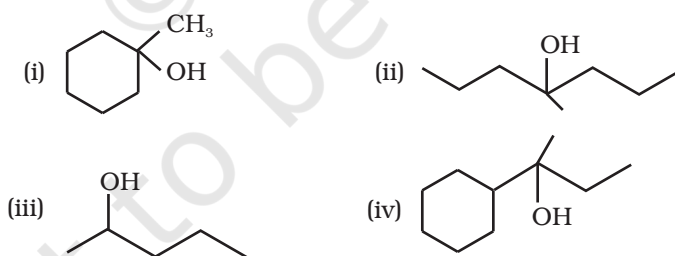
7.29 Explain the fact that in aryl alkyl ethers (i) the alkoxy group activates the benzene ring towards electrophilic substitution and (ii) it directs the incoming substituents to ortho and para positions in benzene ring.

7.30 Write the mechanism of the reaction of HI with methoxymethane.

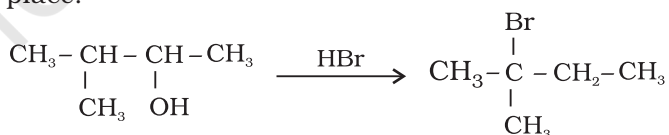
7.31 Write equations of the following reactions:

- (i) Friedel-Crafts reaction – alkylation of anisole.
(ii) Nitration of anisole.
(iii) Bromination of anisole in ethanoic acid medium.
(iv) Friedel-Craft's acetylation of anisole.

7.32 Show how would you synthesise the following alcohols from appropriate alkenes?



7.33 When 3-methylbutan-2-ol is treated with HBr, the following reaction takes place:



Give a mechanism for this reaction.

(Hint : The secondary carbocation formed in step II rearranges to a more stable tertiary carbocation by a hydride ion shift from 3rd carbon atom.

Answers to Some Intext Questions

7.1 Primary alcohols (i), (ii), (iii)

Secondary alcohols (iv) and (v)

Tertiary alcohols (vi)

7.2 Allylic alcohols (ii) and (vi)

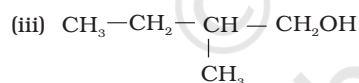
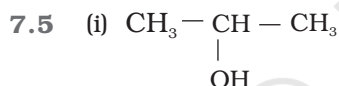
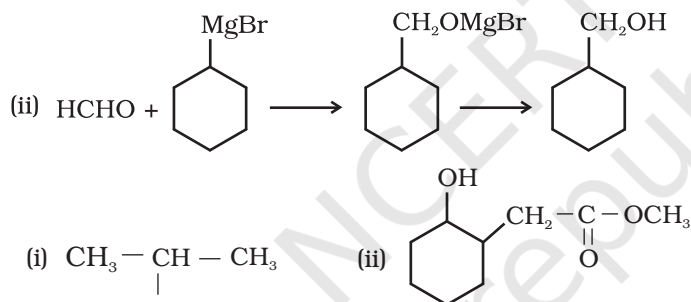
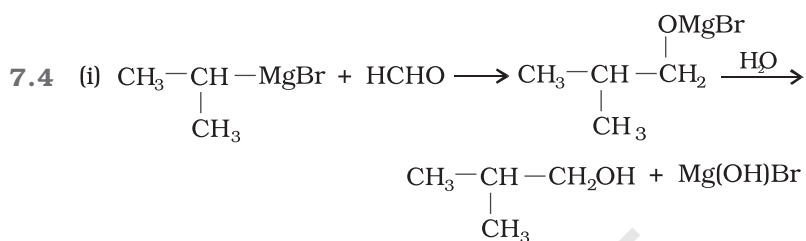
7.3 (i) 4-Chloro-3-ethyl-2-(1-methylethyl)-butan-1-ol

(ii) 2, 5-Dimethylhexane-1,3-diol

(iii) 3-Bromocyclohexanol

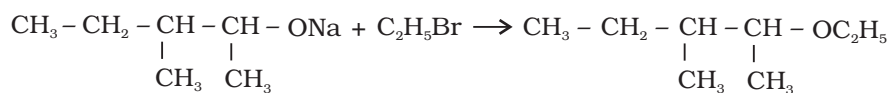
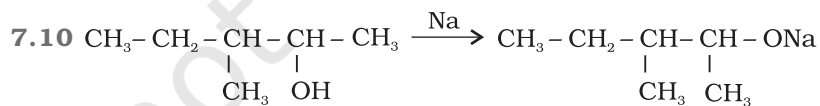
(iv) Hex-1-en-3-ol

(v) 2-Bromo-3-methylbut-2-en-1-ol



7.7 (i) 1-Methylcyclohexene

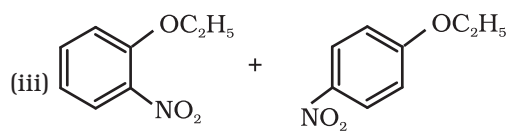
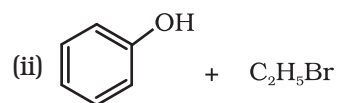
(ii) A Mixture of but-1-ene and but-2-ene. But-2-ene is the major product formed due to rearrangement to give secondary carbocation.



2-Ethoxy-3-methylpentane

7.11 (ii)

7.12 (i) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + \text{CH}_3\text{Br}$



(iv) $(\text{CH}_3)_3\text{C}-\text{I} + \text{C}_2\text{H}_5\text{OH}$



Unit

8

Aldehydes, Ketones and Carboxylic Acids

Objectives

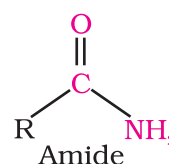
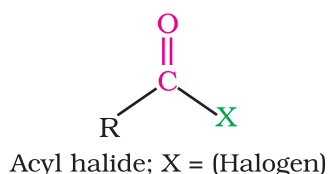
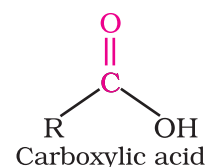
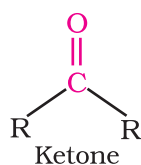
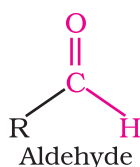
After studying this Unit, you will be able to

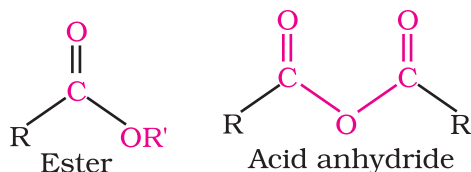
- write the common and IUPAC names of aldehydes, ketones and carboxylic acids;
- write the structures of the compounds containing functional groups namely carbonyl and carboxyl groups;
- describe the important methods of preparation and reactions of these classes of compounds;
- correlate physical properties and chemical reactions of aldehydes, ketones and carboxylic acids, with their structures;
- explain the mechanism of a few selected reactions of aldehydes and ketones;
- understand various factors affecting the acidity of carboxylic acids and their reactions;
- describe the uses of aldehydes, ketones and carboxylic acids.

Carbonyl compounds are of utmost importance to organic chemistry. They are constituents of fabrics, flavourings, plastics and drugs.

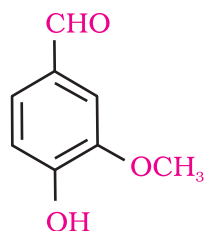
In the previous Unit, you have studied organic compounds with functional groups containing carbon-oxygen single bond. In this Unit, we will study about the organic compounds containing carbon-oxygen double bond ($>C=O$) called carbonyl group, which is one of the most important functional groups in organic chemistry.

In aldehydes, the carbonyl group is bonded to a carbon and hydrogen while in the ketones, it is bonded to two carbon atoms. The carbonyl compounds in which carbon of carbonyl group is bonded to carbon or hydrogen and oxygen of hydroxyl moiety ($-OH$) are known as carboxylic acids, while in compounds where carbon is attached to carbon or hydrogen and nitrogen of $-NH_2$ moiety or to halogens are called amides and acyl halides respectively. Esters and anhydrides are derivatives of carboxylic acids. The general formulas of these classes of compounds are given below:

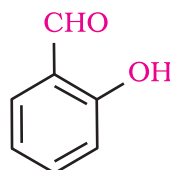




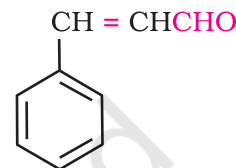
Aldehydes, ketones and carboxylic acids are widespread in plants and animal kingdom. They play an important role in biochemical processes of life. They add fragrance and flavour to nature, for example, vanillin (from vanilla beans), salicylaldehyde (from meadow sweet) and cinnamaldehyde (from cinnamon) have very pleasant fragrances.



Vanillin



Salicylaldehyde



Cinnamaldehyde

They are used in many food products and pharmaceuticals to add flavours. Some of these families are manufactured for use as solvents (i.e., acetone) and for preparing materials like adhesives, paints, resins, perfumes, plastics, fabrics, etc.

8.1 Nomenclature and Structure of Carbonyl Group

8.1.1 Nomenclature

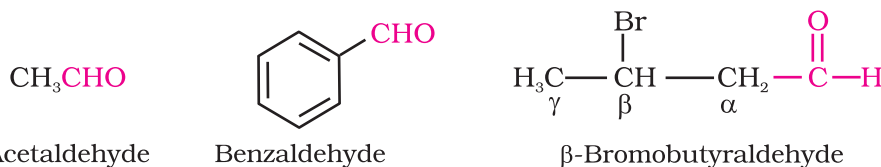
I. Aldehydes and ketones

Aldehydes and ketones are the simplest and most important carbonyl compounds.

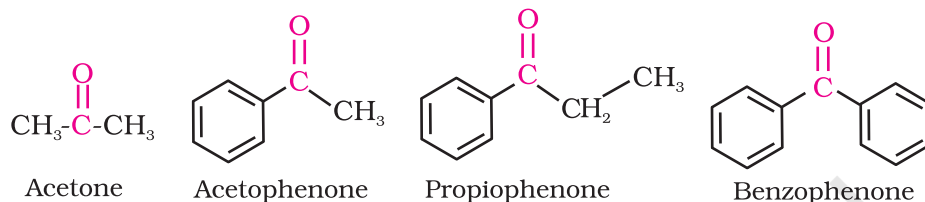
There are two systems of nomenclature of aldehydes and ketones.

(a) Common names

Aldehydes and ketones are often called by their common names instead of IUPAC names. The common names of most aldehydes are derived from the common names of the corresponding carboxylic acids [Section 8.6.1] by replacing the ending *-ic* of acid with aldehyde. At the same time, the names reflect the Latin or Greek term for the original source of the acid or aldehyde. The location of the substituent in the carbon chain is indicated by Greek letters α , β , γ , δ , etc. The α -carbon being the one directly linked to the aldehyde group, β -carbon the next, and so on. For example

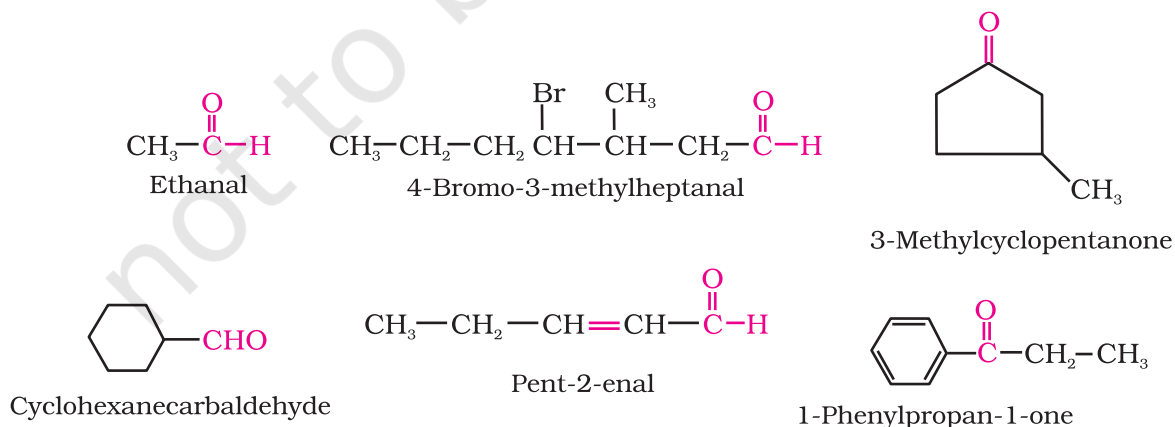


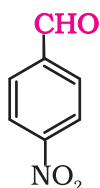
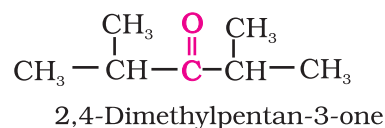
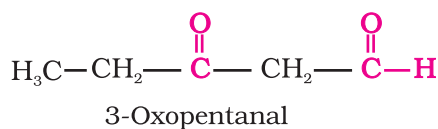
The common names of ketones are derived by naming two alkyl or aryl groups bonded to the carbonyl group. The locations of substituents are indicated by Greek letters, α , α' , β , β' and so on beginning with the carbon atoms next to the carbonyl group, indicated as α , α' . Some ketones have historical common names, the simplest dimethyl ketone is called acetone. Alkyl phenyl ketones are usually named by adding the name of acyl group as prefix to the word phenone. For example



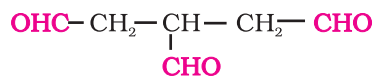
(b) IUPAC names

The IUPAC names of open chain aliphatic aldehydes and ketones are derived from the names of the corresponding alkanes by replacing the ending $-e$ with $-al$ and $-one$ respectively. In case of aldehydes the longest carbon chain is numbered starting from the carbon of the aldehyde group while in case of ketones the numbering begins from the end nearer to the carbonyl group. The substituents are prefixed in alphabetical order along with numerals indicating their positions in the carbon chain. The same applies to cyclic ketones, where the carbonyl carbon is numbered one. When the aldehyde group is attached to a ring, the suffix carbaldehyde is added after the full name of the cycloalkane. The numbering of the ring carbon atoms start from the carbon atom attached to the aldehyde group. The name of the simplest aromatic aldehyde carrying the aldehyde group on a benzene ring is benzenecarbaldehyde. However, the common name benzaldehyde is also accepted by IUPAC. Other aromatic aldehydes are hence named as substituted benzaldehydes.





4-Nitrobenzenecarbaldehyde
or
4-Nitrobenzaldehyde



Propane-1,2,3-tricarbaldehyde

[Note: To give identical treatment to all aldehydic groups, the compound is named as shown above.]

The common and IUPAC names of some aldehydes and ketones are given in Table 8.1.

Table 8.1: Common and IUPAC Names of Some Aldehydes and Ketones

Structure	Common name	IUPAC name
Aldehydes		
HCHO	Formaldehyde	Methanal
CH ₃ CHO	Acetaldehyde	Ethanal
(CH ₃) ₂ CHCHO	Isobutyraldehyde	2-Methylpropanal
	γ-Methylcyclohexanecarbaldehyde	3-Methylcyclohexanecarbaldehyde
CH ₃ CH(OCH ₃)CHO	α-Methoxypropionaldehyde	2-Methoxypropanal
CH ₃ CH ₂ CH ₂ CH ₂ CHO	Valeraldehyde	Pentanal
CH ₂ =CHCHO	Acrolein	Prop-2-enal
	Phthalaldehyde	Benzene-1,2-dicarbaldehyde
	<i>m</i> -Bromobenzaldehyde	3-Bromobenzenecarbaldehyde or 3-Bromobenzaldehyde
Ketones		
CH ₃ COCH ₂ CH ₂ CH ₃	Methyl <i>n</i> -propyl ketone	Pentan-2-one
(CH ₃) ₂ CHCOCH(CH ₃) ₂	Diisopropyl ketone	2,4-Dimethylpentan-3-one
	α-Methylcyclohexanone	2-Methylcyclohexanone
(CH ₃) ₂ C=CHCOCH ₃	Mesityl oxide	4-Methylpent-3-en-2-one

8.1.2 Structure of the Carbonyl Group

The carbonyl carbon atom is sp^2 -hybridised and forms three sigma (σ) bonds. The fourth valence electron of carbon remains in its p -orbital and forms a π -bond with oxygen by overlap with p -orbital of an oxygen. In addition, the oxygen atom also has two non bonding electron pairs. Thus, the carbonyl carbon and the three atoms attached to it lie in the same plane and the π -electron cloud is above and below this plane. The bond angles are approximately 120° as expected of a trigonal coplanar structure (Figure 8.1).

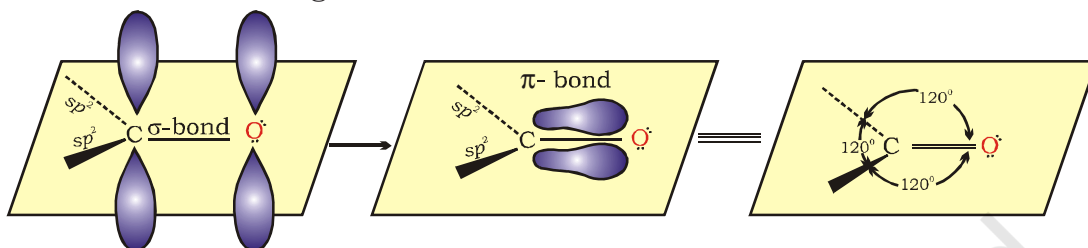
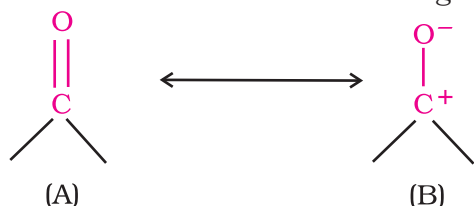


Fig.8.1 Orbital diagram for the formation of carbonyl group

The carbon-oxygen double bond is polarised due to higher electronegativity of oxygen relative to carbon. Hence, the carbonyl carbon is an electrophilic (Lewis acid), and carbonyl oxygen, a nucleophilic (Lewis base) centre. Carbonyl compounds have substantial dipole moments and are polar than ethers. The high polarity of the carbonyl group is explained on the basis of resonance involving a neutral (A) and a dipolar (B) structures as shown.



Intext Questions

8.1 Write the structures of the following compounds.

- | | |
|--|---------------------------|
| (i) α -Methoxypropionaldehyde | (ii) 3-Hydroxybutanal |
| (iii) 2-Hydroxycyclopentane carbaldehyde | (iv) 4-Oxopentanal |
| (v) Di-sec. butyl ketone | (vi) 4-Fluoroacetophenone |

8.2 Preparation of Aldehydes and Ketones

Some important methods for the preparation of aldehydes and ketones are as follows:

8.2.1 Preparation of Aldehydes and Ketones

1. By oxidation of alcohols

Aldehydes and ketones are generally prepared by oxidation of primary and secondary alcohols, respectively (Unit 7, Class XII).

2. By dehydrogenation of alcohols

This method is suitable for volatile alcohols and is of industrial application. In this method alcohol vapours are passed over heavy metal catalysts (Ag or Cu). Primary and secondary alcohols give aldehydes and ketones, respectively (Unit 7, Class XII).

3. From hydrocarbons

- (i) *By ozonolysis of alkenes:* As we know, ozonolysis of alkenes followed by reaction with zinc dust and water gives aldehydes,

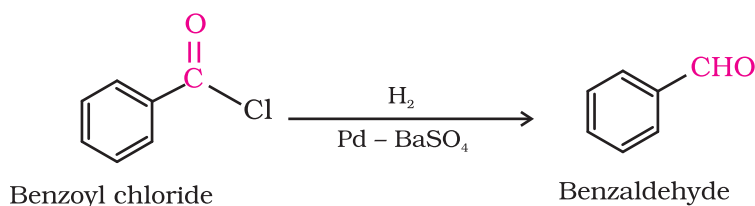
ketones or a mixture of both depending on the substitution pattern of the alkene (Unit 9, Class XI).

- (ii) *By hydration of alkynes:* Addition of water to ethyne in the presence of H_2SO_4 and HgSO_4 gives acetaldehyde. All other alkynes give ketones in this reaction (Unit 9, Class XI).

8.2.2 Preparation of Aldehydes

1. From acyl chloride (acid chloride)

Acyl chloride (acid chloride) is hydrogenated over catalyst, palladium on barium sulphate. This reaction is called **Rosenmund reduction**.



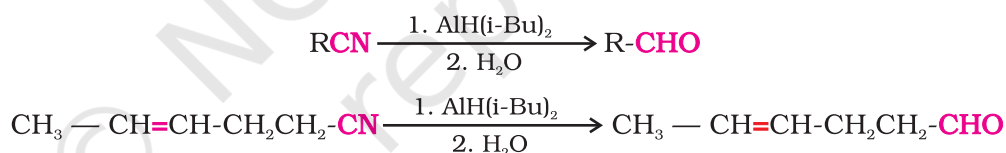
2. From nitriles and esters

Nitriles are reduced to corresponding imine with stannous chloride in the presence of hydrochloric acid, which on hydrolysis give corresponding aldehyde.

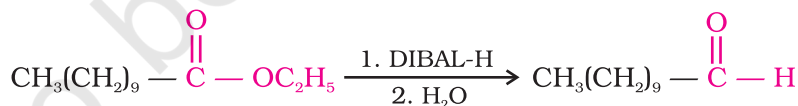


This reaction is called **Stephen** reaction.

Alternatively, nitriles are selectively reduced by diisobutylaluminium hydride, (DIBAL-H) to imines followed by hydrolysis to aldehydes:



Similarly, esters are also reduced to aldehydes with DIBAL-H.



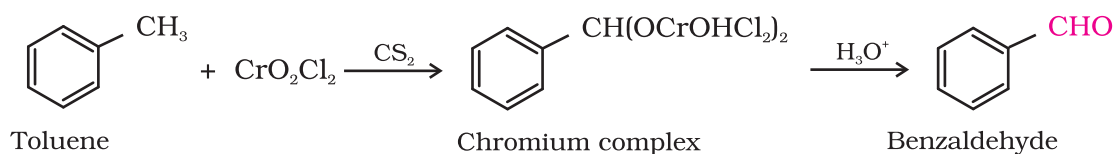
3. From hydrocarbons

Aromatic aldehydes (benzaldehyde and its derivatives) are prepared from aromatic hydrocarbons by the following methods:

(i) By oxidation of methylbenzene

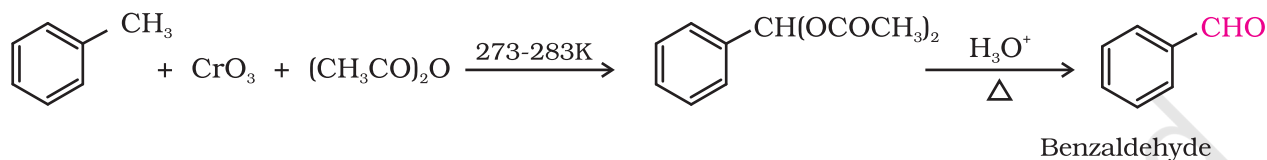
Strong oxidising agents oxidise toluene and its derivatives to benzoic acids. However, it is possible to stop the oxidation at the aldehyde stage with suitable reagents that convert the methyl group to an intermediate that is difficult to oxidise further. The following methods are used for this purpose.

- (a) Use of chromyl chloride (CrO_2Cl_2): Chromyl chloride oxidises methyl group to a chromium complex, which on hydrolysis gives corresponding benzaldehyde.



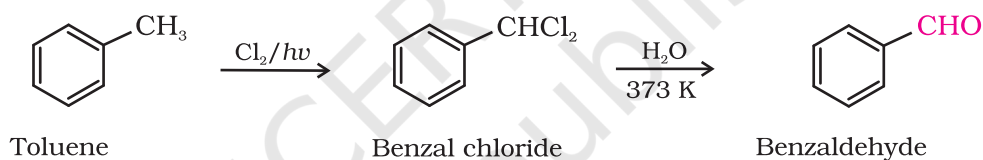
This reaction is called **Etard reaction**.

(b) Use of chromic oxide (CrO_3): Toluene or substituted toluene is converted to benzylidene diacetate on treating with chromic oxide in acetic anhydride. The benzylidene diacetate can be hydrolysed to corresponding benzaldehyde with aqueous acid.



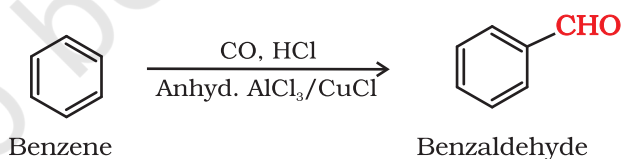
(ii) *By side chain chlorination followed by hydrolysis*

Side chain chlorination of toluene gives benzal chloride, which on hydrolysis gives benzaldehyde. This is a commercial method of manufacture of benzaldehyde.



(iii) *By Gatterman – Koch reaction*

When benzene or its derivative is treated with carbon monoxide and hydrogen chloride in the presence of anhydrous aluminium chloride or cuprous chloride, it gives benzaldehyde or substituted benzaldehyde.

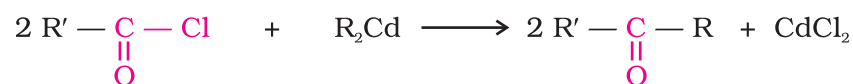


This reaction is known as **Gatterman-Koch** reaction.

8.2.3 Preparation of Ketones

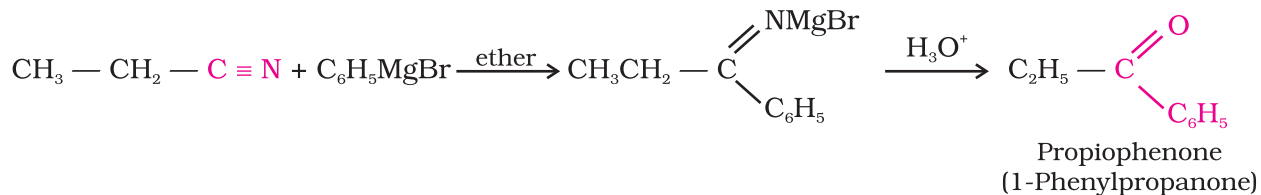
1. From acyl chlorides

Treatment of acyl chlorides with dialkylcadmium, prepared by the reaction of cadmium chloride with Grignard reagent, gives ketones.



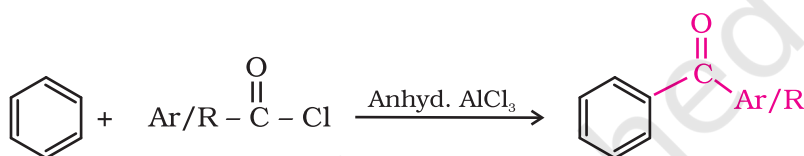
2. From nitriles

Treating a nitrile with Grignard reagent followed by hydrolysis yields a ketone.



3. From benzene or substituted benzenes

When benzene or substituted benzene is treated with acid chloride in the presence of anhydrous aluminium chloride, it affords the corresponding ketone. This reaction is known as **Friedel-Crafts acylation reaction**.



Example 8.1

Give names of the reagents to bring about the following transformations:

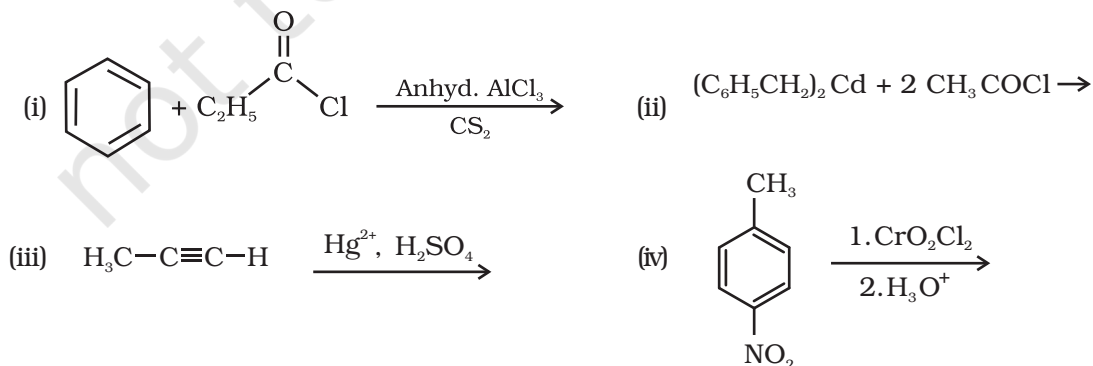
- | | |
|---|------------------------------------|
| (i) Hexan-1-ol to hexanal | (ii) Cyclohexanol to cyclohexanone |
| (iii) <i>p</i> -Fluorotoluene to <i>p</i> -fluorobenzaldehyde | (iv) Ethanenitrile to ethanal |
| (v) Allyl alcohol to propenal | (vi) But-2-ene to ethanal |

Solution

- | | |
|---|---|
| (i) $\text{C}_5\text{H}_5\text{NH}^+\text{CrO}_3\text{Cl}$ (PCC) | (ii) Anhydrous CrO_3 |
| (iii) CrO_3 in the presence of acetic anhydride/
1. CrO_2Cl_2 2. HOH | (iv) (Diisobutyl)aluminium hydride (DIBAL-H) |
| (v) PCC | (vi) $\text{O}_3/\text{H}_2\text{O}$ -Zn dust |

Intext Question

8.2 Write the structures of products of the following reactions;



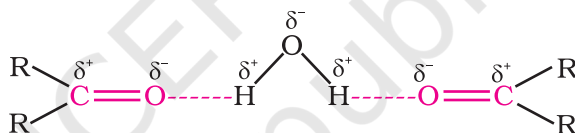
8.3 Physical Properties

The physical properties of aldehydes and ketones are described as follows.

Methanal is a gas at room temperature. Ethanal is a volatile liquid. Other aldehydes and ketones are liquid or solid at room temperature. The boiling points of aldehydes and ketones are higher than hydrocarbons and ethers of comparable molecular masses. It is due to weak molecular association in aldehydes and ketones arising out of the dipole-dipole interactions. Also, their boiling points are lower than those of alcohols of similar molecular masses due to absence of intermolecular hydrogen bonding. The following compounds of molecular masses 58 and 60 are ranked in order of increasing boiling points.

	b.p.(K)	Molecular Mass
n-Butane	273	58
Methoxyethane	281	60
Propanal	322	58
Acetone	329	58
Propan-1-ol	370	60

The lower members of aldehydes and ketones such as methanal, ethanal and propanone are miscible with water in all proportions, because they form hydrogen bond with water.



However, the solubility of aldehydes and ketones decreases rapidly on increasing the length of alkyl chain. All aldehydes and ketones are fairly soluble in organic solvents like benzene, ether, methanol, chloroform, etc. The lower aldehydes have sharp pungent odours. As the size of the molecule increases, the odour becomes less pungent and more fragrant. In fact, many naturally occurring aldehydes and ketones are used in the blending of perfumes and flavouring agents.

Arrange the following compounds in the increasing order of their boiling points:



The molecular masses of these compounds are in the range of 72 to 74. Since only butan-1-ol molecules are associated due to extensive intermolecular hydrogen bonding, therefore, the boiling point of butan-1-ol would be the highest. Butanal is more polar than ethoxyethane. Therefore, the intermolecular dipole-dipole attraction is stronger in the former. *n*-Pentane molecules have only weak **van der Waals forces**. Hence increasing order of boiling points of the given compounds is as follows:



Example 8.2

Solution

Intext Question

8.3 Arrange the following compounds in increasing order of their boiling points.



8.4 Chemical Reactions

Since aldehydes and ketones both possess the carbonyl functional group, they undergo similar chemical reactions.

1. Nucleophilic addition reactions

Contrary to electrophilic addition reactions observed in alkenes, the aldehydes and ketones undergo nucleophilic addition reactions.

(i) Mechanism of nucleophilic addition reactions

A nucleophile attacks the electrophilic carbon atom of the polar carbonyl group from a direction approximately perpendicular to the plane of sp^2 hybridised orbitals of carbonyl carbon (Fig. 8.2). The hybridisation of carbon changes from sp^2 to sp^3 in this process, and a tetrahedral alkoxide intermediate is produced. This intermediate captures a proton from the reaction medium to give the electrically neutral product. The net result is addition of Nu^- and H^+ across the carbon oxygen double bond as shown in Fig. 8.2.

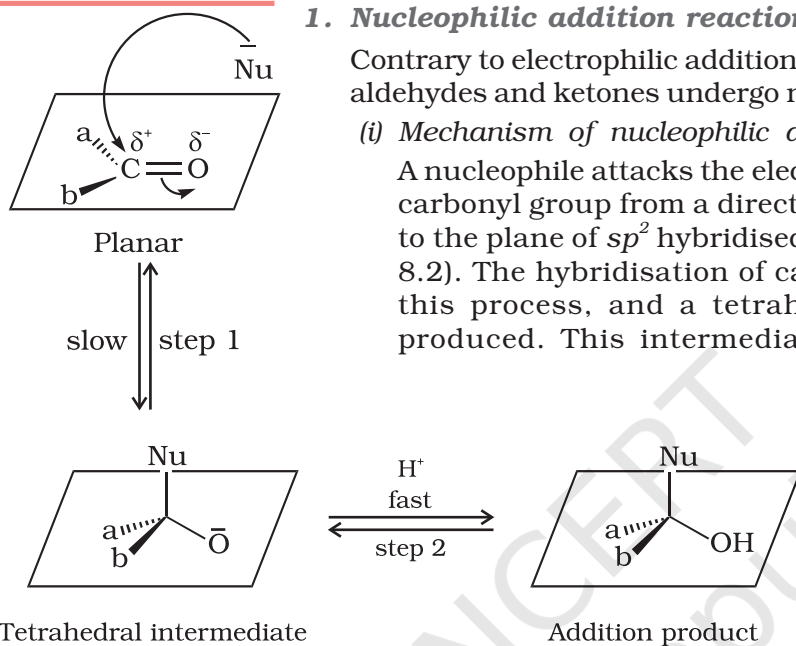


Fig. 8.2: Nucleophilic attack on carbonyl carbon

(ii) Reactivity

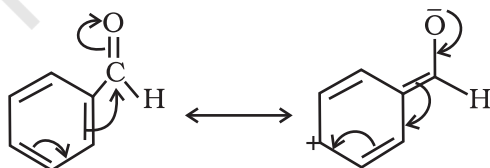
Aldehydes are generally more reactive than ketones in nucleophilic addition reactions due to steric and electronic reasons. Sterically, the presence of two relatively large substituents in ketones hinders the approach of nucleophile to carbonyl carbon than in aldehydes having only one such substituent. Electronically, aldehydes are more reactive than ketones because two alkyl groups reduce the electrophilicity of the carbonyl carbon more effectively than in former.

Example 8.3

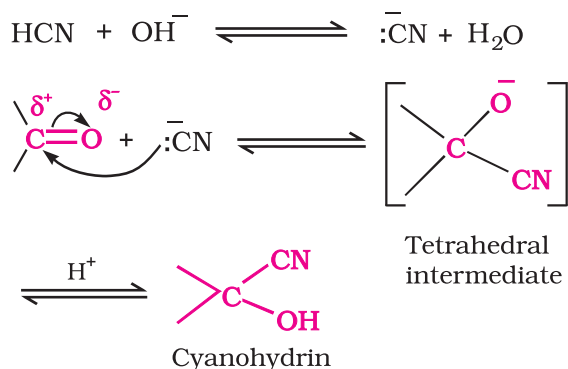
Would you expect benzaldehyde to be more reactive or less reactive in nucleophilic addition reactions than propanal? Explain your answer.

Solution

The carbon atom of the carbonyl group of benzaldehyde is less electrophilic than carbon atom of the carbonyl group present in propanal. The polarity of the carbonyl group is reduced in benzaldehyde due to resonance as shown below and hence it is less reactive than propanal.



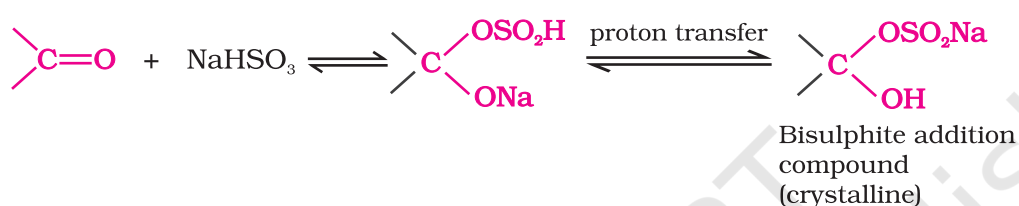
(iii) Some important examples of nucleophilic addition and nucleophilic addition-elimination reactions:



(a) *Addition of hydrogen cyanide (HCN)*: Aldehydes and ketones react with hydrogen cyanide (HCN) to yield cyanohydrins. This reaction occurs very slowly with pure HCN. Therefore, it is catalysed by a base and the generated cyanide ion (CN⁻) being a stronger nucleophile readily adds to carbonyl compounds to yield corresponding cyanohydrin.

Cyanohydrins are useful synthetic intermediates.

(b) *Addition of sodium hydrogensulphite*: Sodium hydrogensulphite adds to aldehydes and ketones to form the addition products.



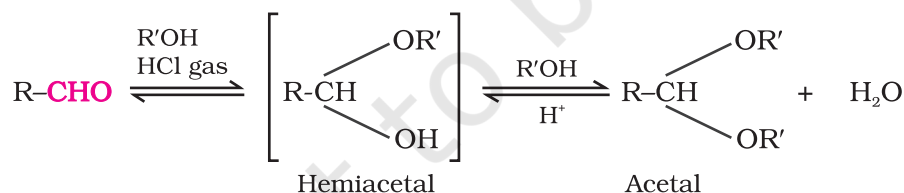
The position of the equilibrium lies largely to the right hand side for most aldehydes and to the left for most

ketones due to steric reasons. The hydrogensulphite addition compound is water soluble and can be converted back to the original carbonyl compound by treating it with dilute mineral acid or alkali. Therefore, these are useful for separation and purification of aldehydes.

(c) *Addition of Grignard reagents*: (refer Unit 7, Class XII).

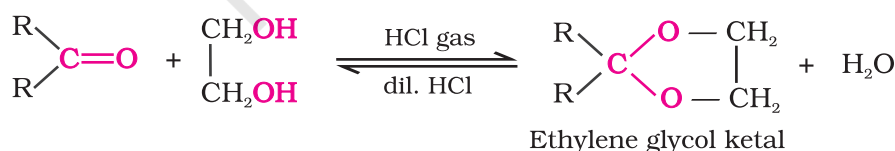
(d) *Addition of alcohols*: Aldehydes react with one equivalent of monohydric alcohol in the presence of dry hydrogen chloride to yield alkoxyalcohol intermediate, known as hemiacetals, which further react with one more molecule of alcohol to

give a *gem*-dialkoxy compound known as acetal as shown in the reaction.



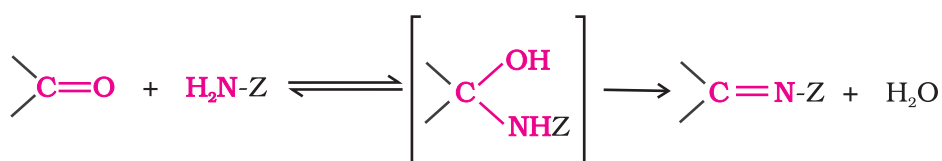
Ketones react with ethylene glycol under similar conditions to form cyclic products known as ethylene glycol ketals.

Dry hydrogen chloride protonates the oxygen of the carbonyl compounds and therefore, increases the electrophilicity of the carbonyl carbon facilitating



the nucleophilic attack of ethylene glycol. Acetals and ketals are hydrolysed with aqueous mineral acids to yield corresponding aldehydes and ketones respectively.

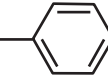
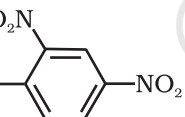
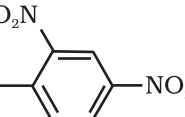
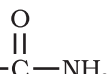
- (e) *Addition of ammonia and its derivatives:* Nucleophiles, such as ammonia and its derivatives $\text{H}_2\text{N-Z}$ add to the carbonyl group of aldehydes and ketones. The reaction is reversible



and catalysed by acid. The equilibrium favours the product formation due to rapid dehydration of the intermediate to form >C=N-Z .

$\text{Z} = \text{Alkyl, aryl, OH, NH}_2, \text{C}_6\text{H}_5\text{NH, NHCONH}_2, \text{etc.}$

Table 8.2: Some N-Substituted Derivatives of Aldehydes and Ketones (>C=N-Z)

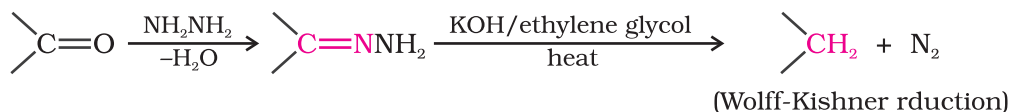
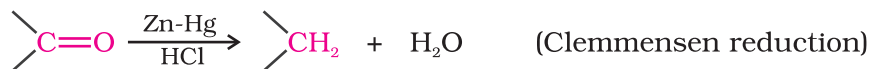
Z	Reagent name	Carbonyl derivative	Product name
-H	Ammonia	>C=NH	Imine
-R	Amine	>C=NR	Substituted imine (Schiff's base)
-OH	Hydroxylamine	>C=N-OH	Oxime
-NH ₂	Hydrazine	>C=N-NH_2	Hydrazone
-HN- 	Phenylhydrazine	>C=N-NH- 	Phenylhydrazone
-HN- 	2,4-Dinitrophenylhydrazine	>C=N-NH- 	2,4 Dinitrophenylhydrazone
-NH- 	Semicarbazide	$\text{>C=N-NH-C(=O)-NH}_2$	Semicarbazone

* 2,4-DNP-derivatives are yellow, orange or red solids, useful for characterisation of aldehydes and ketones.

2. Reduction

- Reduction to alcohols:* Aldehydes and ketones are reduced to primary and secondary alcohols respectively by sodium borohydride (NaBH_4) or lithium aluminium hydride (LiAlH_4) as well as by catalytic hydrogenation (Unit 7, Class XII).
- Reduction to hydrocarbons:* The carbonyl group of aldehydes and ketones is reduced to CH_2 group on treatment with zinc-amalgam and concentrated hydrochloric acid [**Clemmensen**

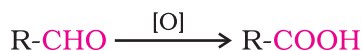
reduction] or with hydrazine followed by heating with sodium or potassium hydroxide in high boiling solvent such as ethylene glycol (**Wolff-Kishner reduction**).



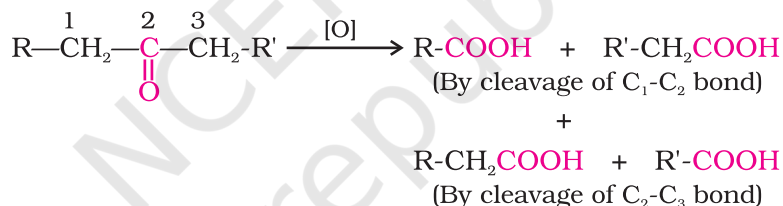
3. Oxidation

Bernhard Tollens
(1841-1918) was a
Professor of Chemistry
at the University of
Gottingen, Germany.

Aldehydes differ from ketones in their oxidation reactions. Aldehydes are easily oxidised to carboxylic acids on treatment with common oxidising agents like nitric acid, potassium permanganate, potassium dichromate, etc. Even mild oxidising agents, mainly Tollens' reagent and Fehlings' reagent also oxidise aldehydes.



Ketones are generally oxidised under vigorous conditions, i.e., strong oxidising agents and at elevated temperatures. Their oxidation involves carbon-carbon bond cleavage to afford a mixture of carboxylic acids having lesser number of carbon atoms than the parent ketone.

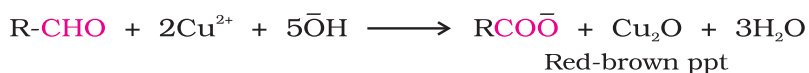


The mild oxidising agents given below are used to distinguish aldehydes from ketones:

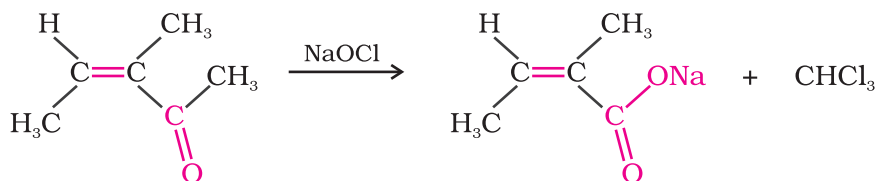
- (i) **Tollens' test:** On warming an aldehyde with freshly prepared ammoniacal silver nitrate solution (Tollens' reagent), a bright silver mirror is produced due to the formation of silver metal. The aldehydes are oxidised to corresponding carboxylate anion. The reaction occurs in alkaline medium.



- (ii) **Fehling's test:** Fehling reagent comprises of two solutions, Fehling solution A and Fehling solution B. Fehling solution A is aqueous copper sulphate and Fehling solution B is alkaline sodium potassium tartarate (Rochelle salt). These two solutions are mixed in equal amounts before test. On heating an aldehyde with Fehling's reagent, a reddish brown precipitate is obtained. Aldehydes are oxidised to corresponding carboxylate anion. Aromatic aldehydes do not respond to this test.



(iii) *Oxidation of methyl ketones by haloform reaction:*
Aldehydes and ketones having at least one methyl group linked to the carbonyl carbon atom (methyl ketones) are oxidised by sodium hypohalite to sodium salts of corresponding carboxylic acids having one carbon atom less than that of carbonyl compound. The methyl group is converted to haloform. This oxidation does not affect a carbon-carbon double bond, if present in the molecule.



Iodoform reaction with sodium hypoiodite is also used for detection of CH_3CO group or $\text{CH}_3\text{CH}(\text{OH})$ group which produces CH_3CO group on oxidation.

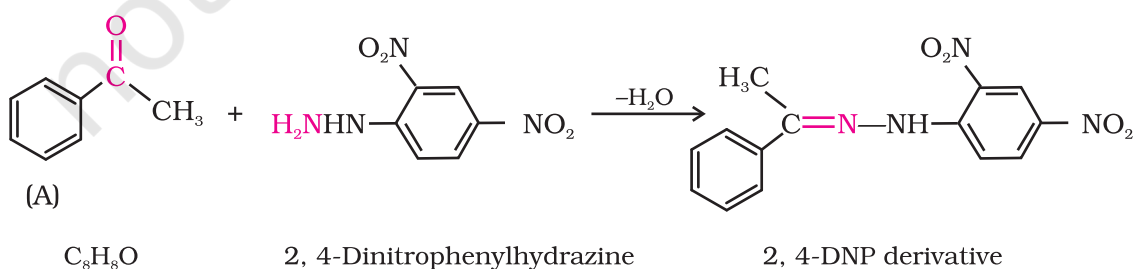
Example 8.4

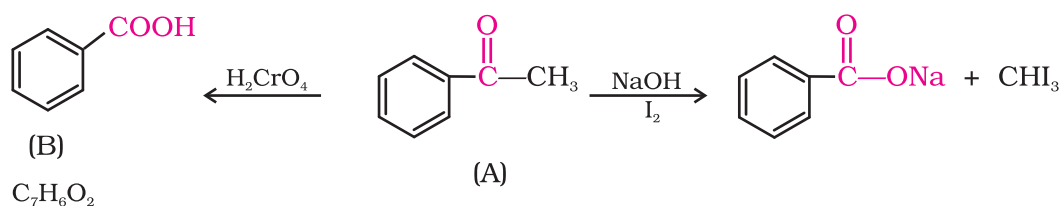
An organic compound (A) with molecular formula $\text{C}_8\text{H}_8\text{O}$ forms an orange-red precipitate with 2,4-DNP reagent and gives yellow precipitate on heating with iodine in the presence of sodium hydroxide. It neither reduces Tollens' or Fehlings' reagent, nor does it decolourise bromine water or Baeyer's reagent. On drastic oxidation with chromic acid, it gives a carboxylic acid (B) having molecular formula $\text{C}_7\text{H}_6\text{O}_2$. Identify the compounds (A) and (B) and explain the reactions involved.

Solution

(A) forms 2,4-DNP derivative. Therefore, it is an aldehyde or a ketone. Since it does not reduce Tollens' or Fehling reagent, (A) must be a ketone. (A) responds to iodoform test. Therefore, it should be a methyl ketone. The molecular formula of (A) indicates high degree of unsaturation, yet it does not decolourise bromine water or Baeyer's reagent. This indicates the presence of unsaturation due to an aromatic ring.

Compound (B), being an oxidation product of a ketone should be a carboxylic acid. The molecular formula of (B) indicates that it should be benzoic acid and compound (A) should, therefore, be a monosubstituted aromatic methyl ketone. The molecular formula of (A) indicates that it should be phenyl methyl ketone (acetophenone). Reactions are as follows:

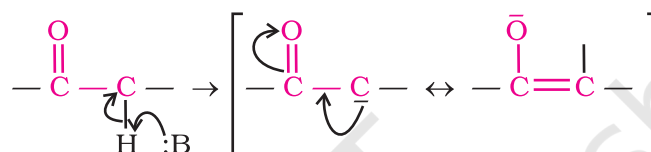




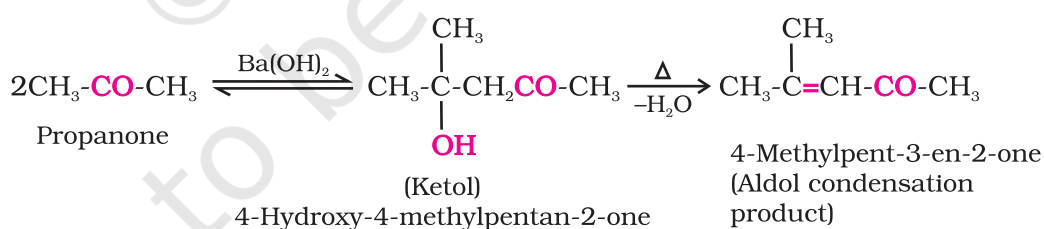
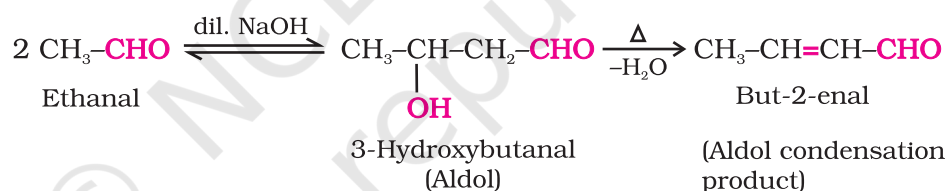
4. Reactions due to α -hydrogen

Acidity of α -hydrogens of aldehydes and ketones: The aldehydes and ketones undergo a number of reactions due to the acidic nature of α -hydrogen.

The acidity of α -hydrogen atoms of carbonyl compounds is due to the strong electron withdrawing effect of the carbonyl group and resonance stabilisation of the conjugate base.

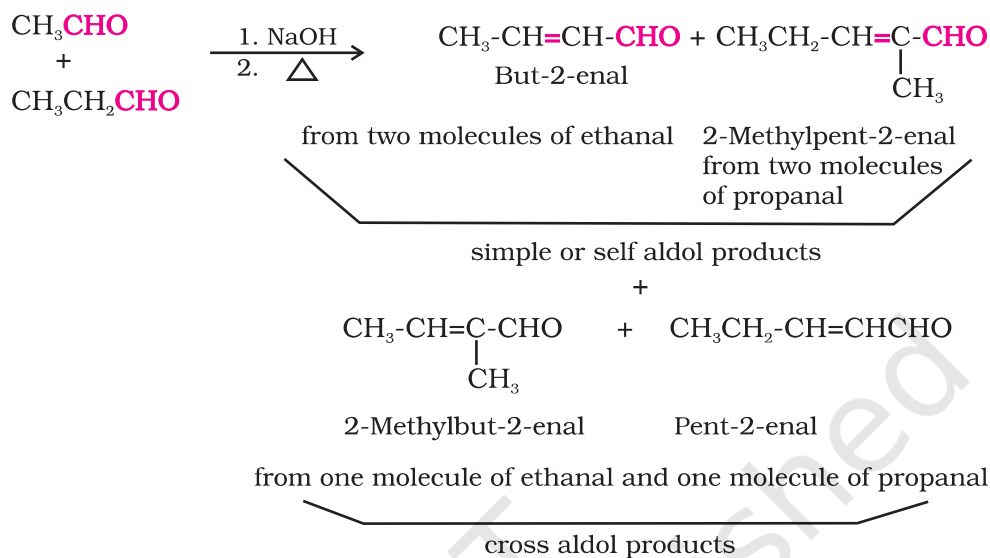


(i) **Aldol condensation:** Aldehydes and ketones having at least one α -hydrogen undergo a reaction in the presence of dilute alkali as catalyst to form β -hydroxy aldehydes (aldol) or β -hydroxy ketones (ketol), respectively. This is known as **Aldol reaction**.

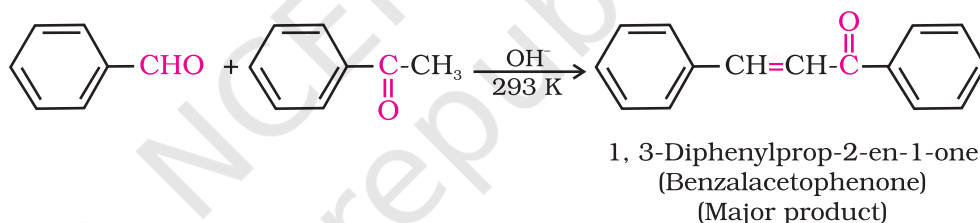


The name aldol is derived from the names of the two functional groups, aldehyde and alcohol, present in the products. The aldol and ketol readily lose water to give α,β -unsaturated carbonyl compounds which are aldol condensation products and the reaction is called **Aldol condensation**. Though ketones give ketols (compounds containing a keto and alcohol groups), the general name aldol condensation still applies to the reactions of ketones due to their similarity with aldehydes.

(ii) **Cross aldol condensation:** When aldol condensation is carried out between two different aldehydes and / or ketones, it is called **cross aldol condensation**. If both of them contain α -hydrogen atoms, it gives a mixture of four products. This is illustrated below by aldol reaction of a mixture of ethanal and propanal.

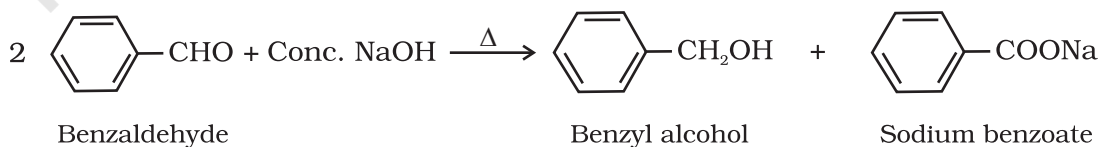
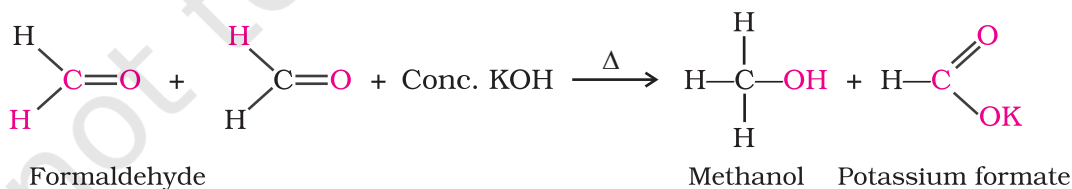


Ketones can also be used as one component in the cross aldol reactions.

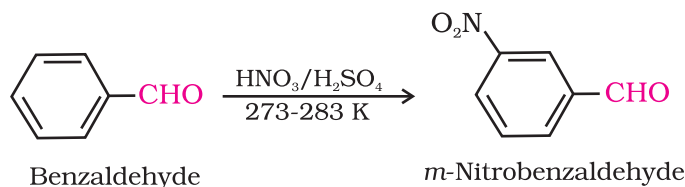


5. Other reactions

(i) **Cannizzaro reaction:** Aldehydes which do not have an α -hydrogen atom, undergo self oxidation and reduction (disproportionation) reaction on heating with concentrated alkali. In this reaction, one molecule of the aldehyde is reduced to alcohol while another is oxidised to carboxylic acid salt.



- (ii) *Electrophilic substitution reaction:* Aromatic aldehydes and ketones undergo electrophilic substitution at the ring in which the carbonyl group acts as a deactivating and *meta*-directing group.



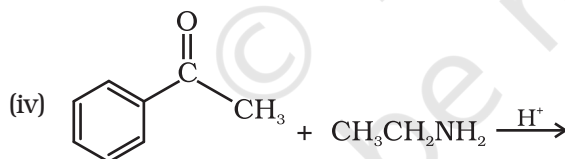
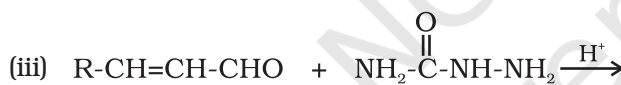
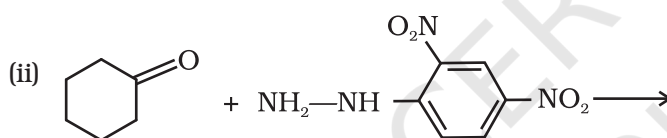
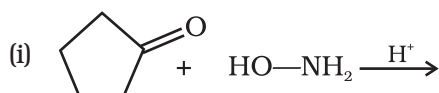
Intext Questions

8.4 Arrange the following compounds in increasing order of their reactivity in nucleophilic addition reactions.

- (i) Ethanal, Propanal, Propanone, Butanone.
 (ii) Benzaldehyde, *p*-Tolualdehyde, *p*-Nitrobenzaldehyde, Acetophenone.

Hint: Consider steric effect and electronic effect.

8.5 Predict the products of the following reactions:



8.5 Uses of Aldehydes and Ketones

In chemical industry aldehydes and ketones are used as solvents, starting materials and reagents for the synthesis of other products. Formaldehyde is well known as formalin (40%) solution used to preserve biological specimens and to prepare bakelite (a phenol-formaldehyde resin), urea-formaldehyde glues and other polymeric products. Acetaldehyde is used primarily as a starting material in the manufacture of acetic acid, ethyl acetate, vinyl acetate, polymers and drugs. Benzaldehyde is used in perfumery and in dye industries. Acetone and ethyl methyl ketone are common industrial solvents. Many aldehydes and ketones, e.g., butyraldehyde, vanillin, acetophenone, camphor, etc. are well known for their odours and flavours.

Carboxylic Acids

Carbon compounds containing a carboxyl functional group, -COOH are called carboxylic acids. The carboxyl group, consists of a *carbonyl* group attached to a *hydroxyl* group, hence its name *carboxyl*. Carboxylic acids may be aliphatic (RCOOH) or aromatic (ArCOOH) depending on the group, alkyl or aryl, attached to carboxylic carbon. Large number of carboxylic acids are found in nature. Some higher members of aliphatic carboxylic acids ($\text{C}_{12} - \text{C}_{18}$) known as **fatty acids**, occur in natural fats as esters of glycerol. Carboxylic acids serve as starting material for several other important organic compounds such as anhydrides, esters, acid chlorides, amides, etc.

8.6 Nomenclature and Structure of Carboxyl Group

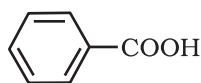
8.6.1 Nomenclature

Since carboxylic acids are amongst the earliest organic compounds to be isolated from nature, a large number of them are known by their common names. The common names end with the suffix *-ic acid* and have been derived from Latin or Greek names of their natural sources. For example, formic acid (HCOOH) was first obtained from red ants (Latin: *formica* means ant), acetic acid (CH_3COOH) from vinegar (Latin: *acetum*, means vinegar), butyric acid ($\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$) from rancid butter (Latin: *butyrum*, means butter).

In the IUPAC system, aliphatic carboxylic acids are named by replacing the ending *-e* in the name of the corresponding alkane with *-oic acid*. In numbering the carbon chain, the carboxylic carbon is numbered one. For naming compounds containing more than one carboxyl group, the alkyl chain leaving carboxyl groups is numbered and the number of carboxyl groups is indicated by adding the multiplicative prefix, *dicarboxylic acid*, *tricarboxylic acid*, etc. to the name of parent alkyl chain. The position of -COOH groups are indicated by the arabic numeral before the multiplicative prefix. Some of the carboxylic acids along with their common and IUPAC names are listed in Table 8.3.

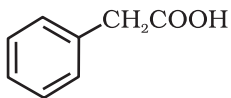
Table 8.3 Names and Structures of Some Carboxylic Acids

Structure	Common name	IUPAC name
HCOOH	Formic acid	Methanoic acid
CH_3COOH	Acetic acid	Ethanoic acid
$\text{CH}_3\text{CH}_2\text{COOH}$	Propionic acid	Propanoic acid
$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$	Butyric acid	Butanoic acid
$(\text{CH}_3)_2\text{CHCOOH}$	Isobutyric acid	2-Methylpropanoic acid
HOOC-COOH	Oxalic acid	Ethanedioic acid
$\text{HOOC-CH}_2\text{-COOH}$	Malonic acid	Propanedioic acid
$\text{HOOC-(CH}_2)_2\text{-COOH}$	Succinic acid	Butanedioic acid
$\text{HOOC-(CH}_2)_3\text{-COOH}$	Glutaric acid	Pentanedioic acid
$\text{HOOC-(CH}_2)_4\text{-COOH}$	Adipic acid	Hexanedioic acid
$\text{HOOC-CH}_2\text{-CH(COOH)-CH}_2\text{-COOH}$	Tricarballic acid or carballic acid	Propane-1, 2, 3- tricarboxylic acid



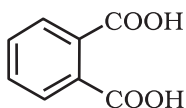
Benzoic acid

Benzenecarboxylic acid
(Benzoic acid)



Phenylacetic acid

2-Phenylethanoic acid

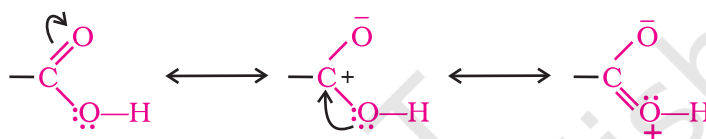


Phthalic acid

Benzene-1, 2-dicarboxylic acid

8.6.2 Structure of Carboxyl Group

In carboxylic acids, the bonds to the carboxyl carbon lie in one plane and are separated by about 120° . The carboxylic carbon is less electrophilic than carbonyl carbon because of the possible resonance structure shown below:

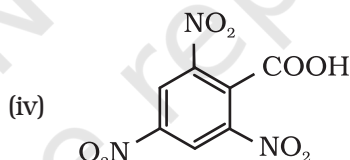
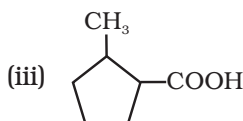


Intext Question

8.6 Give the IUPAC names of the following compounds:

(i) $\text{PhCH}_2\text{CH}_2\text{COOH}$

(ii) $(\text{CH}_3)_2\text{C}=\text{CHCOOH}$

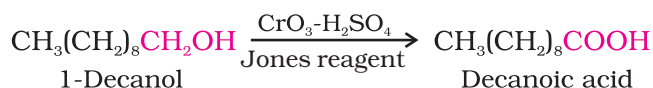
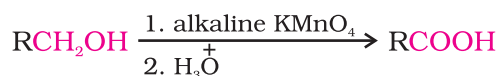


8.7 Methods of Preparation of Carboxylic Acids

Some important methods of preparation of carboxylic acids are as follows.

1. From primary alcohols and aldehydes

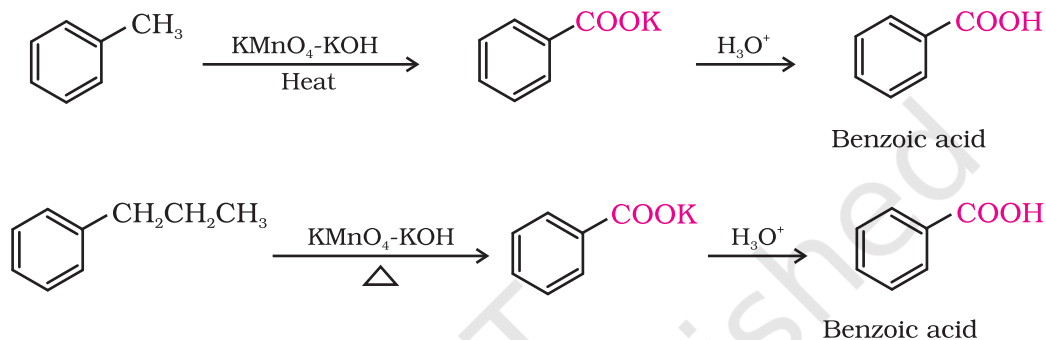
Primary alcohols are readily oxidised to carboxylic acids with common oxidising agents such as potassium permanganate (KMnO_4) in neutral, acidic or alkaline media or by potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) and chromium trioxide (CrO_3) in acidic media (Jones reagent).



Carboxylic acids are also prepared from aldehydes by the use of mild oxidising agents (Section 8.4).

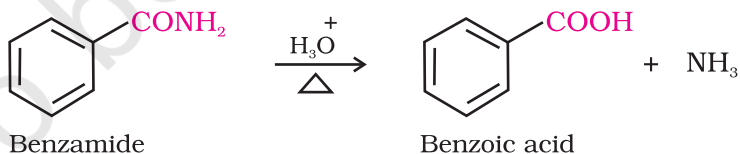
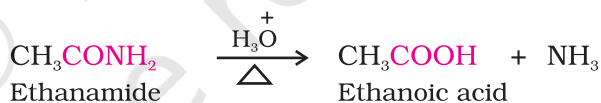
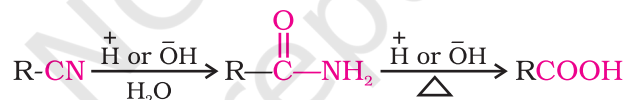
2. From alkylbenzenes

Aromatic carboxylic acids can be prepared by vigorous oxidation of alkyl benzenes with chromic acid or acidic or alkaline potassium permanganate. The entire side chain is oxidised to the carboxyl group irrespective of length of the side chain. Primary and secondary alkyl groups are oxidised in this manner while tertiary group is not affected. Suitably substituted alkenes are also oxidised to carboxylic acids with these oxidising reagents.



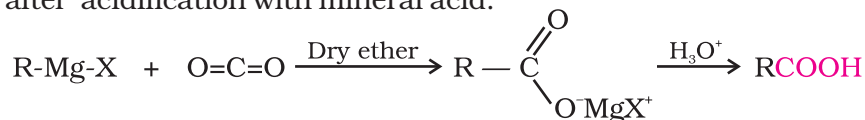
3. From nitriles and amides

Nitriles are hydrolysed to amides and then to acids in the presence of H^+ or OH^- as catalyst. Mild reaction conditions are used to stop the reaction at the amide stage.



4. From Grignard reagents

Grignard reagents react with carbon dioxide (dry ice) to form salts of carboxylic acids which in turn give corresponding carboxylic acids after acidification with mineral acid.

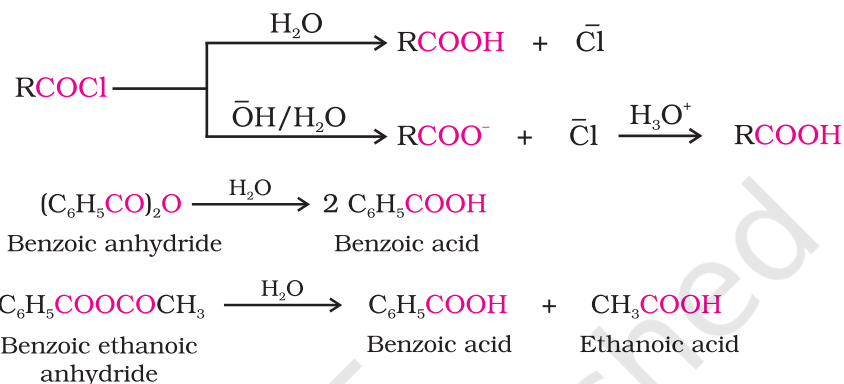


As we know, the Grignard reagents and nitriles can be prepared from alkyl halides (refer Unit 6, Class XII). The above methods

(3 and 4) are useful for converting alkyl halides into corresponding carboxylic acids having one carbon atom more than that present in alkyl halides (ascending the series).

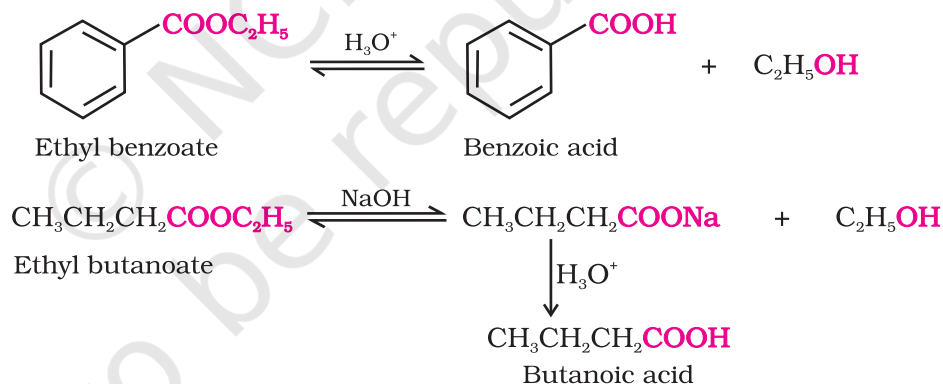
5. From acyl halides and anhydrides

Acid chlorides when hydrolysed with water give carboxylic acids or more readily hydrolysed with aqueous base to give carboxylate ions which on acidification provide corresponding carboxylic acids. Anhydrides on the other hand are hydrolysed to corresponding acid(s) with water.



6. From esters

Acidic hydrolysis of esters gives directly carboxylic acids while basic hydrolysis gives carboxylates, which on acidification give corresponding carboxylic acids.

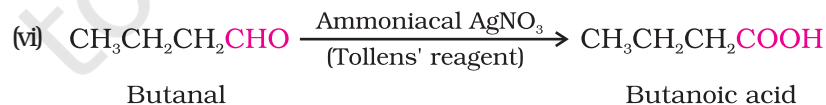
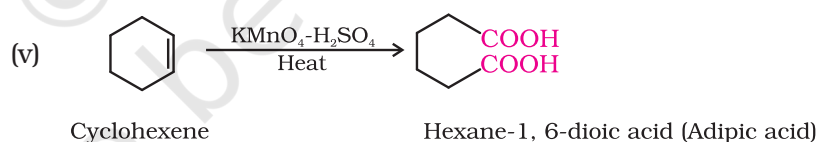
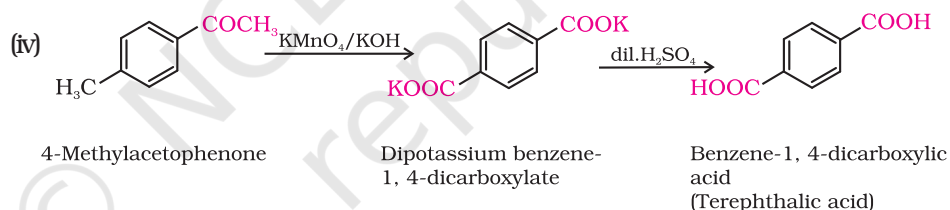
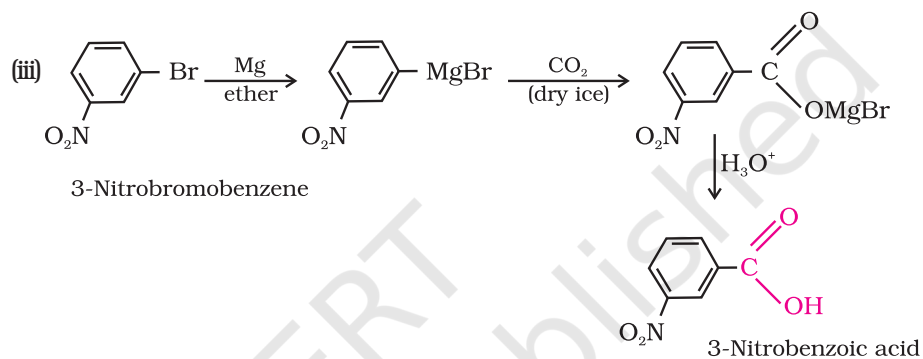
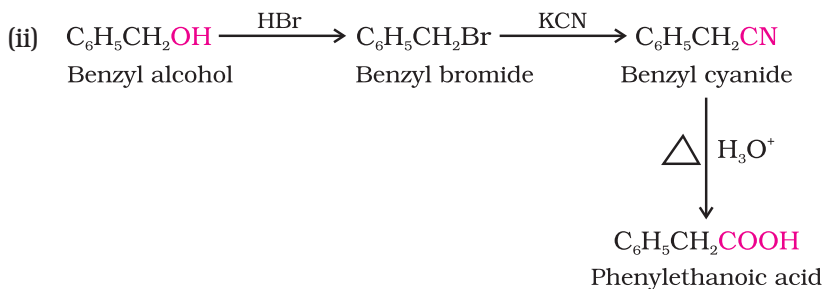


Write chemical reactions to affect the following transformations:

Example 8.5

- Butan-1-ol to butanoic acid
- Benzyl alcohol to phenylethanoic acid
- 3-Nitrobromobenzene to 3-nitrobenzoic acid
- 4-Methylacetophenone to benzene-1,4-dicarboxylic acid
- Cyclohexene to hexane-1,6-dioic acid
- Butanal to butanoic acid.

not



8.7

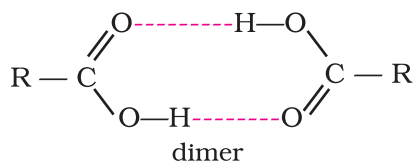
(i) Ethylbenzene

(ii) Acetophenone

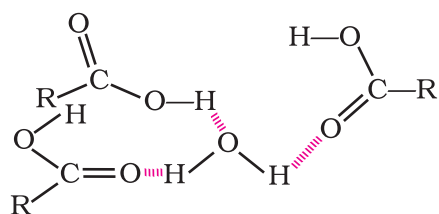
(iii) Bromobenzene

(iv) Phenylethene (Styrene)

8.8 Physical Properties



In vapour state or in aprotic solvent



Hydrogen bonding of RCOOH with H₂O

Aliphatic carboxylic acids upto nine carbon atoms are colourless liquids at room temperature with unpleasant odours. The higher acids are wax like solids and are practically odourless due to their low volatility. Carboxylic acids are higher boiling liquids than aldehydes, ketones and even alcohols of comparable molecular masses. This is due to more extensive association of carboxylic acid molecules through intermolecular hydrogen bonding. The hydrogen bonds are not broken completely even in the vapour phase. In fact, most carboxylic acids exist as dimer in the vapour phase or in the aprotic solvents.

Simple aliphatic carboxylic acids having upto four carbon atoms are miscible in water due to the formation of hydrogen bonds with water. The solubility decreases with increasing number of carbon atoms. Higher carboxylic acids are practically insoluble in water due to the increased hydrophobic interaction of hydrocarbon part. Benzoic acid, the simplest aromatic carboxylic acid is nearly insoluble in cold water. Carboxylic acids are also soluble in less polar organic solvents like benzene, ether, alcohol, chloroform, etc.

8.9 Chemical Reactions

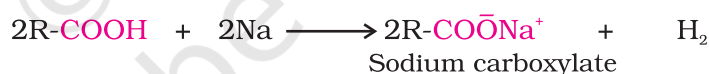
The reaction of carboxylic acids are classified as follows:

8.9.1 Reactions Involving Cleavage of O-H Bond

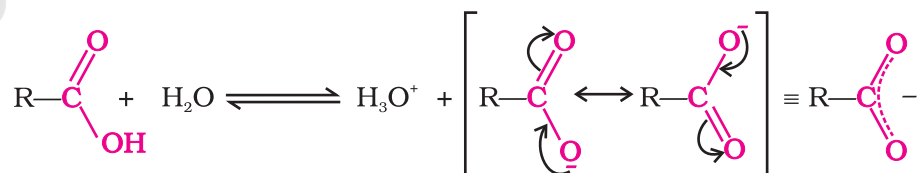
Acidity

Reactions with metals and alkalies

The carboxylic acids like alcohols evolve hydrogen with electropositive metals and form salts with alkalies similar to phenols. However, unlike phenols they react with weaker bases such as carbonates and hydrogencarbonates to evolve carbon dioxide. This reaction is used to detect the presence of carboxyl group in an organic compound.



Carboxylic acids dissociate in water to give resonance stabilised carboxylate anions and hydronium ion.



For the above reaction:

$$K_{eq} = \frac{[\text{H}_3\text{O}^+][\text{RCOO}^-]}{[\text{H}_2\text{O}][\text{RCOOH}]} \quad K_a = K_{eq} [\text{H}_2\text{O}] = \frac{[\text{H}_3\text{O}^+][\text{RCOO}^-]}{[\text{RCOOH}]}$$

where K_{eq} is equilibrium constant and K_a is the acid dissociation constant.

For convenience, the strength of an acid is generally indicated by its $\text{p}K_a$ value rather than its K_a value.

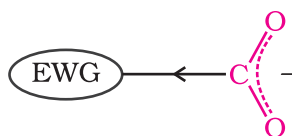
$$\text{p}K_a = -\log K_a$$

The $\text{p}K_a$ of hydrochloric acid is -7.0 , where as $\text{p}K_a$ of trifluoroacetic acid (the strongest carboxylic acid), benzoic acid and acetic acid are 0.23 , 4.19 and 4.76 , respectively.

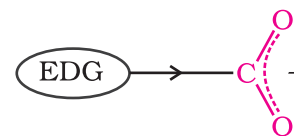
Smaller the $\text{p}K_a$, the stronger the acid (the better it is as a proton donor). Strong acids have $\text{p}K_a$ values < 1 , the acids with $\text{p}K_a$ values between 1 and 5 are considered to be moderately strong acids, weak acids have $\text{p}K_a$ values between 5 and 15 , and extremely weak acids have $\text{p}K_a$ values > 15 .

Carboxylic acids are weaker than mineral acids, but they are stronger acids than alcohols and many simple phenols ($\text{p}K_a$ is ~ 16 for ethanol and 10 for phenol). In fact, carboxylic acids are amongst the most acidic organic compounds you have studied so far. You already know why phenols are more acidic than alcohols. The higher acidity of carboxylic acids as compared to phenols can be understood similarly. The conjugate base of carboxylic acid, a carboxylate ion, is stabilised by two equivalent resonance structures in which the negative charge is at the more electronegative oxygen atom. The conjugate base of phenol, a phenoxide ion, has non-equivalent resonance structures in which the negative charge is at the less electronegative carbon atom. Therefore, resonance in phenoxide ion is not as important as it is in carboxylate ion. Further, the negative charge is delocalised over two electronegative oxygen atoms in carboxylate ion whereas it is less effectively delocalised over one oxygen atom and less electronegative carbon atoms in phenoxide ion (Unit 7, Class XII). Thus, the carboxylate ion is more stabilised than phenoxide ion, so carboxylic acids are more acidic than phenols.

Effect of substituents on the acidity of carboxylic acids: Substituents may affect the stability of the conjugate base and thus, also affect the acidity of the carboxylic acids. Electron withdrawing groups increase the acidity of carboxylic acids by stabilising the conjugate base through delocalisation of the negative charge by inductive and/or resonance effects. Conversely, electron donating groups decrease the acidity by destabilising the conjugate base.

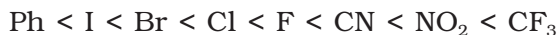


Electron withdrawing group (EWG) stabilises the carboxylate anion and strengthens the acid

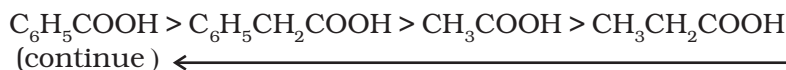
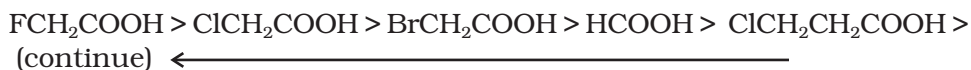


Electron donating group (EDG) destabilises the carboxylate anion and weakens the acid

The effect of the following groups in increasing acidity order is



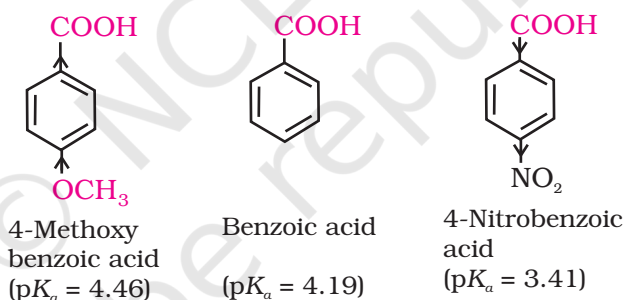
Thus, the following acids are arranged in order of increasing acidity (based on $\text{p}K_a$ values):



Direct attachment of groups such as phenyl or vinyl to the carboxylic acid, increases the acidity of corresponding carboxylic acid, contrary to the decrease expected due to resonance effect shown below:



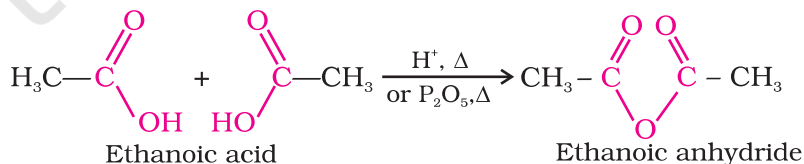
This is because of greater electronegativity of sp^2 hybridised carbon to which carboxyl carbon is attached. The presence of electron withdrawing group on the phenyl of aromatic carboxylic acid increases their acidity while electron donating groups decrease their acidity.



8.9.2 Reactions Involving Cleavage of C-OH Bond

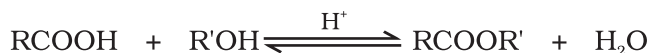
1. Formation of anhydride

Carboxylic acids on heating with mineral acids such as H_2SO_4 or with P_2O_5 give corresponding anhydride.



2. Esterification

Carboxylic acids are esterified with alcohols or phenols in the presence of a mineral acid such as concentrated H_2SO_4 or HCl gas as a catalyst.



Mechanism of esterification of carboxylic acids: The esterification of carboxylic acids with alcohols is a kind of nucleophilic acyl substitution. Protonation of the carbonyl oxygen activates the carbonyl group towards nucleophilic addition of the alcohol. Proton transfer in the tetrahedral intermediate converts the hydroxyl group into $-^+OH_2$ group, which, being a better leaving group, is eliminated as neutral water molecule. The protonated ester so formed finally loses a proton to give the ester.

Carboxylic acid

Tetrahedral intermediate

Proton transfer

Ester

Protonated ester

3. Reactions with PCl_5 , PCl_3 and SOCl_2

The hydroxyl group of carboxylic acids, behaves like that of alcohols and is easily replaced by chlorine atom on treating with PCl_5 , PCl_3 or SOCl_2 . Thionyl chloride (SOCl_2) is preferred because the other two products are gaseous and escape the reaction mixture making the purification of the products easier.

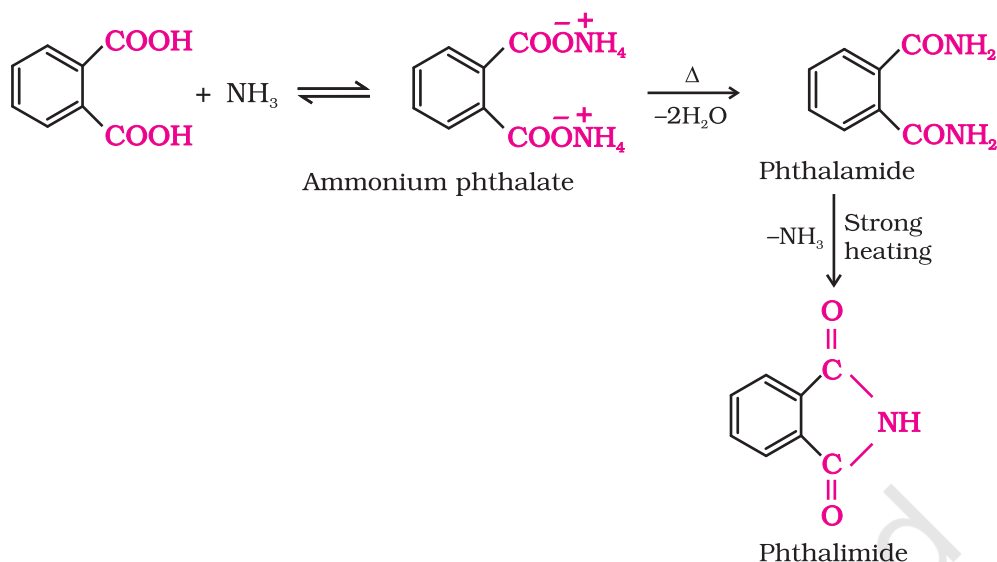
$$\text{RCOOH} + \text{PCl}_5 \longrightarrow \text{RCOCl} + \text{POCl}_3 + \text{HCl}$$
$$3\text{RCOOH} + \text{PCl}_3 \longrightarrow 3\text{RCOCl} + \text{H}_3\text{PO}_3$$
$$\text{RCOOH} + \text{SOCl}_2 \longrightarrow \text{RCOCl} + \text{SO}_2 + \text{HCl}$$
$$\begin{aligned} 3\text{RCOOH} + \text{PCl}_3 &\longrightarrow 3\text{RCOCl} + \text{H}_3\text{PO}_3 \\ \text{RCOOH} + \text{SOCl}_2 &\longrightarrow \text{RCOCl} + \text{SO}_2 + \text{HCl} \end{aligned}$$

4. *Reaction with ammonia*

Carboxylic acids react with ammonia to give ammonium salt which on further heating at high temperature give amides. For example:

$$\text{CH}_3\text{COOH} + \text{NH}_3 \rightleftharpoons \text{CH}_3\text{COO}^-\text{NH}_4^+ \xrightarrow[-\text{H}_2\text{O}]{\Delta} \text{CH}_3\text{CONH}_2$$
 Ammonium acetate Acetamide

$$\text{C}_6\text{H}_5\text{COOH} + \text{NH}_3 \rightleftharpoons \text{C}_6\text{H}_5\text{COO}^-\text{NH}_4^+ \xrightarrow[-\text{H}_2\text{O}]{\Delta} \text{C}_6\text{H}_5\text{CONH}_2$$
 Ammonium benzoate Benzamide



8.9.3 Reactions Involving -COOH Group

1. Reduction

Carboxylic acids are reduced to primary alcohols by lithium aluminium hydride or better with diborane. Diborane does not easily reduce functional groups such as ester, nitro, halo, etc. Sodium borohydride does not reduce the carboxyl group.



2. Decarboxylation

Carboxylic acids lose carbon dioxide to form hydrocarbons when their sodium salts are heated with sodalime (NaOH and CaO in the ratio of 3 : 1). The reaction is known as decarboxylation.

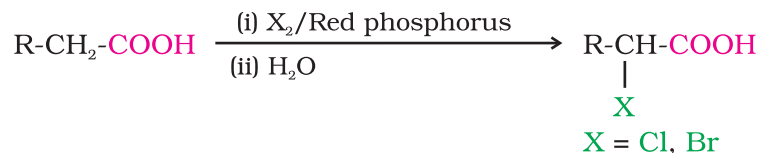


Alkali metal salts of carboxylic acids also undergo decarboxylation on electrolysis of their aqueous solutions and form hydrocarbons having twice the number of carbon atoms present in the alkyl group of the acid. The reaction is known as **Kolbe electrolysis** (Unit 9, Class XI).

8.9.4 Substitution Reactions in the Hydrocarbon Part

1. Halogenation

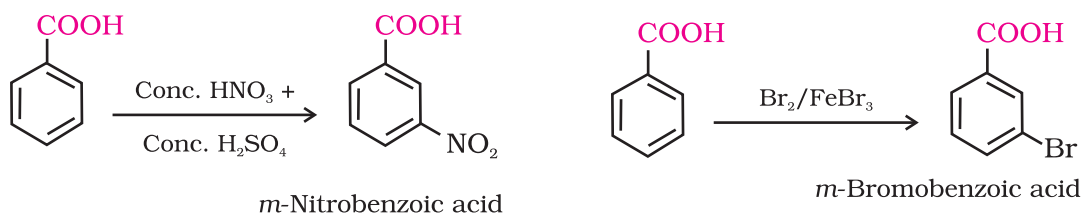
Carboxylic acids having an α -hydrogen are halogenated at the α -position on treatment with chlorine or bromine in the presence of small amount of red phosphorus to give α -halocarboxylic acids. The reaction is known as **Hell-Volhard-Zelinsky reaction**.



α - Halocarboxylic acid

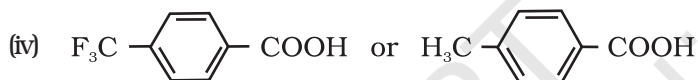
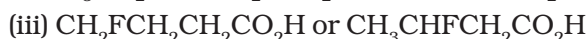
2. Ring substitution

Aromatic carboxylic acids undergo electrophilic substitution reactions in which the carboxyl group acts as a deactivating and meta-directing group. They however, do not undergo **Friedel-Crafts reaction** (because the carboxyl group is deactivating and the catalyst aluminium chloride (Lewis acid) gets bonded to the carboxyl group).



Intext Question

8.8 Which acid of each pair shown here would you expect to be stronger?



8.10 Uses of Carboxylic Acids

Methanoic acid is used in rubber, textile, dyeing, leather and electroplating industries. Ethanoic acid is used as solvent and as vinegar in food industry. Hexanedioic acid is used in the manufacture of nylon-6, 6. Esters of benzoic acid are used in perfumery. Sodium benzoate is used as a food preservative. Higher fatty acids are used for the manufacture of soaps and detergents.

Summary

Aldehydes, ketones and carboxylic acids are some of the important classes of organic compounds containing carbonyl group. These are highly polar molecules. Therefore, they boil at higher temperatures than the hydrocarbons and weakly polar compounds such as ethers of comparable molecular masses. The lower members are more soluble in water because they form hydrogen bonds with water. The higher members, because of large size of hydrophobic chain of carbon atoms, are insoluble in water but soluble in common organic solvents. Aldehydes are prepared by dehydrogenation or controlled oxidation of primary alcohols and controlled or selective reduction of acyl halides. Aromatic aldehydes may also be prepared by oxidation of (i) methylbenzene with chromyl chloride or CrO₃ in the presence of acetic anhydride, (ii) formylation of arenes with carbon monoxide and hydrochloric acid in the presence of anhydrous aluminium chloride, and (iii) cuprous chloride or by hydrolysis of benzal chloride. Ketones are prepared by oxidation of secondary alcohols and hydration of alkynes. Ketones are also prepared by reaction of acyl chloride with dialkylcadmium. A good method for the preparation of aromatic ketones is the **Friedel-Crafts acylation** of aromatic hydrocarbons with acyl chlorides or anhydrides. Both aldehydes and ketones can be prepared by ozonolysis of alkenes. Aldehydes and ketones undergo nucleophilic addition reactions onto the carbonyl group with a number of nucleophiles such as, HCN, NaHSO₃, alcohols (or diols),

ammonia derivatives, and **Grignard reagents**. The α -hydrogens in aldehydes and ketones are acidic. Therefore, aldehydes and ketones having at least one α -hydrogen, undergo **Aldol condensation** in the presence of a base to give α -hydroxyaldehydes (aldol) and α -hydroxyketones (ketol), respectively. Aldehydes having no α -hydrogen undergo **Cannizzaro reaction** in the presence of concentrated alkali. Aldehydes and ketones are reduced to alcohols with NaBH_4 , LiAlH_4 , or by catalytic hydrogenation. The carbonyl group of aldehydes and ketones can be reduced to a methylene group by **Clemmensen reduction** or **Wolff-Kishner reduction**. Aldehydes are easily oxidised to carboxylic acids by mild oxidising reagents such as **Tollens' reagent** and **Fehling's reagent**. These oxidation reactions are used to distinguish aldehydes from ketones. Carboxylic acids are prepared by the oxidation of primary alcohols, aldehydes and alkenes by hydrolysis of nitriles, and by treatment of Grignard reagents with carbon dioxide. Aromatic carboxylic acids are also prepared by side-chain oxidation of alkylbenzenes. Carboxylic acids are considerably more acidic than alcohols and most of simple phenols. Carboxylic acids are reduced to primary alcohols with LiAlH_4 , or better with diborane in ether solution and also undergo α -halogenation with Cl_2 and Br_2 in the presence of red phosphorus (**Hell-Volhard Zelinsky reaction**). Methanal, ethanal, propanone, benzaldehyde, formic acid, acetic acid and benzoic acid are highly useful compounds in industry.

Exercises

8.1 What is meant by the following terms ? Give an example of the reaction in each case.

- | | | |
|-------------------|----------------|-------------------------|
| (i) Cyanohydrin | (ii) Acetal | (iii) Semicarbazone |
| (iv) Aldol | (v) Hemiacetal | (vi) Oxime |
| (vii) Ketal | (viii) Imine | (ix) 2,4-DNP-derivative |
| (x) Schiff's base | | |

8.2 Name the following compounds according to IUPAC system of nomenclature:

- | | |
|--|---|
| (i) $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CHO}$ | (ii) $\text{CH}_3\text{CH}_2\text{COCH}(\text{C}_2\text{H}_5)\text{CH}_2\text{CH}_2\text{Cl}$ |
| (iii) $\text{CH}_3\text{CH}=\text{CHCHO}$ | (iv) $\text{CH}_3\text{COCH}_2\text{COCH}_3$ |
| (v) $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{C}(\text{CH}_3)_2\text{COCH}_3$ | (vi) $(\text{CH}_3)_3\text{CCH}_2\text{COOH}$ |
| (vii) $\text{OHCC}_6\text{H}_4\text{CHO}-p$ | |

8.3 Draw the structures of the following compounds.

- | | |
|--|-------------------------------------|
| (i) 3-Methylbutanal | (ii) <i>p</i> -Nitropropiophenone |
| (iii) <i>p</i> -Methylbenzaldehyde | (iv) 4-Methylpent-3-en-2-one |
| (v) 4-Chloropentan-2-one | (vi) 3-Bromo-4-phenylpentanoic acid |
| (vii) <i>p,p'</i> -Dihydroxybenzophenone | (viii) Hex-2-en-4-ynoic acid |

8.4 Write the IUPAC names of the following ketones and aldehydes. Wherever possible, give also common names.

- | | |
|--|--|
| (i) $\text{CH}_3\text{CO}(\text{CH}_2)_4\text{CH}_3$ | (ii) $\text{CH}_3\text{CH}_2\text{CHBrCH}_2\text{CH}(\text{CH}_3)\text{CHO}$ |
| (iii) $\text{CH}_3(\text{CH}_2)_5\text{CHO}$ | (iv) $\text{Ph}-\text{CH}=\text{CH}-\text{CHO}$ |



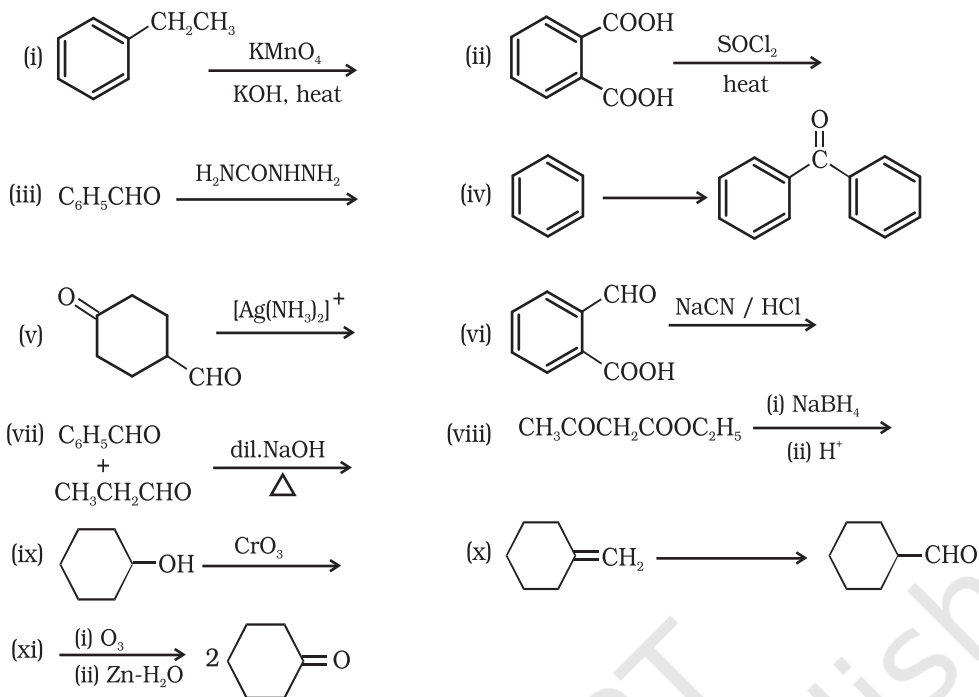
- (vi) PhCOPh

8.5 Draw structures of the following derivatives.

- The 2,4-dinitrophenylhydrazone of benzaldehyde
- Cyclopropanone oxime
- Acetaldehydedimethylacetal
- The semicarbazone of cyclobutanone
- The ethylene ketal of hexan-3-one
- The methyl hemiacetal of formaldehyde

- 8.6** Predict the products formed when cyclohexanecarbaldehyde reacts with following reagents.
- (i) PhMgBr and then H_3O^+
 - (ii) Tollens' reagent
 - (iii) Semicarbazide and weak acid
 - (iv) Excess ethanol and acid
 - (v) Zinc amalgam and dilute hydrochloric acid
- 8.7** Which of the following compounds would undergo aldol condensation, which the Cannizzaro reaction and which neither? Write the structures of the expected products of aldol condensation and Cannizzaro reaction.
- (i) Methanal
 - (ii) 2-Methylpentanal
 - (iii) Benzaldehyde
 - (iv) Benzophenone
 - (v) Cyclohexanone
 - (vi) 1-Phenylpropanone
 - (vii) Phenylacetaldehyde
 - (viii) Butan-1-ol
 - (ix) 2,2-Dimethylbutanal
- 8.8** How will you convert ethanal into the following compounds?
- (i) Butane-1,3-diol
 - (ii) But-2-enal
 - (iii) But-2-enoic acid
- 8.9** Write structural formulas and names of four possible aldol condensation products from propanal and butanal. In each case, indicate which aldehyde acts as nucleophile and which as electrophile.
- 8.10** An organic compound with the molecular formula $\text{C}_9\text{H}_{10}\text{O}$ forms 2,4-DNP derivative, reduces Tollens' reagent and undergoes Cannizzaro reaction. On vigorous oxidation, it gives 1,2-benzenedicarboxylic acid. Identify the compound.
- 8.11** An organic compound (A) (molecular formula $\text{C}_8\text{H}_{16}\text{O}_2$) was hydrolysed with dilute sulphuric acid to give a carboxylic acid (B) and an alcohol (C). Oxidation of (C) with chromic acid produced (B). (C) on dehydration gives but-1-ene. Write equations for the reactions involved.
- 8.12** Arrange the following compounds in increasing order of their property as indicated:
- (i) Acetaldehyde, Acetone, Di-*tert*-butyl ketone, Methyl *tert*-butyl ketone (reactivity towards HCN)
 - (ii) $\text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{COOH}$, $\text{CH}_3\text{CH}(\text{Br})\text{CH}_2\text{COOH}$, $(\text{CH}_3)_2\text{CHCOOH}$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ (acid strength)
 - (iii) Benzoic acid, 4-Nitrobenzoic acid, 3,4-Dinitrobenzoic acid, 4-Methoxybenzoic acid (acid strength)
- 8.13** Give simple chemical tests to distinguish between the following pairs of compounds.
- (i) Propanal and Propanone
 - (ii) Acetophenone and Benzophenone
 - (iii) Phenol and Benzoic acid
 - (iv) Benzoic acid and Ethyl benzoate
 - (v) Pentan-2-one and Pentan-3-one
 - (vi) Benzaldehyde and Acetophenone
 - (vii) Ethanal and Propanal
- 8.14** How will you prepare the following compounds from benzene? You may use any inorganic reagent and any organic reagent having not more than one carbon atom
- (i) Methyl benzoate
 - (ii) *m*-Nitrobenzoic acid
 - (iii) *p*-Nitrobenzoic acid
 - (iv) Phenylacetic acid
 - (v) *p*-Nitrobenzaldehyde.
- 8.15** How will you bring about the following conversions in not more than two steps?
- (i) Propanone to Propene
 - (ii) Benzoic acid to Benzaldehyde
 - (iii) Ethanol to 3-Hydroxybutanal
 - (iv) Benzene to *m*-Nitroacetophenone
 - (v) Benzaldehyde to Benzophenone
 - (vi) Bromobenzene to 1-Phenylethanol
 - (vii) Benzaldehyde to 3-Phenylpropan-1-ol
 - (viii) Benzaldehyde to α -Hydroxyphenylacetic acid
 - (ix) Benzoic acid to *m*-Nitrobenzyl alcohol
- 8.16** Describe the following:
- (i) Acetylation
 - (ii) Cannizzaro reaction
 - (iii) Cross aldol condensation
 - (iv) Decarboxylation

8.17 Complete each synthesis by giving missing starting material, reagent or products



8.18 Give plausible explanation for each of the following:

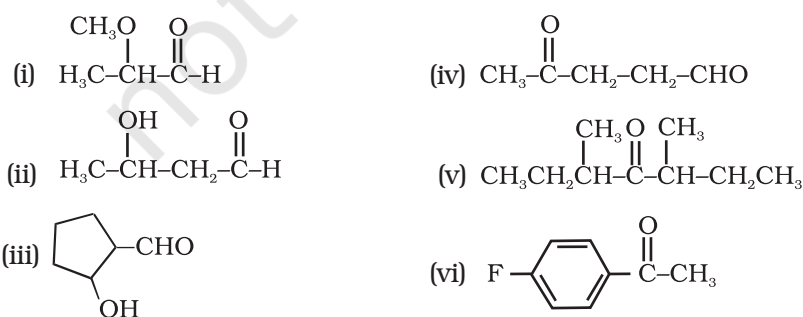
- Cyclohexanone forms cyanohydrin in good yield but 2,2,6-trimethylcyclohexanone does not.
- There are two $-NH_2$ groups in semicarbazide. However, only one is involved in the formation of semicarbazones.
- During the preparation of esters from a carboxylic acid and an alcohol in the presence of an acid catalyst, the water or the ester should be removed as soon as it is formed.

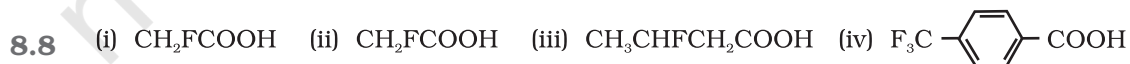
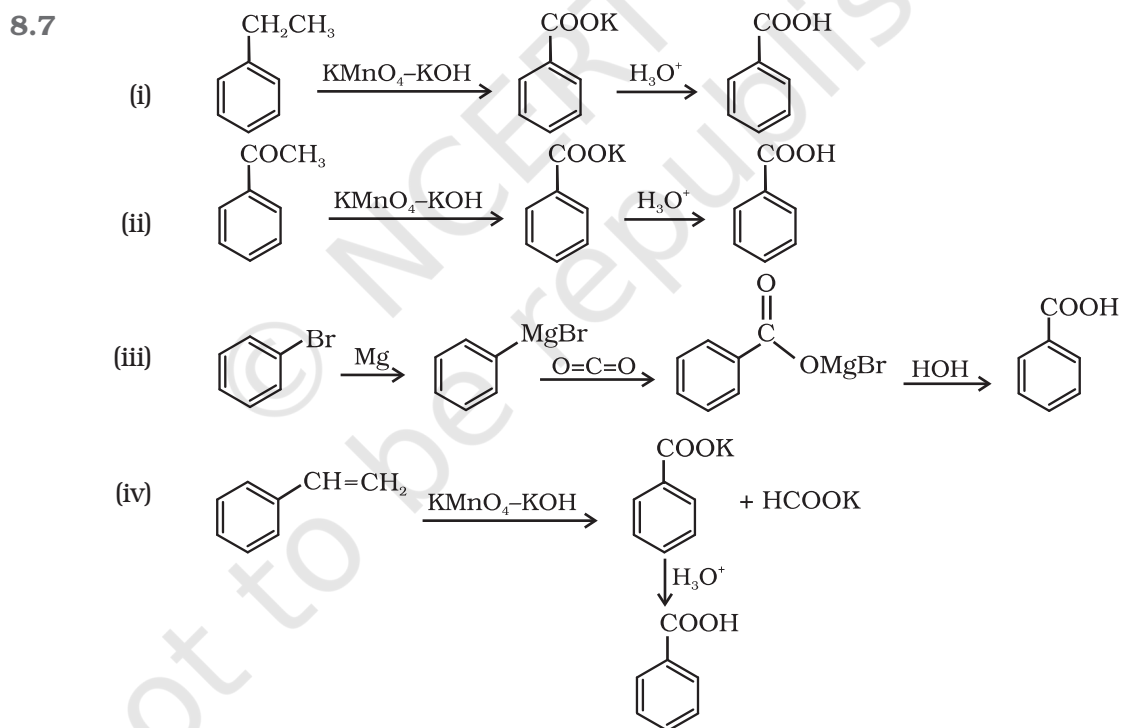
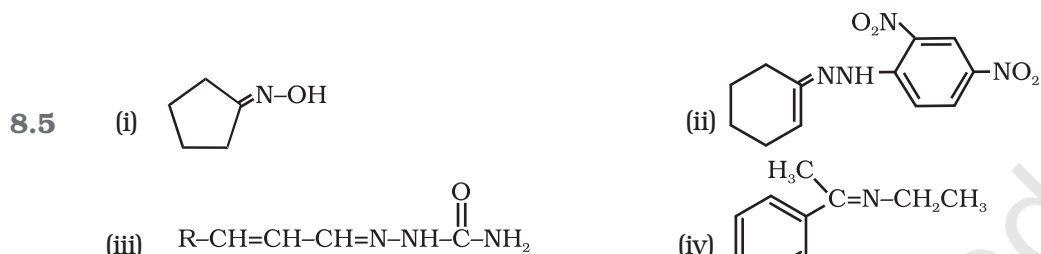
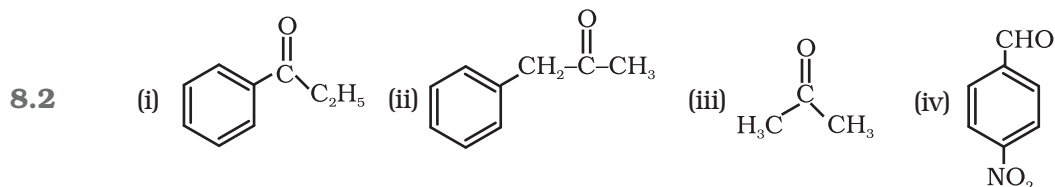
8.19 An organic compound contains 69.77% carbon, 11.63% hydrogen and rest oxygen. The molecular mass of the compound is 86. It does not reduce Tollens' reagent but forms an addition compound with sodium hydrogensulphite and give positive iodoform test. On vigorous oxidation it gives ethanoic and propanoic acid. Write the possible structure of the compound.

8.20 Although phenoxide ion has more number of resonating structures than carboxylate ion, carboxylic acid is a stronger acid than phenol. Why?

Answers to Some Intext Questions

8.1







12086CH13

Unit 9 Amines

Objectives

After studying this Unit, you will be able to

- describe amines as derivatives of ammonia having a pyramidal structure;
- classify amines as primary, secondary and tertiary;
- name amines by common names and IUPAC system;
- describe some of the important methods of preparation of amines;
- explain the properties of amines;
- distinguish between primary, secondary and tertiary amines;
- describe the method of preparation of diazonium salts and their importance in the synthesis of a series of aromatic compounds including azo dyes.

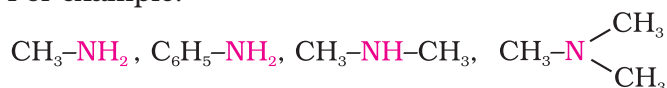
"The chief commercial use of amines is as intermediates in the synthesis of medicines and fibres".

Amines constitute an important class of organic compounds derived by replacing one or more hydrogen atoms of ammonia molecule by alkyl/aryl group(s). In nature, they occur among proteins, vitamins, alkaloids and hormones. Synthetic examples include polymers, dye stuffs and drugs. Two biologically active compounds, namely adrenaline and ephedrine, both containing secondary amino group, are used to increase blood pressure. Novocain, a synthetic amino compound, is used as an anaesthetic in dentistry. Benadryl, a well known antihistaminic drug also contains tertiary amino group. Quaternary ammonium salts are used as surfactants. Diazonium salts are intermediates in the preparation of a variety of aromatic compounds including dyes. In this Unit, you will learn about amines and diazonium salts.

I. AMINES

Amines can be considered as derivatives of ammonia, obtained by replacement of one, two or all the three hydrogen atoms by alkyl and/or aryl groups.

For example:



9.1 Structure of Amines

Like ammonia, nitrogen atom of amines is trivalent and carries an unshared pair of electrons. Nitrogen orbitals in amines are therefore, sp^3 hybridised and the geometry of amines is pyramidal. Each of the three sp^3 hybridised orbitals of nitrogen overlap with orbitals of hydrogen or carbon depending upon the composition of the amines. The fourth orbital of nitrogen in all amines contains an unshared pair of electrons. Due to the presence of unshared pair of electrons, the angle C-N-E, (where E is

C or H) is less than 109.5° ; for instance, it is 108° in case of trimethylamine as shown in Fig. 9.1.

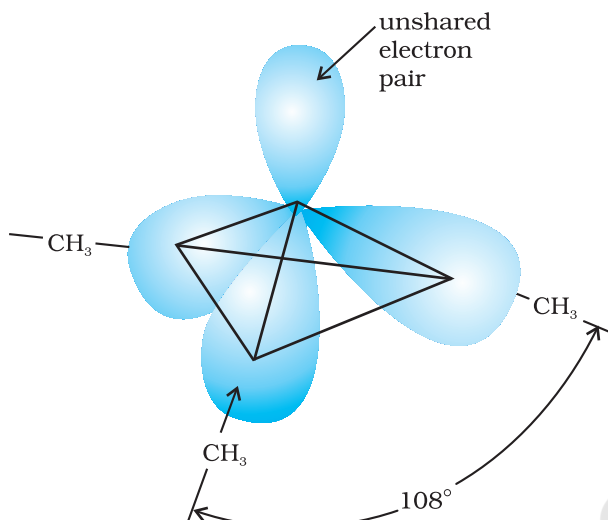
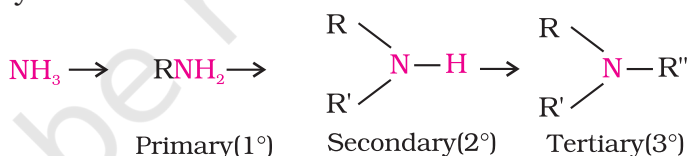


Fig. 9.1 Pyramidal shape of trimethylamine

9.2 Classification

Amines are classified as primary (1°), secondary (2°) and tertiary (3°) depending upon the number of hydrogen atoms replaced by alkyl or aryl groups in ammonia molecule. If one hydrogen atom of ammonia is replaced by R or Ar, we get RNH_2 or ArNH_2 , a primary amine (1°). If two hydrogen atoms of ammonia or one hydrogen atom of R-NH_2 are replaced by another alkyl/aryl (R') group, what would you get? You get $\text{R-NHR}'$, secondary amine. The second alkyl/aryl group may be same or different. Replacement of another hydrogen atom by alkyl/aryl group leads to the formation of tertiary amine. Amines are said to be 'simple' when all the alkyl or aryl groups are the same, and 'mixed' when they are different.



9.3 Nomenclature

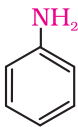
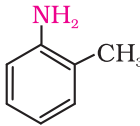
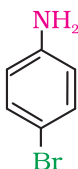
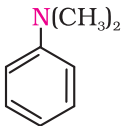
In common system, an aliphatic amine is named by prefixing alkyl group to amine, i.e., alkylamine as one word (e.g., methylamine). In secondary and tertiary amines, when two or more groups are the same, the prefix di or tri is appended before the name of alkyl group. In IUPAC system, primary amines are named as **alkanamines**. The name is derived by replacement of 'e' of alkane by the word amine. For example, CH_3NH_2 is named as methanamine. In case, more than one amino group is present at different positions in the parent chain, their positions are specified by giving numbers to the carbon atoms bearing $-\text{NH}_2$ groups and suitable prefix such as di, tri, etc. is attached to the amine. The letter 'e' of the suffix of the hydrocarbon part is retained. For example, $\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{NH}_2$ is named as ethane-1, 2-diamine.

To name secondary and tertiary amines, we use locant N to designate substituent attached to a nitrogen atom. For example, $\text{CH}_3\text{NHCH}_2\text{CH}_3$ is

named as N-methylethanamine and $(\text{CH}_3\text{CH}_2)_3\text{N}$ is named as N, N-diethylethanamine. More examples are given in Table 9.1.

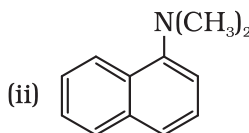
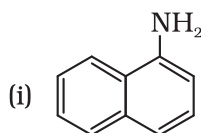
In arylamines, $-\text{NH}_2$ group is directly attached to the benzene ring. $\text{C}_6\text{H}_5\text{NH}_2$ is the simplest example of arylamine. In common system, it is known as aniline. It is also an accepted IUPAC name. While naming arylamines according to IUPAC system, suffix 'e' of arene is replaced by 'amine'. Thus in IUPAC system, $\text{C}_6\text{H}_5-\text{NH}_2$ is named as benzenamine. Common and IUPAC names of some alkylamines and arylamines are given in Table 9.1.

Table 9.1: Nomenclature of Some Alkylamines and Arylamines

Amine	Common name	IUPAC name
$\text{CH}_3-\text{CH}_2-\text{NH}_2$	Ethylamine	Ethanamine
$\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{NH}_2$	<i>n</i> -Propylamine	Propan-1-amine
$\text{CH}_3-\underset{\text{NH}_2}{\text{CH}}-\text{CH}_3$	Isopropylamine	Propan-2-amine
$\text{CH}_3-\underset{\text{H}}{\text{N}}-\text{CH}_2-\text{CH}_3$	Ethylmethanamine	N-Methylethanamine
$\text{CH}_3-\underset{\text{CH}_3}{\text{N}}-\text{CH}_3$	Trimethanamine	N,N-Dimethylmethanamine
$\text{C}_2\text{H}_5-\underset{\text{C}_2\text{H}_5}{\text{N}}-\overset{1}{\text{CH}_2}-\overset{2}{\text{CH}_2}-\overset{3}{\text{CH}_2}-\overset{4}{\text{CH}_3}$	N,N-Diethylbutylamine	N,N-Diethylbutan-1-amine
$\text{NH}_2-\overset{1}{\text{CH}_2}-\overset{2}{\text{CH}}=\overset{3}{\text{CH}_2}$	Allylamine	Prop-2-en-1-amine
$\text{NH}_2-(\text{CH}_2)_6-\text{NH}_2$	Hexamethylenediamine	Hexane-1,6-diamine
	Aniline	Aniline or Benzenamine
	<i>o</i> -Toluidine	2-Methylaniline
	<i>p</i> -Bromoaniline	4-Bromobenzenamine or 4-Bromoaniline
	N,N-Dimethylaniline	N,N-Dimethylbenzenamine

Intext Questions

9.1 Classify the following amines as primary, secondary or tertiary:



9.2 (i) Write structures of different isomeric amines corresponding to the molecular formula, $\text{C}_4\text{H}_{11}\text{N}$.

(ii) Write IUPAC names of all the isomers.

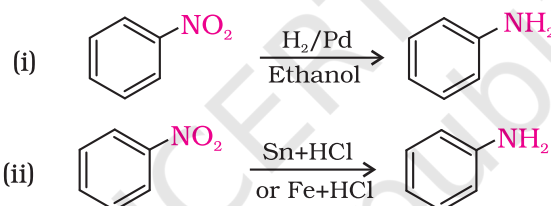
(iii) What type of isomerism is exhibited by different pairs of amines?

9.4 Preparation of Amines

Amines are prepared by the following methods:

1. Reduction of nitro compounds

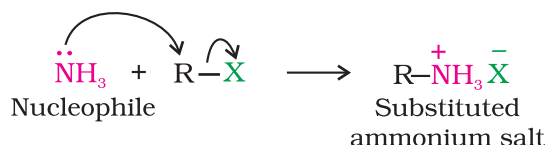
Nitro compounds are reduced to amines by passing hydrogen gas in the presence of finely divided nickel, palladium or platinum and also by reduction with metals in acidic medium. Nitroalkanes can also be similarly reduced to the corresponding alkanamines.

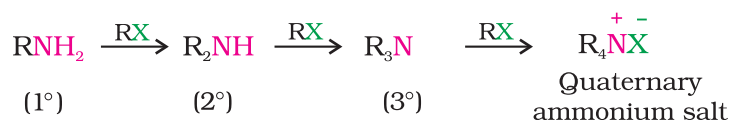


Reduction with iron scrap and hydrochloric acid is preferred because FeCl_2 formed gets hydrolysed to release hydrochloric acid during the reaction. Thus, only a small amount of hydrochloric acid is required to initiate the reaction.

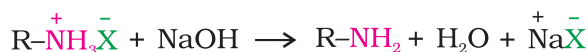
2. Ammonolysis of alkyl halides

You have read (Unit 6, Class XII) that the carbon - halogen bond in alkyl or benzyl halides can be easily cleaved by a nucleophile. Hence, an alkyl or benzyl halide on reaction with an ethanolic solution of ammonia undergoes nucleophilic substitution reaction in which the halogen atom is replaced by an amino ($-\text{NH}_2$) group. This process of cleavage of the $\text{C}-\text{X}$ bond by ammonia molecule is known as **ammonolysis**. The reaction is carried out in a sealed tube at 373 K. The primary amine thus obtained behaves as a nucleophile and can further react with alkyl halide to form secondary and tertiary amines, and finally quaternary ammonium salt.





The free amine can be obtained from the ammonium salt by treatment with a strong base:



Ammonolysis has the disadvantage of yielding a mixture of primary, secondary and tertiary amines and also a quaternary ammonium salt. However, primary amine is obtained as a major product by taking large excess of ammonia.

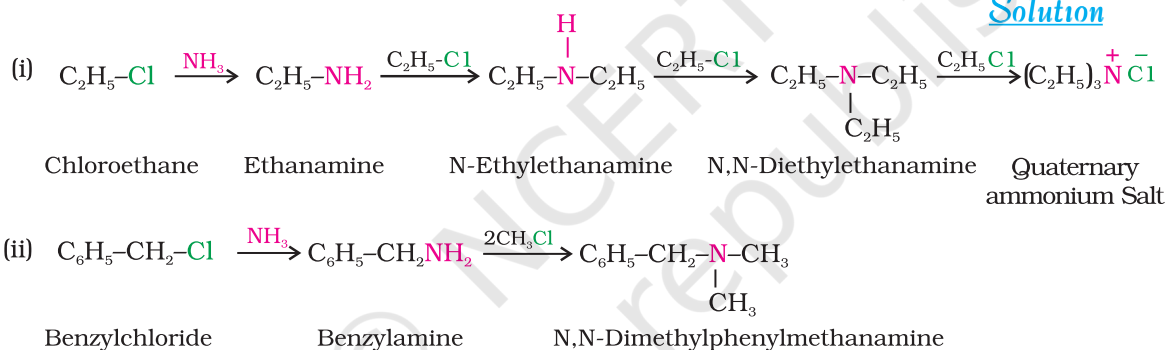
The order of reactivity of halides with amines is $\text{RI} > \text{RBr} > \text{RCl}$.

Write chemical equations for the following reactions:

- Reaction of ethanolic NH_3 with $\text{C}_2\text{H}_5\text{Cl}$.
- Ammonolysis of benzyl chloride and reaction of amine so formed with two moles of CH_3Cl .

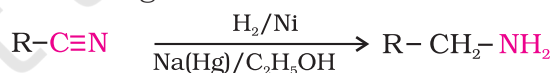
Example Q.1

Solution



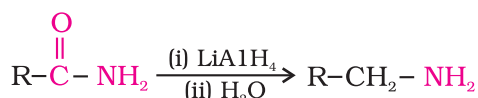
3. Reduction of nitriles

Nitriles on reduction with lithium aluminium hydride (LiAlH_4) or catalytic hydrogenation produce primary amines. This reaction is used for ascent of amine series, i.e., for preparation of amines containing one carbon atom more than the starting amine.



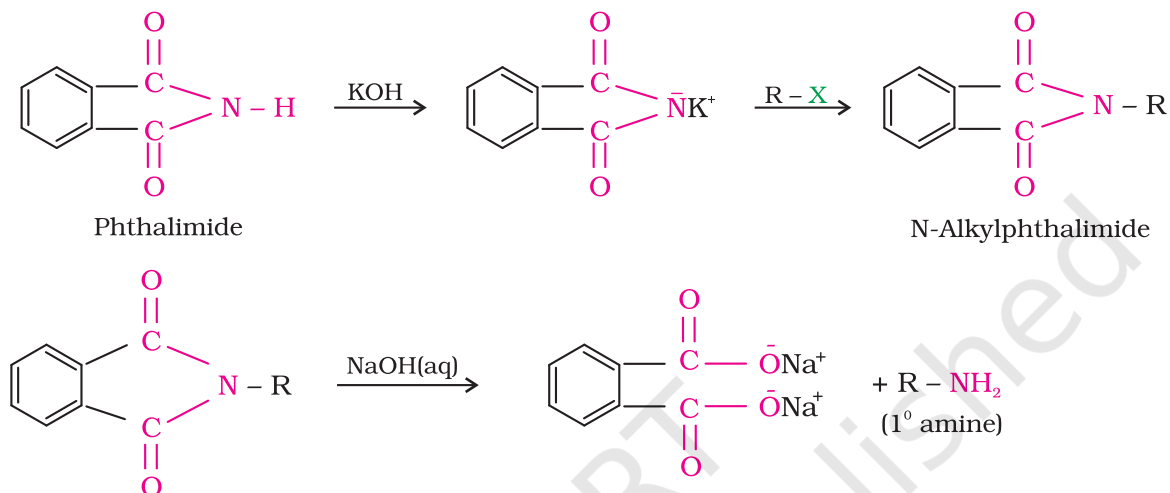
4. Reduction of amides

The amides on reduction with lithium aluminium hydride yield amines.



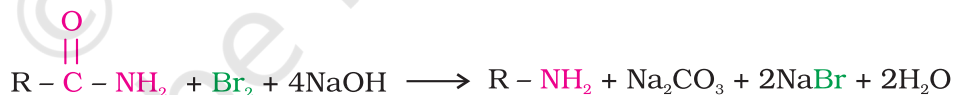
5. Gabriel phthalimide synthesis

Gabriel synthesis is used for the preparation of primary amines. Phthalimide on treatment with ethanolic potassium hydroxide forms potassium salt of phthalimide which on heating with alkyl halide followed by alkaline hydrolysis produces the corresponding primary amine. Aromatic primary amines cannot be prepared by this method because aryl halides do not undergo nucleophilic substitution with the anion formed by phthalimide.



6. Hoffmann bromamide degradation reaction

Hoffmann developed a method for preparation of primary amines by treating an amide with bromine in an aqueous or ethanolic solution of sodium hydroxide. In this degradation reaction, migration of an alkyl or aryl group takes place from carbonyl carbon of the amide to the nitrogen atom. The amine so formed contains one carbon less than that present in the amide.



Example Q.2

Write chemical equations for the following conversions:

- (i) $\text{CH}_3-\text{CH}_2-\text{Cl}$ into $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{NH}_2$
- (ii) $\text{C}_6\text{H}_5-\text{CH}_2-\text{Cl}$ into $\text{C}_6\text{H}_5-\text{CH}_2-\text{CH}_2-\text{NH}_2$

Solution

- (i) $\text{CH}_3-\text{CH}_2-\text{Cl} \xrightarrow{\text{Ethanolic NaCN}} \text{CH}_3-\text{CH}_2-\text{C}\equiv\text{N} \xrightarrow{\text{reduction}} \text{CH}_3-\text{CH}_2-\text{CH}_2-\text{NH}_2$
 Chloroethane Propanenitrile Propan-1-amine
- (ii) $\text{C}_6\text{H}_5-\text{CH}_2-\text{Cl} \xrightarrow{\text{Ethanolic NaCN}} \text{C}_6\text{H}_5-\text{CH}_2-\text{C}\equiv\text{N} \xrightarrow{\text{H}_2/\text{Ni}} \text{C}_6\text{H}_5-\text{CH}_2-\text{CH}_2-\text{NH}_2$
 Chlorophenylmethane Phenylethanenitrile 2-Phenylethanamine
 (Benzyl chloride) (Benzyl cyanide)

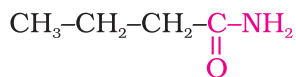
Write structures and IUPAC names of

Example 9.3

- (i) the amide which gives propanamine by Hoffmann bromamide reaction.
(ii) the amine produced by the Hoffmann degradation of benzamide.

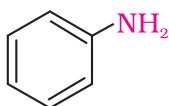
Solution

- (i) Propanamine contains three carbons. Hence, the amide molecule must contain four carbon atoms. Structure and IUPAC name of the starting amide with four carbon atoms are given below:



Butanamide

- (ii) Benzamide is an aromatic amide containing seven carbon atoms. Hence, the amine formed from benzamide is aromatic primary amine containing six carbon atoms.



Aniline or benzenamine

Intext Question

9.3 How will you convert

- (i) Benzene into aniline (ii) Benzene into N, N-dimethylaniline
(iii) $\text{Cl--(CH}_2)_4\text{--Cl}$ into hexan-1,6-diamine?

9.5 Physical Properties

The lower aliphatic amines are gases with fishy odour. Primary amines with three or more carbon atoms are liquid and still higher ones are solid. Aniline and other arylamines are usually colourless but get coloured on storage due to atmospheric oxidation.

Lower aliphatic amines are soluble in water because they can form hydrogen bonds with water molecules. However, solubility decreases with increase in molar mass of amines due to increase in size of the hydrophobic alkyl part. Higher amines are essentially insoluble in water. Considering the electronegativity of nitrogen of amine and oxygen of alcohol as 3.0 and 3.5 respectively, you can predict the pattern of solubility of amines and alcohols in water. Out of butan-1-ol and butan-1-amine, which will be more soluble in water and why? Amines are soluble in organic solvents like alcohol, ether and benzene. You may remember that alcohols are more polar than amines and form stronger intermolecular hydrogen bonds than amines.

Primary and secondary amines are engaged in intermolecular association due to hydrogen bonding between nitrogen of one and hydrogen of another molecule. This intermolecular association is more in primary amines than in secondary amines as there are two hydrogen atoms available for hydrogen bond formation in it. Tertiary amines do not have intermolecular association due to the absence of hydrogen atom available for hydrogen bond formation. Therefore, the order of boiling points of isomeric amines is as follows:

Primary > Secondary > Tertiary

Intermolecular hydrogen bonding in primary amines is shown in Fig. 9.2.

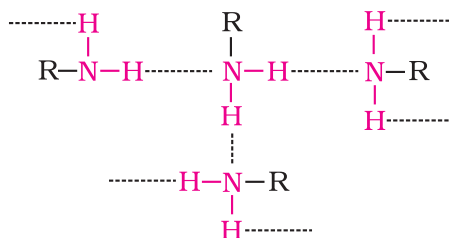


Fig. 9.2 Intermolecular hydrogen bonding in primary amines

Boiling points of amines, alcohols and alkanes of almost the same molar mass are shown in Table 9.2.

Table 9.2: Comparison of Boiling Points of Amines, Alcohols and Alkanes of Similar Molecular Masses

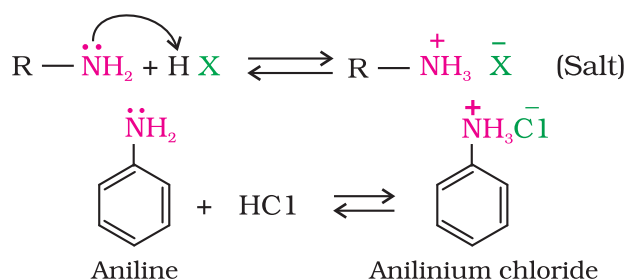
Sl. No.	Compound	Molar mass	b.p./K
1.	$n\text{-C}_4\text{H}_9\text{NH}_2$	73	350.8
2.	$(\text{C}_2\text{H}_5)_2\text{NH}$	73	329.3
3.	$\text{C}_2\text{H}_5\text{N}(\text{CH}_3)_2$	73	310.5
4.	$\text{C}_2\text{H}_5\text{CH}(\text{CH}_3)_2$	72	300.8
5.	$n\text{-C}_4\text{H}_9\text{OH}$	74	390.3

9.6 Chemical Reactions

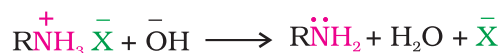
Difference in electronegativity between nitrogen and hydrogen atoms and the presence of unshared pair of electrons over the nitrogen atom makes amines reactive. The number of hydrogen atoms attached to nitrogen atom also decides the course of reaction of amines; that is why primary ($-\text{NH}_2$), secondary (>N-H) and tertiary amines (>N-) differ in many reactions. Moreover, amines behave as nucleophiles due to the presence of unshared electron pair. Some of the reactions of amines are described below:

1. Basic character of amines

Amines, being basic in nature, react with acids to form salts.

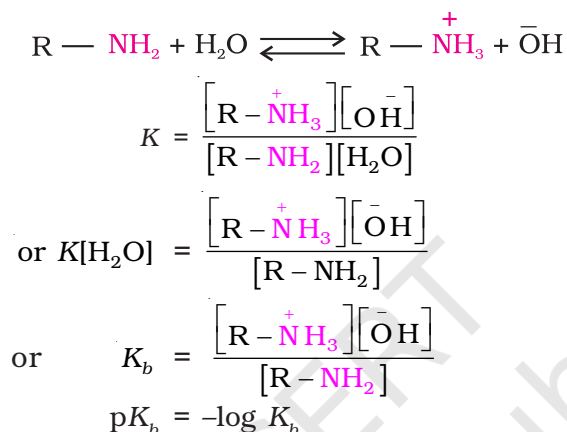


Amine salts on treatment with a base like NaOH, regenerate the parent amine.



Amine salts are soluble in water but insoluble in organic solvents like ether. This reaction is the basis for the separation of amines from the non basic organic compounds insoluble in water.

The reaction of amines with mineral acids to form ammonium salts shows that these are basic in nature. Amines have an unshared pair of electrons on nitrogen atom due to which they behave as **Lewis base**. Basic character of amines can be better understood in terms of their K_b and $\text{p}K_b$ values as explained below:



Larger the value of K_b or smaller the value of $\text{p}K_b$, stronger is the base. The $\text{p}K_b$ values of few amines are given in Table 9.3.

$\text{p}K_b$ value of ammonia is 4.75. Aliphatic amines are stronger bases than ammonia due to +I effect of alkyl groups leading to high electron density on the nitrogen atom. Their $\text{p}K_b$ values lie in the range of 3 to 4.22. On the other hand, aromatic amines are weaker bases than ammonia due to the electron withdrawing nature of the aryl group.

Table 9.3: $\text{p}K_b$ Values of Amines in Aqueous Phase

Name of amine	$\text{p}K_b$
Methanamine	3.38
<i>N</i> -Methylmethanamine	3.27
<i>N,N</i> -Dimethylmethanamine	4.22
Ethanamine	3.29
<i>N</i> -Ethylethanamine	3.00
<i>N,N</i> -Diethylethanamine	3.25
Benzenamine	9.38
Phenylmethanamine	4.70
<i>N</i> -Methylaniline	9.30
<i>N,N</i> -Dimethylaniline	8.92

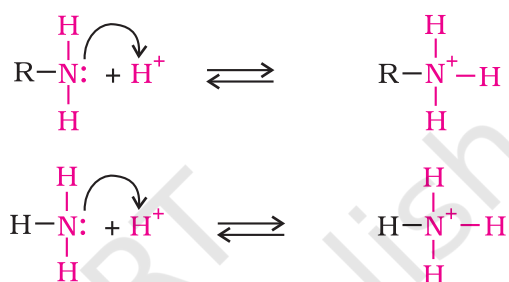
You may find some discrepancies while trying to interpret the K_b values of amines on the basis of +I or -I effect of the substituents present in amines. Besides inductive effect, there are other effects like solvation effect, steric hinderance, etc., which affect the basic strength of amines. Just ponder over. You may get the answer in the following paragraphs.

Structure-basicity relationship of amines

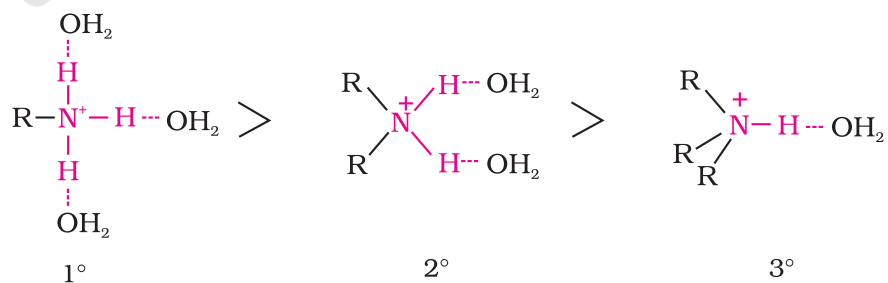
Basicity of amines is related to their structure. Basic character of an amine depends upon the ease of formation of the cation by accepting a proton from the acid. The more stable the cation is relative to the amine, more basic is the amine.

(a) Alkanamines versus ammonia

Let us consider the reaction of an alkanamine and ammonia with a proton to compare their basicity.



Due to the electron releasing nature of alkyl group, it pushes electrons towards nitrogen and thus makes the unshared electron pair more available for sharing with the proton of the acid. Moreover, the substituted ammonium ion formed from the amine gets stabilised due to dispersal of the positive charge by the +I effect of the alkyl group. Hence, alkylamines are stronger bases than ammonia. Thus, the basic nature of aliphatic amines should increase with increase in the number of alkyl groups. This trend is followed in the gaseous phase. The order of basicity of amines in the gaseous phase follows the expected order: tertiary amine > secondary amine > primary amine > NH_3 . The trend is not regular in the aqueous state as evident by their $\text{p}K_b$ values given in Table 9.3. In the aqueous phase, the substituted ammonium cations get stabilised not only by electron releasing effect of the alkyl group (+I) but also by solvation with water molecules. The greater the size of the ion, lesser will be the solvation and the less stabilised is the ion. The order of stability of ions are as follows:



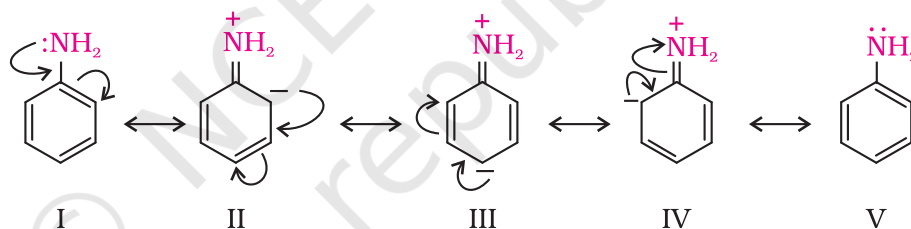
Decreasing order of extent of H-bonding in water and order of stability of ions by solvation.

Greater is the stability of the substituted ammonium cation, stronger should be the corresponding amine as a base. Thus, the order of basicity of aliphatic amines should be: primary > secondary > tertiary, which is opposite to the inductive effect based order. Secondly, when the alkyl group is small, like $-\text{CH}_3$ group, there is no steric hindrance to H-bonding. In case the alkyl group is bigger than CH_3 group, there will be steric hindrance to H-bonding. Therefore, the change of nature of the alkyl group, e.g., from $-\text{CH}_3$ to $-\text{C}_2\text{H}_5$ results in change of the order of basic strength. Thus, there is a subtle interplay of the inductive effect, solvation effect and steric hindrance of the alkyl group which decides the basic strength of alkyl amines in the aqueous state. The order of basic strength in case of methyl substituted amines and ethyl substituted amines in aqueous solution is as follows:

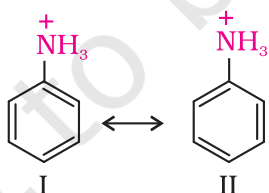


(b) Arylamines versus ammonia

$\text{p}K_b$ value of aniline is quite high. Why is it so? It is because in aniline or other arylamines, the $-\text{NH}_2$ group is attached directly to the benzene ring. It results in the unshared electron pair on nitrogen atom to be in conjugation with the benzene ring and thus making it less available for protonation. If you write different resonating structures of aniline, you will find that aniline is a resonance hybrid of the following five structures.



On the other hand, anilinium ion obtained by accepting a proton can have only two resonating structures (kekule).



We know that greater the number of resonating structures, greater is the stability. Thus you can infer that aniline (five resonating structures) is more stable than anilinium ion. Hence, the proton acceptability or the basic nature of aniline or other aromatic amines would be less than that of ammonia. In case of substituted aniline, it is observed that electron releasing groups like $-\text{OCH}_3$, $-\text{CH}_3$ increase basic strength whereas electron withdrawing groups like $-\text{NO}_2$, $-\text{SO}_3\text{H}$, $-\text{COOH}$, $-\text{X}$ decrease it.

Example 9.4

Arrange the following in decreasing order of their basic strength:

**Solution**

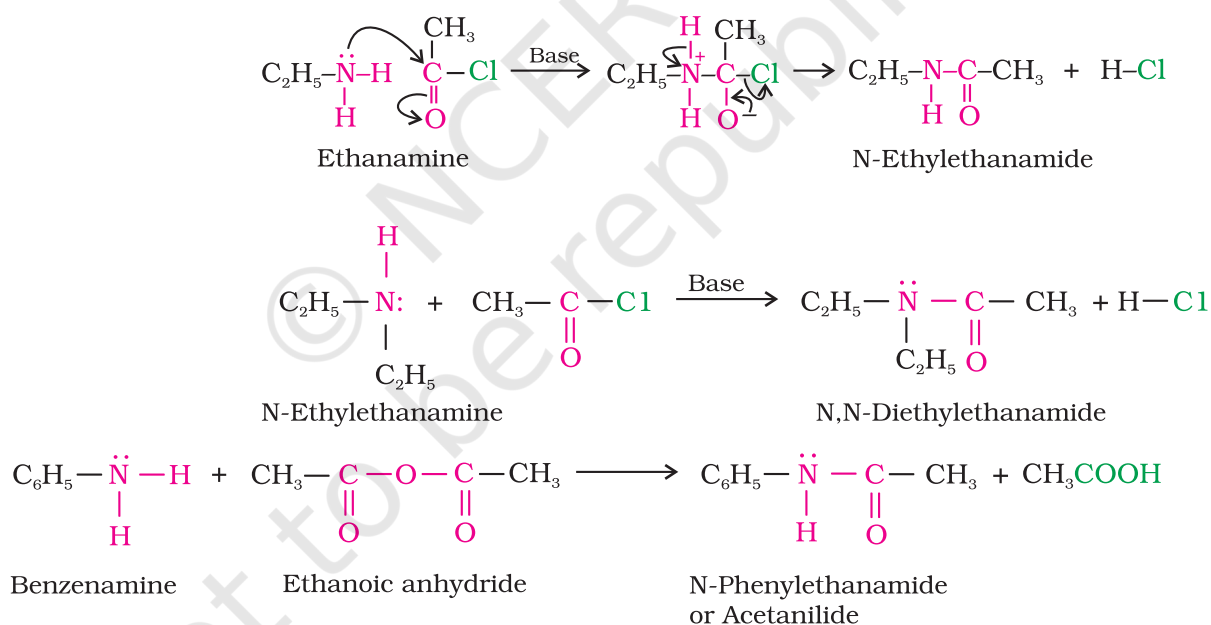
The decreasing order of basic strength of the above amines and ammonia follows the following order:

**2. Alkylation**

Amines undergo alkylation on reaction with alkyl halides (refer Unit 6, Class XII).

3. Acylation

Aliphatic and aromatic primary and secondary amines react with acid chlorides, anhydrides and esters by nucleophilic substitution reaction. This reaction is known as acylation. You can consider this reaction as the replacement of hydrogen atom of $-\text{NH}_2$ or $>\text{N}-\text{H}$ group by the acyl group. The products obtained by acylation reaction are known as amides. The reaction is carried out in the presence of a base stronger than the amine, like pyridine, which removes HCl so formed and shifts the equilibrium to the right hand side.



Amines also react with benzoyl chloride ($\text{C}_6\text{H}_5\text{COCl}$). This reaction is known as benzoylation.



What do you think is the product of the reaction of amines with carboxylic acids? They form salts with amines at room temperature.

4. Carbylamine reaction

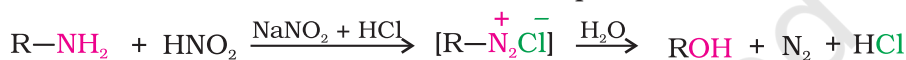
Aliphatic and aromatic primary amines on heating with chloroform and ethanolic potassium hydroxide form isocyanides or carbylamines which are foul smelling substances. Secondary and tertiary amines do not show this reaction. This reaction is known as **carbylamine reaction** or **isocyanide test** and is used as a test for primary amines.



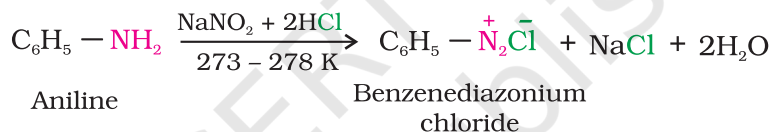
5. Reaction with nitrous acid

Three classes of amines react differently with nitrous acid which is prepared *in situ* from a mineral acid and sodium nitrite.

(a) Primary aliphatic amines react with nitrous acid to form aliphatic diazonium salts which being unstable, liberate nitrogen gas quantitatively and alcohols. Quantitative evolution of nitrogen is used in estimation of amino acids and proteins.



(b) Aromatic amines react with nitrous acid at low temperatures (273-278 K) to form diazonium salts, a very important class of compounds used for synthesis of a variety of aromatic compounds discussed in Section 9.7.

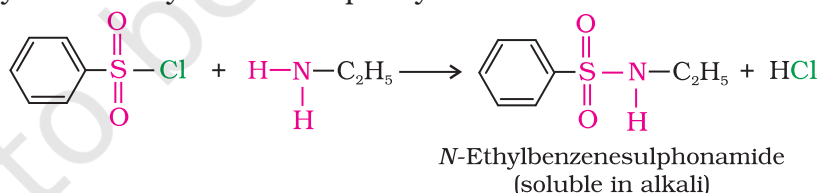


Secondary and tertiary amines react with nitrous acid in a different manner.

6. Reaction with arylsulphonyl chloride

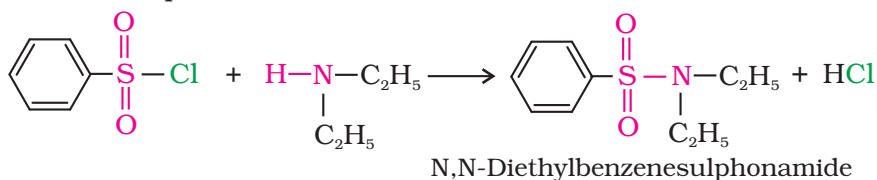
Benzenesulphonyl chloride ($\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$), which is also known as **Hinsberg's reagent**, reacts with primary and secondary amines to form sulphonamides.

(a) The reaction of benzenesulphonyl chloride with primary amine yields N-ethylbenzenesulphonyl amide.



The hydrogen attached to nitrogen in sulphonamide is strongly acidic due to the presence of strong electron withdrawing sulphonyl group. Hence, it is soluble in alkali.

(b) In the reaction with secondary amine, N,N-diethylbenzenesulphonamide is formed.



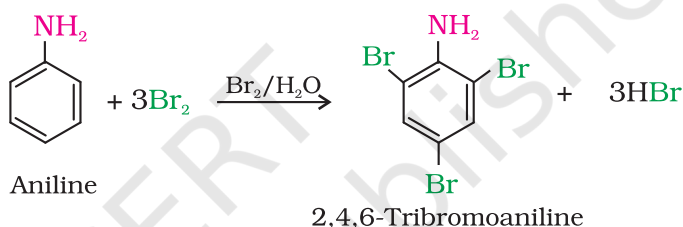
Since N, N-diethylbenzene sulphonamide does not contain any hydrogen atom attached to nitrogen atom, it is not acidic and hence insoluble in alkali.

- (c) Tertiary amines do not react with benzenesulphonyl chloride. This property of amines reacting with benzenesulphonyl chloride in a different manner is used for the distinction of primary, secondary and tertiary amines and also for the separation of a mixture of amines. However, these days benzenesulphonyl chloride is replaced by *p*-toluenesulphonyl chloride.

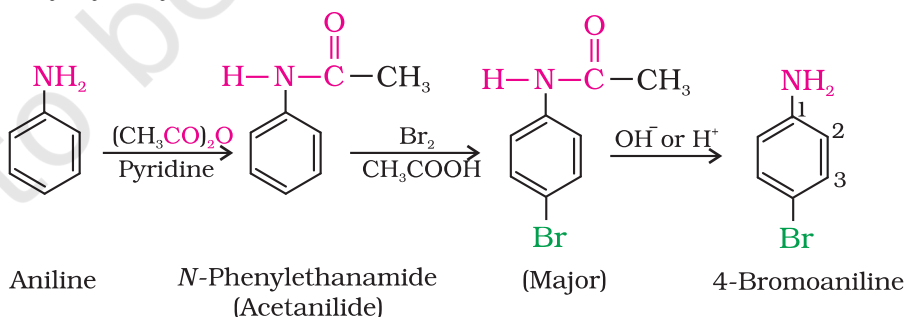
7. Electrophilic substitution

You have read earlier that aniline is a resonance hybrid of five structures. Where do you find the maximum electron density in these structures? *Ortho*- and *para*-positions to the -NH_2 group become centres of high electron density. Thus -NH_2 group is *ortho* and *para* directing and a powerful activating group.

(a) Bromination: Aniline reacts with bromine water at room temperature to give a white precipitate of 2,4,6-tribromoaniline.



The main problem encountered during electrophilic substitution reactions of aromatic amines is that of their very high reactivity. Substitution tends to occur at *ortho*- and *para*-positions. If we have to prepare monosubstituted aniline derivative, how can the activating effect of -NH_2 group be controlled? This can be done by protecting the -NH_2 group by acetylation with acetic anhydride, then carrying out the desired substitution followed by hydrolysis of the substituted amide to the substituted amine.

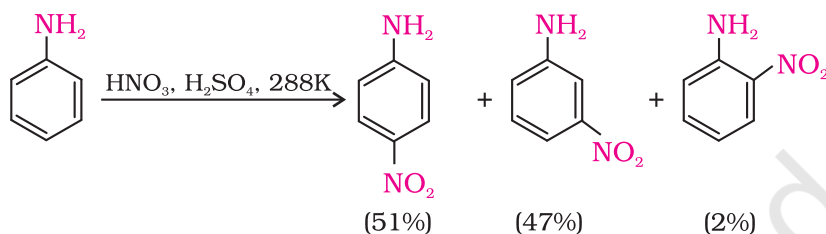


The lone pair of electrons on nitrogen of acetanilide interacts with oxygen atom due to resonance as shown below:

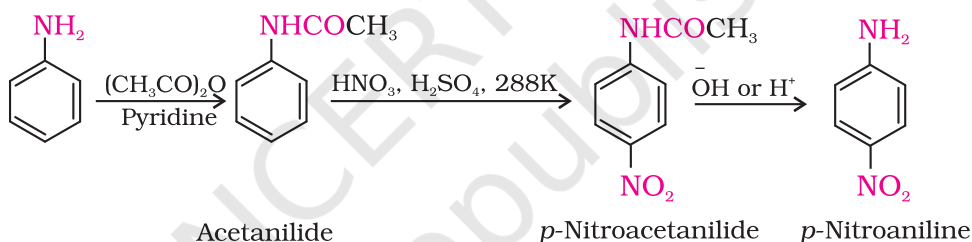


Hence, the lone pair of electrons on nitrogen is less available for donation to benzene ring by resonance. Therefore, activating effect of -NHCOCH_3 group is less than that of amino group.

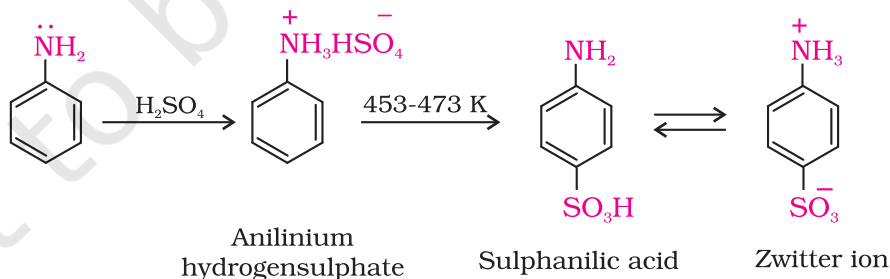
(b) Nitration: Direct nitration of aniline yields tarry oxidation products in addition to the nitro derivatives. Moreover, in the strongly acidic medium, aniline is protonated to form the anilinium ion which is *meta* directing. That is why besides the *ortho* and *para* derivatives, significant amount of *meta* derivative is also formed.



However, by protecting the -NH_2 group by acetylation reaction with acetic anhydride, the nitration reaction can be controlled and the *p*-nitro derivative can be obtained as the major product.



(c) Sulphonation: Aniline reacts with concentrated sulphuric acid to form anilinium hydrogensulphate which on heating with sulphuric acid at 453-473K produces *p*-aminobenzene sulphonic acid, commonly known as sulphanilic acid, as the major product.



Aniline does not undergo Friedel-Crafts reaction (alkylation and acetylation) due to salt formation with aluminium chloride, the Lewis acid, which is used as a catalyst. Due to this, nitrogen of aniline acquires positive charge and hence acts as a strong deactivating group for further reaction.

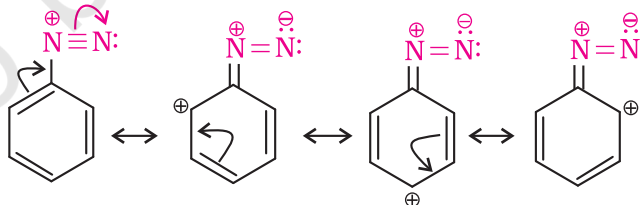
Intext Questions

- 9.4 Arrange the following in increasing order of their basic strength:
(i) $\text{C}_2\text{H}_5\text{NH}_2$, $\text{C}_6\text{H}_5\text{NH}_2$, NH_3 , $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$ and $(\text{C}_2\text{H}_5)_2\text{NH}$
(ii) $\text{C}_2\text{H}_5\text{NH}_2$, $(\text{C}_2\text{H}_5)_2\text{NH}$, $(\text{C}_2\text{H}_5)_3\text{N}$, $\text{C}_6\text{H}_5\text{NH}_2$
(iii) CH_3NH_2 , $(\text{CH}_3)_2\text{NH}$, $(\text{CH}_3)_3\text{N}$, $\text{C}_6\text{H}_5\text{NH}_2$, $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$.
- 9.5 Complete the following acid-base reactions and name the products:
(i) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2 + \text{HCl} \rightarrow$ (ii) $(\text{C}_2\text{H}_5)_3\text{N} + \text{HCl} \rightarrow$
- 9.6 Write reactions of the final alkylation product of aniline with excess of methyl iodide in the presence of sodium carbonate solution.
- 9.7 Write chemical reaction of aniline with benzoyl chloride and write the name of the product obtained.
- 9.8 Write structures of different isomers corresponding to the molecular formula, $\text{C}_3\text{H}_9\text{N}$. Write IUPAC names of the isomers which will liberate nitrogen gas on treatment with nitrous acid.

II. DIAZONIUM SALTS

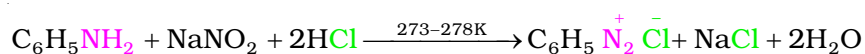
The diazonium salts have the general formula $\text{R}\text{N}_2^+\text{X}^-$ where R stands for an aryl group and X^- ion may be Cl^- , Br^- , HSO_4^- , BF_4^- , etc. They are named by suffixing diazonium to the name of the parent hydrocarbon from which they are formed, followed by the name of anion such as chloride, hydrogensulphate, etc. The N_2^+ group is called diazonium group. For example, $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$ is named as benzenediazonium chloride and $\text{C}_6\text{H}_5\text{N}_2^+\text{HSO}_4^-$ is known as benzenediazonium hydrogensulphate.

Primary aliphatic amines form highly unstable alkyldiazonium salts (refer to Section 9.6). Primary aromatic amines form arenediazonium salts which are stable for a short time in solution at low temperatures (273-278 K). The stability of arenediazonium ion is explained on the basis of resonance.



9.7 Method of Preparation of Diazonium Salts

Benzenediazonium chloride is prepared by the reaction of aniline with nitrous acid at 273-278K. Nitrous acid is produced in the reaction mixture by the reaction of sodium nitrite with hydrochloric acid. The conversion of primary aromatic amines into diazonium salts is known as **diazotisation**. Due to its instability, the diazonium salt is not generally stored and is used immediately after its preparation.



9.8 Physical Properties

Benzenediazonium chloride is a colourless crystalline solid. It is readily soluble in water and is stable in cold but reacts with water when warmed. It decomposes easily in the dry state. Benzenediazonium fluoroborate is water insoluble and stable at room temperature.

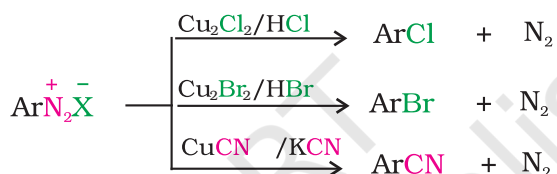
9.9 Chemical Reactions

The reactions of diazonium salts can be broadly divided into two categories, namely (A) reactions involving displacement of nitrogen and (B) reactions involving retention of diazo group.

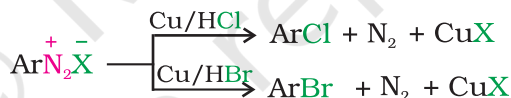
A. Reactions involving displacement of nitrogen

Diazonium group being a very good leaving group, is substituted by other groups such as Cl^- , Br^- , I^- , CN^- and OH^- which displace nitrogen from the aromatic ring. The nitrogen formed escapes from the reaction mixture as a gas.

1. *Replacement by halide or cyanide ion:* The Cl^- , Br^- and CN^- nucleophiles can easily be introduced in the benzene ring in the presence of Cu(I) ion. This reaction is called **Sandmeyer reaction**.

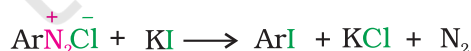


Alternatively, chlorine or bromine can also be introduced in the benzene ring by treating the diazonium salt solution with corresponding halogen acid in the presence of copper powder. This is referred as **Gatterman reaction**.



The yield in Sandmeyer reaction is found to be better than Gattermann reaction.

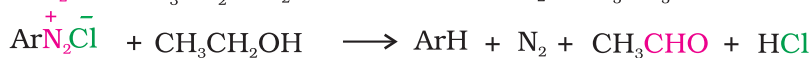
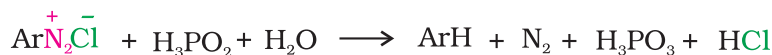
2. *Replacement by iodide ion:* Iodine is not easily introduced into the benzene ring directly, but, when the diazonium salt solution is treated with potassium iodide, iodobenzene is formed.



3. *Replacement by fluoride ion:* When arenediazonium chloride is treated with fluoroboric acid, arene diazonium fluoroborate is precipitated which on heating decomposes to yield aryl fluoride.



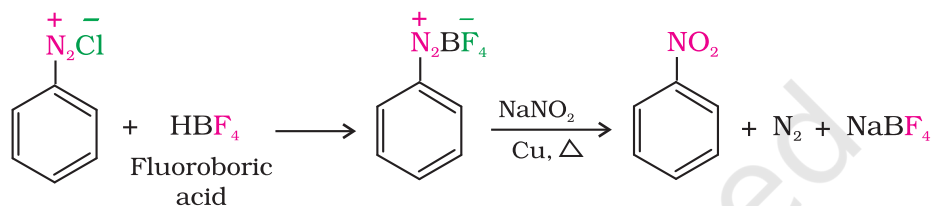
4. *Replacement by H:* Certain mild reducing agents like hypophosphorous acid (phosphinic acid) or ethanol reduce diazonium salts to arenes and themselves get oxidised to phosphorous acid and ethanal, respectively.



5. *Replacement by hydroxyl group:* If the temperature of the diazonium salt solution is allowed to rise upto 283 K, the salt gets hydrolysed to phenol.

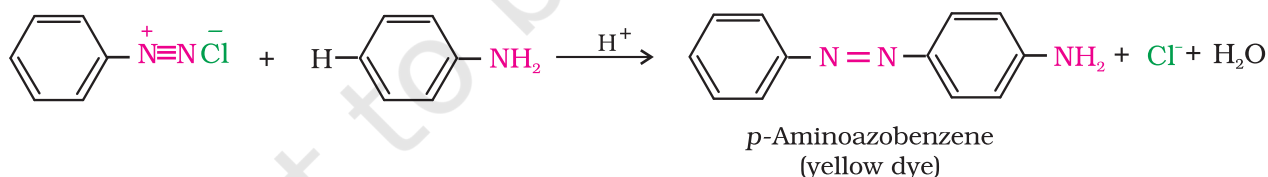
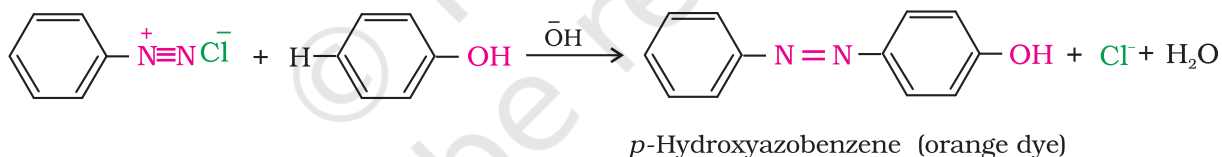


6. *Replacement by $-\text{NO}_2$ group:* When diazonium fluoroborate is heated with aqueous sodium nitrite solution in the presence of copper, the diazonium group is replaced by $-\text{NO}_2$ group.



B. Reactions involving retention of diazo group coupling reactions

The azo products obtained have an extended conjugate system having both the aromatic rings joined through the $-\text{N}=\text{N}-$ bond. These compounds are often coloured and are used as dyes. Benzene diazonium chloride reacts with phenol in which the phenol molecule at its para position is coupled with the diazonium salt to form *p*-hydroxyazobenzene. This type of reaction is known as coupling reaction. Similarly the reaction of diazonium salt with aniline yields *p*-aminoazobenzene. This is an example of electrophilic substitution reaction.



9.10 Importance of Diazonium Salts in Synthesis of Aromatic Compounds

From the above reactions, it is clear that the diazonium salts are very good intermediates for the introduction of $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, $-\text{CN}$, $-\text{OH}$, $-\text{NO}_2$ groups into the aromatic ring.

Aryl fluorides and iodides cannot be prepared by direct halogenation. The cyano group cannot be introduced by nucleophilic substitution of chlorine in chlorobenzene but cyanobenzene can be easily obtained from diazonium salt.

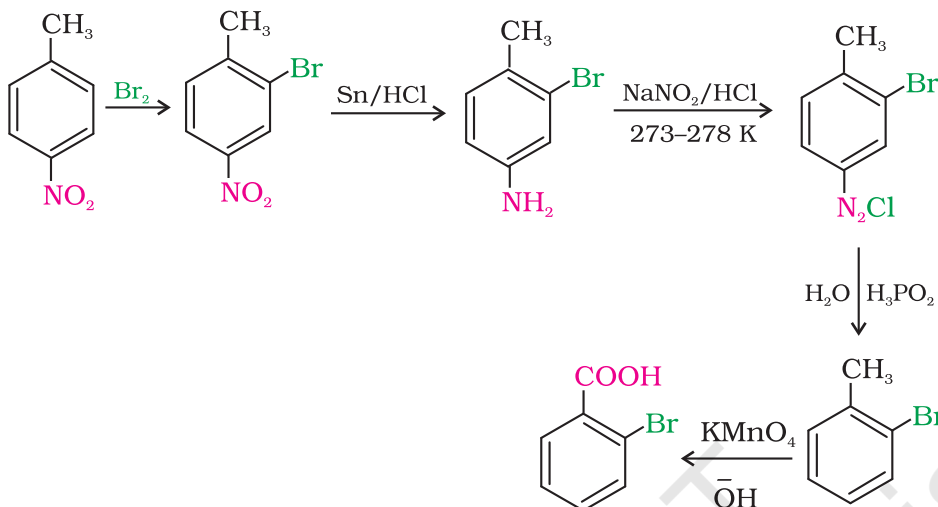
Thus, the replacement of diazo group by other groups is helpful in

preparing those substituted aromatic compounds which cannot be prepared by direct substitution in benzene or substituted benzene.

How will you convert 4-nitrotoluene to 2-bromobenzoic acid ?

Example 9.5

Solution



Intext Question

9.9 Convert

- 3-Methylaniline into 3-nitrotoluene.
- Aniline into 1,3,5-tribromobenzene.

Summary

Amines can be considered as derivatives of ammonia obtained by replacement of hydrogen atoms with alkyl or aryl groups. Replacement of one hydrogen atom of ammonia gives rise to structure of the type R-NH_2 , known as **primary amine**. **Secondary amines** are characterised by the structure R_2NH or $\text{R-NHR}'$ and **tertiary amines** by R_3N , $\text{RNR}'\text{R}''$ or $\text{R}_2\text{NR}'$. Secondary and tertiary amines are known as simple amines if the alkyl or aryl groups are the same and mixed amines if the groups are different. Like ammonia, all the three types of amines have one unshared electron pair on nitrogen atom due to which they behave as **Lewis bases**.

Amines are usually formed from nitro compounds, halides, amides, imides, etc. They exhibit hydrogen bonding which influence their physical properties. In **alkylamines**, a combination of electron releasing, steric and H-bonding factors influence the stability of the substituted ammonium cations in protic polar solvents and thus affect the basic nature of amines. Alkyl amines are found to be stronger bases than ammonia. In **aromatic amines**, electron releasing and withdrawing groups, respectively increase and decrease their basic character. **Aniline** is a weaker base

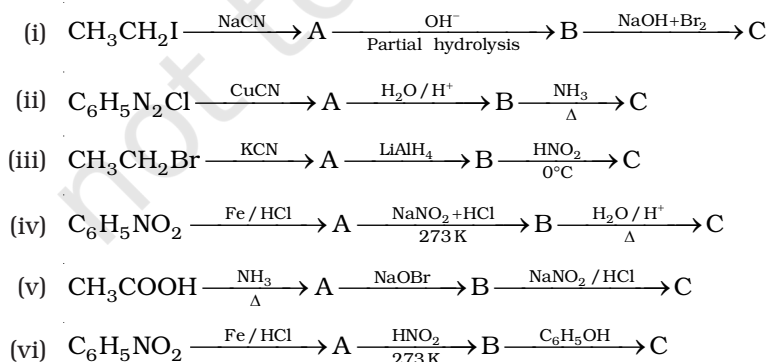
than ammonia. Reactions of amines are governed by availability of the unshared pair of electrons on nitrogen. Influence of the number of hydrogen atoms at nitrogen atom on the type of reactions and nature of products is responsible for identification and distinction between primary, secondary and tertiary amines. *p*-Toluenesulphonyl chloride is used for the identification of primary, secondary and tertiary amines. Presence of amino group in aromatic ring enhances reactivity of the aromatic amines. Reactivity of aromatic amines can be controlled by **acylation** process, i.e., by treating with acetyl chloride or acetic anhydride. Tertiary amines like **trimethylamine** are used as insect attractants.

Aryldiazonium salts, usually obtained from arylamines, undergo replacement of the diazonium group with a variety of nucleophiles to provide advantageous methods for producing aryl halides, cyanides, phenols and arenes by reductive removal of the diazo group. Coupling reaction of aryldiazonium salts with phenols or arylamines give rise to the formation of **azo dyes**.

Exercises

- 9.1** Write IUPAC names of the following compounds and classify them into primary, secondary and tertiary amines.
- (i) $(\text{CH}_3)_2\text{CHNH}_2$ (ii) $\text{CH}_3(\text{CH}_2)_2\text{NH}_2$ (iii) $\text{CH}_3\text{NHCH}(\text{CH}_3)_2$
 (iv) $(\text{CH}_3)_3\text{CNH}_2$ (v) $\text{C}_6\text{H}_5\text{NHCH}_3$ (vi) $(\text{CH}_3\text{CH}_2)_2\text{NCH}_3$
 (vii) $m\text{-BrC}_6\text{H}_4\text{NH}_2$
- 9.2** Give one chemical test to distinguish between the following pairs of compounds.
- (i) Methylamine and dimethylamine (ii) Secondary and tertiary amines
 (iii) Ethylamine and aniline (iv) Aniline and benzylamine
 (v) Aniline and N-methylaniline.
- 9.3** Account for the following:
- (i) $\text{p}K_b$ of aniline is more than that of methylamine.
 (ii) Ethylamine is soluble in water whereas aniline is not.
 (iii) Methylamine in water reacts with ferric chloride to precipitate hydrated ferric oxide.
 (iv) Although amino group is *o*- and *p*- directing in aromatic electrophilic substitution reactions, aniline on nitration gives a substantial amount of *m*-nitroaniline.
 (v) Aniline does not undergo Friedel-Crafts reaction.
 (vi) Diazonium salts of aromatic amines are more stable than those of aliphatic amines.
 (vii) Gabriel phthalimide synthesis is preferred for synthesising primary amines.
- 9.4** Arrange the following:
- (i) In decreasing order of the $\text{p}K_b$ values:
 $\text{C}_2\text{H}_5\text{NH}_2$, $\text{C}_6\text{H}_5\text{NHCH}_3$, $(\text{C}_2\text{H}_5)_2\text{NH}$ and $\text{C}_6\text{H}_5\text{NH}_2$
 (ii) In increasing order of basic strength:
 $\text{C}_6\text{H}_5\text{NH}_2$, $\text{C}_6\text{H}_5\text{N}(\text{CH}_3)_2$, $(\text{C}_2\text{H}_5)_2\text{NH}$ and CH_3NH_2
 (iii) In increasing order of basic strength:
 (a) Aniline, *p*-nitroaniline and *p*-toluidine

- (b) $\text{C}_6\text{H}_5\text{NH}_2$, $\text{C}_6\text{H}_5\text{NHCH}_3$, $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$.
- (iv) In decreasing order of basic strength in gas phase:
 $\text{C}_2\text{H}_5\text{NH}_2$, $(\text{C}_2\text{H}_5)_2\text{NH}$, $(\text{C}_2\text{H}_5)_3\text{N}$ and NH_3
- (v) In increasing order of boiling point:
 $\text{C}_2\text{H}_5\text{OH}$, $(\text{CH}_3)_2\text{NH}$, $\text{C}_2\text{H}_5\text{NH}_2$
- (vi) In increasing order of solubility in water:
 $\text{C}_6\text{H}_5\text{NH}_2$, $(\text{C}_2\text{H}_5)_2\text{NH}$, $\text{C}_2\text{H}_5\text{NH}_2$.
- 9.5** How will you convert:
- Ethanoic acid into methanamine
 - Hexanenitrile into 1-aminopentane
 - Methanol to ethanoic acid
 - Ethanamine into methanamine
 - Ethanoic acid into propanoic acid
 - Methanamine into ethanamine
 - Nitromethane into dimethylamine
 - Propanoic acid into ethanoic acid?
- 9.6** Describe a method for the identification of primary, secondary and tertiary amines. Also write chemical equations of the reactions involved.
- 9.7** Write short notes on the following:
- Carbylamine reaction
 - Diazotisation
 - Hofmann's bromamide reaction
 - Coupling reaction
 - Ammonolysis
 - Acetylation
 - Gabriel phthalimide synthesis.
- 9.8** Accomplish the following conversions:
- Nitrobenzene to benzoic acid
 - Benzene to *m*-bromophenol
 - Benzoic acid to aniline
 - Aniline to 2,4,6-tribromofluorobenzene
 - Benzyl chloride to 2-phenylethanamine
 - Chlorobenzene to *p*-chloroaniline
 - Aniline to *p*-bromoaniline
 - Benzamide to toluene
 - Aniline to benzyl alcohol.
- 9.9** Give the structures of A, B and C in the following reactions:



- 9.10** An aromatic compound 'A' on treatment with aqueous ammonia and heating forms compound 'B' which on heating with Br_2 and KOH forms a compound 'C' of molecular formula $\text{C}_6\text{H}_7\text{N}$. Write the structures and IUPAC names of compounds A, B and C.
- 9.11** Complete the following reactions:
- $\text{C}_6\text{H}_5\text{NH}_2 + \text{CHCl}_3 + \text{alc.KOH} \rightarrow$
 - $\text{C}_6\text{H}_5\text{N}_2\text{Cl} + \text{H}_3\text{PO}_2 + \text{H}_2\text{O} \rightarrow$
 - $\text{C}_6\text{H}_5\text{NH}_2 + \text{H}_2\text{SO}_4 (\text{conc.}) \rightarrow$
 - $\text{C}_6\text{H}_5\text{N}_2\text{Cl} + \text{C}_2\text{H}_5\text{OH} \rightarrow$
 - $\text{C}_6\text{H}_5\text{NH}_2 + \text{Br}_2 (\text{aq}) \rightarrow$
 - $\text{C}_6\text{H}_5\text{NH}_2 + (\text{CH}_3\text{CO})_2\text{O} \rightarrow$
 - $\text{C}_6\text{H}_5\text{N}_2\text{Cl} \xrightarrow[\text{(ii) NaNO}_2 / \text{Cu}, \Delta]{\text{(i) HBF}_4}$
- 9.12** Why cannot aromatic primary amines be prepared by Gabriel phthalimide synthesis?
- 9.13** Write the reactions of (i) aromatic and (ii) aliphatic primary amines with nitrous acid.
- 9.14** Give plausible explanation for each of the following:
- Why are amines less acidic than alcohols of comparable molecular masses?
 - Why do primary amines have higher boiling point than tertiary amines?
 - Why are aliphatic amines stronger bases than aromatic amines?

Answers to Some Intext Questions

- 9.4**
- $\text{C}_6\text{H}_5\text{NH}_2 < \text{NH}_3 < \text{C}_6\text{H}_5\text{CH}_2\text{NH}_2 < \text{C}_2\text{H}_5\text{NH}_2 < (\text{C}_2\text{H}_5)_2\text{NH}$
 - $\text{C}_6\text{H}_5\text{NH}_2 < \text{C}_2\text{H}_5\text{NH}_2 < (\text{C}_2\text{H}_5)_3\text{N} < (\text{C}_2\text{H}_5)_2\text{NH}$
 - $\text{C}_6\text{H}_5\text{NH}_2 < \text{C}_6\text{H}_5\text{CH}_2\text{NH}_2 < (\text{CH}_3)_3\text{N} < \text{CH}_3\text{NH}_2 < (\text{CH}_3)_2\text{NH}$



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Unit 10 Biomolecules

Objectives

After studying this Unit, you will be able to

- explain the characteristics of biomolecules like carbohydrates, proteins and nucleic acids and hormones;
- classify carbohydrates, proteins, nucleic acids and vitamins on the basis of their structures;
- explain the difference between DNA and RNA;
- describe the role of biomolecules in biosystem.

"It is the harmonious and synchronous progress of chemical reactions in body which leads to life".

A living system grows, sustains and reproduces itself. The most amazing thing about a living system is that it is composed of non-living atoms and molecules. The pursuit of knowledge of what goes on chemically within a living system falls in the domain of **biochemistry**. Living systems are made up of various complex biomolecules like carbohydrates, proteins, nucleic acids, lipids, etc. Proteins and carbohydrates are essential constituents of our food. These biomolecules interact with each other and constitute the molecular logic of life processes. In addition, some simple molecules like vitamins and mineral salts also play an important role in the functions of organisms. Structures and functions of some of these biomolecules are discussed in this Unit.

10.1 Carbohydrates

Carbohydrates are primarily produced by plants and form a very large group of naturally occurring organic compounds. Some common examples of carbohydrates are cane sugar, glucose, starch, etc. Most of them have a general formula, $C_x(H_2O)_y$, and were considered as hydrates of carbon from where the name carbohydrate was derived. For example, the molecular formula of glucose ($C_6H_{12}O_6$) fits into this general formula, $C_6(H_2O)_6$. But all the compounds which fit into this formula may not be classified as carbohydrates. For example acetic acid (CH_3COOH) fits into this general formula, $C_2(H_2O)_2$ but is not a carbohydrate. Similarly, rhamnose, $C_6H_{12}O_5$ is a carbohydrate but does not fit in this definition. A large number of their reactions have shown that they contain specific functional groups. Chemically, *the carbohydrates may be defined as optically active polyhydroxy aldehydes or ketones or the compounds which produce such units on hydrolysis*. Some of the carbohydrates, which are sweet in taste, are also called sugars. The most common sugar, used in our homes is named as sucrose whereas the sugar present

in milk is known as lactose. Carbohydrates are also called saccharides (Greek: *sakcharon* means sugar).

Carbohydrates are classified on the basis of their behaviour on hydrolysis. They have been broadly divided into following three groups.

10.1.1 Classification of Carbohydrates

- (i) **Monosaccharides:** A carbohydrate that cannot be hydrolysed further to give simpler unit of polyhydroxy aldehyde or ketone is called a monosaccharide. About 20 monosaccharides are known to occur in nature. Some common examples are glucose, fructose, ribose, etc.
- (ii) **Oligosaccharides:** Carbohydrates that yield two to ten monosaccharide units, on hydrolysis, are called oligosaccharides. They are further classified as disaccharides, trisaccharides, tetrasaccharides, etc., depending upon the number of monosaccharides, they provide on hydrolysis. Amongst these the most common are disaccharides. The two monosaccharide units obtained on hydrolysis of a disaccharide may be same or different. For example, one molecule of sucrose on hydrolysis gives one molecule of glucose and one molecule of fructose whereas maltose gives two molecules of only glucose.
- (iii) **Polysaccharides:** Carbohydrates which yield a large number of monosaccharide units on hydrolysis are called polysaccharides. Some common examples are starch, cellulose, glycogen, gums, etc. Polysaccharides are not sweet in taste, hence they are also called non-sugars.

The carbohydrates may also be classified as either reducing or non-reducing sugars. All those carbohydrates which reduce Fehling's solution and Tollens' reagent are referred to as reducing sugars. All monosaccharides whether aldose or ketose are *reducing sugars*.

10.1.2 Monosaccharides

Monosaccharides are further classified on the basis of number of carbon atoms and the functional group present in them. If a monosaccharide contains an aldehyde group, it is known as an aldose and if it contains a keto group, it is known as a ketose. Number of carbon atoms constituting the monosaccharide is also introduced in the name as is evident from the examples given in Table 10.1

Table 10.1: Different Types of Monosaccharides

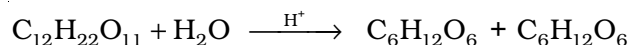
Carbon atoms	General term	Aldehyde	Ketone
3	Triose	Aldotriose	Ketotriose
4	Tetrose	Aldotetrose	Ketotetrose
5	Pentose	Aldopentose	Ketopentose
6	Hexose	Aldohexose	Ketohexose
7	Heptose	Aldoheptose	Ketoheptose

10.1.2.1 Glucose

Glucose occurs freely in nature as well as in the combined form. It is present in sweet fruits and honey. Ripe grapes also contain glucose in large amounts. It is prepared as follows:

Preparation of Glucose

1. **From sucrose (Cane sugar):** If sucrose is boiled with dilute HCl or H_2SO_4 in alcoholic solution, glucose and fructose are obtained in equal amounts.

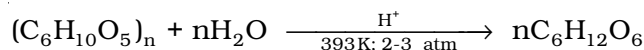


Sucrose

Glucose

Fructose

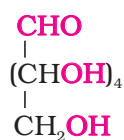
2. *From starch:* Commercially glucose is obtained by hydrolysis of starch by boiling it with dilute H_2SO_4 at 393 K under pressure.



Starch or cellulose

Glucose

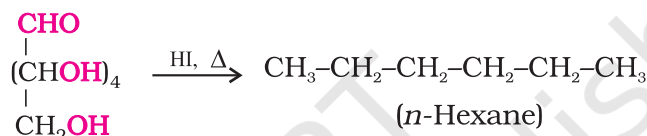
Structure of Glucose



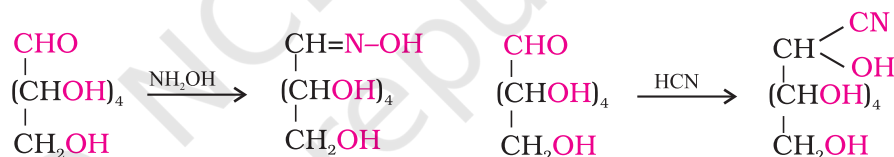
Glucose

Glucose is an aldohexose and is also known as dextrose. It is the monomer of many of the larger carbohydrates, namely starch, cellulose. It is probably the most abundant organic compound on earth. It was assigned the structure given below on the basis of the following evidences:

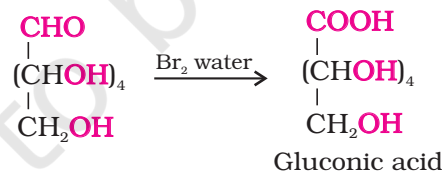
1. Its molecular formula was found to be $\text{C}_6\text{H}_{12}\text{O}_6$.
2. On prolonged heating with HI, it forms n-hexane, suggesting that all the six carbon atoms are linked in a straight chain.



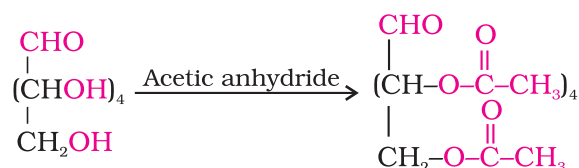
3. Glucose reacts with hydroxylamine to form an oxime and adds a molecule of hydrogen cyanide to give cyanohydrin. These reactions confirm the presence of a carbonyl group ($>\text{C}=\text{O}$) in glucose.



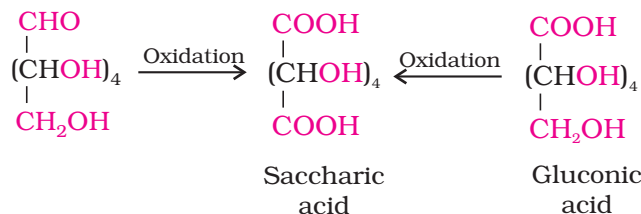
4. Glucose gets oxidised to six carbon carboxylic acid (gluconic acid) on reaction with a mild oxidising agent like bromine water. This indicates that the carbonyl group is present as an aldehydic group.



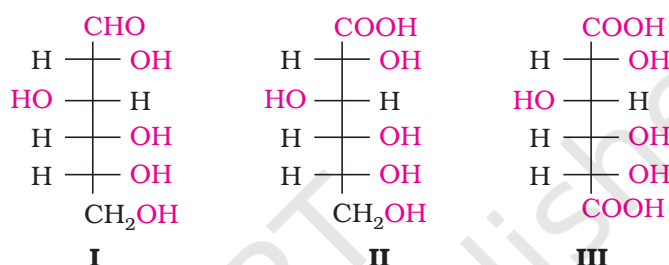
5. Acetylation of glucose with acetic anhydride gives glucose pentaacetate which confirms the presence of five $-\text{OH}$ groups. Since it exists as a stable compound, five $-\text{OH}$ groups should be attached to different carbon atoms.



6. On oxidation with nitric acid, glucose as well as gluconic acid both yield a dicarboxylic acid, saccharic acid. This indicates the presence of a primary alcoholic (–OH) group in glucose.



The exact spatial arrangement of different –OH groups was given by Fischer after studying many other properties. Its configuration is correctly represented as **I**. So gluconic acid is represented as **II** and saccharic acid as **III**.



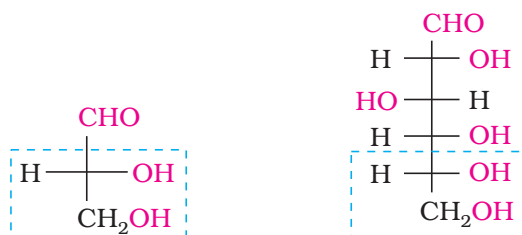
Glucose is correctly named as D(+)-glucose. 'D' before the name of glucose represents the configuration whereas '(+)' represents dextrorotatory nature of the molecule. It should be remembered that 'D' and 'L' have no relation with the optical activity of the compound. They are also not related to letter 'd' and 'l' (see Unit 6). The meaning of D- and L- notations is as follows.

The letters 'D' or 'L' before the name of any compound indicate the relative configuration of a particular stereoisomer of a compound with respect to configuration of some other compound, configuration of which is known. In the case of carbohydrates, this refers to their relation with a particular isomer of glyceraldehyde. Glyceraldehyde contains one asymmetric carbon atom and exists in two enantiomeric forms as shown below.



(+) Isomer of glyceraldehyde has 'D' configuration. It means that when its structural formula is written on paper following specific conventions which you will study in higher classes, the –OH group lies on right hand side in the structure. All those compounds which can be chemically correlated to D (+) isomer of glyceraldehyde are said to have D-configuration whereas those which can be correlated to 'L' (–) isomer of glyceraldehyde are said to have L-configuration. In L (–) isomer –OH group is on left hand side as you can see in the structure. For assigning

the configuration of monosaccharides, it is the lowest asymmetric carbon atom (as shown below) which is compared. As in (+) glucose, —OH on the lowest asymmetric carbon is on the right side which is comparable to (+) glyceraldehyde, so (+) glucose is assigned D-configuration. Other asymmetric carbon atoms of glucose are not considered for this comparison. Also, the structure of glucose and glyceraldehyde is written in a way that most oxidised carbon (in this case —CHO) is at the top.



D- (+) – Glyceraldehyde

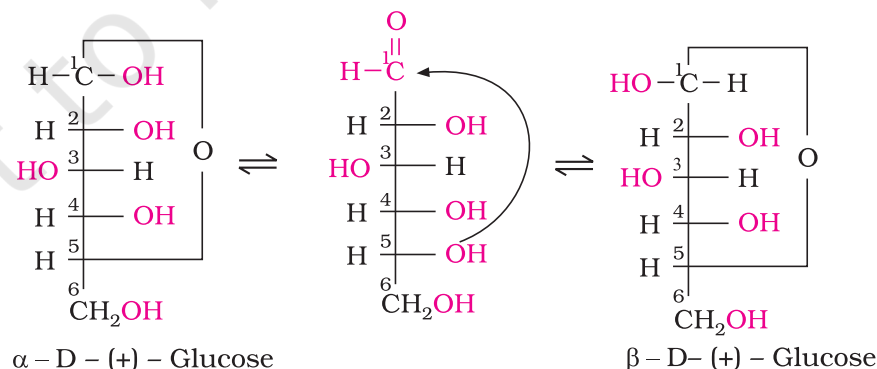
D-(+) – Glucose

Cyclic Structure of Glucose

The structure (I) of glucose explained most of its properties but the following reactions and facts could not be explained by this structure.

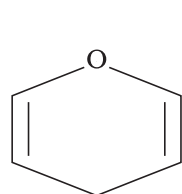
1. Despite having the aldehyde group, glucose does not give Schiff's test and it does not form the hydrogensulphite addition product with NaHSO_3 .
2. The pentaacetate of glucose does not react with hydroxylamine indicating the absence of free —CHO group.
3. Glucose is found to exist in two different crystalline forms which are named as α and β . The α -form of glucose (m.p. 419 K) is obtained by crystallisation from concentrated solution of glucose at 303 K while the β -form (m.p. 423 K) is obtained by crystallisation from hot and saturated aqueous solution at 371 K.

This behaviour could not be explained by the open chain structure (I) for glucose. It was proposed that one of the —OH groups may add to the —CHO group and form a cyclic hemiacetal structure. It was found that glucose forms a six-membered ring in which —OH at C-5 is involved in ring formation. This explains the absence of —CHO group and also existence of glucose in two forms as shown below. These two cyclic forms exist in equilibrium with open chain structure.

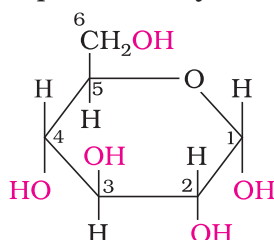


The two cyclic hemiacetal forms of glucose differ only in the configuration of the hydroxyl group at C1, called *anomeric carbon*

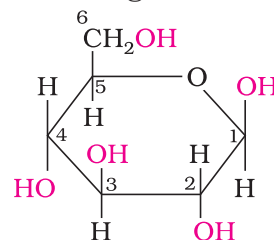
(the aldehyde carbon before cyclisation). Such isomers, i.e., α -form and β -form, are called **anomers**. The six membered cyclic structure of glucose is called **pyranose structure** (α - or β -), in analogy with pyran. Pyran is a cyclic organic compound with one oxygen atom and five carbon atoms in the ring. The cyclic structure of glucose is more correctly represented by Haworth structure as given below.



Pyran



α -D-(+)-Glucopyranose



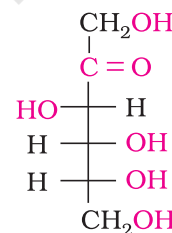
β -D-(+)-Glucopyranose

10.1.2.2 Fructose

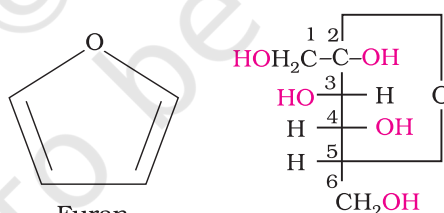
Fructose is an important ketohexose. It is obtained along with glucose by the hydrolysis of disaccharide, sucrose. It is a natural monosaccharide found in fruits, honey and vegetables. In its pure form it is used as a sweetener. It is also an important ketohexose.

Structure of Fructose

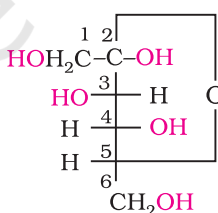
Fructose also has the molecular formula $C_6H_{12}O_6$ and on the basis of its reactions it was found to contain a ketonic functional group at carbon number 2 and six carbons in straight chain as in the case of glucose. It belongs to D-series and is a laevorotatory compound. It is appropriately written as D-(-)-fructose. Its open chain structure is as shown.



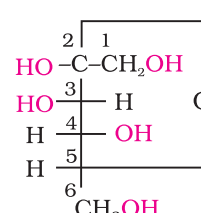
It also exists in two cyclic forms which are obtained by the addition of $-OH$ at C5 to the $(>C=O)$ group. The ring, thus formed is a five membered ring and is named as furanose with analogy to the compound furan. Furan is a five membered cyclic compound with one oxygen and four carbon atoms.



Furan

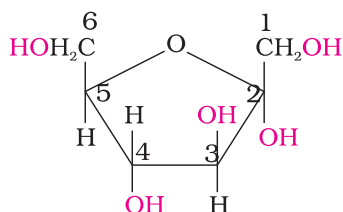


α -D-(-)-Fructofuranose

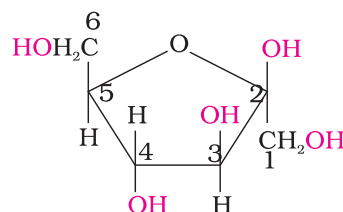


β -D-(-)-Fructofuranose

The cyclic structures of two anomers of fructose are represented by Haworth structures as given.



α -D-(-)-Fructofuranose



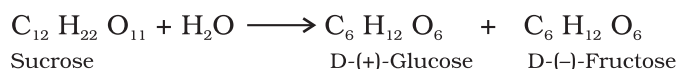
β -D-(-)-Fructofuranose

10.1.3 Disaccharides

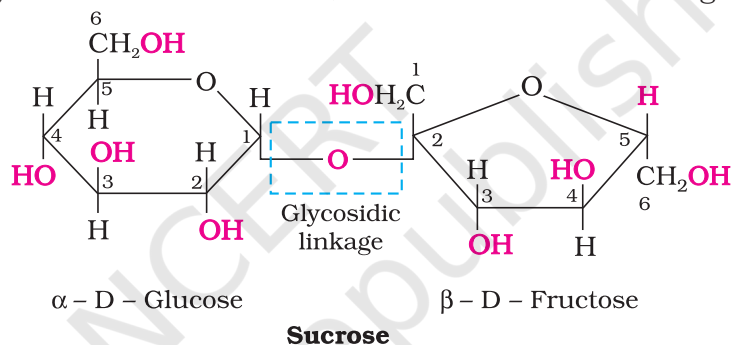
You have already read that disaccharides on hydrolysis with dilute acids or enzymes yield two molecules of either the same or different monosaccharides. The two monosaccharides are joined together by an oxide linkage formed by the loss of a water molecule. Such a linkage between two monosaccharide units through oxygen atom is called *glycosidic linkage*.

In disaccharides, if the reducing groups of monosaccharides i.e., aldehydic or ketonic groups are bonded, these are non-reducing sugars, e.g., sucrose. On the other hand, sugars in which these functional groups are free, are called reducing sugars, for example, maltose and lactose.

- (i) **Sucrose:** One of the common disaccharides is **sucrose** which on hydrolysis gives equimolar mixture of D-(+)-glucose and D-(-) fructose.

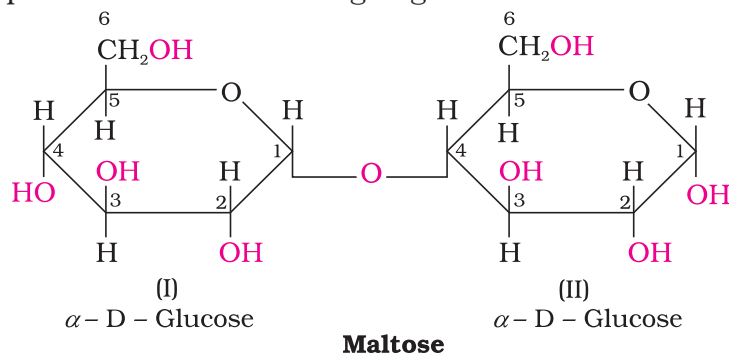


These two monosaccharides are held together by a glycosidic linkage between C1 of α -D-glucose and C2 of β -D-fructose. Since the reducing groups of glucose and fructose are involved in glycosidic bond formation, sucrose is a non reducing sugar.

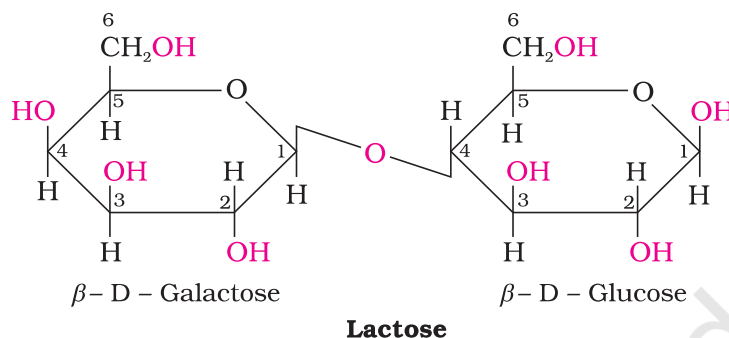


Sucrose is dextrorotatory but after hydrolysis gives dextrorotatory glucose and laevorotatory fructose. Since the laevorotation of fructose (-92.4°) is more than dextrorotation of glucose ($+52.5^\circ$), the mixture is laevorotatory. Thus, hydrolysis of sucrose brings about a change in the sign of rotation, from dextro (+) to laevo (-) and the product is named as **invert sugar**.

- (ii) **Maltose:** Another disaccharide, maltose is composed of two α -D-glucose units in which C1 of one glucose (I) is linked to C4 of another glucose unit (II). The free aldehyde group can be produced at C1 of second glucose in solution and it shows reducing properties so it is a reducing sugar.



(iii) **Lactose:** It is more commonly known as milk sugar since this disaccharide is found in milk. It is composed of β -D-galactose and β -D-glucose. The linkage is between C1 of galactose and C4 of glucose. Free aldehyde group may be produced at C-1 of glucose unit, hence it is also a reducing sugar.

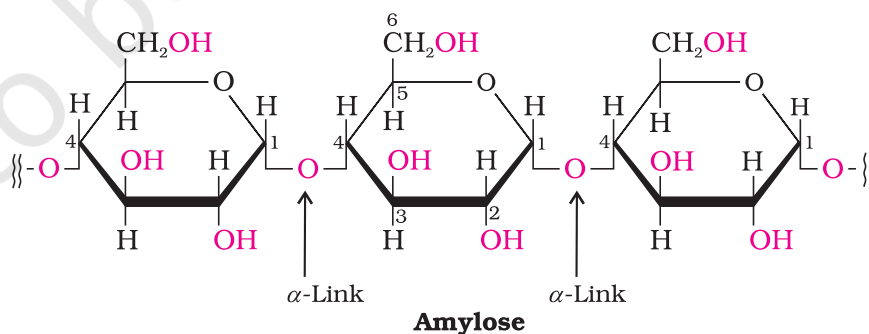


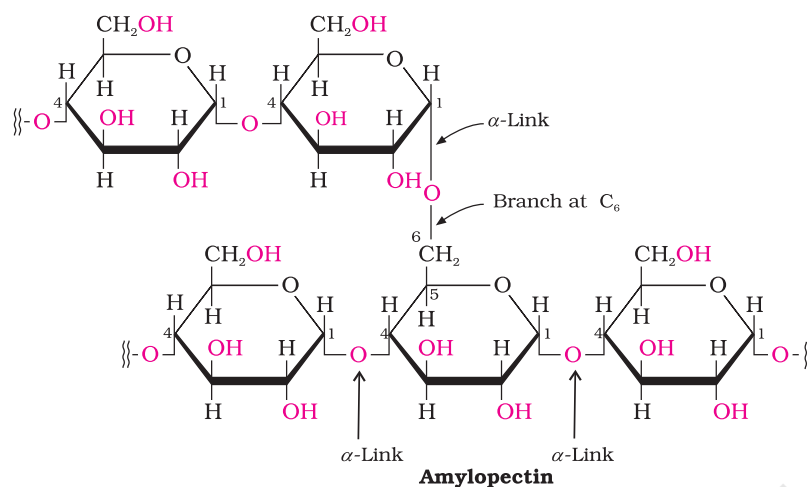
10.1.4 Polysaccharides

Polysaccharides contain a large number of monosaccharide units joined together by glycosidic linkages. These are the most commonly encountered carbohydrates in nature. They mainly act as the food storage or structural materials.

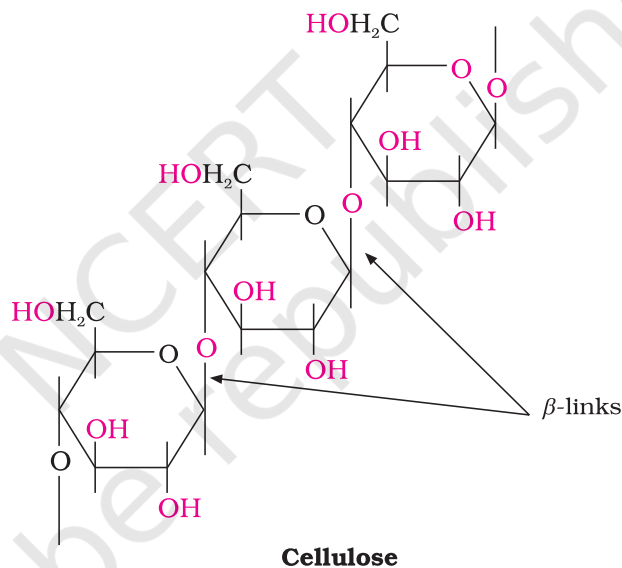
(i) **Starch:** Starch is the main storage polysaccharide of plants. It is the most important dietary source for human beings. High content of starch is found in cereals, roots, tubers and some vegetables. It is a polymer of α -glucose and consists of two components—**Amylose** and **Amylopectin**. Amylose is water soluble component which constitutes about 15-20% of starch. Chemically amylose is a long unbranched chain with 200-1000 α -D-(+)-glucose units held together by C1- C4 glycosidic linkage.

Amylopectin is insoluble in water and constitutes about 80-85% of starch. It is a branched chain polymer of α -D-glucose units in which chain is formed by C1-C4 glycosidic linkage whereas branching occurs by C1-C6 glycosidic linkage.





- (ii) **Cellulose:** Cellulose occurs exclusively in plants and it is the most abundant organic substance in plant kingdom. It is a predominant constituent of cell wall of plant cells. Cellulose is a straight chain



polysaccharide composed only of β -D-glucose units which are joined by glycosidic linkage between C1 of one glucose unit and C4 of the next glucose unit.

- (iii) **Glycogen:** The carbohydrates are stored in animal body as glycogen. It is also known as *animal starch* because its structure is similar to amylopectin and is rather more highly branched. It is present in liver, muscles and brain. When the body needs glucose, enzymes break the glycogen down to glucose. Glycogen is also found in yeast and fungi.

10.1.5 Importance of Carbohydrates

Carbohydrates are essential for life in both plants and animals. They form a major portion of our food. Honey has been used for a long time as an instant source of energy by '**Vaids**' in ayurvedic system of medicine. Carbohydrates are used as storage molecules as starch in plants and **glycogen** in animals. Cell wall of bacteria and plants is made up of cellulose. We build furniture, etc. from cellulose in the form

of wood and clothe ourselves with cellulose in the form of cotton fibre. They provide raw materials for many important industries like textiles, paper, lacquers and breweries.

Two aldopentoses viz. D-ribose and 2-deoxy-D-ribose (Section 10.5.1, Class XII) are present in nucleic acids. Carbohydrates are found in biosystem in combination with many proteins and lipids.

Intext Questions

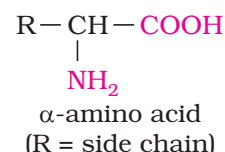
- 10.1** Glucose or sucrose are soluble in water but cyclohexane or benzene (simple six membered ring compounds) are insoluble in water. Explain.
- 10.2** What are the expected products of hydrolysis of lactose?
- 10.3** How do you explain the absence of aldehyde group in the pentaacetate of D-glucose?

10.2 Proteins

Proteins are the most abundant biomolecules of the living system. Chief sources of proteins are milk, cheese, pulses, peanuts, fish, meat, etc. They occur in every part of the body and form the fundamental basis of structure and functions of life. They are also required for growth and maintenance of body. The word protein is derived from Greek word, "**proteios**" which means primary or of prime importance. All proteins are polymers of α -amino acids.

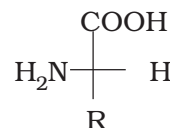
10.2.1 Amino Acids

Amino acids contain amino ($-\text{NH}_2$) and carboxyl ($-\text{COOH}$) functional groups. Depending upon the relative position of amino group with respect to carboxyl group, the amino acids can be classified as α , β , γ , δ and so on. Only α -amino acids are obtained on hydrolysis of proteins. They may contain other functional groups also.

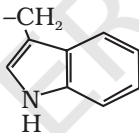
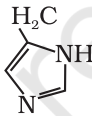
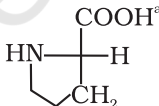


All α -amino acids have trivial names, which usually reflect the property of that compound or its source. Glycine is so named since it has sweet taste (in Greek *glykos* means sweet) and tyrosine was first obtained from cheese (in Greek, *tyros* means cheese.) Amino acids are generally represented by a three letter symbol, sometimes one letter symbol is also used. Structures of some commonly occurring amino acids along with their 3-letter and 1-letter symbols are given in Table 10.2.

Table 10.2: Natural Amino Acids



Name of the amino acids	Characteristic feature of side chain, R	Three letter symbol	One letter code
1. Glycine	H	Gly	G
2. Alanine	$-\text{CH}_3$	Ala	A
3. Valine*	$(\text{H}_3\text{C})_2\text{CH}-$	Val	V
4. Leucine*	$(\text{H}_3\text{C})_2\text{CH}-\text{CH}_2-$	Leu	L

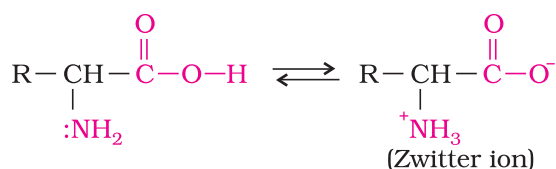
5. Isoleucine*	$\begin{array}{c} \text{H}_3\text{C}-\text{CH}_2-\text{CH}- \\ \\ \text{CH}_3 \end{array}$	Ile	I
6. Arginine*	$\begin{array}{c} \text{HN}=\text{C}-\text{NH}-(\text{CH}_2)_3- \\ \\ \text{NH}_2 \end{array}$	Arg	R
7. Lysine*	$\text{H}_2\text{N}-(\text{CH}_2)_4-$	Lys	K
8. Glutamic acid	$\text{HOOC}-\text{CH}_2-\text{CH}_2-$	Glu	E
9. Aspartic acid	$\text{HOOC}-\text{CH}_2-$	Asp	D
10. Glutamine	$\begin{array}{c} \text{O} \\ \\ \text{H}_2\text{N}-\text{C}-\text{CH}_2-\text{CH}_2- \end{array}$	Gln	Q
11. Asparagine	$\begin{array}{c} \text{O} \\ \\ \text{H}_2\text{N}-\text{C}-\text{CH}_2- \end{array}$	Asn	N
12. Threonine*	$\text{H}_3\text{C}-\text{CHOH}-$	Thr	T
13. Serine	$\text{HO}-\text{CH}_2-$	Ser	S
14. Cysteine	$\text{HS}-\text{CH}_2-$	Cys	C
15. Methionine*	$\text{H}_3\text{C}-\text{S}-\text{CH}_2-\text{CH}_2-$	Met	M
16. Phenylalanine*	$\text{C}_6\text{H}_5-\text{CH}_2-$	Phe	F
17. Tyrosine	$(p)\text{HO}-\text{C}_6\text{H}_4-\text{CH}_2-$	Tyr	Y
18. Tryptophan*		Trp	W
19. Histidine*		His	H
20. Proline		Pro	P

* essential amino acid, a = entire structure

10.2.2 Classification of Amino Acids

Amino acids are classified as acidic, basic or neutral depending upon the relative number of amino and carboxyl groups in their molecule. Equal number of amino and carboxyl groups makes it neutral; more number of amino than carboxyl groups makes it basic and more carboxyl groups as compared to amino groups makes it acidic. The amino acids, which can be synthesised in the body, are known as **non-essential amino acids**. On the other hand, those which cannot be synthesised in the body and must be obtained through diet, are known as **essential amino acids** (marked with asterisk in Table 10.2).

Amino acids are usually colourless, crystalline solids. These are water-soluble, high melting solids and behave like salts rather than simple amines or carboxylic acids. This behaviour is due to the presence of both acidic (carboxyl group) and basic (amino group) groups in the same molecule. In aqueous solution, the carboxyl group can lose a proton and amino group can accept a proton, giving rise to a dipolar ion known as *zwitter ion*. This is neutral but contains both positive and negative charges.

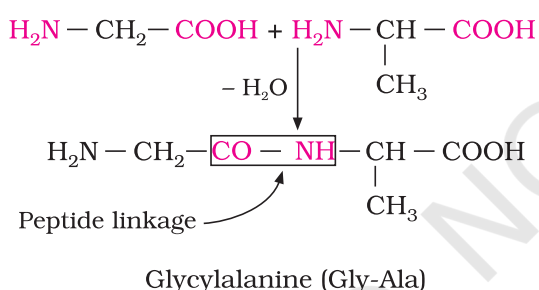


In zwitter ionic form, amino acids show amphoteric behaviour as they react both with acids and bases.

Except glycine, all other naturally occurring α -amino acids are optically active, since the α -carbon atom is asymmetric. These exist both in 'D' and 'L' forms. Most naturally occurring amino acids have L-configuration. L-Aminoacids are represented by writing the $-\text{NH}_2$ group on left hand side.

10.2.3 Structure of Proteins

You have already read that proteins are the polymers of α -amino acids and they are connected to each other by **peptide bond** or **peptide linkage**. Chemically, peptide linkage is an amide formed between $-\text{COOH}$ group and $-\text{NH}_2$ group. The reaction between two molecules of similar or different amino acids, proceeds through the combination of the amino group of one molecule with the carboxyl group of the other. This results in the elimination of a water molecule and formation of a peptide bond $-\text{CO}-\text{NH}-$. The product of the reaction is called a dipeptide because it is made up of two amino acids. For example, when carboxyl group of glycine combines with the amino group of alanine we get a **dipeptide**, glycylalanine.



If a third amino acid combines to a dipeptide, the product is called a **tripeptide**. A tripeptide contains three amino acids linked by two peptide linkages. Similarly when four, five or six amino acids are linked, the respective products are known as **tetrapeptide**, **pentapeptide** or **hexapeptide**, respectively. When the number of such amino acids is more than ten, then the products are called **polypeptides**. A polypeptide with more than hundred amino acid residues, having molecular mass higher than 10,000u is called a protein. However, the distinction between a polypeptide and a protein is not very sharp. Polypeptides with fewer amino acids are likely to be called proteins if they ordinarily have a well defined conformation of a protein such as insulin which contains 51 amino acids.

Proteins can be classified into two types on the basis of their molecular shape.

(a) Fibrous proteins

When the polypeptide chains run parallel and are held together by hydrogen and disulphide bonds, then fibre-like structure is formed. Such proteins are generally insoluble in water. Some common examples are keratin (present in hair, wool, silk) and myosin (present in muscles), etc.

(b) Globular proteins

This structure results when the chains of polypeptides coil around to give a spherical shape. These are usually soluble in water. Insulin and albumins are the common examples of globular proteins.

Structure and shape of proteins can be studied at four different levels, i.e., primary, secondary, tertiary and quaternary, each level being more complex than the previous one.

(i) *Primary structure of proteins:* Proteins may have one or more polypeptide chains. Each polypeptide in a protein has amino acids linked with each other in a specific sequence and it is this sequence of amino acids that is said to be the primary structure of that protein. Any change in this primary structure i.e., the sequence of amino acids creates a different protein.

(ii) *Secondary structure of proteins:* The secondary structure of protein refers to the shape in which a long polypeptide chain can exist. They are found to exist in two different types of structures viz. α -helix and β -pleated sheet structure. These structures arise due to the regular folding of the backbone of the polypeptide chain due to hydrogen bonding between $\text{C}=\text{O}$ and $-\text{NH}-$ groups of the peptide bond.

α -Helix is one of the most common ways in which a polypeptide chain forms all possible hydrogen bonds by twisting into a right handed screw (helix) with the $-\text{NH}$ group of each amino acid residue hydrogen bonded to the $\text{C}=\text{O}$ of an adjacent turn of the helix as shown in Fig.10.1.

In β -pleated sheet structure all peptide chains are stretched out to nearly maximum extension and then laid side by side which are held together by intermolecular hydrogen bonds. The structure resembles the pleated folds of drapery and therefore is known as β -pleated sheet.

(iii) *Tertiary structure of proteins:* The tertiary structure of proteins represents overall folding of the polypeptide chains i.e., further folding of the secondary structure. It gives rise to two major molecular shapes viz. fibrous and globular. The main forces which stabilise the 2° and 3° structures of proteins are hydrogen bonds, disulphide linkages, van der Waals and electrostatic forces of attraction.

(iv) *Quaternary structure of proteins:* Some of the proteins are composed of two or more polypeptide chains referred to as sub-units. The spatial arrangement of these subunits with respect to each other is known as quaternary structure.

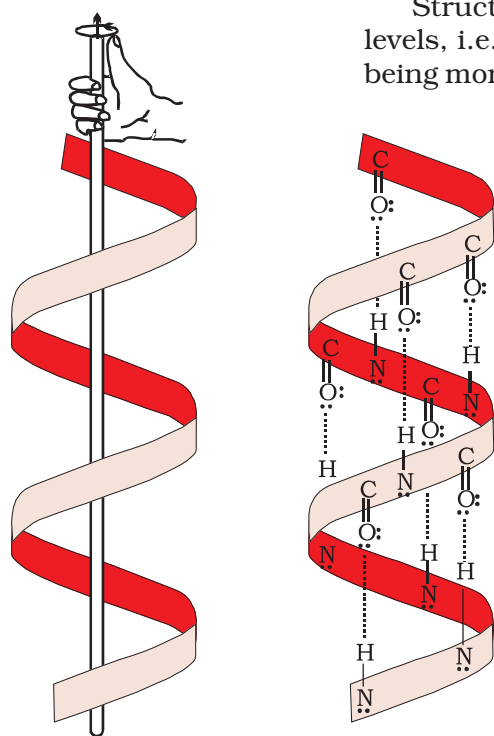


Fig. 10.1: α -Helix structure of proteins

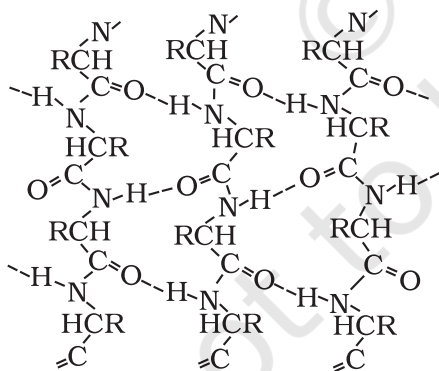


Fig. 10.2: β -Pleated sheet structure of proteins

A diagrammatic representation of all these four structures is given in Figure 10.3 where each coloured ball represents an amino acid.

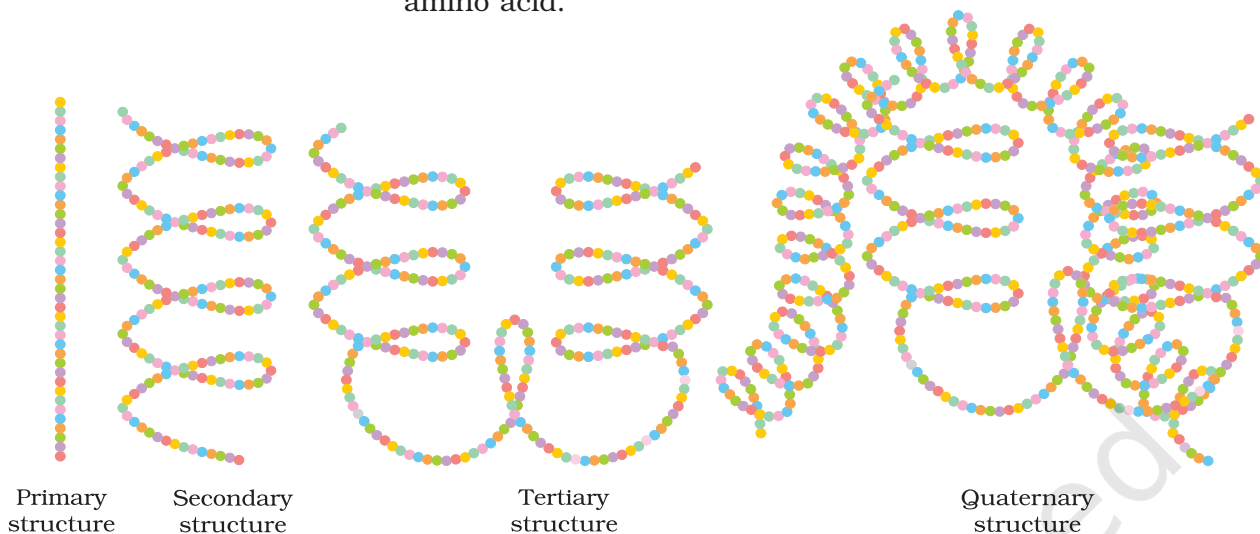


Fig. 10.3: Diagrammatic representation of protein structure (two sub-units of two types in quaternary structure)

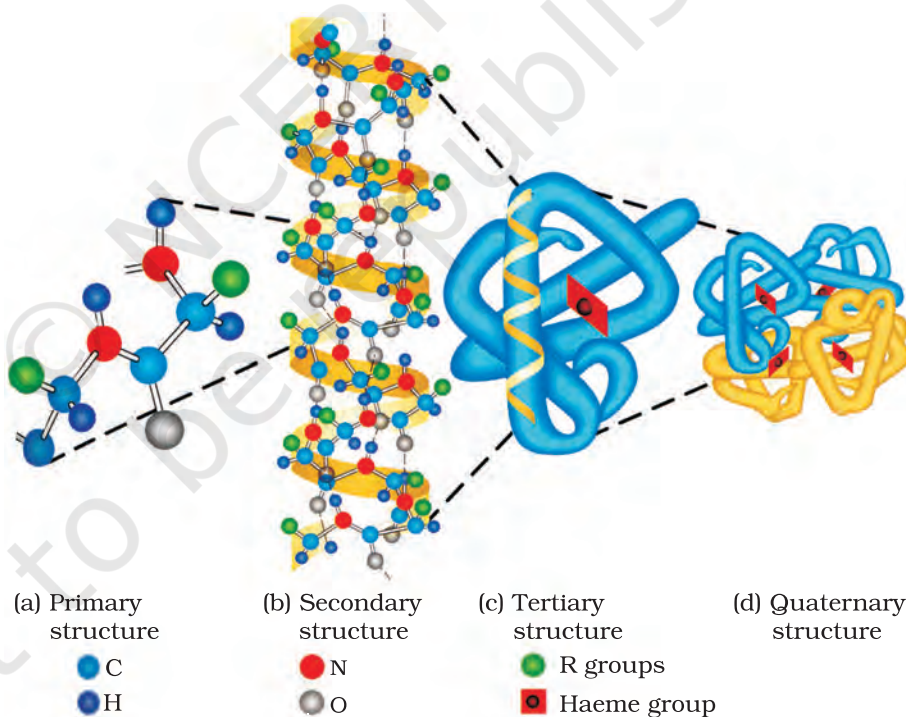


Fig. 10.4: Primary, secondary, tertiary and quaternary structures of haemoglobin

10.2.4 Denaturation of Proteins

Protein found in a biological system with a unique three-dimensional structure and biological activity is called a native protein. When a protein in its native form, is subjected to physical change like change in temperature or chemical change like change in pH, the hydrogen bonds are disturbed. Due to this, globules unfold and helix get uncoiled and protein loses its biological activity. This is called **denaturation** of

protein. During denaturation secondary and tertiary structures are destroyed but primary structure remains intact. The coagulation of egg white on boiling is a common example of denaturation. Another example is curdling of milk which is caused due to the formation of lactic acid by the bacteria present in milk.

Intext Questions

- 10.4** The melting points and solubility in water of amino acids are generally higher than that of the corresponding halo acids. Explain.
- 10.5** Where does the water present in the egg go after boiling the egg?

10.3 Enzymes

Life is possible due to the coordination of various chemical reactions in living organisms. An example is the digestion of food, absorption of appropriate molecules and ultimately production of energy. This process involves a sequence of reactions and all these reactions occur in the body under very mild conditions. This occurs with the help of certain biocatalysts called **enzymes**. Almost all the enzymes are globular proteins. Enzymes are very specific for a particular reaction and for a particular substrate. They are generally named after the compound or class of compounds upon which they work. For example, the enzyme that catalyses hydrolysis of maltose into glucose is named as *maltase*.



Sometimes enzymes are also named after the reaction, where they are used. For example, the enzymes which catalyse the oxidation of one substrate with simultaneous reduction of another substrate are named as **oxidoreductase** enzymes. The ending of the name of an enzyme is **-ase**.

10.3.1 Mechanism of Enzyme Action

Enzymes are needed only in small quantities for the progress of a reaction. Similar to the action of chemical catalysts, enzymes are said to reduce the magnitude of activation energy. For example, activation energy for acid hydrolysis of sucrose is 6.22 kJ mol^{-1} , while the activation energy is only 2.15 kJ mol^{-1} when hydrolysed by the enzyme, sucrase. Mechanism for the enzyme action has been discussed.

10.4 Vitamins

It has been observed that certain organic compounds are required in small amounts in our diet but their deficiency causes specific diseases. These compounds are called **vitamins**. Most of the vitamins cannot be synthesised in our body but plants can synthesise almost all of them, so they are considered as essential food factors. However, the bacteria of the gut can produce some of the vitamins required by us. All the vitamins are generally available in our diet. Different vitamins belong to various chemical classes and it is difficult to define them on the basis of structure. They are generally regarded as **organic compounds required in the diet in small amounts to perform specific biological functions for normal maintenance of optimum growth**

and health of the organism. Vitamins are designated by alphabets A, B, C, D, etc. Some of them are further named as sub-groups e.g. B₁, B₂, B₆, B₁₂, etc. Excess of vitamins is also harmful and vitamin pills should not be taken without the advice of doctor.

The term “**Vitamine**” was coined from the word vital + amine since the earlier identified compounds had amino groups. Later work showed that most of them did not contain amino groups, so the letter ‘e’ was dropped and the term **vitamin** is used these days.

10.4.1 Classification of Vitamins

Vitamins are classified into two groups depending upon their solubility in water or fat.

- (i) *Fat soluble vitamins:* Vitamins which are soluble in fat and oils but insoluble in water are kept in this group. These are vitamins A, D, E and K. They are stored in liver and adipose (fat storing) tissues.
- (ii) *Water soluble vitamins:* B group vitamins and vitamin C are soluble in water so they are grouped together. Water soluble vitamins must be supplied regularly in diet because they are readily excreted in urine and cannot be stored (except vitamin B₁₂) in our body.

Some important vitamins, their sources and diseases caused by their deficiency are listed in Table 10.3.

Table 10.3: Some important Vitamins, their Sources and their Deficiency Diseases

Sl. No.	Name of Vitamins	Sources	Deficiency diseases
1.	Vitamin A	Fish liver oil, carrots, butter and milk	Xerophthalmia (hardening of cornea of eye) Night blindness
2.	Vitamin B ₁ (Thiamine)	Yeast, milk, green vegetables and cereals	Beri beri (loss of appetite, retarded growth)
3.	Vitamin B ₂ (Riboflavin)	Milk, eggwhite, liver, kidney	Cheilosis (fissuring at corners of mouth and lips), digestive disorders and burning sensation of the skin.
4.	Vitamin B ₆ (Pyridoxine)	Yeast, milk, egg yolk, cereals and grams	Convulsions
5.	Vitamin B ₁₂	Meat, fish, egg and curd	Pernicious anaemia (RBC deficient in haemoglobin)
6.	Vitamin C (Ascorbic acid)	Citrus fruits, amla and green leafy vegetables	Scurvy (bleeding gums)
7.	Vitamin D	Exposure to sunlight, fish and egg yolk	Rickets (bone deformities in children) and osteomalacia (soft bones and joint pain in adults)

8. Vitamin E	Vegetable oils like wheat germ oil, sunflower oil, etc.	Increased fragility of RBCs and muscular weakness
9. Vitamin K	Green leafy vegetables	Increased blood clotting time

10.5 Nucleic Acids

Every generation of each and every species resembles its ancestors in many ways. How are these characteristics transmitted from one generation to the next? It has been observed that nucleus of a living cell is responsible for this transmission of inherent characters, also called **heredity**. The particles in nucleus of the cell, responsible for heredity, are called chromosomes which are made up of proteins and another type of biomolecules called **nucleic acids**. These are mainly of two types, the **deoxyribonucleic acid (DNA)** and **ribonucleic acid (RNA)**. Since nucleic acids are long chain polymers of **nucleotides**, so they are also called polynucleotides.

James Dewey Watson

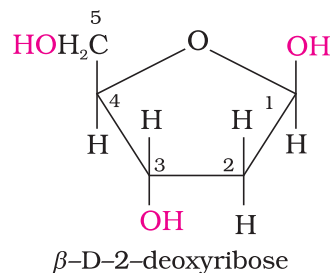
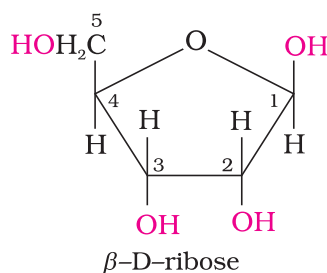


Born in Chicago, Illinois, in 1928, Dr Watson received his Ph.D. (1950) from Indiana University in Zoology. He is best known for his discovery of the structure of DNA for which he shared with Francis Crick and Maurice Wilkins the 1962 Nobel prize in Physiology and Medicine. They proposed that DNA molecule takes the shape of a double helix, an elegantly simple structure that resembles a gently twisted ladder. The rails of the ladder are made of alternating units of phosphate and the sugar deoxyribose;

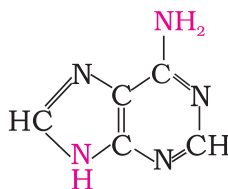
the rungs are each composed of a pair of purine/ pyrimidine bases. This research laid the foundation for the emerging field of **molecular biology**. The complementary pairing of nucleotide bases explains how identical copies of parental DNA pass on to two daughter cells. This research launched a revolution in biology that led to modern recombinant DNA techniques.

10.5.1 Chemical Composition of Nucleic Acids

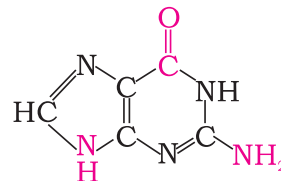
Complete hydrolysis of DNA (or RNA) yields a pentose sugar, phosphoric acid and nitrogen containing heterocyclic compounds (called bases). In DNA molecules, the sugar moiety is β -D-2-deoxyribose whereas in RNA molecule, it is β -D-ribose.



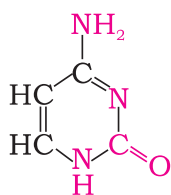
DNA contains four bases viz. adenine (A), guanine (G), cytosine (C) and thymine (T). RNA also contains four bases, the first three bases are same as in DNA but the fourth one is uracil (U).



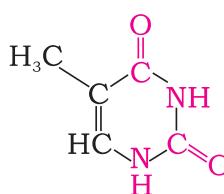
Adenine (A)



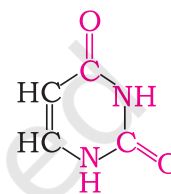
Guanine (G)



Cytosine (C)



Thymine (T)



Uracil (U)

10.5.2 Structure of Nucleic Acids

A unit formed by the attachment of a base to 1' position of sugar is known as **nucleoside**. In nucleosides, the sugar carbons are numbered as 1', 2', 3', etc. in order to distinguish these from the bases (Fig. 10.5a). When nucleoside is linked to phosphoric acid at 5'-position of sugar moiety, we get a nucleotide (Fig. 10.5).

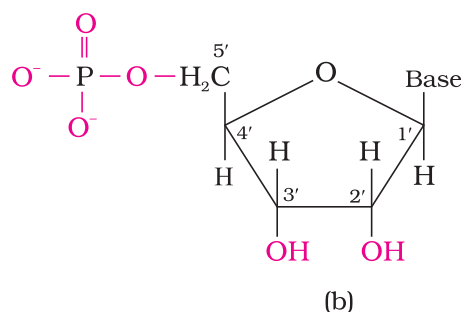
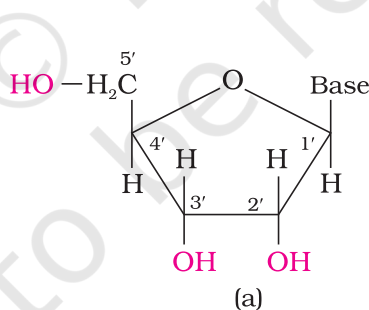


Fig. 10.5: Structure of (a) a nucleoside and (b) a nucleotide

Nucleotides are joined together by phosphodiester linkage between 5' and 3' carbon atoms of the pentose sugar. The formation of a typical dinucleotide is shown in Fig. 10.6.

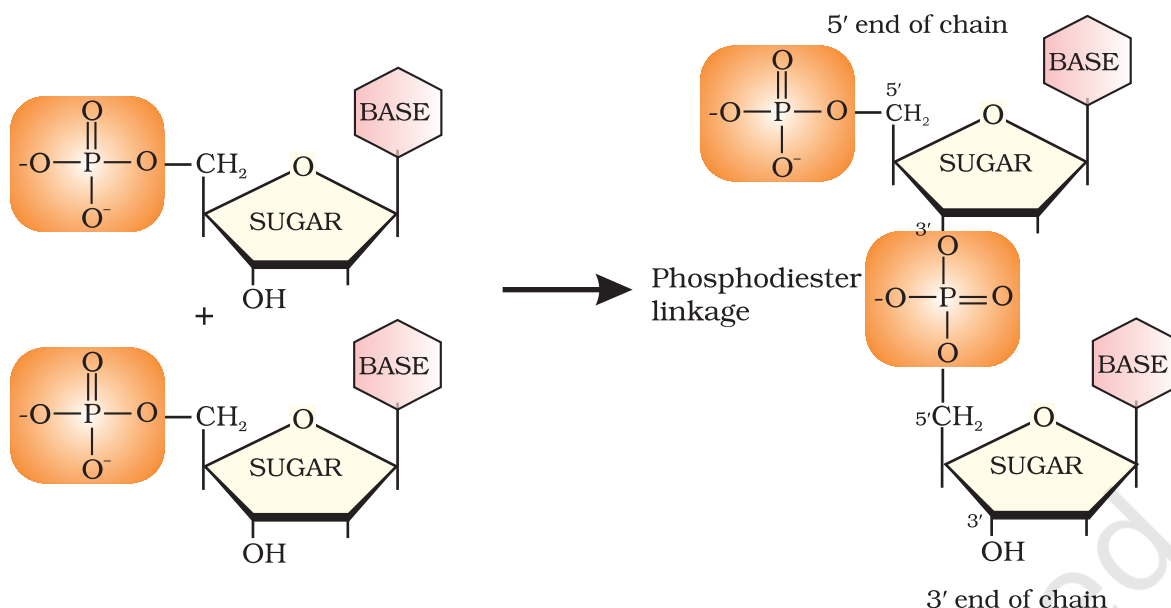
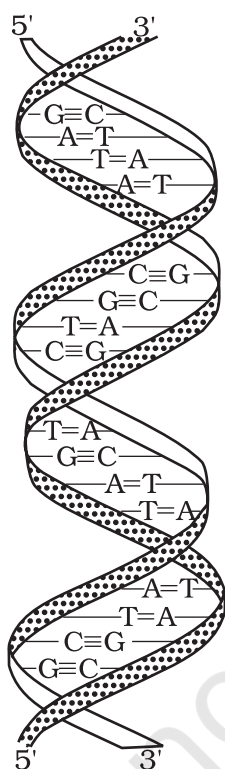
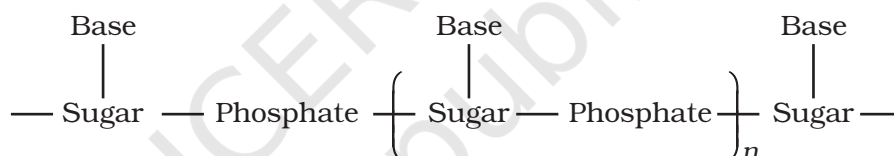


Fig. 10.6: Formation of a dinucleotide



A simplified version of nucleic acid chain is as shown below.



Information regarding the sequence of nucleotides in the chain of a nucleic acid is called its primary structure. Nucleic acids have a secondary structure also. James Watson and Francis Crick gave a double strand helix structure for DNA (Fig. 10.7). Two nucleic acid chains are wound about each other and held together by hydrogen bonds between pairs of bases. The two strands are complementary to each other because the hydrogen bonds are formed between specific pairs of bases. Adenine forms hydrogen bonds with thymine whereas cytosine forms hydrogen bonds with guanine.

In secondary structure of RNA single stranded helix is present which sometimes folds back on itself. RNA molecules are of three types and they perform different functions. They are named as **messenger RNA (m-RNA)**, **ribosomal RNA (r-RNA)** and **transfer RNA (t-RNA)**.

Fig. 10.7: Double strand helix structure for DNA



Har Gobind Khorana

Har Gobind Khorana, was born in 1922. He obtained his M.Sc. degree from Punjab University in Lahore. He worked with Professor Vladimir Prelog, who moulded Khorana's thought and philosophy towards science, work and effort. After a brief stay in India in 1949, Khorana went back to England and worked with Professor G.W. Kenner and Professor A.R.Todd. It was at Cambridge, U.K. that he got interested in both proteins and nucleic acids. Dr Khorana shared the Nobel Prize for Medicine and Physiology in 1968 with Marshall Nirenberg and Robert Holley for cracking the genetic code.

DNA Fingerprinting

It is known that every individual has unique fingerprints. These occur at the tips of the fingers and have been used for identification for a long time but these can be altered by surgery. A sequence of bases on DNA is also unique for a person and information regarding this is called DNA fingerprinting. It is same for every cell and cannot be altered by any known treatment. DNA fingerprinting is now used

- (i) in forensic laboratories for identification of criminals.
- (ii) to determine paternity of an individual.
- (iii) to identify the dead bodies in any accident by comparing the DNA's of parents or children.
- (iv) to identify racial groups to rewrite biological evolution.

10.5.3 Biological Functions of Nucleic Acids

DNA is the chemical basis of heredity and may be regarded as the reserve of genetic information. DNA is exclusively responsible for maintaining the identity of different species of organisms over millions of years. A DNA molecule is capable of self duplication during cell division and identical DNA strands are transferred to daughter cells. Another important function of nucleic acids is the protein synthesis in the cell. Actually, the proteins are synthesised by various RNA molecules in the cell but the message for the synthesis of a particular protein is present in DNA.

10.6 Hormones

Hormones are molecules that act as intercellular messengers. These are produced by endocrine glands in the body and are poured directly in the blood stream which transports them to the site of action.

In terms of chemical nature, some of these are steroids, e.g., estrogens and androgens; some are poly peptides for example insulin and endorphins and some others are amino acid derivatives such as epinephrine and norepinephrine.

Hormones have several functions in the body. They help to maintain the balance of biological activities in the body. The role of insulin in keeping the blood glucose level within the narrow limit is an example of this function. Insulin is released in response to the rapid rise in blood glucose level. On the other hand hormone glucagon tends to increase the glucose level in the blood. The two hormones together regulate the glucose level in the blood. Epinephrine and norepinephrine mediate responses to external stimuli. Growth hormones and sex hormones play role in growth and development. Thyroxine produced in the thyroid gland is an iodinated derivative of amino acid tyrosine. Abnormally low level of thyroxine leads

to hypothyroidism which is characterised by letharginess and obesity. Increased level of thyroxine causes hyperthyroidism. Low level of iodine in the diet may lead to hypothyroidism and enlargement of the thyroid gland. This condition is largely being controlled by adding sodium iodide to commercial table salt ("Iodised" salt).

Steroid hormones are produced by adrenal cortex and gonads (testes in males and ovaries in females). Hormones released by the adrenal cortex play very important role in the functions of the body. For example, glucocorticoids control the carbohydrate metabolism, modulate inflammatory reactions, and are involved in reactions to stress. The mineralocorticoids control the level of excretion of water and salt by the kidney. If adrenal cortex does not function properly then one of the results may be Addison's disease characterised by hypoglycemia, weakness and increased susceptibility to stress. The disease is fatal unless it is treated by glucocorticoids and mineralocorticoids. Hormones released by gonads are responsible for development of secondary sex characters. Testosterone is the major sex hormone produced in males. It is responsible for development of secondary male characteristics (deep voice, facial hair, general physical constitution) and estradiol is the main female sex hormone. It is responsible for development of secondary female characteristics and participates in the control of menstrual cycle. Progesterone is responsible for preparing the uterus for implantation of fertilised egg.

Intext Questions

- 10.6 Why cannot vitamin C be stored in our body?
- 10.7 What products would be formed when a nucleotide from DNA containing thymine is hydrolysed?
- 10.8 When RNA is hydrolysed, there is no relationship among the quantities of different bases obtained. What does this fact suggest about the structure of RNA?

Summary

Carbohydrates are optically active polyhydroxy aldehydes or ketones or molecules which provide such units on hydrolysis. They are broadly classified into three groups — **monosaccharides**, **disaccharides** and **polysaccharides**. Glucose, the most important source of energy for mammals, is obtained by the digestion of starch. Monosaccharides are held together by glycosidic linkages to form disaccharides or polysaccharides.

Proteins are the **polymers** of about twenty different **α -amino acids** which are linked by peptide bonds. Ten amino acids are called essential amino acids because they cannot be synthesised by our body, hence must be provided through diet. Proteins perform various structural and dynamic functions in the organisms. Proteins which contain only α -amino acids are called simple proteins. The **secondary** or **tertiary structure of proteins** get disturbed on change of pH or temperature and they are not able to perform their functions. This is called **denaturation of proteins**. Enzymes are **biocatalysts** which speed up the reactions in biosystems. They are very specific and selective in their action and chemically majority of **enzymes** are proteins.

Vitamins are accessory food factors required in the diet. They are classified as fat soluble (A, D, E and K) and water soluble (B group and C). Deficiency of vitamins leads to many diseases.

Nucleic acids are the polymers of nucleotides which in turn consist of a base, a pentose sugar and phosphate moiety. Nucleic acids are responsible for the transfer of characters from parents to offsprings. There are two types of nucleic acids — **DNA** and **RNA**. DNA contains a five carbon sugar molecule called **2-deoxyribose** whereas RNA contains ribose. Both DNA and RNA contain adenine, guanine and cytosine. The fourth base is thymine in DNA and uracil in RNA. The structure of DNA is a double strand whereas RNA is a single strand molecule. DNA is the chemical basis of heredity and have the coded message for proteins to be synthesised in the cell. There are three types of RNA — mRNA, rRNA and tRNA which actually carry out the protein synthesis in the cell.

Exercises

- 10.1 What are monosaccharides?
- 10.2 What are reducing sugars?
- 10.3 Write two main functions of carbohydrates in plants.
- 10.4 Classify the following into monosaccharides and disaccharides.
Ribose, 2-deoxyribose, maltose, galactose, fructose and lactose.
- 10.5 What do you understand by the term glycosidic linkage?
- 10.6 What is glycogen? How is it different from starch?
- 10.7 What are the hydrolysis products of
(i) sucrose and (ii) lactose?
- 10.8 What is the basic structural difference between starch and cellulose?
- 10.9 What happens when D-glucose is treated with the following reagents?
(i) HI (ii) Bromine water (iii) HNO_3
- 10.10 Enumerate the reactions of D-glucose which cannot be explained by its open chain structure.
- 10.11 What are essential and non-essential amino acids? Give two examples of each type.
- 10.12 Define the following as related to proteins
(i) Peptide linkage (ii) Primary structure (iii) Denaturation.
- 10.13 What are the common types of secondary structure of proteins?
- 10.14 What type of bonding helps in stabilising the α -helix structure of proteins?
- 10.15 Differentiate between globular and fibrous proteins.
- 10.16 How do you explain the amphoteric behaviour of amino acids?
- 10.17 What are enzymes?
- 10.18 What is the effect of denaturation on the structure of proteins?
- 10.19 How are vitamins classified? Name the vitamin responsible for the coagulation of blood.
- 10.20 Why are vitamin A and vitamin C essential to us? Give their important sources.
- 10.21 What are nucleic acids? Mention their two important functions.
- 10.22 What is the difference between a nucleoside and a nucleotide?
- 10.23 The two strands in DNA are not identical but are complementary. Explain.
- 10.24 Write the important structural and functional differences between DNA and RNA.
- 10.25 What are the different types of RNA found in the cell?